Mole concept

(T: Theory, P: Problem)

T-1 **Density: (Mass density)** $= \frac{\text{Mass}}{\text{volume}}$, S.I. unit kg/m^3, CGS unit: gm/ml, Temp: dependent

T-2. **Relative density :** $\frac{\text{Density of 1}^{st}\text{ substance}}{\text{Denisty of 2}^{nd}\text{ substance}}$

For solid/liquid: Relative density (Specific gravity) $= \frac{\text{Density of solid/Liquid (S/L)}}{\text{Density of water at 4°C}}$

$= \frac{\text{Density S/L (gm/ml)}}{1\text{gm/ml}}$, **unitless**

T-3. **For gas**: $d = \frac{PM}{RT}$, $(d \alpha M)$, $\left(d_{gas}\right)_{\substack{STP \\ Or \\ NTP}} = \frac{GMM_{gas}}{22.4} \times g/L$

Relative density $= \frac{\text{Density of gas}_1}{\text{Density of gas}_2}$ at same T,P.

T-4. Special case of relative density is vapor density (VD) $= \frac{d_{gas}}{d_{H_2}} = \frac{M_{gas}}{2}$

T-5. **Determination of percentage composition of given compound**

Mass percentage of an element $= \frac{\text{mass of an element}}{\text{mass of compound}} * 100$

T-6. **Concept of minimum molar mass:** compound should contain atleast one atom of that element whose percentage is given.

$$\%\text{ of an Element} = \frac{\text{Mass of one atom of an Element}}{\text{MM of compound}} \times 100 \ ;$$

T-7. **Empirical Formula (EF) & Molecular Formula (MF)**

EF: Chemical formula contains the simplest whole number ratio of various atoms present in a compound.

MF: Chemical formula containing the actual number of atoms of each element in a molecule

Relation b/w MF & BF:

$MF = (EF)_n$ & $MM_{(MF)} = n \times MM_{(EF)}$ (MM = Molar mass, n= natural number)

T-8. **Mole-mole analysis :**

$aA + bB \rightarrow cC + dD$ (**Balanced chemical equation**)

Reactant Product a, b, c, d – Stoichiometric coefficients

$$\frac{(n_A)\text{Reacted}}{a} = \frac{(n_B)\text{Reacted}}{b} = \frac{(n_C)\text{formed}}{c} = \frac{(n_D)\text{formed}}{d}$$

T-9. **Limiting reagent & Excess reagent:** (LR & ER)

L.R: Reactant which is present in the limit and consumed first in the reaction

E.R: Reactant which is present in the excess and left in the end of the reaction.

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Note: Concept of LR & ER arises when mole ratio of reactants & Stoichiometry coefficient ratio of reactants is not same.

Determination of LR:

(1) Balance the chemical reaction.
(2) Calculate mole of each reactant.
(3) Divide the given mole of reactant with their respective Stoichiometry coefficients.
(4) Reactant for which this ratio is minimum will be L.R.

T-10. Percentage Yield of Reaction:

$$\% \text{ Yield} = \frac{\text{Obtained mass / mole of product}}{\text{calculated mass / mole of product}} \times 100$$

T-11 Principle of Atom Conservation (P.O.A.C):

NOTE: (1) P.O.A.C. can apply only on that element. Which is 100% consumed. (or on limiting reagent)
(2) If Reaction 100% completed then POAC can apply on each element of reaction.
(3) While applying POAC no need to balance the chemical equation.

T-12. % Composition of Mixture

Mixture A & B (having same nature or non-reacting in nature)

T-13. Percentage purity of sample:

100 gm sample of NaOH having 80% purity
i.e. W_{NaOH} = 80 gm, $W_{impurity}$ = 20 gm

P-2. Specific gravity of H_2SO_4 is 1.84 i.e. density of H_2SO_4 is 1.84 gm/ml

P-3.1 Calculate Relative density of SO_2 gas w.r.t. CH_4 gas?

Sol. Relative density = $\frac{M_{SO_2}}{M_{CH_4}} = \frac{64}{16} = 4$

P-4 Calculate Relative density(V.D) of a sample containing 5 mole of O_2 & 5 mole of N_2 w.r.t. to He?

Sol. $M_{mix} = \frac{n_1M_1 + n_2M_2}{n_1 + n_2} = \frac{5\times 28 + 5\times 32}{10} = 30$ vapor Density = $\frac{M_{mix}}{M_{He}} = \frac{30}{4} = 7.5$

P-3.2 Relative density. of gas, if its density is 0.5 gm/Ltr at NTP , is?

Sol. $d = \frac{PM}{RT}$, at NTP $\left(P = 1 \text{ atm}, T = 273 \text{ k}, R = \frac{1}{12}\frac{\text{atm Ltr.}}{\text{mol .k.}}\right)$

$\frac{1}{2} = \frac{1\times M}{\frac{1}{12}\times 273} \Rightarrow M = 11.2$ gm, Vapor Density = $\frac{MM}{2} = \frac{11.2}{2} = 5.6$

P-5. Calculate percentage of H & C in C_2H_6?

Sol. Percentage of hydrogen = $\frac{w_H}{MM}\times 100 = \frac{6}{30}\times 100 = 20\%$,

Percentage of carbon = $\frac{w_c}{MM}\times 100 = \frac{24}{30}\times 100 = 80\%$

P-6. Insulin contain 0.0016 % of sulphur. Calculate minimum molar mass of insulin?

Sol. % of S = $\frac{\text{Mass of 1'S'}}{\text{MM of compound}}$; $0.0016 = \frac{32}{MM}\times 100$ $\therefore$ MM = 2×10^6 g

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P-7 Hydrocarbon contain 80% carbon. If molecular mass of hydrocarbon is 30 u. Then calculate Empirical formula and Molecular formula.

Sol.

Symbol of element	% of element (by wt)	No of mole = $\frac{(\%\text{ by wt})}{GAM}$	Simplest atomic ratio= $\frac{\text{no of mole}}{\text{Lowest value}}$	Simplest whole no. atomic ratio
'C'	80	80/2 = 20/3	$\frac{20/3}{20/3}=1$	1
'H'	20	20/1 = 20	$\frac{20}{20/3}=3$	3

$EF: CH_3$, $MM_{(EF)} = 12 + 3 = 15$, $n = \frac{MM_{(MF)}}{MM_{(EF)}} = \frac{30}{15} = 2$, $MF = n(EF) = C_2H_6$

Name Of Compound	Molecular Formula	Empirical formula	n
Hydrogen peroxide	H_2O_2	HO	2
Acetic acid	CH_3COOH	CH_2O	2

P-8 Calculate mass of Oxygen required to burn completely 54 gm of Aluminum & also calculate mole, mass of product formed? ($4Al + 3O_2 \longrightarrow 2Al_2O_3$)

Sol. $4Al + 3O_2 \longrightarrow 2Al_2O_3$, mole Al = 54/27 = 2

$$\frac{2}{4}=\frac{n(Al)}{4}=\frac{n(O_2)}{3}=\frac{n(Al_2O_3)}{2} \Rightarrow \frac{n(O_2)}{3}=\frac{1}{2} \Rightarrow n_{O_2}=\frac{3}{2} \Rightarrow w_{O_2}=\frac{3}{2}\times 32 = 48gm$$

$$\Rightarrow \quad \frac{n(Al_2O_3)}{2}=\frac{1}{2} \Rightarrow n(Al_2O_3)=1 \Rightarrow W_{Al_2O_3}=1\times 102 = 102gm$$

P-9. 5 Mole of SO_2 & 5 mole O_2 are reacted to form SO_3. Determine L.R., weight of E.R. left and amount of SO_3 formed?

Sol:

$$2SO_2 + O_2 \longrightarrow 2SO_3$$

t= 0 : 5 , 5

$$\frac{\text{Mole}}{\text{Sto.coeff.}} \quad \left(\frac{5}{2} < \frac{5}{1}\right)$$

LR ER

$$\frac{n_{SO_2}}{2}=\frac{n_{O_2}}{1}=\frac{n_{SO_3}}{2} \text{ (According to L.R.)}$$

$$\frac{5}{2}=\frac{(n_{O_2})_{Reacted}}{1}=\frac{(n_{SO_3})_{Formed}}{2} \Rightarrow (n_{O_2})_{Reacted} = 5/2,\ (n_{SO_3})_{Formed} = 5\text{ mol},\ W_{SO_3} = 5 \times 80 = 400\text{ gm}$$

No of mole O_2 (left) = 5 – 5/2 = 2.5, W_{O_2} (left) = $\frac{5}{2}\times 32 = 80gm$

P-10. 16.8 gm $MgCO_3$ on heating give what volume of CO_2 at STP, if Yield of reaction is 75% ?

Sol: $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

$\frac{16.8}{84} = 0.2$ Mole 0.2 mole (Calculated)

$$\%\text{ Yield} = \frac{(n_{CO2})_{obtained}}{(n_{CO2})_{Calculated}}\times 100$$

$$75 = \frac{(M_{CO_2})_{Obtained}}{0.2}\times 100 \Rightarrow n_{CO2} = \frac{3}{4}\times\frac{1}{5}=\frac{3}{20} \Rightarrow V_{CO2}\ (STP) = \frac{3}{20}\times 22.4 = 3.36Ltr.$$

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P-11. 3 gm of carbon reacted completed with 5 gm of oxygen. Calculated mole of CO & CO_2 formed?

Sol. $C_{(s)} + O_{2(g)} \longrightarrow CO_{(g)} + CO_{2(g)}$

$\downarrow \quad \downarrow$

Mole $\frac{3}{12} \quad \frac{5}{32} \quad x \quad y$ mole formed

POAC on 'C' $(n_C) \times 3 + (n_{O_2}) \times 0 = (n_{CO}) \times 1 + (n_{CO_2}) \times 1$

$\frac{3}{12} = x \times 1 + y + 2$ ..... (1)

POAC on 'O' $(n_C) \times 0 + (n_{O_2}) \times 2 = (n_{CO}) \times 1 + (n_{CO_2}) \times 2$

$\frac{5}{32} \times 2 = x \times 1 + y \times 2$ ..... (2) x = 3/16 Mole , y = 1/16 mole

P-12. 10 mole of mixture of HCl & H_2SO_4. required 16 mole of NaOH for complete neutralization. Calculate percentage of HCl in mixture?

Sol: Let mole of HCl = x, mole of H_2SO_4 = 10 – x

mole of H^+ = mole of OH^- **(for complete neutralization)**

$x \times 1 + (10 - x) \times 2 = 16 \Rightarrow x = 4$ % of HCl = $\frac{n_{HCl}}{n_T} \times 100 = \frac{4}{100} \times 100 = 40\%$

P-13 50 gm sample of $CaCO_3$ on heating given 17.6 gm CO_2. Calculate % purity of sample?

Sol. $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

n_i $\frac{17.6}{44} = 0.4$ mole $n_{CaCO_3=0.4}$, $(W_{CaCO_3})_{Reacted} = 0.4 \times 100 = 40$ gm

Percentage purity = $\frac{w_{reacted}}{w_{given}} \times 100 = \frac{40}{50} \times 100 = 80\%$

T-14. **Oxidation number / state (ON/OS):** It is an imaginary or apparent charge developed over atom of an element when it goes from its elemental free state to combined state in molecules.

Note: O.S. may be +ve, –ve, zero or fractional in a compound.

Rules governing oxidation number

(1) **In elemental from**: Oxidation number of each atom in I_2, Na , Al, O_2, O_3 F_2, N_2, C_{60}, S_8, P_4 is Zero

(2) **In combine state:**

(1) O.N. of F is always **-1**

(2) O.N. of O is always **-2** (except: In peroxide in H_2O_2, Na_2O_2 is "**–1**")

In superoxide KO_2 is "$\frac{-1}{2}$" , In ozonide KO_3 is "$\frac{-1}{3}$" , in O_2F_2 is "**+1**" , in OF_2 is "**+2**"

(3) O.N. of Al is **+3** always

(4) ON of hydrogen: With metal "**–1**" e.g. CaH_2, NaH

: With Non metal "**+1**" e.g. NH_3, HNO_3, H_2SO_4

(5) O.N. of alkali metal = "**+1**", O.N. of alkali earth metal = "**+2**" always

(6) Maximum O.S. of any element is "+8" e.g. Xe in XeO_4

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(7) Max O.S. of an element = No of electron in valence shell (for p-block)
(i) For boron family = "+3"
(ii) For carbon family = "+4"
(iii) For Nitrogen family = "+5"
(iv) For oxygen family = "+6" (except oxygen = +2)
(v) For Halogen family = "+7" (except F = 0)
(8) Minimum O.S. of any element = no of e^- in outermost shell –8 **(except metal)**
(9) Summation of O.N. of all atoms = 0. (In case of compounds)
= Net changes on ion (In case of ions)

T-15. **Oxidation**: The process in which (i) Electron is lost (ii) O.S. increases

T-17. **Reduction** : The process in which (i) Electron is gained (ii) O.S. decreases

T-18. **Oxidizing agent (O.A.)** : Self undergoes reduction in redox reaction and oxidizes others
e.g. $O_2, O_3, F_2, K_2Cr_2O_7$ $KMnO_4, HNO_3$, conc.H_2SO_4

T-19. **Reducing agent (R.A.) :** Self undergoes oxidation in redox reaction and reduces others
eg: metals, H_2, metal hydride, $H_2C_2O_4$, H_2S, KI , $Na_2S_2O_3$

T-20. **Both OA & RA**: An element should contain present in intermediate oxidation state i.e. lie in between maximum and minimum oxidation state
ex: $\overset{-1}{H_2}\overset{}{O_2}$, $\overset{+4}{S}O_2$, $\overset{+4}{Mn}O_2$

T-21 Type of redox Reaction:

(a) **Disproportionation Reaction**: if Same element from a single OS in same molecule undergoes oxidation as well as reduction.

$X_2 + OH^-$ (Cold & dil.) $\longrightarrow X^- + \overset{+1}{X}O^-$

$X_2 + OH^-$ (Hot & conc.) $\longrightarrow X^- + \overset{+5}{X}O_3^-$ (X = Cl, Br, I)

Note: F_2 does not undergo disproportionation as it cannot show +ve O.S.

(b). **Comproportionation reaction**: Reverse of disproportionation is called Comproportionation reaction
$Br^- + BrO_3^- \longrightarrow Br_2$

P-14. Calculates O.S. of the following underline element?
1. $K_2\underline{Cr}_2O_7$: $\Rightarrow 2 + 2x - 14 = 0, \Rightarrow x = +6$
2. $H_2\underline{S}O_4$: $\Rightarrow 2 + x - 8 = 0, \Rightarrow x = +6$
3. $CaO\underline{Cl}_2 \Rightarrow Ca^{2+} + OCl_2^- \longrightarrow Ca^{2+} + Cl^- + OCl^-$ (O.S. of one Cl is +1, of other Cl is –1)
4. $\underline{Fe}_3O_4$: $\Rightarrow$ $3 \times x - 8 = 0 \Rightarrow x = 8/3$ (average O.S.), $\overset{+2}{Fe}O + \overset{+3}{Fe}_2O_3$
5. $\underline{Cr}O_5$: $\Rightarrow$ $x - 10 = 0 \Rightarrow x = 10$ (Not possible since greater than maximum) O.S. of Cr = +6

T-22 Various Concentration Terms

Solution: Solution is a homogenous mixture of two or more substances in which individual components will lose their identity. Solvent + solute = Solution.
Solvent: generally in large amount, **Solute**: generally in small amount.
e.g.: aqueous solution of salt. i.e. solute is salt & Solvent is water.
dilute solution (dil): Solute → very less or solvent → very large
concentration solution (conc): solute → very large (Relatively) or solvent→ very less
concentration of solution : it is always defined w.r.t. solute.

$$\text{Concentration} = \frac{\text{mass / mole / volume of solute}}{\text{mass / volume of solvent / solution}}$$

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Percentage calculation:

T-23. % weight by volume (% w/v): wt(in g) of solute present in 100 ml of solution.

$$\%w/v = \frac{\text{mass of solate(gm)}}{\text{volume of solution(ml)}} \times 100$$ **Temp dependent.**

T-24. % weight by weight (%w/w): wt(in g) of solute present in 100 gm of solution

$$\%\ w/w = \frac{\text{mass of solate}}{\text{mass of solution}} \times 100$$ ($w_{solute} + w_{solvent} = w_{solution}$) **Temp independent.**

T-25. Molarity (M): Number of moles of a solute present in 1 L of the solution.

.

$$M = \frac{\text{No of mole of solute}}{\text{volume of solution(Ltr)}} = \frac{w_{solute} \times 1000}{M_{solute} \times v_{solution}\,(ml)}$$, Deci molar = $\frac{1}{10}$ M , mili molar = $\frac{1}{1000}$ M

unit :mole/Ltr or **molar** or **M**. & **Temperature Dependent**

T-27. molality (m): Number of moles of solute dissolved in1000 g of a solvent

$$m = \frac{\text{No of mole of solute}}{\text{mass of solvent(kg)}} = \frac{n_{solute}}{w_{solvent}\,(kg)}$$

unit :mole/kg or molal or m. & **Temperature Independent**

T-28. Mole fraction (X) : ratio of no of mole of substance to the total number of mole.

$$X_{sub} = \frac{\text{mole of substance}}{\text{Total mole}}$$, **unit less**, **Temperature independent.**

Let a solution : moles of solute = A & moles of solvent = B

$$X_{(solute)} = \frac{A}{A+B}, \quad X_{(solvent)} = \frac{B}{A+B}, \quad X(A) + X(B) = 1$$

(1) $\sum X_i = 1$ (2) $0 < X < 1$,

T-30. Concentration in ppm:

No of ppm: $\frac{\text{mass of solute(in gm)}}{\text{mass of solution(in gm)}} \times 10^6$ **Unit less**

T-31 . Some special formula:

(a) % w/v = % w/w × d_s $d_s \rightarrow$density of solution in **gm/ml**

(b) $M = \frac{\%w/v \times 10}{MM_{Solute}} = \frac{\%w/w \times d_s \times 10}{MM_{Solute}}$

(c) $m = \frac{x_{Solute}}{1 - X_{Solute}} \times \frac{1000}{MM_{solvent}}$

(d) $m = \frac{1000}{1000d_s - M(MM_{Solute})}$ $m \rightarrow$ molality, $M \rightarrow$ Molarity, $d_s \rightarrow$ solution density (gm/ml)

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T-32. ***Dilution formula:*** Process of addition of water in any solution is known as dilution.

On dilution (i) no of mole of solute constant (ii) Volume of solution increase

(iii) Molarity of Solution decrease.

Let a solution: M_1, V_1 add water V_2. then final molarity(M_2) of solution

$M_1V_1 = M_2 (v_1 + v_2)$ **NOTE**: at very large dilution molarity of solution equal to molality.

P-23. 100 ml of 30% w/v aqueous NaOH solution, having Density 1.5 gm/ml. then mass of solute, Solvent & Solution will be ?

Sol. 30 gm solute present in 100 ml of solution.

Mass of solute (NaOH) = 30 gm

Mass of solution = $v_{solution} \times d_{solution} = 1.5 \times 100 = 150$ gm

Mass of solvent (H_2O) = $w_{solution} - w_{solute} = 150 - 30 = 120$ gm.

P-24 50g of 40% w/w NaOH (aq) solution. Then mass of solute, Solvent & Solution will be ?

Sol. 20 gm NaOH present in 50 gm of solution.

Mass of solute (NaOH) = 20 gm, Mass of solvent (H_2O) = 30 gm, Mass of solution = 50 gm

P-26. 90 gm glucose dissolved in sufficient amount of water to make 500 ml of solution. Calculate molarity?

Sol. No of mole of glucose = 90/180 = 1/2

Volume of solution = $\frac{500}{1000} = \frac{1}{2}$ Ltr $\qquad$ $M = \frac{n}{V(Ltr)} = \frac{1/2}{1/2} = 1$

P-27. 360 gm of glucose added in sufficient water to produce 1.36 kg of solution calculate molality of solution?

Sol. mole of glucose = 360/180 =2

Mass of solvent = $w_{solution} - w_{solute} = 1.36 - 0.36 = 1$ kg $\Rightarrow m = \frac{n}{w(kg)} = 2/1 = 2$ m

P-29. 30 gm urea is mixed with 45 gm of water determine mole fraction of urea and water?

Sol. No of mole of solute (urea) = $\frac{30}{60} = \frac{1}{2} = 0.5$

no of mole of solvent (water) = $\frac{45}{18} = 2.5$, Total mole = 2.5 + 0.5 = 3.0

X(area) = $\frac{0.5}{3.0} = \frac{1}{6}$, $\qquad$ X(water) = $\frac{2.5}{3.0} = \frac{5}{6} = 1 - X(urea) = 1 - 1/6 = 5/6$

P-30. 10 gm of $CaCl_2$ is present in 100 kg water. calculate ppm?

Sol. no of ppm = $\frac{mass\,of\,solute(gm)}{mass\,of\,solution(in\,gm)} = \frac{10}{100 \times 1000} \times 10^6 = 100$ ppm.

P-31.1 Mole fraction of I_2 in benzene 0.2. Calculate molality of solution?

Sol. X_{solute} (I_2) = 0.2, $\qquad$ solvent is benzene (MM = 78)

$m = \frac{x_s}{1 - x_s} \times \frac{1000}{MM_{solvent}} = \frac{0.2}{0.8} \times \frac{1000}{78} = \frac{250}{78} = 3.29$

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EQUIVALENT CONCEPT

T-1. Equivalent mass of element : Number of parts by mass of an element which reacts or displaces from a compound 1.008 parts by mass of hydrogen, 8 parts by mass of oxygen and 35.5 parts by mass of chlorine, is known as the equivalent weight of that element.

Equivalent weight (E) :

In general, Eq. wt. (E) = $\frac{\text{Atomic weight or Molecular weight}}{\text{valency factor(v.f)}} = \frac{\text{Mol.wt.}}{\text{n-factor}} = \frac{M}{x}$

Number of Equivalents = $\frac{\text{mass of species}}{\text{eq. wt. of that species}}$

For a solution, Number of equivalents = N_1V_1, where N is the normality and V is the volume in litres

- Equivalent mass is a pure number which, when expressed in gram, is called gram equivalent mass.
- The equivalent mass of a substance may have different values under different conditions.
- There is no hard and fast rule that equivalent weight will be always less than the molecular mass.

T-2. Valency factor calculation :

- **For Elements :** Valency factor = valency of the element.
- **For Acids :** Valency factor = number of replaceable H^+ ions per acid molecule.
- **For Bases :**

  Valency factor = number of replacable OH^- ions per base molecule.
- **Acid-base reaction :**

  In case of acid base reaction, the valence factor is the actual number of H^+ or OH^- replaced in the reaction. The acid or base may contain more number of replaceble H^+ or OH^- than actually replaced in reaction.
  - **v. f.** for base is the number of H^+ ion from the acid replaced by each molecule of the base
  - **v. f.** for acid is the number of OH^- replaced from the base by each molecule of acid.
- **Salts :**

(a) In non-reacting condition

- **Valency factor =** Total number of positive charge or negative charge present in the compound.

Note : In case of hydrated salt, positive/negative charge of water molecule is not counted.

(b) In reacting condition

(c) Equivalent weight of oxidising / reducing agents in a redox reaction

In case of redox change , **v.f. =** Total change in oxidation number per molecule .

T-3. Normality :

Normality of a solution is defined as the number of equivalents of solute present in one litre (1000 mL) solution.

Let V mL of a solution is prepared by dissolving W g of solute of equivalent weight E in water.

- Number of equivalents of solute = $\frac{W}{E}$

  VmL of solution contain $\frac{W}{E}$ equivalents of solute

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$\therefore$ 1000 mL solution will contain $\frac{W \times 1000}{E \times V}$ equivalents of solute.

- **Normality (N) =** $\frac{W \times 1000}{E \times V}$
- **Relations between Normality and Molarity :**

  Normality (N) = Molarity x Valency factor

  or $N \times V$ (in mL) $= M \times V$ (in mL) $\times n$

  or **milliequivalents = millimoles × n**

T-4. Law of Equivalence :

The law states that one equivalent of an element combine with one equivalent of the other. In a chemical reaction, equivalents and milli equivalents of reactants react in equal amount to give same number of equivalents or milli equivalents of products separately.

Accordingly

(i) $aA + bB \rightarrow mM + nN$; meq of A = meq of B = meq of M = m.eq. of N

(ii) In a compound M_xN_y ; meq of M_xN_y = meq of M = meq of N

T-5.

- **Normality of any solution depends on reaction while molarity does not.**
- **The concept of equivalents is handy, but it should be used with care.** One must

T-6. Titrations :

Titration is a procedure for determining the concentration of a solution by allowing its carefully measured volume to react with a standard solution of another substance, whose concentration is known.

Standard solution - It is a solution whose concentration is known and is taken in burette. It is also called **Titrant.**

Titrate : Solution consisting of substance to be estimated, its generally taken in a beaker .

Equivalence point : It is the point when number of equivalents of titrant added becomes equal to number of equivalents of titrate.

At equivalence point :

$$n_1V_1M_1 = n_2V_2M_2$$

Indicator : An auxiliary substance added for physical detection of completion of titration at equivalence point. It generally show colour change on completion of titration.

T-7. Calculation of Available Chlorine from a sample of Bleaching Powder :

The weight of available Cl_2 released from the given sample of bleaching powder on reaction with dilute acids or CO_2 is called available chlorine.

where W = amount of belaching powder taken in g.

or **% of available Cl_2 =** $\frac{\mathbf{3.55 \times M \times V}}{\mathbf{W}}$

T-8. Hydrogen peroxide (H_2O_2)

H_2O_2 can behave both like oxidising and reducing agent in both the mediums (acidic and basic).

- **Oxidising agent : ($H_2O_2 \rightarrow H_2O$)**

  (a) Acidic medium : $2e^- + 2H^+ + H_2O_2 \longrightarrow 2H_2O$

  v.f. = 2

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(b) Basic medium : $2e^- + H_2O_2 \longrightarrow 2OH^-$

v.f = 2

- **Reducing agent : ($H_2O_2 \rightarrow O_2$)**

(a) Acidic medium : $H_2O_2 \longrightarrow O_2 + 2H^+ + 2e^-$

v.f = 2

(b) Basic medium : $2OH^- + H_2O_2 \longrightarrow O_2 + 2H_2O + 2e^-$

v.f = 2

Volume strength of H_2O_2 : Strength of H_2O_2 is represented as 10V , 20 V , 30 V etc.

20V H_2O_2 means **one litre** of this sample of H_2O_2 on decomposition gives **20L of O_2** gas at **STP.**

Decomposition of H_2O_2 is given as :

$$H_2O_2 \longrightarrow H_2O + \frac{1}{2} O_2$$

1 mole $\quad \frac{1}{2} \times 22.4$ L O_2 at STP

= 34g $\quad$ = 11.2 L O_2 at STP

- **Molarity of H_2O_2 (M) = $\frac{\text{Volume strength of } H_2O_2}{11.2}$**

T-9. Hardness of water (Hard water does not give lather with soap)

Temporary hardness - due to bicarbonates of Ca & Mg

Permanent hardness - due to chlorides & sulphates of Ca & Mg. There are some method by which we can soften the water sample.

(a) By boiling : $2HCO_3^- \longrightarrow H_2O + CO_2 + CO_3^{2-}$ **or**

By Slaked lime : $Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow CaCO_3 + 2H_2O$

$Ca^{2+} + CO_3^{2-} \longrightarrow CaCO_3$

(b) By Washing Soda : $CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

(c) By ion exchange resins : $Na_2R + Ca^{2+} \longrightarrow CaR + 2Na^+$

(d) By adding chelating agents like $(PO_3^-)_3$ etc.

Measurement of Hardness :

Hardness is measured in terms of ppm (parts per million) of $CaCO_3$ or equivalent to it.

$$\text{Hardness in ppm} = \frac{\text{mass of } CaCO_3}{\text{Total mass of solution}} \times 10^6$$

T-10. Strength of Oleum :

Oleum is SO_3 dissolved in 100% H_2SO_4. Sometimes, oleum is reported as more than 100% by weight, say y% (where y > 100). This means that (y – 100) grams of water, when added to 100 g of given oleum sample, will combine with all the free SO_3 in the oleum to give 100% sulphuric acid.

Hence, weight % of free SO_3 in oleum = 80(y –100)/18

* * * * * *

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Periodic Table & Gaseous State

1. Shielding Effect and Effective nuclear charge (Z_{eff})

Definition: The effect in which **inner electrons shield valence electron** from the **attraction due to nucleus** is called shielding effect or screening effect.
Screening effect decreases as **s-orbital > p-orbital > d-orbital > f-orbital**

Definition: Effective nuclear charge (Z_{eff}) is the **net positive charge** due to which **attraction is experienced** by **valence electron**. It can be approximated by the equation :

$$Z_{eff} = Z - \sigma$$

Z_{eff} = Effective nuclear charge
Z = Atomic number
σ = **Screening constant** or **shielding constant**

Variation of Z_{eff} :

Along the period (Left to Right)	**Increases**	**Reason : n is constant but proton number increases in nucleus**
Down the group	**Almost constant**	**Reason : n increased and protons also increased**

d-orbital & **f-orbital** electrons **shield** nuclear charge **poorly** leading to exceptions in periodic properties.

2. Atomic radius & Ionic radius

Definition: Atomic radius is distance between **outermost electron** and **nucleus**.

Type of atomic radius depends on the **type of chemical bond between atoms**. These are:
1. Covalent radius 2. Metallic radius 3. van der Waal's radius

A. Covalent radius :
Definition: Covalent radius is **one-half of the internuclear distance** between **two singly bonded nuclei** of like atoms (or bond length).

B. Metallic radius :
Definition: It is **one-half of the internuclear distance** between **two adjacent metallic atoms in crystalline lattice structure**.

C. van der Waal's radius :
Definition: It is **one-half of the internuclear distance** of adjacently placed atoms **in solid state of a noble gas**.

Order : van der Waal's radius > Metallic Radius > Covalent Radius

FACTORS AFFECTING ATOMIC SIZE		
(i) Effective nuclear charge	Atomic radius $\propto \frac{1}{\text{Effective nuclear charge}}$	Eg. Decreasing order of 2nd period elements : **Li > Be > B > C > N > O > F**
(ii) Number of shells	Atomic radius $\propto$ **No. of shells**	Eg. Increasing order of 1st group elements : **Li < Na < K < Rb < Cs**
(iii) Screening effect	Atomic radius $\propto$ **Screening effect**	

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Variation of Atomic radius :

Variation in a Period (left to right) :	**Decreases**	**Reason :** Z_{eff} increases but n remains constant.
Variation in a Group (top to bottom) :	**Increases**	**Reason:** n increases and Z_{eff} almost remains constant.
Irregularities :	(1) In first transition series, the covalent radii of the elements **decrease from left to right across a row, in the middle the radii becomes almost constant and near the end the radii increases slightly.**	**Reason:** $\underbrace{Sc\ Ti\ V\ Cr}_{\because Z_{eff}>\sigma}$ $\underbrace{Mn\ Fe\ Co\ Ni}_{\because Z_{eff}\approx\sigma}$ $\underbrace{Cu\ Zn}_{\because Z_{eff}<\sigma}$
	(2) $r_{4d} \approx r_{5d}$ Eg. $r_{Zr} \approx r_{Hf}$ $r_{Nb} \approx r_{Ta}$	**Reason :** Lanthanide contraction (poor shielding of nuclear charge by 4f electrons)
	(3) $r_{Al} \geq r_{Ga}$	**Reason :** d-contraction
	(4) In general, noble gases have **maximum** radius in their respective periods.	**Reason :** van der Waal radius is measured for them.

Ionic radius :

(i) Cationic radius	When neutral gaseous atom loses electron it converts into cation. **Atomic radius > Cationic radius** Size of cation $\propto \frac{1}{\textbf{Magnitude of + ve charge or } Z_{eff}}$
	Example : $Fe > Fe^{+2} > Fe^{+3}$
(ii) Anionic radius	When neutral gaseous atom gains electron it converts into anion. **Anionic radius > Atomic radius** Size of anion $\propto$ **Magnitude of –ve charge** **Example :** $I^- > I$

3. Ionisation energy

Ionisation energy (IE) is defined as the **amount of energy required** to **remove the most loosely bound electron** from **an isolated gaseous atom** to form a cation.

$$M(g) \xrightarrow{(IE_1)} M^+(g) + e^- \quad ; \quad M^+(g) + IE_2 \longrightarrow M^{2+}(g) + e^-$$

$$M^{2+}(g) + IE_3 \longrightarrow M^{+3}(g) + e^-$$

$IE_1 < IE_2 < IE_3$ …. (for every element, **independent** of electronic configuration)

- **Units : kJ mol^{-1}, eV (electron volt).**

Ionisation potential **(in V)** has same numerical value as ionization energy **(in eV)**

Factors Affecting Ionisation Energy	
(i) Effective nuclear charge	IE $\propto$ **Z_{eff}**
(ii) Number of shells	IE $\propto \frac{1}{\textbf{No. of shells}}$
(iii) screening effect	IE $\propto \frac{1}{\sigma\textbf{(Screeing constant)}}$
(iv) Electronic configuration	**Half-filled and full-filled** subshell have exceptionally **high** IE.
(v) Magnitude of +ve charge	IE $\propto$ **Magnitude of positive charge**
(vi) Magnitude of –ve charge	IE $\propto \frac{1}{\textbf{Magnitude of negative charge}}$
(vii) Penetration effect of the electron	Ionisation energies are in the order **s > p > d > f.**

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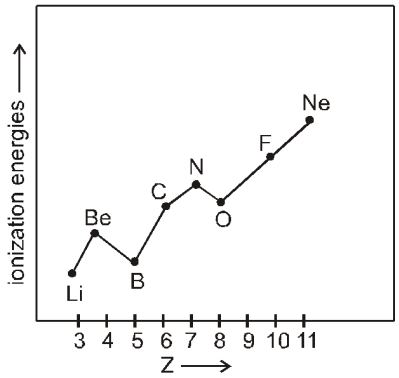
Variation of Ionisation energy :

Variation in a Period (left to right)	**Increases**	**Reason :** Z_{eff} increases but n remains constant.
Variation in a Group (top to bottom)	**Decreases**	**Reason:** n increases and Z_{eff} almost remains constant.
Irregularities :	**(i)** IE_1 order of second period is **Li < B < Be < C < N < O < F < Ne**	**Reason :** Be > B Be has stable $2s^2$ configuration. N > O N has half filled p^3 configuration.
	(ii) IE_1 **Tl > In & Ga > Al**	**Reason : Lanthanide contraction & d-contraction.**
	(iii) IE_1 **Au > Cu > Ag**	**Reason : Lanthanide contraction.**

Following graph shows variation of ionization energies (kJ/mole) with atomic number in second period (Li-Ne).



Max IE in periodic table : **He** ; Min IE in periodic table : **Cs**

4. Electron gain enthalpy / Electron affinity

Electron gain enthalpy ($\Delta_{eg}H$) is the enthalpy change when **an electron is added** to the **valence shell** of an **isolated gaseous atom.**

Electron affinity is conventionally defined as the **energy released** when **an electron is added** to the **valence shell** of an **isolated gaseous atom.**

$F(g) + e^- \longrightarrow F^-(g) \Rightarrow$ 328 kJ of energy is released per mole of F.

$\therefore$ E_a = **328 kJ mol^{-1}** ; $\Delta_{eg}H$ = **–328 kJ mol^{-1}**

- Now a days, both electron affinity and electron gain enthalpy terms are used.
  E.A. = +ve (**Exothermic**) $\Rightarrow$ $\Delta_{eg}H < 0$
  E.A. = –ve (**Endothermic**) $\Rightarrow$ $\Delta_{eg}H > 0$
- Electron gain **may be either exothermic or endothermic** depending on the **element**, but second electron gain enthalpy is always **+ve.**

Factors Affecting Electron Affinity	
(i) Effective nuclear charge	EA $\propto$ Z_{eff}
(ii) Number of shells	EA $\propto \frac{1}{\text{No. of shells}}$
(iii) screening effect	EA $\propto \frac{1}{\sigma\text{(Screeing constant)}}$
(iv) Electronic configuration	**Half-filled and fully-filled subshell have +ve $\Delta_{eg}H$ (endothermic)**

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Variation of Electron Affinity :

Variation in a Period (left to right)	**Increases**	**Reason:** Z_{eff} increases but n remains constant.
Variation in a Group (top to bottom)	**Decreases**	**Reason:** n increases and Z_{eff} almost remains constant.
Irregularities	**(i)** EA of second period is F > O > C > Li > B > **N > Be > Ne**	**Reason:** N has half filled p^3 configuration but Be and Ne has stable full filled configuration.
	(ii) EA (17^{th} grp) **Cl > F > Br > I** (16^{th} grp) **S > Se > Te > O**	**Reason:** electron goes to the larger n = 3 energy level and consequently occupies a larger region of space leading to much less electron-electron repulsion.
	(iii) Alkali metals have **–ve $\Delta_{eg}H$ (exothermic)**	

Max EA in periodic table : **Cl** ; II Max EA in periodic table : **F**

$EA(A_+) = IE(A)$; $EA(A) = IE(A_-)$

4. Electronegativity

Electronegativity is a measure of the **tendency of an element** to **attract shared electrons towards itself** in a **covalently bonded molecule**.

Units : Unitless

Factors Affecting Electronegativity	
(i) Effective nuclear charge	$EN \propto Z_{eff}$
(ii) Number of shells	$EN \propto \frac{1}{\text{No. of shells}}$
(iii) screening effect	$EN \propto \frac{1}{\sigma(\text{Screeing constant})}$
(iv) Higher oxidation state	**With increase in the oxidation state of element, its electronegativity also increases.**
(v) % s-character	**The electronegativity also increases as the s-character in the hybrid orbitals increases.** Hybrid orbital sp^3 sp^2 sp **s-character 25% 33% 50%** → Electronegativity increases

Variation of Electronegativity :

Variation in a Period (left to right)	**Increases**	**Reason:** Z_{eff} increases but n remains constant.
Variation in a Group (top to bottom)	**Decreases**	**Reason:** n increases and Z_{eff} almost remains constant.
Irregularities	**In Boron family, on moving from top to bottom, EN initially decreases, then increases marginally**	**Reason:** Poor shielding of nuclear charge by d and f-electrons.

Max. EN in periodic table : **F** ; Min. EN in periodic table : **Cs**

Scales of electronegativity :

(a) Pauling's scale :

$$\Delta = X_A - X_B = \mathbf{0.208}\sqrt{E_{A-B} - \sqrt{E_{A-A} \times E_{B-B}}}$$

E_{A-B} / E_{A-A} / E_{B-B} = **Bond energy** of A–B / A–A / B–B bond.
(All bond energies are in **kCal / mol)**

Electronegativity of some elements on Pauling scale:

Elements	H	Li	Be	B	C	N	O	F	Si	P	S	Cl	Br	I
Electronegativity	2.1	1.0	1.5	2.0	2.5	3.0	3.5	4.0	1.8	2.1	2.5	3.0	2.8	2.5

(b) Mulliken's scale :

$$\chi_M = \frac{\mathbf{IE + EA}}{\mathbf{2}} \text{ (both expressed in } \mathbf{eV})$$

Relation between values of both scales :

$$EN_P \approx \frac{EN_M}{2.8}$$

Inert pair effect

The **outer shell 's' electrons (ns^2)** penetrate to **(n – 1)d electrons** and thus become closer to nucleus and are **more effectively pulled towards the nucleus**. This results in **less availability** of ns^2 electron pair for bonding or ns^2 electron pair becomes **inert**. The inert pair effect begins after **n ≥ 4** and **increases** with increasing value of n.

(i) Dominance of **lower oxidation state** on moving down the group in **boron, carbon and nitrogen** family may also be explained by the inert pair effect.

(ii) The inert pair effect is not the explanation of why monovalency, bivalency and trivalency in group 13, 14 and 15. It merely describe what happens, i.e. two electrons do not participate in bonding. The reason that they do not take part in bonding is **energy**.

Order of stability of oxidation state according to Inert pair effect

$Tl^{+} > Tl^{+3}$ $Pb^{+2} > Pb^{+4}$ $Bi^{+3} > Bi^{+5}$	$Ga^{+} < In^{+} < Tl^{+}$ $Ga^{+3} > In^{+3} > Tl^{+3}$	$Ge^{+2} < Sn^{+2} < Pb^{+2}$ $Ge^{+4} > Sn^{+4} > Pb^{+4}$	$As^{+3} < Sb^{+3} < Bi^{+3}$ $As^{+5} > Sb^{+5} > Bi^{+5}$

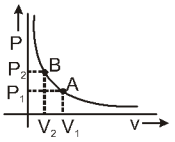
T: Theory , P: Problem

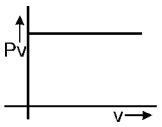
Gas Laws:

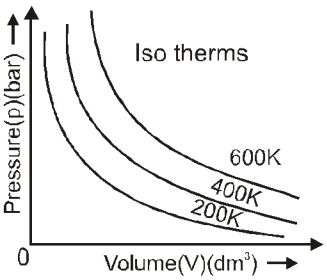
T-1 **Boyle's law:** $V \propto \frac{1}{P}$ (n, T= Constant) , Hence PV = constant (K). (K=nRT)

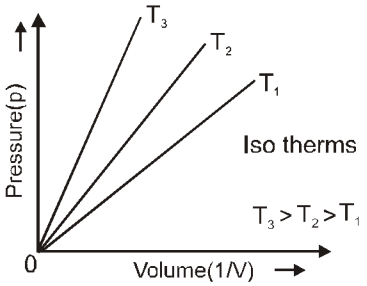
: For the two points 'A' and 'B', $P_1V_1 = K$ & $P_2V_2 = K$ $\therefore$ $P_1V_1 = P_2V_2$

Graph:









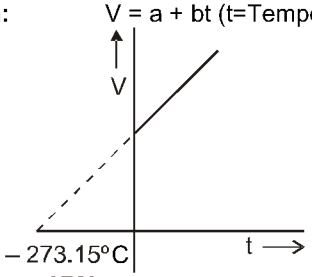
T-2 **Charles law:** $V \propto T$ or $V = kT$

$\frac{V}{T}$ = constant (K) $\left(K = \frac{nR}{P}\right)$ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ T = Temperature on absolute scale,

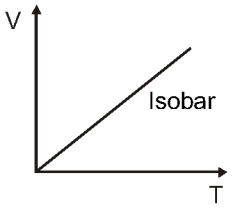
Relation between temperature & Volume

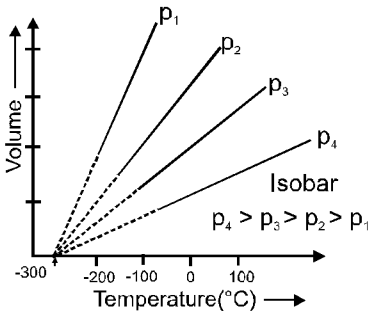
$V_t = V_0\left(1 + \frac{t}{273}\right)$ V_t = Volume of gas at t°C, V_0 = Volume of gas at 0°C, $\frac{1}{273}$ is temperature coefficient.

Graph: V = a + bt (t=Temperature on centigrade scale)



Relation : T = t + 273





T-3 **Gay-lussac's law :** $P \alpha T$ (n,V is Constant)

$\frac{P}{T}$ = Constant (K) , $\left(K = \frac{nR}{V}\right)$ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ ($T \rightarrow$ temperature on absolute scale)

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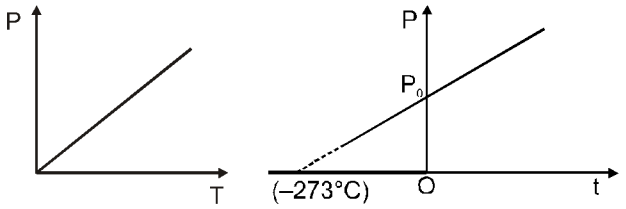
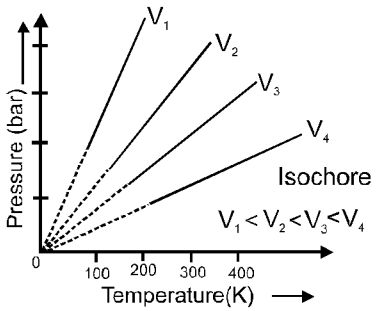
Graph:





(t=Temperature on centigrade scale)

T-4 Avogadro's Hypothesis :
For same values of pressure & temperature, equal number of molecules or moles of different gases will occupy same volume.

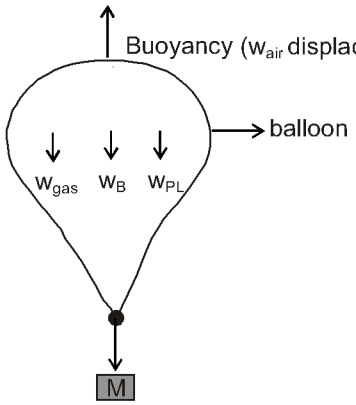
$V \alpha n$ (T, P is Constant), V/n=constant (K), $\left(K = \frac{RT}{P}\right)$

T-5 Combined gas law & ideal gas equation : $\frac{PV}{T} = K$ (n is constant). $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

PV= nRT (R= universal gas constant, independent of nature of gas)

T-6 Calculation of pay load:

Pay load is defined as the maximum weight that can be lifted by a gas filled balloon.



For maximum weight that can be lifted, (at constant velocity or Zero acceleration) applying force balance

$W_{air} = W_{gas} + W_{ballon} + W_{Payload}$

$W_{air} = d_{air} \times V_{balloon}$, $V_{balloon} = \left(\frac{4}{3}\pi r^3\right)$ $W_{balloon} = \left(\frac{PVM}{RT}\right)$

P = Pressure of gas (atm), V = Volume of balloon (Ltr), T = Temperature of gas (K), r= Radius of spherical balloon (m)

Measurement of Pressure

T-7 Barometer: A barometer is an instrument that is used for the measurement of atmospheric pressure.

$P_{atm} = \rho gh$, ρ= Density of liquid filled in tube (Kg/m^3), h=vertical height of liquid in column (m), g= gravitational Acceleration(m/s^2) = 9.8

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Reason for use of Mercury In barometer: Highest Density liquid , less Volatile in nature , Non sticky in nature.

Normal atmospheric pressure, which we call 1 atmosphere (1 atm), is defined as the pressure exerted by the atmosphere at mean sea level. It comes out to be corresponding to height 760 mm of Hg = 76 cm of Hg of column.

$$1 \text{ atm} = \rho.gh = (13.6 \times 10^3) \times 9.8 \times 0.76 = 1.013 \times 10^5 \text{ Pascal.}$$

Replacing height of any liquid with mercury liquid $H_\ell = \dfrac{\rho_{Hg} \times H_{hg}}{\rho_\ell}$

Note: If water used in barometer then height of water column: $H_{water} = \dfrac{13.6 \times 0.76}{1} = 10.33\,m$

T-8 Faulty Barometer: Space above the mercury gets filled by air(gas) instead of having vacuum. Such type of barometer is known as faulty barometer. The reading of such a barometer will be less than the true pressure.

For such a faulty barometer

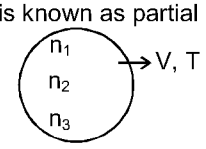
$P_0A = Mg + P_{gas}\,A$, $P_0 = \rho gh + P_{gas}$ or $\rho gh = P_0 - P_{gas}$

(P_0 = True atmospheric pressure) ; (ρgh = Measured atmospheric pressure)

Dalton's law of partial pressure :

T-9 Partial pressure :

In a mixture of non-reacting gases, pressure exerted by an individual gas when present alone in the same container at same temperature is known as partial pressure of gas.



Partial pressure of first component gas

$$P_1 = \frac{n_1RT}{v}; \qquad P_2 = \frac{n_2RT}{v}; \qquad P_3 = \frac{n_3RT}{v}$$

T-10 Dalton's law :

For a non-reacting gaseous mixture, total pressure of the mixture is the summation of partial pressure of the different component gases.

$$P_{Total} = P_1 + P_2 + P_3 = \frac{(n_1 + n_2 + n_3)RT}{v}$$

$$\frac{P_1}{P_T} = \frac{n_1}{n_T} = X_1 \text{ (= mole fraction of first component gas)}$$

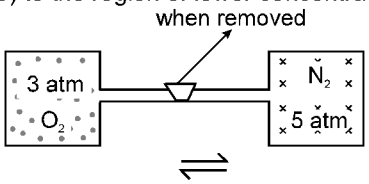
$$\frac{P_2}{P_T} = \frac{n_2}{n_T} = X_2 \text{ (= mole fraction of second component gas)}$$

T-11 Mixing of Gases (Connecting vessel problems) (gases are non-reacting in nature)

1. Total no of mole remain constant. (An equation should be obtained from here considering 2nd point)
2. Final pressure in all connected vessels is same.

T-12 Graham's law of diffusion/effusion

Diffusion: Net spontaneous flow of gaseous molecules from region of high concentration (higher **partial pressure**) to the region of lower concentration **or lower partial pressure**



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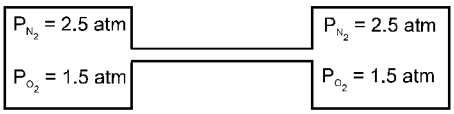
Flow will be from both sides, N_2 will try to equalize its partial Pressure in both the vessels, and so will O_2 till both have same partial pressure in both bulb.



T-13 Graham's Law :

"Under same conditions of temperature and pressure (partial pressure), the rate of diffusion of gas into vacuum is inversely proportional to square root of the density of gas."

Rate of diffusion, $r \propto \frac{1}{\sqrt{d}}$ d = density of gas (P,T= constant)

$\frac{r_1}{r_2} = \frac{\sqrt{d_2}}{\sqrt{d_1}} = \frac{\sqrt{M_2}}{\sqrt{M_1}} = \sqrt{\frac{V.D_2}{V.D_1}}$ V.D is vapour density (P,T= constant)

Rate of Diffusion:

r = volume flow rate = $\frac{dV_{out}}{dt}$, r = mole flow rate = $\frac{dn_{out}}{dt}$, $r = \frac{\text{drop in pressure}}{\text{time taken}} = \frac{(dP)_{drop}}{dt}$

r = distance travelled by gaseous molecule in a thin glass tube per unit time = $\frac{dx}{dt}$

The general form of the grahams law of diffusion can be stated as follows, when one or all of the parameters are varied.

rate $\propto \frac{P}{\sqrt{TM}} A$; P – Pressure, A – area of hole, T – Temp. , M – mol. wt. of gas

When P, A and T are constant for two different gases : $\frac{r_2}{r_1} = \sqrt{\frac{M_1}{M_2}}$

If partial pressure of gases are not equal or gas mixture is diffusing out from a container :

Rate of diffusion is found to be proportional to partial pressure & inversely proportional to square root of molecular mass.

$r \propto P$, $r \propto \frac{1}{\sqrt{M}}$; $\frac{r_1}{r_2} = \frac{P_1}{P_2}\sqrt{\frac{M_2}{M_1}}$ $\frac{P_1}{P_2} = \frac{X_1}{X_2} = \frac{n_1}{n_2} = \frac{w_1}{w_2}.\frac{M_2}{M_1}$

Successive diffusion: $\left(\frac{n_B}{n_A}\right)_{Final} = \left(\frac{n_B}{n_A}\right)_{Initial} \times \left(\sqrt{\frac{M_A}{M_B}}\right)^n$, n = Number of steps of diffusion

T-14 Kinetic Theory of Gases:

Postulates / assumptions of KTG:

2. The volume occupied by the molecules is negligible in comparison to the total volume of the gas. For an ideal gas, volume of gas molecule ≈ 0.
3. Molecular attraction forces are negligible. Infact, for an ideal gas, attractive or repulsive forces are equal to zero.
4. The average K.E. of gaseous molecules is proportional to the absolute temperature of the gas.

   $\frac{1}{2} m(\overline{u^2}) \propto T$ (bar is for average)
5. Effect of force of gravity on gas molecules is negligible (Due to very small mass).
6. Gas molecules are not at rest but possess rapid motion in straight line in all directions.
7. Collisions are perfectly elastic in nature (momentum and kinetic energy remain conserved).

T-15 Kinetic gas Equation : $PV = \frac{1}{3} mNU^2_{rms}$

P = Pressure of gas(Pa), V = Volume of container (m^3), m = mass of one molecule (Kg), N = Number of molecule, U_{rms} = root mean square speed (m/s)

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Average translational kinetic energy (K.E) = $\frac{3}{2}$nRT (For n mole of gas)

Average translational KE= $\frac{3}{2}$RT (For 1 mole of gas)

(only dependent on temperature, not on nature of the gas)

Average translational K.E = $\frac{3}{2}$kT (Per Molecule of gas) $K = \frac{R}{N_A}$ known as Boltzmann Constant

T-16 Mean Free path: Average Distance travelled by gas molecules between two successive collisions.

Formula: $\lambda = \frac{RT}{\sqrt{2}\pi d^2 . N_A P}$ d = diameter of partical, p = pressure of gas, T = temperature of gas

Note: No. of collision per unit volume per unit time is known as collision frequency.

T-17 Root mean square speed: Root of mean of square of all molecular speeds

$$U_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{d}} \text{ m/s}$$

M=molar mass (Kg), T=temperature (K), R= $\frac{25}{3}\frac{J}{mol.K}$, P=pressure of gas (Pa), d=density of gas $\left(\frac{Kg}{m^3}\right)$

$$U_{rms} = \sqrt{\frac{U_1^2 + U_2^2 + ... + U_n^2}{n}} \text{ m/s}$$

T-18 Average speed: Average of all molecular speeds

$$U_{av} = \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8P}{\pi d}} \text{ m/s} \qquad U_{avg} = \frac{U_1 + U_2 + ... + U_n}{n} \text{ m/s}$$

T-19 Most probable speed: The speed possessed by maximum fraction of molecules at the given temperature

$$U_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{d}} \text{ m/s}$$

T-20 Relationship between all speeds for a given gas at temperature T of gas

$U_{mps} < U_{avg} < U_{rms}$

$U_{mps} : U_{avg} : U_{rms} :: \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3} :: 1 : 1.12 : 1.22$

T-21 Maxwell's distributions of molecular speeds :

Postulates/Assumptions of speed distributions

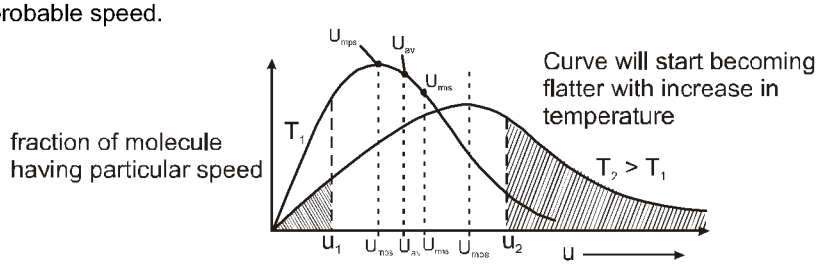
(1) The area under the curve will denote fraction of molecules having speeds between zero and infinity

(2) Total area under the curve will be constant and will be **unity** at all temperatures.

(3) On increasing temperature peak of the curve shifts towards right side and in downward direction. Curve becomes flatter (broader).

(4) Fraction of molecule corresponding to very low / very high speed is very less.

(5) Maximum fraction of molecule pocess a common velocity which represents peak of curve is known as most probable speed.



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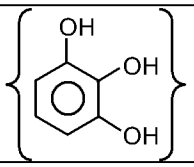
T-22 Eudiometry:

The analysis of gaseous mixtures by measuring volume change during the physical & chemical process.
At constant P,T we can apply Volume- volume Analysis(or Mole- mole analysis) for gases only.

Gas	Absorbed by
CO_2	KOH(aq.), NaOH (aq.)
O_2	Alkaline Pyrogallol 
O_3	Turpentine oil
NH_3	Acidic Solution
SO_2	Basic Solution
HCl	Water, Basic solution
H_2O	Conc H_2SO_4 (Dehydrating Agent)

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T: Theory, F Formula

T-1 **Real Gas:** A gas which does not follow ideal gas laws at **all temperatures** & **pressures.**. No gas occurring in nature is ideal, all are just real gases.
Real gases deviate from ideal behavior because
(a) Real gas molecules have a finite volume.
(b) Inter molecular attractive forces between real gas molecules is not zero.

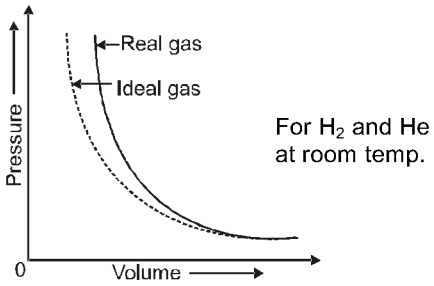
T-2 **Compressibility factor: (Z)** Factor representing type & extent of deviation of real gases from ideal behavior.

F1 $Z = \frac{P_r V_r}{n_r R T_r} = \frac{PV_m}{RT} = \frac{PM}{dRT}$, **F2** $Z = \frac{V_{real}}{V_{ideal}} \left(\text{if } P_r = P_i, T_r = T_i \text{ \& } n_r = n_i \right)$

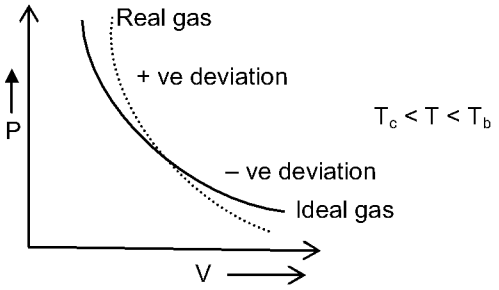
F3 $\mathbf{Z =} \frac{P_{real}}{P_{ideal}} \left(\text{if } V_r = V_i, T_r = T_i \text{ \& } n_r = n_i \right)$

V_m is volume of one mole of gas or molar volume.
$Z = 1 \Rightarrow$ ideal gas ; $Z < 1 \Rightarrow$ Real gas showing negative deviation ; $Z > 1 \Rightarrow$ Real gas showing positive deviation.

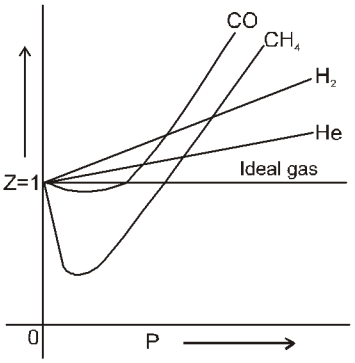
T-3 PV Graph for real Gas:



For H_2 and He at room temp.



T-4 Variation of Z with pressure at constant temperature:



Conclusion From this Graph:

1. As pressure tends to Zero, Z Approaches 1.
2. In low pressure region, most of gases show Negative Deviation.(except H_2 and He)
3. At High pressure region, all gases show Positive Deviation.

T-5 Variation of Z with pressure at different temperatures (for same gas):

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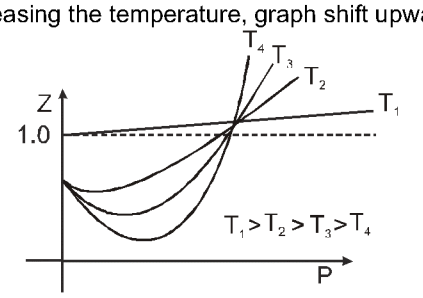
On increasing the temperature, graph shift upwards (Dip of Curve Decrease)



Conclusions:

(1) Z = 1 for ideal gas ;

(2) Z > 1 at all pressures for He/H_2 (if T > T_b)

(3) Z < 1 at low pressure (for all gases if T < T_b)

(4) Z > 1 at high pressure (for all gases)

.....

T-6 Vander Waal Equation of real gases:

The ideal gas equation does not consider the effect of attractive forces and molecular volume.

Vander Waal's corrected the ideal gas equation by taking the effect of

(a) Molecular attraction **(b)** Molecular volume

(a) Molecular attraction: Real gas has some **intermolecular Force of attraction**. So pressure of real gas obeying Vander Waal equation is less than Ideal gas.

$P_i = P_R + \frac{an^2}{v^2}$, where: **"a" is Vander waal constant for force of attraction.** **Unit of a:** $\frac{atm\,Ltr^2}{mole^2}$

Comparison of value of a:

(i) $a_{\text{polar gas}} > a_{\text{Non polar gas}}$

(ii) If gas have same nature, then "a" α **Molar mass of gas**

(iii) If **Hydrogen bonding** take place, then "a" has higher value.

(iv) For isomers, value of "a" α **Size of Molecule**.

Higher "a" means:

1. Higher interparticle Attraction
2. Higher T_c
3. Higher Liquefaction tendency
4. Lower Volatile tendency
5. Higher Boiling point & less vapor Pressure
6. Less pressure needed for liquefaction

(b) Molecular volume : Free space available will be less for real gas than ideal gas in same container.

$V_{real} > V_{ideal}$

$V_i = \mathbf{V_{container} - nb}$, n= no of mole of gas.

b= Excluded Volume or Co-volume. (It take into account finite Size of molecule)

Excluded volume per mole =b $= 4\left\{\frac{4}{3}\pi r^3\right\}N_A$ **Unit of b**: $\frac{Ltr}{Mole}$

Comparison of value of b: **b α Molecular Size**

F-4 Vander waal's equation : $\left(P + \frac{an^2}{v^2}\right)$ **(V – nb) = nRT for n mole of real gas.**

Note: Value of a, b is independent of temperature, only Depends upon gas.

.....

T-7 Explanation of Z from Vander Waal gas equation:

(i) At Low pressure (Temp Moderate):

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F-5 $Z = 1 - \frac{a}{V_m RT}$

(**Z < 1** , i.e. Attractive force Dominates (**Negative deviation**)

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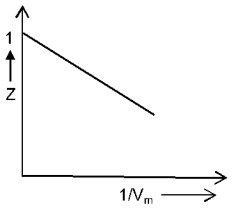


(ii) At High Pressure (temp Moderate)

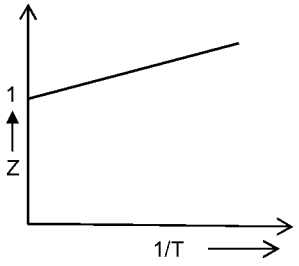
F-6 $Z = \frac{Pb}{RT} + 1$

(Z > 1 , ie. Molecular Size dominate(positive Deviation))

Note: **(a)** For **H_2 or He, a ≈ 0**, $Z = 1 + \frac{Pb}{RT}$

(b) "a" is responsible for **–ve deviation & "b"** responsible for **+ve deviation.**

(c) For real gas with V.large molar volume, PV = nRT



T-8. Vander waals' equation in virial form:

Virial Equation is a generalized equation of gaseous state in which Z is expressed in form of an infinite series of powers of $\frac{1}{V_m}$ or P .

$$Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + ...$$

All other equations can be written in the form of virial equation of state.

$$Z = 1 + \frac{1}{V_m}\left(b - \frac{a}{RT}\right) + \frac{b^2}{V_m^2} + \frac{b^3}{V_m^3} + ... \text{ (Vander Waal equation in virial form)}$$

On comparing Vander Waals equation with virial equation:

2nd Virial Coefficient: $B = b - \frac{a}{RT}$, **3rd Virial Coefficient**: $C = b^2$, **4th Virial Coefficient:** $D = b^3$

At low pressure: V_m will be larger. Hence $\frac{1}{V_m^2}, \frac{1}{V_m^3}$ ..... can be neglected

$$Z = 1 + \frac{1}{V_m}\left(b - \frac{a}{RT}\right) \qquad \text{If } \left(b - \frac{a}{RT}\right) = 0 \Rightarrow \text{ at } T = \frac{a}{Rb}, \quad Z = 1$$

F-7 So at $T_b = \frac{a}{Rb}$, gas will behave as an ideal gas (or follows Boyles law)

T-9 **Boyles Temperature:(T_b)**: temperature at which real gas follow ideal gas behavior for low pressure is called Boyles' temperature.

Note: at Boyles' temperature, effect produced by intermolecular attraction is cancelled by effect produced by molecular Size.

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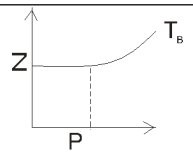
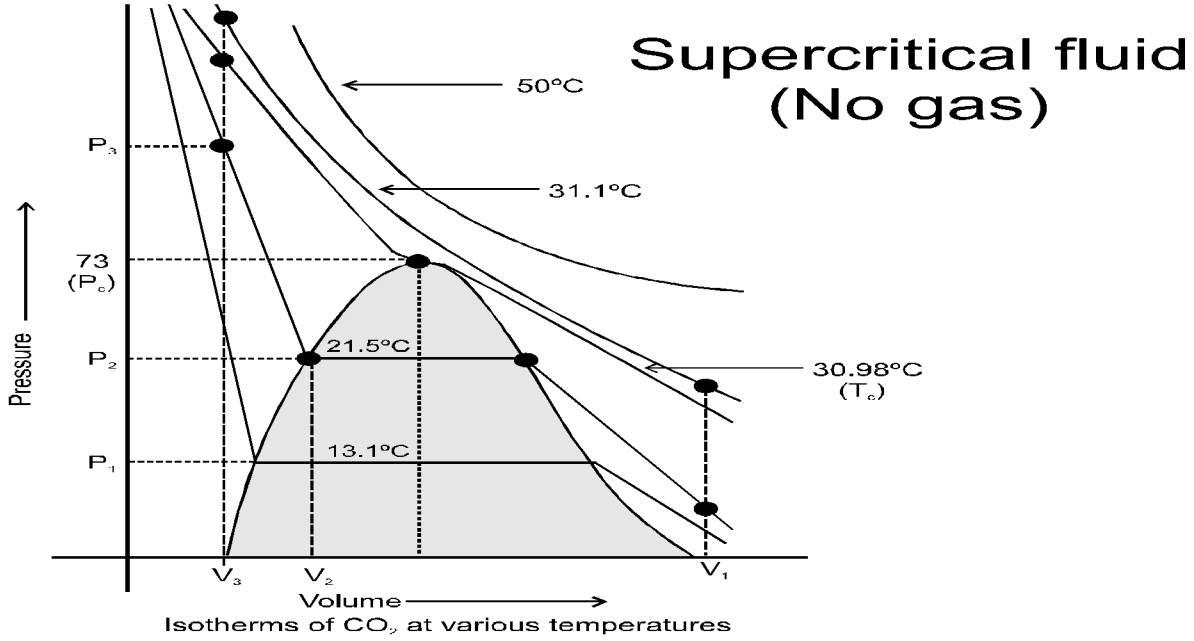
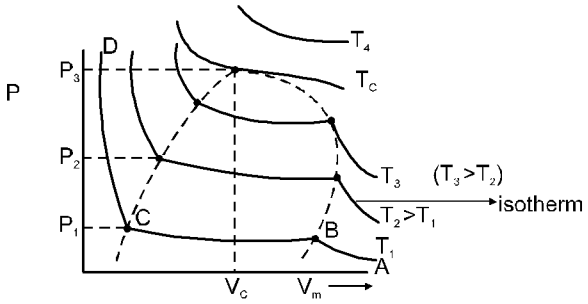
Relationship between T_b & T_c:	Graph:
$T_c = \frac{8}{27} T_b \quad (T_b > T_c)$	

T-10 Critical Phenomena (Liquefaction of gas)





Conclusion from this Graph:

$T_4 > T_c > T_3 > T_2 > T_1$

point B: Liquefaction start at T_1

point C: Liquefaction ends at T_1

AB → gas , BC → vapour + liquid , CD → liquid

Pressure	temperature	Physical State
$P > P_c$	$T > T_c$	Super Critical fluid
P_c	T_c	transition state
$P > P_c$	$T < T_c$	liquid State
$P < P_c$	$T < T_c$	only Liquid or only vapor or Both

Note: VGE is not applicable in Liquid–vapor region.
Solution of VGE equation is:
(a) -ve Slope : 1 real & 2 imagnary Roots
(b) Zero Slope : 3 unequal & real roots

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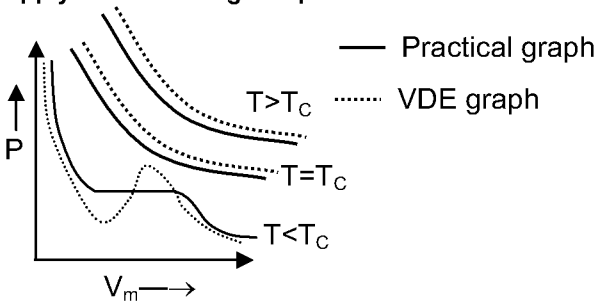
(c) at CP: All three roots are real & equal. $(V_m - V_c)^3 = 0$

Apply Vander waal gas equation on these curve	**Chatarctristics of graph.**
	(i) It has only negative or zero slope. (ii) Slope is zero (maximum value) at critical point (iii) Critical point is a point of inflection at which $\left(\frac{\partial P}{\partial V_m}\right) = 0$ and $\frac{\partial}{\partial V_m}\left(\frac{\partial P}{\partial V_m}\right) = 0$ (iv) At critical point (a) Enthalpy of vaporization is zero. ($\Delta H_{vap} = 0$) (b) Surface tension is zero.

T-11 Critical point: At this point, all the physical properties of liquid phase will be equal to physical properties of vapour phase, such as density of liquid = density of vapour

Critical constant using vander waals' equations:

Critical temperature (T_c): it is maximum temp above which real gas can't be liquefying by applying any pressure.

(For a gas, If T<Tc, then gas is called vapor) . $\mathbf{T_c = \frac{8a}{27Rb}}$

Critical Pressure (P_c): it is minimum pressure required to liquefy the gas at critical temperature.

$$\mathbf{P_C = \frac{a}{27b^2}}$$

Critical Volume (V_c): It is Volume occupied by 1 mole gas at critical pressure & temperature.

$\mathbf{V_c = 3b}$

Note: 1. If all $\mathbf{T_c}$, $\mathbf{P_c}$ **&** $\mathbf{V_c}$ are given, then use P_c & T_c to calculate the value of a & b, because V_c can't be measured correctly

2.Compressibily factor at critical Point: $Z = \frac{P_c V_c}{R T_c} = \frac{3}{8}$ (It is fixed for all gases at Critical point)

T-12 Reduced Equation of State:

Reduced Temp: Temperature of any gas with respect to critical temp of the gas: $T_r = \frac{T}{T_C}$

Reduced Pressure: pressure of gas with respect to critical pressure of the gas: $P_r = \frac{P}{P_C}$

Reduced Volume: Volume of gas with respect to critical volume of the gas: $V_r = \frac{V_m}{V_C}$

F-8 $\left(P_r + \frac{3}{V_r^2}\right)(3V_r - 1) = 8\,T_r$ (Reduced equation of state; obtained from Vander Waal equation by substitution of P, V_m & T from above)

Note: 1. Above equation is independent from a, b and R, so will be followed by each and every gas, independent of its nature.

2. If two gases have same value of P_r & V_r, then they will have same value of T_r.

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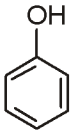
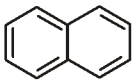
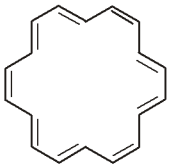
IUPAC & Isomerism

1.	Alkanes	(Saturated hydrocarbon)	$CH_3 - CH_2 - CH_3$	Propane
2.	Alkenes	(Unsaturated hydrocarbon)	$CH_3 - CH = CH_2$	Propene
3.	Alkynes	(Unsaturated hydrocarbon)	$HC \equiv CH$	Ethyne
4.	Aliphatic compounds with other functional groups		$CH_3–CH_2–OH$	Ethanol (absolute alcohol)
			$CH_3–CH_2–COOH$	Propanoic acid
5.	Homo alicyclic compound			Cyclohexane
6.	Heteroalicyclic compound			Tetrahydropyran
7.	Heterocyclic aromatic compound			Pyridine
8.	Homo cyclic benzenoid aromatic compounds			Phenol
				Napthalene
9.	Homo cyclic Non-benzenoid aromatic compound			[18] annulene

1. Organic compounds and functional groups

Number of known organic compounds is much more than inorganic compounds but, it has been possible to group them into classes or families based on their structural features. This has given organic chemistry a logical and systematic shape. Examples are as follows :

1.1 Alkanes

These are open–chain aliphatic saturated compounds which have no functional groups. These are also called paraffins. These are represented by the general formula, C_nH_{2n+2} where n = 1, 2, 3, ...

The names of the alkanes for n = 1, 2, ..., 10 must be committed to memory since all organic nomenclature is related directly or indirectly to the name of alkane C_nH_{2n+2}

1.2 Alkyl groups

When we remove a hydrogen from alkane, we get an alkyl group :

General formula $C_nH_{2n+2} \xrightarrow{-H} C_nH_{2n+1}$ $\left(\text{alkane} \xrightarrow[+\text{yl}]{-\text{ane}} \text{alkyl}\right)$

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1.3 **Alkenes**

Alkenes are open chain unsaturated hydrocarbons and having carbon–carbon double bonds (C = C). They have the general formula C_nH_{2n} where n = 2, 3, 4 ... etc. These are also called **alkylenes or olefins**. The first three members are generally named by their common names.

1.4 **Alkenyl groups**

Like alkyl groups (alkane – H), there are three commonly encountered alkenyl groups which are given common names.

1.5 **Alkynes**

Unsaturated aliphatic hydrocarbons containing a carbon–carbon triple bond are called alkynes. In the common system, they are called acetylenes, after the name of the first member of this family. (i.e. acetylene)

General formula : C_nH_{2n-2}

where n = 2, 3, 4..... etc.

IUPAC names : Alkane – ane + yne = Alkyne

The position of the triple bond on the parent chain is designated by lowest possible arabic numerals

1.6 **Alkynyl groups**

$HC \equiv CH \xrightarrow{-H}$ $HC \equiv C -$

(-e + yl = Ethynyl)

$H_3C - C \equiv CH \xrightarrow{-H}$ $H_3C - C \equiv C -$

(-e + yl = propynyl)

1.7 **Functional group and residue**

The characteristic group of atom which decided the physical and chemical properties of an organic molecule is called functional group.

Functional group is that portion of molecule which is highly reactive and takes part in chemical reactions. Rest of the molecule is called Residue.

Some common names of hydrocarbon alkyl groups

(A) Iso alkyl group

A compound having $-\underset{\displaystyle CH_3}{\overset{|}{CH}}-CH_3$ group is called iso alkyl group

e.g. $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-$ Iso propyl; $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-CH_2-$ Iso butyl; $CH_3-\underset{\displaystyle CH_3}{\overset{|}{CH}}-CH_2-CH_3$ Iso pentane

Note: Iso alkyl group name can be used in IUPAC Nomenclature and its first letter 'I' is consider for alphabetical seniority

e.g. $\overset{1}{CH_3}-\overset{2}{\underset{\displaystyle CH_3}{\overset{|}{CH}}}-\overset{3}{\underset{\displaystyle CH}{\overset{|}{CH}}}-\overset{4}{CH_2}-\overset{5}{CH_2}-\overset{6}{CH_3}$ (with CH_3 and $CH-CH_3$ branches) 3–Isopropyl–2–methylhexane

(B) Neo alkyl group

Compound having $\left(\begin{array}{c} CH_3 \\ | \\ CH_3-C-CH_2- \\ | \\ CH_3 \end{array}\right)$ group is called neo alkyl group.

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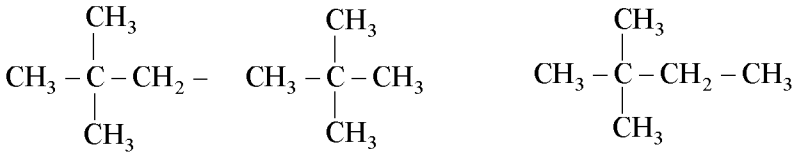
e.g.



Neopentyl Neopentane Neohexane

(C) Types of carbon and hydrogen atoms in alkanes

The carbon atoms in an alkane molecule may be classified into four types as primary (1°), secondary (2°), tertiary (3°) and quaternary (4°) as follows

(a) A carbon atom attached to one another (or no other) carbon atom is called a primary carbon atom and is designated as 1° carbon

e.g. (i) $\overset{1^\circ}{C}H_3 - \overset{1^\circ}{C}H_3$ (ii) $\overset{1^\circ}{C}H_3 - CH_2 - \overset{1^\circ}{C}H_3$

(b) A carbon atom attached to two other carbon atoms is called a secondary carbon atom and is designated as 2° carbon

e.g. (i) $CH_3 - \overset{2^\circ}{C}H_2 - CH_3$ (ii) $CH_3 - \overset{2^\circ}{C}H_2 - \overset{2^\circ}{C}H_2 - CH_3$

(c) A carbon atom attached to three other carbon atoms is called a tertiary carbon atom and is designated as 3° carbon.

e.g. (i) $CH_3 - \underset{3^\circ}{\overset{\displaystyle CH_3}{\overset{|}{C}}}H - CH_3$ (ii) $CH_3 - \underset{3^\circ}{\overset{\displaystyle CH_3}{\overset{|}{C}}}H - CH_2 - CH_3$

(d) A carbon atom attached to four other carbon atoms is called a quaternary carbon atom and is designated as 4° carbon

e.g. (i) $CH_3 - \overset{4^\circ}{\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}} - CH_3$ (ii) $CH_3 - \overset{4^\circ}{\underset{\displaystyle CH_2 - CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}} - CH_2 - CH_3$

The hydrogen atoms attached to 1°, 2° and 3° carbon atoms are called primary (1°) secondary (2°) and tertiary (3°) hydrogen atoms. It may be noted here that there is nothing like quaternary hydrogen atom. Since a quarternary carbon does not carry any hydrogen.

2. IUPAC system of nomenclature

The basic criteria for naming a structure by IUPAC system is choice of a parent name or the basic carbon skeleton.

Nomenclature of alkanes is fundamental, to naming whole class of organic compounds because it helps us to identify the basic carbon skeleton.

General rules for IUPAC nomenclature

The IUPAC system is the most rational and widely used system of nomenclature in organic chemistry. The most important feature of this system is that any given molecular structure has only one IUPAC name and any given IUPAC name denotes only one molecular structure.

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The IUPAC name of any organic compound consists of five parts, i.e.,

1. Word root 2. Primary suffix 3. Secondary suffix 4. Primary prefix 5. Secondary prefix

Thus, a complete IUPAC name of an organic compound consists of the following parts in the given sequence:

Secondary prefix + Primary prefix + Word root + Primary suffix + Secondary suffix

2.1 Word root :

It is the basic unit of the name. It denotes the number of carbon atoms present in the principal chain (the longest possible continuous chain of carbon atoms including the functional group and based upon the common names of alkanes) of the organic molecules.

Chain length	Word root	Chain length	Word root	Chain length	Word root
C_1	Meth	C_9	Non	C_{20}	Icos
C_2	Eth	C_{10}	Dec	C_{30}	Triacont
C_3	Prop	C_{11}	Undec	C_{40}	Tetracont
C_4	But	C_{12}	Dodec	C_{50}	Pentacont
C_5	Pent	C_{13}	Tridec	C_{60}	Hexacont
C_6	Hex	C_{14}	Tetradec	C_{70}	Heptacont
C_7	Hept	C_{15}	Pentadec	C_{80}	Octacont
C_8	Oct	C_{16}	Hexadeca	C_{100}	Cent & Hect

2.2 Primary Suffix.

A primary suffix is always added to the word root to indicate whether the carbon chain is saturated or unsaturated. The three basic primary suffixes are given below :

Type of carbon chain	Primary suffix	General name
(a) Saturated	– ane	Alkane
(b) Unsaturated with one double bond	– ene	Alkene
(c) Unsaturated with one triple bond	– yne	Alkyne

If the parent carbon chain contains two, three or more double or triple bonds, numerical prefix such as di (for two), tri (for three), tetra (for four) etc. are added to the primary suffix. For example,

Type of carbon chain	Primary suffix	General name
(a) Unsaturated with two double bonds	(a) + diene	Alkadiene
(b) Unsaturated with two triple bonds	(a) + diyne	Alkadiyne
(c) Both double and triple bonds	enyne	Alkenyne

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2.3 Secondary suffix :

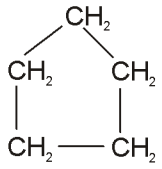
The following examples illustrate the use of word root, primary suffix and secondary suffix in naming of organic compounds.

Organic compounds	Word root	Primary suffix	Secondary suffix	IUPAC name
CH_3CH_2OH	Eth	an(e)	ol	Ethanol
$CH_3CH_2CH_2NH_2$	Prop	an(e)	amine	Propanamine
$CH_3CH_2CH_2COOH$	But	an(e)	oic acid	Butanoic acid
CH_3CH_2CN	Prop	an(e)	nitrile	Propanenitrile
$CH_2 = CHCHO$	Prop	en(e)	al	Propenal
$HC \equiv CCOOH$	Prop	yn(e)	oic acid	Propynoic acid

2.4 Primary prefix :

A primary prefix is used simply to distinguish cyclic from acyclic compounds.

For example, in case of carbocyclic compounds, (cyclic compounds containing only carbon atoms in the ring), a primary prefix, **cyclo** is used immediately before the word root. Thus.

e.g. 

Cyclo + pent + ane = Cyclopentane

Primary prefix | Word root | Primary suffix | IUPAC name

If the prefix cyclo is not used, it simply indicates that the compound is acyclic or open chain.

2.5 Secondary prefix :

Substituent group	Secondary prefix	Substituent group	Secondary prefix
– F	Fluoro	$-OCH_3$ (– OMe)	Methoxy
– Cl	Chloro	$-OC_2H_5$(–OEt)	Ethoxy
– Br	Bromo	– R	Alkyl
– I	Iodo	$-CH_3$ (– Me)	Methyl
$-NO_2$	Nitro	$-C_2H_5$ (– Et)	Ethyl
– NO	Nitroso	$-CH_2CH_2CH_3$(n-Pr)	n-Propyl
$-\overset{\oplus}{N} \equiv N$	Diazo	$-CH(CH_3)_2$ (– iPr)	Isopropyl
– OR	Alkoxy	$-C(CH_3)_3$ (t-Bu)	t-Butyl

Organic compounds	Secondary prefix	Word root	Primary suffix	IUPAC name
$CH_3CH_2 – Br$	Bromo	eth	ane	Bromoethane
$CH_3 – NO_2$	Nitro	meth	ane	Nitromethane
$C_2H_5 – OC_2H_5$	Ethoxy	eth	ane	Ethoxyethane

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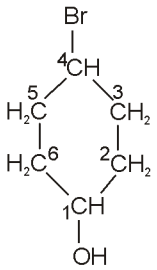
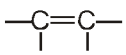
e.g. 

4-Bromo + cyclo + hex + an (e) + 1-ol

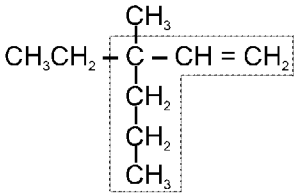
Secondary prefix | Primary prefix | Word root | Primary suffix | Secondary suffix

3. IUPAC nomenclature of alkenes/alkynes

3.1 Alkenes :

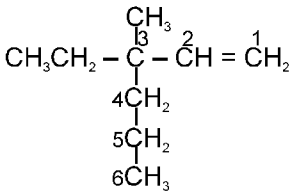
Functional group : 

(1) Select the longest chain containing carbon–carbon double bond. This need not be the longest chain in the compound as a whole. Parent name will be alkene corresponding to number of carbon atoms in the longest chain.

e.g. 

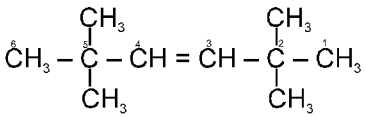
Longest chain is as shown above. It has 6 atoms ⇒ hexene = parent name

(2) Carbon atoms in the longest chain is numbered so that doubly bonded carbon atom gets the lowest number. The position of double bond is indicated by the smaller of the numbers assigned to two carbon atoms of double bond.

∴ The above example can be rewritten as, 

Position of double bond will be indicated as no. 1.

Hence, name will be, 3–Ethyl–3–methylhex–1–ene

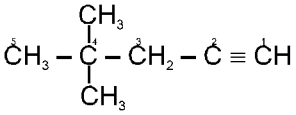
e.g. 

2, 2, 5, 5-Tetramethylhex-3-ene

3.2 Alkynes

Parent chain selection and numbering of longest chain is exactly same as that for alkenes.

e.g. $CH_3C{\equiv}C{-}CH_3$ But-2-yne



4,4–Dimethylpent–1–yne

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4. **IUPAC nomenclature of hydrocarbons containing both double and triple bonds**

(i) Such hydrocarbon is named as alkenyne (not alkynene).

(ii) Numbering is done in a manner that double and triple bonds get the lowest possible number. If double bond and triple bond both have same number then double bond is given preference our triple bond.

$HC \equiv C - CH_2 - CH = CH_2$

(wrong) **1 2 3 4 5 (numbering is done from alkyne)**

(write) **5 4 3 2 1 (numbering is done from alkene)**

5. **IUPAC nomenclature of alicyclic compounds**

(1) The names of alicyclic compounds are obtained by adding the prefix "cyclo"

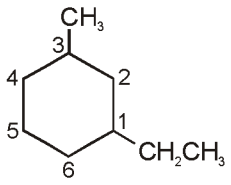
e.g.

Cyclobutane

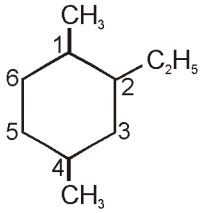
Cyclopentene

(2) The numbering of the carbon atoms in the ring is done in such a way that the substituent which comes first in the alphabetical order is given the lowest possible number and it does not violate the lowest set of locants rule.

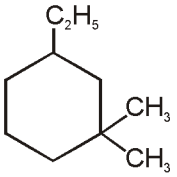
e.g.



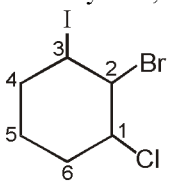
1 – Ethyl – 3 – methyl cyclohexane



2 – Ethyl – 1, 4 – dimethyl cyclohexane



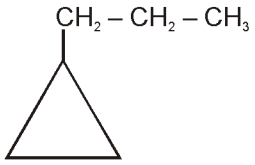
3 – Ethyl – 1, 1 – dimethyl cyclohexane



2 – Bromo – 1 – chloro – 3 – iodocyclohexane

(3) When the ring contains more or equal number of carbon atoms than the alkyl group attached to it, then it is named as a derivative of cycloalkane and the alkyl group is treated as substituent

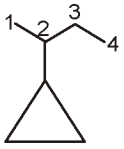
e.g.



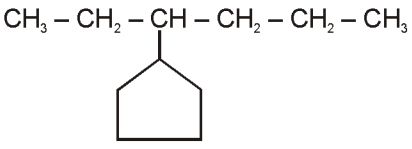
Propylcyclopropane

(4) The alkane chain contains greater number of carbon atoms than present in the ring, then the compound is considered as the derivative of alkane and the ring is designated as substituent.

e.g.



2 – Cyclopropylbutane



3 – Cyclopentylhexane

(5) If ring has unsaturation and side chain is saturated then ring is selected as parent chain.

If side chain has unsaturation and ring is saturated then side chain is selected as parent chain.

If both have unsaturation the chain with maximum unsaturation has selected as parent chain.

If equal unsaturation then longest chain is selected as parent chain.

If unsaturation and number of carbon atoms both are equal then ring is selected as parent chain.

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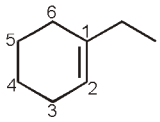
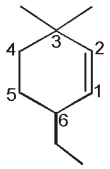
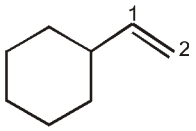
e.g.



1-Ethylcyclohex-1-ene



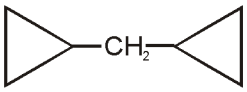
6-Ethyl-3, 3 -dimethylcyclohex-1-ene



Cyclohexyl ethene

(6) If more than one alicyclic ring is attached to a single chain then the compound is named as a derivative of alkane and the ring are treated as a substituent group.

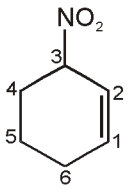
e.g.



Dicyclopropylmethane

(7) If a multiple bond and some other substitutents are present in the ring, the numbering is done in such a way that the multiple bond gets the lowest number

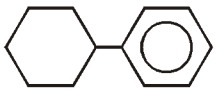
e.g.



3 – Nitrocyclohex – 1 – ene

(8) If a compound contains an alicyclic ring directly linked to the benzene ring. It is named as a derivative of benzene.

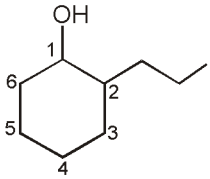
e.g.



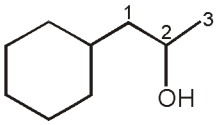
Cyclohexylbenzene

(9) If functional group is present in cyclic compounds than the main chain is taken in which principal functional lie's, if the principal functional group is present in ring also then main chain will be taken for the maximum no. of carbon atoms.

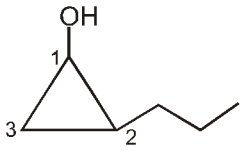
e.g.



2-Propylcyclohexan-1-ol



1-Cyclohexylpropan-2-ol



2-Propyl cyclopropan-1-ol

(10) When chain terminating functional group is directly attached with ring then ring is taken as parent chain & special suffix is used for this functional group.

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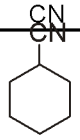
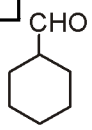
Functional Group	Suffix
CHO	Carbaldehyde
COOH	Carboxylic Acid
COX	Carbonyl halide
COOR	Alkyl Carboxylate
$CONH_2$	Carboxamide
CN	Carbonitrile

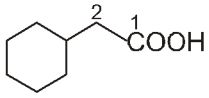
e.g.



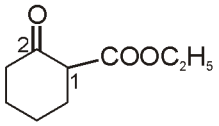
Cyclohexanecarbonitrile



Cyclohexanecarbaldehyde



2-Cyclohexyl ethanoic acid



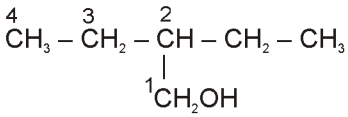
Ethyl 2–oxocyclohexane–1–carboxylate

6. IUPAC nomenclature of compounds containing functional groups

6.1 Rules for non chain terminating functional groups

(1) Parent chian : Select the longest possible chain with maximum functional group and maximum unsaturation without caring whether it also denotes the longest possible chain or not.

e.g.



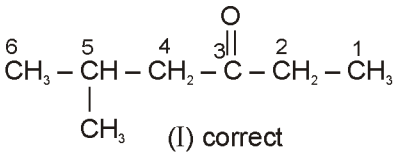
2-Ethyl butan-1-ol

(Parent chain contains four rather than five carbon atoms)

(2) Lowest number for the functional group

Numbering is done from that side of the chain which gives lowest locant to the principle functional group followed by double and triple bonds.

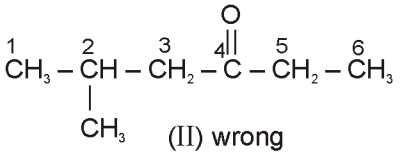
e.g.



(I) correct

5-Methyl hexan-3-one

($>C=O$ group gets lowest number 3)

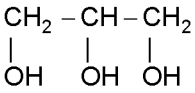


(II) wrong

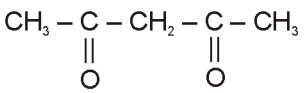
($>C=O$ group gets number 4 which is not lowest)

(3) If a compound contains two or more like groups, the numerical prefixes di, tri, tetra etc. are used

e.g.



Propane – 1,2,3 – triol



Pentane-2, 4-dione

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Class	Name	Suffix	Prefix
1. $R-COOH$	Alkanoic Acid	– oic acid (carboxylic acid)	carboxy
2. $R-SO_3H$	Alkane sulhonic Acid	– sulphonic acid	sulpho
3. $R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R$	Alkanonic Anhydride	– oic anhydride (carboxylic anhydride)	————
4. $R-COOR$	Alkyl alkanoate	– oate (carboxylate)	alkoxy carbonyl or alkanoyl oxy
5. $R-\overset{\overset{O}{\|}}{C}-X$	Alkanoyl halide	–oyl halide (carbonyl halide)	halo carbonyl
6. $R-\overset{\overset{O}{\|}}{C}-NH_2$	Alkanamide	– amide (carboxamide)	carbamoyl
7. $R-C\equiv N$	Alkanenitrile	– nitrile (carbonitrile)	cyano
8. $R-\overset{\overset{O}{\|}}{C}-H$	Alkanal	– al (carbaldehyde)	formyl / oxo
9. $R-\overset{\overset{O}{\|}}{C}-R$	Alkanone	– one	oxo
10. $R-OH$	Alkanol	– ol	hydroxy
11. $R-SH$	Alkanethiol	– thiol	mercapto
12. $R-NH_2$	Alkanamine	– amine	amino

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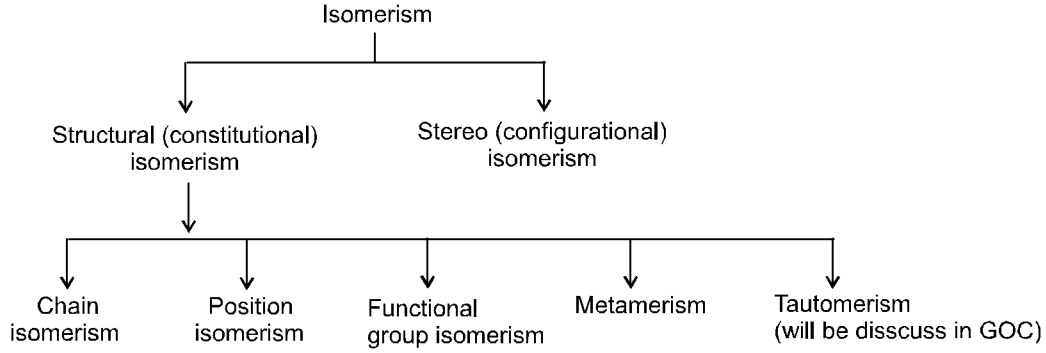
ISOMERISM

1. Isomerism :

The phenomenon of existence of 2 or more compounds possessing the same molecular formula but different properties is known as isomerism. Such compounds are known as isomers.

2. Classification of isomerism



3. Structural isomerism :

When two or more organic compounds have same molecular formula but different structural formula, (i.e., they differ in connectivity of atoms) then they are called **structural isomers** and the phenomenon is called structural isomerism

Structural isomers have always different IUPAC name

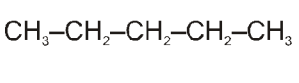
Ex. $\underset{\text{But}-2-\text{ene}}{CH_3-CH=CH-CH_3}$, $\underset{\text{But}-1-\text{ene}}{CH_3CH_2CH=CH_2}$ structural Isomers

4. Various types of structural isomers are :

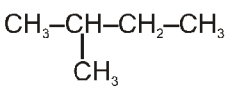
(a) Chain isomerism : Compounds having same molecular formula but different carbon skeletons (either difference in main chain or side chain) are known as chain isomers & phenomenon is known as chain isomerism.

Condition : They have same nature of locants.

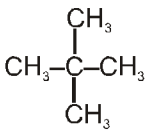
Ex.



(i) Pentane
(Main chain of 5C)



(ii) Isopentane
(Main chain of 4 C)



(iii) Neopentane
(Main chain of 3C)

(i), (ii) & (iii) are chain isomers :

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5. **(b) Position isomerism :** Compounds have same size of main chain & side chain along with same nature of locants but having different position of locants are known as position isomers & phenomenon is position isomerism.

Ex. $CH_3CH_2CH_2OH$ & $CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$

Propan-1-ol Propan-2-ol

Difference only in position of –OH group

6. **(c) Functional isomerism :** Compounds having same molecular formula but different functional group are known as functional isomers & phenomenon is functional isomerism.

Ex. CH_3-O-CH_3 & CH_3-CH_2-OH

Methoxymethane Ethanol

7. If one isomer has ring & another has aliphatic chain, also known as **ring chain isomerism**

8. 1°, 2°, 3° amines are functional isomers.

9. 1°, 2°, 3° amides are functional isomers.

10. Alcohol attached to sp^2 C is chemically different from alcohol attached to sp^3 C.

11. **Metamerism :** Compounds having same nature of functional groups but different nature of alkyl groups along that polyvalent functional group are known as metamers & phenomenon is metamerism.

Ex. $CH_3-CH_2-O-CH_2-CH_2$ & $CH_3-O-CH_2-CH_2-CH_3$

Ethoxy ethane

[Ethyl groups on either sides of O.]

Methoxy propane

[Methyl & propyl groups on either sides of O.]

12. Following compounds don't exist at room temperature therefore not considered as a structural isomer.

Sol. (i) $-\underset{|}{C}=\underset{|}{C}-OH$ **Vinylic alcohol**

(ii) $-\underset{\displaystyle NH_2}{\underset{|}{C}}-NH_2$ **Gemdimine**

(iii) $-\underset{\displaystyle OH}{\underset{|}{C}}-OH$ **Gemdiol**

(iv) $-\underset{\displaystyle OR}{\underset{|}{C}}-OH$ **Hemiacetal**

(v) R–O–O–R **Any peroxy compound**

(vi) $-C=\underset{\displaystyle NH_2}{\underset{|}{C}}-$ **Vinylic Amine**

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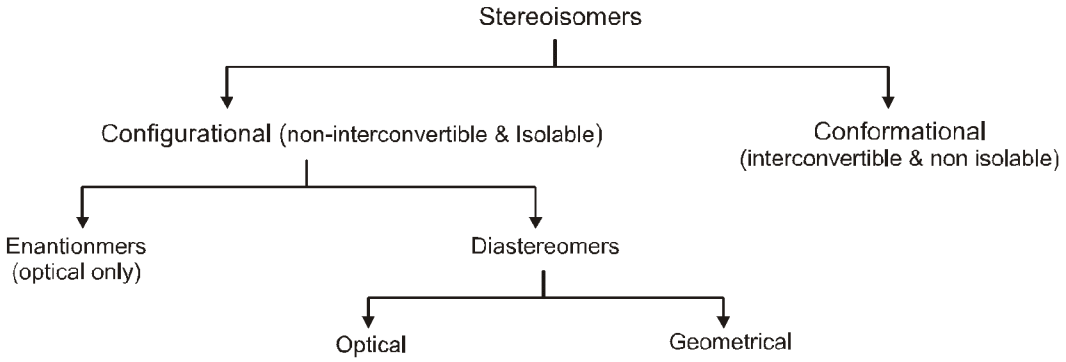
Introduction :

The particular kind of isomers that are different from each other only in the way the atoms are oriented in space are called stereoisomers. These isomers have same **connectivity** of atoms and groups. Stereoisomers have remarkably different physical, chemical and biological properties.

Classification of Stereoisomers :



1. Configurational Isomers :

(I) These isomers differ in the configuration (The spatial arrangement of atoms that characterises a particular stereoisomer is called its configuration).

(II) Configurational isomerism arises due to noninterconvertibility at room temperature. Since these are non interconvertible, therefore can be separated by physical or chemical methods.

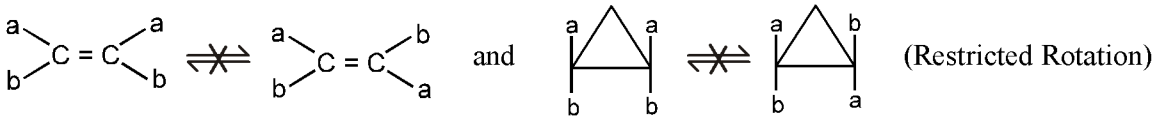
1.1 Geometrical isomerism :

D-1 Isomers which possess the same **molecular and structural formula** but differ in the arrangement of atoms or groups in space due to **restricted rotation** are known as geometrical isomers and the phenomenon is known as geometrical isomerism.

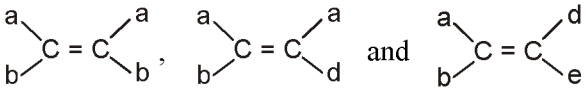
Conditions of geometrical isomerism :

(I) Geometrical isomerism arises due to the presence of a double bond or a ring structure (i.e. $>C=C<$, $>C=N-$, $-N=N-$ or ring structure)

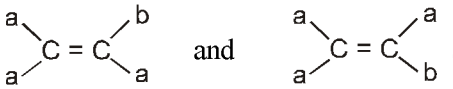
Due to the rigidity of double bond or the ring structure the molecules exist in two or more orientations. This **rigidity to rotation** is described as restricted rotation / hindered rotation / no rotation.



(II) Different groups should be attached at each doubly bonded atom.



On the other hand, following types of compounds can not show geometrical isomerism :



(III) Groups responsible to show geometrical isomerism must be nearly in the same plane.

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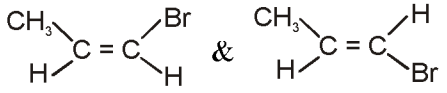
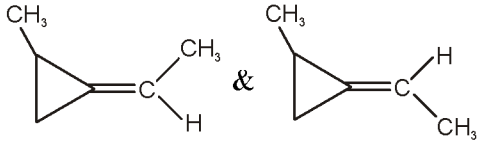
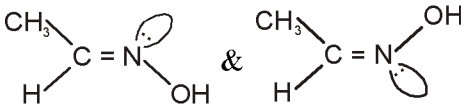
Examples of Geometrical isomers :

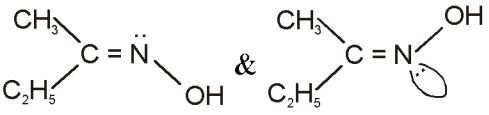
(I) Along $>C=C<$ bond

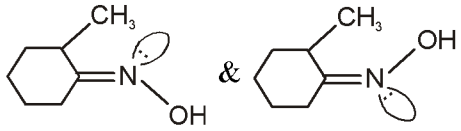
(a)  &

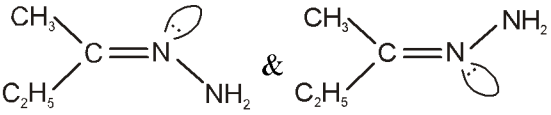
(b)  &

(II) Along $>C=N-$ bond

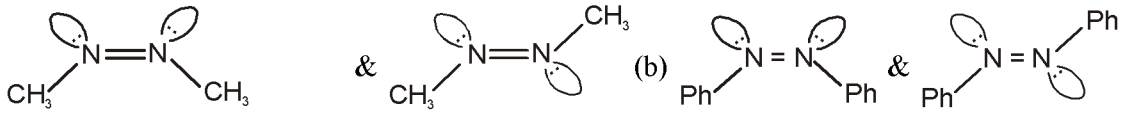
(a)  &

(b)  &

(c)  &

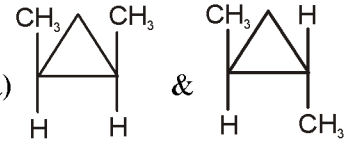
(d)  &

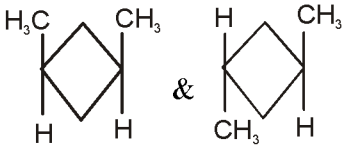
(III) Along –N=N– bond

(a) 

(b)

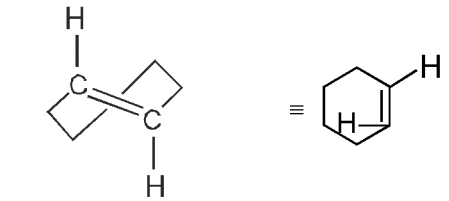
(IV) Along σ bond of cycloalkane

(a) 

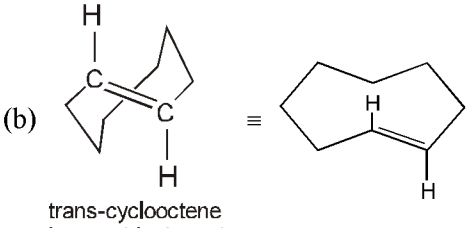
(b) 

(V) Along $>C=C<$ in ring structures :

Usually in cycloalkenes double bond has cis configuration. Their trans isomers do not exist due to large angle strain. But if the ring is large enough then the trans stereoisomer is also possible. The **smallest trans cycloalkene** that is stable enough to be isolated & stored is **trans-cyclooctene.**

(a) 

trans-cyclohexene
highly unstable (not exist)

(b) 

trans-cyclooctene
less stable than cis
(isolable at – 90°C)

Configurational nomenclature in geometrical isomerism :

Configuration	Criteria	Remarks
cis / trans	Similarity of groups	If the two similar groups are on same side of restricted bond the configuration is cis otherwise trans.
E/Z	Seniority of groups	If the two senior groups are on same side of restricted bond the configuration is Z (Z = zusammen = together) otherwise E (E = entgegen = opposite).

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Sequence rules (Cahn - Ingold - Prelog sequence rules) :

For deciding the seniority of groups following (CIP) rules are applied :

Rule I : The group with the first atom having higher atomic number is senior. According to this rule the seniority of atoms is :

$I > Br > Cl > S > F > O > N > C > H$

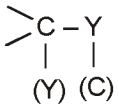
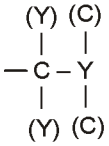
Rule II : The higher mass isotope is senior.

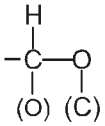
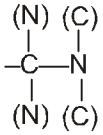
Thus (a) $-T > -D > -H$. (b) $-C^{14}H_3 > -C^{12}H_3$

Rule III : If the first atom of group is identical then second atom is observed for seniority.

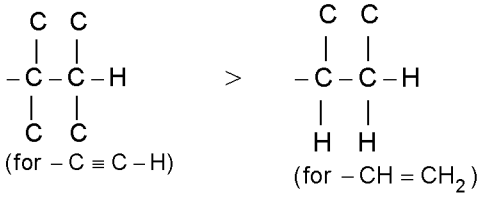
(a) $-CH_2Cl > -CH_2OH > -CH_2NH_2 > -CH_2CH_3 > -CH_3$

Rule IV : Groups containing double or triple bonds are assigned seniority as if both atoms were duplicated or triplicated that

>C=Y is written hypothetical as  & –C≡Y is written hypothetical as 

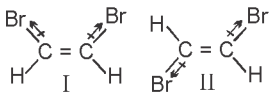
is written hypothetical as  & –C≡N is written hypothetical as 

Ex. To decide seniority of multiple bonded groups ($-C \equiv CH$, $-CH = CH_2$) their hypothetical equivalents are compared.



Rule V : Bond pair is senior to lone pair.

Physical Properties of Geometrical Isomers :

Physical properties		Remarks
Dipole moment	I > II	cis-isomer has resultant of dipoles while in trans isomer dipole moments cancel out
Boiling point	I > II	Molecules having higher dipole moment have higher boiling point due to larger intermoleculer force of attraction
Solubility (in H_2O)	I > II	More polar molecules are more soluble in H_2O.
Melting point	II > I	More symmetric isomers have higher melting points due to better packing in crystalline lattice & trans isomers are more symmetric than cis.
Stability	II > I	The molecule having more vander waal strain are less stable. In cis isomer the bulky groups are closer they have larger vander waals strain.

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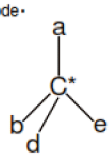
Chiral Carbon

If a carbon-atom is attached with four different group then It is known as asymmetric carbon atom or chiral carbon, which is represented as $*C_{abde}$.



Dissymmetry or Chirality:
The molecules or objects in which minimum three elements of symmetry (Centre of symmetry, Plane of symmetry and alternating axis of symmetry) are absent, called dissymetric molecules/objects.

Asymmetry:
When all the elements of symmetry are absent the molecule is said to be asymmetric.

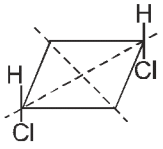
Dissymmetry, Chirality and Optical Activity:
Dissymmetry and Chirality is the minimum and sufficient condition for a molecule to be optically active.

Note : If a molecule contains only one asymmetric carbon atom, then the molecule as a whole becomes chiral and optically active and show optical isomerism.

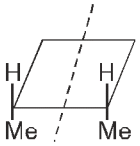
Symmetry elements :-

(1) Plane of symmetry (P.S.)
(2) Centre of symmetry (C.S.)
(3) Axis of symmetry
(4) Alternating axis of symmetry

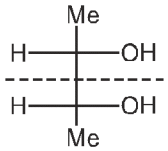
(1) Plane of symmetry : The imaginary plane which divides a molecule into two equal halves which are related as mirror image of each other is known as plane of symmetry



Plane of symmetry



Plane of symmetry



Plane of symmetry

Note : All the planar molecules always have plane of symmetry

(2) Centre of symmetry (COS) :
It is a point of rotary where if we draw the lines in opposite direction these lines meet to the same group at same distance .

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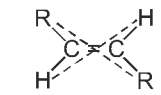
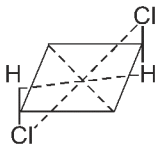
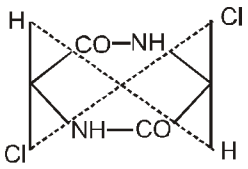


Centre of symmetry



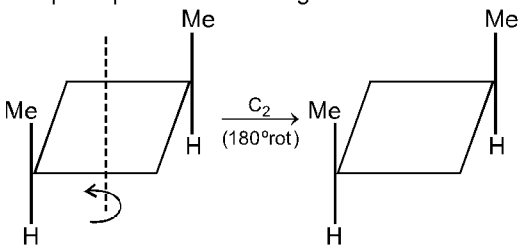
Centre of symmetry



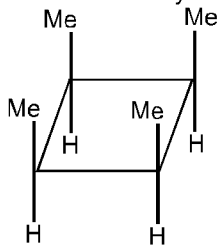
Centre of symmetry

(3) Axis of symmetry (C_n) : When the molecule is rotated by $\frac{360^\circ}{n}$ angle, and if the rotated molecule is superimposable on its original then the molecule is said to have n-fold axis of symmetry.

Ex.

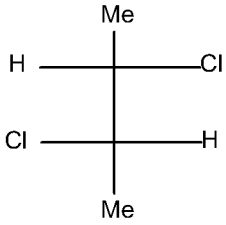


Ex.



(C_4 axis)

Ex.



(C_2 axis)

Configuration nomenclature in Optical Isomers :
(1) Absolute configuration(R/S Nomenclature)
(2) D- L System (Relative configuration)

(1) Absolute configuration(R/S Nomenclature) :

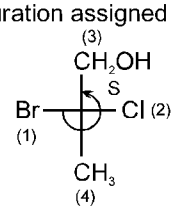
R/S Configuration in Fisher Projection Formula : R → Rectus S → Sinister

Step-1 : The priorities of groups which are attached with the asymmetric C-atom are assigned by sequence rule.

Step-2 : The lowest priority group is brought to the bottom of Fischer projection by two or even simultaneous exchanges.

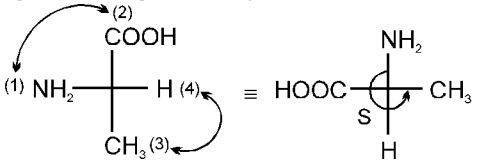
Step-3 : Then an arrow is drawn from first priority group to second priority group to third priority group. If the arrow is clockwise the configuration assigned to the projection is **R** & If it is anticlockwise the configuration assigned is **S**.

e.g.



Q. The configuration of given compounds is

(i)



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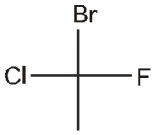
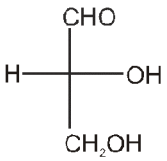
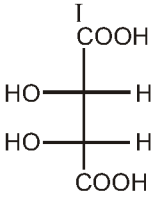
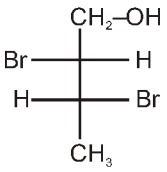
(ii) 

(iii) 

(iv) 

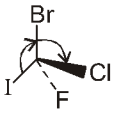
(v) 

Answers :

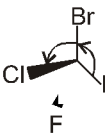
(i) S (ii) S (iii) R (iv) (2S, 3R) (v) (2R, 3R)

(II) **R/S Configuration in wedge–dash :**

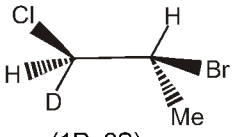
Step 1 : Decide the priority of groups by sequence rule.
Step 2 : Bring the lowest prior group to dash by even simultaneous exchanges.
Step 3 : Draw an arrow from first prior group to second prior group till third prior group.
Step 4 : If the direction of arrow is clockwise the configuration is R and if anticlockwise it is S.

Ex-1. 

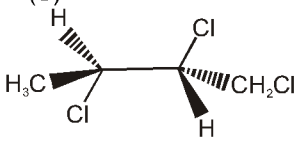
(R)

Ex-2. 

(S)

Ex-3. 

(1R, 2S)

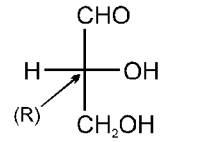
Ex-4. 

(2R, 3S)

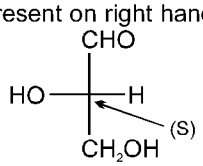
(2) D- L System (Relative configuration) :

(i) It is decided on correct Fisher Projection Formula Only.

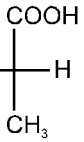
(ii) If any group except hydrogen are present on right hand side then nomenclature is **D,** Otherwise **L.**



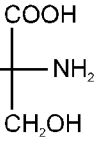
D-(+)-glyceraldehyde
– OH group on right side



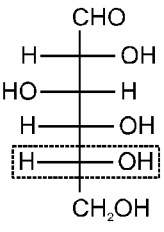
L-(–)-glyceraldehyde
– OH group on left side

Eg. **(i)** 

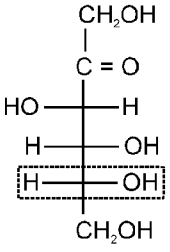
L-Alanine

(ii) 

D(+)-serine

(iii) 

D-(+)-Glucose

(iv) 

D-Fructose

Enantiomers:- The non super impossable **mirror-image stereoisomers** are called enantiomers.

Two types :-

(a) Dextrorotatory (b) Laevorotatory

Every enantiomer is optically active.

The two enantiomers can rotate the plane-polarised light with equal magnitude and opposite signs.

They have similar physical properties except the sign of optical rotation.

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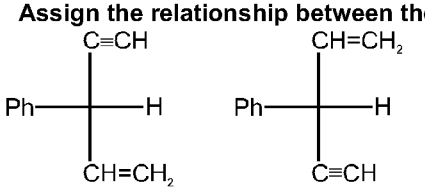
They have similar physical properties except the sign of optical rotation, so the can't seprate by fractional distillation.

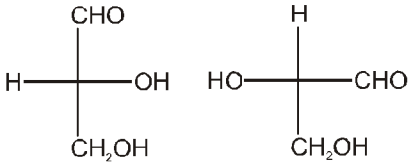
Non-superimposability of enantiomers: The dissymmetric molecules are always non-superimposable
on their mirror-image orientations.

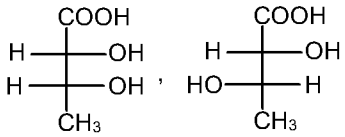
Diastereomers:- The optical isomer which are neither mirror image nor superimpossible to each other are called optical diastereomers.

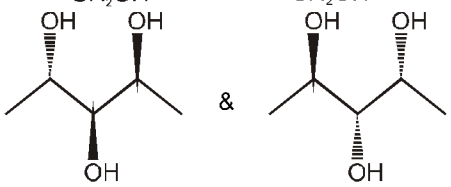
Optical Diastereomers have different physical properties, so these can be separated by normal physical methods of separation.

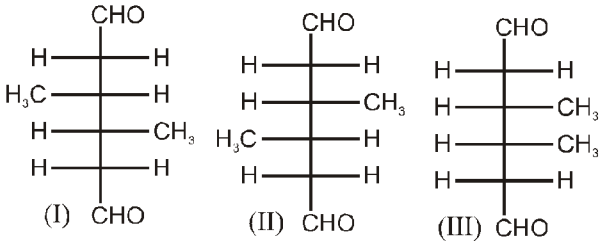
Q. **Assign the relationship between the following compounds.**

(i) 

(ii) 

(iii) 

(iv) 

(v) 

I & II.....

I & III.....

II & III.....

Answers:

(i) Enantiomer (ii) Identical (iii) Diastereomer (iv) Enantiomers

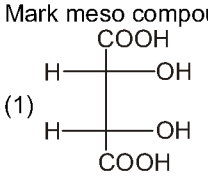
(v) I & II Enantiomers, I & III Diastereomers, II & III Diastereomers

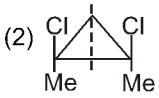
Meso compound:

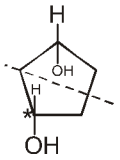
An optical inactive molecule whose at least one diastereomer is optically active.

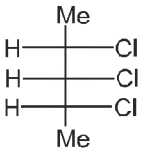
* Mirror image of meso compound is superimposable over each other & non resolvable.
* Molecule contains chiral centers & symmetry but optically inactive.

Q. Mark meso compound among followings

(1) 

(2) 

(3) 

(4) 

Ans. (1) (2) (3) and (4)

Separation of Racemic Mixture:

The enantiomers in a d ℓ pair have identical physical properties, so these cannot be separated by normal physical methods of separation. The special method is used for separation of d pair known as **optical resolution**.

→ **General scheme of resolution:**

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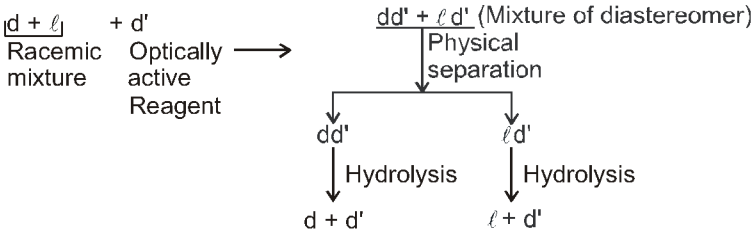


Optical purity and Enantiomeric excess :

$\therefore$ optical purity = enantiomeric excess

$\Rightarrow$ Optical purity = $\frac{\text{observed optical rotation}}{\text{optical rotation of pure enantiomer}} \times 100$

enantiomeric excess = $\frac{|d-\ell|}{d+\ell} \times 100 = \frac{\text{excess of one enantiomer over other}}{\text{entire mixture}} \times 100$

Q. If we have some 2-butanol with observed rotation of + 9.72, we compare this rotation with + 13.5 rotation of the pure (+) enantiomer.

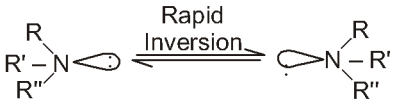
optical purity = $\frac{9.72}{13.5} \times 100 = 72\%$.

That means 72% is pure (+) 2-Butanol and 28% is (± mixture)
Total (+) isomer = 72 + 14 = 86%, (–) isomer = 14%

Ans. **14**

Optically active compounds without chiral carbon:-

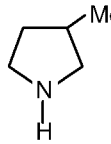
(1) **Asymmetric N* : -**



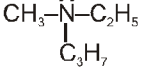
→ A nitrogen atom attached with 3 different groups (sp^3) is chiral, asymmetric.

→ In case of acyclic compounds, the **Nitrogen**, having trigonal pyramidal shape undergoes rapid inversion at room temperature. So at room temperature, everytime a racemic mixture of 'd' and '' forms exists. This racemic mixture can never be resolved at room temperature. Nitrogen converts into its enantiomer.

→ So, acyclic molecules with chiral nitrogen are chiral but optically inactive ($[\alpha] = 0$) and are non-resolvable. If asymmetric Nitrogen atom is present in ring structure, then inversion does not take place and such molecules are optically active $([\alpha] \neq 0)$.

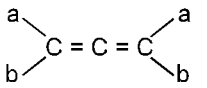
Eg. (1) 

$[\alpha] \neq 0$

(2) 

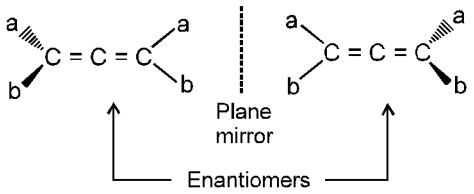
$[\alpha] = 0$

(2) **Case of allene :**

(a) Cumulene with even no. of π bonds : 

(allene)

Since the groups at the end of allene are in perpendicular plane, it will not show geometrical isomerism. The molecule lacks centre of symmetry as well as plane of symmetry. Overall the structure has molecular dissymmetry which is the sufficient condition for optical activity. The molecule will exist in two enantiomeric forms.



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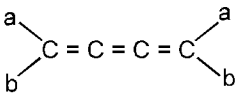
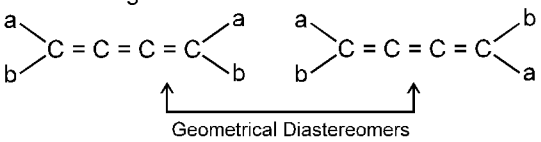
(b) Cumulene with odd no. of π bonds :



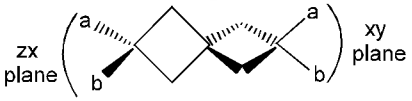
The groups at the end of cumulene structure lie in same plane (ZX plane). Therefore it will have a plane of symmetry (ZX plane). The molecules lacks molecular dissymmetry & it will not show optical activity. But the compound will exist in two geometrical diastereomeric forms.



(3) Case of spiranes :

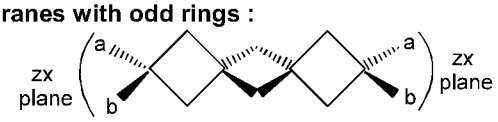
A similar case like cumulene is observed in spiranes. The spiranes with even rings and different groups at terminal carbons show optical activity & optical isomerism, while the spiranes with odd rings shows geometrical isomerism.

(a) spiranes with even rings :



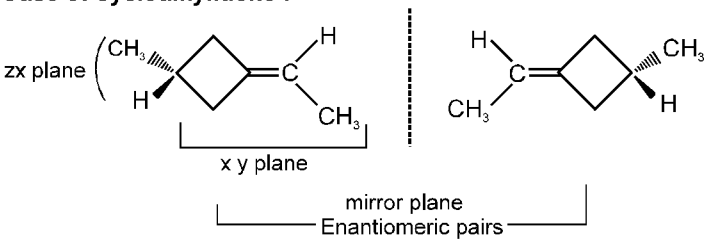
shows optical isomerism.

(b) spiranes with odd rings :



shows geometrical isomerism.

(4) Case of cycloalkylidene :



Calculation of total number of stereoisomers :

Case I : Molecules is not symmetrical

Total stereoisomers = 2^n

n is a stereocentre

Ex-1. $CH_3-\underset{Cl}{CH}-\underset{Cl}{CH}-CH_2-CH_3$

Total stereoisomers = $2^n = 2^2 = 4$

Ex-2. $CH_3-\underset{Cl}{CH}-\underset{Cl}{CH}-\underset{Cl}{CH}-CH_2-CH_3$

Sol. n = 3,

$2^3 = 8$

Case II : Molecules is symmetrical

(a) When even number of stereocentre are present

Total stereoisomers = $2^{n-1} + 2^{\frac{n}{2}-1}$

optical active isomers = 2^{n-1}

Meso isomers = $2^{\frac{n}{2}-1}$

Ex-1 $CH_3-\underset{OH}{CH}-\underset{OH}{CH}-CH_3$

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n = 2

Number of Optical stereoisomers = $2^{n-1} + 2^{\frac{n}{2}-1} = 3$

(b) When odd number of stereocentre are present

(i) When middle centre is pseudo chiral

Total stereoisomers = 2^{n-1}

optical active isomers = $(2)^{\frac{n-1}{2}}$

Meso isomers = $(2)^{\frac{n-1}{2}}$

(ii) Middle centre is geometrical centre

Total stereoisomers = $2^{n-1} + (2)^{\frac{n-1}{2}}$

Optical active isomers = 2^{n-1}

Meso isomers = $(2)^{\frac{n-1}{2}}$

Ex.2 $CH_3-\overset{*}{C}H(Cl)-\overset{*}{C}H(Cl)-\overset{*}{C}H(Cl)-CH_3$

n = 3

Number of stereoisomers = $2^{n-1} = 4$

Ex.3 $CH_3-CH(OH)-CH=CH-CH(OH)-CH_3$

Number of stereoisomers = 6

Optical active = 4

Meso = 2

CONFORMATIONAL ISOMERISM:

1. **Conformers / Rotamers or conformations:** The infinite number of spatial oreintation of molecule arises due to free rotation along a sigma covalent bond
2. **Conformational Isomers:** Those conformations which are most stable and have minimum P.E. are defined as conformational isomers. These are not true isomers and can never be isolated.
3. **Strains :**

   (i) Torsional Strain (eclipsing strain): It is defined as the electronic repulsion between the bond-pairs electrons of two adjacent eclipsed bonds. It is active at torsional angles 0°, 120°, 240° in which the molecule has eclipsed conformation. It is considered almost zero in the staggered conformation. (D.H.A. = 60°, 180°, 300°)

   (ii) Vander Waal strain: It is the repulsion between the group attached at adjacent bonds. The vander waal strain is maximum in eclipsed conformation, and minimum in anti conformation, while intermediate in gauche conformations.

   It is almost zero for H-atom since the sum of vander waal radii is less than internuclear distance between two H atom.

   (iii) Angle Strain: It arises due to distortion in normal bond-angles. In acyclic compounds, there is no distortion of bond-angle due to free rotation. So, all conformations have zero angle strain in acyclic compound.

4. **Conformational Analysis of Ethane : CH_3-CH_3**

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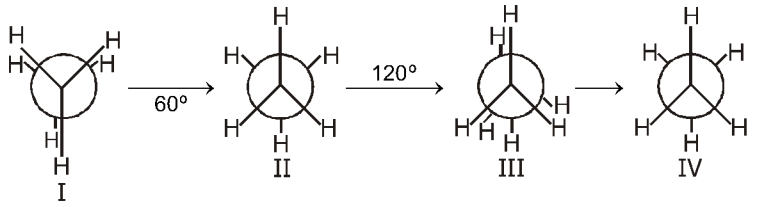
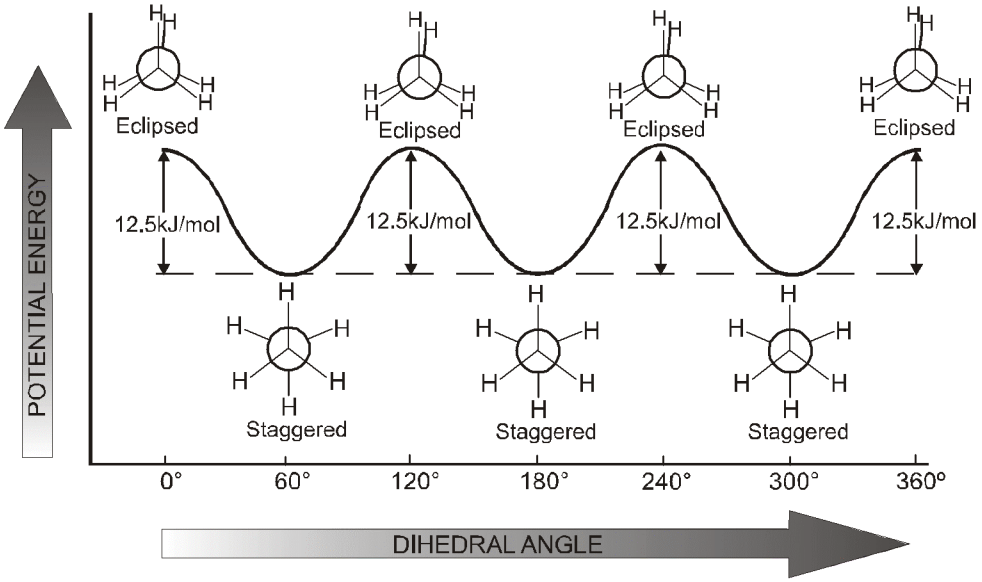
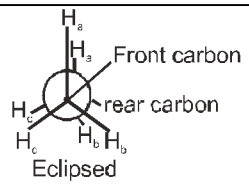
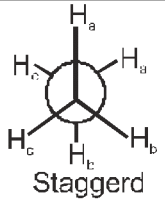


Eclipsed **Staggered** **Eclipsed** **Staggered**

Energy - Level Diagram of ethane



	CH_3–CH_3	Torsional Strain	Vander Waal strain	Stability
(a)		**Maximum**	**Zero**	**Staggered > Eclipsed**
(b)		**Zero**	**Zero**	

5. Conformational Analysis of Butane : $CH_3–CH_2–CH_2–CH_3$ ($C_2–C_3$)

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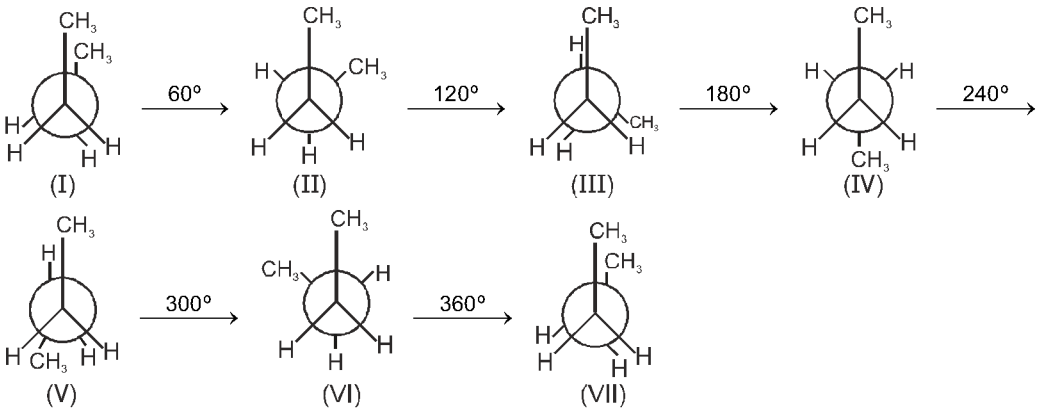


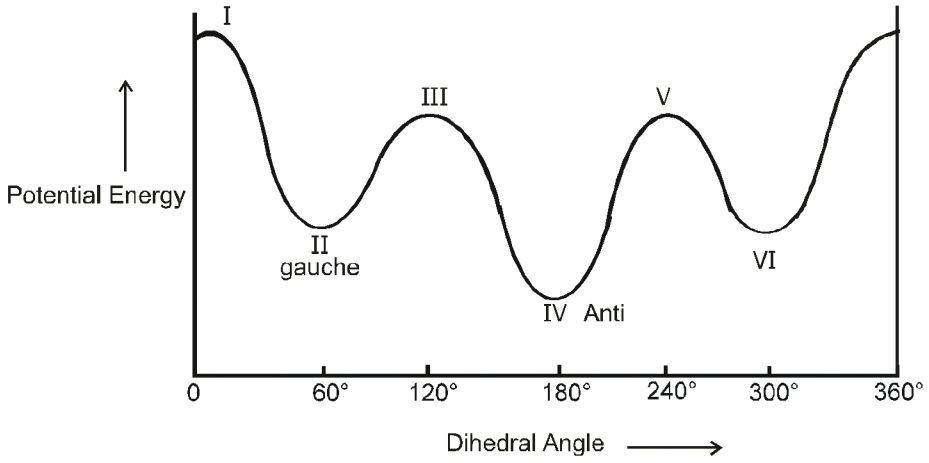
I/VII = Fully eclipsed II/IV = Gauche form

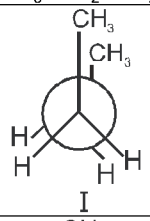
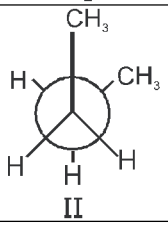
III/V = Partially eclipsed IV = Anti form

Stability Order : IV > II > III > I Anti > Gauche > Partially eclipsed > Fully eclipsed

P.E. Order : IV < II < III < I

Potential Energy Diagram of butane :



	Example	Torstional Strain	Vander Waal strain	Stability
(I)	$CH_3–CH_2–CH_2–CH_3$			Anti > Gauche > Partial eclipsed > Eclipsed
(a)	 Fully eclipsed I	Maximum	Maximum	
(b)	 Gauche form II	Absent	Present between two CH_3 group	

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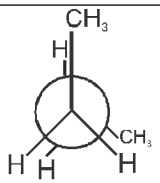
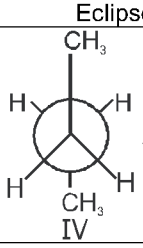
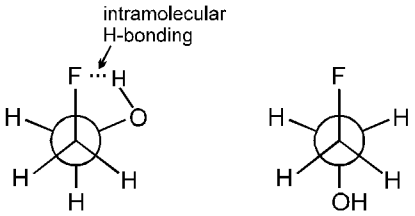
(c)	 Partially (III) Eclipsed	Maximum	Present between CH_3 and H group	
(d)	 Anti form IV	Absent	Absent	

6. Conformational Analysis in case of intramolecular hydrogen bonding :

In case of G–CH_2–CH_2–OH, where G=–OH, –NH_2, –F, –NR_2, –NO_2, –COOH, –CHO the Gauche form is more stable than the anti form due to intramolecular hydrogen bonding i.e.

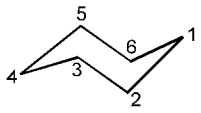


Gauche form anti form

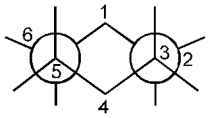
Stability : Gauche form > anti form.

7. Conformational analysis of cyclohexane :

(I) Chair form: The most stable conformation of cyclohexane ring is the chair conformation. In this non-planar structure the C–C bonds angles are all 109.5°. This conformation is free from all the strains (like angle strain and torsional strain.)

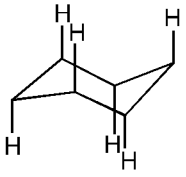


chair conformation

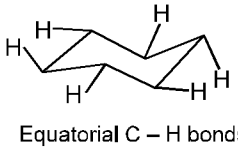


chair form (all staggered bonds)

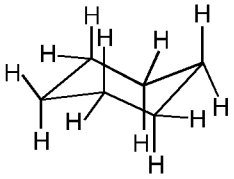
Axial and equatorial bonds in chair form of cyclohexane :



Axial C – H bonds



Equatorial C – H bonds



Axial & equatorial bonds together

The 12 hydrogen atoms of chair conformation of cyclohexane can be divided into two groups. Six of the hydrogens, called axial hydrogens, hence their bonds parallel to a vertical axis that passes through the rings centre. These axial bonds are directed up & down on adjacent carbons. The second set of six hydrogens called equatorial hydrogens are located approximately along the equator of the molecule.

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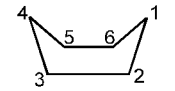
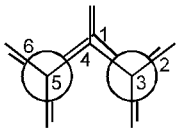
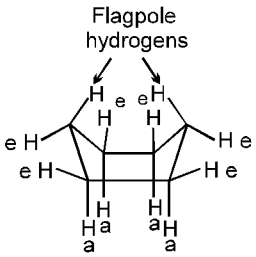
(II) Boat form: Another conformation which is known as boat conformation has exactly eclipsed conformations.



Boat conformation



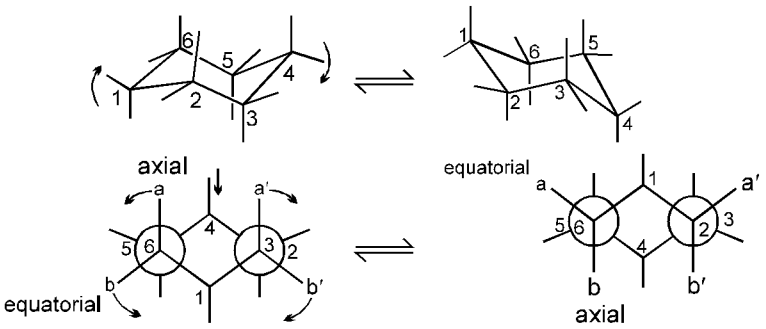
Boat conformation
(all eclipsed bonds)



In boat form of cyclohexane **6 hydrogens are equatorial**, **4 hydrogens are axial** and **two hydrogens are flagpoles.** It is an **unstable conformation** of cyclohexane due to **torsional strain** among axial hydrogens and due to **van der waals strain** caused by crowding between the "flagpole" hydrogens.

Conformational inversion (Ring flipping) in cyclohexane :

Like alkanes cyclohexane too is conformationally mobile. Through a process known as ring inversion, chair-chair interconversion, or more simply ring flipping one chair conformation is converted to another chair through the half chair, twist boat and boat form.



By ring flipping all axial bonds convert to equatorial and vice-versa. The activation energy for cyclohexane ring inversion is 45 kJ/mol. It is a very rapid process with a half-life of about 10^{-5} sec. at 25°C.

The relative energy profile of various conformations of cyclohexane

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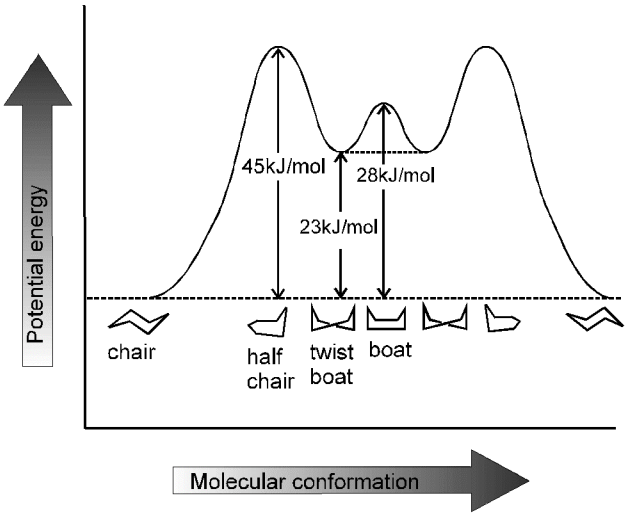
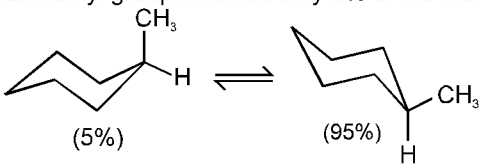


because of the greater stability of the chair form, more than 99% of the molecules are estimated to be in a chair conformation of any given moment. **Twist boat form of cyclohexane is chiral.**

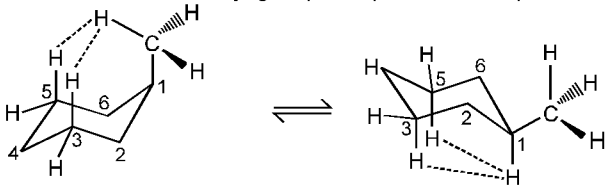
Conformational analysis of monosubstituted cyclohexanes :

In ring inversion in methylcyclohexane the two chair conformations are not equivalent. In one chair the methyl group is axial ; in the other it is equatorial. At room temperature 95% of the methylcyclohexane exist in **equatorial** methyl group whereas only 5% of the molecule have an axial methyl group.



1, 3-diaxial repulsion :

A methyl group is less crowded when it is equatorial than when it is axial. The distance between the axial methyl groups at C-1 and two hydrogens at C-3 & C-5 is less than the sum of their vander waal radii which causes vander waal strain in the axial conformation this type of crowding is called 1, 3-diaxial repulsions. When the methyl group is equatorial, it experience no significant crowding.

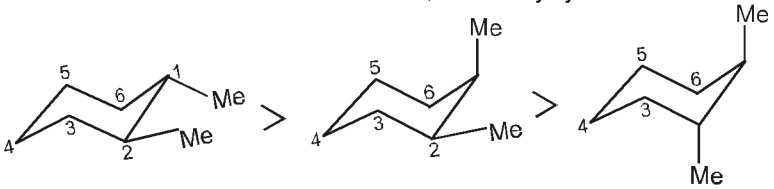


Vander waals strain between hydrogen of axial CH_3 and axial hydrogens at – C_3 and C_5

No vander waals strain between axial hydrogen at C–1 and axial hydrogens at C–3 & C-5

Q. Draw the most stable conformation of 1, 2-dimethylcyclohexane.

Ans.



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CHEMICAL BONDING

Octet rule : "Tendency of atoms to have eight electrons in their outermost shell is known as Lewis octet rule". To achieve inert gas configuration, atoms lose, gain or share electrons.

To write the Lewis dot structure, following steps are to be followed :

(i) The total number of electrons are obtained by adding the valence electrons of combining atoms.
(ii) For Anions, we need to add one electron for each negative charge.
(iii) For cations, we need to subtract one electron for each positive charge.
(iv) After then, the central atom is decided.

To decide central atom, following steps are followed :

(1) In general the least electronegative atom occupies the central position in the molecule/ion.
(2) Generally the atom which is/are less in number acts as central atom.
(3) Generally central atom is the atom which can form maximum number of bonds (which is generally equal to the number of electrons present in the valence shell of the atom).
(4) Atom of highest atomic number or largest atom generally acts as central atom.
(5) Fluorine and hydrogen can never act as central atoms.
(6) After accounting for the shared pairs of electrons for single bonds, the remaining electron pairs are either utilized for multiple bonding or remain as the lone pairs.

Formal charge

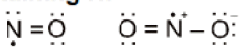
Formal Charge : The formal charge of an atom in a molecule or ion is defined as the difference between the number of valence electrons of that atom in an isolated or free state and the number of electrons assigned to that atom in the lewis structure.

$$\text{Formal charge (F.C.) On an atom in a Lewis structure} = \left[\text{Total number of valence electron in the free atom}\right] - \left[\text{Total number of non bonding (lone pair) electrons}\right] - (\tfrac{1}{2})\left[\text{Total number of bonding (shared) electrons}\right]$$

Limitations of Octet Rule with examples :

a. **Incomplete octet of central atom :**
Eg. BeF_2, BF_3, $AlCl_3$

b. **Odd-electron molecules :**
Eg. NO, NO_2, ClO_2, ClO_3
Draw LDS of those containing N.



c. **Expanded octet :**
Eg. PF_5, SF_6, PCl_5, SO_3, SO_2, H_2SO_4 etc.
Draw LDS of compounds of P & F, S & F (with no LP on S) and oxyacid of S with 12 e around S atom.

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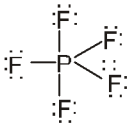
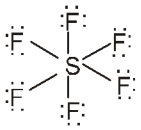
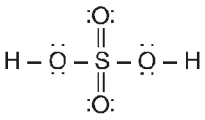


PF_5



SF_6



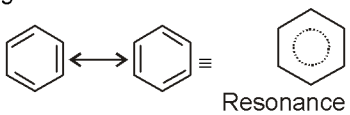
H_2SO_4

d. **Noble gases compounds:**
Eg. XeF_2, KrF_2, $XeOF_4$ etc.

Concept of Resonance :
According to the concept of resonance, whenever a single Lewis structure cannot describe a molecule accurately, a number of structures with similar energy, positions of nuclei, bonding and non-bonding pairs of electrons are taken as the canonical structures of the hybrid which describes the molecule accurately.

Definition of Resonance : Resonance may be defined as the phenomenon in which two or more structures involving identical position of atom can be written for a particular compound.

Resonance Hybrid : It is the actual structure of all different possible structures that can be written for the molecule without violating the rules of covalence maxima for the atoms.



Resonance hybrid

Bond order in oxoanions and corresponding acids :

- Bond order = $\frac{\text{Total No. of bonds formed between two atoms in all structures}}{\text{Total No. of resonating structures}} = \frac{\sigma+\pi}{\sigma}$

VBT, overlapping of orbitals
Orbital Overlap Concept

- Acc. to orbital overlap concept, the formation of a covalent bond between two atoms results from overlapping of valence shell orbitals of atoms leading to pairing of electrons present in the valence shell having opposite spins.
- The extent of overlap decides the strength of a covalent bond.
- Greater the overlap, stronger is the bond formed between two atoms.

Types of Overlapping and Nature of Covalent Bonds
The covalent bond may be classified into two types depending upon the types of overlapping :
(i) sigma (σ) bond, (ii) pi (π) bond, (iii) delta (δ) bond

Sigma (σ) bond : This type of covalent bond is formed by the end to end (head-on) overlap of bonding orbitals along the internuclear axis. This is called as head on overlap or axial overlap. This can be formed by any one of the following types of overlapping of s/p atomic orbitals.

● **s-s overlapping :** In this case, there is overlap of two half filled s-orbitals along the internuclear axis as shown below :

s–orbital + s–orbital → s–s overlapping

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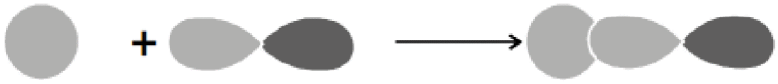
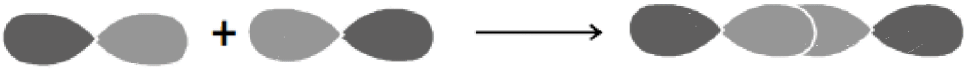
● **s-p overlapping:** This type of overlap occurs between half filled s-orbital of one atom and half filled p-orbital of another atom.



s–orbital p–orbital s–p orbital

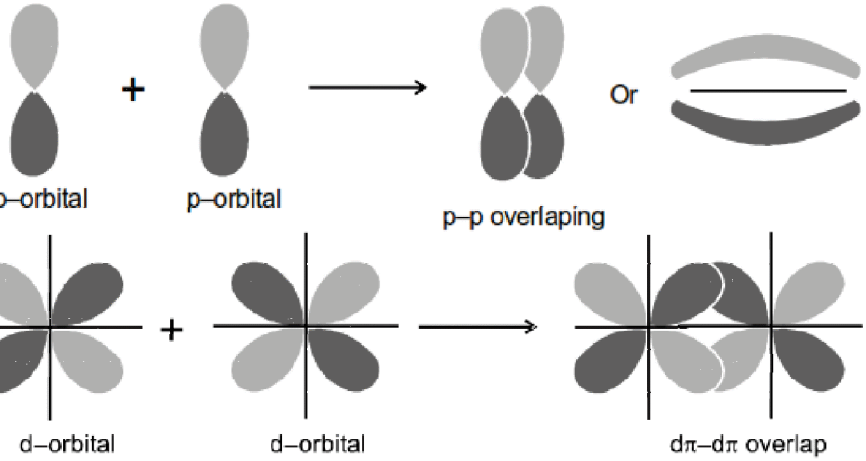
● **p-p overlapping** : This type of overlap takes place between half filled p-orbitals of the two approaching atoms.

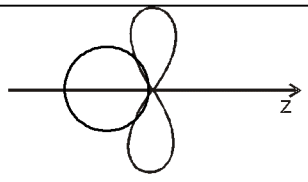
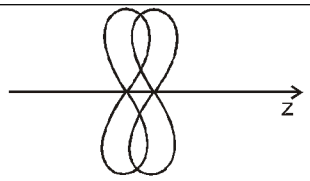


p–orbital p–orbital p–p overlaping

. **pi(π) bond :** In the formation of π bond, the atomic orbitals overlap in such a way that their axes remain parallel to each other and perpendicular to the internuclear axis. The orbitals formed due to sidewise overlapping consists of two saucer type charged clouds above and below the plane of the participating atoms.

Eg. p-p overlapping, p-d overlapping, d-d overlapping.



S. No.	Atomic orbitals involved in overlap	Structure of orbital overlap (INA = Z-axis)	Nature of Bond (σ / π / δ) / no bond
1.	s and p_x		no bond
2.	p_x and p_x		π bond

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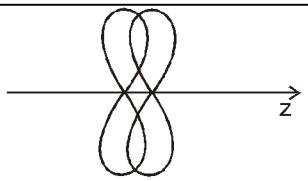
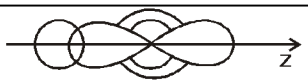
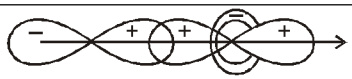
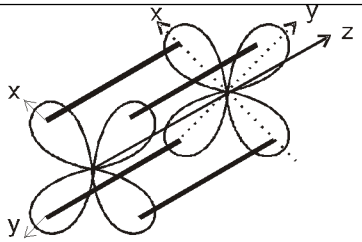
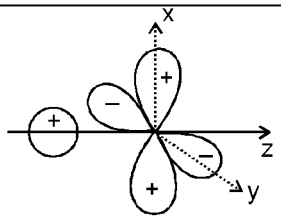
3.	p_y and p_y		π bond
4.	p_z and p_z		σ bond
5.	s and d_{z^2}		σ bond
6	p_z and d_{z^2}		σ bond
7.	$d_{x^2-y^2}$ and $d_{x^2-y^2}$		δ bond
8.	s and $d_{x^2-y^2}$		no overlap

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1. The shape of a molecule depends upon the **number of valence shell electron pairs** around it.

2. The valence shell electron pairs tend to occupy those positions which **minimize repulsion.**

3. Give three examples of each type of shape

S.No.	Shape	Example
1.	Linear	$BeCl_2$, NO_2^+, XeF_2, ICl_2^-
2.	Bent/Angular	SO_2, NO_2^-, H_2O
3.	Trigonal Planar	BF_3, CH_3^+, CO_3^{-2}, NO_3^-
4.	Tetrahedral	CCl_4, CH_4, SiH_4
5.	Trigonal Pyramidal	NH_3, $SnCl_3^-$, SO_3^{-2}
6.	Trigonal bipyramidal	PCl_5, SbF_5, PF_2Cl_2, XeO_3F_2
7.	See-saw	SF_4, SF_2Cl_2, XeO_2F_2
8.	T- shaped	ClF_3, XeF_3^+, BrF_3, $XeOF_2$
9.	Octahedral	SF_6, $TeCl_6$, XeO_6^{-4}
10.	Square Pyramidal	IF_5, XeF_5^+, $XeOF_4$
11.	Square Planar	XeF_4, $[ICl_4]^-$

4. XeF_6 is found in **distorted octahedral** geometry.

5. In PF_5 all five bond lengths are equal because of **pseudo-rotation.**

6. In PCl_5 axial bond lengths are **longer** than equatorial bond lengths.

7. Define Hybridization.

   The **intermixing of atomic orbitals of nearly same energy** to produce new orbitals of same energy and shape is known as hybridization.

8. The hybrid orbitals are always equivalent **in energy and shape.**

9. The hybrid orbitals are **more** effective in formation of **stable** bonds than pure atomic orbitals.

10. Which d-orbitals are involved in sp^3d, sp^3d^2 and sp^3d^3 hybridisation?

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Sol: sp^3d - d_z^2

sp^3d^2 - d_z^2, $d_{x^2-y^2}$

sp^3d^3 - d_z^2, $d_{x^2-y^2}$, d_{xy}

11. sp^3d hybridization is considered to be a combination of **sp^2 and pd** hybridization.

12. Steric number is the sum **of number of lone pairs and bond pairs** on a particular atom.

13. Steric number is used to calculate **hybridization.**

14. Steric number of following species is

ClF_3, XeF_2, XeF_4, CO_2, IF_7.

Ans: 5,5,6,2, and 7.

15. Consider the following table:

S.No.	l.p. + σ bond	Total no. of l.p.	Shape
1.	5	**3**	Linear
2.	**5**	1	See-saw
3.	4	**2**	Bent shape
4.	**6**	2	Square Planar
5.	5	**2**	Bent "T" Shape

16. Identify planar species among the following:

SF_4, BrF_3, XeF_2, IF_5, SbF_4^-, SF_5^-, SeF_3^+, CH_3^+, PCl_4^+.

Sol: BrF_3, XeF_2, CH_3^+.

17. What are the factors affecting bond angle?

Ans: (i) Hybridisation (ii) No. of lone pairs on central atom

(iii) Electronegativity of central atom

(iv) Electronegativity of terminal atom

(v) Drago's rule

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19. According to Bent's rule higher electronegative atoms prefer to be in orbitals having less **s- characters.**

20. The bond angles of NH_3, NH_4^+ and NH_2^- are in the order:

Sol: $NH_4^+ > NH_3 > NH_2^-$.

21. The bond angles of NO_2, NO_2^+ and NO_2^- are in the order:

Sol: $NO_2^+ > NO_2 > NO_2^-$

22. The bond angles of BF_3, PF_3 and ClF_3 are in the order:

Sol: $BF_3 > PF_3 > ClF_3$

23. The bond angles of NF_3, NCl_3 and NH_3 are in the order:

Sol: $NCl_3 > NH_3 > NF_3$.

24. The bond angles of PCl_3, NCl_3 and BCl_3 are in the order:

Sol: $BCl_3 > NCl_3 > PCl_3$.

25. The bond angles of OF_2, OH_2 and OCl_2 are in the order:

Sol: $OCl_2 > H_2O > F_2$.

26. $F - Xe - F$ bond angle in XeO_2F_2 is approximately **180°**.

27. In hydrocarbon C_3Hm, X-ray diffraction (which does not see H atoms) finds a $C - C - C$ Bond angle of about 120°. The value of 'm' must be **6.**

28. H2O2 is found in **open book** type structure.

29. What are the factors affecting the bond length?

Ans: (i) Size and radius of atom

(ii) Electronegativity

(iii) Bond order

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1. When in a species an atom has a l.p. and its adjacent atom has appropriate vacant orbital then **coordinate** bond is formed, which is known as **backbonding.**

2. Back bonding increases the **bond strength** and decreases **bond length.**

3. The extent of back bonding is higher if orbitals involved are of **smaller** size.

4. Among boron halides, extent of back bonding is $BF_3 > BCl_3 > BBr_3$.

5. Among the atoms of third period, with no lone pair on them, the order of extent of back bonding follows the order : **Si < P < S < Cl**

6. The electron deficient species which can accept l.p. are known as **Lewis acid** whereas l.p. donor species is known as **Lewis Base.**

7. The extent of overlapping $\propto \frac{1}{Lewis\ acid\ charcter}$.

8. Between BF_3 and $[BF_4]^-$, the B-F bond length of BF_3 is **less** because of **back bonding.**

9. Trisilylamine i.e. $(SiH_3)_3N$ is a planar molecule with sp2 hybridisation, whereas $(CH_3)_3N$ is a pyramidal molecule with sp3 hybridisation.

10. In following species, no. of $d\pi - p\pi$ bonds is equal to :

    (i) SO_2 (ii) SO_3 (iii) H_2SO_4 (iv) $HClO_4$

    (v) H_3PO_4

Sol: (i) 1 (ii) 2 (iii) 2 (iv) 3 (v) 1.

11. Between PH_3 and NH_3, **NH_3** is better base.

12. Anhydrous $AlCl_3$ exist in a **dimer** form as **Al_2Cl_6**. The hybridisation of Al in anhydrous $AlCl_3$ is **sp^3**.

13. Solid **$BeCl_2$** exist in polymer form as **$(BeCl_2)_n$** . The hybridisation of Be in solid $BeCl_2$ is **sp^3**.

14. Solid ICl_3 exist in a **dimer** form as **I_2Cl_6**. The hybridisation of 'I'atom in solid ICl_3 is **sp^3d^2**.

15. Electron deficient bonding has **3c-2e** type bonds.

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16. B_2H_6 has **four** 2c-2e type bonds and **two** 3c-2e type bonds.

17. In B_2H_6, the bond angle between terminal 'H' atoms is **greater** than bridging 'H' atoms.

18. Solid BeH_2 is found as polymer chain as $(BeH_2)_n$, linked by **banana bond** or **electron deficient** bond.

19. $Al(CH_3)_3$ exist as $Al_2(CH_3)_6$, joined by **banana bond** or **electron deficient** bond.

20. Intermolecular H- bonding **increases** boiling point.

21. Among HF, H_2O and NH_3, the strongest H-Bonding is shown by **HF.**

22. Between dimethyl ether and ethyl alcohol, **ethyl alcohol** has higher boiling point.

23. Between D_2O and H_2O, **D_2O** is more viscous because of H-bonding.

24. Between ice and water, **ice** has lower density because of H-bonding.

25. Water can form maximum **four** H-bonds.

26. Among the following, which has intramolecular H-bonding :

(i) ortho-Nitrophenol (ii) para-nitrophenol (iii) H_2O (iv) Boric acid

(v) Persulphate ion (vi) Salicylaldehyde (vii) HCN (viii) NH_3

(ix) CH_3OH (x) Chloral Hydrate

Ans: (i), (v), (vi), (x).

27. Between Cl_2 and HCl, **HCl** has higher boiling point because of **dipole-dipole** attraction.

28. If type is same, then strength of van der Waal force is directly proportional to **Molecular mass.**

29. Among H_2O, H_2S, H_2Se and H_2Te, the order of boiling point is **$H_2O > H_2Te > H_2Se > H_2S$.**

30. Among HF, HCl, HBr and HI, the order of boiling point is **HF > HI > HBr > HCl.**

31. Among NH_3, PH_3, AsH_3 and SbH_3, the order of boiling point is **$SbH_3 > NH_3 > AsH_3 > PH_3$.**

32. The intermolecular force in alkanes is **dispersion** forces or London forces.

33. When NaCl is dissolved in water, then interaction between Na^+ ion and water is **ion-dipole** attraction.

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1. According to MOT, formation of molecular orbitals is based upon **Linear Combination of Atomic Orbitals (LCAO)** method.

2. According to MOT, the atomic orbitals of **comparable** energy and proper **symmetry** combine to give molecular orbitals.

3. An atomic orbital is monocentric whereas a molecular orbital is **multicentric/polycentric.**

4. Bonding molecular orbitals are of **lower** energy than atomic orbitals whereas antibonding molecular orbitals are of **higher** energy than atomic orbitals.

5. Those orbitals which have centre of symmetry (COS) are known **gerade** orbitals whereas those which do not have COS are known as **ungerade** orbitals.

6. Sigma (σ) molecular orbitals are symmetrical around **the bond-axis.**

7. Give the order of energy of atomic orbitals for species like Be_2, B_2, C_2, N_2 ?

Ans: $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2P_x = \pi 2P_y) < \sigma 2P_z < (\pi^* 2P_x = \pi^* 2P_y) < \sigma^* 2P_z$

8. Give the order of energy of atomic orbitals for species like O_2, F_2 ?

Ans: $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2P_z < (\pi 2P_x = \pi 2P_y) < (\pi^* 2P_x = \pi^* 2P_y) < \sigma^* 2P_z$

9. Which of the following orbitals are gerade?

$\sigma 1s, \pi 2P_x, \pi^* 2P_x, \sigma 2P_z, \sigma^* 2P_z, d_{xy,}\ P_x, 1s.$

Ans: $\sigma 1s, \pi^* 2P_x, \sigma 2P_z, d_{xy,}\ 1s.$

10. The molecular orbitals like the atomic orbitals are filled according to the **Aufbau principle**, **Pauli's exclusion principle** and **Hund's rule of maximum multiplicity.**

11. According to MOT, Bond order is defined as

$$Bond\ \ Order = \frac{n_b - n_a}{2}$$

Where, n_b = **Number of electrons in bonding orbitals**

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n_a = **Number of electrons in antibonding orbitals**

12. Among following species, how many have fractional bond order:

(i) O_2^+ (ii) CO (iii) B_2 (iv) O_2^- (v) NO^+ (vi) He_2^+

(vii) C_2^{+2} (viii) CN^- (ix) N_2^-

Ans: (i), (iv), (vi), (ix)

13. If Hund's rule is violated, then which of the following species would be diamagnetic?

$B_2, O_2, N_2^-, C_2, NO, OF, N_2^{-2}, BN.$

Ans: $B_2, O_2, C_2, N_2^{-2}, BN.$

14. Find LUMO of following species:

C_2, N_2, He_2^+ and $NO.$

Ans: $\sigma 2P_z, (\pi^* 2P_x \text{ or } \pi^* 2P_y), \sigma 2s, \sigma^* 2P_z.$

15. Although O_2 is paramagnetic, still it is colorless gas. Explain.

Sol: Because of large energy gap between HOMO and LUMO, absorption in UV region. Therefore it is colorless.

16. Give the order of bond energy for following

$N_2^+, N_2^{-2}, N_2^-, N_2.$

Ans: $N_2 > N_2^+ > N_2^- > N_2^{-2}.$

17. Which of the following species show sp mixing?

$B_2, O_2^+, C_2, C_2^{-2}, F_2.$

Sol: $B_2, C_2, C_2^{-2}.$

18. Arrange the following compounds in the increasing order of O-O bond length:

(i) O_2 (ii) $O_2[BF_4]$ (iii) KO_2

Ans: (ii) < (i) < (iii)

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19. Find number of nodal planes in each of following orbital

$\sigma 1s, \pi 2P_x, \sigma^* 2P_z, \pi^* 2P_y$.

Ans: 0, 1, 2 and 3 respectively.

20. Which of the following statements are incorrect :

S_1 : Bond order of O_2^- is more than O_2^+.

S_2 : NO^+ is more stable than N_2^+.

S_3 : Bond length in N_2^+ is less than N_2.

S_4 : O_2 has lower dissociation energy than O_2^+.

S_5 : Bond length B_2 is more than B_2^-.

S_6 : He_2^+ has the same energy as two isolated He atoms.

Ans: S_1, S_3 and S_6.

21. Why Br_2 is coloured?

Ans: The difference in energy between HOMO and LUMO is such that lies in visible region.

22. Assuming 2s-2p mixing is not there, then which of the following species are paramagnetic?

Be_2, B_2, C_2, N_2, O_2, NO, F_2.

Ans: C_2, O_2, NO.

23. Which of the following species has bond order ≥1.5?

(a) NO_3^- (b) CO_3^{-2} (c) F_2 (d) Cl_2

(e) Br_2 (f) O_2^{-2} (g) O_2^- (h) N_2^-

(i) O_2^{+2} (j) Li_2^+

Ans: (g), (h), (i)

24. Give the increasing order of $O-O$ bond length among following:

H_2O_2, O_2 and O_3.

Ans: $O_2 < O_3 < H_2O_2$.

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25. Which among the following have identical bond order:

(i) CN^- (ii) O_2^- (iii) NO^+ (iv) CN^+

Ans: (i) and (iii).

26. What is the increasing order of bond length among the H_2, H_2^+, H_2^-.

Ans: $H_2 < H_2^+ < H_2^-$.

27. According to band theory, the **highest energy** electrons of metallic crystals occupy either a **partially filled valence band** or a **filled valence band** that overlaps with an empty conduction band.

28. A band within which (or into which) electrons must move to allow electrical conduction is known as **conduction band.**

29. In an insulator, band gap between **conduction band and valence band** is high.

30. According to electron sea model, the valence electrons of the interacting metals atoms **delocalize**. In other words, instead of orbiting their respective metal atoms, they form a "sea" of electrons that surrounds th**e positively charged kernels** of interacting metal ions.

31. Write the factors affecting the strength of metallic bond?

Ans: 1. Effective nuclear charge; metallic bonding $\propto$ effective nuclear charge

2. Number of delocalized electrons per atom;

metallic bonding $\propto$ number of delocalized electrons

3. The size of ion. metallic bonding $\propto \frac{1}{size}$

32. Between Alkali metals and alkaline earth metals, which have higher melting point and why?

Ans: Alkaline earth metals should have higher melting point as they contain 2 electrons for metallic bonding whereas alkali metals contain only one. Beside that alkaline earth metals are smaller in size also.

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33. Why transition metals have high heat of atomization?

Ans: Transition metals may use inner unpaired d-electrons along with the outer s-electrons for bonding , as (n-1)d and ns have nearly same energy. So they have a large number of metallic bonds per atoms (always more than two).

34. Why iron is harder than sodium?

Ans: Iron has more number of metallic bonds and because of smaller size its metallic bonds are stronger also.

35. What is the trend and reason of trend of melting point in transition series?

Ans: As the number of unpaired d-electrons initially increase, melting point also increases. It faces a dip in between (due to half-filled stable configuration) and finally decreases as number of unpaired d-electrons decreases.

36. Give four properties of metallic bonding.

Ans: 1. Metallic bond is non-directional in nature.

2. Metallic bonds are weaker than covalent bond.

3. The valence electrons are mobile in a metallic bond.

4. Metallic bond strength is inversely proportional to size of an atom

37. How metallic conductors and semiconductors are affected by heating?

Ans: As temperature increases metallic conduction decreases because of vibrations in metallic kernels but conduction in semiconductors increases because of jumping of more number of electrons from valence band to conduction band.

38. Give the name of metals which are liquid at 30° C ?

Ans: Hg, Cs and Ga.

39. Which metal has the highest melting point?

Ans: Among all metals, tungsten has the highest melting point (3422°C).

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1. Ionic or Electrovalent Bond :

It is the electrostatic attraction in between cation and anion.

2. Conditions for the formation of ionic compounds :

(i) Electronegativity difference between two combining elements must be larger.

(ii) Ionization enthalpy ($M(g) \longrightarrow M^+(g) + e^-$) of electropositive element must be low.

(iii) Negative value of electron gain enthalpy ($X(g) + e^- \longrightarrow X^-(g)$) of electronegative element should be high.

(iv) Lattice enthalpy ($M^+(g) + X^-(g) \longrightarrow MX(s)$) of ionic solid must be high.

3. Lattice Enthalpy :

The lattice enthalpy of an ionic solid is defined as the enthalpy change to completely separate one mole of a solid ionic compound into gaseous constituent ions.

4. General properties of ionic compounds :

General characteristic of bond	Ionic bond or Electrovalent bond
(1) Definition	Bond formed by the transfer of electron(s) from electropositive (metal) to electronegative (non-metal) atoms. e.g. NaCl, KCl, K_2SO_4, etc.
Physical properties (1) Crystalline nature	Electrovalent compounds are usually crystalline in nature.
(2) Constituents particles	(+ve) and (–ve) ions are the constituent particles.
(3) Melting & boiling points	Due to strong electrostatic forces of attraction, the ions are held tightly in their positions in the crystal lattice. Thus electrovalent compounds possess high melting and boiling poins. eg. NaCl (m.p = 801°C), CaF_2(m.p.=1423°C)

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(4) Mechanical properties	Ionic compounds are hard in nature Reason: Due to strong forces of attraction between oppositely charged ions, which keep them in their allotted positions.
(5) Solubility	Ionic compounds are fairly soluble in polar solvents and insoluble in non-polar solvents.
(6) Electrical conductivity	Do not conduct electricity in solid state due to absence of free ions. Conduct electricity in molten state & in solution.
(7) Directional properties	Non-directional.

5. **Isomorphism :** Simple ionic compounds do not show isomerism but isomorphism is their important characteristic. Crystals of different ionic compounds having similar crystal structures are known to be isomorphs to each other and the phenomenon is known as isomorphism.

e.g., $FeSO_4 .7H_2O$ | $MgSO_4 . 7H_2O$

6. **Conditions for isomorphism**

(i) The two compounds must have the same formula type e.g., $MgSO_4$ & $ZnSO_4$; $BaSO_4$ & $KMnO_4$ are isomorphous because they have same formula type. All alums are isomorphous because they have same general formula :

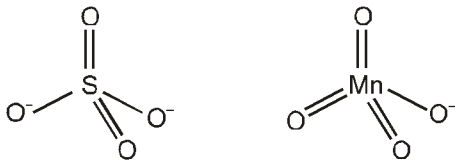
$$M_2SO_4.M_2'(SO_4)_3.24H_2O$$

M = monovalent ; M′ = trivalent

(ii) The respective structural units, atoms or ions need not necessarily be of same size in the two compounds but their relative size should be little different.

(iii) The cations of both compound should be of similar shape or structure (isostructural). Similarly anions of both compounds should be isostructural.

(a) SO_4^{2-} and MnO_4^- have same shape i.e. tetrahedral, so isomorphous.



(iv) The respective structural units should have same polarisation property.

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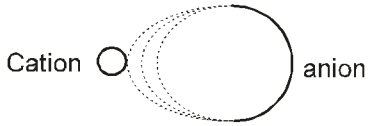
7. Fajan's Rule

Covalent character in ionic compounds (Fajan's rule) :

When anion and cation approach each other, the valence shell of anion is pulled towards cation nucleus and thus shape of anion is deformed. This phenomenon of deformation of anion by a cation is known as polarisation and the ability of cation to polarize a near by anion is called as polarizing power of cation.



Fajan's pointed out that greater is the polarization of anion in a molecule, more is covalent character in it.

More distortion of anion, more will be polarisation then covalent character increases.

8. Factors affecting covalent characters, according to Fajan's rule.

(i) Size of cation : Smaller is the cation more is its polarizing power and thus more will be the polarisation of anion. Hence more will be covalent character in compound.

Size of cation α 1 / polarisation.

e.g. $BeCl_2$ $MgCl_2$ $CaCl_2$ $SrCl_2$ $BaCl_2$

$\longrightarrow$

Size of cation increases Polarisation decreases Covalent character decreases

(ii) Size of anion :

Size of anion α polarisation

e.g., LiF LiCl LiBr LiI

$\longrightarrow$

– Size of anion increases

– Polarisation increases

– Covalent character increases

(iii) Charge on cation :

Charge on cation α polarisation.

e.g., NaCl $MgCl_2$ $AlCl_3$

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Na^+ Mg^{2+} Al^{3+}

⟶

– Charge of cation increases

– Polarisation increases

– Covalent character increases

(iv) Charge on anion :

Charge on anion α polarisation.

e.g., AlF_3 Al_2O_3 AlN

F^- , O^{2-} , N^{3-}

⟶

– Charge on anion increases

– Polarisation increases

– Covalent character increases

(v) **Pseudo inert gas configuration of cation :**

Cu^+ = $[Ne]\ 3s^2\ p^6\ d^{10}$ $Na^+ = 1s^2\ 2s^2\ p^6$

$18e^-$ $8e^-$

Pseudo inert gas configuration Inert gas configuration

(poor shielding of d-electrons) (more shielding of s and p electrons)

9. Why Ag_2S is less soluble than Ag_2O in H_2O?

Ans. Because Ag_2S is more covalent due to bigger S^{2-} ion.

10. Why $Fe(OH)_3$ is less soluble than $Fe(OH)_2$ in water

Ans. Because Fe^{3+} is smaller than Fe^{2+} and thus charge is more.

Therefore, $Fe(OH)_3$ is more covalent than $Fe(OH)_2$.

11. Why AgCl is white AgBr, AgI, Ag_2CO_3 are yellow. Similarly, $SnCl_2$ is white but SnI_2 is black. $PbCl_2$ is white but PbI_2 is yellow.

Ans. The colour of some compounds can be explained on the basis of polarisation of their bigger negative ions.

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12. Melting point of following

$BeCl_2$, $MgCl_2$, $CaCl_2$, $SrCl_2$, $BaCl_2$

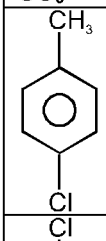
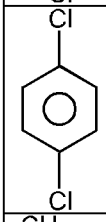
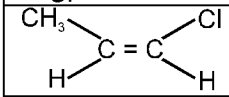
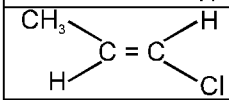
——————>——————>–

CaF_2, $CaCl_2$, $CaBr_2$, CaI_2

——————>——————>–

13. Dipole moment.

❍ **Following compounds have zero dipole moment :**

COMPOUND	POLAR	NON POLAR
HF	**Polar**	
BF_3		**Non Polar**
CO_2		**Non Polar**
HCl	**Polar**	
CCl_4		**Non Polar**
CH_2Cl_2	**Polar**	
SF_6		**Non Polar**
XeF_2		**Non Polar**
PCl_5 (g)		**Non Polar**
CH_3Cl	**Polar**	
SO_3		**Non Polar**
	**Polar**	
		**Non Polar**
	**Polar**	
	**Polar**	

14. Some important points about dipole moment :

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(i) A polyatomic molecule having polar covalent bonds but zero dipole moment indicates the symmetrical structure of the molecule. e.g. B–F bonds are polar in BF_3 but BF_3 has $\mu=0$ due to its symmetrical geometry.

(ii) If molecule have $\mu = 0$, then it should be linear or having symmetrical geometry.

e.g. linear – CO_2, CS_2, $BeCl_2(g)$; symmetrical geometry – BF_3, CH_4, PCl_5, SF_6, IF_7, XeF_4.

(iii) If molecule has $\mu \neq 0$ then it should be angular or having unsymmetrical geometry.

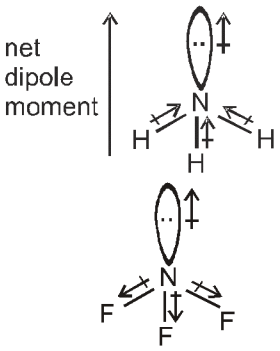
$SnCl_2$, $PbCl_2$, SO_2 , angular molecular geometry.

NH_3, H_2O, NF_3, SF_4, H_2S, unsymmetrical molecular geometry.

15. % Ionic character = $\frac{\mu_{Experimental}}{\mu_{Theoretical}} \times 100 = \frac{\mu_{(Observed)}}{\mu_{(100\%\ Ionic\ compound)}} \times 100$

16. Why NH_3 is having more dipole moment than NF_3.

Sol. In $\ddot{N}H_3$ molecule Nitrogen is more electronegative than Hydrogen. So the net dipole moment is towards. Nitrogen atom but in NF_3 molecule.



Fluorine is more electronegative than. Nitrogen so the net dipole moment is towards fluorine atoms. In NH_3 the bond pair moments and lone pair moments are in the same direction while in NF_3 the lone pair moment and bond pair moments are in opposite direction.

17. Acidic oxides

(a) Solution in water will be acidic in nature (b) will react with base but not with an acid

Ex. $SO_2 \xrightarrow{H_2O} H_2SO_3$ Sulphurous acid $\Rightarrow$ $SO_3 \xrightarrow{H_2O} H_2SO_4$ sulphuric acid

$CO_2 \xrightarrow{H_2O} H_2CO_3$ Carbonic acid

Generally non-metallic oxides are acidic oxides

18. Basic oxides

(a) Solution in water will be basic in nature (b) will react with an acid but not with a base

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Ex. $Na_2O \xrightarrow{H_2O} 2NaOH$ $CaO \xrightarrow{H_2O} Ca(OH)_2$

Generally, metallic oxides are basic oxides

19. Amphoteric oxides

Can react with an acid as well as with a base.

Generally, metalloids or elements close to metalloids can form amphoteric oxides.

Oxides of Be, Zn, Al, Ga, Sn, Pb, As(+3 os), Sb (+3 os) and Cr (+2, +3 os) are amphoteric in nature.

BeO $\Rightarrow$ $BeO + 2HCl \longrightarrow BeCl_2 + H_2O$, $BeO + 2NaOH \longrightarrow NaAlO_2$ (Sodium meta aluminate)

ZnO $\Rightarrow$ $ZnO + HCl \longrightarrow ZnCl_2$, $ZnO + NaOH \longrightarrow Na_2ZnO_2$ (sodium zincate)

20. Neutral oxides : will not react with acids or bases

CO, N_2O, NO, OF_2

21. Amphiprotic oxide : which can accept and release H^+ ions

Ex. $\Rightarrow H_2O$

22. Periodicity in nature of oxides

(a) L $\rightarrow$ R, metallic character $\downarrow$ and non-metallic character $\uparrow$. So, basic character of oxides $\downarrow$ and acidic character $\uparrow$

(b) T $\rightarrow$ B, metallic character$\uparrow$ so, basic character of oxide will $\uparrow$

Basic character increases down the group

(c) If the same element is forming oxides in diff oxidation states, then greater the oxidation no greater will be acidic nature

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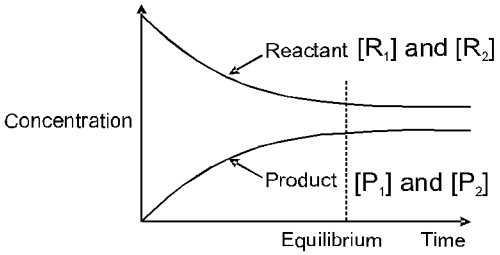
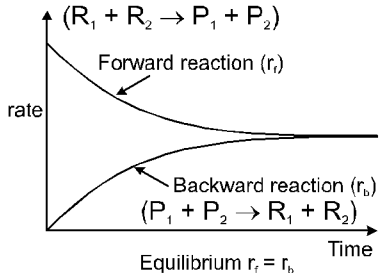
Surface Chemistry & Chemical Equilibrium

1. Equilibrium is a state in which there is not **observable change** as time goes by.
2. Chemical equilibrium is **dynamic in nature. Both forward and backward reactions/processes do not stop at equilibrium.**
3. At equilibrium, Gibb's **energy of reaction $\Delta G = 0$**
4. Equilibrium can be achieved from **both directions**. However, stoichiometric constraints will be there.
5. At equilibrium, concentration of reactants & products becomes constant with time.



6. At equilibrium, rate of **forward reaction** becomes equal to rate of **backward reaction. Both have a non-zero value.**



7. Active Mass= **Molar concentration or molarity. (Active mass is unitless)**
8. Law of mass action:

   The rate of a chemical reaction at a given temperature is proportional to the product of active mass of reactants raised to the powers of their stoichiometric coefficients. Not always true.

9. $aA_{(g)} + bB_{(g)} \rightleftharpoons cC_{(g)} + dD_{(g)}$

$$K_c = \frac{[C]_{eq}^c [D]_{eq}^d}{[A]_{eq}^a [B]_{eq}^b} ; [C]_{eq} = \frac{n_{Ceq} RT}{V}$$

$$K_p = \frac{P_C^d P_D^d}{P_A^a P_B^b}$$; All are partial pressures of various species at equilibrium; $P_C = \left(\frac{n_C}{n_{Tot}}\right)_{eq} \times P_{eq} = [C]_{eq} RT$

10. Relation between K_p & K_c is

$$K_p = K_c(RT)^{\Delta ng} ; \Delta ng = (n_{Tot})_{g,\ products} - (n_{Tot})_{g,\ reactants}$$

11. Relation between K_p & K_x is

$$K_p = K_x(P)^{\Delta ng} ; \quad P = \text{total pressure at equilibrium}$$

12. If $\Delta n_g = 0$, then $K_p = K_c = K_x$ (For gaseous phase reaction)

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13.

Sr.No.	Reactions	Types of equilibrium Homogeneous / Heterogeneous	Expression of K_P in terms of partial pressure	Expression of K_C in terms of concentration	Relation between K_P & K_C
1.	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	Homogeneous	$K_P = \frac{P_{HI}^2}{P_{H_2}.P_{I_2}}$	$K_C = \frac{[HI]^2}{[H_2][I_2]}$	$K_P = K_C(RT)^0$
2.	$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	Homogeneous	$K_P = \frac{P_{NH_3}^2}{P_{N_2}.P_{H_2}^3}$	$K_C = \frac{[NH_3]^2}{[N_2][H_2]^3}$	$K_P = K_C(RT)^{-2}$
3.	$N_2O_5(g) \rightleftharpoons 2NO_2(g) + \frac{1}{2} O_2(g)$	Homogeneous	$K_P = \frac{(P_{O_2})^{1/2}.P_{NO_2}^2}{P_{N_2O_5}}$	$K_C = \frac{[O_2]^{1/2}.[NO_2]^2}{[N_2O_5]}$	$K_P = K_C(RT)^{3/2}$
4.	$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	Homogeneous	$K_P = \frac{P_{Cl_2}.P_{PCl_3}}{P_{PCl_5}}$	$K_C = \frac{[PCl_3].[Cl_2]}{[PCl_5]}$	$K_P = K_C(RT)^1$
5.	NH_4COONH_2 (s) $\rightleftharpoons 2NH_3(g) + CO_2(g)$	Heterogeneous	$K_P = P_{CO_2} . P_{NH_3}^2$	$K_C = [CO_2][NH_3]^2$	$K_P = K_C(RT)^3$
6.	$CaCO_3$ (s) $\rightleftharpoons$ CaO(s) + $CO_2(g)$	Heterogeneous	$K_P = P_{CO_2}$	$K_C = [CO_2]$	$K_P = K_C(RT)^1$
7.	NH_4HS (s) $\rightleftharpoons NH_3(g) + H_2S(g)$	Heterogeneous	$K_P = P_{H_2S} \cdot P_{NH_3}$	$K_C = [NH_3][H_2S]$	$K_P = K_C(RT)^2$

14.

S. No.	Reaction & Equilibrium constant	Relation between Equilibrium constant
1.	$H_2 + I_2 \rightleftharpoons 2HI$ …K_1	Give Relation between K_1 & K_2 **Ans.** $K_2 = (K_1)^2$
	$2H_2 + 2I_2 \rightleftharpoons 4HI$ …K_2	K_1 and K_3 **Ans.** $K_3 = (K_1)^{½}$
	$\frac{1}{2} H_2 + \frac{1}{2} I_2 \rightleftharpoons HI$ ...K_3	K_2 and K_3 **Ans.** $K_3 = (K_2)^{¼}$
2.	$A + B \longrightarrow C + D$ …K_1	Give Relation between K_1 & K_2 **Ans.** $K_2 = \frac{1}{k_1}$
	$C + D \longrightarrow A + B$ …K_2	K_2 and K_3 **Ans.** $K_3 = k_2^2$
	$2C + 2D \longrightarrow 2A + 2B$ …K_3	K_1 and K_3 **Ans.** $K_3 = \left(\frac{1}{k_1}\right)^2$
3.	$A \rightleftharpoons B$ …K_1	Express K_4 in term of K_1, K_2 and K_3
	$B \rightleftharpoons C$ ...K_2	

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	$C \rightleftharpoons D$ ...K_3 $A \rightleftharpoons D$ ...K_4	**Ans.** $K_4 = K_1 \times K_2 \times K_3$
4.	$S(s) + O_2(g) \rightleftharpoons SO_2(g)$ $K_1 = 10^{52}$ $2S(s) + 3O_2(g) \rightleftharpoons 2SO_2(g)$ $K_2 = 10^{129}$ $2SO_2(s) + O_2(g) \rightleftharpoons 2SO_3(g)$ $K_3 = ?$	Find out K_3 **Ans.** $K_3 = \frac{k_2}{(k_1)^2} = \frac{10^{129}}{(10^{52})^2} = 10^{25}$
5.	$2P(g) + 3Cl_2(g) \rightleftharpoons 2PCl_3(g)$ K_1 $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$ K_2 $2P(g) + 5Cl_2(g) \rightleftharpoons 2PCl_5(g)$ $K_3 = ?$ (in terms of K_1 & K_2)	Express K_3 in term of K_1 & K_2 **Ans.** $K_3 = K_1(K_2)^2$

15. Reaction quotient (Q_c or Q_p) : It has same expression as that of equilibrium constant (K_c or K_p) but it is defined at all moments instead of equilibrium only.

16. If $Q > K_{eq}$ $\Rightarrow$ Reaction will net move in backward direction from given state to achieve equilibrium.

17. If $Q < K_{eq}$ $\Rightarrow$ Reaction will net move in forward direction from given state to achieve equilibrium.

18. **Properties of equilibrium constant**

K depends only upon the **temperature** & **stoichiometry coefficients** of the reaction.

(a) Effect of Stoichiometry coefficients:

(i) Multiplying by any number:

$A(g) \rightleftharpoons B(g)$, K_1

$nA(g) \rightleftharpoons nB(g)$, $K'' = (K_1)^n$, n can be fraction, integer or negative number

(ii) Adding two reactions:

$A(g) \rightleftharpoons B(g)$ K_1

$B(g) \rightleftharpoons C(g)$ K_2

On adding, $A(g) \rightleftharpoons C(g)$ $K = K_1 . K_2$

(iii) Subtracting two reactions:

$A(g) \rightleftharpoons B(g)$ K_1

$B(g) \rightleftharpoons C(g)$ K_2

On Subtracting, $A(g) + C(g) \rightleftharpoons 2B(g)$ $K = K_1 / K_2$

(b) Effect of Temperature:

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303\, R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$; ΔH = Enthalpy of reaction(SI unit : Joule/mole)

R= universal gas constant (SI unit value : 25/3 J/mole-K) or 2 cal/mol-K

Note: 1. The unit of R and $\Delta H/T$ should be same

2. If $T_2 > T_1$, then $K_2 > K_1$ if ΔH = +ve (endothermic reaction)

$K_2 < K_1$ if ΔH = –ve (exothermic reaction)

3. ln(x)=2.303Log(x)

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19. Degree of dissociation (α) and vapour density

It is the fraction of one mole dissociated into the products.

$$\alpha = \frac{\text{no. of moles dissociated}}{\text{Initial no. of moles taken}}$$

Note : % dissociation = α × 100

Let a gas A_n dissociates to give n moles of A as follows-

	$A_n(g)$	$\rightleftharpoons$	$nA(g)$	
t = 0	a		0	
$t = t_{eq}$	a – x		n.x	$\alpha = \frac{x}{a} \Rightarrow x = a\alpha.$
	a – a α = a(1–α)		n a α	

Total no. of moles = a – a α + n a α = [1 + (n – 1) α] a

	$5A(g)$	$\rightleftharpoons$	$3B(g)$	+ $2C(g)$	+ $4D(g)$	
t = 0	a		0	0	0	
$t = t_{eq}$	a – 5aα		3aα	2aα	4aα	**(Wrong)**
$t = t_{eq}$	a – aα		3aα/5	2aα/5	4aα/5	**(Right)**

Significance of n $n = \frac{\text{sum of stoichiometric coefficients of products}}{\text{sum of stoichiometric coefficients of reactants}}$

(i) for $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (n = 2/1 = 2)

(ii) for $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$ (n = 4/2 = 2)

20. Observed molecular weight and Observed Vapour Density of the mixture

Observed molecular weight of $A_n(g)$ = $\frac{\text{weight of } A_n(g)}{\text{total no. of moles at equilibrium}} = \frac{a.M_{th}}{a(1+(n-1)\alpha)}$

$\therefore$ $M_{obs} = \frac{M_{th}}{[1+(n-1)\alpha]}$ where M_{th} = theoretical molecular weight (n = atomicity)

$M_{mixture} = \frac{M_{A_n}}{[1+(n-1)\alpha]}$, M_{A_n} = Molar mass of gas A_n

Vapour density (V.D). : Density of the gas relative to density of hydrogen under same temp & pressure.

D = vapour density without dissociation = $\frac{M_{A_n}}{2}$

d = vapour density of mixture = observed/experimental v.d. = $\frac{M_{mix}}{2}$

D= absolute density without dissociation (of Reactant) = $\frac{PM_0}{RT}$ gm/ltr ,

d= Absolute density of mixture = observed/Experimental = $\frac{PM_{mix}}{RT}$ gm/ltr

$\frac{D}{d} = 1 + (n-1)\alpha$ $\therefore$ $\alpha = \frac{D-d}{(n-1)\times d} = \frac{M_T - M_o}{(n-1)M_0}$

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M_0 = Observed molecular wt. or molecular wt. of the mixture at equilibrium.

Note : **1.** Relation of α is not applicable for n = 1.

2. Some Standard Expression of K_p : (i) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ $K_P = \frac{\alpha^2 . P}{1-\alpha^2}$

(ii) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ $K_P = \frac{4.\alpha^2 . P}{1-\alpha^2}$

where P is total pressure at equilibrium and α is Degree of Dissociation of reactant

Note: Standard expressions will give correct result if equilibrium has been established starting with only reactants.

21. Thermodyanamics of equilibrium

For a general reaction, $mA + nB \rightleftharpoons pC + qD$, ΔG is given by-

$\Delta G = \Delta G^\circ + 2.303\ RT \log_{10} Q$

where ΔG = Gibb's energy of reaction

ΔG° = Standard Gibb's energy of reaction

Q = reaction quotient

Standard State:

For pure Solid & Liquid: **Pressure: 1 atm**

For Gas: **Partial Pressure: 1 atm**

For Solution (Solid/liquid/gas)**: Concentration : 1M**

Temperature**: Any Fixed Value (Normally 25^0 C)**

At equilibrium, Q = K , $\Delta G = 0$

$\Delta G^\circ = -2.303\ RT \log_{10} K$

Now since, $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ where ΔH° = Standard enthalpy change of the reaction

ΔS° = Standard entropy change

$$\log_{10} K = -\frac{\Delta H^\circ}{2.303} \cdot \frac{1}{RT} + \frac{\Delta S^\circ}{2.303R}$$

Graph:

If plot of ln K vs $\frac{1}{T}$ is plotted then it is a straight line with slope = $-\frac{\Delta H^\circ}{R}$ and intercept = $\frac{\Delta S^\circ}{R}$

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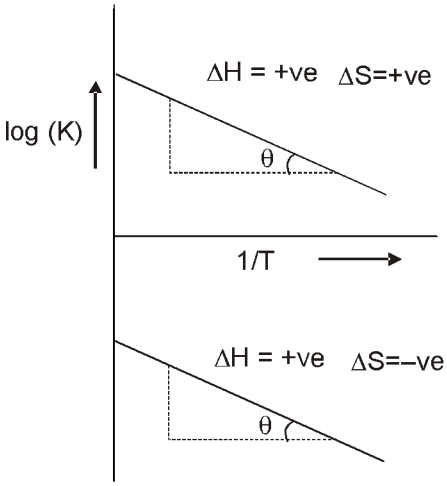
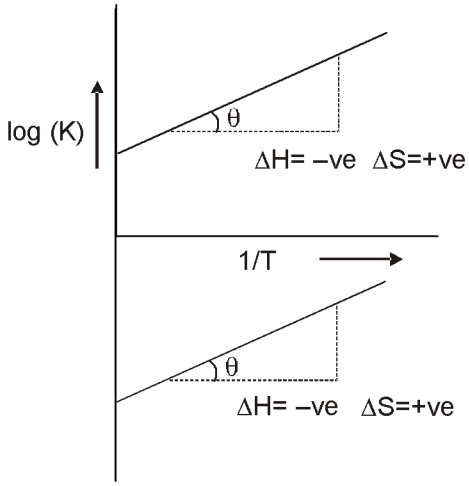


Endothermic reaction



Exothermic reaction

In above graphs, Slope = $\frac{-\Delta H^\circ}{2.303R}$ = tanθ , y intercept = $\frac{\Delta S^\circ}{2.303R}$

If at temperature T_1, equilibrium constant is K_1 and at T_2, it is K_2, then

$$\log_{10}K_1 = \frac{-\Delta H^\circ}{2.303R}\cdot\frac{1}{T_1} + \frac{\Delta S^\circ}{2.303R} \quad \text{..... (i)}$$

$$\log_{10}K_2 = \frac{-\Delta H^\circ}{2.303R}\cdot\frac{1}{T_2} + \frac{\Delta S^\circ}{2.303R} \quad \text{..... (ii)}$$

[Assuming ΔHº and ΔSº remain constant in this temperature range, though they do not remain constant]

$$\log\left(\frac{K_1}{K_2}\right) = \frac{\Delta H^\circ}{2.303R}\left(\frac{1}{T_2}-\frac{1}{T_1}\right)$$ **Vant Hoff equation**

Note :
- ΔH should be substituted with sign.
- Unit of ΔH/T and gas constant R should be same.
- For endothermic (ΔH > 0) reactions, value of the equilibrium constant increases with the rise in temperature
- For exothermic (ΔH < 0) reaction, value of the equilibrium constant decreases with increase in temperature

Condition for spontaneity:

ΔG < 0 for spontaneous process or reaction. (**ΔG^0 may be +ve, -ve or Zero)**

ΔG > 0 for Non spontaneous process or reaction.

ΔG = 0 for process or reaction at equilibrium

Equilibrium at Standard Condition:

Q=1, ΔG = 0 , $\Delta G^0 = 0$

ΔG=RTln $\frac{Q}{K}$, if Q>K, then **ΔG > 0, Reaction is non Spontaneous & will go in backward direction**

if Q < K, then **ΔG < 0, Reaction is Spontaneous & will go in forward direction**

if Q= K, then **ΔG = 0, Reaction is at Equilibrium**

T-1 Heterogeneous equilibrium

For pure solid and pure liquid, active mass is taken to be unity i.e. 1 as their concentration remains constant throughout the reaction.

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ $\quad K_P = (P_{CO_2})_{eq}$, $\quad K_C = [CO_2(g)]_{eq}$

$$[CaCO_3(s)] = \frac{\text{moles}}{\text{volume}} = \frac{\frac{W_{CaCO_3}}{M_{CaCO_3}}}{V} = \frac{\text{density } CaCO_3}{M_{CaCO_3}} = \text{constant (Depends upon Temperature)}$$

$H_2O(\ell) \rightleftharpoons H_2O(g)_{eq}$ $\quad K_P = (P_{H_2O(g)})_{eq}$, $K_C = [H_2O(g)]_{eq}$

T-2 Le-chatelier's principle:

It explains effect of change of pressure, temperature, volume, concentration etc. on equilibrium.
According to this, if any change is made at equilibrium, then equilibrium will shift in that direction by which effect on change can be minimized.

Effect of change of amount at equilibrium:

If the **amount** of a component is increased, reaction shifts in a direction which tends to decrease its **amount**.

$A(g) \rightleftharpoons B(g)$

Change at equilibrium	Direction of equilibrium shift
A is added	Forward
B is removed	
A is Removed	Backward
B is added	

Note : The addition (in small amount) of any solid component does not affect the equilibrium.

Effect of change in pressure & volume at equilibrium:

If $\Delta n_g = 0$, then no shift in equilibrium.

If $\Delta n_g \neq 0$, (i) On Increasing pressure or decreasing volume at equilibrium**,** the equilibrium will shift in that direction where no. of mole (gaseous) will decrease.

(ii) On Increasing volume or decreasing pressure at equilibrium**,** the equilibrium will shift in that direction where no. of mole (gaseous) will increase.

Effect of change in Temperature at equilibrium:

$$\log\left(\frac{K_1}{K_2}\right) = \frac{\Delta H^o}{2.303R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

Endothermic	**Exothermic**
A+Heat $\rightleftharpoons$ B	**A –Heat $\rightleftharpoons$ B**
A $\rightleftharpoons$ B –Heat	**A $\rightleftharpoons$ B +Heat**
A $\rightleftharpoons$ B **$\Delta H>0$**	A $\rightleftharpoons$ B **$\Delta H<0$**
$T\uparrow$, $K_{eqbm}\uparrow$ $\Rightarrow$ (Forward shift) **$T\downarrow$, $K_{eqbm}\downarrow$ $\Rightarrow$ (backward shift)**	**$T\uparrow$, $K_{eqbm}\downarrow$ $\Rightarrow$ (backward shift)** **$T\downarrow$, $K_{eqbm}\uparrow$ $\Rightarrow$ (Forward shift)**

Effect of catalyst addition at equilibrium:

No shift in equilibrium

1. Rate of forward & backward increase by same extent but K_{eqbm} remains same.
2. Catalyst can change kinetic parameters but not thermodynamic parameters (**K_{eqbm}, ΔH, ΔS, ΔG)**
3. Catalyst can't change concentration but can change time to reach equilibrium.

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Effect of adding of Inert gas at constant pressure & volume at equilibrium:

Inert gas: it could be any gas which does not react with any components of Reaction.

1. **At constant volume:** No shift in Chemical Equilibrium (since Q_C will still remain same) but Physical Equilibrium may get affected.
2. **At constant pressure:** Total no. of gaseous mole increase. To maintain pressure constant, volume increase & concentration decrease. So Equilibrium will shift in that direction where no. of gaseous mole increase.

Reaction	Nature of Rxn	Favorable Condition for Formation of product		
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	**$\Delta H < 0$**	Temp: **Low**	Pressure : **High**	Volume **: low**
$N_2(g) + O_2(g) \rightleftharpoons 2\,NO(g)$	**$\Delta H > 0$**	Temp: **High**		
$N_2(g) + 3\,H_2(g) \rightleftharpoons 2NH_3(g)$	**$\Delta H < 0$**	Temp: **Low**	Pressure : **High**	Volume **: low**

Note: In industry for manufacturing ammonia, process is carried out at high pressure & temperature in presence of catalyst to increase the rate of formation of ammonia.

Physical Equilibrium:
If pressure increases at equilibrium, then Equilibrium will shift in that direction where Density increases.

Reaction	Nature of Rxn	Description	Density	Favorable condition of products
$H_2O(s) \rightleftharpoons H_2O\ (\ell)$	ΔH = + ve	Melting of ice	d(s) < d **(ℓ)**	Temp: **High** , Pressure: **High**
$H_2O(\ell) \rightleftharpoons H_2O(g)$	ΔH = + ve	Boiling of water	d(g) < d **(ℓ)**	Temp: **High** , Pressure: **Low**
$C_{(graphite)} \rightleftharpoons C_{(diamond)}$	ΔH = + ve	Formation of diamond	$d_{(diamond)} > d_{(graphite)}$	Temp: **High** , Pressure: **High**
$CO_{2(g)} \rightleftharpoons CO_{2(aq)}$	ΔH = - ve	Solubility of gas in water	d(g) < d **(aq)**	Temp: **low** , Pressure: **High**
$NaCl_{(s)} \rightleftharpoons NaCl_{(aq)}$	ΔH = + ve	Solubility of solid in water	d(s) > d **(aq)**	Temp: **High,** Pressure**: No effect**

Note: For most substances, density of liquid is less than that of solid.

……

T-3 Vapor pressure and Relative Humidity

- It is the pressure exerted by the vapors over its liquid when it is in equilibrium with the liquid. Vapour pressure of water is also called **aqueous tension**.

$$H_2O\ (\ell) \rightleftharpoons H_2O\ (g);\ \Delta H = +\text{ ve} \quad , \quad K_P = P_{H_2O} = \textbf{constant at fixed temperature}$$

Note: **1.** V.P. of liquid is **independent** of pressure, volume concentration and Area change. It depends upon only nature of liquid and **temperature** & its value increases with increase in temperature. (But **not linearly**)

2. At 25°C, vapour pressure of water $\approx$ 24 mm of Hg

3. If any liquid is in equilibrium with its vapour, then its pressure will not change on changing the volume of container (keeping temperature constant).

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- **Adsorption :** The phenomenon of attracting and retaining the molecules of a substance on the surface of a liquid or a solid resulting into a higher concentration of the molecules on the surface is called adsorption. As a result of adsorption, there is a decrease of surface energy. The process of removal of an adsorbed substance from the surface on which it is adsorbed is called desorption. It is the reverse of adsorption and can be brought about by heating or by reducing the pressure.
- **Adsorbent and adsorbate :**The substance on the surface of which adsorption occurs is known as adsorbent. The substances that get adsorbed on the solid surface due to intermolecular attractions are called adsorbate.Charcoal, silica, gel, alumina gel are good adsorbents because they have highly porous structures and have large surface area. Colloids on account of their extremely small dimensions possess enoromous surface area per unit mass and are, therefore, also good adsorbents.
- **Sorption :** When both adsorption and absorption take place simultaneously.

Difference between adsorption and absorption :

The important points of distinction between adsorption and absorption

Absorption	Adsorption
It is the phenomenon in which the particles of gas or liquid get uniformly distributed throughout the body of	It is the phenomenon of higher concentration of gas or liquid on the surface than in the bulk of the solid.
The concentration is the same throughout the material.	The concentration on the surface of the adsorbent is different (has higher concentration) from that in the bulk.
It is a bulk phenomenon.	it is a surface phenomenon.
Absorption occurs at uniform rate.	Adsorption is rapid in the beginning and its rate slowly decreases
It is a slow process	*It is a fast process*

- **Thermodynamics of adsorption**

1. Adsorption involves attracting molecules of adsorbate on surface of the adsorbent. Due to this energy is released and thus heat of adsorption is negative i.e. adsorption is always exothemic

2. The molecules of adsorbate are held on surface of the solid adsorbent due to this entropy decreases i.e. ΔS is negative.

3. $\Delta G = \Delta H - T\Delta S$, Therefore adsorption will occur only when ΔG is negative and this is possible only if $|\Delta H| > |T\Delta S|$

4. As adsorption proceeds ΔH becomes less and less negative hence ΔH becomes equal to $T\Delta S$ and ΔG becomes zero This is the state at which equilibrium is attained.

- **Enthalpy of adsorption $\Delta H_{adsorption}$:** It is the amount of the heat released when 1 mole of an adsorbate gets adsorbed on a particular adsorbent at adsorption equilibrium. It depends upon the nature of both the adsorbate as well as adsorbent.

Types of adsorption : The adsorption is classified into two types :

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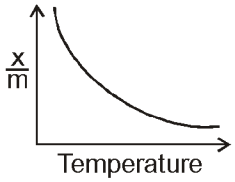
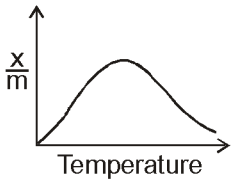
Differences between Physical and Chemical Adsorption		
Property	**physical Adsorption**	**Chemical Adsorption**
Nature of adsorption	Weak	Strong
Enthalpy of adsorption	Low	High
Reversibility of adsorption	Reversible and occur rapidly	Irreversible and occurs slowly
Temp. at which adsorption is more pronounced	Low temp.	High temp.
Effect of change in temp.	Decrease with rise in temp.	Increase with rise in temp.
Specificity of adsorption	Not specific generally take place on all surface	Highly specific take place on specific surface
Nature of adsorbate layer	Multi-layered	Mono-layered
Effect of pressure on adsorption	Increase with pressure of adsorbate gas and attain limiting value	Negative adsorption
Energy of activation	Very low	significant relative high
Dependence on nature of adsorbate and adsorbent	depends on adsorbate (gas) only easily liquefiable gases are more readily adsorbed	Depends on adsorbate as well as adsorbent No correlation be can be achieved
Easy of desorption	easy since Vander Waal force involved	Not easy since chemical forces on involved
Graph		

Adsorption of gases on solids :

The extent of adsorption of a gas on a solid surface is affected by the following factors:

- **The nature of the gas (i.e. nature of the adsorbate).** The easily liquefiable gases such as HCl, NH_3, Cl_2 etc. are adsorbed more than the permanent gases such as H_2, N_2 and O_2.

  extent of adsorption $\propto$ ***critical temperature of gas*** $\propto$ ***ease of liquification***

Gas	*H_2*	*CH_4*	*CO_2*	*SO_2*
T_c	*33*	*190 K*	*304 K*	*330K*

- **Nature of adsorbent.** The extent of adsorption of a gas depends upon the nature of adsorbent. Activated charcoal (i.e. activated carbon), metal oxides (silica gel and aluminium oxide) and clay can adsorb gases which are easily liquified. Gases such as H_2, N_2 and O_2 are generally adsorbed on finely divided transition metals Ni and Co.

  extent of adsorption $\propto$ surface area of solid.

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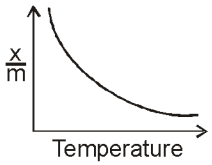
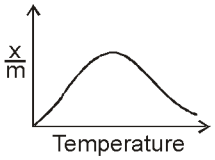
- **Activation of adsorbent :**
  (a) Metallic adsorbents are activated by mechanical rubbing or by subjecting it to some chemical reactions.
  (b) To increase the adsorbing power of adsorbents, they are sub-divided into smaller pieces. As a results, the surface area is increased and therefore, the adsorbing power increases.

- **Effect of temperature :**Mostly the process of adsorption is exothermic and the reverse process or desorption is endothermic. . Physical adsorption decreases continuously with increase in temperature whereas chemisorption increases initially, shows a maximum in the curve and then it decreases continuously.





- **Effect of pressure.** The extent of adsorption of a gas per unit mass of adsorbent depends upon the pressure of the gas.

$$\frac{x}{m} \propto P$$

Where x = mass of adsorbate
M = mass of adsorbent

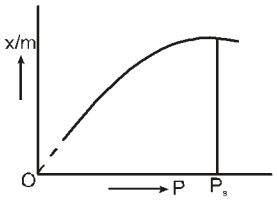


Figure-1

It is clear from the figure-1 that extent of adsorption (x/m) increases with pressure and becomes maximum corresponding to pressure P_s called saturation pressure.

Enzyme catalyst: Enzyme are complex nitrogenous organic compound which of emzyme Cl produced by living plants and animal the chracterstics of enzyme catalyst are .

1. Most highly efficient
2. Highly specific in nature
3. Highly active under optimum temperature
4. Highly active under optimum Ph.
5. Increase activity in presence of coenzyme
6. Influence of induce and poison

Applications of adsorption :

1. **In gas masks :** Activated charcoal is generally used in gas masks to adsorb poisonous and toxic gases from air. These masks are commonly used by the miners because there are poisonous gases like CO, CH_4 etc. in the atmosphere in coal mines.
2. **In dyeing of cloths :** Mordants such as alums are used in dyeing of cloths. They adsorb the dye particles which, otherwise, do not stick to the cloths.
3. **In dehumidizers :** Silica gel is commonly used to adsorb humidity or moisture from air.
4. **Removal of colouring matter :** Many substances such as sugar, juice and vegetable oils are coloured due to the presence of impurities. They can be decolourised by placing them in contact with adsorbents like activated charcoal or fuller's earth.
5. **Heterogeneous catalysis :** The phenomenon of adsorption is useful in the heterogeneous catalysis. The metals such as Fe, Ni, Pt, Pd, etc. are used in the manufacturing processes such as Contact process, Haber process and the hydrogenation of oils. Their use is based upon the phenomenon of adsorption.
6. **Refining Petroleum :** Silica gel is used as adsorbent in petroleum refining.
7. **Chromatography :** It is a method for separation of component and is based on preferential adsorption column is very common device used.
8. **Creating vacuum :** High vacuum can be created by removing gas by adsorption.
9. **Adsorption Indicators :** In volumetric analysis, adsorption indicator is used Surface of certain precipitates such as silver halide have the property of adsorbing some dye like eosin, fluorescein, etc In the case of precipitation titration ($AgNO_3$ vs NaCl) of the indicator is adsorbed at the end point producing a characteristic colour on the precipitate.
10. **In froth floatation process :** (in metallurgy).
11. **Softening of hard water :** Ion exchange resins used for softening of hard water is based upon selective and competive adsorption of ions on resins.

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$Na_2Z + Ca^{+2} \longrightarrow CaZ + 2\,Na^+$

The organic polymers containing groups like –COOH, $–SO_3H$ and $–NH_2$ etc. possess the property of selective adsorption of ions from solution. These are quite useful in the softening of water.

- **Catalysts :** Berzillus in 1835 used the word catalyst first time for some substance which alter rate of chemical reaction and themselves remain chemically and quantitatively unchanged after the reaction and the phenomenon is known as catalysis.

  Eg : Potassium chlorate when heated at 653K to 873K, it gives O_2, When MnO_2 is used in this reaction the O_2 is quickly at the low temperature hence MnO_2 is a catalyst

  $2KClO_3 \rightarrow 2KCl + 3O_2$

- **Homogeneous Catalysis :** When catalysts and reactants are in same phase then the process is said to be homogeneous catalysis and

  Eg : (i) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

  (ii) $CH_3COOCH_3(\ell) \xrightarrow{HCl(\ell)} CH_3COOH(aq)$

  (iii) $C_{12}H_{22}O_{11}(aq.) + H_2O(\ell) \xrightarrow{H_2SO_4(\ell)} \underset{\text{Glucose}}{C_6H_{12}O_6(aq.)} + \underset{\text{Fructose}}{C_6H_{12}O_6(aq.)}$

- **Heterogenous Catalysis :** When catalysts and reactants are in different phases, then process a know as heterogenous catalysis and catalyst is called heterogeneous catalyst

  Eg : (i) $2SO_3(g) + O_2(g) \xrightarrow{Pt(s)} 2SO_3(g)$

  (ii) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

  (iii) $4NH_3(g) + 5O_2(g) \xrightarrow{Pt(s)} 4NO(g) + 6H_2O(g)$

  (iv) Vegetable oils $(\ell) + H_2(g) \xrightarrow{Ni(s)}$ Vegetable ghee (s).

- **Types of Catalysis**

Type of Catalysis			
Positive	**Negative**	**Auto**	**Induced**
Substance which increase the rate of chemical reaction $2SO_2 + O_2 \xrightarrow{Pt} 2SO_3$	Substance which decrease the rate of chemical reaction $2H_2O_2 \xrightarrow[\text{negative catalyse}]{H_2O_2} 2H_2O^+$	One of Product behave as catalyst for that reaction and increase the rate of reaction, than this phenomena is called autocatalysis $CH_3COOC_2H_5 + H_2O$ $\Downarrow$ $CH_3COOH + C_2H_5OH$	When one reaction catalyst another reaction, than this phenomena is called as induced catalyses $Na_2SO_3 \rightarrow Na_2SO_4$

- **Promotors/Activators :** Substance which are not catalyst themselves but its presence can increase the catalytic activity of catalyst. A promotors increase the number of active sites on the surface Eg :

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(i) $N_2 + 3H_2 \xrightarrow[Mo(promotor)]{Fe(catalyst)} 2NH_3$

(ii) Vegetable Oil + $H_2 \xrightarrow[Cu(promotor)]{Ni(catalyst)}$ Vegetable ghee.

(iii) $CO + 2H_2 \xrightarrow[Cr_2O_3(promotor)]{ZnO(catalyst)} CH_3OH$

- **Catalytic Poisons/ Anti catalysts :** Substance which are not catalyst themselves but whose presence decrease the activity of the catalyst. Poisoning is due to preferential adsorption of poison on the surface of the catalyst.

(i) $N_2 + 3H_2 \xrightarrow[CO/H_2S(catalytic\ \ poisons)]{Fe(catalyst)} 2NH_3$

(ii) $2SO_2 + O_2 \xrightarrow[As_2S_3(catalytic\ \ poisons)]{Platinised\ \ asbestos\ \ (catalyst)} 2SO_3$

(iii) Rosenmund Reactions : $RCOCl + H_2 \xrightarrow[BaSO_4(\ poisons\ catalytst)]{Pd(catalyst)} RCHO + HCl$

- **Catalyst Inhibitor:** Those substances which reduce the rate of a chemical reactions are known as inhibitor. e.g. H_3PO_4 and glycerol decrease the rate of decomposition of hydrogenperoxide.

- **Characteristics of Catalysis :**

(i) A Catalyst remains unchanged in mass and chemical compositions at the end of reactions. However its physical state can be changed. Eg :
Granular MnO_2 during decomposition of $KClO_3$ is left as powder at the end of the reaction.

(ii) **Highly efficient :** Finely devided state of catalyst is more efficient for the reactions because surface area increases and more adsorption take place.

(iii) A catalyst cannot initiate the reaction. But some times the activation energy is so large that practically a reaction may not start until a catalyst lowers the activation energy significantly. For example, mixture of hydrogen and oxygen do not react at room temperature but the reaction occurs very rapid in presence of Pt black.

$H_2 + O_2 \xrightarrow{room\ temperature}$ No reaction

$H_2 + O_2 \xrightarrow{Pt\ black} H_2O.$

(iv) **Highly specific :** Catalyst are generally specific in nature. A substance which act as a catalyst in a particular reaction, fails to catalyse other reaction.

(vi) Catalyst cannot change equilibrium state but it helps to attain equilibrium quickly.

(vii) A catalyst does not change the enthalpy, entropy and free energy of a reaction.

(viii) **Highly active under optimum temperature :** There is a particular temperature at which the efficiency of a catalyst is maximum, this temperature is known as optimum temperature. On either side of the optimum temperature, the activity of catalyst decreases.(optimum range : 298 to 310 K).

(ix) **Highly active under optimum pH :** Range is pH 5 to 7.

(x) Influence of Inhibitor or poison.

- **Adsorption Theory of Heterogeneous Catalyst :** This theory explains the mechanism of heterogeneous catalyst. This theory is combination of two theory, intermediate compound formation theory and the old adsorption theory, the catalytic activity is localised on the surface on the catalyst. The mechanism involvs 5 steps.
(i) Diffusion of reactant to the surface of the catalyst of.
(ii) Adsorption of reactant molecules on the surface of the catalyst.
(iii) Formation of activated intermediate.
(iv) Formation of reactions product on the catalyst surface.
(v) Diffusion of reactions product from the catalyst surface or desorption.

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Some Industrial Catalytic reactions

S.N.	Process	Catalyst
1.	Haber's Process for the manufacture of ammonia $N_2 + 3H_2 \rightleftharpoons 2NH_3$	Finely divided iron, Molybdenum as promoter. Conditions : 200 atmospheric pressure and 450-500°C temperature.
2.	Ostwald's process for the manufacture of nitric acid. $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$; $2NO + O_2 \rightarrow 2NO_2$ $4NO + 2H_2O + O_2 \rightarrow 4HNO_3$.	Platinised asbestos Temperature 300°C
3.	Lead chamber process for the manufacture of sulphuric acid $2SO_2 + O_2 \rightleftharpoons 2SO_3$; $SO_3 + H_2O \rightarrow H_2SO_4$.	Nitric oxide.
4.	Contact process for the manufacture of sulphuric acid. $2SO_2 + O_2 \rightleftharpoons 2SO_3$; $SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$	Platinised asbestos or vanadium pentoxide(V_2O_5). Temperature 400–450°C.

- **Colloid Solution :**

Colloid State : A substance is said to be in colloidal state when the size of the particle of disperse phase is greater than particle of true solution and less than that of suspension solution particle, their range of diameters lie between 1 and 1000 nm (10^{-9} to 10^{-6} m).

Colloid solution : It is a heterogeneous system consisting of 2 phase :

(1) Disperse Phase (D.P) : The phase which is dispersed through the medium is called dispersed phase or discontinuous phase or internal phase.

(2) Dispersion Medium (D.M) : A medium in which colloidal particles are dispersed is called dispersion medium. It is also known as continuous phase or outer phase or external phase.

Colloidal solution = D.P. + D.M.

Ex. In Gold sol, Gold is D.P and water is D.M.

Classification of colloids :

1. **On the basis of physical state of D.P. and D.M.**

On the bases of physical state of D.P. and D.M. colloidal solution may be divided into eight system.

D P	D M	Type of colloid	Examples
Solid	Solid	Solid Sol	Some coloured glasses, and gem stones
Solid	Liquid	Sol	Paints, cell fluids
Solid	Gas	Aerosol	Smoke, dust
Liquid	Solid	Gel	Cheese, butter, jellies
Liquid	Liquid	Emulsion	Milk, hair cream
Liquid	Gas	Liquid Aerosol	Fog, mist, cloud, insecticide sprays
Gas	Solid	Solid Sol	Pumice stone, foam rubber
Gas	Liquid	Foam	Froth, whipped cream, soap lather.

* Solution of gas in gas is not a colloidal system because it forms homogeneous mixture.

2. On the basis of D.M. : Colloidal solution are classified as

D.M.	Name of colloidal system
Water	Hydro sol or aqua sol
Alcohol	Alco sols
Benzene	Benzo sols
Air	Aero sols

* Aquadag & oildag are colloidal solution of graphite in water & oil respectively.

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* Colloidal solution are often termed as sol.

3. On the Basis of interaction of D.P. for D.M. : There are two types-

DISTINCTION BETWEEN LYOPHILIC AND LYOPHOBIC COLLOIDS

S.No.	Property	Lyophilic colloids	Lyophobic colloids
1	Ease of preparation	There are easily formed by direct mixing.	These are formed only by special methods
2	Reversible or irreversible nature	These are reversible in nature	These are irreversible in nature.
3	Particles nature	The particles of colloids are true molecules but are big in size	The particles are aggregates of many molecules
4	Stability	These are very stable	These are unstable and require traces of stabilizers
5	Action of electrolytes	No effect	The addition of small amount of electrolytes causes precipitation (called coagulation) of colloidal solution.

S.No.	Property	Lyophilic colloids	Lyophobic colloids
6	Charge on particles	*The particles do not carry any charge.*	The particles move in a specific direction either towards anode or cathode depending upon their charge in an electric field
7	Hydration	The particles of colloids are heavily hydrated due to the attraction for the solvent.	The particles of colloids are not appreciably hydrated.
8	*Viscosity*	*It is higher than that of dispersion medium*	*It is nearly same as that of dispersion medium*
9	*Tyndall effect*	*They do not show tyndall effect*	*They show tyndall effect.*
10	*Surface tension*	*Lower than dispersion medium*	*It is nearly same as that of dispersion medium*
11	*Coagulation of precipitation*	*Precipitated by high concentration of electrolyte*	*Precipitated by low concentration of electrolyte*
12	*Migration in electric field*	*May or may not migrate as they may not carry charge*	*Migrate toward anode or cathode as these particles carry charge*
13	*Example*	*Mostly organic nature; Starch and Gelatin*	*Inorganic nature; Transition metal salt in water, gold etc., Metal solution*

4. On the basis of chemical composition:

Inorganic Colloids

(i) Metal sols : Cu, Ag, Au, Pt Sols.

(ii) Non Metal sols : S, I_2, Graphite

(iii) Sol of oxide and hydroxide : SnO_2, TiO_2, Fe_2O_3, $Fe(OH)_3$, $Al(OH)_3$, $Cr(OH)_3$

(iv) Salt Sol - AgBr, AgI, As_2S_3, etc.

Organic Colloids

(i) Homopolar sol - In this type of colloid, particles carry similar type of charge. eg. Sol of rubber in benzene which contain - ve charge colloidal particle of latex.

(ii) Hydroxy Sol- Starch sol

5. On the basis of charge on particles

(i) positive Sol

(a) Metal Oxide & Hydroxide - SnO_2, TiO_2, Fe_2O_3, $Al(OH)_3$, $Fe(OH)_3$, $Cr(OH)_3$.

(b) Basic Dyes Methylene blue, vismark brown.

(ii) negative Sol -

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(a) Metal sol - Ag, Au, Pt, Cu
(b) Acidic dye - congo red, eosin
(c) Sulphide Sol- CdS, HgS, As_2S_3, Sb_2S_3.
(d) Natural sol - Blood, clay, charcoal, latex rubber, dust particle in water, starch carbon particle in smoke, gum.

6. Other type of colloids : Multimolecular, macromolecular and associated colloids

- **Multimolecular colloids :** In this type, the particles consist of an aggregate of atoms or small molecules size less than 1 nm. For example, sols of gold atoms and sulphur (S_8) molecules. In these colloids, the particles are held together by van der Waal's forces.
- **Macromolecular colloids :** In this types, the particles of the dispersed phase are sufficiently big in size (macro) to be of colloidal dimensions. These macromolecules forming the dispersed phase are generally polymers having very high molecular masses. These colloids are quite stable and resemble true solutions in many respects. Naturally occuring macromolecules are starch, cellulose, proteins, enzymes, gelatin, etc.
- **Associated colloids (Micelles):** These are the substances which behave as normal strong electrolytes at low concentration but behave as colloidal particles at higher concentration. These associated particles are also called micelles. Ex. Soap.

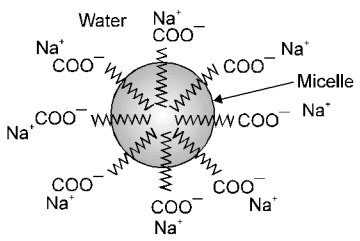
Micelles :

⇒ Micelles are relatively small, spherical structures composed of any where for a few to few thousand molecule that attract one another to reduce surface tension within a membrane of cell.
The formation of micelles takes place only at particular temp, that temperature is called **Kraft temperature**. The concentration above which micelle formation becomes appreciable is termed is **critical micelles concentration**. its value depend upon natures of D.P. and D.M. eg. Surface active agent (surfactants, which decrease the surface tension) like soaps and detergents form micelle beyond CMC (~10^{-3} mol/litre for soaps).
* Usually longer the hydrophobic chain, smaller is its CMC.
* Also CMC increase with decreasing polarity of the D.M.
* The micelles 'formation takes place only above a particular temperature called as Kraft Temperature (T_k).
*At CMC, the micelles are spherical in shape, but that start flattening with increase in concentration and ultimately form sheet or film like structures which have a thickness of two molecules. These are called **lamelar micelles** or **McBain Micelles.**

Mechanism of micelle formation: Let us take the example of soap solutions. Soap is sodium salt of a higher fatty acid and may be represented as $RCOO^-Na^+$ e.g., sodium stearate viz.$CH_3(CH_2)_{16}COO^-\ Na^+$ which is a major component of many bar soaps. When dissolved in water, it dissociates into RCOO- and Na+ ions. The $RCOO^-$ ions, however, consist of two parts i.e., long hydrocarbon chain R (also called non-polar 'tail') which is hydrophobic (water repelling) and the polar group COO^- (also called polar-ionic 'head') which is hydrophilic (water loving). The $RCOO^-$ Ions are, therefore, present on the surface with their COO^- groups in water and the hydrocarbon chains R staying away from it, and remain at the surface, but at higher concentration these are pulled into the bulk of the solution and aggregate in a spherical form with their hydrocarbon chains pointing towards the centre with COO^- part remaining outward on the surface. An aggregate thus formed is known as **'Ionic micelle'**. These micelles may contain as many as upto 100 such ions.

Aggregation of $RCOO^-$ ions to form an ionic micelle.

Similarly, in case of detergents, e.g., sodium lauryl sulphate viz. $CH_3(CH_2)_{11}SO_4^-\ Na^+$, the polar group is $-SO_4^-$ along with the long hydrocarbon chain. Hence, the mechanism of micelle formation is



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same as that of soaps.

Example of micelles :

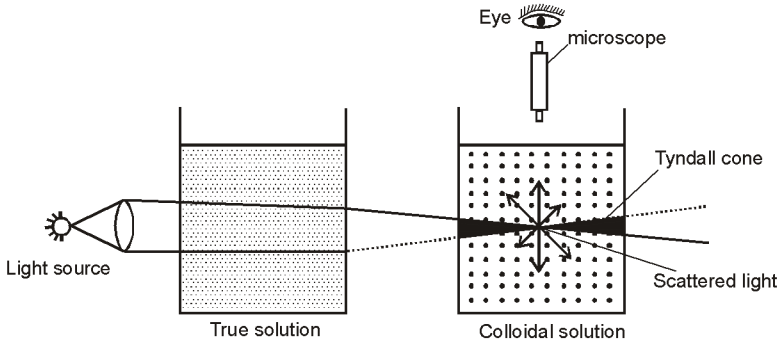
(i) Sodium stearate $C_{17}H_{35}COO^-Na^+$(Soap).

(ii) Sodium lauryl sulphate $CH_3[CH_2]_{11}SO_4^-Na^+$ (Detergent).

(iii) Cetyl trimethyl ammonium bromide (Detergent). $CH_3(CH_2)_{15}N^+(CH_3)_3Br^-$.

Important properties of colloidal sols :

Tyndall effect : Tyndall, in 1869, observed that if a strong beam of light is passed through a colloidal sol placed in a dark place, the path of the beam gets illuminated. This phenomenon is called Tyndall effect, which is due to the scattering of light by the colloidal particles. The illuminated path of beam is called Tyndall cone. This phenomenon is due to scattering of light from the surface of colloidal particles. In a true solution there are no particles of sufficiently large diameter to scatter light & hence the beam is invisible.



*The intensity of scattered light depends on the difference between the refractive indice of the D.P and D.M., In lyophobic colloids, this difference is appreciable and therefore the tyndal effect is quite well defined but in lyophilic sols the difference is very small and the tyndal effect is very weak. Thus in sols of silicic acid, blood serum, albumin, etc. there is little or no tyndal effect.

Example of Tyndall Effect

→ Blue colour of sky and sea water.

→ Visibility of tail of comets.

→ Light thrown from a projector in cinema hall.

→ Appearance of dust particle in a semi darked room.

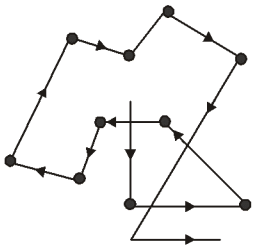
- **Application of Tyndall Effect :**

(i) In making ultramicroscopes.

(ii) In finding heterogenity of solution.

- **Mechanical Properties :**

Brownian movement: Robert Brown, a botanist, discovered in 1827 that pollen grains placed in water do not remain at rest but move about continuously and randomly. Later on, this phenomenon was observed in case of colloidal particles when they were seen under an ultramicroscope. The particles were seen to be in constant zig-zag motion as shown in fig. This zig-zag motion is called Brownian movement.



(Fig. Brownian movement)

Brownian movement arises because of the impact of the molecules of the dispersion medium with the colloidal particles.

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Dependence : *(i) Size of colloidal particles*

$$Mobility \propto \frac{1}{\text{Size of the particle}}$$

(ii) Viscosity of solution

$$Mobility \propto \frac{1}{\text{Viscosity}}$$

(iii) Temperature

$$Mobility \propto Temperature$$

- **Factors Affecting Brownian Movement :**

(i) If particles is large then brownian movement becomes less.
(ii) Brownian movement increases with increasing temperature.
(iii) The brownian movement does not change with time & remains same for months or even for a year.

- **Important :**

(i) In confirmation of kinetic energy.
(ii) Determination of Avogadro numbers.
(iii) Stability of colloidal solution : Brownian movement does not allow the colloidal particles to settle down to gravity & thus is responsible for their stability.

● **Coagulation/Flocculation : This process of aggregation of colloidal particles into an insoluble precipitate by the addition of some suitable electrolyte is known as coagulation.** At lower concentration of electrolytes, the aggregation of particles is called flocculation that can be reversed on shaking while at higher concentration of electrolyte, coagulation takes place and the same cannot be reversed simply by shaking. The stability of the lyophobic colloids is due to presence of charge on colloidal particles. If, somehow, the charge is removed, the particles will come near to each other to form aggregates and settle down under the force of gravity.

Coagulation value or Flocculation value : It needs to be noted that the coagulation of a colloidal solution by an electrolyte does not take place until the added electrolyte has certain minimum concentration in the solution. **The minimum concentration of electrolyte in millimoles required to cause coagulation of one litre of colloidal solution is called coagulation value.** It is express in terms of millimoles/litre.

$$Coagulation\ value = \frac{\text{millimoles of electrolyte}}{\text{volume of sol in litre}}$$

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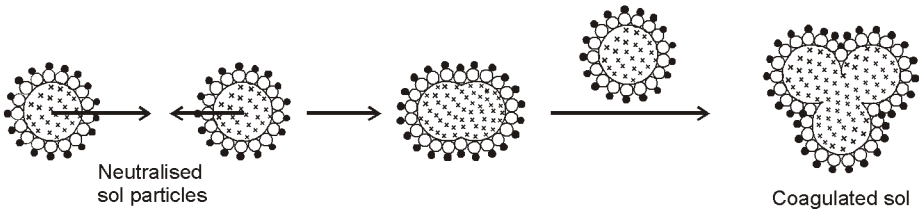

Coagulation of lyophobic sols can be carried out by the following methods.

(i) By electrophoresis

(ii) By mutual precipitation : It is a process in which oppositely charged sol are mixed in proper proportion to neutralise the charge of each other causing coagulation of both the sol.

Example : Positively charged $Fe(OH)_3$ and negatively charged As_2S_3 colloidal particle containing sol on mixing get coagulated.

(iii) By Prolonged Dialysis : On prolonged dialysis, traces of the electrolyte present in the sol are removed almost completely and the colloidal unstable and ultimately coagulate.

(iv) By Boiling : Sols such as sulphur and silver halides dispersed in water may be coagulated by boiling because increased collisions between sol particle and the water molecule removed the adsorbed electrolytes. This takes away the charge from the particles and helps them to coagulate.

(v) By cooling : Certain sol can also be coagulated by lowering temperature. For example, accumulation of cream on the surface of milk on cooling. This is because at lower temperature the dispersion medium molecules do not exert sufficient force on to the dispersed particles and hence the Brownian motion becomes less effective.

(vi) By the addition of electrolyte : When excess of an electrolyte is added, the colloidal particles are precipitated.

- **Comparision of relative coagulating power of two electrolyte for the same colloidal solution :**

The coagulation value decrease with increase in charge of the coagulating ion.

$$\text{Coagulating power} \propto \frac{1}{\text{coagulation value}}.$$

$$\frac{\text{coagulating power of electrolyte A}}{\text{coagulating power of electrolyte B}} = \frac{\text{coagulation value B}}{\text{coagulation value A}}.$$

- **Factor-Affecting Coagulations : (i) Nature of sols :** The lyophobic colloid can easily coagulate because it is a less stable colloid, but lyophilic colloids coagulate hardly by the addition of electrolyte due to protective layer of D.M. surrounding the colloidal particle.

(ii) Nature of electrolyte : In equimolar electrolyte, strong electrolyte have greater coagulating power than weak electrolyte. Example : 0.1M NaCl > 0.1M CH_3COOH.

- **Hardy-Schulze Rule :** According to this rule greater is the valency of coagulating ion, greater its power to cause precipitation. This is known as Hardy-Schulze rule.

In case of positive charged sol, the coagulating power of anion is in the order of $[Fe(CN)_6]^{4-} > PO_4^{3-} > SO_4^{2-} > Cl^-$

In case of negative charged sol, the coagulating power of cation is in the order of $Al^{3+} > Ba^{2+} > Na^+$.

- The coagulating power of bivalent ion is 20-80 times higher than monovalent ion and coagulating power of trivalents is many times more than bivalent.

Gold Number : Zpsigmondy (1901) introduce a term called gold number it is defined as ''the minimum amount of the protective colloid in milligrams which when added to 10 ml of a standard gold sol is just sufficient to prevent a colour change from red to blue on the addition of 1 ml of 10% sodium chloride solution. It may be noted that smaller of the gold number, greater will be protecting power of the protective colloid.

$$\text{Protecting power} \propto \frac{1}{\text{gold number}}.$$

$$\text{Gold Number} = \frac{\text{weight of lyophilic sol in mg} \times 10}{\text{volume of gold sol in mL}}$$

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STRUCTURAL IDENTIFICATION & POC

Introduction :

The main objective of an organic chemist is the determination of the structure of a new organic compound which has been obtained in pure state either from a natural source or synthesised in the laboratory.

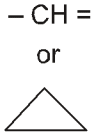
In order to establish the correct structure of an organic compound, it is necessary to detect skeleton of compound, elements and functional groups present in the organic compound.

1. Hydrogen Deficiency Index :

The presence of double bonds or rings within a molecule is indicated by a quantity called hydrogen deficiency index.

Applications : To indentify the no. of π bonds or rings and also helpful in determining the structure of the molecule.

Definition : Deficiency of 2H atoms with respect to fully saturated acyclic hydrocarbon is equivalent to One HDI It is also known as Double Bond Equivalence (DBE) or degree of unsaturation (DU).

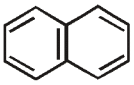
$H_3C-H_2C-CH_3$ (DU = O) $\xrightarrow{-2H}$ [$CH_3-CH=CH_2$ or  (DU = 1)] $\xrightarrow{-2H}$ $CH_3-C\equiv CH$ or $CH_2=C=CH_2$ or (DU = 2)

$$\text{Degree of unsaturation (D.U.)} = \frac{(2n+2)-(\text{No. of H atoms}+\text{No. of X atoms}-\text{No. of N atoms})}{2}$$

Where n = number of carbon atoms in the molecule

Note : Total no. of cyclic rings + double bonds will give us degree of unsaturation.
One double bond = one DU
One ring = one DU
One triple bond = two DU

Ex. (i) $CH_2 = CH_2$ DU = $\frac{(2\times2+2)-4}{2}$ = 2/2 = 1 (ii) DU = 2

(iii) DU = 4 (iv)  DU = 7

(v) C_6H_6O DU = $\frac{(2n+2)-6}{2} = \frac{14-6}{2}$ = 4 (vi) C_6H_5Cl D.U. = $\frac{14-(5+1)}{2}$ = 4

2. Monohalogenation :

When an alkane or a cycloalkane is treated with halogen (Cl_2, Br_2, F_2, I_2), a photochemical reaction takes place, in which a C–H bond cleaves and a C–X bond is formed. In such reactions if one H-atom is substituted by one halogen atom, then this is known as monohalogenation reaction.

Applications : If a molecule has more than one type of H-atoms, then on monochlorination, it forms a mixture of monochloroisomers. **All these products (structures) are position isomers.**

Conclusion : Hence, it can be concluded that the total no. of position isomers (structural) of monochloro compounds is equal to the number of different types of H-atoms present in the reactant. The different type of H-atoms are also known as non-identical Hydrogens or non-equivalent Hydrogens or chemically different Hydrogens.

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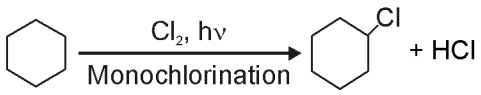
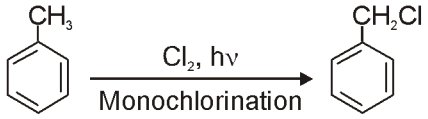
Note : In aromatic hydrocarbons, the hydrogen atoms of the side-chain are chlorinated, but H-atoms of Benzene ring are stable.

Ex. (i) $CH_4 \xrightarrow[\text{Monochlorination}]{Cl_2,\ h\nu} CH_3Cl + HCl$ ½1–Monochloroproduct½

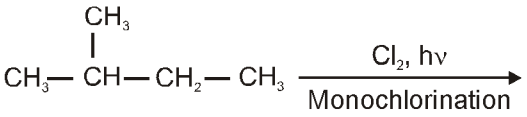
(ii)  ½1–Monochloroproduct½

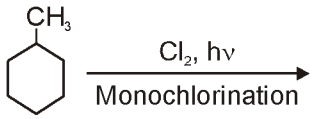
(iii)  ½1–Monochloroproduct½

Note : Only one monochloro product is formed because aromatic H atoms are inert towards this reaction.

(iv) $CH_3–CH_2–CH_2–CH_3 \xrightarrow[\text{Monochlorination}]{Cl_2,\ h\nu}$ 2 Products (structure isomers)

(v) $CH_3–CH_2–CH_2–CH_2–CH_3 \xrightarrow[\text{Monochlorination}]{Cl_2,\ h\nu}$ 3 Products (structure isomers)

(vi)  4 Products (structure isomers)

(vii)  5 Products (structure isomers)

3. Catalytic Hydrogenation :

Alkenes, Alkynes, polyenes or polyynes can be hydrogenated by using catalysts Ni/Pt/Pd at room temperature. All Carbon–Carbon π bonds(C=C, C≡C) get hydrogenate. The reaction can't be stopped at any intermediate stage.

Note : **(i)** Aromatic π bonds are stable at room temperature but can be hydrogenated at high temperature.

(ii) It can be concluded that the hydrogenation product of an alkene or alkyne or any unsaturated compound is always a saturated compound.

(iii) The no. of moles of H_2 consumed by 1 mole of compound is equal to the no. of π bonds.

(iv) During catalytic hydrogenation *carbon skelton* does not change.

Application : This reaction gives an information about molecule that the molecule is saturated or unsaturated.

General reaction :

(a) $R–CH=CH–R + H_2 \xrightarrow{Ni} R–CH_2–CH_2–R$

(b) $R–C\equiv C–R + 2H_2 \xrightarrow{Ni/Pt/Pd} R–CH_2–CH_2–R$

$\downarrow H_2$

$R–CH=CH–R \xrightarrow{H_2} R–CH_2–CH_2–R$
(Not isolated)

(c) $CH_2=CH–CH=CH_2 \xrightarrow{2H_2/Ni} CH_3–CH_2–CH_2–CH_3$

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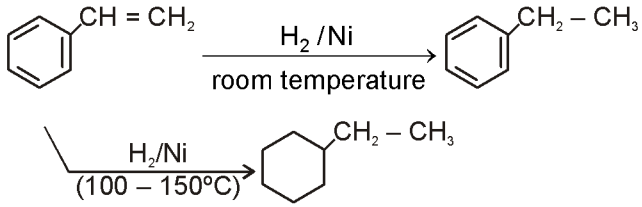
(d) 

4. Ozonolysis :

Ozonolysis reaction is used to determine the position of C=C, C≡C in a molecule. In this reaction Alkene, Alkyne and polyalkene on ozonolysis undergo oxidative cleavage. It is of two types.

(i) Reductive ozonolysis

Reagents are : (1) O_3 (ozone) (2) Zn or $(CH_3)_2S$ and H_2O or CH_3COOH

The products are carbonyl compounds (aldehydes or ketones).

(ii) Oxidative ozonolysis

Reagents are : (1) O_3 (ozone) (2) H_2O_2 or H_2O

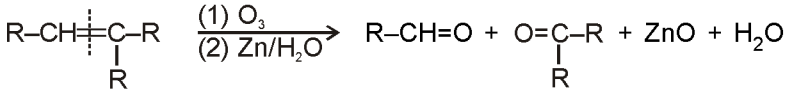
The products are ketones and/or acids.

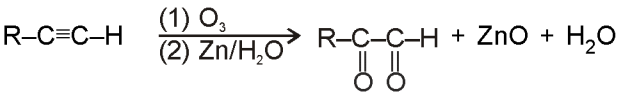
Note : (i) Ozonolysis does not interfere with other functional groups.

(ii) At higher temperature, the aromatic double bonds can also undergo ozonolysis reaction.

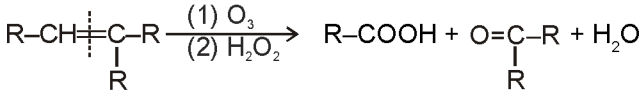
General Reaction :

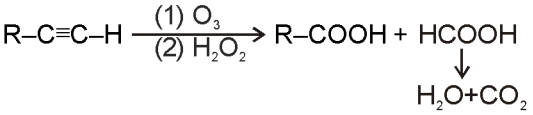
(i) Reductive ozonolysis





(ii) Oxidative ozonolysis



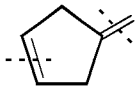


Ex. (a) $CH_2=CH_2 \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} CH_2=O + CH_2=O$

(b) $CH_3–CH_2–CH=CH_2 \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} CH_3–CH_2–CH=O + O=CH_2$

(c) $CH_2=CH–CH_2–CH=CH–CH_3 \xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3} CH_2=O + O=CH–CH_2–CH=O + O=CH–CH_3$

(d) $\xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3}$ 2 OHC–CH_2–CHO (Propandial)

(e)  $\xrightarrow[(2)\ Zn/H_2O]{(1)\ O_3}$ O=CH–CH_2–C(=O)–CH_2–CH=O + O=CH_2

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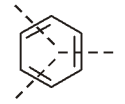
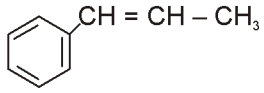
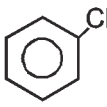
(f)  $\xrightarrow[Zn,H_2O]{O_3(\Delta)}$ 3 O = C(H) – C(H) = O (Ethanedial) or (glyoxal)

(g)  $\xrightarrow{(i)O_3}$ $\xrightarrow{(ii)Zn,\ H_2O}$  + $CH_3CH{=}O$

$\xrightarrow{(i)O_3,\Delta}$ $\xrightarrow{(ii)Zn,\ H_2O}$ $2\ O{=}CH{-}CH{=}O + O{=}CH{-}\underset{\underset{O}{\|}}{C}{-}CH{=}O + CH_3CH{=}O$

5. Practical Organic Chemistry (POC) :

When we are looking for the information regarding the functional nature of the compound then we go for the following lab test.

(A) Test for acidic/active hydrogen :

When any compound release H_2 gas after reaction with sodium or potassium or alkali metals or sodamide then this reflects the presence of acidic hydrogen.

Active H : Those H atoms which is attached with more electronegative atoms like O,N,S,X, C_{sp}.

General reaction : $Z-\textcircled{H} + Na \rightarrow \bar{Z}\overset{+}{Na} + \frac{1}{2}H_2\uparrow$

Ex. **(a)** $R-NH_2 \xrightarrow{Na} R-NHNa + \frac{1}{2}H_2\uparrow$ **(b)** $R-SH \xrightarrow{Na} RSNa + \frac{1}{2}H_2\uparrow$

(c) $R-COOH \xrightarrow{Na} R-COONa + \frac{1}{2}H_2\uparrow$ **(d)** $PhOH \xrightarrow{Na} PhONa + \frac{1}{2}H_2\uparrow$

(e) $R{-}OH + Na \rightarrow R{-}O^-Na^+ + \frac{1}{2}H_2\uparrow$ **(f)** $R{-}SO_3H + Na \rightarrow R{-}SO_3^-Na^+ + \frac{1}{2}H_2\uparrow$

(g) $R{-}C{\equiv}CH + Na \rightarrow R{-}C{\equiv}C^-Na^+ + \frac{1}{2}H_2\uparrow$ **(h)** $R-CH=CH_2 + Na \rightarrow$ No reaction.

(i) $R-CH_2-O-CH_3 + Na \rightarrow$ No reaction. **(j)** $HCl \xrightarrow{Na} NaCl + \frac{1}{2}H_2\uparrow$

(B) Test for unsaturation :

(i) Bromine water test (Br_2 + H_2O, Red-brown solution) :

This is used to distinguish between saturated (alkane) and unsaturated (alkene/alkyne) hydrocabon.

$R{-}CH_2{-}CH_3 \xrightarrow{Br_2,H_2O}$ no reaction

$R{-}CH=CH_2 \xrightarrow{Br_2,H_2O} R{-}\underset{Br}{\underset{|}{CH}}{-}\underset{Br}{\underset{|}{CH_2}}$ Red-brown colour disappears

$R{-}C{\equiv}CR \xrightarrow{Br_2,H_2O} R{-}\underset{Br}{\underset{|}{\overset{Br}{\overset{|}{C}}}}{-}\underset{Br}{\underset{|}{\overset{Br}{\overset{|}{C}}}}{-}R$ Red-brown colour disappears

Note : Benzene does not give this test, whereas phenol and aniline gives this test.

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(ii) Baeyer Reagent (Cold dil. alkaline $KMnO_4$ Pink/purple solution) :

This is also used to distinguish between saturated (alkane) and unsaturated (alkene/alkyne) compounds.

$$R–CH_2–CH_3 \xrightarrow{\text{Cold, dil.KMnO4}} \text{no reaction}$$

$$R–CH=CH_2 \xrightarrow{\text{Cold, dil.KMnO4}} R–\underset{OH}{CH}–\underset{OH}{CH_2} \quad \text{Purple colour disappears}$$

$$R–C\equiv CR \xrightarrow{\text{Cold, dil.KMnO4}} R–\overset{O}{\overset{\|}{C}}–\overset{O}{\overset{\|}{C}}–R \quad \text{Purple colour disappears}$$

$$2KMnO_4 \longrightarrow K_2O + 2MnO_2 + 3[O]$$

↑ Brown solution

(C) Test for terminal Alkyne

(i) Tollen's Reagent [$AgNO_3 + NH_4OH$ or $\{Ag(NH_3)_2\}^+OH^{\ominus}$] :

Tollen's Reagent gives white precipitate (Silver alkynide) with terminal alkyne.

$$R–C\equiv CH \xrightarrow{AgNO_3+NH_4OH} R–C\equiv CAg + NH_4NO_3$$

White precipitate

(ii) Ammonical cuprous chloride ($Cu_2Cl_2 + NH_4OH$) :

It gives red precipitate with terminal alkyne.

$$2\,R–C\equiv CH \xrightarrow{Cu_2Cl_2+NH_4OH} 2\,R–C\equiv CCu\downarrow + 2NH_4Cl + 2H_2O$$

Red precipitate

(D) Test for alcohols :

(i) Lucas Reagent test (Conc. HCl + anhydrous $ZnCl_2$)

* It gives white turbidity or cloudiness with alcohols (OH groups attached with sp^3 hybridised carbon).

* Lucas Reagent is used to distinguish between 1°, 2°, 3° alcohol because 1°, 2°, 3° alcohols react with different rates.

(a) 1° alcohol $R–CH_2–OH \xrightarrow{\text{Conc. HCl + anhydrous } ZnCl_2} R–CH_2–Cl$

(does not give appreciable reaction or gives white turbidity in 30 min.)

(b) 2° alcohol $R_2CH–OH \xrightarrow{\text{Conc. HCl + anhydrous } ZnCl_2} R_2CH–Cl$ (gives white turbidity in 5 min.½

(c) 3° alcohol $R_3C–OH \xrightarrow{\text{Conc. HCl + anhydrous } ZnCl_2} R_3C–Cl$ (gives white turbidity immediately)

*** Phenols and enols do not give Lucas test.**

(ii) Victor Mayer test :

1° Alcohol%

$$RCH_2–OH \xrightarrow{P+I_2} RCH_2–I \xrightarrow{AgNO_2} RCH_2–NO_2 \xrightarrow{HNO_2} R–\overset{NO_2}{\overset{|}{C}}=N–OH \xrightarrow{base} \text{Blood Red colour}$$

2° Alcohol% $R_2CH–OH \xrightarrow{P+I_2} R_2CH–I \xrightarrow{AgNO_2} R_2CH–NO_2 \xrightarrow{HNO_2} R_2\overset{N=O}{\overset{|}{C}}–NO_2 \xrightarrow{base}$ Blue colour

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3° Alcohol $\xrightarrow{}$ $R_3C-OH \xrightarrow{P+I_2} R_3C-I \xrightarrow{AgNO_2} R_3C-NO_2 \xrightarrow{HNO_2}$ no reaction $\xrightarrow{base}$ no colour

(iii) Cerric Ammonium Nitrate test : (Group reagent for alcohols)

Alcohols(1°,2°,3°) also give characteristic red colour with cerric ammonium nitrate $[(NH_4)_2Ce(NO_3)_6]$ solution.

(E) Test for phenol or enol :

Neutral $FeCl_3$ test :

It forms coloured complex with phenol or enol (OH groups attached with sp^2 carbon).

$$6\,PhOH + FeCl_3 \longrightarrow \left[\left(PhO\right)_6 Fe\right]^{3-} + 3HCl + 3H^+$$

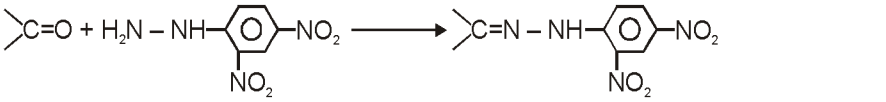
(Coloured complex)

*** It does not give positive test with alcohol.**

(F) Test for carbonyl compounds :

2, 4-DNP ½2, 4– Dinitrophenyl hydrazine+½Test :

Carbonyl compouds (all aldehydes and ketones) give yellow–orange precipitate with 2,4–DNP. It is also knwon as **Brady's reagent**.



2, 4-DNP (yellow–orange precipitate of Hydrazone)

(G) Test for Aldehydes :

(a) Tollen's Reagent [$AgNO_3 + NH_4OH$ or $\{Ag(NH_3)_2\}^+ OH^{\ominus}$] :

Tollen's Reagent gives silver mirror or Black precipitate with aldehydes.

$$R-CH=O \xrightarrow{AgNO_3+NH_4OH} R-\overset{O}{\overset{\|}{C}}-O^{\ominus} + Ag\downarrow$$

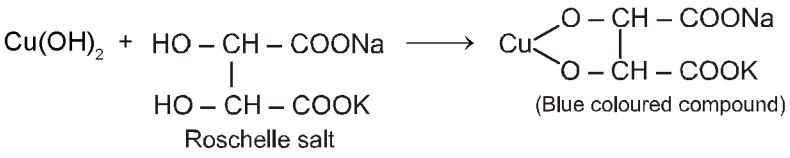
silver mirror

Note : HCOOH also gives this test.

(b) Fehling's solution :

It is an alkaline solution of cupric ion complexed with sodium potassium tartarate.
There are two solutions in Fehling solution
Solution (A) : $CuSO_4$ solution and
Solution (B) : Alkaline solution of sodium potassiumtartarate.
When these two solutions are mixed we get deep blue coloured solution.

$$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 + Na_2SO_4$$



Roschelle salt (Blue coloured compound)

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Equal volume of both the solutions are heated with aldehyde to give red brown precipitate of cuprous oxide (Cu_2O) which confirms the presence of aldehyde.

$$R-CHO + \underset{\text{Blue}}{2CuO} \longrightarrow RCOOH + Cu_2O \text{ (Red ppt)}$$

$$RCHO + 2Cu^{2+} + 3OH^{\ominus} \longrightarrow RCOO^{\ominus} + \underset{\text{(Red ppt)}}{2Cu^{\oplus}} + 2H_2O$$

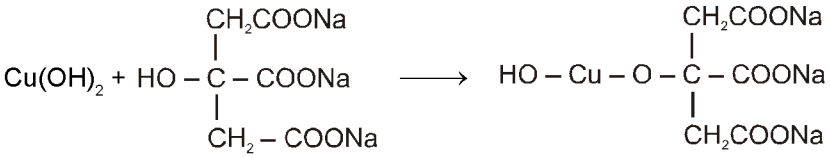
(c) Benedict solution :

It also consists of two solutions.

Solution (A) : $CuSO_4$ solution and

Solution (B) : Alkaline solution of sodium Citrate.

$$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 + Na_2SO_4$$



(Blue colour)

Aldehyde gives positive test with Benedict solution.

$$RCH=O + \underset{\text{(blue)}}{2Cu^{2+}} + 3OH^{\ominus} \longrightarrow RCOO^{\ominus} + \underset{\text{(Red ppt.)}}{2Cu^{\oplus}} + 2H_2O$$

(d) Schiff's reagent :

It is dilute solution of rosaniline hydrochloride whose pink colour has been discharged by passing SO_2. Aldehyde restores pink colour when treated with schiff's reagent (Magenta solution in H_2SO_3).

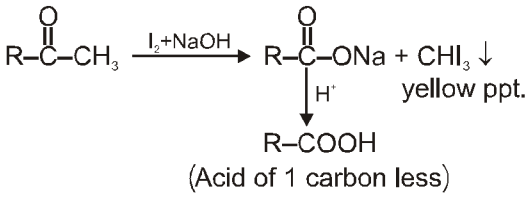
Benzaldehyde does not gives Fehling, Bendict & Schiff's test

*** Aromatic aldehydes (**Benzaldehyde**) do not gives** Fehling and Bendict **test.**

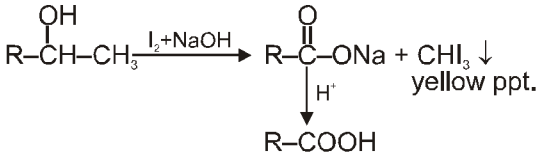
(H) Iodoform Test :

Reagents : I_2 + NaOH or NaOI ¼Where R = H, alkyl, aryl group½

Acetaldehyde, all methyl ketones & ethyl alcohol give Iodoform test.



(Acid of 1 carbon less)



(I) Test for acids/esters/amides :

(i) Sodium bicarbonate test ($NaHCO_3$) %

All the acids which are stronger than H_2CO_3 give CO_2 gas with $NaHCO_3$.

$$HCl + NaHCO_3 \longrightarrow NaCl + H_2CO_3$$

$$\longrightarrow H_2O + CO_2 \uparrow$$

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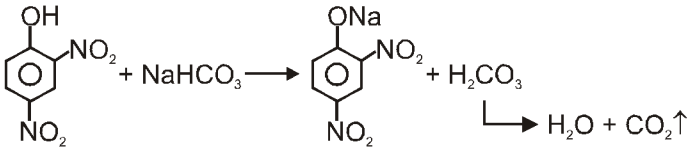
$$RCOOH + NaH\overset{*}{C}O_3 \longrightarrow R{-}COONa + H_2\overset{*}{C}O_3 \longrightarrow H_2O + \overset{*}{C}O_2\uparrow$$

$$RSO_3H + NaHCO_3 \longrightarrow RSO_3Na + H_2CO_3 \longrightarrow H_2O + CO_2\uparrow$$



Note : If electron withdrawing group (NO_2) is present at ortho or para position of phenol then it gives positive test with sodium bicarbonate.

(ii) Litmus test ⁘ Acid converts blue litmus into red litmus while base converts red litmus into blue.

(iii) Acid amide gives smell of ammonia when heated with alkali.

$$R{-}\underset{\underset{O}{\|}}{C}{-}NH_2 \xrightarrow[\Delta]{NaOH} R{-}\underset{\underset{O}{\|}}{C}{-}ONa + NH_3$$

(iv) Esters are sweet (fruity) smelling liquids.
When react with NaOH & Phenophthalein then pink colour will disappear on heating.

RCOOR' + NaOH + Phenophthalein (Pink) $\xrightarrow{\Delta}$ R COOH + R' OH (Colourless solution)

(J) Test for Amines :

(i) $NaNO_2$ + aqueous HCl test ⁘ it is used to distinguish between 1°, 2° and 3° amines and also distinguish between aliphatic and aromatic primary amines.

1° Aliphatic Amine ⁘ $R{-}CH_2{-}NH_2 \xrightarrow{NaNO_2 + aq.\ HCl} R{-}CH_2{-}OH + N_2\uparrow$

1° Aromatic Amine : $Ph{-}NH_2 \xrightarrow[(0\text{–}5°C)]{NaNO_2 + aq.\ HCl} Ph{-}\overset{+}{N}_2\overset{\ominus}{Cl}$ ¼diazonium salt)

2° Amine ⁘ $R - NH - R \xrightarrow{NaNO_2 + aq.\ HCl} R{-}\underset{\underset{N=O}{|}}{N}{-}R$ (N -Nitroso amine) yellow oily liquid

3° Amine : $R_3N \xrightarrow{NaNO_2 + aq.\ HCl}$ unstable nitrite

(ii) Carbyl amine test ($CHCl_3$ + KOH)

1° Amine (Aliphatic or Aromatic) ⁘

$$R{-}CH_2{-}NH_2 \xrightarrow{CHCl_3 + KOH} R{-}CH_2{-}N \equiv C + 3KCl + 3H_2O$$

(unpleasent smell of isocyanide)

$$Ph - NH_2 \xrightarrow{CHCl_3 + KOH} Ph - N \equiv C$$

*** 2° Amine and 3° Amine do not give this test.**

(iii) Hofmann mustard oil test

It is a test for 1° amine and aniline. Primary amine reacts with carbon disulphide to form dithioalkyl carbamic acids which decompose on heating with mercuric chloride ($HgCl_2$) to give alkyl isothiocyanates having smell like mustard oil .

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(a) $R-NH_2 + S=C=S \longrightarrow R-NH-C(=S)-SH \xrightarrow[\Delta]{HgCl_2} R-N=C=S + HgS + 2HCl$

Dithioethylcarbamic acid — Alkylisothiocyanate

(b) $CH_3CH_2NH_2 + S=C=S \xrightarrow{\Delta} CH_3CH_2NH-C(=S)-SH \xrightarrow[\Delta]{HgCl_2} CH_3CH_2-N=C=S + HgS + 2HCl$

(1° amine) — Ethyl isothiocyanate

Similarly aniline gives phenlyisothiocyanate

(c) $C_6H_5NH_2 \xrightarrow[(ii)\ HgCl_2/]{(i)\ S=C=S} C_6H_5N=C=S$

Phenylisothiocyanate

*** 2° Amine and 3° Amine do not give this test.**

(iv) Hinsberg reagent ($C_6H_5SO_2Cl$) :

This is used to distinguish between 1°,2° & 3° amine.

1° Amine : $R-NH_2 + PhSO_2Cl \xrightarrow{Pyridine} R-NH-SO_2-Ph \xrightarrow{Base}$ Compund is **soluble** in base.

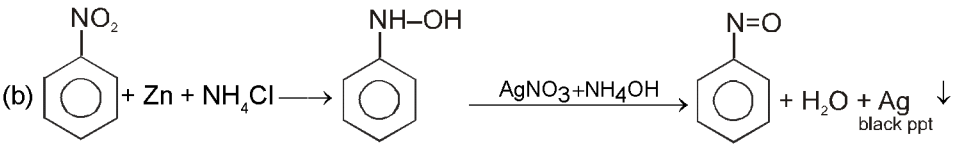
2° Amine : $R_2NH + PhSO_2Cl \xrightarrow{Pyridine} R_2N-SO_2-Ph \xrightarrow{Base}$ Compund is **insoluble** in base.

3° Amine : $R_3N\ PhSO_2Cl \xrightarrow{Pyridine}$ No reaction.

(K) Test of Nitro group : (Mulliken- Barker's Test)

Nitroalkane & nitrobenzene give black ppt on reduction with Zn and ammonium chloride followed by treating with Tollen's reagent. **This is also said to be Mulliken's test.**

(a) $R-NO_2 \xrightarrow[\Delta]{Zn-NH_4Cl} R-NHOH \xrightarrow[T.R.]{AgNO_3 + NH_4OH} Ag\downarrow$

(b) 

(L) Test for Carbohydrates & Sugars

Molisch test :

This is a general test for carbohydrates. One or two drops of alcoholic solution of α-naphthol is added to 2 ml glucose solution. 1 ml of conc. H_2SO_4 solution is added carefully along the sides of the test tube. The formation of a violet ring at the junction of two liquids confirms the presence of a carbohydrate or sugar.

(M) Test for Amino acids and Proteins :

Ninhydrin Test : This is used to detect the presence of amino acids & proteins.

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6. Elements detection :

(A) Detection of Carbon and Hydrogen :

Carbon and hydrogen are detected by heating the compound with copper(II) oxide. Carbon present in the compound is oxidised to carbon dioxide (tested with lime-water, which develops turbidity) and hydrogen to water (tested with anhydrous copper sulphate, which turns blue).

$$C + 2CuO \xrightarrow{\Delta} 2Cu + CO_2$$

$$2H + CuO \xrightarrow{\Delta} Cu + H_2O$$

$$CO_2 + Ca(OH)_2 \xrightarrow{\Delta} CaCO_3 \downarrow + H_2O$$

$$5H_2O + CuSO_4 \xrightarrow{\Delta} CuSO_4.5H_2O$$

White Blue

(B) Detection of Other Elements :

Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by "Lassaigne's test". The elements present in the compound are converted from their covalent form to their ionic form by fusing the organic compound with sodium metal.
Following reactions take place: C, N, S and X come from organic compound.
Cyanide, sulphide and halide of sodium so formed on sodium fusion are extracted from the fused mass by boiling it with distilled water. This extract is known as sodium fusion extract or Lassaigne solution.

$$Na + C + N \xrightarrow{\Delta} NaCN$$

$$2Na + S \xrightarrow{\Delta} Na_2S$$

$$Na + X \xrightarrow{\Delta} NaX$$

$$Na + C + N + S \xrightarrow{\Delta} NaSCN$$

(i) Test for Nitrogen :

The sodium fusion extract is boiled with iron(II) sulphate and then acidified with dilute sulphuric acid. The formation of Prussian blue or green colour confirms the presence of nitrogen. Alternatively $FeCl_3$ and dil. HCl may be added.

$$6NaCN + FeSO_4 \longrightarrow Na_4[Fe(CN)_6]$$

$$Na_4[Fe(CN)_6] + 4Fe^{3+} \xrightarrow{xH_2O} Fe_4[Fe(CN)_6]_3.\ xH_2O$$

Ferric ferrocyanide (Prussian blue)

(ii) Test for Sulphur

(a) The sodium fusion extract is acidified with acetic acid and lead acetate is added to it. A black precipitate of lead sulphide indicates the presence of sulphur.

$$Na_2S + (CH_3COO)_2Pb \longrightarrow PbS$$

Black

(b) On treating sodium fusion extract with sodium nitroprusside, appearance of a violet colour indicates the presence of sulphur.

$$Na_2S + Na_2[Fe(CN)_5\ NO] \longrightarrow Na_4[Fe(CN)_5NOS]$$

Sodium nitroprusside Sodium thionitroprusside (Violet/Purple)

(c) In case, nitrogen and sulphur both are present in an organic compound, then sodium thiocyanate is formed. It gives blood red colour with neutral $FeCl_3$.

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$$Na + C + N + S \longrightarrow NaSCN$$

$$\text{Neutral } FeCl_3 + NaSCN \longrightarrow \underset{\text{Blood red}}{Fe(SCN)_3}$$

(iii) Test for Halogens :

The sodium fusion extract is acidified with nitric acid and then treated with silver nitrate. A white precipitate, soluble in ammonium hydroxide shows the presence of chlorine, a yellowish precipitate, sparingly soluble in ammonium hydroxide shows the presence of bromine and a yellow precipitate, insoluble in ammonium hydroxide shows the presence of iodine.

Note : The sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test. These ions would otherwise interfere with silver nitrate test for halogens.

$$NaX + AgNO_3 \longrightarrow AgX \downarrow$$

(iv) Test for Phosphorus :

The compound is heated with an oxidising agent (sodium peroxide). The phosphorus present in the compound is oxidised to phosphate. The solution is boiled with nitric acid and then treated with ammonium molybdate. A yellow colouration or precipitate indicates the presence of phosphorus.

$$Na_3PO_4 + 3HNO_3 \longrightarrow H_3PO_4 + 3NaNO_3$$

$$\underset{\substack{\text{Ammonium}\\\text{molybdate}}}{H_3PO_4 + 12(NH_4)_2MoO_4} + 21\ HNO_3 \longrightarrow \underset{\substack{\text{Ammonium}\\\text{Phosphomolybdate}}}{(NH_4)_3PO_4.\ 12\ MoO_3} + 21\ NH_4NO_3 + 12H_2O$$

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IONIC EQUILIBRIUM

Equilibrium established between **dissociated part & undissociated part** is known as **ionic equilibrium**

$AB(aq) \rightleftharpoons A^+(aq) + B^-(aq)$

T-1

Classification of substances	
Electrolyte: ($0 < \alpha \leq 1$)	**Non Electrolyte :($\alpha \approx 0$)**
Gives ions, conducts electricity **in aqueous & molten** state and has **ionic/Polar covalent bond**	does not give ions & does not Conduct electricity and has **non Polar covalent bond**.
e.g. acid, base & salt	e.g. urea, glucose, sucrose, CCl_4, acetone, ether, alkane.

T-2

Classification of Electrolyte: On the basis of **degree of dissociation (α)** electrolyte can be classified as.	
Strong electrolyte (S.E.):	**Weak electrolyte (WE):**
Electrolyte which are 100% ionized. **(α = 100% or 1)**	Electrolyte which are negligible dissociated into their ions. **(α << 1 (unknown))**
(a) **Strong Acid:** $HClO_4$, HI, HBr, HCl, H_2SO_4, HNO_3	**(a) Weak acid:** $HClO_2$, HClO, H_2SO_3, H_2SO_5, HNO_2, HF, H_2CO_3, H_2S, CH_3COOH, HCOOH, Ph-COOH ,H_3PO_4 etc **(All organic acid)**
(b) **Strong base:** NaOH, KOH, RbOH, CsOH, $Sr(OH)_2$, $Ba(OH)_2$	(b) **Weak Base:** NH_3, NH_4OH, Ph $\ddot{N}H_2$ (Aniline), $C_5H_5\ddot{N}$ (Pyridine), N_2H_4. **(all organic base),** LiOH, $Be(OH)_2$, $Mg(OH)_2$, **d-block hydroxide**
(c) **Salt:** NaCl, (**SASB**), CH_3COONa (**WASB**), NH_4Cl (**WBSA**), CH_3COONH_4 (**WAWB**)	(c) **All Sparingly soluble salt**: AgCl, AgBr, AgI, $BaSO_4$, CuS, HgS etc

Theories Defining acids and bases

(i) **Arrhenius acid - base Theory :**

T-3 Arrhenius acid (AA): Any substance which give **H^+ Ion in water**

Type of acid	No of **H^+**	Examples	General Notation of Acid
Mono basic / mono protic acid:	**1 H^+**	HCl, HBr, HI, H_3PO_2	**HA**
Dibasic/ diprotic acid	**2 H^+**	H_2S, H_3PO_3, H_2CO_3,	**H_2A**
Tribasic/ triprotic acid	**3 H^+**	H_3PO_4, H_3AsO_4 H_3As	**H_3A**

Arrhenius base (AB): any substance which give **OH^- ion in water.**

Type of base	No of **OH^-**	Example	General Notation of Base
Mono acidic Base	**1 OH^-**	NaOH, KOH, NH_4OH or $NH_3.H_2O$	**BOH**
Di acidic Base	**2 OH^-**	$Ca(OH)_2$, $Ba(OH)_2$, $Sr(OH)_2$	**$B(OH)_2$**
Tri acidic base	**3 OH^-**	$Al(OH)_3$, $Fe(OH)_3$	**$B(OH)_3$**

NOTE: Acidity: defined for **base** and **equal to no of OH^-** ion per molecule.

Basicity: defined for **acid** and **equal to no of H^+** ion per molecule.

T-4 (ii) Bronsted-Lowry concept (Conjugate acid-base concept / Protonic concept)

H^+ Donor is Bronsted acid & **H^+ acceptor** is Bronsted base in **any solvent**.

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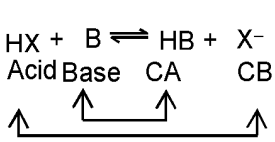
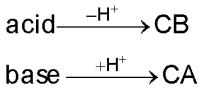
Conjugate acid base pair: acid base pair having **difference of one H^+ ion**.





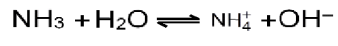
e.g. (NH_3,NH_4^+), (HCl, Cl^-), (H_2O, OH^-), (H_3O^+, H_2O)

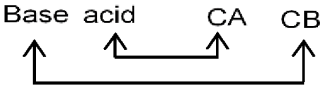
P-1 Write Conjugate Acid & Base of the following:

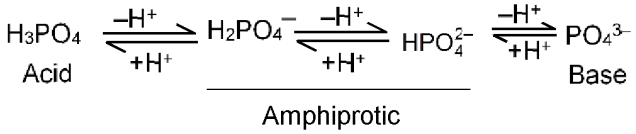
Compound	CA	CB
HS^-	H_2S	S^{2-}
HCO_3^-	H_2CO_3	CO_3^{2-}
$H_2Po_4^{--}$	H_3PO_4	HPO_4^{2-}

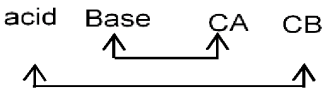
Amphiprotic Species : Can **donate as well as accept** the H^+ ion (proton)

e.g. HS^-,HCO_3^-,H_2O









H_2O as an amphiprotic species

NOTE: (1) Acid base reaction generally proceeds from **stronger (acid / base)** towards weaker (acid / base)

(2) Strong acid / base have **very weak** conjugate base / acid.

(3) Weak acid / base have **weak conjugate** base / acid.

T-5 (iii) Lewis Acid/Base Theory:

Lewis acid: Lone pair **acceptor**	**Lewis base:** Lone pair **donor**
e.g. (1) BF_3, $AlCl_3$, $BeCl_2$ (**Incomplete octet**) (2) SiF_4, PCl_5, SO_3 (Species in which central atom is surrounded by **more EN atom** & has Vacant **d orbital**). (3) Mg^{2+}, Al^{3+}, Fe^{2+}, Zn^{2+} (**cation**). **Note :** Na^+, K^+, Rb^+ Cs^+. have **quite lesser tendency** to act as Lewis acid	e.g. (1) NH_3, H_2O, ROH, $R\ddot{N}H_2$ (Molecule in which central atom has **atleast one lone pair** (with **Complete octet**) surrounded By **less EN atom.** (2) Anion: Cl^-, F^-, OH^-, NH_2^-
Note: (1) All anions don't act as Lewis base: e.g. BH_4^-, AlH_4^-, $AlCl_4^-$, CH_2 (2) All cations don't act as Lewis acid: e.g. NH_4^+ (3) Boric acid behaves as **Lewis acid not Arrhenius / Bronsted acid.** $B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+$ (From Water)	

T-6 **Degree of dissociation (α): Fraction of 1 mole of electrolyte** that has dissociated under given condition.

$$\left[\alpha = \frac{\text{No of mole dissocated}}{\text{No of mole intially taken}}\right] \qquad (0 < \alpha \leq 1)$$

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Ostwald Dilution Law: (Applicable for **weak electrolyte only**)

$AB(aq) \rightleftharpoons A^+(aq) + B^-(aq)$ C = concentration (mole/Ltr)

t = 0 C

t = eq^m C-Cα Cα Cα

$K_i = \frac{[A^+][B^-]}{[AB]} = \frac{C\alpha^2}{1-\alpha}$, K_i is **ionization/ Dissociation** constant (Depends **only on temperature**)

If α is small w.r.t 1(unity) $1-\alpha \approx 1$ then $K_i = C\alpha^2 \Rightarrow \alpha = \sqrt{\frac{K_i}{C}}$ When (α < 10% or 0.1)

$[ion] = [A^+] = [B^-] = C\alpha = \sqrt{K_i C}$

Factor affecting Degree of Dissociation (α):

(i) Nature of Solute: (1) For strong electrolyte: **α = 1 or 100% (known).**

(2) For weak electrolyte: **α < 1 (unknown).**

(ii) Nature of solvent: Solvent should be in **polar in nature** and its dielectric constant should be **high**.

Note: water is a **universal solvent**.

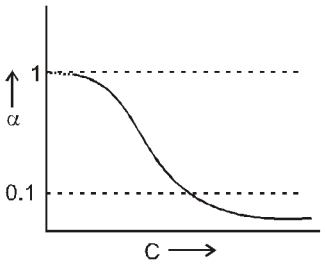
(iii) Concentration of Electrolyte:

(i) For strong electrolyte: **α = 1**, **always** at **any concentration** of strong electrolyte.

(ii) For weak electrolyte: 'α' $\alpha \frac{1}{\sqrt{Conc}}$. $V\downarrow, C\uparrow, \alpha\downarrow$ or $V\uparrow, C\downarrow, \alpha\uparrow$

At **infinite dilution**: $V \to \infty$, $C \to 0$, $\alpha \to 1$

Note: At infinite dilution: **weak electrolyte** behaves as **strong electrolyte**.



(iv) Temperature: Generally, **all Dissociation reaction** have **$\Delta H > 0$** (**endothermic**).

$HA \rightleftharpoons H^+ + A^-$, $\Delta H > 0$, $T\uparrow$, $K_{eq.}\uparrow$, Reaction move forward, $\alpha\uparrow$, $[ion] = C\alpha\uparrow$

(v) Common ion effect (CIE): Decreases of α of Weak electrolyte due to presence of common ion produced from other electrolyte (Strong/Weak) is known as **common ion effect.**

When **minimum two** electrolyte having a common ion are present in the solution.

Strong electrolyte: $AC(aq) \longrightarrow A^+(aq) + C(aq)$

Weak electrolyte: $AB(aq) \rightleftharpoons A^+(aq) + B^-(aq)$ $[A^+]\uparrow$,Reaction will go backwards, $\alpha\downarrow$,$[B^-] = \downarrow$

NOTE :

1. C.I.E. will operate **only on weak electrolyte.**
2. **Higher** is the concentration of common ion, **more** will be the C.I.E. so **less will be** α of weak Electrolyte.

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T-7 Properties of water:

(1) **Acid Base nature**: Water can be Arrhenius acid/base, Bronsted acid/base and Lewis base. But **not Lewis acid.**

(2) **Molar concentration**: $[H_2O] = 55.55\ M = \frac{1000}{18}$ mole/Ltr.

(3) **Ionic product of water or Self-ionization of water or Auto protolysis Constant of water**.

$H_2O \rightleftharpoons H^+ + OH^-$, $\Delta H > 0$, $K_w = [H^+][OH^-]$ at 25°C $[H^+] = [OH^-] = 10^{-7}$ M, $K_W = 10^{-14}$.

T↑, $[H^+] = [OH^-] = \uparrow$, $P^{OH} = P^H = \downarrow$, $K_W\uparrow$, $P_{KW}\downarrow$ but **water remains neutral**

(4) **Absolute dissociation constant of water (K_a or K_b of water) :**

$H_2O \rightleftharpoons H^+ + OH^-$, $K_a = K_b = \frac{[H^+][OH^-]}{[H_2O]} = \frac{K_W}{[H_2O]} = \frac{10^{-14}}{1000} \times 18 =$ **1.8×10^{-16}**

or P^{ka} or P^{kb} of water = 15.74 (**at 25°C**)

(5) **Degree of Dissociation of water at 25°C:** α_{H_2O} **$= 1.8 \times 10^{-9} = 1.8 \times 10^{-7}\%$**

Upon **addition to any acid / Base** to water α_{H_2O} **decrease** due to Common ion effect

T-8 p^H & p^{OH} of Solution :

$p^H = -\log [H^+] = \log \left(\frac{1}{[H^+]}\right)$, $p^{OH} = -\log [OH^-] = \log \left(\frac{1}{[OH^-]}\right)$

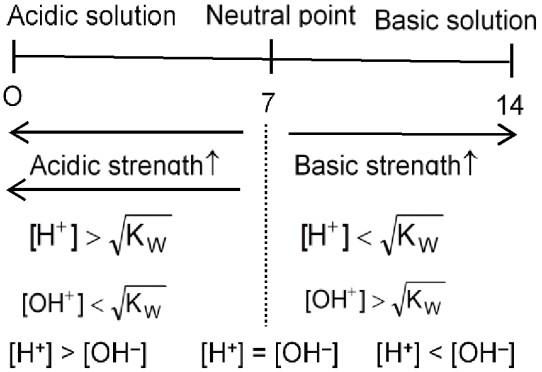
Relation between p^H & p^{OH}:

In any Aqueous solution: $[H^+][OH^-]=K_w$ & $P^H + P^{OH} = P^{KW}$ K_W is ionic product of water.

at 25°C $[H^+][OH^-] = K_W = 10^{-14}$ & $P^H + P^{OH} = 14$

Note: P^H can be **negative** or **more than 14** for very **concentrated acidic** & **basic** solutions respectively.

P^H scale: (at 25°C)



Neutral Point: $\frac{P^{Kw}}{2} = 7$ (at 25°C)

Note: At **very large dilution**, P^H of solution will tend **towards 7**. (Neutral point).

Relation b/w K_a & K_b (Applicable Only for Conjugate acid base pair)

$K_a.K_b = K_W$ & $P_{ka} + P_{kb} = P_{kw}$

Note (1). $(K_a)_A \cdot (K_b)_{CB} = K_W$ & $(K_a)_{CA} \cdot (K_b)_B = K_W$

(2) $P^{ka}(H_3O^+) = P^{kb}(OH^-) =$ **-1.74**

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T-4 PH calculation:

Acidic solution		Basic solution	
(i) **Strong acid**		(i) **strong base**	
$[H^+]_{SA} > 10^{-6}$ M	**Neglect** H^+ from **water**.	$[OH^-]_{SB} > 10^{-6}$ .M	Neglect OH^- from water.
10^{-8} M$\leq [H^+]_{SA} < 10^{-6}$ M	**Consider** H^+ from water.	10^{-8} M$\leq [OH^-]_{SB} <$ 10^{-6} M	Consider OH^- from water.
$H^+]_{SA} \leq 10^{-9}$ M	Neglect H^+ from strong acid. ($P^H \approx 7$)	$[OH^-]_{SB} \leq 10^{-9}$ M	Neglect OH^- from strong base. ($P^H \approx 7$)

Table:

[HCl]	Source of H^+	$[H^+]$	P^H	[NaOH]	Source of OH^-	$[H^-]$	P^{OH}
10^{-6} M	HCl	10^{-6} M	**6**	10^{-6} M	NaOH	10^{-6} M	6
10^{-7} M	HCl + H_2O	1.62×10^{-7} M	**6.79**	10^{-7} M	NaOH + H_2O	1.62×10^{-7} M	6.79
10^{-8} M	HCl + H_2O	1.05×10^{-7} M	**6.97**	10^{-8} M	NaOH + H_2O	1.05×10^{-7} M	6.97
10^{-9} M	H_2O	10^{-7} M	**7**	10^{-9} M	H_2O	10^{-7} M	7

(ii) **Mixing of strong acid**	(iii) **Mixing of strong base**
$[H^+] = \frac{\text{total mole of } H^+}{\text{total volume of solution(L)}}$, $[OH^-] = \frac{K_W}{[H^+]}$	$[OH^-] = \frac{\text{total mole of } OH^-}{\text{total volume of solution(L)}}$, $[H^+] = \frac{K_W}{[H^+]}$
(iv) **Weak acid : (K_a, C)**	**(iii) Weak base : (K_b, C)**
$HA \rightleftharpoons H^+ + A^-$ t = 0 C t = eqbm C–Cα Cα Cα $K_a = \frac{[H^+][A^-]}{[HA]} = \frac{C\alpha^2}{1-\alpha}$ ……(1)	$BOH \rightleftharpoons B^+ + OH^-$ t = 0 C t = eqbm C–Cα Cα Cα $K_b = \frac{[OH^-][B^+]}{[BOH]} = \frac{C\alpha^2}{1-\alpha}$ ……(2)
$\alpha = \sqrt{\frac{K_a}{C}}$, If α< 10% or 0.1 , then 1–α ≈1 $P^H = \frac{1}{2}(P_{K_a} - \log C)$ If α< 10% or 0.1 If **α > 10% or 0.1** , Solve Quadratic (1), calculate α , then $[H^+] = C\alpha$	$\alpha = \sqrt{\frac{K_b}{C}}$, If α< 10% or 0.1 , then 1–α ≈1 $P^{OH} = \frac{1}{2}(P_{K_b} - \log C)$, If α< 10% or 0.1 $P^H = 14 - P^{OH}$ If **α > 10% or 0.1** Solve Quadratic (2), calculate α, then $[OH^-] = C\alpha$, $[H^+] = \frac{K_W}{[OH^-]}$

T-11 (4) Mixing of Strong acid & Strong base:

Calculate Total mole of $H^+ = n_{H^+}$ & Total mole of $OH^- = n_{OH^-}$

If $n_{H^+} > n_{oH^-}$ then **Acidic** solution & $[H^+]_f = \frac{n_{H^+} - n_{OH^-}}{\text{Total volume(L)}}$

If $n_{oH^-} > n_{H^+}$ then **Basic** solution & $:[OH^-]_f = \frac{n_{OH^-} - n_{H^+}}{\text{Total volume(L)}}$, then $[H^+]_f = \frac{kw}{[OH^-]_f}$

If $n_{H^-} = n_{oH^-}$ then solution will be **Neutral** $P^H = \frac{p^{kw}}{2} = 7$ at 25°C

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T-12 **Salt :** When acid & base are mixed in **equal proportion**, final solution contains salt.

Acid + base ⟶ salt + water

Type of salt	Nature	example
SASB	Neutral	NaCl, KNO_3
SAWB	Acidic	NH_4Cl, $ZnCl_2$
WASB	Basic	CH_3COONa, NaCN
WAWB	(acidic / basic / Neutral)	CH_3COONH_4, NH_4CN.

T-13 **Salt hydrolysis:** phenomenon of interaction of **cation** or **anion** or both or **none** of the salt with water to produce to acidic, basic or neutral solution is known as **salt hydrolysis**.

$HA + BOH \rightleftharpoons AB + H_2O$,

If forward; $\Delta H < 0$, acid base **neutralization Reaction** $(K_{Acid\ base})_{New}$.

If Backward; $\Delta H > 0$, **Hydrolysis Reaction** $(K_{Hydrolysis})$

$$(K_{AB})_{Neutralization} = \frac{1}{K_{Hydrolysis}}$$

$HA \longrightarrow B^+ + A^-$

$B^+ + H_2O \rightleftharpoons BOH + H^+$, **Cationic Hydrolysis**, acidic solution, $P^H < 7$.

$A^- + H_2O \rightleftharpoons HA + OH^-$ **Anionic Hydrolysis**, basic solution, $P^H > 7$

Note: (1) **Anion of strong acid** will not go hydrolysis: $ClO_4^-, SO_4^{2-}, NO_3^-, Cl^-, Br^-, I^-$,

(2) **Cation of strong base** will Not go hydrolysis ; Na^+, K^+, Rb^+, Cs^+, Ca^{2+}, Ba^{2+}, Sr^{2+}

T-14 **P^H calculation of salt solution**

(1) **Salt of SASB:** e.g. NaCl in water. **No hydrolysis**. **Neutral Solution,** P^H = 7 at 25°C

(2) Salt of **SAWB, WASB, WAWB**

Salt of SAWB e.g. NH_4Cl	Salt of WASB e.g. CH_3COONa	Salt of WAWB e.g CH_3COONH_4
$NH_4Cl \longrightarrow \underset{C}{\overset{\checkmark}{NH_4}} + \underset{C}{\overset{\boxtimes}{Cl^-}}$	$CH_3COONa \longrightarrow \underset{C}{\overset{\checkmark}{CH_3COO^-}} + \underset{C}{\overset{\boxtimes}{Na^+}}$	$CH_3COONH_4 \longrightarrow \underset{C}{\overset{\checkmark}{CH_3COO^-}} + \underset{C}{\overset{\boxtimes}{NH_4^+}}$
$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$ C-Ch Ch Ch	$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$ C-Ch Ch Ch	$CH_3COO^- + NH_4^+ \rightleftharpoons$ C-Ch C-Ch $CH_3COOH + NH_3$ Ch Ch
$K_h = \frac{Ch^2}{1-h} = \frac{K_w}{K_b}$ .....(1)	$K_h = \frac{Ch^2}{1-h} = \frac{K_w}{K_a}$ .....(2)	$K_h = \left(\frac{h}{1-h}\right)^2 = \frac{K_w}{K_b.K_a}$
$h = \sqrt{\frac{k_h}{c}}$	$h = \sqrt{\frac{k_h}{c}}$	$\sqrt{K_H} = \left(\frac{h}{1-h}\right) = \sqrt{\frac{K_w}{K_b.K_a}}$
If h < 10%. then $P^H = \frac{1}{2}(p^{kw} - P^{kb} - \log c)$	If h <10 % then $P^H = \frac{1}{2}(p^{kw} + P^{Ka} + \log c)$	$P^H = \frac{1}{2}(p^{kw} + P^{ka} - p^{kb})$
If h > 10% then solve Quadratic(1) & calculate h , $[H^+] = C.h.$	If h > 10% then solve Quadratic(2) & Calculate h , $[OH^-] = C.h.$, $[H^+] = \frac{kw}{[OH^-]}$	

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Note :	Note :	Note :
(1) C is concentration of **cation** (2) K_h is Hydrolysis constant (3) h is Degree of Hydrolysis (0 < h < 1) (4) Degree of hydrolysis depends upon concentration of cation.	(1) C is concentration of **anion** (2) K_h is Hydrolysis constant (3) h is Degree of Hydrolysis (0 < h < 1) (4) Degree of hydrolysis depends upon concentration of anion.	(1) P^H of solution is **independent of concentration of salt**. (2) It is **assuming degree of hydrolysis** of cation & anion **as same** (3) Degree of hydrolysis is **independent of** concentration of salt.

T-15 Buffer Solution:

A solution which **resists** a P^H change on the addition of **small amount of acid & base or by dilution** is known as **buffer solution**.

Type of Buffer Solution:

Acidic Buffer Solution	**Basic buffer Solution**	**Salt Buffer**
Weak acid + Its salt with **Strong base**. e.g. $CH_3COOH + CH_3COONa$	**Weak base** + Its salt with **Strong acid**. e.g. $NH_4OH + NH_4Cl$	Salt of **WAWB.** **e.g. CH_3COONH_4**
Method of preparation of acidic buffer	**Method of preparation of Basic buffer**	
(a) $CH_3COOH + CH_3COO^-$ (Direct method) (C_1) (C_2) (Indirect method) (b) $CH_3COOH + NaOH(LR) \rightarrow CH_3COONa + H_2O$ M_1V_1 M_2V_2 0 0 $\left(\frac{M_1V_1 - M_2V_2}{V_1V_2}\right)$ 0 $\left(\frac{M_2V_2}{V_1+V_2}\right)$ – (c) WASB + SA ⟶ WA + SASB $CH_3COONa + HCl(LR) \rightarrow CH_3COOH + NaCl$	(a) $NH_3 + NH_4^+$ (Direct method) (C_1) (C_2) (Indirect method) (b) $NH_4OH + HCl(LR) \rightarrow NH_4Cl + H_2O$ M_1V_1 M_2V_2 0 0 $\left(\frac{M_1V_1 - M_2V_2}{V_1V_2}\right)$ 0 $\left(\frac{M_2V_2}{V_1+V_2}\right)$ – (c) WBSA + SB(LR) ⟶ WB + SASB $NH_4Cl + NaOH(LR) \longrightarrow NH_4OH + NaCl$	
$P^H = P^{Ka} + \log\left(\frac{CB}{WA}\right)$, $\frac{CB}{WA} = \frac{n_{CB}}{n_{WA}} = \frac{[CB]}{[WA]}$	$P^{OH} = P^{Kb} + \log\left(\frac{CA}{WB}\right)$, $\frac{CA}{WB} = \frac{n_{CA}}{n_{WB}} = \frac{[CA]}{[WB]}$	
For buffer action : $\frac{1}{10} \le \left[\frac{CB}{WA}\right] \le 10$	**For buffer action :** $\frac{1}{10} \le \left[\frac{CA}{WB}\right] \le 10$	
Buffer range : (P^H) $(P^{Ka}-1) \le$ Buffer range $\le (P^{Ka}+1)$	**Buffer range : (P^{OH})** $(P^{Kb}-1) \le$ Buffer range $\le (P^{Kb}+1)$	
Best buffer condition : $\left[\frac{CB}{WA}\right] = 1$, $P^H = P^{Ka}$	**Best buffer condition :** $\left[\frac{CA}{WB}\right] = 1$, $P^{OH} = P^{Kb}$	

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Buffer action :

Let 'a' mole of SA is added to an acidic buffer.

LR

$CH_3COO^-Na^+ + HCl \longrightarrow CH_3COOH + NaCl$

t = 0	C_1	a	C_2
t = end	C_1-a	0	$C_2 + a$

$$P^H = P^{Ka} + \log\left(\frac{C_1 - a}{C_2 + a}\right)$$

Let 'b' mole of SB is added to an acidic buffer.

$CH_3COOH + NaOH \longrightarrow CH_3COO^-NO^+ + H_2O$

t = 0	C_2	b	C_1
t = end	C_2	o	$C_1 + b$

$$P^H = P^{Ka} + \log\left(\frac{C_1 - b}{C_2 + b}\right)$$

Buffer action :

Let 'a' mole of SA is added to a basic buffer.

(LR)

$NH_3 + HCl \longrightarrow NH_4^+Cl^-$

t = 0	C_1	a	C_2
t = end	C_1-a	0	$C_2 + a$

$$P^{OH} = P^{Kb} + \log\left(\frac{C_2 + a}{C_1 - a}\right)$$

Let 'b' mole of SB is added to a basic buffer.

LR

$NH_4^+ + NaOH \longrightarrow NH_4OH + Na^+$

t = 0	C_2	b	C_1
t = end	C_2-b	0	$C_1 + b$

$$P^{oH} = P^{Kb} + \log\left(\frac{C_2 - b}{C_1 + b}\right)$$

T-16 Ionic Equilibrium 3 (Solubility Equilibrium)

Solubility's: The **maximum mole** of solute present in **1L Solution** to form **saturated solution**

Solubility product (K_{sp}): is a type of **equilibrium constant**, it dependents only on **temperature** for a particular salt.

Let a sparingly soluble salt A_xB_y be dissolved in water. Let its solubility in H_2O = '**s**' M.

$A_xB_y \rightleftharpoons xA^{y+} + yB^{-x}$

t_{eqm} – xs ys

$K_{sp} = (xs)^x (ys)^y = x^x.y^y.(s)^{x+y}$

E.g. for AB type salt : $K_{sp} = s^2$
for AB_2 or A_2B salt : $K_{sp} = 4s^3$
for AB_3 or A_3B salt : $K_{sp} = 27s^4$

Note : $Hg_2Cl_2 \rightleftharpoons Hg_2^{2+} + 2Cl^-$

T-17 Condition of precipitation :

. Consider ionic product (Q_{IP}) similar to reaction quotient Q in chemical equilibrium

Relation between Q_{IP} & K_{SP}	Nature of Solution	PPT will form
$Q_{IP} > K_{SP}$	Super Saturated	Yes
$Q_{IP} < K_{SP}$	Unsaturated	No
$Q_{IP} = K_{SP}$	Saturated	Just start ppt formation

NOTE: Modify the concentration of the precipitating ions because of volume change occurring upon mixing both solutions.

T-18 Common Ion effect on solubility :

1. Because of the presence of **common** ion in solution, the solubility of the **sparingly soluble salt** generally **decreases**.
2. **Neglect** the concentration of common ion coming from **sparingly soluble salt** with respect to that coming from **completely soluble salt.**

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CORDINATION COMPOUNDS

1. Addition salts can be classified into two categories: **(i) Double salts (ii) Complex salts**.

2. Define double salts and give two examples.

Ans: These salts contain two or more simple salts in stoichiometric proportion. These salts on dissolution give test of all the ions. eg, $KCl.MgCl_2.6H_2O$(carnalite), $(NH_4)_2SO_4.FeSO_4.6H_2O$ (Mohr's salt).

3. Define coordination compounds or complexes and give two examples.

Ans: In coordination compounds or complexes, metal atom/ion is surrounded by neutral molecules or ions through coordinate bonds. These salts on dissolution do not give test of all the ions. eg, $K_4[Fe(CN)_6]$ (Potassium ferrocyanide), $[Fe(H_2O)_5(NO)]SO_4$ (Brown ring complex)

4. Define ligands. Classify ligands & central metal atom as Lewis acid or Lewis base.

Ans: The neutral molecules, anions or cations directly linked with central metal atom or ion by coordinate bond are known as ligands. Ligands are lone pair donors (Lewis base) while central metal atom is lone pair acceptor (Lewis acid).

5. Define coordination sphere and ionisation sphere.

Ans: **Coordination sphere:** The central atom/ion and the ligands attached to it are collectively known as coordination sphere. These are enclosed in square bracket.

Ionisation Sphere: The remaining ions i.e. outside the coordination sphere are known as ionisation sphere.

6. Define denticity.

Ans: It is the number of coordinate σ −bonds formed by a given ligand to given central atom.

7. Give two examples each of symmetric and unsymmetric bidentate ligands.

Ans: Symmetric bidentate ligands: (i) ethyelenediamine(en) (ii) oxalate(ox^{-2})

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Unsymmetric bidentate ligands: (i) propylenediamine (pn) (ii) glycinate (gly^-)

8. Define chelation and chelate rings.

Ans: When a polydentate ligand is attached with a central atom, then rings are formed. These rings are known as chelate rings and this phenomenon is known as chelation.

9. If a hexadentate ligand such as $EDTA^{-4}$ is attached with central atom, then how many chelate rings are formed?

Ans: 5.

10. Define flexidentate and ambidentate ligands. Give two examples of each.

Ans: Flexidentate: Polydentate ligand which can have different denticity in different compounds are known as flexidentate ligands. eg, CO_3^{-2}, SO_4^{-2}.

Ambidentate: These ligands have two or more donating sites but at a time donation cannot be done by all the sites. Hence, they happen to have more than one way of donation. eg, CN^-, NO_2^-

11. Define Coordination number.

Ans: It is the number of coordinate σ −bonds by which a given central atom is attached with ligands.

12. Define homoleptic and heteroleptic complexes.

Ans: Complexes in which all the ligands are identical are known as homoleptic complexes; whereas complexes having more than one type of ligands are known as a heteroleptic complexes.

13. Write the IUPAC name of the following complexes:

(i) $[Ni(NH_3)_4][NiCl_4]$

(ii) $[Fe(H_2O)_5(NO)]SO_4$

(iii) $[NiCl_2\{P(Et)_2(C_6H_5)\}_2]$

(iv) $K_3[Fe(CN)_6]$

(v) $[Cr(\eta^6 - C_6H_6)_2]$

(vi) $[Fe(\eta^5 - C_5H_5)_2]$

(vii) $K[PtCl_3(\eta^2 - C_2H_4)]$

(viii) $K_2[Cr^{VI}NH_3(CN)_2O_2(O_2)]$

Ans: (i) tetraamminenickel(II) tetrachloridonickelate(II)

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(ii) pentaaquanitrosoniumiron(I) sulphate

(iii) dichloridobis-(diethylphenylphosphine)nickel(II)

(iv) potassium hexacyanido-C-ferrate(III)

(v) bis($benzene$)chromium(0)

(vi) bis(cyclopentadienyl)iron(II)

(vii) Potassium trichlorido(ethene)platinate(II)

(viii) potassium amminedicyanido-C-dioxidoperoxidochromate(VI)

14. $EDTA^{-4}$ is ethylenediaminetetraacetate ion. The total number of N-Co-O bond linkages in $[Co(EDTA)]^{-}$ complex ion is:

Ans: 8

15. In the complex acetylbromidodicarbonylbis(triehylphosphine)iron(II), the number of Fe-C bond(s) is :

Ans: 3

16. According to Werner's theory, metals exhibit have two types of valencies-**(i) primary valency (ii) secondary valency.**

17. Primary valency is also known as **principal** valency. It corresponds to **Oxidation state** of central metal atom.

18. Secondary valency equals to **coordination number** of central metal atom. Secondary valency is non-ionisable.

19. A solution containing 0.319g of a complex $CrCl_3.6H_2O$ was passed through cation exchanger and the solution given out was neutralized by 28.5 mL of 0.125 M $NaOH$. What would be the formula of the complex.[Molecular weight of complex =266.5u]

Ans: $[Cr(H_2O)_6]Cl_3$

Sol: $Cl^{-} = HCl = NaOH$

Moles of complex = $\frac{0.319}{266.} = 0.0012$; Moles of $NaOH = \frac{28.5\times0.125}{1000} = 0.0036$

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1 mol of complex $= \frac{0.0036}{0.0012} = 3$ moles of HCl

Hence, complex is $[Cr(H_2O)_6]Cl_3$.

20. On treatment of 100 mL of 0.1 M solution of $CoCl_3.6H_2O$ with excess of $AgNO_3$; 1.2×10^{22} ions are precipitated. Identify the complex:

Ans: $[Co(H_2O)_5Cl]Cl_2.H_2O$

21. Give the order of molar conductance of following with reason :

(I) $[Co(NH_3)_6]Cl_3$ (II) $[Co(NH_3)_3Cl_3]$

(III) $[Co(NH_3)_4Cl_2]Cl$ (IV) $[Co(NH_3)_5Cl]Cl_2$

Ans: I > IV > III > II ; Greater the number of total charged units produced by complex, greater is molar conductance.

22. Define effective atomic number (EAN) of the central metal atom. State EAN rule.

Ans: EAN = Number of electrons in metal or ion + number of electrons donated by ligands to metal.

According to EAN rule, for stable complexes, EAN of central metal atom should be equal to atomic number of noble gas lying next to it.

23. Calculate the effective atomic number of the metal atoms in the following complexes/complex ions.

(i) $[Cr(CO)_6]$ (ii) $[Fe(CN)_6]^{-3}$ (iii) $[Co(CN)_6]^{-4}$ (iv) $[Ni(NH_3)_6]^{+2}$

(v) $K_2[HgI_4]$ (vi) $[Ag(NH_3)_2]^+$ (vii) $K_2[PtCl_4]$

Ans: (i) 36 (ii) 35 (iii) 37 (iv) 38

(v) 86 (vi) 50 (vii) 84.

24. EAN rule is most commonly followed by **metal carbonyls** & metal nitrosyls.

25. State the d-orbitals involved and geometry of the dsp^2, dsp^3 and sp^3d^2 hybridisations.

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Ans:	Hybridisation	geometry	d-orbitals used
	dsp^2	square planar	$d_{x^2-y^2}$
	dsp^3	square pyramidal	$d_{x^2-y^2}$
	dsp^3	trigonal bipyramidal	d_{z^2}
	sp^3d^2	octahedral	$d_{x^2-y^2}$, d_{z^2}

26. Fill the following table

	Complex	**Hybridisation**	**Magnetic property**	**Shape**
(i)	$K_4[Fe(CN)_6]$		diamagnetic	
(ii)	$[Fe(H_2O)_6]^{+2}$	sp^3d^2		
(iii)	$[Ag(NH_3)_2]^+$			
(iv)	$[NiCl_4]^{-2}$		paramagnetic	
(v)	$[Ni(CN)_4]^{-2}$	dsp^2		
(vi)	$Na_2[Fe(CN)_5NO]$		diamagnetic	
(vii)	$[Ni(NH_3)_6]^{+2}$			
(viii)	$[Cr(CN)_6]^{-3}$			

Ans:	**Complex**	**Hybridisation**	**Magnetic property**	**Shape**
(i)	$K_4[Fe(CN)_6]$	d^2sp^3	Diamagnetic	octahedral
(ii)	$[Fe(H_2O)_6]^{+2}$	sp^3d^2	Paramagnetic	octahedral
(iii)	$[Ag(NH_3)_2]^+$	sp	Diamagnetic	linear
(iv)	$[NiCl_4]^{-2}$	sp^3	Paramagnetic	tetrahedral
(v)	$[Ni(CN)_4]^{-2}$	dsp^2	Diamagnetic	square planar
(vi)	$Na_2[Fe(CN)_5NO]$	d^2sp^3	Diamagnetic	octahedral
(vii)	$[Ni(NH_3)_6]^{+2}$	sp^3d^2	Paramagnetic	octahedral
(viii)	$[Cr(CN)_6]^{-3}$	d^2sp^3	Paramagnetic	octahedral

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27. Write four major drawbacks of VBT.

Ans: 1. It could not explain colour of complexes.

2. It does not give a quantitative interpretation of the thermodynamic or kinetic stabilities of coordination compounds.

3. It does not differentiate between strong and weak ligands.

4. It couldn't give exact prediction regarding the tetrahedral and square planar complexes.

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General Organic Chemistry

Inductive effect → Polarisation of sigma bonds due to electronegative difference between the bonded atoms in the molecule or ion is called inductive effect.

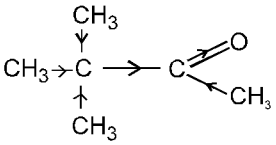
+I Groups → Alkyl groups and anions are electron releasing (+I) groups.
Decreasing +I order → $-O^{\ominus} > -COO^{\ominus} > -C(CH_3)_3 > -CH(CH_3)_2 > -CH_2-CH_3 > -CH_3 > -T > -D > -H$
–I Groups → Atoms more electronegative than carbon and cations are electron withdrawing (–I) groups.
Decreasing –I order → $-\overset{\oplus}{N}R_3 > -\overset{\oplus}{S}R_2 > -\overset{\oplus}{N}H_3 > -NO_2 > -SO_2R > -CN > -COOH >$ $-F > -Cl > -Br > -I > -OR > -OH > -C\equiv CR > Ar > -CH=CR_2$

Electron displacement direction

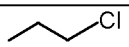
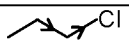
1. $^{\ominus}OOC \rightarrow \overset{\delta-}{C}H_2 \rightarrow \overset{\delta\delta-}{C}H_2 - CH_3$

2. $HOOC \leftarrow \overset{\delta+}{C}H_2 \leftarrow \overset{\delta\delta+}{C}H_3$

3. $CH_3 \rightarrow \overset{+\delta\delta}{C}H = \overset{+\delta}{C}H \rightarrow \overset{-\delta}{N}O_2$

4. 

5.

S.No	Compound	Arrow representation
1.		
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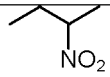
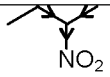
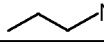
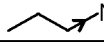
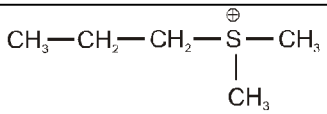
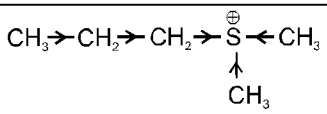
3.		
4.	$CH_3-CH_2-CH_2-\overset{\oplus}{N}H_3$	$CH_3 \rightarrow CH_2 \rightarrow CH_2 \rightarrow \overset{\oplus}{N}H_3$
5.	$CH_3-CH_2-CH_2-O^{\ominus}$	$CH_3 \leftarrow CH_2 \leftarrow CH_2 \leftarrow O^{\ominus}$
6.	$Cl-CH=CH_2$	$Cl \leftarrow CH=CH_2$
7.		
8.	$CH_3-CH_2-\overset{\ominus}{C}H_2$	$CH_3 \leftarrow CH_2 \leftarrow \overset{\ominus}{C}H_2$
9.		

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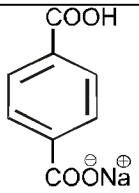
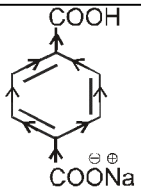
10.		

Resonance effect

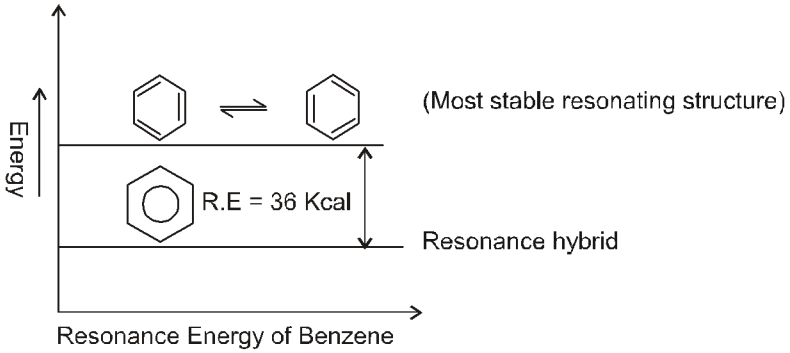
Def → It is the delocalisation of p-orbital or π bond electron in the conjugated system.
Resonance is the phenomenon in which more than one lewis structures can be drawn for a molecule or ion.

 # Rules(1) Resonance structures are more than one lewis structure having the same placement of atoms but a different arrangement of electrons.
(2) Resonance structures must have same number of unpaired electrons.
(3) No 2nd period element can have more than 8 electrons.
(4) Resonance structures are not real.
(5) Resonance structures are not in equilibrium.
(6) Resonance structures are not isomers.
(7) A better resonance structure is one that has more bonds and fewer charges.

Resonance hybrid → (1) It is the composite of all possible resonance structures.
(2) Resonance hybrid is more stable than any resonance structure.

Resonance Energy:
The P.E. difference between the most stable resonating structure and resonance hybrid is called resonance energy.

Fig.2



Condition of Resonance :

1. Conjugation between C=C and C=C
2. Conjugation between +ve charge and C=C
3. Conjugation between lone pair and C=C
4. Conjugation between odd electron and C=C
5. Conjugation between –ve charge and C=C

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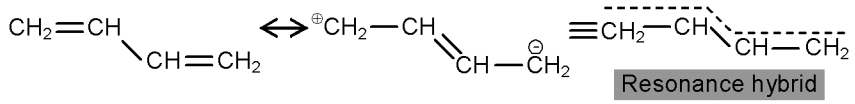
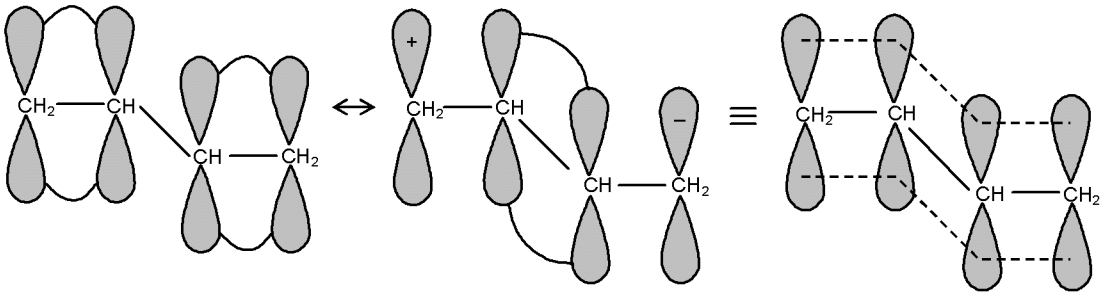
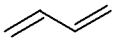
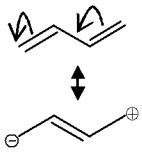
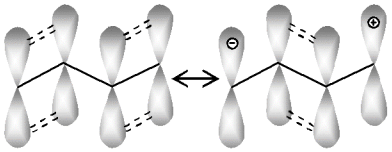
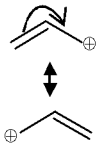
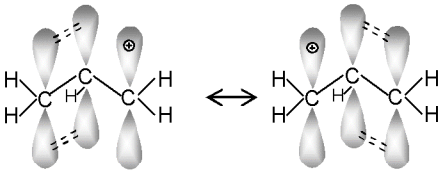
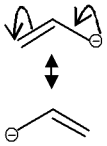
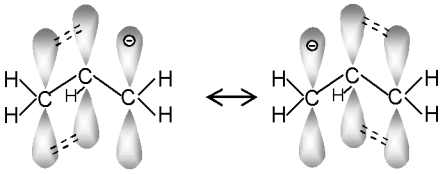
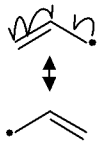
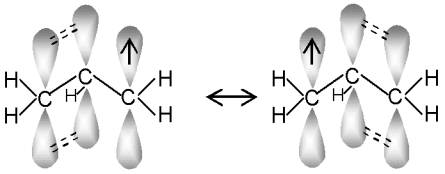
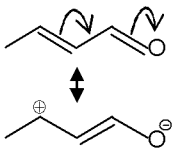
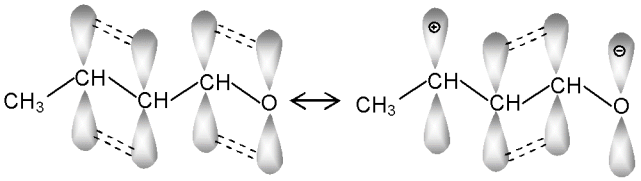
Ex.





Compound	Resonance structures	Orbital diagram
(1) 		
(2) $CH_2 = CH - \overset{\oplus}{C}H_2$		
(3) $CH_2 = CH - \overset{\ominus}{C}H_2$		
(4) $CH_2 = CH - \dot{C}H_2$		
(5) **$CH_3 - CH = CH - CHO$**		

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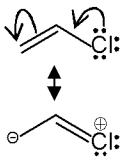
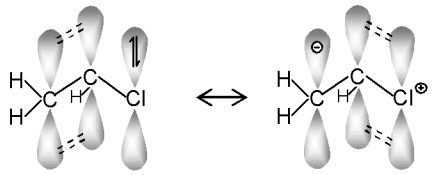
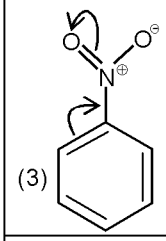
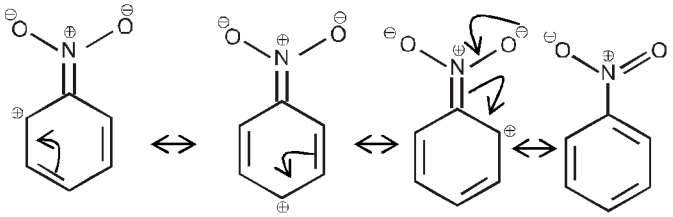
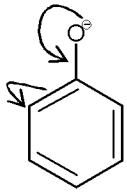
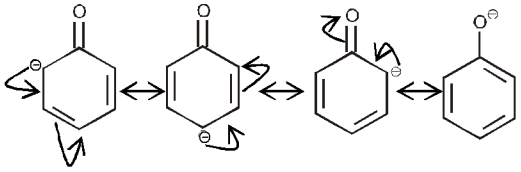
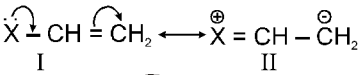
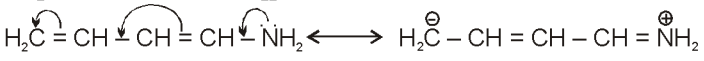
(6) $CH_2 = CH - \ddot{Cl}:$		

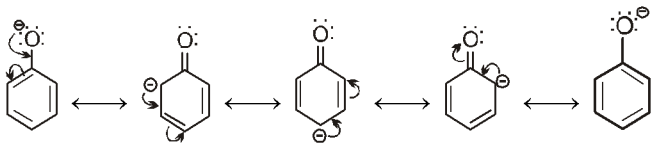
Compound	Resonance structures
(1) $CH_2 = C = O$	$\ominus CH_2 - C \equiv O^{\oplus}$
(2) $CH_2 = CH - C \equiv N$	$\overset{\oplus}{C}H_2 - CH = C = \overset{\ominus}{N}$
(3) 	
(4) 	

Relative order of +M groups :

$-\ddot{O}^{\ominus} > -NH_2 > -NHR > -NR_2 > -OH > -OR > -NHCOR > -OCOR > -Ph > -F > -Cl > -Br > -I$

Ex. (a) (I) 

(II) 

(III) 

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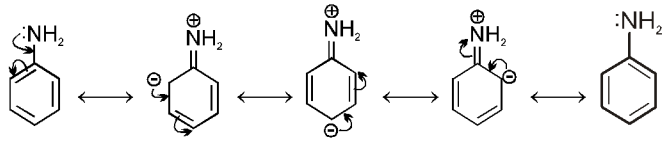
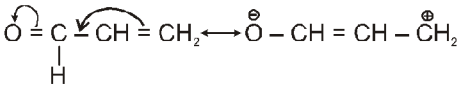
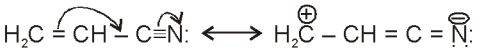
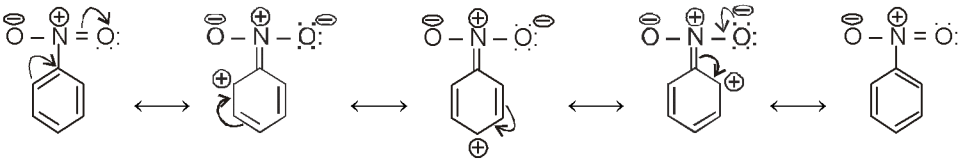
(IV) 

Relative order of –M groups :

$-NO_2 > -CN > -CHO > >C=O > -\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R > -\overset{\overset{O}{\|}}{C}-O-R > -COOH > -CONH_2 > -\overset{\overset{O}{\|}}{C}-O^-$

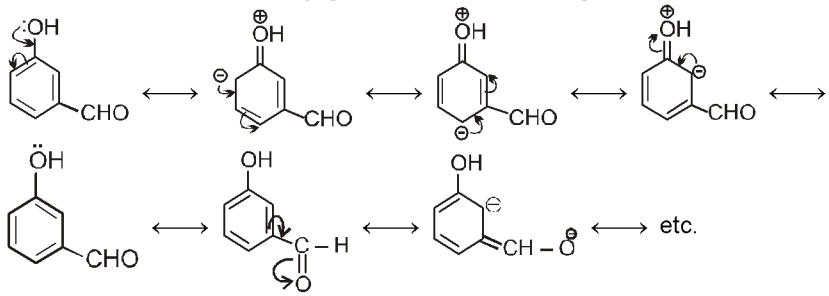
Ex. (b) (I) 

(II) 

(III) 

Note :

1. When a +M group and –M group are at meta-positions with respect to each other then they are not in conjugation with each other, but conjugation with benzene ring exists.



2. +M group increases electron density in benzene ring, called activating group while –M group decreases electron density in the benzene ring, called deactivating group.

Hyperconjugation

It is delocalisation of C–H sigma bond electron with p-orbital. Also known as σ–π-conjugation or no bond resonance. It may take place in alkenes, alkynes, carbocations, free radicals and benzene nucleus.

Necessary Condition : Presence of at least one hydrogen at saturated carbon which is α with respect to alkenes, alkynes, carbocations, free radicals and benzene nucleus.

(i) Hyperconjugation in alkenes

Delocalisation of σ bond electrons in π* orbital.

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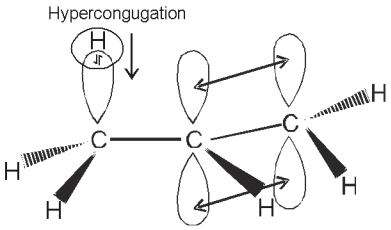
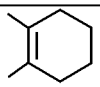
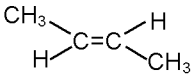
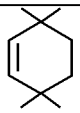
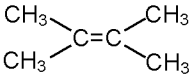
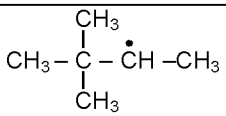
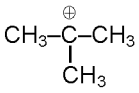
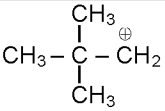
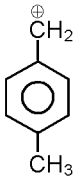


Orbital Diagram showing hyperconjugation in propene

S.No.	Species	Number of Hyperconjugative Hydrogen	S.No.	Species	Number of Hyperconjugative Hydrogen
1.	$CH_3–CH=CH_2$	3	8.		10
2.		6	9.		0
3.		12	10.		0
4.	$CH_3 – C \equiv C – CH_3$	6	11.	$CD_3 – \overset{\oplus}{C}H – CD_3$	6
5.		9	12.		0
6.	$CH_3 – \overset{\bullet}{C}H – CH_2 – CH_3$	5	13.		3
7.	$CH=CH–CH=CH_2$	0	14.		0

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Comparision between electronic effects :

Inductive effect	Mesomeric effect	Hyperconjugative effect
(1) It is found in saturated and unsaturated compounds.	(1) It is found in unsaturated compounds especially having conjugated system.	(1) It is found in carbocation, carbon free radical and unsaturated compounds.
(2) It involves partial shifting of sigma electrons.	(2) It involves complete shifting of pi-electrons of pi-bonds or lone pair of electrons.	(2) It involves partial shifting of sigma-electrons into adjacent p-orbital.
(3) The electron pair is slightly displaced from its position and thus partial charges are developed.	(3) The electron pair is completely transferred and thus full positive and negative charges are developed.	(3) The electron pair is partially transferred.
(4) It is transmitted over a quite short distance. The effect becomes negligible after third atom in the chain (distance dependent effect).	(4) It is transmitted from one end to other end of the chain provided conjugation is present. It is distance independent effect.	(4) It is transmitted from one end to other end of the chain provided conjugation is present. It is distance independent effect.

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Reaction intermediates

Reactive, shortlived, high energy, unstable species, those are formed in the course of organic reactions are called **reaction intermediates**.

(A) Carbanions :

A carbon intermediate which contain three bond pair and a negative charge on it, is called carbanion.

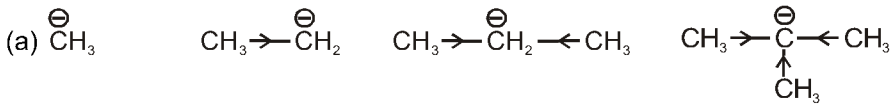
Characteristics :
(i) Carbanions are electron rich (8 electrons at carbon).
(ii) Carbanions act as Lewis base and nucleophiles.
(iii) It is diamagnetic in nature due to paired e^-.
(iv) Carbanions may be sp^3 or sp^2 or sp hybridized.
(v) Carbanions donot undergo rearrangement generally.

Stability of carbanion $\propto$ electron withdrawing effect (–M, –I)
$\propto$ electron negativity of carbon.
$\propto$ delocalisation of negative charge

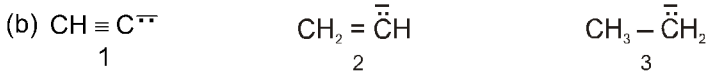
☞

Hybridisation	Examples
sp^3	$\overset{\ominus}{C}H_3$, $CH_3-\overset{\ominus}{C}H_2$, $CH_3\overset{\ominus}{C}HCH_3$, $(CH_3)_3\overset{\ominus}{C}$,
sp^2	$H_2C=\overset{\ominus}{C}H$, $CH_2=CH-\overset{\ominus}{C}H_2$,
sp	$HC\equiv\overset{\ominus}{C}$

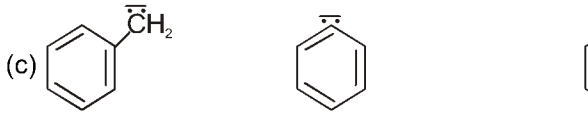
Q.1. Stability order of carbanions :

(a) 

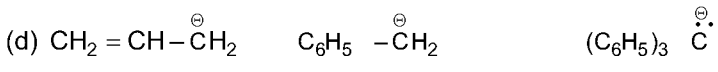
1 2 3 4

(b) 

1 2 3

(c) 

1 2 3

(d) 

1 2 3

Answers. 1 : (a) 1>2>3>4 (b) 1 > 2 > 3 (c) 1 > 2 > 3 (d) 1 < 2 < 3

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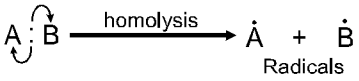
(B) Carbon free radicals :

Homolysis of covalent bond results into free radical intermediates possess the **unpaired electrons**.



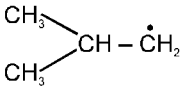
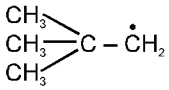
It is generated in presence of sun light, peroxides or high temperature.

Characteristics : (i) It is neutral species with odd e^- (7 electrons at carbon).. (ii) It is paramagnetic in nature due to odd e^-. (iii) Carbon atom having odd electron is in sp^2 hybridised state (iv) Rearrangement is not observed generally.
Stability of free radical $\propto$ Resonance, hyperconjugation and +I effects.

Q.2. Stability of carbon free radicals :

(a) $(C_6H_5)_3\dot{C}$ $(C_6H_5)_2\dot{C}H$ $C_6H_5-\dot{C}H_2$ $CH_2=CH-\overset{\dot{}}{\underset{CH_3}{C}}-CH_3$ $H_3C-\dot{C}H-CH_3$ $\dot{C}H_3$

(b) $(H_3C)_3\dot{C}$ $H_3C-\dot{C}H-CH_3$ $H_3C-\dot{C}H_2$ $\dot{C}H_3$

(c) $CH_3-\dot{C}H_2$ $CH_3-CH_2-\dot{C}H_2$  

(Due to resultant of inductive effect and hyperconjugation, both operates in same direction)

Answers 2. (a) 1 > 2 > 3 > 4 > 5 > 6 (b) 1> 2 > 3 > 4 (c) 1 > 2 > 3 > 4

(C) Carbocations :

A carbon intermediate which contains three bond pair & a positive charge on it is called carbocation.

Characteristics : (i) Carbocations are electron deficient (6 electrons at carbon).. (ii) Carbocations act as Lewis acids and electrophiles. (iii) It is diamagnetic in nature. (iv) Carbocation may be sp^2 or sp hybridized. (v) Carbocations can rearrange into more stable cations.
Stability of carbocations $\propto$ electron releasing effect (+M,+H,+I) $\propto$ delocalisation of positive charge

Hybridisation	**Example**
sp^2	$\overset{\oplus}{C}H_3$, $CH_3\overset{\oplus}{C}H_2$, $CH_3\overset{\oplus}{C}HCH_3$, $(CH_3)_3\overset{\oplus}{C}$
sp	$H_2C=\overset{\oplus}{C}H$, $HC\equiv\overset{\oplus}{C}$

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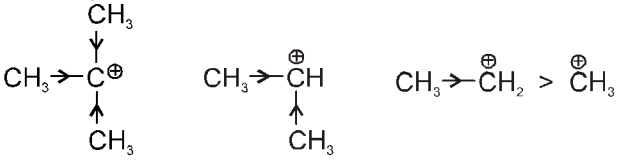
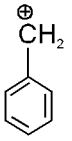
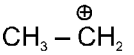
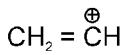
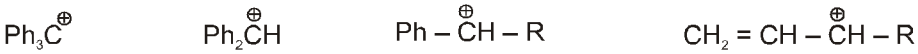
Q.3. Stability order of carbocations:

(a) 

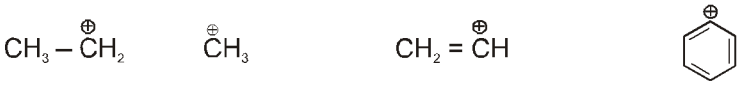
(b) 
Benzylic cation


Ethyl cation


Vinyl cation

(c) 

(d)

(e) 

Answers 3.

(a) 1 > 2 > 3 > 4 (b) 1 > 2 > 3 (c) 1 > 2 > 3 > 4 (d) 1 > 2 > 3 > 4 (e) 1 > 2 > 3 > 4

Rearrangement of carbocations:

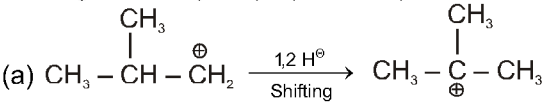
Only those carbocation will rearrange which can produce more stable carbocation. It can be done either by

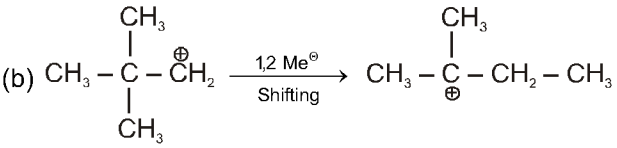
(i) Shifting of H, alkyl, aryl, bond (1, 2-shifting)

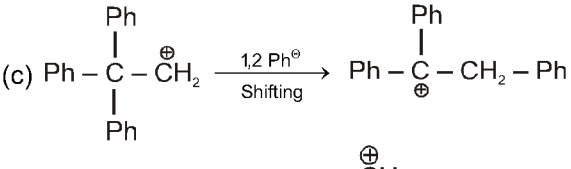
(ii) Ring expansion (more strained ring to less strained ring)

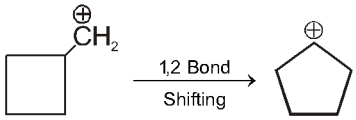
(iii) Ring contraction

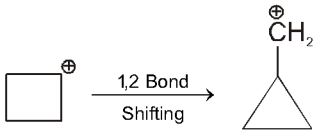
Shifting of H, alkyl, aryl, (1, 2-shift)

(a) 

(b) 

(c) 

(ii) Ring expansion: 

(iii) Ring contraction: 

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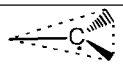
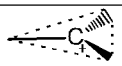
Q.4.

	Carbon free radical	Carbocation	Carbanion
			
Shape	trigonal planar	trigonal planar	Pyramidal
Hybridisation	sp^2	sp^2	sp^3
No. of electrons in outermost shell	7	6	8

Q.5. Hybridisation of charged atoms in the following species is :

(a) $H_2C=\overset{\ominus}{C}H$ (b) $CH_2=CH-\overset{\ominus}{C}H_2$ (c) (d)

(e) (f) $Ph\overset{\oplus}{C}H_2$ (g) $CH_3-\overset{\oplus}{C}H_2$

Answers 5.

(a) sp^2 (b) sp^2 (c) sp^2 (d) sp^3 (e) sp^2 (f) sp^2 (g) sp^2

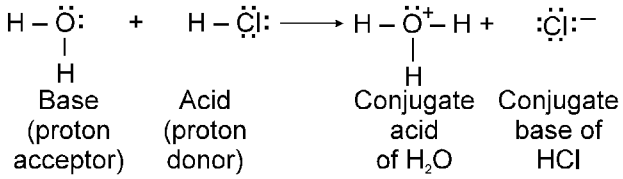
(D) Basic strength :

Definition

(a) Arhenius base: Base is a substance that can donate OH^- ions in H_2O.

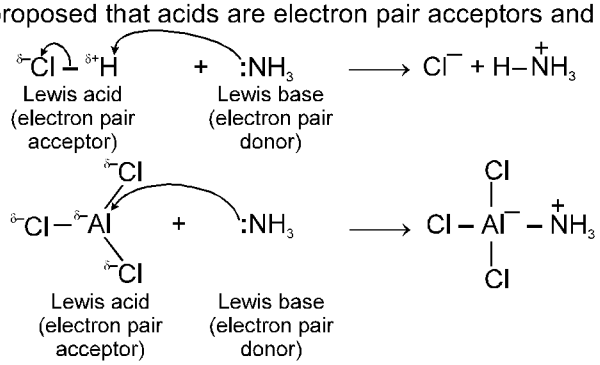
(b) The Bronsted Lowry definition of acids and bases:

An acid is a substance that can donate (or loose) a proton, and a base is a substance that can accept a proton.



(c) The Lewis definition of acids and bases:

Lewis proposed that acids are electron pair acceptors and bases are electron pair donors.



Basic strength(K_b):

$K_b \propto$ proton acceptor tendency

$K_b \propto$ lone pair donor tendency

$K_b \propto$ stability of conjugate acid

$K_b \propto$ +I, +M

$K_b \propto \frac{1}{-M,-I}$

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Basic strength of aliphatic bases:

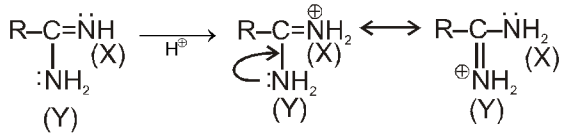
(1) On the basis of +I effect basic strength of amines should be $3^o > 2^o > 1^o > NH_3$
But this order is applicable only when the amines are in gaseous state or in case of non-polar aprotic solvent.

(2) Solvent effect : In aqueous solution basicity of aliphatic amines is

$R_2NH > RNH_2 > R_3N > NH_3$ (if R = CH_3) ..... (1)

$R_2NH > R_3NH > RNH_2 > NH_3$ (if R = C_2H_5) .....(2)

(3) Amidines are more basic in nature because their conjugate acid and are more stable due to resonance.



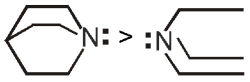
Nitrogen (X) is more basic than nitrogen (Y).

(4) Basic strength of nitrogeneous compound depends upon the hybridised state of nitrogen

$R–CH_2–CH_2–NH_2 > R–CH_2–CH=NH > R–CH=CH–NH_2 > R–C\equiv N$

sp^3 (localised) sp^2 (localised) sp^2 (delocalised) sp

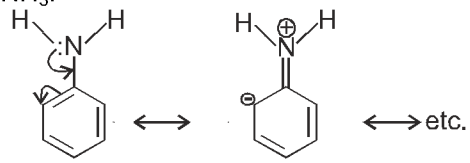
(5) Cyclic amines are more basic than acyclic amines of same nature



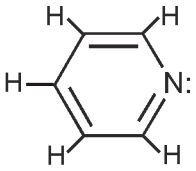
Th3. Basic strength of aromatic amines and substituted anilines:

(a) Aniline:

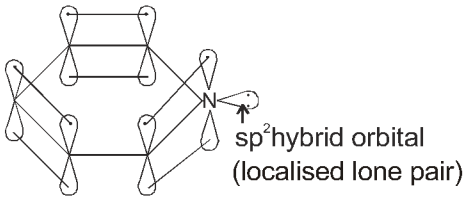
Lone pair of aniline lies in conjugation with a multiple bond, it resides in '2p' atomic orbital, so that it can get resonance stabilisation and hence, basic strength decreases. So, Aniline is a weaker base than NH_3.



(b) Pyridine (C_5H_5N) : 6-π electrons, aromatic

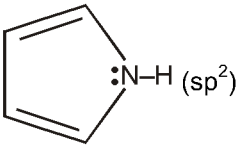


A.O. Diagram:

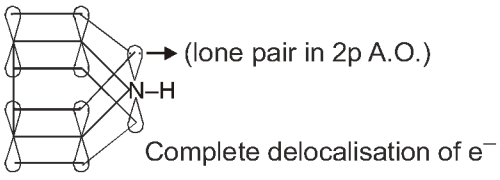


Lone pair of N in pyridine is localised so it is more basic than aniline.

(c) Pyrrole (C_4H_5N:) : 6-π electrons, aromatic



A.O. Diagram:



Lone pair of N in pyrrole is delocalised in the aromaticity so it is very less basic than aniline.

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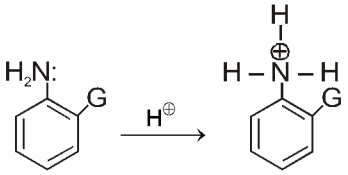
(d) Ortho substituted anilines :

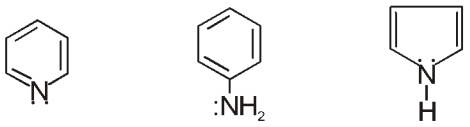


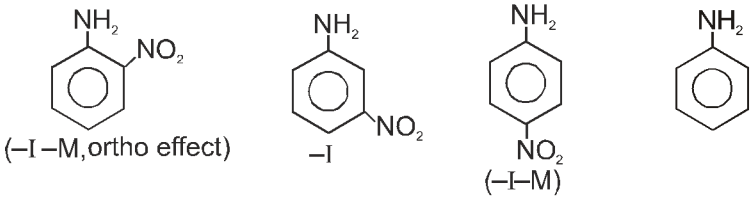
(conjugate acid)

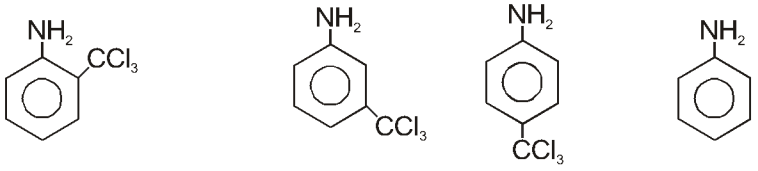
(i) Ortho-substituted anilines are mostly weaker bases than aniline itself.

(ii) Ortho-substituent causes steric hinderance to solvation in the product (conjugate acid i.e. cation).

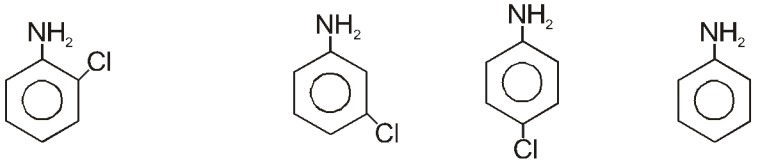
Q.6. Basic strength order of the following compounds is :

(a) NH_3 

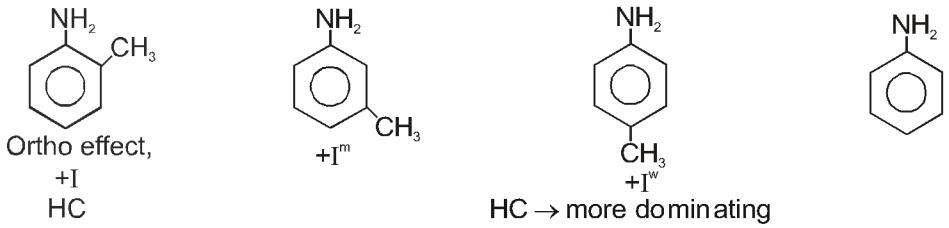
(b) 

(c) 

Only (–I) decides the order.

(d) 

Only (–I) decides the order.

(e) 

Answers 6.

(a) 1 > 2 > 3 > 4 (b) 4 > 2 > 3 > 1 (c) 4 > 3 > 2 > 1 (d) 4 > 3 > 2 > 1 (e) 3 > 2 > 4 > 1

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(E) Acidic strength :

Definition

(a) Arhenius acid: An acid is a substance that can donate (or loose) a proton in H_2O.

(b) The Bronsted Lowry definition of acids and bases:

An acid is a substance that can donate (or loose) a proton, and a base is a substance that can accept a proton.

Acidic strength(K_a):

$K_a \propto$ proton donor tendency

$K_a \propto$ lone pair acceptor tendency

$K_a \propto$ stability of conjugate base

$K_a \propto$ –I, –M

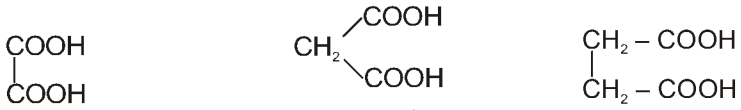
$K_a \propto \frac{1}{+M, +H, +I}$

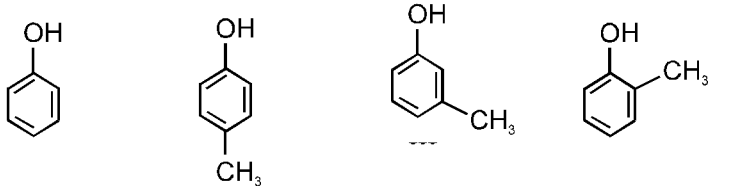
Thus, the following acids are arranged in order of decreasing acidity (based on pK_a values as per NCERT) :

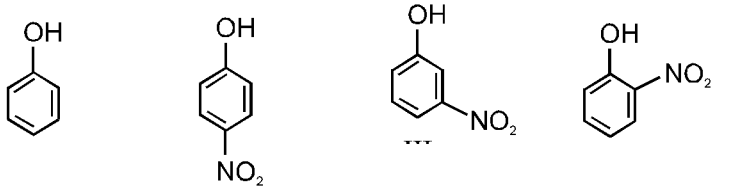
CF_3COOH > CCl_3COOH > $CHCl_2COOH$ > NO_2CH_2COOH > $NC–CH_2COOH$ > FCH_2COOH > $ClCH_2COOH$ > $BrCH_2COOH$ > $HCOOH$ > $ClCH_2CH_2COOH$ > C_6H_5COOH > $C_6H_5CH_2COOH$ > CH_3COOH > CH_3CH_2COOH

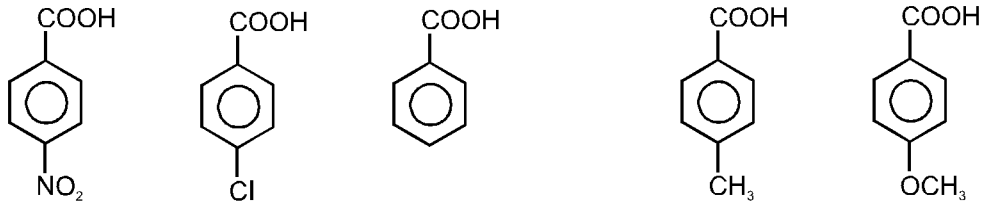
Q.7. Acidic strength order of the following compounds is :

(a) $F–CH_2–COOH$ $Cl–CH_2COOH$ $Br–CH_2COOH$ $I–CH_2COOH$

(b) 

(c) 

(d) 

(e) 

Answers 7.

(a) 1 > 2 > 3 > 4 (b) 1 > 2 > 3 (c) 1 > 3 > 2 > 4 (d) 2 > 4 > 3 > 1 (e) 1 > 2 > 3 > 4 > 5

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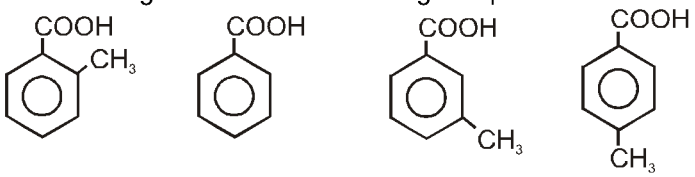
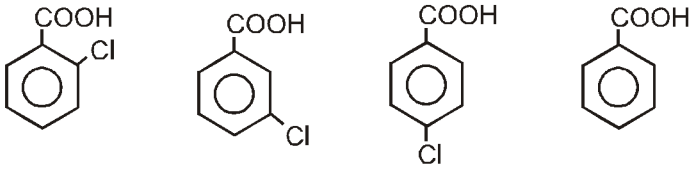
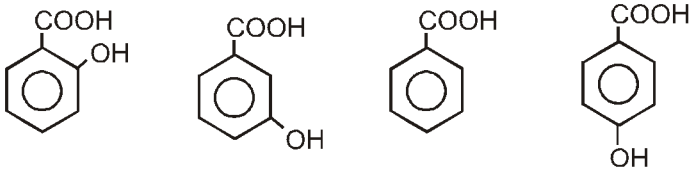
Ortho effect :

It is common observation that ortho substituted benzoic acids are more acidic as compared to benzoic acid itself. This is called ortho effect (which is combined effect of steric hindrance, crowding & electronic effect) in substituted benzoic acid. However exceptions are seen.

Q.8. Acidic strength order of the following compounds is :

(a) 

(b) 

(c) 

Answers 8. (a) 1 > 2 > 3 > 4 (b) 1 > 2 > 3 > 4 (c) 1 > 2 > 3 > 4

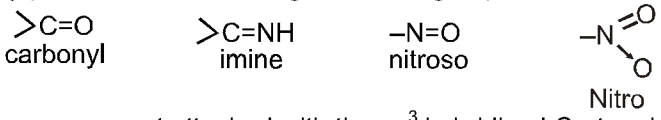
(G) Tautomerism :

Definition : Tautomerism is a phenomenon by which a single compound exists in two or more readily interconvertible structures that differ in the relative positions of at least one atomic nucleus, generally hydrogen.

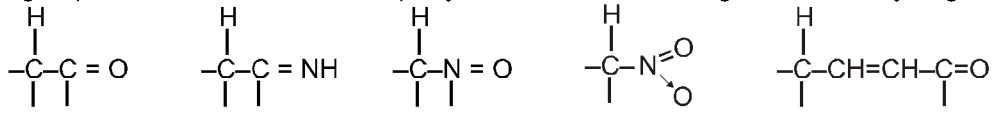
These two isomers remain in dynamic equilibrium. These can be isolated and also give different lab tests.

Conditions :

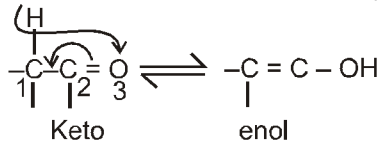
1. Usually present in the following functional groups



2. These groups must attached with the sp^3 hybridised C-atom having atleast one hydrogen atom as –

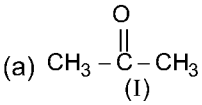


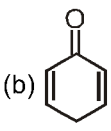
To get tautomer of above structures α-hydrogen atom is shifting to more electronegative atom attached to double bond (i.e. hydrogen atom from 1st atom to 3rd atom) and double bond is developed between 1,2-atom from 2,3-atom. This can be represented as:

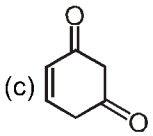


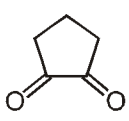
These two forms (remain in equilibrium) are called tautomers of each other. The interconvertibility of tautomers is a chemical reaction which involves making and breaking of bonds.

Q.9. Write the tautomers of following :

(a) 

(b) 

(c) 

(d) 

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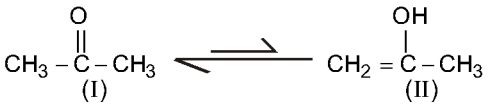
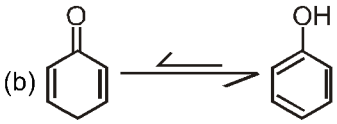
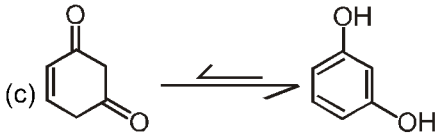
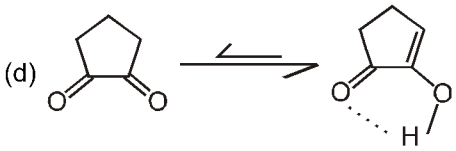
Answers 9. (a) 

(b) 

(c) 

(d) 

% Enol content in the carbonyl compounds :

(1) For monocarbonyl % enol is very less

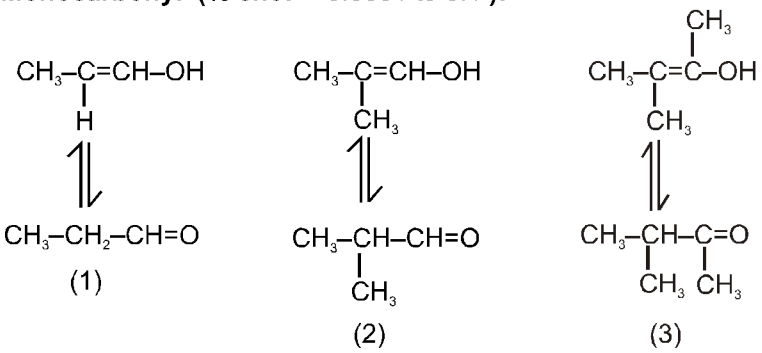
(2) Enol content increases with increase in the stability of enol by resonance, hyperconjugation, hydrogen bonding etc.

(3) For a carbonyl compound having active methylene group percentage of enol content is more because enolic form has intramolecular H-bonding and stabilised by resonance.

If active methylene group is more acidic then enol content will be further increases.

Examples

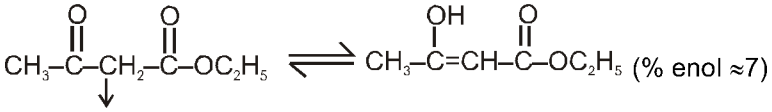
(a) Monocarbonyl (% enol ≃ 0.0001 to 0.1):



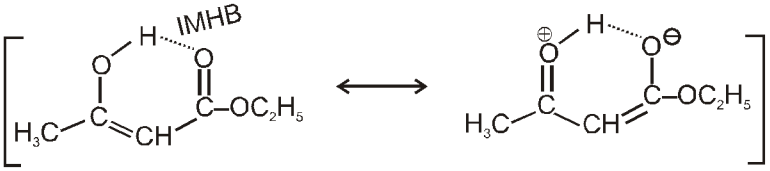
(1) (2) (3)

% Enol content for above carbonyl compounds is : 3 > 2 > 1.

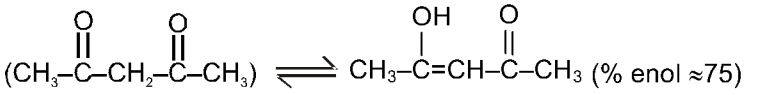
(b) Acetoacetic ester :

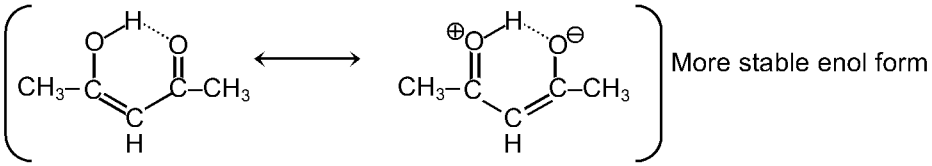


(active methylene)
acetoacetic ester



(c) Acetylacetone:





More stable enol form

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ELECTROCHEMISTRY

1.1 Basic Terms of Electrochemistry

D-1 Electrochemistry is the area of chemistry concerned with the Interconversion of chemical energy and electrical energy.

D-2 Electrochemical cell: It is device in which interconversion between **chemical Energy** and **electrical energy takes place.**

D-3 Types of electrochemical cell

Electrochemical cell are of two types

Galvanic cells or Voltaic cell
- A spontaneous chemical reaction **generates** an electric current.

Eg. Dry cell, Fuel cell

Electrolytic cell
- An electric current **drives** a nonspontaneous reaction.

Eg. Rechargeable battery (during charging)

Thus the two types of cells are reverse of each other.

D-4 Electrode or Half cell:

Whenever a metal strip (or a conducting surface) is put in an electrolyte such that electron loss/ gain can take place in system, such a combination is known as **electrode.**

D-5 Anode:

Electrode at which **oxidation** occurs is called anode.

Eg. Zn rod is placed in $ZnSO_4$ solution.

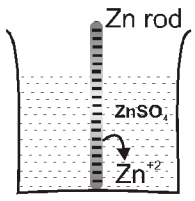

Figure -1

Zn atom will move in the solution to form Zn^{+2} while electrons get accumulated on rode. After some time following equilibrium will be established:

$Zn(s) \rightleftharpoons Zn^{2+}(aq.) + 2e^-$

- The electrode potential is represented by $E(Zn(s) \mid Zn^{2+}(aq))$

Anode of Galvanic or Voltaic cell	Anode of Electrolytic cell
oxidation will take place	**oxidation** will take place.
It acts as source of **electrons**.	It acts as sink of **electrons**.
It is of **negative** polarity.	It is of **positive** polarity.

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D-6 Cathode:

Electrode at which **reduction** occurs is called anode.

Eg. Cu rod is placed in $CuSO_4$ solution.

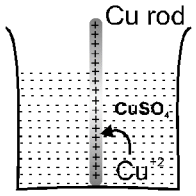

Figure -2

The following equilibrium will be established:

$Cu^{2+}(aq.) + 2e^- \rightleftharpoons Cu(s)$.

- Their electrode potential can be represented by $E(Cu^{2+}(aq)|Cu(s))$

Cathode of Galvanic Or Voltaic cell	**Cathode of Electrolytic cell**
reduction will take place.	**reduction** will take place.
Acts as **sink of electron.**	Acts as **source of electron.**
Positive polarity will be developed.	**Negative** polarity will be developed.

Note: Some electrodes need a conducting surface providing material for electron flow. Inert electrodes like Pt, graphite etc (which themselves do not undergo chemical change) are used for above purpose.

P-1

	Type of Electrode	**Electrode reaction or half cell reaction**	**Representation**
1	Metal-metal soluble salt electrode (Eg. Zn rod is placed in $ZnSO_4$ solution)	Reduction : $Zn^{2+}(aq.) + 2e^- \rightarrow Zn(s)$	$E_{Zn^{2+}/Zn(s)}$
		Oxidation : $Zn(s) \rightarrow Zn^{2+}(aq.) + 2e^-$	$E_{Zn(s)/Zn^{2+}}$
2	Gas ion electrode (Eg. Hydrogen electrode)	Reduction : $2e^- + 2H^+(aq.) \rightarrow H_2(g)$	E_{H^+/H_2}
		Oxidation : $H_2(g) \rightarrow 2H^+(aq) + 2e^-$	E_{H_2/H^+}
3	Redox electrode (Eg. Permagnate-Mn^{2+} electrode)	Reduction : $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	$E_{MnO_4^-/Mn^{2+}}$
4	Metal-metal insoluble salt electrode (Eg. AgCl electrode)	Reduction : $AgCl(s) + e^- \rightarrow Ag(s) + Cl^-$	$E_{Cl^-/AgCl(s)/Ag(s)}$
		Oxidation : $Ag(s) + Cl^- \rightarrow AgCl(s) + e^-$	$E_{Ag(s)/AgCl(s)/Cl^-}$

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1.2 Construction and Working Principle of Galvanic Cell :

D-7

- It has **two half-cells,** each having a beaker containing a metal strip that dips in its **aqueous salt solution**.
- The metal strips are called **electrodes** and are connected by an conducting wire.
- Two solutions are connected by a **salt bridge.**
- The oxidation and reduction half reactions occur at a separate electrodes and electric current flows through the wire. (in external circuit)

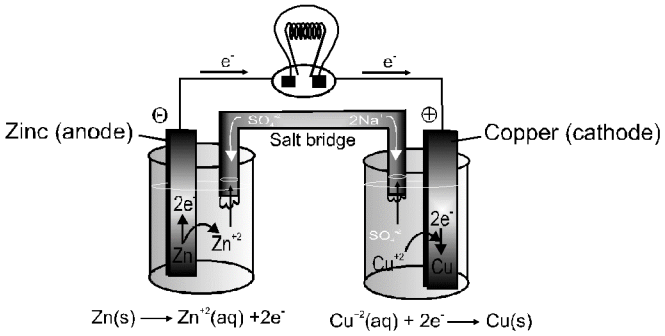

Figure -3

D-8 Salt Bridge:

- A salt bridge is a U–shaped inverted tube that contains a **gel** permeated with an **inert electrolyte**.
- Generally tube is filled with a paste of **agar-agar** powder with a natural electrolyte/generally not common to anodic/cathodic compartment with **porous plugs** at each mouth of tube.

P-2 Function of Salt Bridge :

- It connects the solution of two half-cell to **complete** the circuit.
- It maintains the **electrical neutrality** of the solution in order to give continuous flow of generation of current.
- If the salt bridge is removed, then voltage drops to **zero**.
- The ions of the **inert electrolyte** do not **react** with other ion in the solution and the ions are not oxidised or reduced at the electrodes.

P-3 Selection of electrolyte for Salt Bridge :

- The electrolyte in salt bridge should be such that speed of its cation **equals speed (mobility)** of its anion in electrical field.
- For that purpose charge and sign of the ions should be almost equal.
  Mobility of cation = Mobility of anion
  KCl is generally preferred but KNO_3 or NH_4NO_3 can also be used.

❍ If **Ag^+, Hg_2^{2+}, Pb^{2+}, Tl^+** ions are present in a solution, then in salt bridge, **KCl** is not used because there can be formation of precipitate of **AgCl, Hg_2Cl_2, $PbCl_2$ or TlCl** at mouth of tube which will prevent the migration of ions and its functioning will stop.

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1.3 Rule for Shorthand Notation for Galvanic Cells :

P-4

- The anode half-cell is always written on the **left** followed on the **right** by cathode half-cell.
- The separation of **two phases** (gen. state of matter) is shown by a **vertical line**.
- The various materials present in the same phase are shown together using **commas**.
- The salt bridge is represented by a **double slash (||).**
- The significant features of the substance viz. pressure of a gas, concentration of ions etc. are indicated in **brackets** immediately after writing the substance.
- Surface providing materials should be written at extremes (before Anode or after Cathode)
- Ions if involved should be preferably written adjacent to salt bridge (eg. **Pt H_2/H^+ or H^+/H_2 Pt**)

D-9 Electrode Potential (E) and Standard electrode potential (E^o)

- The potential difference developed between **metal electrode** and **its ions** in solution in known as electrode potential.
- Electrode potential depends upon :
  - Concentration of the solution.
  - Nature of the metal.
  - Nature of the electrolyte.
  - Pressure temperature conditions.
- The potential difference developed between metal electrodes and the solution of its ions at **1 M concentration** at 1 bar partial pressure and at a particular fixed temperature is known as **standard electrode potential**.

P-5

Oxidation Potential (O.P.)	**Reduction Potential (R.P.)**
The electrode potential for **oxidation half reaction.**	The electrode potential for **reduction half reaction**.
Tendency to get **oxidised.**	Tendency to get **reduced.**
Greater the O.P. then greater will be tendency to **get oxidized or reduce** others.	Greater the R.P. greater will be tendency to get **reduced or oxidize** others.

D-10 Reference electrode :

- The potential of a single electrode cannot be determined, however the potential difference between two electrodes can be accurately measured using a voltmeter.
- An electrode is chosen as a reference with respect to which all other electrodes are valued.
- **Standard Hydrogen Electrode (SHE)** is taken as standard reference electrode. Its electrode potential is arbitrarily assumed to be **0.00 volt (at all temperatures).**
- Standard Hydrogen Electrode **(SHE)** consists of a platinum electrode in contact with H_2 gas and aqueous H^+ ions at standard state conditions (**1 atm H_2 gas**, 1 M H^+ (aq)).

$2H^+(aq, 1M) + 2e^- \rightarrow H_2(g, 1\ atm)$ $E° = 0$ V

$H_2(g, 1atm) \rightarrow 2H^+(aq, 1M) + 2e^-$ $E° = 0$ V

1.4 Calculation of cell potential :

D-11 Cell potential (E_{cell}) and Standard cell potential (E^{o}_{cell})

The difference in electrode potentials of the **two half cell** reactions (oxidation half cell and reduction half cell) is known as emf of the cell or cell potential.

E_{cell} = Reduction potential of cathode – Reduction potential of anode

= $E_{cathode} - E_{anode}$ (both are reduction potential).

or E_{cell} = reduction potential of cathode + oxidation potential of anode

= E_{red}(cathode) + E_{ox}(anode)

- The standard cell potential E° is the cell potential when both reactants and products are in their standard states – solutes at **1 M** concentration, gases at a partial pressure of 1 atm, solids and liquids in pure from, with all at a specified temperature, usually 25°C.
- E is **intensive property** so on multiplying / dividing cell reaction by any number, the E value would not change.

P-6 SHE

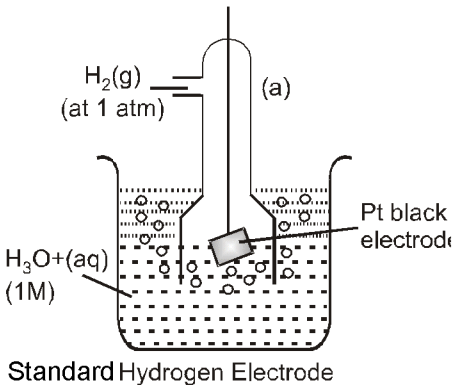

Figure -4

At Anode

$H_2(g) \rightleftharpoons 2H^+ + 2e^-$

- Oxidation potential
  O.P. = E $H_2(g)/H^+(aq)$
- Under standard state
- E°$H_2(g)/H^+(aq)$ = SOP

At Cathode

$2H^+ + 2e^- \rightleftharpoons H_2(g)$

- Reduction Potential
  R.P = E $H^+/ H_2(g)$
- Under standard state.
- E°$H^+/ H_2(g)$ = SRP

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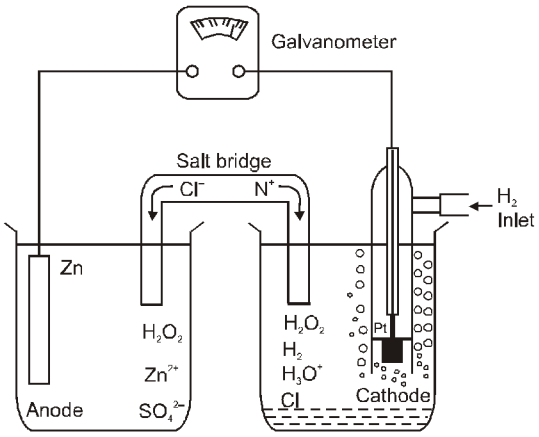
P-7 To calculate standard potential of any other electrode a cell is formed by coupling with standard hydrogen electrode (SHE) and its potential difference is measured. That gives the value of electrode potential of that electrode.



A galvanic cell measuring the Zn^{2+} |Zn half-cell potential

Figure -5

Anode : Zinc electrode (standard)

Cathode : SHE

Cell : Zinc electrode || SHE

Cell potential :

$E^{\circ}_{cell} = E^{\circ}_{H^+/H_2(g)} - E^{\circ}_{Zn^{2+}/Zn}$

= 0.76 V (at 298 K experimentaly)

So, $E^{\circ}_{Zn^{2+}/Zn}$ = – 0.76 V (SRP)

or $E^{\circ}_{Zn/Zn^{2+}(aq)}$ = 0.76 V(SOP)

- So, w.r.t. H_2, Zn has greater tendency to get **oxidised**.

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1.5 Electrochemical Series :

D-12 If SRP value of different electrode are calculated with reference electrode like SHE at 298 K and then arranged in a series [in increasing order], then such series is known as **electrochemical series.** Metal reactivity series is a subset of electrochemical series.

Electrochemical Series :

Increasing strength of reducing agent ↑

Increasing strength of oxidising agent ↓

Electrode	SRP (at 298 K)
*Li	– 3.05 V
K	– 2.93 V
Ba	
Ca	– 2.87 V
Na	– 2.71 V
Mg	– 2.37 V
Al	
* Electrolytes (H_2O) : $H_2O(l) + e^- \rightarrow \frac{1}{2}H_2 + OH^-$	– 0.828 V
*Zn	– 0.76 V
Cr	– 0.74 V
*Fe	– 0.44 V
Cd	– 0.40 V
Co	
Ni	– 0.24 V
Sn	– 0.14 V
Pb	– 0.13 V
D_2	–0.01 V
*H_2	0.00 V
Cu	0.34 V
I_2	
Fe	0.77 V
Hg	0.79 V
Ag	
Hg	
Br_2	1.06 V
* Electrolytes : $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O(\ell)$	1.23 V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{+3} + 7H_2O$	1.33 V
$Cl_2 + 2e^- \rightarrow 2\,Cl^-$	1.36 V
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	1.51 V
$F_2 + 2e^- \rightarrow 2F^-$	2.87 V

- Li is best reducing agent in aqueous solution, not $Li^+(aq)$.
- F_2 is best oxidising agent in aqueous solution, not $F^-(aq)$.

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Application of Electrochemical Series

P-1. Greater the value of S.R.P. greater will be tendency to get reduced, so, the element or compound will be stronger oxidising agent.

Smaller the value of SRP (higher the value of SOP), greater will be the tendency to get oxidised hence compound or elements will be stronger reducing agent.

Eg. $\overset{+7}{Mn}O_4^- \longrightarrow Mn^{2+}$ $E^\circ = 1.51$ V

$\overset{+6}{Cr_2}O_7^{2-} \longrightarrow Cr^{3+}$ $E^\circ = 1.40$ V

In MnO_4^- and $Cr_2O_7^{2-}$, better oxidising agent is MnO_4^- (because MnO_4^- has higher SRP)

P-2. Anode – Electrode higher up in the electrochemical series

Cathode – any electrode lower in electrochemical series in comparison to anode

P-3. Metals which are higher up in the electrochemical series can displace metal ions which are lower in the series from their salt solution

$CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu\downarrow$

$ZnSO_4 + Cu \longrightarrow X$ (no reaction)

P-4. On moving down the electrochemical series, the electropositive nature of the metals or elements decreases and also the reactivity of metals decreases.

(Alkali or alkaline earth metals) + cold water $\rightarrow H_{2(g)}\uparrow$

(Fe, Ni, Co, Pb, Sn) + Steam $\rightarrow H_{2(g)}\uparrow$

(Cu, Ag, Au) + Cold water or Steam $\rightarrow$ No reaction.

P-5. Metals placed above hydrogen can release $H_{2(g)}\uparrow$ on reaction with dilute acid solution.

$Mg + HCl \longrightarrow MgCl_2 + H_{2(g)}\uparrow$

$Cu + HCl \longrightarrow$ cannot release $H_{2(g)}$

P-6. **Thermal stability of metal oxide : It d**epends upon electropositive nature of metal. As electropositive nature decreases from top to bottom, the thermal stability of the oxide also decrease from top to bottom in electrochemical series.

$BaO, Na_2O, Al_2O_3 \xrightarrow{\Delta}$ No decomposition.

(a) $Ag_2O \xrightarrow{\Delta} 2Ag + 1/2\ O_2$; (b) $2HgO \xrightarrow{\Delta} 2Hg + O_2$

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2.3 Calculation of SRP of an electrode from given SRP value of other electrode.

Gibb's Free energy change

F-1 (a) For cell reaction

$$\Delta G^\circ = -nFE^\circ_{cell}$$

where

(a) F = Faraday's constant = 96500 C ≈ (96485 C)

= change on 1 mole of e^-

= $(1.6 \times 10^{-19}\ C) \times 6.022 \times 10^{23} = 96500\ C$

(b) n = no. of e^- involved in cell reaction

F-2 (b) For electrode reaction

$$\Delta G^\circ_{electrode} = -nFE^\circ_{electrode}$$

n = no. of e^- transferred in electrode reaction

2.4

P-7. Factor affecting the electrode potential of an electrode :

(i) It is an intensive property, so it is not an additive property.

So we can not directly add or subtract E° value of two electrode to determine E° value of 3rd electrode.
* By using two electrode, if a cell is constructed then their E° values are additive.

(ii) It does not depend on mass or size of electrode & also does not change by multiplying or dividing electrode reaction by any no.

P-8. Factor affecting the Gibb's free energy of an electrode :

(i) It is an extensive property so we can add or subtract ΔG value of two reaction:

It changes by multiplying or dividing electrode reaction

1. $Zn^{2+} + 2e^- \longrightarrow Zn(s)$ $\quad E^\circ = -0.76\ V$

$\Delta G^\circ = -2 \times 96500 \times (-0.76)$

$2Zn^{2+} + 4e^- \longrightarrow 2Zn(s)$ $\quad E^\circ = -0.76\ V$

$\Delta G^\circ = -4 \times 96500 \times (-0.76)$

2. $Zn \longrightarrow Zn^{2+}_{(aq)} + 2e^-$ $\quad E^\circ_{Zn\,|\,Zn^{2+}} = 0.76\ V$

$\Delta G^\circ = -nFE^\circ$

$\Delta G^\circ = -2 \times 96500 \times (0.76)$

(ii) ΔG° = The max. amount of useful non-PV work which can be obtained from a given cell

$\Delta G^\circ = -nFE^\circ_{cell}$

For electrode $\Delta G^\circ = -nFE^\circ_{electrode}$

$\Delta G^\circ = -RT\ln K_{eq}$

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Nernst equation & its Applications

Nernst Equation:

P-1 Cell potential depends on temperature and on the composition of the reaction mixture (upon the concentration of the solutions and the partial pressure of the gases, if any). The dependence can be derived from thermodynamics as follows:

From thermodynamics

F-1

$\Delta G = \Delta G^\circ + RT \ln Q$

$$-nFE = -nFE^\circ + 2.303\, R\, T \log Q$$

$$E = E^\circ - \frac{2.303RT}{nF} \log Q$$

Take T = 298 K, R = 8.314 J/mol K, F = 96500 C

Now we get, $E = E^\circ - \frac{0.059}{n} \log Q$

Where n = number of transferred electron, Q = reaction quotient (includes molarity of aq. species and partial pressures of gases in atm involved in cell reaction)

Nernst equation can be applied to half cell reactions also.

Nernst Equation for various electrodes (half cells) at 25°C

P-2 Metal-metal soluble salt electrode

Eg. $Zn^{2+} + 2e^- \rightarrow Zn(s)$

$$E_{Redn} = E^0_{red} - \frac{0.059}{2} \log\left(\frac{1}{Zn^{+2}}\right)$$

P-3 Gas-ion electrode

Eg. $2H^+ + 2e^- \rightarrow H_2(g)$

$$E_{Red^n} = E^0_{Red^n} - \frac{0.059}{2} \log\left(\frac{P_{H_2}}{[H^+]^2}\right)$$

P-4 Redox electrode

Eg. $4H_2O + Mn^{2+} \rightarrow MnO_4^- + 8H^+ + 5e^-$

$$E_{OX} = E^0_{ox} - \frac{0.059}{5} \log \frac{[MnO_4^-][H^+]^8}{[Mn^{+2}]}$$

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E_{Redn} in (ii) case is more, so stronger oxidising agent

Note : Nernst equation is applied to a galvanic cell (combination of two electrodes) like :

$Cu(s) \mid Cu^{2+}(aq.) \parallel Ag^{+}(aq.) \mid Ag(s)$ as follows :

$$E_{cell} = E^{\circ}_{cell} - \frac{0.059}{n} \log Q \text{ at } 25^{\circ}C$$

Where, $E^{\circ}_{cell} = E^{o}_{Ag^{+}/Ag} - E^{o}_{Cu^{2+}/Cu}$

and n = number of electrons cancelled while making cell reaction = 2

$$Q = \left(\frac{[Cu^{2+}]}{[Ag^{+}]^2} \right)$$

(according to cell reaction : $Cu(s) + 2Ag^{+}(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$)

Equilibrium in electrochemical cell and Work done by a cell

F-3 For a general cell reaction

$aA + bB \rightleftharpoons cC + dD$

From thermodynamics: $\Delta G = \Delta G^{0} + RT\ell nQ$

From electrochemistry: $\boxed{E_{cell} = E^{0}_{cell} - \frac{RT}{nF} \ln Q}$

At equilibrium, $\Delta G = 0$ and $E_{cell} = 0$ (cell will stop functioning)

F-4 So, $\Delta G^{0} = -RT\ell n K_{eq}$

at equilibrium, $-nF E^{\circ}_{cell} = -2.303 RT \log (K_{eq})$

$$\log K_{eq} = \frac{nF}{2.303\ RT} E^{\circ}_{cell}$$

at 298 K and R = 8.314 J/mol K,

$$\boxed{\log K_{eq} = \frac{n}{0.059} E^{\circ}_{cell}}$$

D-1 Work done by a cell :

Let 'n' faraday charge flows in a cell of EMF 'E'; then maximum work that can be obtained from the cell will be calculated as: work = Charge × Potential = nFE

In other words, magnitude of –ve value of ΔG gives the maximum amount of electrical work (non PV work) that can be obtained from cell.

Concentration cells:

D-1 A concentration cell consists of two electrodes of the same material, each electrode dipping in a solution of its own ions and the solutions being at different concentrations. $E^{\circ}_{cell} = 0$ for a concentration cell. The two solutions are separated by a salt bridge.

e.g. $Ag(s) \mid Ag^{+}(a_1) \parallel Ag^{+}(a_2) \mid Ag(s) \longrightarrow (a_1 < a_2)$ a_1 & a_2 are concentrations of Ag^{+} of each half cell

at LHS electrode anode : $Ag(s) \longrightarrow Ag^{+}(a_1) + e^{-}$

at RHS electrode cathode : $Ag^{+}(a_2) + e^{-} \longrightarrow Ag(s)$

The net cell reaction is : $Ag^{+}(a_2) \longrightarrow Ag^{+}(a_1)$

The nernst eq. is

$$\boxed{E_{cell} = -\frac{0.059}{n} \log \frac{a_1}{a_2}}$$ (Here n = 1, Temp. 298 K)

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P-2 Likewise, the e.m.f. of the cell consisting of two hydrogen electrodes operating at different pressure P_1 and P_2 ($P_1 > P_2$) and dipping into a solution HCl is :

$$E_{cell} = \frac{0.059}{2} \log \frac{P_1}{P_2} \quad \text{(at 298 K)}$$

Solubility product and E° (Metal-Metal Insoluble Salt Electrode) :

P-3 A half-cell containing metal M, its sparingly soluble salt MA in contact with a solution of a soluble salt NaA of the same anion, i.e. NaA | MA(s) | M(s) is set up at standard conditions ([NaA = 1 M]).

Since $K_{sp} = [M^+]\,[A^-]$,

$\therefore \quad [M^+] = K_{sp}$

So, above electrode will behave like a metal-metal soluble salt electrode

M^+ (aq., K_{sp}) | M(s)

$\therefore \quad E^\circ_{A^-/MA/M} = E_{M^+/M}$

Value of RHS can be obtained by applying Nernst equation as follows:

$$E^\circ_{A^-/MA/M} = E^\circ_{M^+/M} - \frac{0.059}{1} \log \frac{1}{K_{sp}}$$

$\therefore$

$$E^\circ_{A^-/MA/M} = E^\circ_{M^+/M} + \frac{0.059}{1} \log K_{sp}$$

P-4 Complex electrode

Eg. $[Cu(NH_3)_4]^{2+}(aq) + 2e^- \rightarrow Cu(s) + 4NH_3(aq)$

$$E_{redn} = E^\circ_{red^n} - \frac{0.059}{2} \log \frac{[NH_3]^4}{[Cu(NH_3)_4^{2+}]}$$

Also, similar to above:

$$E^\circ_{[Cu(NH_3)_4]^{2+}|NH_3+Cu} = E^\circ_{Cu^{2+}/Cu} - \frac{0.059}{2} \log K_f[Cu(NH_3)_4]^{2+}$$

Note :

(i) SRP value of a metal-metal insoluble salt electrode as well as a complex electrode is always is always less than that of same metal's metal-metal soluble salt electrode.

(ii) A non-standard metal-metal insoluble salt electrode can be converted into a metal-metal soluble salt electrode by a way similar to above using K_{sp} relation of salt.

Eg. Ag | AgCl | Cl^- (c_1) | Ag^+ (c_2) | Ag

$E^\circ cell = E^\circ_{Ag^+/Ag} - E^\circ_{Cl^-/AgCl/Ag} \neq 0$, so not a conc. cell.

For calculating E_{cell}, a new cell can be made (conc. cell) equivalent to the above one as follows:

Ag | Ag^+ (K_{sp} / c_1) | Ag^+ (K_{sp} / c_2) | Ag

$$E_{cell}\ (original) = E_{cell}\ (new) = -\frac{0.059}{1} \log_{10} \left(\frac{K_{sp}/c_1}{c_2} \right)$$

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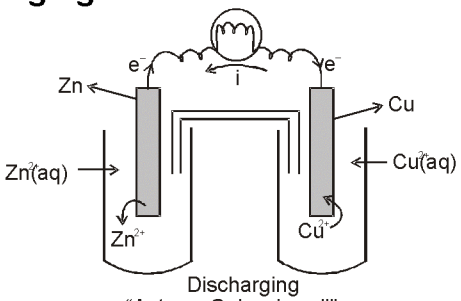
5.1 Discharging & charging of electrochemical cell

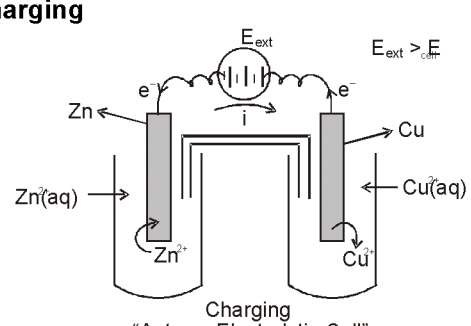
P-1 (i) Discharging : Acts as galvanic cell
P-2 (ii) Charging : Acts as electrolytic cell

Discharging



Discharging
"Acts as Galvanic cell"

Charging



Charging
"Acts as Electrolytic Cell"
E_{cell} is forced by E_{ext} to do reverse reaction

$Zn + Cu^{2+}(aq) \rightleftharpoons Zn^{2+}(aq) + Cu$
$Cu \rightarrow Cu^{2+} + 2e^-$ (Anode)
$Zn^{2+} + 2e^- \rightarrow Zn$ (Cathode)

5.2 Electrolysis & Electrolytic cell

D-1 Definition of electrolysis & electrolytic cell

Electrolysis:

Electrolyte is a combination of cations and anions which in fused state or in aqueous solution can conduct electricity.

This is possible due to the movement of ions from which it is made of.

The process of using an electric current to bring about a chemical change is called electrolysis.

Electrolysis is a process of oxidation and reduction due to current in the electrolytic solution.

P-3 The products obtained during electrolysis depend on following factors:

- The nature of the electrolyte
- The concentration of electrolyte

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- The charge density flowing during electrolysis.
- The nature of the electrode

Active vs Inactive electrodes :

P-4 Those electrodes in the cell are active in which the substances themselves are components of the half reactions.

Eg. As the Daniel cell operates, the mass of the zinc electrode gradually decreases, and the $[Zn^{2+}]$ in the anode half – cell increases. At the same time, the mass of the copper electrode increases and the $[Cu^{2+}]$ in the cathode half – cell decreases; we say that the Cu^{2+} "plates out" on the electrode.

P-5 For many redox reactions, however, there are no reactants or products capable of serving as electrodes. Inactive electrodes are used then, most commonly rods of graphite or platinum, materials that conduct electrons into or out of the cell but do not take part in the half-reactions.

Eg. In a voltaic cell based on the following half reactions, for instance, the species cannot act as electrodes:

$2I^-(aq) \longrightarrow I_2(s) + 2e^-$ [anode ; oxidation]

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \longrightarrow Mn^{2+}(aq) + 4H_2O(\ell)$ [cathode ; reduction]

Therefore, each half – cell consists of inactive electrodes immersed in an electrolyte solution that contains all the species involved in that half -reaction. In the anode half-cell, I^- ions are oxidized to solid I_2. The electrons released flow into the graphite anode, through the wire, and into the graphite cathode. From there, the electrons are consumed by MnO_4^- ions as they are reduced to Mn^{2+} ions.

5.3 Reactions that occur during Electrolysis

1. In molten state electrolysis, generally only one cation & one anion are available which move to oppositely charged electrodes (cation to cathode (–) & anion to anode(+)) and get discharged there undergoing reduction & oxidation respectively.

In aqueous state electrolysis, along with cation & anion coming from electrolyte, two additional reactions compete :

(a) Reduction of water, competing with cations present for reaction at cathode :

$2e^- + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^-(aq)$; $E^\circ = -0.83V$

(b) Oxidation of water, competing with anions present for reaction at anode :

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$; $E^\circ = -1.23\ V$

2. In general, among all the reduction reactions competing at cathode, the one having maximum SRP will be most favourable. Similarly, among all the oxidation reactions competing at anode, the one having maximum SOP will be most favourable.

3. Non-oxidizable anions are not oxidized. In such a case, solvent is oxidized:

$2H_2O \longrightarrow 4H^+ + O_2\uparrow + 4e^-$

Non Oxidizable:

$SO_4^{2-}, NO_3^-, PO_4^{3-}, CO_3^{2-}, HCO_3^-, CrO_4^{2-}, Cr_2O_7^{2-}, MnO_4^-$

Note: Whenever $Cl^-(aq)$ is oxidized to Cl_2, small quantity of H_2O is also oxidized to O_2.

Reaction (rare):

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$2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$ (Requires very high E_{ext})
(persulphate)

4. Metal ions below hydrogen in electrochemical series are reduced to metal which is deposited at cathode (nearly exclusively).
   $Bi^{3+} + 3e^- \longrightarrow Be$
   $Cu^{2+} + 2e^- \longrightarrow Cu$
   $Ag^+ + e^- \longrightarrow Ag$

- From Li^+ to Al^{3+}, aqueous metal ions are not reduced, instead, solvent H_2O is reduced entirely to H_2.
  $2H_2O + 2e^- \longrightarrow H_2\uparrow + 2(OH^-)$
- The remaining metal ions ($Mn^{2+} \rightarrow Pb^{2+}$) are deposited at cathode with liberation of H_2.

5. If reactive electrodes are used, oxidation of metal anode should first be considered. Except Au & Pt, all metal electrodes are oxidized in preference to water from solute.

5.4 Examples of electrolysis using Inert electrode (Pt or graphite)

Possible reactions at Anode & Cathode in various electrolytic cells:

P-6 Electrolysis of molten $PbBr_2$

Cathode : $Pb^{2+} + 2e^- \rightarrow Pb(s)$ $E^\circ = -0.13V$

Anode : $2Br^- \rightarrow Br_2 + 2e^-$ $E^\circ = -1.09\ V$

$E_{ext} > 1.22\ V\ (1.09 - (-0.13))$ should be applied

P-7 Electrolysis of aq $CuSO_4$

Cathode : $Cu^{2+} + 2e^- \rightarrow Cu(s)$ $E^\circ = 0.34\ V$

$2e^- + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^-(aq)$ $E^\circ = -0.83V$

First has greater SRP, so more favorable.

Anode : $2\,SO_4^{2-} \rightarrow S_2\,O_8^{2-} + 2e^-$ $E^\circ = -2.05\ V$

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $E^\circ = -1.23\ V$

Second has greater SOP, so more favorable.

Overall reaction : $2CuSO_4 + 2H_2O \rightarrow 2Cu + O_2 + 2H_2SO_4$
(at cathode) (at anode)

P-8 Electrolysis of aq NaBr

Cathode : $Na^+(aq) + e^- \rightarrow Na(s)$ $E^\circ = -2\ V$

$2e^- + 2H_2O(\ell) \rightarrow H_2 + 2OH^-$ $E^\circ = -0.83\ V$

Second has greater SRP, so more favorable.

Anode : $2Br^- \rightarrow Br_2 + 2e^-$ $E^\circ = -1.08V$

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $E^\circ = -1.23V$

First has greater SOP, so more favorable.

Overall reaction: $2NaBr + 2H_2O \rightarrow 2NaOH + H_2 + Br_2$
(at cathode) (at anode)

P-9 Electrolysis of aq NaCl

Cathode : $Na^+ + e^- \rightarrow Na$ $E^\circ = -2V$

$2e^- + 2H_2O(\ell) \rightarrow H_2(g) + 2OH^-$ $E^\circ = -0.83\ V$

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Second has greater SRP, so more favorable.

Anode : $2Cl^- \rightarrow Cl_2 + 2e^-$ $E°=-1.30V$

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $E°=-1.23V$

Second has greater SOP, so more favorable.

Note : According to thermodynamics, oxidation of H_2O to produce O_2 should take place on anode but experimentally (experiment from chemical kinetics) the rate of oxidation of water is found to be very slow. To increase its rate, the greater potential difference is applied called over voltage or over potential but because of this oxidation of Cl^- ions also become feasible and this takes place on anode significantly.

Overall reaction: $2NaCl + 2H_2O \rightarrow \underset{\text{(at cathode)}}{2NaOH} + H_2 + \underset{\text{(at anode)}}{Cl_2}$ (+ some O_2)

P-10 Electrolysis of aq CH_3COONa

Cathode : $2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$

Anode : $2CH_3COO^- \rightarrow 2CO_2 + C_2H_6 + 2e^-$

Overall reaction: $2CH_3COONa + 2H_2O \rightarrow \underset{\text{(at anode)}}{2CO_2 + C_2H_6} + \underset{\text{(at cathode)}}{H_2} + 2NaOH$

5.5 Electrolysis using Active electrodes

P-11 Electrolysis of $CuSO_4(aq)$ using Cu electrodes.

Cathode (reduction) :

$Cu^{2+} + 2e^- \rightarrow Cu$ $E° = +0.34$ V

$2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-$ $E° = -0.83$ V

First has greater SRP, so more favorable.

Anode (oxidation) :

$SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$ $E°_{OX} = -2.05$ V

$2H_2O(\ell) \rightarrow O_2 + 2H^+ + 4e^-$ $E° = 1.23$ V

$Cu(s) \rightarrow Cu^{2+} + 2e^-$ $E° = -0.34$ V

Third has greatest SOP, so most favorable.

Cu anode gets thinner, Cu cathode gets thicker, while conc. of Cu^{2+} in solution remains same.

Used in electrolytic refining of Cu in metallurgy.

P-12 $AgNO_3(aq)$ using Cu cathode & Ag anode.

Cathode : $Ag^+ + e^- \rightarrow Ag(s)$ $E° = 0.8$ V

$2H_2O(\ell) + 2e^- \rightarrow H_2(g) + 2OH^-$ $E° = -0.83$ V

First has greater SRP, so more favorable.

Anode : $NO_3^- \rightarrow X$ (No reaction)

$2H_2O(\ell) \rightarrow O_2 + 4H^+ + 4e^-$ $E° = -1.23$ V

$Ag(s) \rightarrow Ag^+(aq) + e^-$ $E° = -0.80$ V

Third has greatest SOP, so most favorable.

Used in electroplating (Cu cathode gets coated by Ag)

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HYDROCARBON

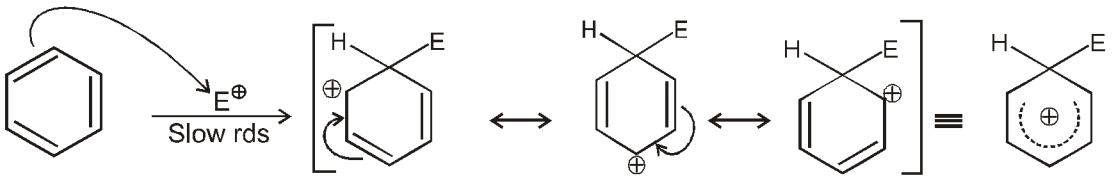
I. Electrophilic substitution reaction of Aromatic compounds :-

Like an alkene, benzene has cloud of pi electrons above and below its sigma bond framework. Although benzene's pi electrons are in a stable aromatic system still they are available to attack with a strong electrophile to give a carbocation. This resonance-stabilized carbocation is called a **sigma complex** because the electrophile is joined to the benzene ring by a new sigma bond.

The sigma complex (also called an arenium ion) is not aromatic because the sp^3 hybrid carbon atom interrupts the ring of p orbitals. This loss of aromaticity contributes to the highly endothermic nature of thus first step. The sigma complex regains aromaticity either by a reversal of the first step (returning to the reactants) or by loss of the proton on the tetrahedral carbon atom, leading to the substitution product.

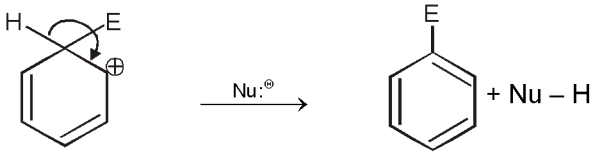
The overall reaction is the substitution of an electrophile $(\overset{\oplus}{E})$ for a proton $(\overset{\oplus}{H})$ on the aromatic ring so it is called **electrophilic aromatic substitution**.

Step 1 : Attack of an electrophile on benzene ring forms the sigma complex

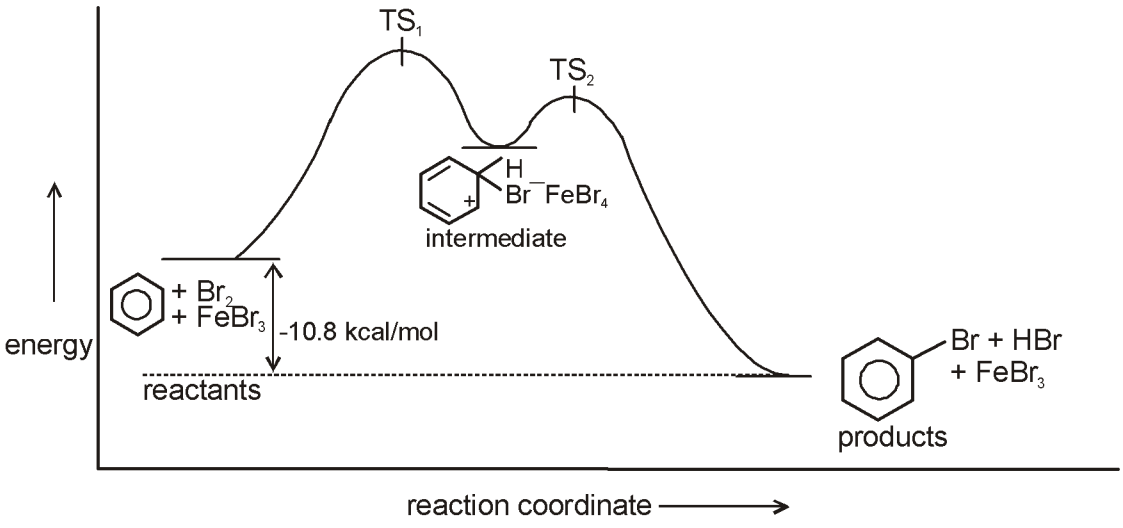


Resonance hybrid
[σ – complex]
Arenium Ion

Step 2 : Loss of a proton gives the substitution product.



Energy diagram : For bromination reaction, electrophile is bromonium ion (Br^+)



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For an electrophilic aromatic substitution reaction to overcome the high activation energy that charaterizes the first step, the electrophile must be a fairly reactive one. Many of the electrophilic reagents that react rapidly with alkenes do not react at all with benzene. For example peroxy acids and diborane, fall into this category, others such as bromine react with benzene only in presence of catalysts that increases their electrophilicity.

Effect of substituent groups in monosubstituted benzene :

(A) Ortho-para directing and activating groups : All electron releasing groups (+m, +I) are ortho-para directing groups and activating towards electrophilic reactions.

(B) Ortho para directing but deactivating groups : Halogens are deactivating but ortho-para directing groups.

Reactivity of benzene decreases by –I effect of halogens and ortho-para directing nature is decided by +m effect of halogens.

(C) Meta directing and deactivating groups : Mostly electron withdrawing groups (–m, –I) are meta directing groups and deactivating towards electrophilic reactions.

	Substituent groups	Reactivity (effect on rate)	Directing nature (effect on orientation)
1.	$-O^- > -NH_2 > -NHR > -NR_2 > -OH$	Very strongly activating	Ortho-para directing
2.	$-OR > -NH-\overset{O}{\overset{\|\|}{C}}-R > -O-\overset{O}{\overset{\|\|}{C}}-R$	Strongly activating	Ortho-para directing
3.	$-R, -Ar, -CH=CH_2$	Activating	Ortho-para directing
4.	$-X(F, Cl, Br, I), -N=O, -CH_2X, -CHX_2$	Deactivating	Ortho-para directing
5.	$-CHO, -\overset{O}{\overset{\|\|}{C}}-R, -COOH, -COOR, -COCl, -C\equiv N, -SO_3H$	Strongly deactivating	Meta directing
6.	$-\overset{+}{N}R_3, -\overset{+}{S}R_2, -CF_3, -NO_2$	Very strongly deactivating	Meta directing

Effect of substituent groups in disubstituted benzene :

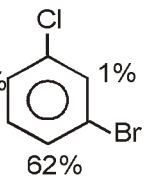
(1) If activating and deactivating both groups are present in a system then position of electrophile will be determined by activating group.

(2) If both groups present in a system are deactivator then position of electrophile will be determined by stronger deactivator.

(3) If both the groups are activating group then position of electrophile will be determined by stronger activator.

(4) There is often little substitution between two groups that are meta to each other.

e.g Nitration product of

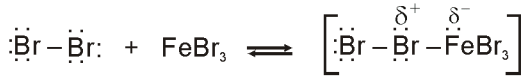


(% nitro product)

(i) Halogenation

Chlorine and bromine in presence of lewis acid (like $AlCl_3$, $FeCl_3$) react with benzene. Fluorination and iodination of benzene and other arenes rarely performed. Fluorine is so reactive that its reaction with benzene is difficult to control. Iodination is too slow and has an unfavourable equilibrium constant.

Step 1 : Formation of a stronger electrophile.



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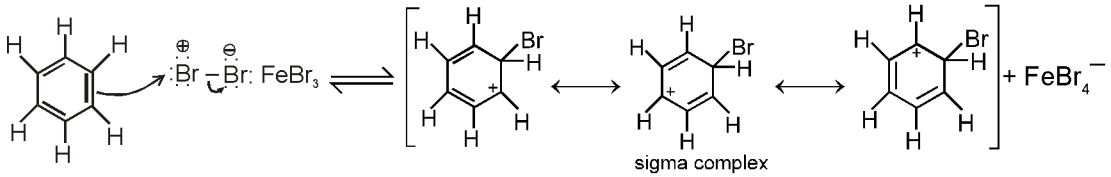
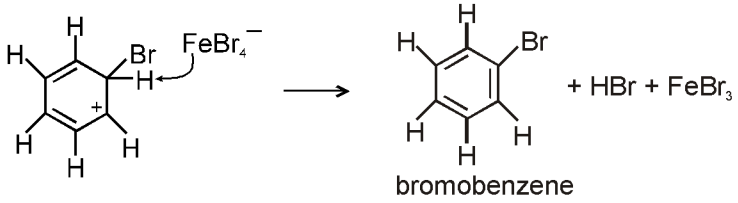
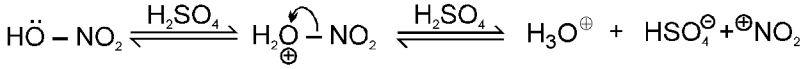
Step 2 : Electrophilic attack and formation of the sigma complex.

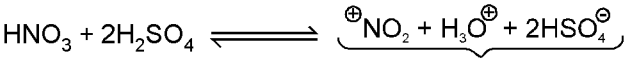


Step 3 : Loss of a proton gives the products.

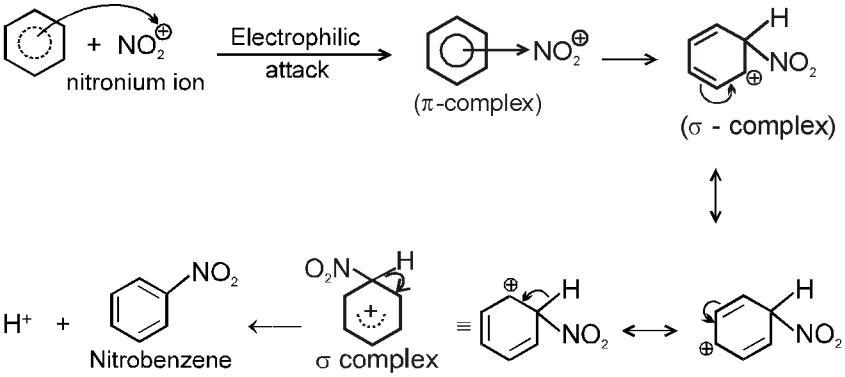


(ii) Nitration





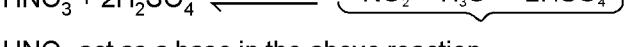
HNO_3 act as a base in the above reaction.

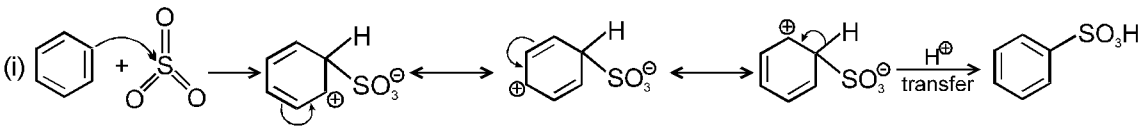


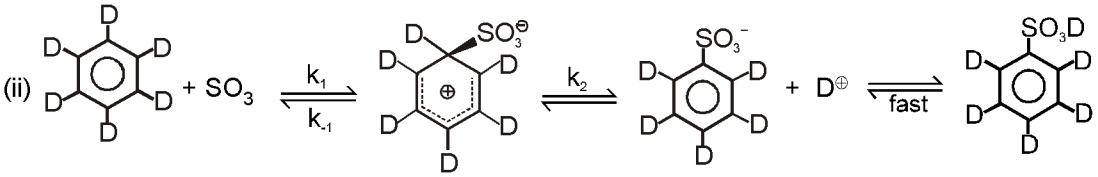
* Purpose of sulphonic acid in the reaction is to increase the concentration of nitronium ion (NO^+_2)

(iii) Sulphonation

Sulphur trioxide in sulphuric acid is used as the sulphonation agent.







Sulphonation, is **reversible** and takes place in concentrated sulphuric acid. ($K_{-1} \simeq K_2$).

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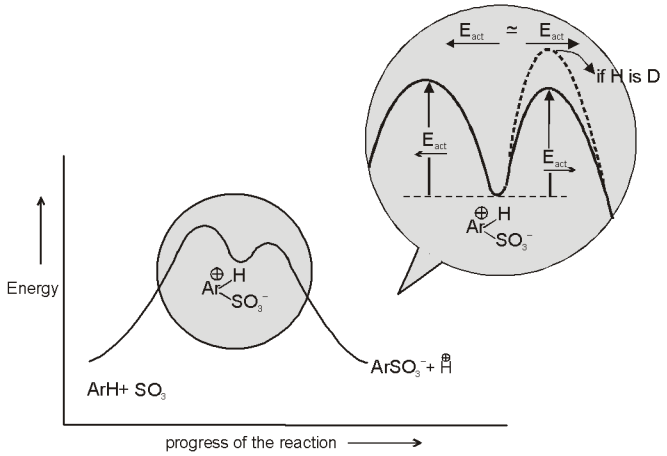
Energy Diagram

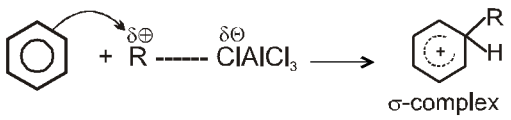


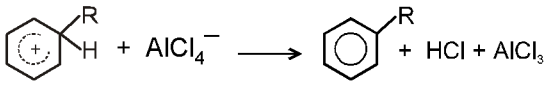
Some $Ar\overset{\oplus}{H}SO_3^-$ or $Ar\overset{\oplus}{D}SO_3^-$ go on to product, some revert to the starting material and decrease the rate of reaction. This effect is known as isotope effect.

(iv) Friedel Craft's reaction

(a) Alkylation : The mechanism for Friedel Craft's reaction involves the following steps.

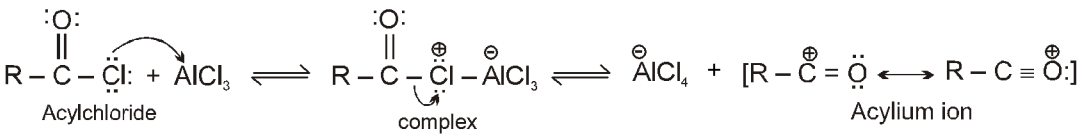
(i) $R-\ddot{Cl}: + AlCl_3 \longrightarrow \overset{\delta+}{R} \text{---- } Cl \text{ ---- } \overset{\delta-}{AlCl_3}$

(ii) 

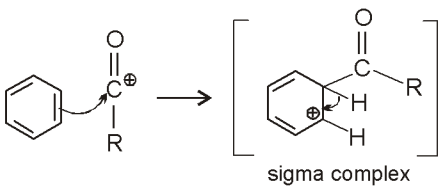
(iii) 

(b) Acylation : Acylation of benzene may be brought about with acid chlorides or anhydrides in presence of Lewis acids.

Step 1 : Formation of an acylium ion.



Step 2 : Electrophilic attack.



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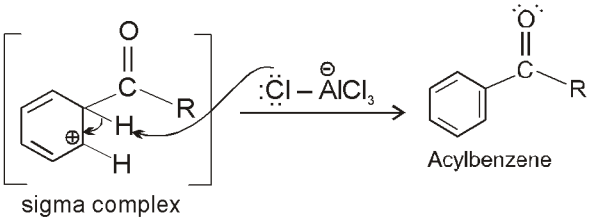
Step 3 : Loss of a proton. Complexation of the product.



Limitations of Friedel craft's reactions :

(i) Highly reactive rings like anililine and highly deactivated rings like nitrobenzene, cyanobenzene do not give friedel craft reactions.

(ii) Poly alkylation is possible but poly acylation is not possible.

(iii) Rearranged products may form in the friedel craft alkylation reactions.

(iv) Diketone also undergoes the Friedel Craft reaction with benzene.

II. Free radical substitution reaction of Alkane :-

Characteristic reaction of alkanes is **free radical substitution** reaction, these reactions are generally chain reactions which are completed in three steps :

(i) chain initiation (ii) chain propagation. (iii) chain termination

Halogenation : $R–H + X_2 \xrightarrow[250° - 400°C]{\text{UV Light or temp}} R–X + HX$

Mechanism :

(i) Chain initiation : $X_2 \xrightarrow[250° - 400°C]{\text{UV or temp.}} \dot{X} + \dot{X}$

(ii) Chain propagation : $\dot{X} + R - H \xrightarrow{rds} \dot{R} + HX$

$\dot{R} + X - X \longrightarrow R - X + \dot{X}$

(iii) Chain termination : $\dot{X} + \dot{X} \longrightarrow X_2$

$\dot{R} + \dot{R} \longrightarrow R - R$

$\dot{R} + \dot{X} \longrightarrow R - X$

Reactivity of X_2 : $F_2 > Cl_2 > Br_2 > I_2$

Reactivity of H : 3°H > 2°H > 1° H

Remarks : With F_2 alkanes react so vigorously that even in the dark and at room temperature, reactant is diluted with an Inert gas. Iodination is reversible reaction, since HI is formed as a by product and It is a strong reducing agent and reduces alkyl iodide back to alkane. Hence iodination can be done only in presence of strong oxidising agent like HIO_3, HNO_3 or HgO.

eg. $CH_4 \xrightarrow[h\upsilon]{Cl_2} \underset{+\ HCl}{CH_3Cl} \xrightarrow[h\upsilon]{Cl_2} \underset{+\ HCl}{CH_2Cl_2} \xrightarrow[h\upsilon]{Cl_2} \underset{+\ HCl}{CHCl_3} \xrightarrow[h\upsilon]{Cl_2} \underset{+\ HCl}{CCl_4}$

In a chain reaction following reagents are involved -

(i) Initiators : They initiate the chain reaction, Initiators are peroxide (R_2O_2), Perester's etc.

(ii) Inhibitors : A substance that slows down or stops the reaction is known as inhibitors

For example O_2 is a good inhibitor

$\dot{R} + O_2 \longrightarrow R - O - \dot{O} + \dot{R} \longrightarrow R - O - O - R$

all reactive alkyl free radicals are consumed so reaction stops for a period of time.

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Halogenation of higher alkane :

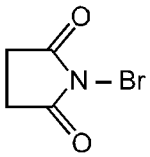
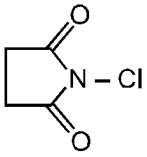
(a) $CH_3–CH_2–CH_3 \xrightarrow[\text{light, 25°C}]{Cl_2} CH_3-\underset{Cl}{\underset{|}{CH}}-CH_3 + CH_3-CH_2-CH_2-Cl$

(b) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_3 \xrightarrow[\text{light, 25°C}]{Cl_2} CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2-Cl + CH_3-\underset{Cl}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-CH_3$

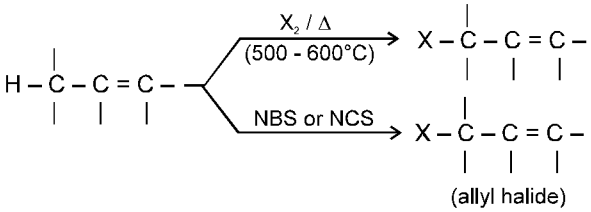
(c) $CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_3 \xrightarrow[\text{heat, 127°C}]{Br_2} \underset{\text{trace}}{CH_3-\underset{CH_3}{\underset{|}{CH}}-CH_2Br} + \underset{\text{over 99\%}}{CH_3-\underset{CH_3}{\underset{|}{\overset{CH_3}{\overset{|}{C}}}}-Br}$

III. Free radical Allylic and benzylic substitution reaction :-

Reagents: $Cl_2/h\nu$ or Δ , $Br_2/h\nu$ or Δ , NBS or NCS

NBS = N-Bromosuccinimide  ; NCS = N-Chlorosuccinimide 

General Reaction



$X_2 = Cl_2, Br_2$

It is free radical substitution reaction

IV. Free Radical Addition reaction of Alkene & Alkyne :-

Reagents : HBr / R_2O_2, hν [Kharasch Effect or Peroxide Effect]

General Reaction :

$$R-CH=CH_2 + HBr \xrightarrow{R_2O_2/h\nu} R-CH_2-CH_2-Br$$

Mechanism :

(1) $R-O-O-R \xrightarrow{h\nu} 2RO^{\bullet}$

(2) $RO^{\bullet} + HBr \longrightarrow ROH + Br^{\bullet}$

.....Chain initiation steps

(3) $R-CH=CH_2 + Br^{\bullet} \longrightarrow \underset{\text{(major)}}{R-\dot{C}H-CH_2-Br\ (2^{\circ}\dot{R})} + \underset{\text{(minor)}}{R-\underset{Br}{\underset{|}{CH}}-\dot{C}H_2\ (1^{\circ}\dot{R})}$

(4) $R-\dot{C}H-CH_2-Br + HBr \longrightarrow R-CH_2-CH_2-Br + Br^{\bullet}$

$Br^{\bullet}$ repeats step (3).

.....Chain propagating steps

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Remarks :

(1) When HBr is added to an unsymmetrical alkene in presence of sunlight and peroxide. Then an **Anti Markovnikov's Addition Product is obtained.**

(2) It is a free radical chain reaction.

(3) In presence of peroxide and sunlight $Br^{\bullet}$ is formed in chain initiation step.

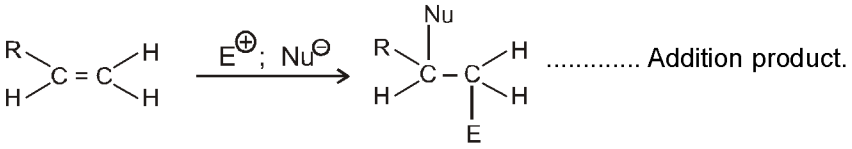
(4) $Br^{\bullet}$ forms more stable alkyl radical by homolysis of C=C π bond.

(5) In the last step alkyl radical abstracts $H^{\bullet}$ from HBr and Anti markownikov's product is obtained.

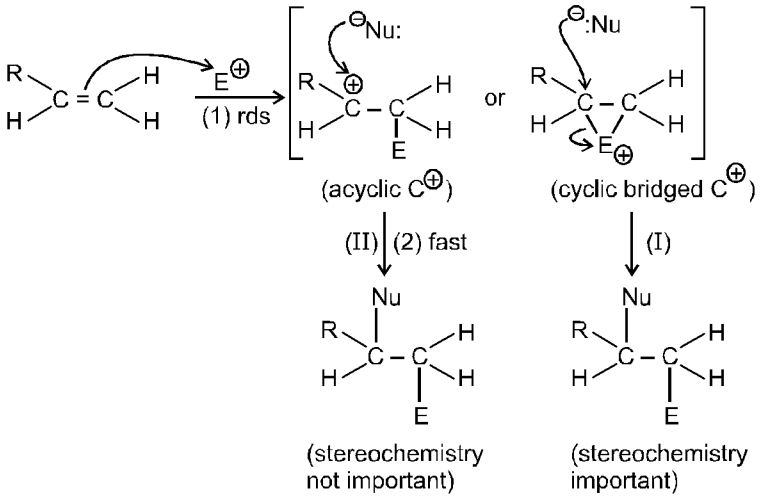
V. Electrophilic addition reaction of Alkenes :-

Due to presence of weak π electrons in alkene and alkyne, it will go for electrophilic addition reaction.
In electrophilic addition reactions one weak π-bond (251 KJ mol^{-1}) is broken and two strong σ bonds ($2 \times 347 = 694$ KJ mol^{-1}) are formed. The overall reaction is accompanied by a release of about 694-251 = 443KJmol^{-1} of energy.

General Reaction of electrophilic addition :



The overall reaction mechanism can be visualised as followed :



Anti Addition

Remarks :

Alkenes, Alkynes and Alkadienes are electron rich species. So they function as $Nu:^{\ominus}$ species (due to loosely bound π-electrons)

These compounds mainly give electrophilic addition reactions. Due to nucleophilic nature of alkenes/alkynes having affinity for $E^{\oplus}$.

The reaction is initiated by an attack of $E^{\oplus}$.

Rate of reaction $\propto$ stability of carbocation

Reactivity of an Alkene :

(1) Presence of e^{-} releasing groups (+m, +I) at C = C increases nucleophilicity and reactivity.

(2) Presence of ERG stabilises the intermediate carbocation. (3) More stable $C^{\oplus}$, more is reactivity.

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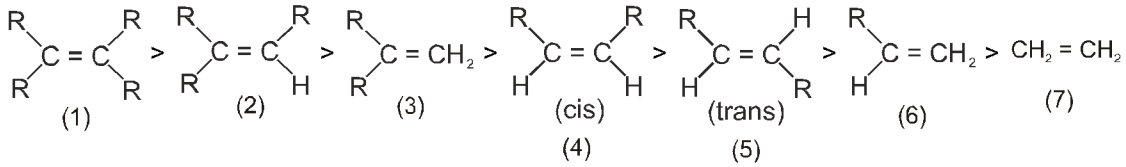
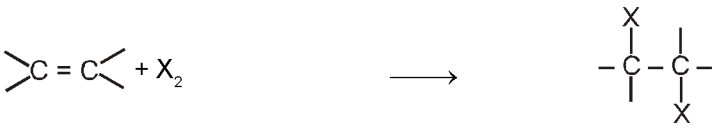
Examples of Reactivity Orders :-

(i) 

(ii) ERG – CH = CH_2 > CH_2 = CH_2 > EWG – CH = CH_2

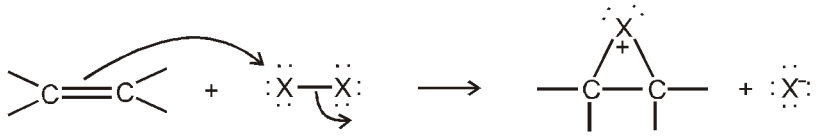
(i) ADDITON OF HALOGEN (HALOGENATION) :

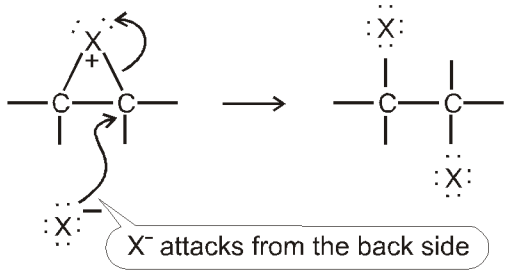


(X_2 = Cl_2, Br_2, but not F_2 and I_2)

usually anti addition

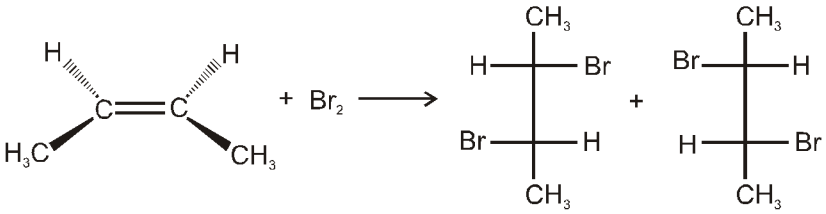
Mechanism

Step 1 : 

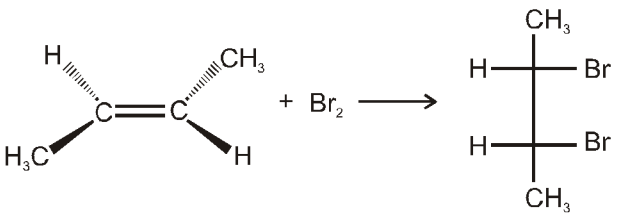
Step 2 : 

Remarks : Anti addition ; No rearrangement ; Bridged Carbocation ; Markovnikoff's additions

The addition of bromine to cyclopentene is a stereospecific **anti addition.**

Ex. 

Cis-2-butene (±)-2,3-dibromobutane

Ex. 

Trans-2-butene Meso-2,3-dibromobutane

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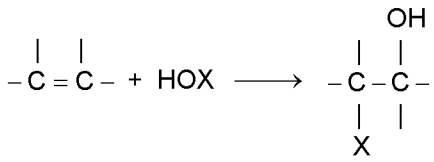
(ii) ADDITION OF HOX :

Reagents : (i) $X_2 + H_2O \rightarrow HOX + HX$ or (ii) HOX , H^+

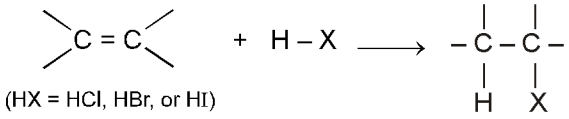
Electrophile : X^+, Halonium ion ;



Remarks : Anti addition ; No rearrangement ; Bridged Carbocation ; Markovnikoff's additions

(iii) ADDITION OF HYDROGEN HALIDES (+ HX) :

Reagents : HX; **Electrophile :** H^+

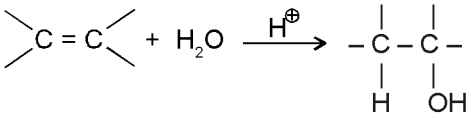


(HX = HCl, HBr, or HI)

Remarks : Markovnikoff additon ; Intermediate carbocation ; Rearrangement may take place

(iv) ADDITION OF H_2O (HYDRATION REACTION)

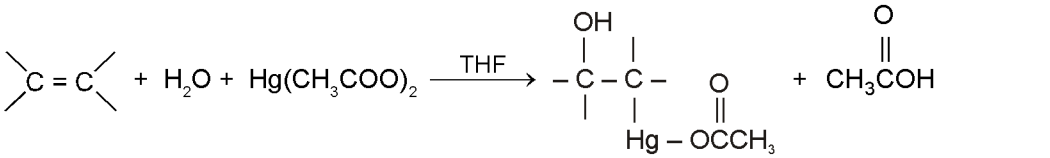
(a) Acid-Catalyzed Hydration of Alkenes : **Reagents :** Dilute H_2SO_4 or H_3O^+

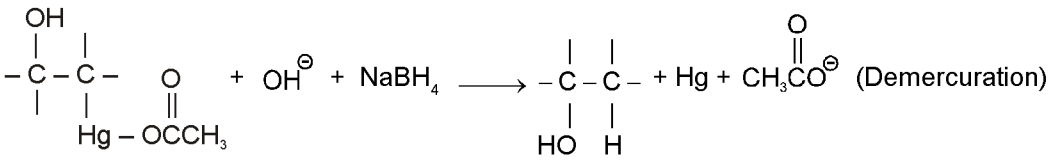


Remarks : Markovnikoff additon ; Intermediate carbocation ; Rearrangement may take place

(b) Oxymercuration-Demercuration : **Reagents :** (i) $Hg(OAc)_2$, H_2O (ii) $NaBH_4$

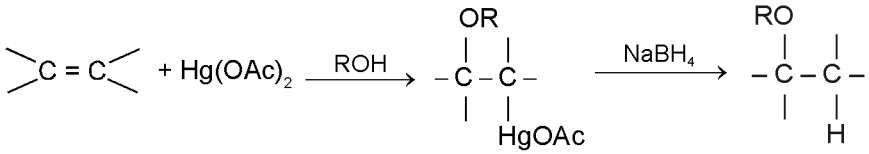
Electrophile : $\overset{+}{H}gOAC$



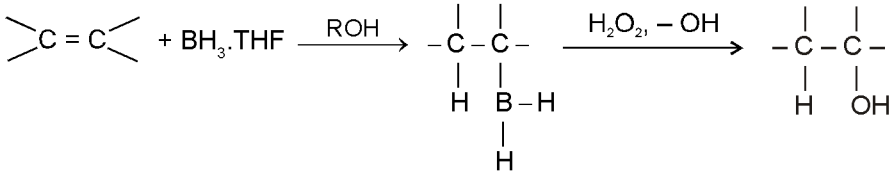


Remarks : No rearrangement ; Bridged Carbocation ; Markovnikoff's additions

Alkoxymercuration - demercuration



(c) Hydroboration-oxidation : **Reagents :** BH_3,THF/H_2O_2,^-OH



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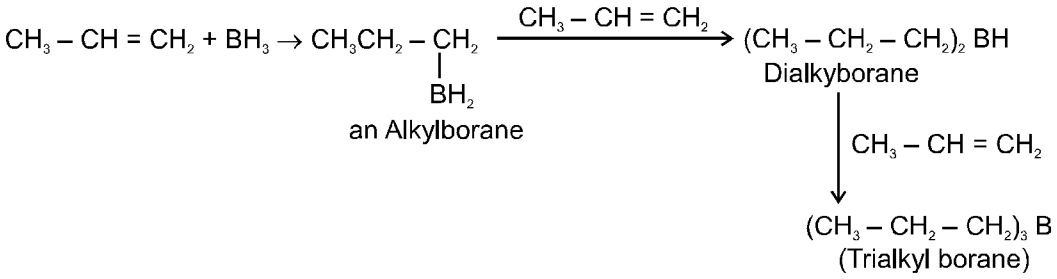
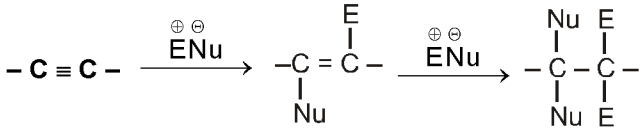
Mechanism :



Remarks : Anti Markovnikov's addition ; Syn addition.

VI. Electrophilic Addition to Alkynes :-

Many of the reactions of alkynes are similar to the corresponding reactions of alkenes. Like the pi bond of an alkene, the pi bonds of an alkyne are electron-rich, and they readily undergo addition reaction. The bond energy of the alkyne triple bond is about 226 kJ (54 kcal) more than the bond energy of an alkene double bond.
Since sigma bonds are generally stronger than pi bonds, the reaction is usually exothermic. Alkynes have two pi bonds, so upto two molecules can add across the triple bond.
We must consider the possibility of a double addition whenever a reagent adds across the triple bond of an alkyne. Some conditions may allow the reaction to stop after a single addition, while other conditions give double addition.

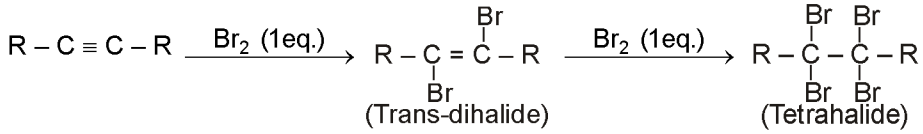
General Reaction : 

Common Reagents :

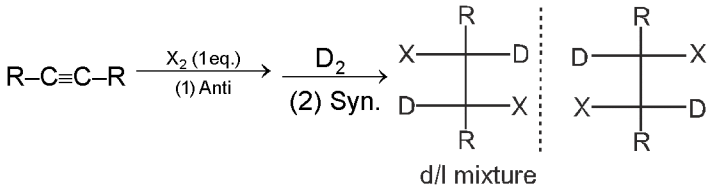
(i) + X_2 ($X^{\oplus}$) (ii) + HOX ($X^{\oplus}$)
(iii) + HX ($H^{\oplus}$) (iv) + H_2O/H^+ ($H^{\oplus}$)

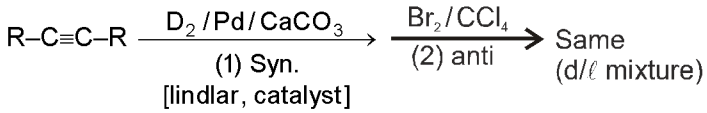
Markovnikov's rule of addition : In an electrophilic addition reaction to alkenes and alkynes the electrophile attacks in such a way that a more stable carbocation intermediate is formed on which nucleophile attacks in the next step.

(i) ADDITON OF HALOGEN (HALOGENATION) :



Remark : - Reaction is Anti in both step





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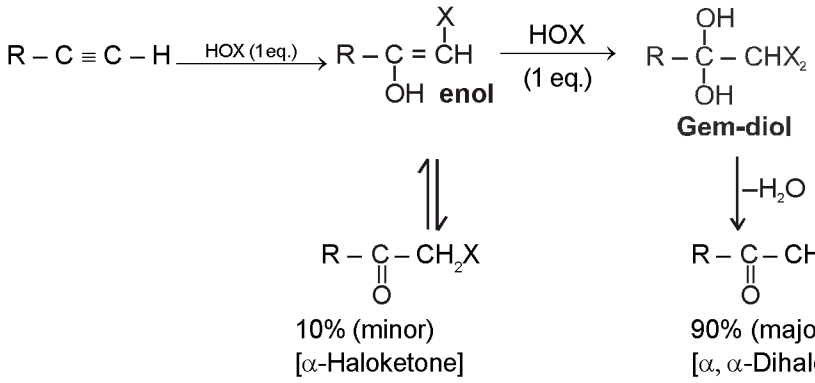
(ii) ADDITION OF HOX :

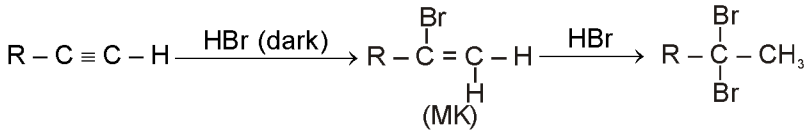
Alkynes + HOX ⟶ α, α-dihaloketone + α-haloketone
(90% major) (10% minor)



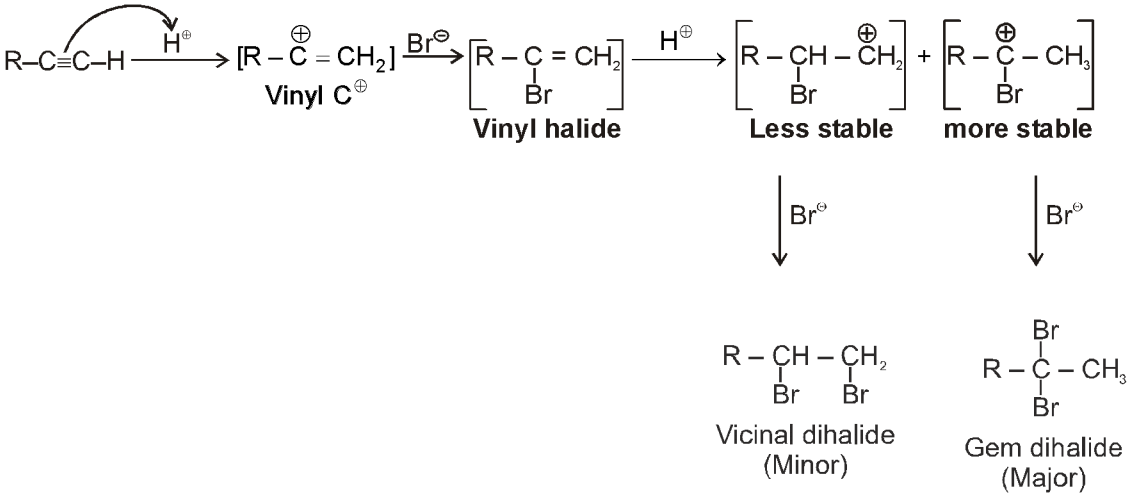
Remarks :

Two molecules of HOX can be added, the end product is α, α-Dihaloketone.
The intermediate product is an enol which gives a minor product α-haloketone.

(iii) ADDITION OF HYDROGEN HALIDES (+ HX) :



Mechanism :



Remarks : (1) Markovnikov's Addition in both steps.
(2) If two moles of HX are added the final product is Gemdihalide.
(3) **Electrophilic addition to terminal alkyne is regioselective.**

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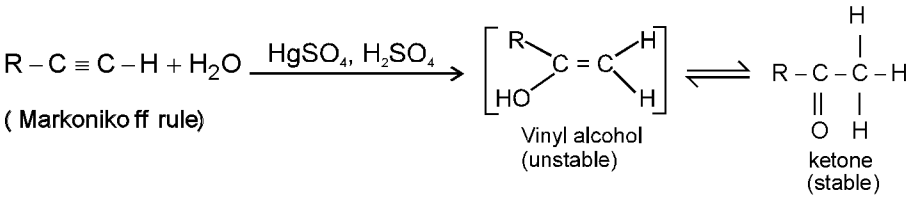
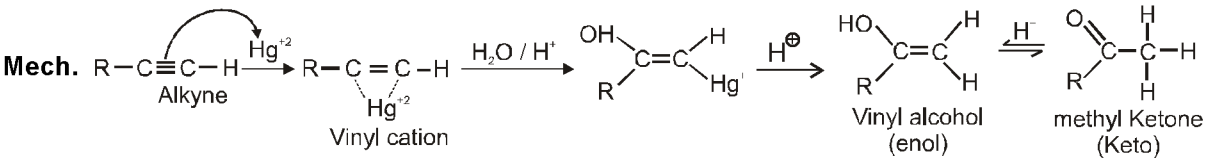
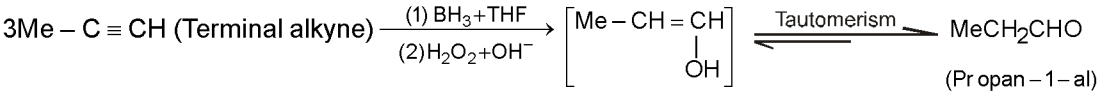
(iv) ADDITION OF H_2O (HYDRATION REACTION) :

(a) Mercuric ion catalyzed hydration :

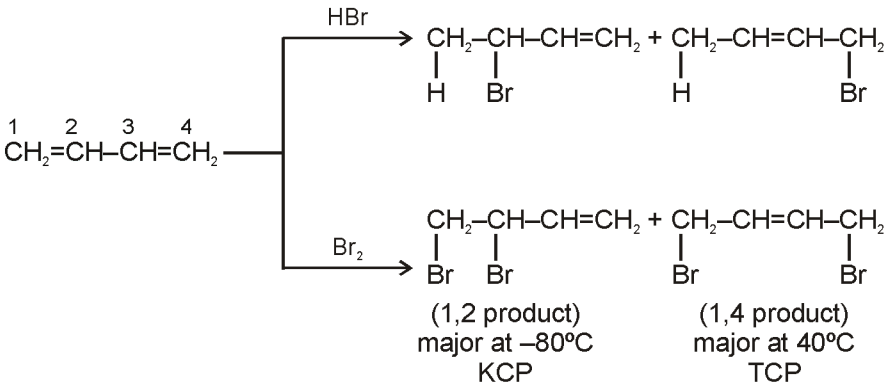




(b) Hydroboration Oxidation of Alkynes :



Alkadienes :



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N and O family (70 Marks)

1. Fill in the blanks:

(i) Nitrolim is a mixture of carbon and **calcium cyanamide ($CaCN_2$)**. Its hydrolysis products are $CaCO_3$ and NH_3. It is prepared by heating CaC_2 with N_2.

(ii) In **Ostwald's** process, nitric acid is obtained by oxidation of ammonia. First, NH_3 is oxidized to NO by O_2. Then, NO is oxidized to NO_2 by O_2. Then, water is added to NO_2 to produce HNO_3 & NO (which is reused in 2^{nd} step).

(iii) **White** phosphorus is reactive (catches fire even at room temperature) because of its highly strained tetrahedral structure. Red phosphorus has linear polymeric structure and is less reactive. Black phosphorus is thermodynamically the most stable allotropic form of phosphorus. White phosphorus has ΔH_f° at 298 K, red and black have –ve value. White phosphorus is stored in water. All forms of phosphorus are insoluble in water. White phosphorus is soluble in CS_2 but red is not.

(iv) Dehydration of HNO_3 by P_4O_{10} yields N_2O_5 and HPO_3. Dehydration of conc. H_2SO_4 by P_4O_{10} yields SO_3 and HPO_3.

(v) Phosphorus pentaoxide reacts with PCl_5 to form $POCl_3$, which on hydrolysis produces H_3PO_4 and HCl.

(vi) **Nitrogen** is obtained in lab by heating a mixture of ammonium chloride and sodium nitrite and is collected by downward displacement of water as it is water insoluble. However, ammonia is collected by downward displacement of air as it is lighter than air.

(vii) Laughing gas (N_2O) is obtained by heating ammonium **nitrate.** It is a supporter of combustion (better than air). Its mixture with O_2 is used for anesthesia by dentists.

(viii) White **Phosphorus** is obtained when calcium phosphate is heated with silica and carbon in electric furnace. By-products are $CaSiO_3$ and CO. White phosphorus when heated in atmosphere of CO_2 or coal gas at 573 K produces red phosphorus. α-black phosphorus is obtained by heating red phosphorus in a sealed tube at 803 K. β-black phosphorus is obtained by heating white phosphorus at 473 K under pressure.

(ix) Complete hydrolysis of PCl_5 gives HCl and H_3PO_4. PCl_5 on reaction with SO_2 forms $POCl_3$ and $SOCl_2$ (thionyl chloride). $SOCl_2$ on hydrolysis produces H_2SO_3 and HCl.

(x) In P_4O_{10}, the number of oxygen atoms bonded to each phosphorus atom is **four** and number of $p\pi$-$p\pi$ bonds is 0. It is a very good dehydrating agent but can not dry NH_3 gas. It combines with CaO forming $Ca_3(PO_4)_2$ (Thomas Slag) during steel formation.

(xi) The substance used in the torch of Holme's light of ship along with calcium carbide is **calcium phosphide**. PH_3 when burnt in O_2 forms H_3PO_4. PH_3 has smell of rotten fish or garlic like odour.

(xii) The water softner, calgon, is **sodium hexametaphosphate** (a polyphosphate). It forms soluble complexes with calcium ions and thus remove hardness of water.

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(xiii) Ammonia is made dry by use of **quick lime** (CaO). Conc. H_2SO_4 cannot be used for drying H_2S, HBr and HI gases because it will under go redox reaction with these forming S, Br_2 and I_2 along with SO_2 and H_2O in each case. H_2 gas cannot be dried using conc. H_2SO_4 as process is exothermic and H_2 gas may catch fire. However, HCl gas can be dried using conc. H_2SO_4.

(xiv) N_2O_3 is anhydride of HNO_2 acid. N_2O_5 is anhydride of HNO_3 acid. P_4O_6 is anhydride of phosphorous acid. P_4O_{10} is anhydride of orthophosphoric acid. NO_2 / N_2O_4 is mixed anhydride of nitrous acid and nitric acid.

2. Fill in the blanks:

(i) The catalyst used in Contact process is V_2O_5. The catalyst used in Ostwald process is Pt gauge. The catalyst used in the Haber's process is Fe + Mo (promoter). H_2S gas is usually prepared in lab from Kipp's apparatus.

(ii) SO_2 reacts with Cl_2 in presence of sunlight and moisture to form HCl and H_2SO_4. SO_2 is a good reducing agent and produces green solution with acidified $K_2Cr_2O_7$ forming Cr^{3+} and SO_4^{2-}. SO_2 gas also turns lime water milky forming $CaSO_3$ but excess passage of gas makes milkiness disappear forming Calcium bisulphite.

(iii) Conc. H_2SO_4 chars paper, wood, starch and sugar because it removes **water** from them behaving like a good dehydrating agent. With sugar, it forms sugar charcoal. It is also a moderate oxidizing agent (better than conc. H_3PO_4 but not than conc. HNO_3).

(iv) Sulphur acts as **cross linking** agent in vulcanization of rubber. Sulphur has maximum catenation tendency in family due to strong S–S bond.

(v) Sulfur dioxide bleaches colouring matter by **reduction**, so is temporary in nature. When brought in air, colouring matter again gets oxidized retaining colour. H_2O_2 and O_3 bleaches colouring matter by **oxidation**, so is permanent in nature.

(vi) Oxygen is absorbed by **alkaline pyragallol**. CO is absorbed by ammonical cuprous chloride.

(vii) Ozone is absorbed in **turpentine oil**. O_3 is a blue gas, liquefies to form blue liquid and solidifies to form violet black crystals. It strongly absorbs UV radiations.

(viii) Ozone oxidises H_2S to S, H_2O and O_2, and Hg to Hg_2O and O_2 leading to Tailing of mercury.

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N and O family

1. What happens when :

(i) Gold is treated with aqua-regia and chromium is treated with Conc. HNO_3.

(ii) Zinc is treated with cold dilute nitric acid and Mg is treated with 2% HNO_3.

(iii) Phosphine is treated with an acidified $CuSO_4$ solution.

(iv) Iodine reacts with concentrated nitric acid and copper reaction with dilute HNO_3.

(v) Disodium hydrogen phosphate is added to ammoniacal solution of magnesium sulphate and Microcosmic salt is heated.

Ans: (i) Gold get dissolved by forming chloro complex.

$$2Au + 3HNO_3 + 11HCl \longrightarrow 2HAuCl_4 + 3NOCl + 6H_2O$$

Chromium develops an oxide coating on its surface which prevents the metal from further reaction (passivation).

(ii) Ammonium nitrate is formed.

$$4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$$

$Mg + 2HNO_3 \quad Mg(NO_3)_2 + H_2$

(iii) Cupric phosphide ppt is formed, which decomposes to produce black ppt of Cu.

$$3CuSO_4 + 2PH_3 \longrightarrow Cu_3P_2 + 3H_2SO_4$$

(iv) Iodic acid is formed.

$$I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$$

$3Cu + 8HNO_3 \quad 3Cu(NO_3)_2 + 2NO + 4H_2O$

(v) Magnesium ammonium phosphate (white ppt) is formed.

$$Na_2HPO_4 + MgSO_4 + NH_4OH \longrightarrow MgNH_4PO_4 + Na_2SO_4 + H_2O$$

$Na(NH_4)HPO_4 \quad NaPO_3 + NH_3 + H_2O$

$\longrightarrow$

2. Explain the following with relevant reasons:

(i) Nitrate and nitrite ions cannot be distinguished by the action of concentrated H_2SO_4 but can be distinguished by dilute H_2SO_4.

(ii) NO_2 cannot be dried by means of NaOH solution.

(iii) Ammonia is used for refrigeration and N_2 is used to provide inert atmosphere

(iv) White phosphorus can be made to burn in an atmosphere of nitrous oxide gas.

(v) Urea is a better nitrogenous fertilizer than ammonium sulphate.

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(vi) Black spots in white colour of old paintings can be remove by rubbing a cloth dipped in H_2O_2 solution.

Ans: (i) Because both liberate brown fumes of NO_2 on heating with conc. H_2SO_4 but only NO_2^- can liberate NO_2 gas with dilute H_2SO_4.

(ii) NO_2 is an acidic gas. It is absorbed in NaOH solution producing $NaNO_2$, $NaNO_3$ and H_2O(neutralization as well as disproportionation).

(iii) Ammonia has high heat of vapourisation. N_2 is very less reactive due to strong N≡N bond.

(iv) N_2O is a supporter of combustion.

$P_4 + 10N_2O \longrightarrow P_4O_{10} + 10N_2$

(v) Urea on hydrolysis gives

NH_3 and CO_2. NH_3 is assimilated by plants and CO_2 goes to atmosphere. Ammonium sulphate on hydrolysis gives NH_3 and H_2SO_4. NH_3 is assimilated while H_2SO_4 is retained in soil making it acidic.

(vi) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$
This property is utilised in restoring the white colour in old paintings which turns black due to the formation of PbS by the action of atmospheric H_2S.

3. What happens when :

Write balanced reactions if involved

(i) Hydrogen sulphide is bubbled through an aqueous solution of sulphur dioxide.

(ii) Hydrogen sulphide is passed through acidified ferric chloride solution.

(iii) Ozone reacts with potassium ferrocyanide solution.

(iv)Sulphur dioxide is bubbled through an aqueous solution of copper sulphate in presence of potassium thiocyanate.

(v) Sulphur dioxide gas, water vapour and air are passed over heated sodium chloride. (vi) Dry I_2 is made to react with ozone and product is made to react with water.

(vii) Potassium iodide is heated with conc. H_2SO_4 .

(viii) Copper metal reacts with conc. H_2SO_4

Ans: (i) colloidal sulphur (white) is formed.

$Moist\ SO_2 + 2H_2S \longrightarrow 3S + 2H_2O$ (Comproportionation reaction)

(ii) $2FeCl_3 + H_2S \longrightarrow 2FeCl_2 + 2HCl + S$

green solution white turbidity

(iii) Ferrocyanide get oxidised into ferricyanaide.

$2K_4[Fe(CN)_6] + O_3 + H_2O \longrightarrow 2K_3[Fe(CN)_6] + 2KOH + O_2$

(iv) Cupric sulphate get reduced into cuprous thiocyanate (white ppt)

$2CuSO_4 + SO_2 + 2KSCN + 2H_2O \longrightarrow 2CuSCN + K_2SO_4 + 2H_2SO_4$

(v) Initially H_2SO_4 is formed which reacts with $NaCl$ to form Na_2SO_4. (Hargreaves process)

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$$2SO_2 + O_2 + 2H_2O \longrightarrow 2H_2SO_4$$

$$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$$

(vi) $2I_2 + 9[O_3] \longrightarrow \underset{\text{yellow solid}}{I_4O_9} + 9O_2$

$I_4O_9 + 2H_2O \longrightarrow HIO_2 + 3HIO_3$

(vii) Iodine is formed.

$$2KI + 2H_2SO_4 \longrightarrow K_2SO_4 + I_2 + SO_2 + 2H_2O$$

(viii) $Cu + 2H_2SO_4(conc.) \longrightarrow CuSO_4 + SO_2 + 2H_2O$

4. Explain the following:

(i) Sulphur dioxide is more powerful reducing agent in an alkaline medium than in acidic medium.

(ii) Dry SO_2 does not bleaches dry flowers.

(iii) NH_3 cannot be dried by passing through anhy. $CaCl_2$ or P_4O_{10}.

(iv) In the manufacturing of H_2SO_4 by the contact process, sulphur trioxide is not directly dissolved in water.

(v) Sodium metal is made to react with N_2O_5 and Hydroxyl amine is made to react with H_2O_2.

(vi) $KMnO_4$ should not be dissolved in conc. H_2SO_4.

(vii) Nitrates are less abundant in earth's crust as compared to phosphates.

(viii) Hypo is used as an antichlor.

Ans: (i) The reducing property is represented as

$$SO_2 + 2OH^- \longrightarrow SO_4^{-2} + 2H^+ + 2e^-$$

Addition of acid favours reverse reaction whereas presence of OH^- favours the forward reaction.

(ii) Only in presence of moisture, the nascent hydrogen responsible for bleaching (reduction) is produced.

(iii) $CaCl_2(anhy) + 8NH_3 \longrightarrow CaCl_2.8NH_3$

$NH_3 + P_4O_{10} + H_2O \longrightarrow (NH_4)_3PO_4$

(iv) SO_3 reacts with water to form H_2SO_4 which is not easily condensed and escapes in the atmosphere as fog(harmful due to its corrosive nature). Hence, it is absorbed in conc. H_2SO_4 to form oleum; which may be diluted to form H_2SO_4.

(v) $N_2O_5 + Na \longrightarrow NaNO_3 + NO_2$

$2NH_2OH + H_2O_2 \longrightarrow N_2 + 4H_2O$

(vi) Mn_2O_7 is formed which is unstable and explosive in nature as follows:

$$Mn_2O_7 \longrightarrow 2MnO_2 + \frac{3}{2}O_2$$

(vii) Because of following two reasons, the nitrates are less abundant in earth crust.

(i) Nitrates have greater solubility in water.

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(ii) Nitrates are prone to microbial action.

(viii) $Na_2S_2O_3 + 4Cl_2 + 5H_2O \longrightarrow 2NaHSO_4 + 8\ HCl$

It destroys the chlorine present in fibres during dyeing. That is why the $Na_2S_2O_3$ is called as antichlor.

5. Aqueous solution of how many of the following species turn blue litmus red?

(i) SF_4 (ii) PCl_5 (iii) N_2O (iv) NO_2Cl

(v) SF_6 (vi) SeF_6 (vii) $AsCl_3$ (viii) $POCl_3$

(ix) SO_2 (x) SO_2Cl_2 (xi) $SOCl_2$ (xii) $COCl_2$

(xiii) CCl_4

Ans: 10 [Except (iii),(v) and (xiii)]

iii is a neutral oxide, v is structurally very stable, so does not undergoes hydrolysis & xiii does not undergoes hydrolysis at room temperature due to non-availability of vacant d-orbitals on C to accept lone pair from water.

6. Complete the following reactions:

(i) $P_4(white) + sulphuryl\ chloride \longrightarrow$

(ii) $P_4(white) + thionyl\ chloride \longrightarrow$

(iii) $NH_3 + NaOCl \xrightarrow{\text{gelatin}}$

Ans: (i) $P_4 + 10SO_2Cl_2 \rightarrow 4PCl_5 + 10SO_2$

(ii) $P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 4SO_2 + S_2Cl_2$

(iii) $NH_3 + NaOCl \longrightarrow NaOH + NH_2Cl$

$NH_2Cl + NH_3 \longrightarrow N_2H_4 + NH_4Cl$

(Hydrazine)

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P-BLOCK ELEMENTS (HALOGEN & NOBLE GASES)

GROUP 17 ELEMENTS : THE HALOGEN FAMILY

Introduction :

The five representative elements fluorine (F), chlorine (Cl), bromine (Br), iodine (I) and astatine (At) are members of Group 17. These are collectively known as the **halogens.** The halogens are highly reactive non-metallic elements. At is radioactive.

Occurrence :

Element	Source
F	• Main Source : Fluorspar (CaF_2) or Fluorite • Another Source : Fluoroapatite [$3Ca_3(PO_4)_2.CaF_2$] (Mainly source of P) Cryolite : Na_3AlF_6
Cl	• Most abundant compound of Cl : NaCl (Sea water) Carnalite : $KCl.MgCl_2.6H_2O$.

Element Preparation

Fluorine

S.No.	Method	Process	Comments
1.	Moisson's Method (Electrolytic)	$CaF_2 + H_2SO_4$ (Conc.) $\rightarrow CaSO_4 + 2HF$	
		$KF + HF \rightarrow K[HF_2]$	
		Fluorine cell : Electrodes : Anode : Carbon Cathode : Steel Electrolyte : KHF_2 (1 part) + HF (5 part) Temperature of the both : –24°C **On electrolysis :** At cathode : $K^+ + e^- \rightarrow K$ $K + HF \rightarrow KF + H$ $2H \rightarrow H_2(g)$ At Anode : $F^- \rightarrow F + e^-$ $2F \rightarrow F_2$	**Difficulties:** • HF is corrosive & also very toxic • H_2O should not be present. Otherwise F_2 will oxidize H_2O to O_2 $2F_2 + 2H_2O \rightarrow 4HF + O_2$ • Graphite anodes must not be used, since graphite reacts with fluorine, forming a polymeric substance known as graphite fluoride.
2.	Chemical Method	$K_2[MnF_6]+2\ SbF_5 \longrightarrow 2K\ [SbF_6] + MnF_3 + F_2$ (via $MnF_4 \longrightarrow MnF_3 + \frac{1}{2}F_2$)	• The stronger Lewis acid SbF_5 displaces the weaker one, MnF_4 from its salt.

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Chlorine

S.No.	Method Process		Comments
1.	Laboratory Preparation	$H_2SO_4 + NaCl \rightarrow HCl + NaHSO_4$ $4HCl + MnO_2 \rightarrow MnCl_2 + 2H_2O + Cl_2\uparrow$	• HCl is dried by passing through conc. H_2SO_4.
2.	Commercially:		
(i)	By electrolysis of aq. NaCl solution in the manufacture of NaOH	$2NaCl + 2H_2O \xrightarrow{\text{Electrolysis}} 2NaOH + Cl_2 + 2H_2$	
(ii)	Deacon's process	$4HCl + O_2 \underset{440°C}{\overset{CuCl_2\ \text{Catalyst}}{\rightleftharpoons}} 2Cl_2 + 2H_2O$	
3.	Miscellaneous Reactions	**(a)** $Ca\begin{matrix}OCl\\Cl\end{matrix} + 2HCl \rightarrow CaCl_2 + Cl_2 + H_2O$	
		(b) $2KMnO_4+16HCl\rightarrow 2KCl+2MnCl_2+5Cl_2+8H_2O$	
		(c) $PbO_2 + 4\ HCl \rightarrow PbCl_2 + Cl_2 + 2\ H_2O$	

Bromine:

S.No.	Method	Process
1.	Extracted from sea water	• Cl_2 gas passed through solution to oxidize Br^- to Br_2 $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$ ↓ Removed by steam of air & passed to solution of Na_2CO_3. **Reactions :** $3Br_2 + 3Na_2CO_3 \rightarrow 5NaBr + NaBrO_3 + 3CO_2$ $5NaBr + NaBrO_3 + 2H_2SO_4 \rightarrow 5HBr + HBrO_3 + 3Na_2SO_4$ $5HBr + HBrO_3 \rightarrow 3Br_2 + 3H_2O$ ↓ Pure Bromine
2.	Common method	$2\ NaBr + 3H_2SO_4 + MnO_2 \xrightarrow{\Delta} Br_2 + MnSO_4 + 2NaHSO_4 + 2H_2O$ (conc.)

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Iodine:

S.No.	Method	Process	Comments
1.	Source : Chile saltpetre	• Pure $NaNO_3$ is obtained by dissolving salt petre in H_2O & crystallizing $NaNO_3$ • Iodate residues thus accumulated & concentrate in mother liquor • Concentrate → divided into 2 part 1st part reduced with $NaHSO_3$ $2IO_3^- + 6HSO_3^- \rightarrow 2I^- + 6SO_4^{2-} + OH^-$ 2nd part is mixed $3I^- + IO_3^- + 6H^+ \rightarrow 3I_2 + 3H_2O$	• Chile salt petre is mainly $NaNO_3$ • Contain traces of $NaIO_3$ (Sodium iodate) & $NaIO_4$ (Sodium periodate) • Purified by sublimation
2.	Source: Natural blue	$2I^- + Cl_2 \rightarrow I_2 + 2Cl^-$ ↓ blown out by air	
3.	From sea-weeds:	$2NaI + MnO_2 + 3H_2SO_4 \longrightarrow 2NaHSO_4 + MnSO_4 + I_2 + 2H_2O$	
4.	Other Method	$2CuSO_4 + 4KI \rightarrow 2K_2SO_4 + 2CuI\downarrow + I_2$ (white ppt.)	• This I_2 gets dissolved into KI forming KI_3, since I_3^- ions are yellow, therefore solution develops yellow colour.

Periodic Trends, Properties of Elements and Chemical Bonding

Atomic & Physical Properties :

S.No.	Property	F	Cl	Br	I
1.	Electron gain enthalpy /(kJ mol^{-1}) *(Cl > F > Br > I)*	• *Maximum negative electron gain enthalpy in the corresponding period.*			
2.	Distance X -X/pm *($F_2 < Cl_2 < Br_2 < I_2$)*				
3.	Enthalpy of dissociation (X_2)/kJ mol^{-1} *(Cl – Cl > Br – Br > F – F > I –I)*	• *Smaller enthalpy of dissociation of F_2 is due to relatively larger electron-electron repulsion among the lone pairs in F_2 molecule.*			
4.	Electronegativity *(F > Cl > Br > I)*	4	3.0	2.8	2.5
5.	$\Delta_{Hyd}H$ (X^-) / kJ mol^{-1} *(F > Cl > Br > I)*	• *Smaller the ion, higher is the hydration energy.*			
6.	$E^\ominus$ / V (SRP) ; X_2 (g)+ $2e^- \rightarrow 2X^-$ *(aq)* *($F_2 > Cl_2 > Br_2 > I_2$)* *(Order of oxidising power)*	• *More the value of the SRP, more powerful is the oxidising agent.*			
7.	Colour (at room temp.)	Pale green yellow gas.	Greenish-yellow gas	Reddish brown liquid	Dark violet solid
8.	Oxidization state	–1	–1, +1, +3, +5, +7	–1, +1, +3, +5, +7	–1, +1, +3, +5, +7

CHEMICAL PROPERTIES :

Oxidation states and trends in chemical reactivity

(1) The fluorine atom has no d orbitals in its valence shell and therefore cannot expand its octet. Being the most electronegative, it exhibits only –1 oxidation state.

(2) All the halogens are highly reactive. They react with metals and non-metals to form halides. The reactivity of the halogens decreases down the group.

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(3) The reactions of iodine with water is non- spontaneous I^- can be oxidised by oxygen in acidic medium; just the reverse of the reaction observed with fluorine.

$2F_2(g) + 2H_2O(\ell) \longrightarrow 4H^+(aq) + 4F^-(aq) + O_2(g)$

$X_2(g) + H_2O(\ell) \longrightarrow HX(aq) + HOX(aq)$; (where X = Cl or Br)

$4I^-(aq) + 4H^+(aq) + O_2(g) \longrightarrow 2I_2(s) + 2H_2O(\ell)$

Anomalous behaviour of fluorine

(1) The anomalous behaviour of fluorine is due to its small size, highest electronegativity, low F–F bond dissociation enthalpy, and non availability of d orbitals in valence shell.

(2) Most of the reactions of fluorine are exothermic (due to the small and strong bond formed by it with other elements).

(3) It forms only one oxoacid while other halogens form a number of oxoacids.

(4) Hydrogen fluoride is liquid (b.p. 293 K) due to strong hydrogen bonding. Other hydrogen halides are gases.

Reactivity towards oxygen :

Oxidation State	F	Cl	Br	I
–1	OF_2, O_2F_2	–	–	–
+1		Cl_2O	Br_2O	
+4		ClO_2	BrO_2	I_2O_4
+5		–	–	I_2O_5
+6		Cl_2O_6, ClO_3	–	
+7		Cl_2O_7		I_2O_7

(1) Fluorine forms two oxides OF_2 and O_2F_2. However, only OF_2 is the thermally stable at 298 K.

(2) ClO_2 is used as a bleaching agent for paper pulp and textiles and in water treatment.

(3) I_2O_5 is very good oxidising agent and is used in the estimation of carbon monoxide.

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GROUP 18 ELEMENTS : (THE Noble Gases / ZERO GROUP FAMILY)

Group 18 consists of six elements: helium, neon, argon, krypton , xenon and radon . All these are gases and chemically unreactive. They form very few compounds . Because of this they are termed noble gases.

Occurrence

All the noble gases except radon occur in the atmosphere. Their atmospheric abundance in dry air is ~ 1% by volume of which argon is the major constituent. Helium and sometimes neon are found in minerals of radioactive origin e.g., pitchblende, monazite, cleveite. The main commercial source of helium is natural gas. Xenon and radon are the rarest elements of the group. Radon is obtained as a decay product of ^{226}Ra.

$$^{226}_{88}Ra \longrightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$$

- Most abundant element in air is Ar. Order of abundance in the air is Ar > Ne > Kr > He > Xe.

Electronic Configuration :

All noble gases have general electronic configuration ns^2np^6 except helium which has $1s^2$. Many of the properties of noble gases including their inactive nature are ascribed to their closed shell structures.

Ionisation Enthalpy :

Due to stable electronic configuration these gases exhibit very high ionisation enthalpy. However, it decreases down the group with increases in atomic size.

Atomic Radii :

Atomic radii increase down the group with increase in atomic number.

Electron Gain Enthalpy :

Since noble gases have stable electronic configurations, they have no tendency to accept the electron and therefore, have larger positive values of electron gain enthalpy.

Chemical Properties :

In general, noble gases are least reactive. Their inertness to chemical reactivity is attributed to the following reasons:

(i) The noble gases except helium ($1s^2$) have completely filled $ns^2\ np^6$ electronic configuration in their valence shell.

(ii) They have high ionisation enthalpy and more positive electron gain enthalpy.

- If Helium is compressed and liquified it forms He(I) liquid at 4.2 K. This liquid is a normal liquid like any other liquid. But if it is further cooled then He(II) is obtained at 2.2 K, which is known as super fluid, because it is a liquid with properties of gases. It climbs through the walls of the container & comes out. It has very high thermal conductivity & very low viscosity.

Clatherate compounds :

Inert gas molecules get trapped in the cages formed by the crystal structure of water.

During the formation of ice Xe atoms will be trapped in the cavities (or cages) formed by the water molecules in the crystal structure of ice. Compounds thus obtained are called clatherate compounds.

There are no chemical bonds. They do not possess an exact chemical formula but approx it is 6 water molecules : 1 inert gas molecule. The cavity size is just smaller than the atom of the noble gas. Such compounds are also formed by the other organic liquids like dihydroxybenzene (for example quinol). The smaller noble gases He and Ne do not form clathrate compounds because the gas atoms are small enough to escape from the cavities. Clathrate provides a convenient means of storing radioactive isotopes of Kr and Xe produced in nuclear reactors.

COMPOUNDS OF XENON :

(i) XENON FLUORIDES :

PREPARATION :

Xenon reacts directly with fluorine when the gases are heated at 300-600°C in a sealed nickel vessel and the products formed depend on the Xe / F_2 ratio.

$$Xe + F_2 \begin{cases} \xrightarrow{2:1\ \text{mixture},\ 400^\circ C,\ 1\ atm} XeF_2 \\ \xrightarrow{1:5\ \text{mixture},\ 600^\circ C,\ 7\ atm} XeF_4 \\ \xrightarrow{1:20\ \text{mixture},\ 600^\circ C,\ 7\ atm} XeF_6 \end{cases}$$

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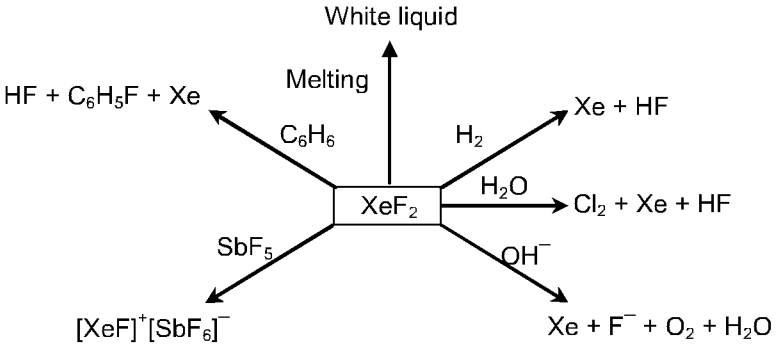
PROPERTIES :

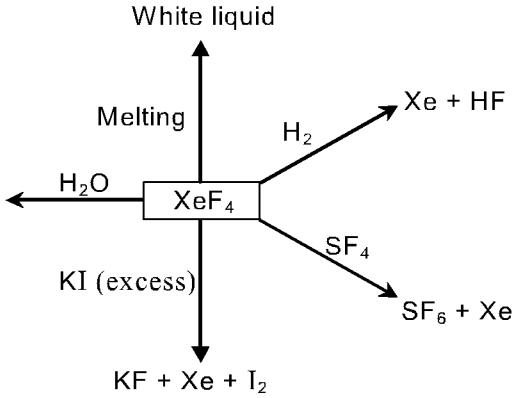
These are colourless (white) solids. They can be sublimed at room temperature and can be stored indefinitely in nickel or monel (an alloy of nickel) containers.

Chemical properties:

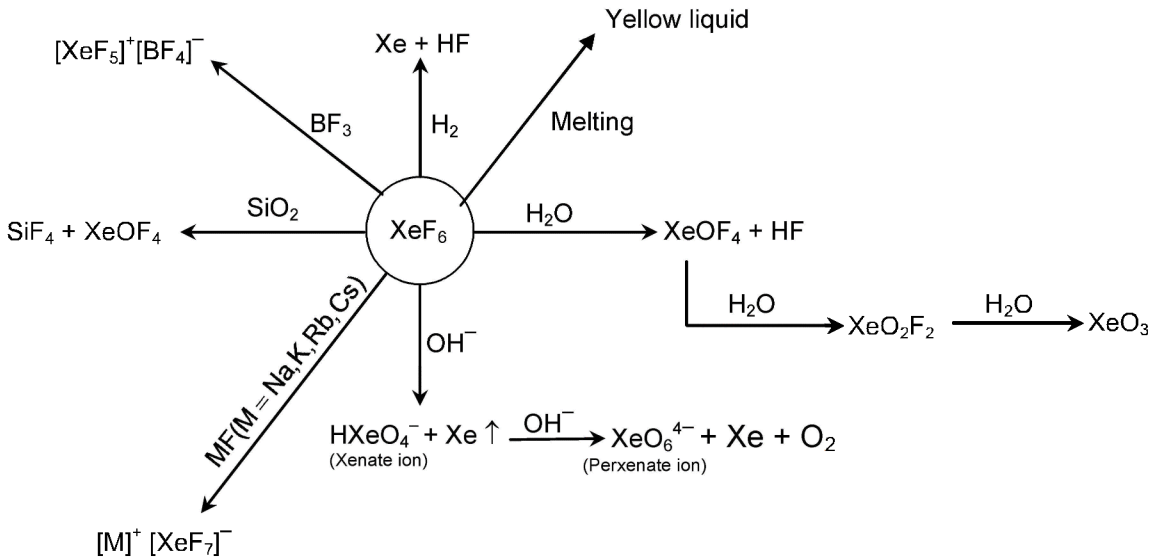
1. **XeF_2 :**



2. **XeF_4 :** $Xe + XeO_3 + HF + O_2$



3. **XeF_6 :**



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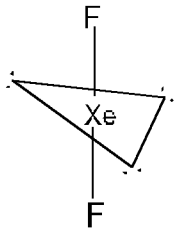
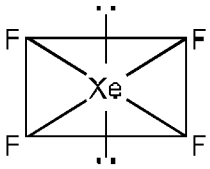
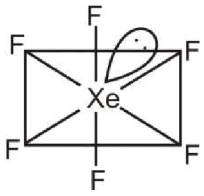
Compounds	Structures	Hybridisation	Geometry	Shape
XeF_2		sp^3d	trigonal bipyramidal	linear
XeF_4		sp^3d^2	octahedral	square planar
XeF_6			pentagonal bipyramidal	distorted octahedron

(ii) XENON–OXYGEN COMPOUNDS :

Hydrolysis of XeF_4 and XeF_6 with water gives XeO_3.

$6\ XeF_4 + 12\ H_2O \longrightarrow 4\ Xe + 2\ XeO_3 + 24\ HF + 3\ O_2$

$XeF_6 + 3\ H_2O \longrightarrow XeO_3 + 6\ HF$

Partial hydrolysis of XeF_6 gives oxyfluorides, $XeOF_4$ and XeO_2F_2.

$XeF_6 + H_2O \longrightarrow XeOF_4 + 2\ HF$

$XeF_6 + 2\ H_2O \longrightarrow XeO_2F_2 + 4\ HF$

XeO_3 is a colourless explosive solid and has a pyramidal molecular structure. $XeOF_4$ is a colourless volatile liquid and has a square pyramidal molecular structure.

USES :

1. (a) Helium is a non–inflammable and light gas. Hence, it is used in filling balloons for meteorological observations.
   (b) It is also used in gas–cooled nuclear reactors.
   (c) Liquid helium (b.p.4.2 K) finds use as cryogenic agent for carrying out various experiments at low temperatures.
   (d) It is used to produce and sustain powerful superconducting magnets which form an essential part of modern NMR spectrometers and Magnetic Resonance Imaging (MRI) systems for clinical diagnosis.
   (e) It is used as a diluent for oxygen in modern diving apparatus because of its very low solubility in blood.
2. (a) Neon is used in discharge tubes and fluorescent bulbs for advertisement display purposes.
   (b) Neon bulbs are used in botanical gardens and in green houses.
3. (a) Argon is used mainly to provide an inert atmosphere in high temperature metallurgical processes (arc welding of metals or alloys) and for filling electric bulbs.
   (b) It is also used in the laboratory for handling substances that are air–sensitive.
4. Xenon and Krypton are used in light bulbs designed for special purposes.

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SOLUTION & COLLIGATIVE PROPERTIES

OSMOTIC PRESSURE :

(i) $\pi = \rho gh$ Where, ρ = density of soln., h = equilibrium height.

(ii) **Vont – Hoff Formula** (For calculation of O.P.)

$$\pi = CST$$

$$\pi = CRT = \frac{n}{V} RT \text{ (just like ideal gas equation)}$$

$\therefore$ C = total conc. of all types of particles.

$$= C_1 + C_2 + C_3 + ...$$

$$= \frac{(n_1 + n_2 + n_3 + ...)}{V}$$

Note : If V_1 mL of C_1 conc. + V_2 mL of C_2 conc. are mixed.

$$\pi = \left(\frac{C_1V_1 + C_2V_2}{V_1 + V_2}\right) RT \quad ; \qquad \pi = \left(\frac{\pi_1V_1 + \pi_2V_2}{RT}\right)$$

Type of solutions :

(a) Isotonic solution – Two solutions having same O.P.

$\pi_1 = \pi_2$ (at same temp.)

(b) Hyper tonic– If $\pi_1 > \pi_2$. $\Rightarrow$ Ist solution is hypertonic solution w.r.t. 2nd solution.

(c) Hypotonic – IInd solution is hypotonic w.r.t. Ist solution.

Abnormal Colligative Properties : (In case of association or dissociation)

VANT HOFF CORRECTION FACTOR (i) :

$$i = \frac{\text{exp/observed/actual/abnormal value of colligative property}}{\text{Theoritical value of colligative property}}$$

$$= \frac{\text{exp./observed no. of particles / conc.}}{\text{Theoritical no. of particles}} = \frac{\text{observed molality}}{\text{Theoritical molality}}$$

$$= \frac{\text{theoretical molar mass (formula mass)}}{\text{experimental/observed molar mass (apparent molar mass)}}$$

- $i > 1 \Rightarrow$ dissociation.
  $i < 1 \Rightarrow$ association.

- $$i = \frac{\pi_{exp.}}{\pi_{theor}}$$

$\therefore$ $\pi = iCRT$

$\pi = (i_1C_1 + i_2C_2 + i_3C_3.....) RT$

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Relation between i & α (degree of dissociation) :

$i = 1 + (n - 1)\alpha$ Where, $n = x + y$.

Relation b/w degree of association β & i.

$$i = 1 + \left(\frac{1}{n} - 1\right)\beta$$

RELATIVE LOWERING OF VAPOUR PRESSURE (RLVP) :

Vapour pressure : $P_{Soln.} < P$

Lowering in VP $= P - P_S = \Delta P$

Relative lowering in vapour pressure $RLVP = \frac{\Delta P}{P}$

Raoult's law : (For non – volatile solutes)

Experimentally relative lowering in V.P = mole fraction of the non volatile solute in solutions.

$$RLVP = \frac{P - P_s}{P} = X_{Solute} = \frac{n}{n+N}$$

$$\boxed{\frac{P - P_s}{P_s} = \frac{n}{N}}$$

$$\boxed{\frac{P - P_s}{P_s} = (\text{molality}) \times \frac{M}{1000}}$$ (M = molar mass of solvent)

If solute gets associated or dissociated

$$\frac{P - P_s}{P_s} = \frac{i.n}{N}$$

$$\frac{P - P_s}{P_s} = i \times (\text{molality}) \times \frac{M}{1000}$$

- According to Raoult's law

(i) $p_1 = p_1^0 X_1$. where X_1 is the mole fraction of the solvent (liquid).

(ii) An alternate form $\rightarrow$ $\frac{p_1^0 - p_1}{p_1^0} = X_2$.

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Elevation in Boiling Point :

$$\Delta T_b = i \times K_b m$$

$$K_b = \frac{RT_b^2}{1000 \times L_{vap}} \quad \text{or} \quad K_b = \frac{RT_b^2 M}{1000 \times \Delta H_{vap}}$$

$$L_{vap} = \left(\frac{\Delta H_{vap}}{M}\right)$$

Depression in Freezing Point :

$$\therefore \Delta T_f = i \times K_f . m.$$

$$K_f = \text{molal depression constant} = \frac{RT_f^2}{1000 \times L_{fusion}} = \frac{RT_f^2 M}{1000 \times \Delta H_{fusion}} .$$

Raoult's Law for Binary (Ideal) mixture of Volatile liquids :

$\mathbf{P_A = X_A P_A^\circ}$ $\quad\therefore\quad$ $\mathbf{P_B = X_B P_B^\circ}$

if $P_A^\circ > P_B^\circ$ $\quad\therefore\quad$ A is more volatile than B

$\quad\therefore\quad$ B.P. of A < B.P. of B

$\quad\therefore\quad$ According to Dalton's law

$$\mathbf{P_T = P_A + P_B = X_A P_A^0 + X_B P_B^0}$$

x_A' = mole fraction of A in vapour above the liquid / solution.

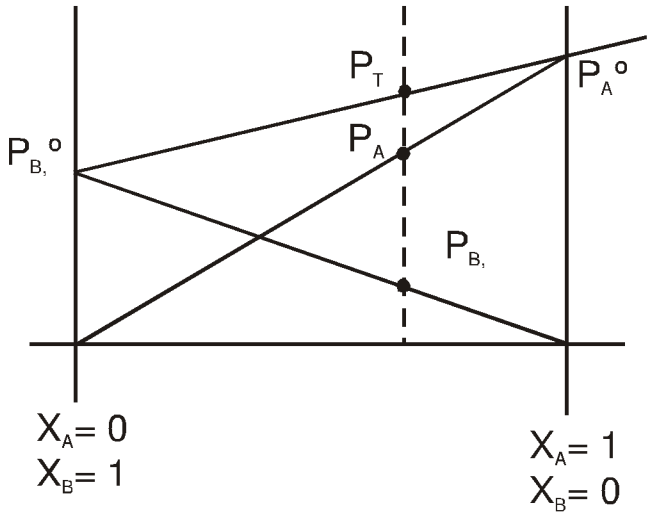
x_B' = mole fraction of B

$$\mathbf{P_A = X_A P_A^\circ = X_A' P_T}$$

$$\mathbf{P_B = X_B' P_T = X_B P_B^o}$$

$$\frac{1}{P_T} = \frac{x_A'}{P_A^o} + \frac{x_B'}{P_B^o} .$$

Graphical Representation :



A more volatile than B $(P_A^\circ > P_B^\circ)$

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Ideal solutions (mixtures) :

Mixtures which follow Raoul'ts law at all temperature.

A ------ A $\Rightarrow$ A -------- B,
B ----- B

$\Delta H_{mix} = 0$: $\Delta V_{mix} = 0$:
ΔS_{mix} = + ve as for process to proceed : ΔG_{mix} = – ve

eg. (1) Benzene + Toluene.
(2) Hexane + heptane.
(3) $C_2H_5Br + C_2H_5I$.

Non-Ideal solutions : Which do not obey Raoult's law.

(a) Positive deviation : –

(i) $P_{T,exp} > (X_A P^o_A + X_B P_B{}^o)$

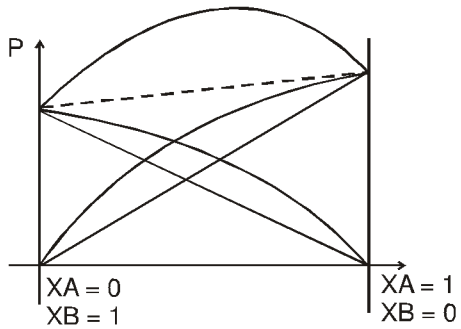
(ii) $\begin{matrix} A - - - - A \\ B - - - - B \end{matrix} >$ A ---- B

↓

Force of attraction

(iii) ΔH_{mix} = +ve energy absorbed
(iv) ΔV_{mix} = +ve (1L + 1L > 2L)
(v) ΔS_{mix} = +ve
(vi) ΔG_{mix} = –ve

eg. $H_2O + CH_3OH$.
$H_2O + C_2H_5OH$
C_2H_5OH + hexane
C_2H_5OH + cyclohexane.
$CHCl_3 + CCl_4 \rightarrow$ dipole dipole interaction becomes weak.



$P^0A > P^0B$

(b) Negative deviation

(i) P_T exp < $X_A P_A{}^o + X_B P^o{}_B$

(ii) $\begin{matrix} A - - - - A \\ B - - - - B \end{matrix} <$ A ------ B.

strength of force of altraction.

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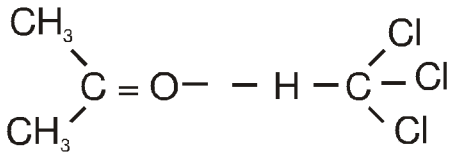
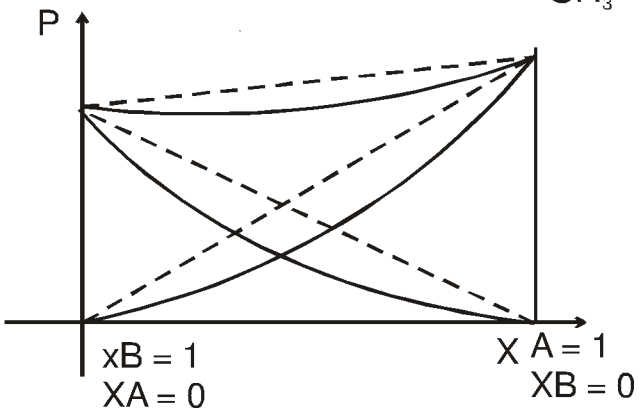
(iii) $\Delta H_{mix} = -ve$ (iv) $\Delta V_{mix} = -ve$ (1L + 1L < 2L)

(v) $\Delta S_{mix} = +ve$ (vi) $\Delta G_{mix} = -ve$

eg. $H_2O + HCOOH$
$H_2O + CH_3COOH$
$H_2O + HNO_3$

$CHCl_3 + CH_3OCH_3 \Rightarrow$





$P^0A > P^0B$

Immiscible Liquids :

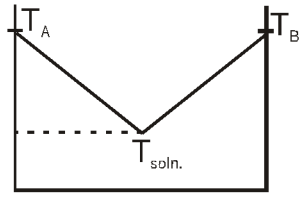
(i) $P_{total} = P_A + P_B$

(ii) $P_A = P_A^0 X_A = P_A^0$ [Since, $X_A = 1$].

(iii) $P_B = P_B^0 X_B = P_B^0$ [Since, $X_B = 1$].

(iv) $P_{total} = P_A^0 + P_B^0$ (v) $\frac{P_A^0}{P_B^0} = \frac{n_A}{n_B}$ (vi) $\frac{P_A^0}{P_B^0} = \frac{W_A M_B}{M_A W_B}$

$$P_A^0 = \frac{n_A RT}{V} \quad ; \quad P_B^0 = \frac{n_B RT}{V}$$



B.P. of solution is less than the individual B.P.'s of both the liquids.

Henry Law :

This law deals with dissolution of gas in liquid i.e. mass of any gas dissolved in any solvent per unit volume is proportional to pressure of gas in equilibrium with liquid.

$m \alpha p$

$m = kp$

$$m \rightarrow \frac{\text{weight of gas}}{\text{Volume of liquid}}$$

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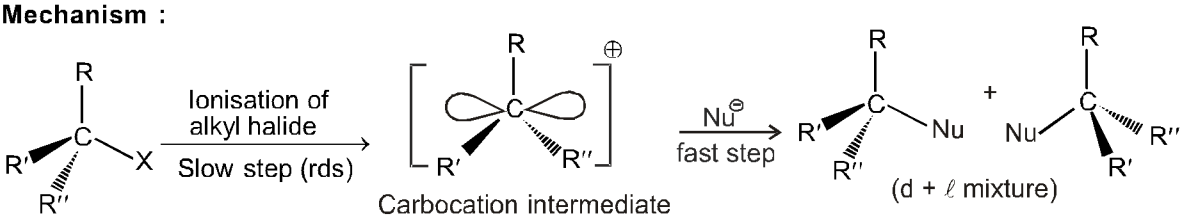
Haloalkanes and Haloarenes

S_N1 Reaction of Alkyl halide

Mechanism :



Characteristics of S_N1 reactions :

1. It is unimolecular, two step process.
2. Carbocation intermediate is formed so rearrangement is possible in S_N1 reaction.
3. It is first order reaction
4. **Kinetics of the reaction** Rate $\propto$ [Alkyl halide]

Rate of S_N1 reaction is independent of concentration and reactivity of nucleophile.

5. Energetics of the S_N1

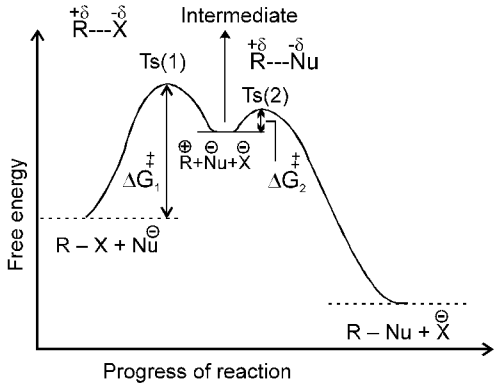


Figure : Free energy diagram for the S_N1 reaction.

6. Factors affecting the rates of S_N1

(i) The structure of the substrate : The rds of the S_N1 reaction is ionization step, a carbocation is formed in this step. This ionisation is strongly endothermic process, rate of S_N1 reaction depends strongly on carbocation stability because carbocation is the intermediate of S_N1 reaction which determines the energy of activation of the reaction.

Reactivity of $S_N1 \propto$ stability of carbocation.

S_N1 reactivity : $3° > 2° > 1° > CH_3 - X$

(ii) Concentration and reactivity of the nucleophile $\rightarrow$ The rate of S_N1 reaction is unaffected by the concentration and nature of the nucleophile

Weak, neutral, mostly solvents (protic) itself functions as nucleophiles in S_N1 reaction. So S_N1 reaction is termed as solvolysis reaction.

water $\rightarrow$ hydrolysis ; $C_2H_5OH \rightarrow$ ethanolysis

$CH_3COOH \rightarrow$ acetolysis ; $NH_3 \rightarrow$ ammonolysis

(iii) Effect of the solvent : (Ionising ability of the solvent)

The use of a polar protic solvent will greatly increase the rate of ionisation of an alkyl halide in any S_N1 reaction because it solvate cations and anions so effectively and stabilises the transition state leading to the intermediate carbocation and halide ion, thus the energy of activation is lower.

$R - X \rightleftharpoons \overset{\oplus}{R} + \overset{\ominus}{X}$ (Solvolysis)

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Solvated ions

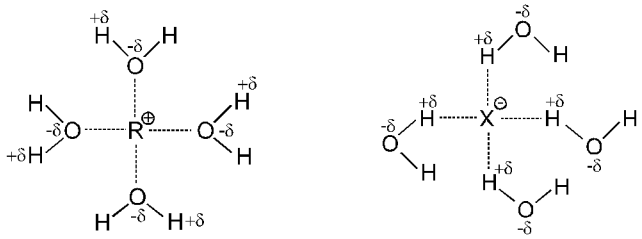


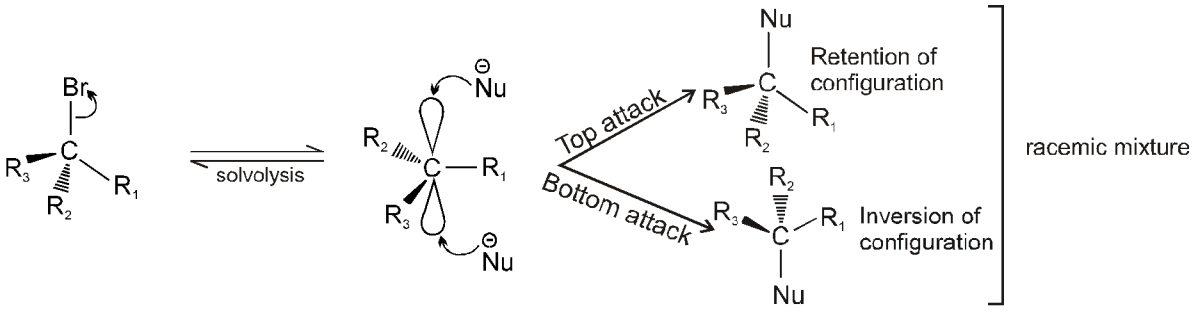
Table - : Dielectric constants (ϵ) and ionisation rates of t-Butylchloride in few common solvents

Solvent	ϵ	Relative rate
H_2O	80	8000
CH_3OH	33	1000
C_2H_5OH	24	200
$(CH_3)_2CO$	21	1
CH_3CO_2H	6	–

(iv) The nature of the leaving group → In the S_N1 reaction the leaving group begins to acquire a negative charge as the transition state is reached` stabilisation of this developing negative charge at the leaving group stabilises the transition state and this lowers the free energy of activation and there by increases the rate of reaction. Leaving ability of halogen is $F^- < Cl^- < Br^- < I^-$

7. Stereochemistry of S_N1 reactions → In the S_N1 mechanism, the carbocation intermediate is sp^2 hybridized and planar, A nucleophile can attack on the carbocation from either face, if reactant is chiral than after attack of nucleophile from both faces gives both enantiomers as the product, which is called racemisation.

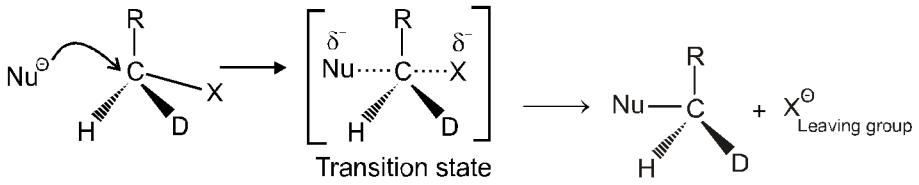
Mechanism of racemisation (S_N1) →



Reagents for alkyl halide are : H_2O , RCOOH , ROH & RSH

S_N2 Reaction of Alkyl halide :

Mechanism :



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Characteristic of S_N2

1. It is bimolecular, one step concerted process
2. It is second order reaction because in the rds both species are involved
3. **Kinetics of the reaction :** rate $\propto$ [alkyl halide] [nucleophile]
   rate = k[alkyl halide] [nucleophile]

If the concentration of alkyl halide in the reaction mixture is doubled, the rate of the nucleophilic substitution reaction is double. If the concentration of nucleophile is doubled the rate of reaction is also double. If the concentration of both are doubled then the rate of the reaction quadriples.

4. **Energetics of the reaction :**

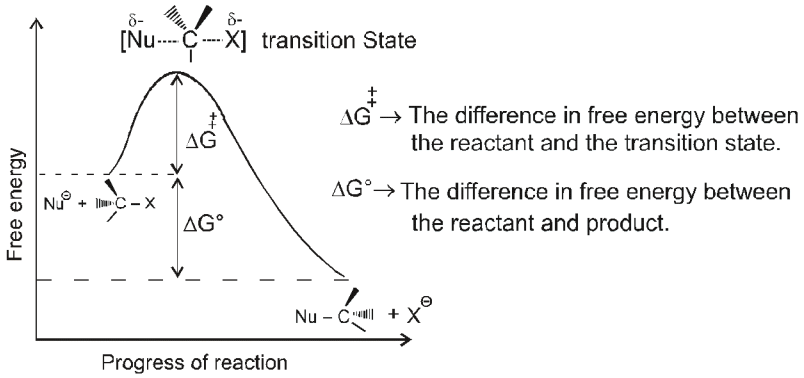
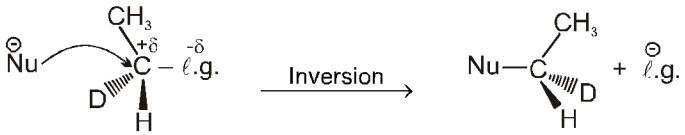


Figure : A free energy diagrams for S_N2 reaction

5. No intermediates are formed in the S_N2 reaction, the reaction proceeds through the formation of an unstable arrangment of atoms or group called transition state.
6. **The stereochemistry of S_N2 reactions** → As we seen earlier, in an S_N2 mechanism the nucleophile attacks from the back side, that is from the side directly opposite to the leaving group. This mode of attack causes an inversion of configuration at the carbon atom that is the target of nucleophilic attack. This inversion is also known as **Walden inversion**.



7. **Factor's affecting the rate of S_N2 reaction :** Number of factors affect the relative rate of S_N2 reaction, the most important factors are

(i) Effect of the structure of the substrate :

S_N2 reactivity CH_3 > 1° > 2° >> 3° (unreactive)

The important reason behind this order of reactivity is a steric effect. Very large and bulky groups can often hinder the formation of the required transition state and crowding raises the energy of the transition state and slows down reaction.

Table : Relative rate of reactions of alkyl halide in S_N2 reaction.

Substituent	Compound	Relative rate
Methyl	CH_3X	30
1°	CH_3CH_2X	1
2°	$(CH_3)_2CHX$	0.02
Neopentyl	$(CH_3)_3CCH_2X$	0.00001
3°	$(CH_3)_3CX$	~ 0

(ii) Concentration and reactivity of the nucleophile

– As nucleophilicity of nucleophile increases rate of S_N2 increases.

– Anionic nucleophiles mostly give S_N2 reaction

– A stronger nucleophile attacks upon α-carbon with faster rate than the rate of departing of leaving group.

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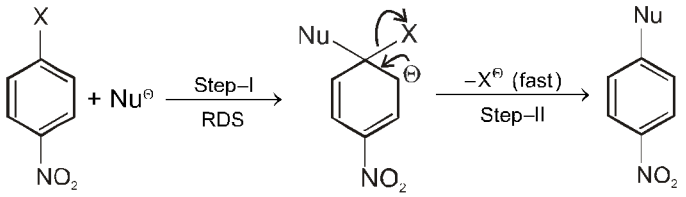
(iii) The effect of the solvent: Polar aprotic solvent have crowded positive centre, so they do not solvate the anion appreciably therefore the rate of S_N2 reactions increased when they are carried out in polar aprotic solvent.

(iv) The nature of the leaving group : Weaker bases are good leaving groups. A good leaving group always stabilise the transition state and lowers its free energy of activation and there by increases the rate of the reaction. Order of leaving ability of halide ion $F^- < Cl^- < Br^- < I^-$

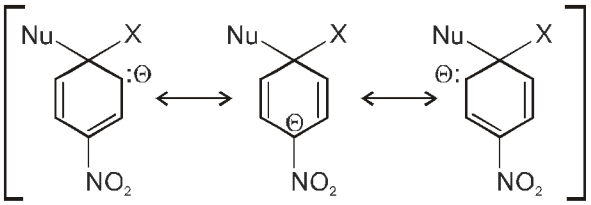
Reagents for alkyl halide are : OH^- , SH^- , I^- ,CN^-, NH_3 ..... **(strong anionic nucleophile)**

Bimolecular nucleophilic substitution in aromatic compound (S_N2Ar) Reaction :

This is the characteristic reaction of arylhalides with ortho or para electron withdrawing substituent. The reaction mechanism can be visualised as :

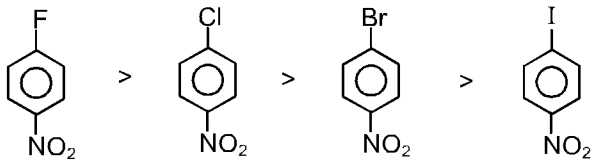


Intermediate ion is stabilized by resonance.



EWG ↑ Rate of reaction ↑

Reactivity order towards (S_N2Ar)



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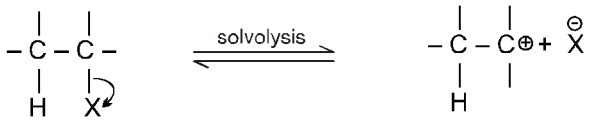
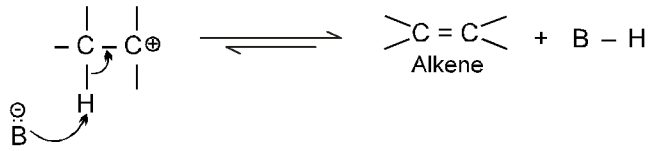
E1 Reaction of Alkyl halide :

Mechanism :

Step 1 :



Step 2 :



Characteristics of E1 reaction :

(i) It is unimolecular, two step process.

(ii) It is a first order reaction.

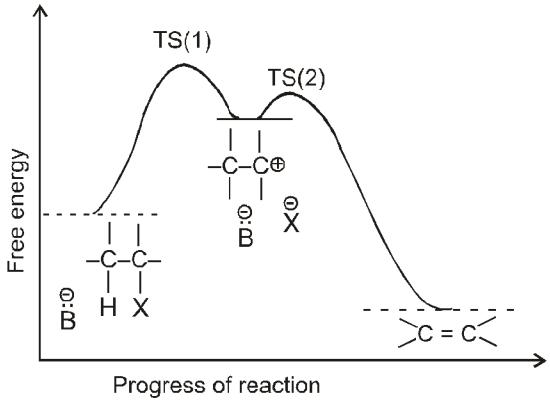
(iii) Reaction intermediate is carbocation, so rearrangment is possible

(iv) In the second step, a base abstracts a proton from the carbon atom adjacent to the carbocation, and forms alkene.

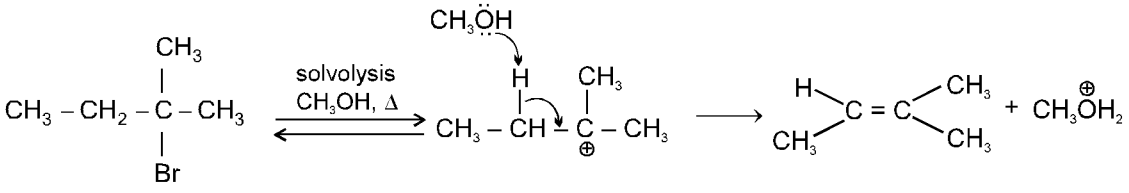
Rate of E1 = 3° Alcohol > 2° Alcohol > 1° Alcohol.
Rate of reaction ∝ stability of carbocation

(v) Kinetics : Rate ∝ [Alkylhalide]
Rate = k [Alkylhalide]

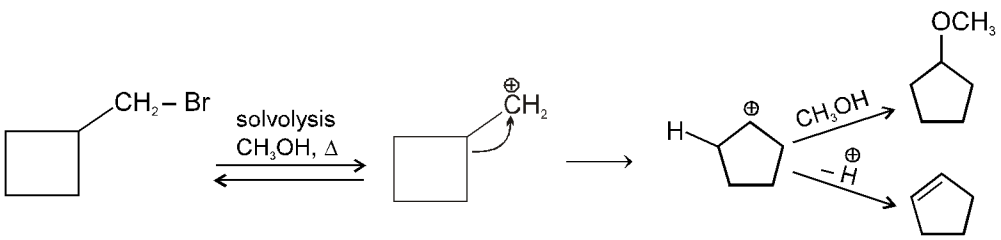
(vi) Energetics : The free energy diagram for the E1 reaction is similar to that for the S_N1 reaction.



e.g.



e.g.



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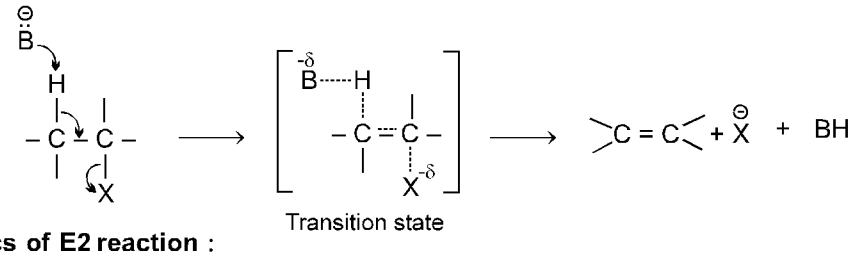
E2 Reaction of Alkyl halide :

Dehydrohalogenation is the elimination of a hydrogen and a halogen from an alkyl halide to form an alkene. Dehydrohalogenation can take place by E1 and E2 mechanism.

Reagent

(i) Hot alcoholic solution of KOH or EtO^- / EtOH (ii) $NaNH_2$ (iii) t-BuO^-K^+ in t-BuOH

Mechanism :



Characteristics of E2 reaction :

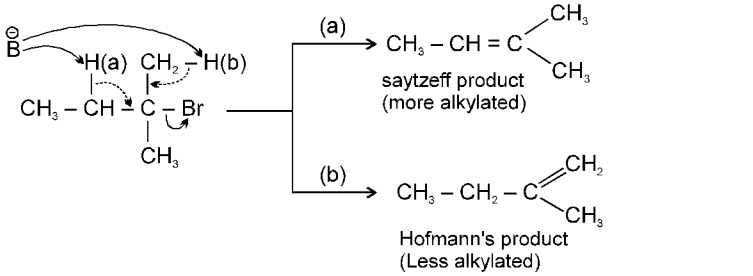
1. This is a single step, bimolecular reaction
2. It is a second order reaction
3. Kinetics → Rate $\propto$ [R – X] [Base]

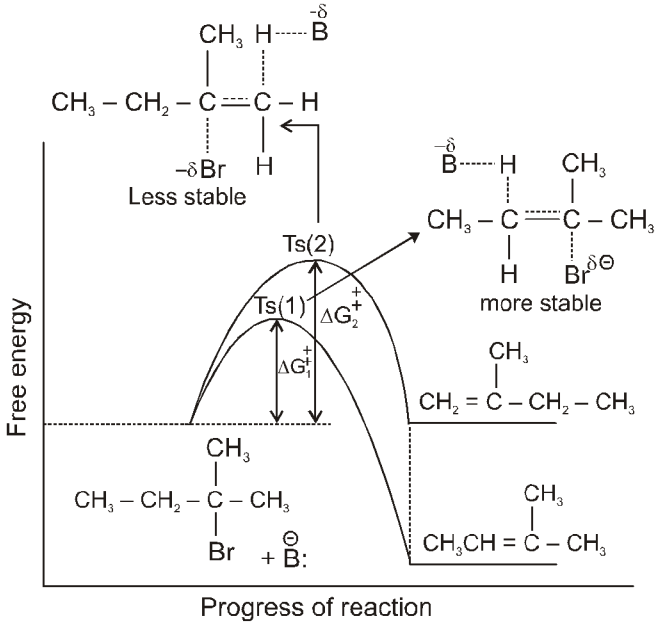
   Rate = k [R – X] $[\overset{\ominus}{B}]$
4. Rearrangment is not possible
5. For the lower energy of activation, transition state must be stable
6. E2 follows a concerted mechanism
7. The orientation of proton & leaving group should be antiperiplanar.
8. Here β – H is eliminated by base hence called **β elimination**
9. Positional orientation of elimination → In most E1 and E2 eliminations where there are two or more possible elimination products, the product with the most highly substituted double bond will predominate. This rule is called the saytzeff or zaitsev rule.

Reactivity towards E2 → R – I > R – Br > R – Cl > R – F

Rate of E2 reaction = 3° Alkylhalide > 2° Alkylhalide > 1° Alkylhalide

For example : Dehydrohalogenation of 2-bromo-2-methylbutane can yield two products.





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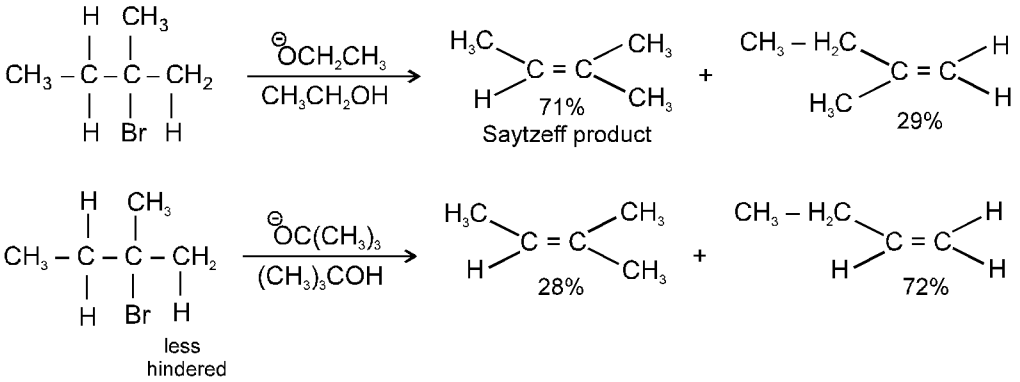
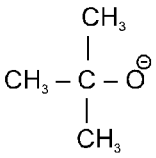
Formation of the Hoffmann product

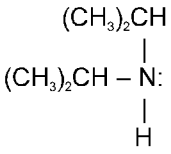
Bulky bases can also accomplish dehydrohalogenations that do not follow the saytzeff rule. Due to steric hindrance, a bulky base abstracts a less hindered proton, often the one that leads to formation of the least highly substituted product, called the Hoffmann product.



Bulky base :



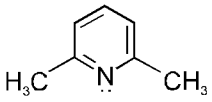
t-Butoxide



Di-isopropylamine

$(CH_3CH_2)_3N$:

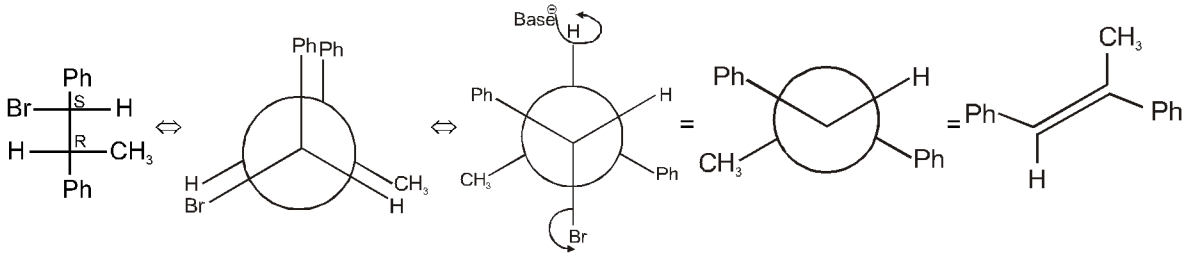
Triethylamine



2, 6-dimethyl pyridine

Stereospecific E2 reactions

The E2 is stereospecific because it normally goes through an anti periplanar transition state. The products are alkene.



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Alcohols, Phenol and Ether

S_N1 Reaction of Alcohols

(i) Reaction with hydrogen halides

A common method is to treat the alcohol with a hydrohalic acid, usually HI or HBr. These acids are used to convert alcohols into the corresponding alkyl halides.

(i) In acidic solution, an alcohol is in equilibrium with its protonated form. Protonation converts the hydroxy group from a poor leaving group $(OH)^{\ominus}$ into a good leaving group (H_2O). If the alcohol is protonated all the usual substitution and elimination reactions are feasible, depending on the structure (1°, 2°, 3°) of the alcohol.

(ii) Halides are anions of strong acids, so they are weak bases. Solutions of HBr and HI contain nucleophilic $Br^{\ominus}$ and $I^{\ominus}$ ions.

(iii) Concentrated hydrobromic acid rapidly converts t-Butyl alcohol to t-Butyl bromide. The strong acid protonates the hydroxyl group, converting it into a good leaving group. The hindered tertiary carbon atom cannot undergo S_N2 displacement, but it can ionise to a tertiary carbocation. Attack by bromide ion gives the alkyl bromide. The mechanism is similar to S_N1 mechanism.

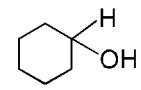
(iv) 1-Butanol reacts with sodium bromide in concentrated sulfuric acid to give 1-Bromobutane by an S_N2 displacement.

$$\underset{\text{1-butanol}}{CH_3(CH_2)_2-CH_2OH} \xrightarrow{NaBr,\ H_2SO_4} \underset{\text{1-bromobutane (90\%)}}{CH_3(CH_2)_2-CH_2Br}$$

Protonation converts the hydroxy group to a good leaving group, but ionization to a primary carbocation is unfavourable. The protonated unbranched primary alcohol is well suited for the S_N2 displacement.

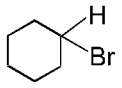
(v) Secondary alcohols also react with HBr to form alkyl bromides usually by the S_N1 mechanism.

e.g.



Cyclohexanol

$\xrightarrow{HBr}$



Bromocyclohexane (80%)

(vi) HCl (Hydrochloric acid) reacts with alcohols in much the same way that as the hydrobromic acid.

(vii) Chloride ion is a weaker nucleophlile than bromide ion because it is smaller and less polarizable. Lewis acid, such as $ZnCl_2$, is sometimes necessary to promote the reaction of HCl with primary and secondary alcohols.

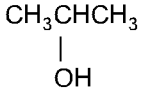
Mechanism :

$$R-OH \underset{}{\overset{H^{\oplus}}{\rightleftharpoons}} R-\overset{\oplus}{O}H_2 \xrightarrow[-H_2O]{RDS} \overset{\oplus}{R} \xrightarrow{X^{\ominus},\ Fast} R-X$$

Reactivity of HX : HI > HBr > HCl

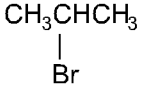
Reactivity of ROH : 3° > 2° > 1°

e.g.



Isopropyl alcohol

$\xrightarrow[\text{or } NaBr,\ H_2SO_4]{\text{conc. HBr}}$
reflux



Isopropylbromide

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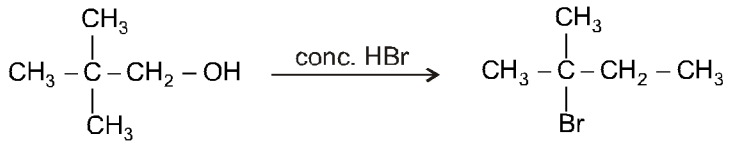
e.g. 

Lucas Reagent

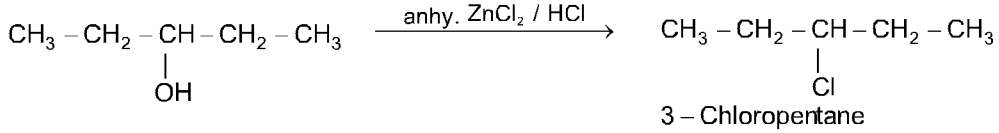
(i) A mixture of concentrated hydrochloric acid and zinc chloride is called the Lucas reagent.

(ii) Whether an alcohol is primary, secondary or tertiary is identified by the Lucas test, which is based upon the difference in reactivity of the three classes of alcohol towards hydrogen halides.

(iii) Alcohol (of not more than six carbons in their molecule) are soluble in the Lucas reagent. The corresponding alkyl chlorides are insoluble.

(iv) Formation of a chloride from an alcohol is indicated by the cloudiness that appears when the chloride separates from the solution hence, the time required for cloudiness to appear is a measure of the reactivity of the alcohol.

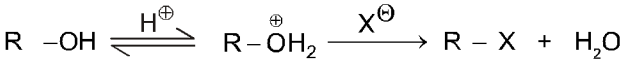
(v) A tertiary alcohol reacts immediately with the Lucas reagent, a secondary alcohol reacts within five minutes and a primary alcohol does not react appreciably at room temperature.

e.g. 
3 – Chloropentane

S_N2 Reaction of Alcohol

(i) Reaction with HX : The protonated β unbranched primary alcohol is well suited for the S_N2 reaction.

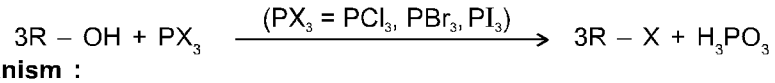
Mechanism :



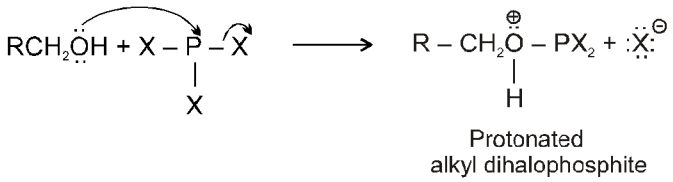
(ii) Reaction with phosphorus trihalides

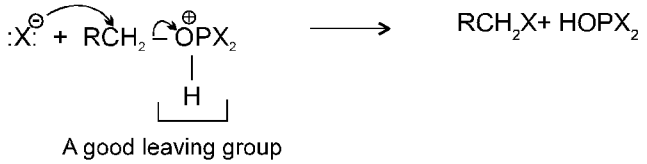
Several phosphorus halides are useful for converting alcohols to alkyl halides. PBr_3, PCl_3, & PCl_5 work well and are commercially available.

Phosphorus halides produce good yields of most primary and secondary alkyl halides, but none works well with tertiary alcohols. The two phosphorus halides used most often are PBr_3 and the P_4/I_2 combination.



Mechanism :

Step : 1 
Protonated alkyl dihalophosphite

Step : 2 
A good leaving group

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Remarks

The mechanism for the reaction involves attack of the alcohol group on the phosphorus atom, displacing a halide ion and forming a protonated alkyl dihalophosphite

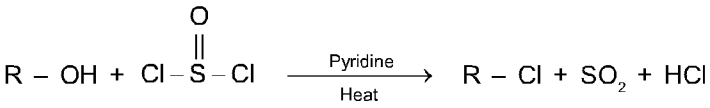
In second step a halide ion acts as nucleophile to displace $HOPX_2$, a good leaving group due to the electronegative atoms bonded to the phosphorus.

(iii) Reaction with PCl_5

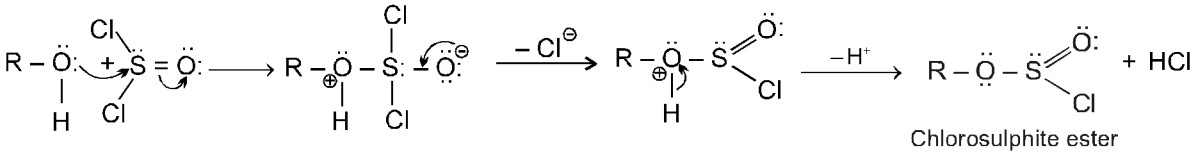
$$R{-}OH + PCl_5 \xrightarrow{S_N2} R{-}Cl + HCl + POCl_3$$

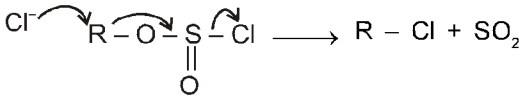
(iv) Reaction with thionyl chloride in presence of pyridine

Thionyl chloride ($SOCl_2$) is often the best reagent for converting an alcohol to an alkyl chloride. The by products (gaseous SO_2 and HCl) leave the reaction mixture and ensure that there can be no reverse reaction.



Mechanism :





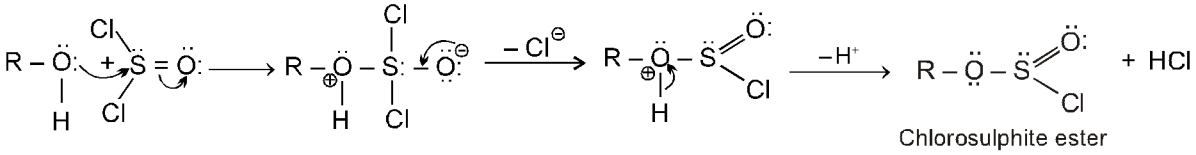
In the first step, the nonbonding electrons of the hydroxy oxygen atom attack the electrophilic sulphur atom of thionyl chloride. A chloride ion is expelled a proton and gives test of chloro sulphite ester. Second step is an S_N2 mechanism.

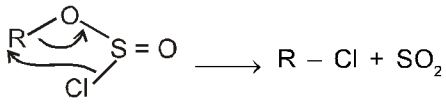
S_Ni Reaction :

In S_Ni mechanism an internal nucleophile attacks from the same side of leaving group, means retension of configuration. It is an S_Ni mechanism, where i means internal

$$ROH + SOCl_2 \longrightarrow RCl + SO_2 + HCl$$

Mechanism :





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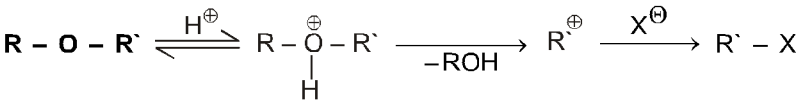
Nucleophilic Substitution Reaction of Ethers :

S_N1 Reaction of Ethers

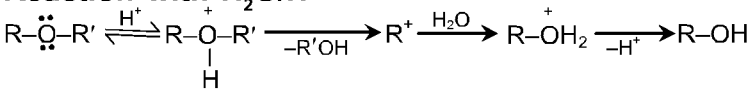
(i) Reaction with HX

Ethers are unreactive towards most bases, but they can react under acidic conditions. A protonated ether can undergo substitution or elimination with the expulsion of an alcohol. Ethers react with conc. HBr and HI because these reagents are sufficiently acidic to protonate the ether, while bromide iodide are good nucleophiles for the substitution.
If R or R' is 3° then mechanism will be S_N1 otherwise S_N2 .

Mechanism :



(ii) Reaction with H_2O/H^+

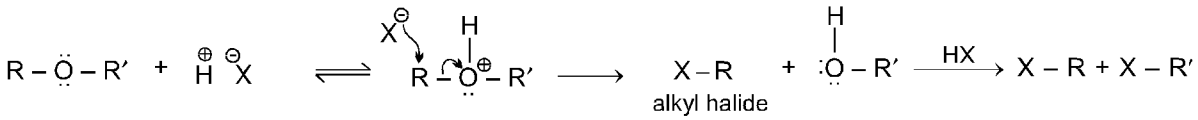


S_N2 Reaction of Ethers

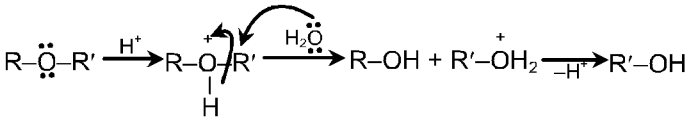
(i) Reaction with HX

A protonated ether can undergo substitution reaction. Ether react with conc. HBr and HI because these reagents are sufficiently acidic to protonate the ether. If R or R' is 3° then mechanism will be S_N1 otherwise S_N2.

Mechanism :



(ii) Reaction with H_2O/H^+



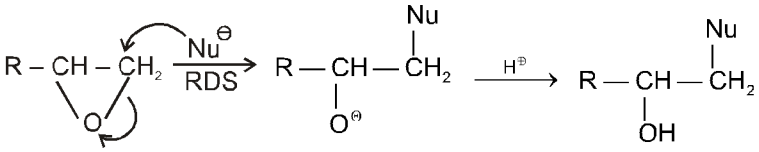
(Steric crowding R > R′)

Nucleophilic Substitution Reaction of Epoxide :

Epoxides are much more reactive than ether because of angle strain in three membered ring therefore epoxide readily undergo nucleophilic substitution reaction.

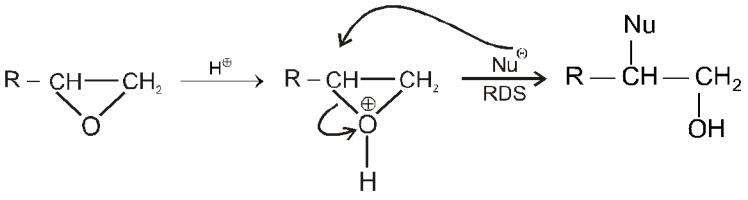
In basic medium mechanism is S_N2. Nucleophile atacks on less hindered carbon.

Mechanism :



In acidic medium mechanism is S_N1 type. Nucleophilic attacks on more substituted carbon.

Mechanism :



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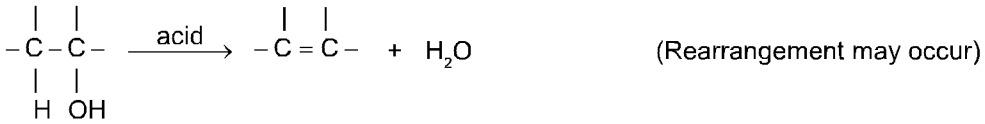
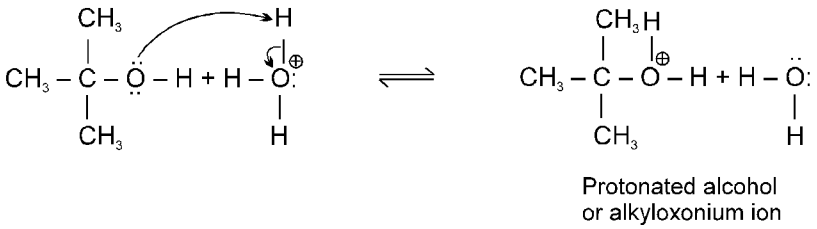
E1 Reaction of Alcohol

Dehydration requires an acidic catalyst to protonate the hydroxyl group of the alcohol and convert it to a good leaving group. Loss of water, followed by loss of proton, gives the alkene. An equilibrium is established between reactants and products. For E1 mechanism reagents are (i) H_3PO_4/Δ
(ii) H_2SO_4 / 160°

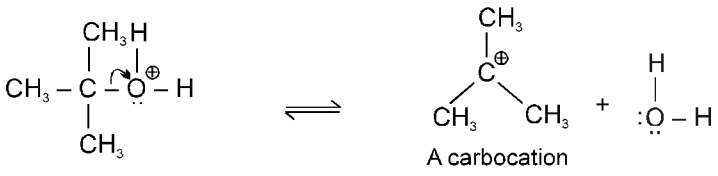


Mechanism :

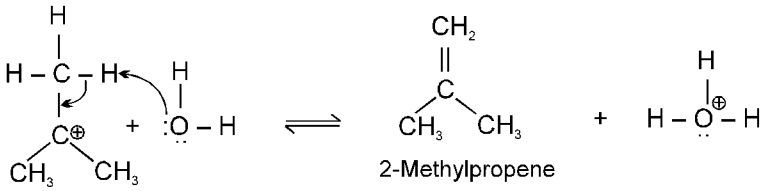
Step 1 :



Step 2 :



Step 3



Remarks

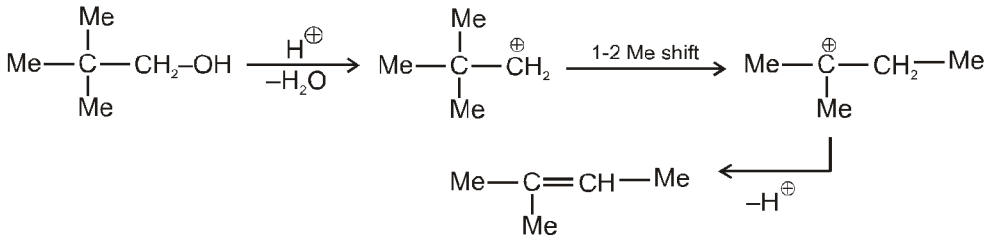
In first step, an acid-base reaction a proton is rapidly transferred from the acid to one of the unshared electon pairs of the alcohol.

In second step the carbon oxygen bond breaks. The leaving group is molecule of water :

Finally,in third step the carbocation transfers a proton to a molecule of water. The result is the formation of a hydronium ion and an alkene.

Reactivity of ROH : 3° > 2° > 1°

e.g.



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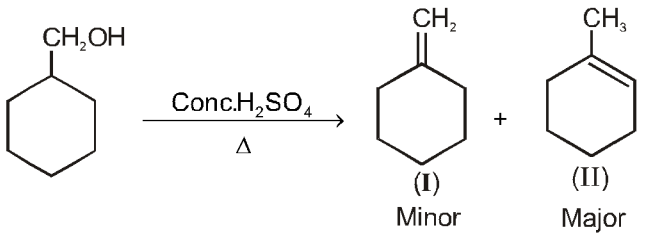
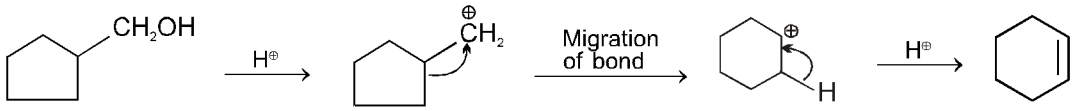
e.g.



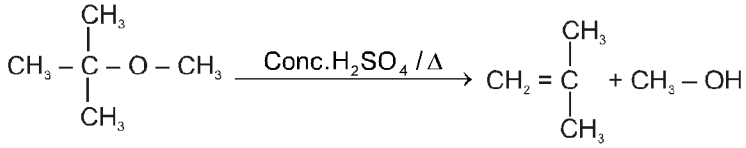
e.g.



E1 Reaction of Ether :

Elimination is not a favourable reaction for ether, but however few reactions have been observed.
E1 Elimination takes place via formation of stable carbocation.
Ether undergoes dehydration reaction in the presence of conc. H_2SO_4 / Δ .

e.g.



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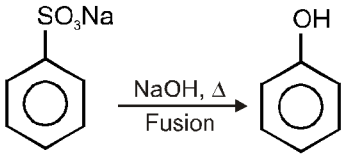
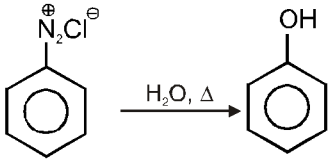
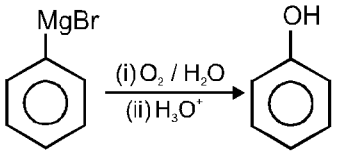
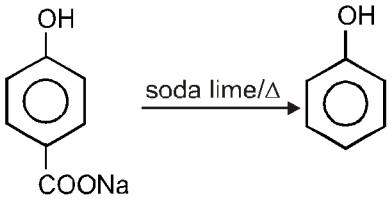
1. Phenol (ArOH) :

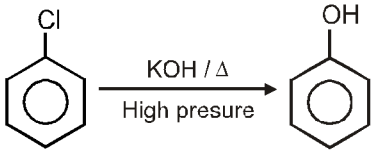
(A) Preparation of ArOH

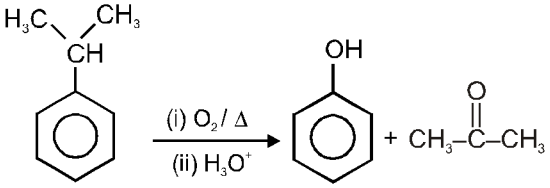
(1) 

(2) 

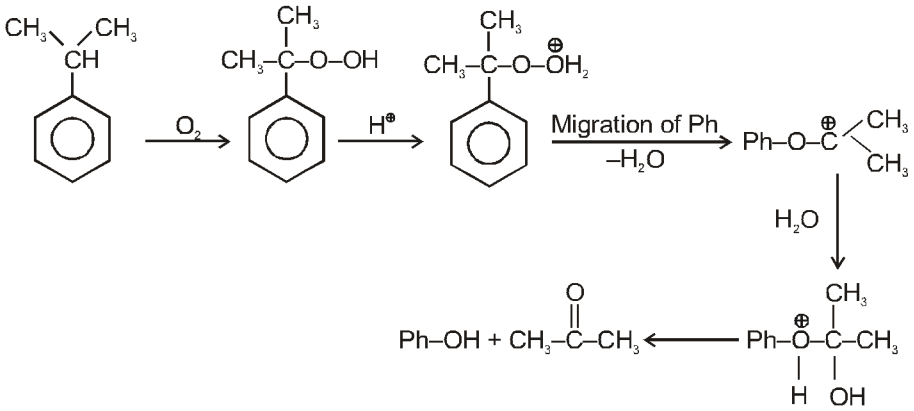
(3) 

(4) 

(5) 

(6) 

Mechanism (Cumene hydroperoxide method) :



(B) Properties of phenol :

Phenol is a colourless crystalline solid, m.p. 43°C, b.p. 182°C, which turns pink on exposure to air and light. Phenol is used as an antiseptic and disinfectant and in the preparation of dyes, drugs, bakelite, etc.

(C) Chemical Reactions of Phenols :

Phenol generally gives electrophilic substitution with electrophilic reagents.

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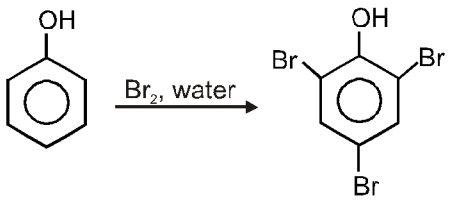
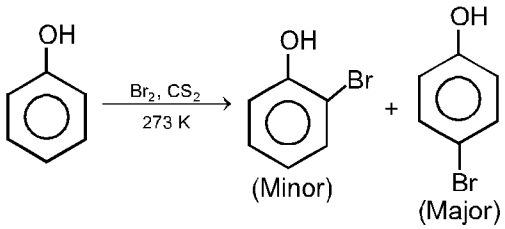
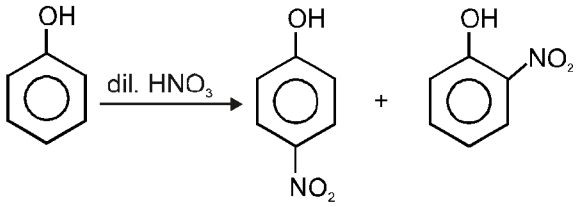
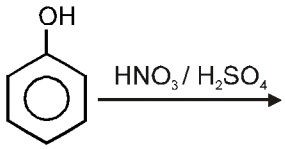
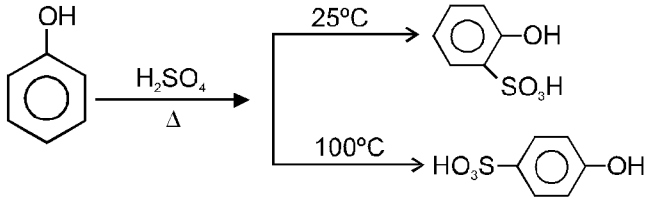
(1) 

(2) 

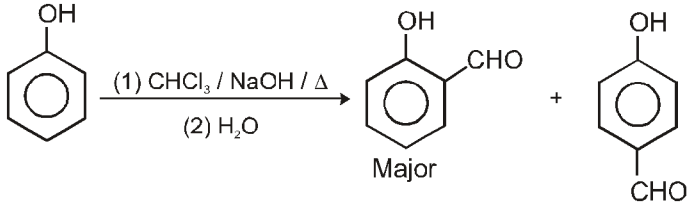
(3) 

(4) 

(Major is oxidative cleavage of ring) + Picric acid (small amount)

(5) 

(6) **Reimer-Tiemann formylation reaction :**



Mechanism :

Step (I) : $CHCl_3 + NaOH \longrightarrow \overset{\ominus}{C}Cl_3 \longrightarrow : CCl_2$

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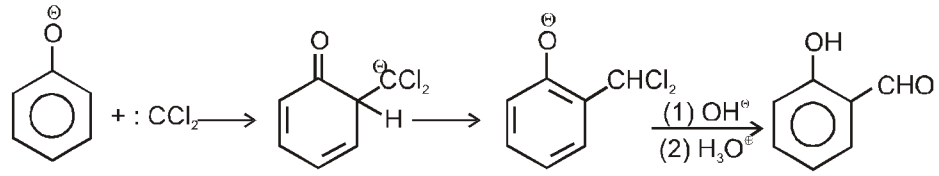
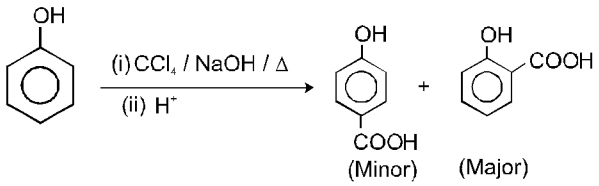
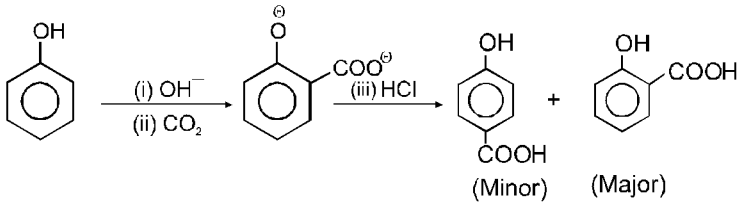
Step (II) :



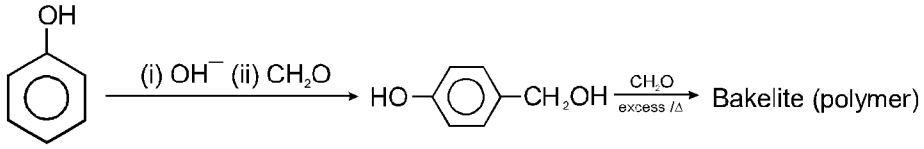
(7) Reimer-Tiemann carboxylation reaction :



(8) Kolbe reaction :



(9)



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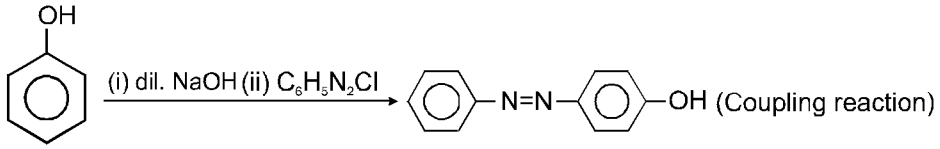

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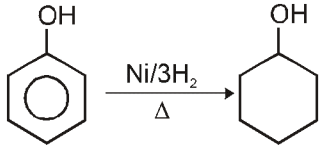

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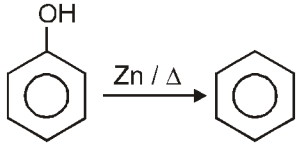

(12) Williamson ether synthesis

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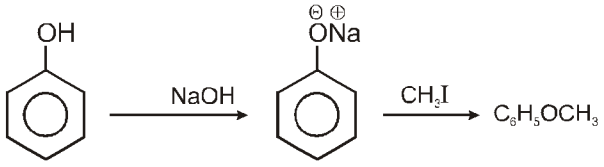
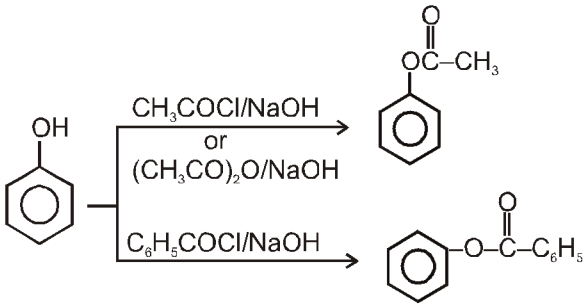
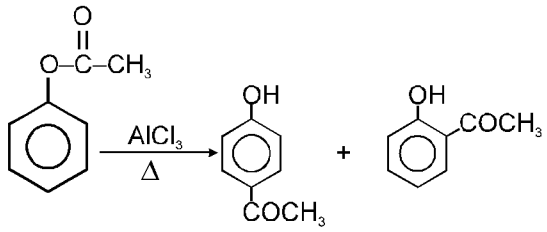


(13) Schotten-Bauman reaction. Protection of –OH and –NH_2 group.

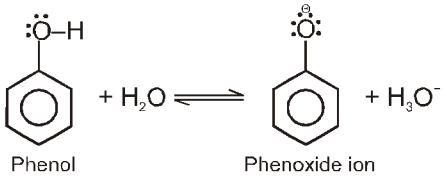


(14) Fries Rearrangement :



(D) Test of Phenol :

(i) Phenols turn blue litmus red. Phenols behave as acid and ionise in aqueous solution to give H^+ ions.



(ii) Reaction with Ferric Chloride : Phenol gives a violet-coloured water soluble complex with neutral ferric chloride.

$$6C_6H_5OH + FeCl_3 \longrightarrow [Fe(OC_6H_5)_6]^{-3} + 3H^+ + 3HCl$$

Violet complex

(iii) Libermann's test : When a sodium nitrite ($NaNO_2$) is added to a phenol dissoved in conc. sulphuric acid, red or brown colour is produced and changes to blue by the addition of a strong alkali. This test can be used to distinguish between phenols from alcohols.

$$2NaNO_2 + H_2SO_4 \longrightarrow 2HNO_2 + Na_2SO_4$$

Sodium nitrite Nitrous acid

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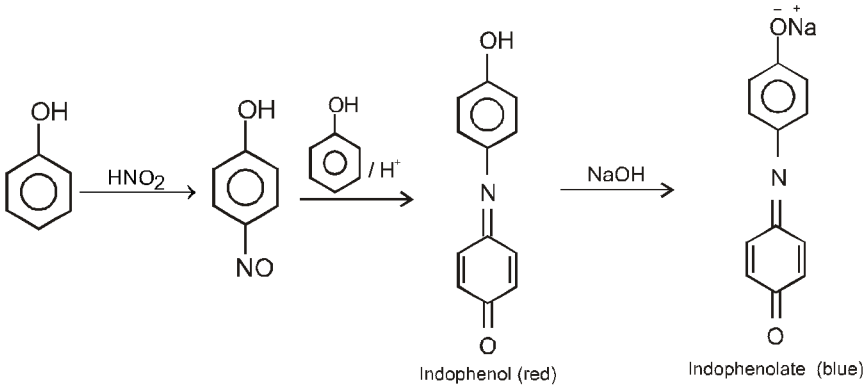

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CHEMICAL KINETICS & REDIOACTIVITY

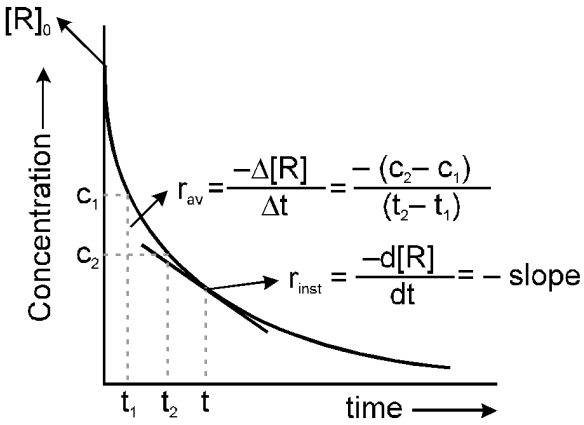
RATE/VELOCITY OF CHEMICAL REACTION :

$$\text{Rate} = \frac{\Delta c}{\Delta t} = \frac{\text{mol/lit.}}{\text{sec}} = \text{mol lit}^{-1}\ \text{time}^{-1} = \text{mol dm}^{-3}\ \text{time}^{-1}$$

Types of Rates of chemical reaction :

For a reaction $R \longrightarrow P$

$$\textbf{Average rate} = \frac{\text{Total change in concentration}}{\text{Total time taken}}$$



$$R_{\text{instantaneous}} = \lim_{t\to 0}\left[\frac{\Delta c}{\Delta t}\right] = \frac{dc}{dt} = -\frac{d[R]}{dt} = \frac{d[P]}{dt}$$

RATE LAW (DEPENDENCE OF RATE ON CONCENTRATION OF REACTANTS) :

Rate = K (conc.)^{order} – differential rate equation or rate expression

Where K = Rate constant = specific reaction rate = rate of reaction when concentration is unity

unit of K = $(\text{conc})^{1-\text{order}}\ \text{time}^{-1}$

Order of reaction :

$m_1A + m_2B \longrightarrow$ products.

$R \propto [A]^p [B]^q$ Where p may or may not be equal to m_1 & similarly q may or may not be equal to m_2.

p is order of reaction with respect to reactant A and q is order of reaction with respect to reactant B and (p + q) is **overall order of the reaction.**

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INTEGRATED RATE LAWS :

C_0 or 'a' is initial concentration and C_t or a – x is concentration at time 't'

(a) zero order reactions :

Rate = k [conc.]° = constant

$$\text{Rate} = k = \frac{C_0 - C_t}{t} \quad \text{or} \quad C_t = C_0 - kt$$

Unit of K = mol lit^{-1} sec^{-1}, Time for completion = $\frac{C_0}{k}$

at $t_{1/2}$, $C_t = \frac{C_0}{2}$, so $kt_{1/2} = \frac{C_0}{2} \Rightarrow$ $\mathbf{t_{1/2} = \frac{C_0}{2k}}$ $\therefore t_{1/2} \propto C_0$

(b) First Order Reactions :

(i) Let a 1st order reaction is, A $\longrightarrow$ Products

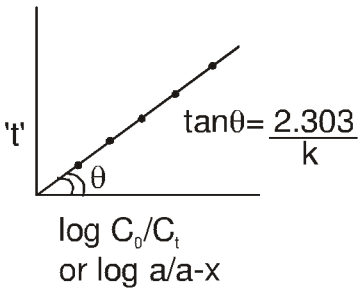
$$\mathbf{t = \frac{2.303}{k} \log \frac{a}{a-x}} \quad \text{or} \quad k = \frac{2.303}{t} \log \frac{C_0}{C_t}$$

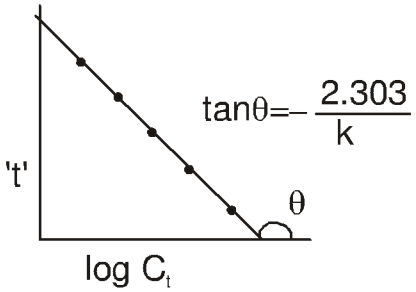
$$\Rightarrow \quad t_{1/2} = \frac{\ell n 2}{k} = \frac{0.693}{k} = \text{Independent of initial concentration.}$$

$$t_{Avg.} = \frac{1}{k} = 1.44\, t_{1/2}.$$

Graphical Representation :

$$t = -\frac{2.303}{k} \log C_t + \frac{2.303}{k} \log C_0$$





(c) Second order reaction :

2nd order Reactions

Two types

A + A $\longrightarrow$ products

a a

(a – x) (a –x)

$$\therefore \frac{dx}{dt} = k\,(a-x)^2$$

$$\Rightarrow \frac{1}{(a-x)} - \frac{1}{a} = kt$$

A + B $\longrightarrow$ products.

a b 0

a – x b – x

$$\frac{dx}{dt} = k\,(a-x)\,(b-x)$$

$$\mathbf{k = \frac{2.303}{t(a-b)} \log \frac{b(a-x)}{a(b-x)}}$$

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METHODS TO DETERMINE ORDER OF A REACTION

(a) Initial rate method :

$r = k [A]^a [B]^b [C]^c$ if [B] = constant
[C] = constant

then for two different initial concentrations of A we have

$$r_{0_1} = k [A_0]_1^a \quad , \quad r_{0_2} = k [A_0]_2^a$$

$$\Rightarrow \quad \frac{r_{0_1}}{r_{0_2}} = \left(\frac{[A_0]_1}{[A_0]_2} \right)^a$$

(b) Using integrated rate law : It is method of trial and error.

(c) Method of half lives :

for n^{th} order reaction $\quad t_{1/2} \propto \frac{1}{[R_0]^{n-1}}$

(d) Ostwald Isolation Method :

rate $= k [A]^a [B]^b [C]^c = k_0 [A]^a$

METHODS TO MONITOR THE PROGRESS OF THE REACTION :

(a) Progress of gaseous reaction can be monitored by measuring total pressure at a fixed volume & temperature or by measuring total volume of mixture under constant pressure and temperature.

$$\therefore \quad \mathbf{k = \frac{2.303}{t} \log \frac{P_0 (n-1)}{nP_0 - P_t}}$$

{Formula is not applicable when n = 1, the value of n can be fractional also.}

(b) By titration method :

1. $\therefore \; a \propto V_0 \qquad a - x \propto V_t \quad \Rightarrow \quad k = \frac{2.303}{t} \log \frac{V_0}{V_t}$

2. Study of acid hydrolysis of an easter.

$$k = \frac{2.303}{t} \log \frac{V_\infty - V_0}{V_\infty - V_t}$$

(c) By measuring optical rotation produced by the reaction mixture :

$$k = \frac{2.303}{t} \log \left(\frac{\theta_0 - \theta_\infty}{\theta_t - \theta_\infty} \right)$$

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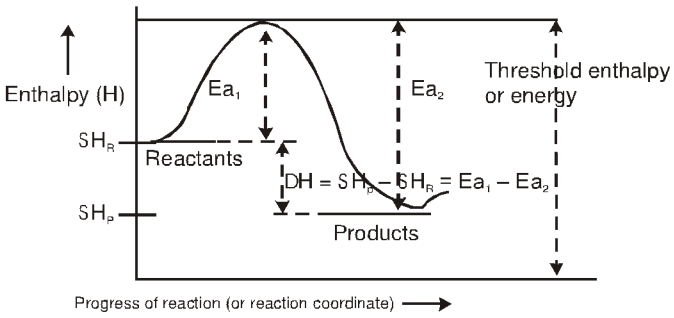
EFFECT OF TEMPERATURE ON RATE OF REACTION.

$$T.C. = \frac{K_{t+10}}{K_t} \approx 2 \text{ to } 3 \text{ (for most of the reactions)}$$

Arhenius theroy of reaction rate.



SH_R = Summation of enthalpies of reactants
SH_P = Summation of enthalpies of reactants
DH = Enthalpy change during the reaction
Ea_1 = Energy of activation of the forward reaction
Ea_2 = Energy of activation of the backward reaction

$E_P > E_r$ $\rightarrow$ endothermic
$E_P < E_r$ $\rightarrow$ exothermic

$\Delta H = (E_P - E_r)$ = enthalpy change
$\Delta H = E_{af} - E_{ab}$
$E_{threshold} = E_{af} + E_r = E_b + E_p$

Arhenius equation

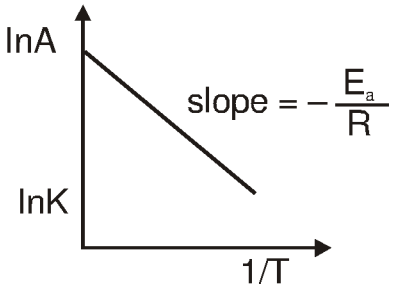
$k = Ae^{-E_a RT}$ $\qquad$ $r = k\,[conc.]^{order}$

$$\frac{d\ln k}{dT} = \frac{E_a}{RT^2} \qquad \log k = \left(-\frac{Ea}{2.303\ R}\right)\frac{1}{T} + \log A$$

If k_1 and k_2 be the rate constant of a reaction at two different temperature T_1 and T_2 respectively, then we have

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303\ R}.\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

❖ $\ln k = \ln A - \frac{E_a}{RT}$



$E_a \geq O$

❖ $T \rightarrow \infty$, $K \rightarrow A$.

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METALLURGY, d & f block

The compound of a metal found in nature is called a mineral. The minerals from which metal can be economically and conveniently extracted are called **ores**. An ore is usually contaminated with earthy or undesired materials known as gangue.

(a) **Native ores** contain the metal in free state. Silver, gold, platinum etc, occur as native ores.

(b) **Oxidised ores** consist of oxides or oxysalts (e.g. carbonates, phosphates, sulphates and silicates) of metals.

(c) **Sulphurised ores** consist of sulphides of metals like iron, lead, zinc, mercury etc.

(d) **Halide ores** consist of halides of metals.

Metal	Ores	Composition
Aluminium	Bauxite	$AlO_X(OH)_{3-2X}$ [where 0 < X < 1] Al_2O_3
	Diaspore	$Al_2O_3.H_2O$
	Corundam	Al_2O_3
	Kaolinite (a form of clay)	$[Al_2 (OH)_4 Si_2O_5]$
Iron	Haematite	Fe_2O_3
	Magnetite	Fe_3O_4
	Siderite	$FeCO_3$
	Iron pyrite	FeS_2
	Limonite	$Fe_2O_3.3H_2O$
Copper	Copper pyrite	$CuFeS_2$
	Copper glance	Cu_2S
	Cuprite	Cu_2O
	Malachite	$CuCO_3.Cu(OH)_2$
	Azurite	$2CuCO_3.Cu(OH)_2$
Zinc	Zinc blende or Sphalerite	ZnS
	Calamine	$ZnCO_3$
	Zincite	ZnO
Lead	Galena	PbS
	Anglesite	$PbSO_4$
	Cerrusite	$PbCO_3$
Magnesium	Carnallite	$KCl.MgCl_2$ $6H_2O$ (K_2MgCl_4 $.6H_2O$)
	Magnesite	$MgCO_3$
	Dolomite	$MgCO_3$ $CaCO_3$
	Epsomsalt (Epsomite)	$MgSO_4$ $7H_2O$
	Langbeinite	$K_2Mg_2(SO_4)_3$
Tin	Cassiterite (Tin stone)	SnO_2
Silver	Silver glance (Argentite)	Ag_2S
	Chlorargyrite (Horn silver)	AgCl

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Metallurgy :

The scientific and technological process used for the extraction/isolation of the metal from its ore is called as metallurgy.

The isolation and extraction of metals from their ores involve the following major steps:

(A) Crushing and Grinding : The ore is first crushed by jaw crushers and ground to a powder.

(B) Concentration :
The removal of unwanted useless impurities from the ore is called **dressing, concentration or benefaction of ore**.

(i) Hydraulic washing or Gravity separation or Levigation method :
It is based on the difference in the densities of the gangue and ore particles. This method is generally used for the concentration of oxide and native ores.

(ii) Electromagnetic separation :
It is based on differences in magnetic properties of the ore components. Chromite ore($FeO.Cr_2O_3$) is separated from non–magnetic silicious impurities and cassiterite ore(SnO_2) is separated from magnetic Wolframite ($FeWO_4 + MnWO_4$).

(iii) Froth floatation process. This method is commonly used for the concentration of the low grade sulphide ores like galena, PbS (ore of Pb); copper pyrites $Cu_2S.Fe_2S_3$ or $CuFeS_2$ (ore of copper) ; zinc blende, ZnS (ore of zinc) etc., and is based on the fact that gangue and ore particles have different degree of wettability with water and pine oil; the gangue particles are preferentially wetted by water while the ore particles are wetted by oil. In this process one or more chemical frothing agents are added.

(iv) Leaching : Leaching is often used if the ore is soluble in some suitable solvent, e.g, acids, bases and suitable chemical reagents.

(C) Extraction of crude metal from concentrated ore :
The isolation of metals from concentrated ore involves two major steps as given below.

(i) Conversion to oxide :
Calcination. It is a process of heating the concentrated ore strongly in a limited supply of air or in the absence of air. The process of calcination brings about the following changes :
(a) The carbonate ore gets decomposed to form the oxide of the metal.
(b) Water of crystallisation present in the hydrated oxide ore gets lost as moisture.
(c) Organic matter, if present in the ore, gets expelled and the ore becomes porous. Volatile impurities are removed.

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Roasting :

It is a process of heating the concentrated ore (generally sulphide ore) strongly in the excess of air or O_2 below its melting point. Roasting is an exothermic process once started it does not require additional heating.

Smelting :

Slag formation : In many extraction processes, an oxide is added deliberately to combine with other impurities and form a stable molten phase immiscible with molten metal called a slag. The process is termed smelting.

The principle of slag formation is essentially the following :
Nonmetal oxide (acidic oxide) + Metal oxide (basic oxide) $\longrightarrow$ Fusible (easily melted) slag

Removal of unwanted basic and acidic oxides: For example, FeO is the impurity in extraction of Cu from copper pyrite.

$$2CuFeS_2 + 4O_2 \longrightarrow Cu_2S + 2FeO + 3SO_2$$

$$Cu_2S + \underbrace{FeO + SiO_2}_{\text{(roasted pyrite)}} \longrightarrow \underset{\text{(upper layer)}}{FeSiO_3\ \text{(Fusible slag)}} + \underset{\text{(lower layer)}}{Cu_2S\ \text{(matte)}}$$

Matte also contains a very small amount of iron(II) sulphide.
To remove unwanted acidic impurities like sand and P_4O_{10}, smelting is done in the presence of limestone.

$$CaCO_3 \longrightarrow CaO + CO_2$$

$$CaO + SiO_2 \longrightarrow CaSiO_3\ \text{(fusible slag)}$$

$$6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2\ \text{(fusible slag - Thomas slag)}$$

(ii) Reduction of a metal oxide :

The free metal is obtained by reduction of a compound, using either a chemical reducing agent or electrolysis.

Chemical reduction method :

Reduction with carbon :

$$PbO + C \longrightarrow Pb + CO\ \text{(extraction of lead)}$$

Reduction with CO : In some cases CO produced in the furnace itself is used as a reducing agent.

$$Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$$

Reduction by other metals :

Metallic oxides (Cr and Mn) can be reduced by a highly electropositive metal such as aluminium that liberates a large amount of energy (1675 kJ/mol) on oxidation to Al_2O_3. The process is known as Goldschmidt or

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aluminothermic process and the reaction is known as thermite reaction.

$$Cr_2O_3 + Al \rightarrow 2Cr\ (\ell) + Al_2O_3$$

Magnesium reduction method : Magnesium is used in similar way to reduce oxides. In certain cases where the oxide is too stable to reduce, electropositive metals are used to reduce halides.

$$TiCl_4 + 2\,Mg \xrightarrow[1000-1150°C]{\text{Krollprocess}} Ti + 2\,MgCl_2$$

$$TiCl_4 + 4Na \xrightarrow{\text{IMI process}} Ti + 4\,NaCl$$

Self-reduction method :

This method is also called auto-reduction method or air reduction method. If the sulphide ore of some of the less electropositive metals like Hg, Cu, Pb, Sb, etc. are heated in air, a part of these is changed into oxide or sulphate then that reacts with the remaining part of the sulphide ore to give its metal and SO_2.

$$Cu_2S + 3O_2 \longrightarrow 3Cu_2O + 2\,SO_2$$

$$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$$

Electrolytic reduction :

It presents the most powerful method of reduction and gives a very pure product. As it is an expensive method compared to chemical methods, it is used either for very reactive metals such as magnesium or aluminum or for production of samples of high purity.

1. **In aqueous solution :** Electrolysis can be carried out conveniently and cheaply in aqueous solution that the products do not react with water. Copper and zinc are obtained by electrolysis of aqueous solution of their sulphates.

2. **In fused melts :** Aluminum is obtained by electrolysis of a fused mixture of Al_2O_3 and cryolite $Na_3[AlF_6]$.

Extraction of Aluminium : It involves the following processes

(a) Purification of bauxite :

(i) Bayer's Method

(used for red bauxite containing Fe_2O_3 and silicates as impurities)

$$Al_2O_3 . 2H_2O + 2NaOH \xrightarrow[8\ atm]{190°C} 2NaAlO_2\ (soluble) + 3H_2O$$

Fe_2O_3 (insoluble) separated as red mud by filteration solution is diluted with water and seeded with freshly prepared $Al(OH)_3$. It induces the precipitation of $Al(OH)_3$. $Al(OH)_3$ is filtered leaving behind silicates in solution.

$$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3\downarrow$$

$$2Al(OH)_3 \xrightarrow{1473K\ \Delta} Al_2O_3 + 3H_2O$$

(ii) Hall's Method

(used for red bauxite containing Fe_2O_3 and silicates as impurities)

$$Al_2O_3 . 2H_2O + Na_2CO_3 \xrightarrow{\text{Fuse}} 2NaAlO_2\ (soluble) + CO_2 + 2H_2O$$

$$2NaAlO_2 + 3H_2O + CO_2 \xrightarrow{60°C} 2Al(OH)_3\downarrow + Na_2CO_3$$

$$2Al(OH)_3 \xrightarrow{1473K\ \Delta} Al_2O_3 + 3H_2O$$

(iii) Serpeck's Method

(used for white bauxite containing silica as impurities)

$$Al_2O_3 . 2H_2O + 3C + N_2 \xrightarrow[\text{Electric furnace}]{1800°C} 2AlN + 3CO + 2H_2O$$

$$2AlN + 3H_2O \rightarrow Al(OH)_3\downarrow + NH_3$$

$$SiO_2 + 2CO \rightarrow 2CO_2 + Si$$

Silicone volatilises at this temp.

$$2Al(OH)_3 \xrightarrow{1473K\ \Delta} Al_2O_3 + 3H_2O$$

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(b) Electrolytic reduction (Hall-Heroult process) :

$$2Al_2O_3 + 3C \longrightarrow 4Al + 3CO_2$$

Cathode : Al^{3+} (melt) + $3e^- \longrightarrow$ Al(l)

Anode : $C(s) + O^{2-}$ (melt) $\longrightarrow$ CO(g) + $2e^-$

$C(s) + 2O^{2-}$ (melt) $\longrightarrow$ CO_2 (g) + $4e^-$

Metallurgy of some important metals

1. Extraction of iron from ore haematite :

Reactions involved :

At 500 – 800 K (lower temperature range in the blast furnace)

$$3\,Fe_2O_3 + CO \longrightarrow 2\,Fe_3O_4 + CO_2$$

$$Fe_3O_4 + CO \longrightarrow 3Fe + 4\,CO_2$$

$$Fe_2O_3 + CO \longrightarrow 2FeO + CO_2$$

At 900 – 1500 K (higher temperature range in the blast furnace):

$C + CO_2 \longrightarrow 2\,CO$; $FeO + CO \longrightarrow Fe + CO_2$

Limestone is also decomposed tom CaO which removes silicate impurity of the ore as slag. The slag is in molten state and separates out from iron.

$CaCO_3 \longrightarrow CaO + CO_2$; $CaO + SiO_2 \longrightarrow CaSiO_3$

2. Extraction of copper :

From copper glance / copper pyrite (self reduction) :

$$2CuFeS_2 + 4O_2 \longrightarrow Cu_2S + 2FeO + 3SO_2$$

$Cu_2S + FeO + SiO_2 \longrightarrow FeSiO_3$ (fusible slag) + Cu_2S (matte)

$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$; $FeO + SiO_2 \longrightarrow FeSiO_3$

$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$;

$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2$ (self reduction)

3. Extraction of lead :

(i) $2PbS(s) + 3O_2(g) \xrightarrow{\Delta} 2PbO(s) \xrightarrow[\Delta]{+C} 2Pb(\ell) + CO_2(g)$

(ii) $3PbS(s) \xrightarrow[\text{air}]{\text{heat in}} PbS(s) + 2PbO(s) \xrightarrow[\text{absence of air}]{\text{Heat in}} 3Pb(\ell) + SO_2(g)$

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4. Extraction of zinc from zinc blende :

The ore is roasted in presence of excess of air at temperature 1200 K.

$$2\,ZnS + 3O_2 \longrightarrow 2\,ZnO + 2SO_2$$

The reduction of zinc oxide is done using coke.

$$ZnO + C \xrightarrow{\text{Coke, 1673 K}} Zn + CO$$

5. Extraction of tin from cassiterite :

The concentrated ore is subjected to the electromagnetic separation to remove magnetic impurity of Wolframite.

SnO_2 is reduced to metal using carbon at 1200–1300°C in an electric furnace. The product often contains traces of Fe, which is removed by blowing air through the molten mixture to oxidise FeO which then floats to the surface.

$$SnO_2 + 2C \longrightarrow Sn + 2CO$$
$$2Fe + O_2 \longrightarrow 2FeO$$

6. Extraction of Magnesium :

From Sea water (Dow's process) :

Sea water contains 0.13% magnesium as chloride and sulphate. It involves following steps.

(a) Precipitation of magnesium as magnesium hydroxide by slaked lime.

(b) Preparation of hexahydrated magnesium chloride.

The solution on concentration and crystallisation gives the crystals of $MgCl_2.6H_2O$.

(c) Preparation of anhydrous magnesium chloride.

(d) Electrolysis of fused anhydrous $MgCl_2$ in presence of NaCl.

$$MgCl_2 \rightleftharpoons Mg^{2+} + 2Cl^-$$

At cathode : $Mg^{2+} + 2e^- \longrightarrow Mg$(99% pure) ;

At anode : $2Cl^- \longrightarrow Cl_2 + 2e^-$

7. Extraction of gold and silver (MacArthur-Forrest cyanide process) :

(a) From native ores : Extraction of gold and silver involves leaching the metal with CN^-.

$$4Au/Ag\,(s) + 8CN^-(aq) + 2H_2O(aq) + O_2(g) \longrightarrow 4[Au/Ag\,(CN)_2]^-(aq) + 4OH^-(aq)$$

$$2[Au/Ag\,(CN)_2]^-(aq) + Zn(s) \longrightarrow 2Au/Ag\,(s) + [Zn(CN)_4]^{2-}\,(aq)$$

(b) From argentite ore :

$$Ag_2S\ (\text{conc. ore}) + 2NaCN \underset{\text{Air}}{\rightleftharpoons} 2AgCN + Na_2S.$$
$$4Na_2S + 5O_2 + 2H_2O \longrightarrow 2Na_2SO_4 + 4NaOH + 2S$$
$$AgCN + NaCN \longrightarrow Na[Ag(CN)_2]\ (\text{soluble complex})$$
$$2Na[Ag(CN)_2] + Zn\ (\text{dust}) \longrightarrow 2Ag\downarrow + Na_2[Zn(CN)_4].$$

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(D) Purification or Refining of metals :

Physical methods : These methods include the following processes:

(I) Liquation process : This process is used for the purification of the metal, which itself is readily fusible, but the impurities present in it are not, used for the purification of Sn and Zn, and for removing Pb from Zn-Ag alloy.

(II) Fractional distillation process : This process is used to purify those metals which themselves are volatile and the impurities in them are nonvolatile and vice-versa. Zn, Cd and Hg are purified by this process.

(III) Zone refining method (Fractional crystallisation method) : This process is used when metals are required in very high purity, for specific application. For example pure Si and Ge are used in semiconductors

Chemical methods : These methods include the following methods:

(I) OXIDATIVE REFINING :

This method is usually employed for refining metals like Pb, Ag, Cu, Fe, etc. In this method the molten impure metal is subjected to oxidation by various ways.

(II) POLING PROCESS :

This process is used for the purification of copper and tin which contains the impurities of their own oxides.

$$\text{Green wood} \rightarrow \text{Hydrocarbons} \rightarrow CH_4$$

$$4CuO + CH_4 \rightarrow 4Cu \text{ (pure metal)} + CO_2 + 2H_2O$$

(III) ELECTROLYTIC REFINING :

Some metals such as Cu, Ni, and Al are refined electrolytically.

(IV) VAPOR PHASE REFINING :

(i) Extraction of Nickel (Mond's process) : The sequence of reaction is

$$H_2O(g) + C \longrightarrow CO(g) + H_2$$

$$Ni(s) + 4\,CO(s) \xrightarrow{50°C} [Ni(CO_4)]\,(g)$$

$$[Ni\,(CO)_4](g) \xrightarrow{200°C} Ni + 4CO(g)$$

(ii) Van Arkel–De Boer process :

$$\text{Impure Ti} + 2I_2 \xrightarrow{50-250°C} TiI_4 \xrightarrow[\text{Tungsten filament}]{1400°C} Ti + 2I_2$$

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CBT d & f block

1. The most abundant metal and transition metal in earth crust is Aluminium & **Iron** respectively

2. The highest oxidation state exhibited by a transition element is **+8 (shown by Os & Ru)**

3. Mn^{+2} ion shows maximum paramagnetic character among the bivalent ions of first transition series with $\mu = 5.92$ BM.

4. Chromite ore has the formula $FeO.Cr_2O_3$ and is the starting material for preparation of $K_2Cr_2O_7$.

5. The chromate ion in acidic medium changes to **dichromate** and vice versa in basic medium.

6. Photographic films and plates are coated with $AgBr$.

7. $AgCl$, AgBr and AgI dissolve in excess of $NaCN$ and $Na_2S_2O_3$ to give the complex compounds $Na[Ag(CN)_2]$ and $Na_3[Ag(S_2O_3)_2]$

8. Lunar caustic is $AgNO_3$.

9. Lithopone is a mixture of $BaSO_4$ and ZnS.

10. White zinc used in paints, is chemically ZnO.

11. Concentrated nitric acid renders aluminium, chromium and iron **passive.**

12. Hydrated ferrous sulphate is known as **green** vitriol which on heating loses 7 water molecules and turns white forming $FeSO_4$(anhy)

13. Philosopher's wool has the formula ZnO and it is yellow when hot and white when cold.

14. Pyrolusite ore has the formula MnO_2.

15. Which elements have unusual configuration in 3d and 5d series ?

Ans: In 3d series Cr and Cu have unusual configuration. In 5d series W and Au have unusual configuration.

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16. Which elements have unusual configuration in 4d series ?

Ans: Except Y, Zr *and* Cd, all others have unusual configuration.

17. Metals of second and third d-series in general have **greater** enthalpies of atomisation than the corresponding elements of the first series with third greater than second.

18. 4d and 5d series elements have greater tendency to form metal-metal bonding as compared to 3d series elements with third greater than second.

19. Write the reaction of charcoal and $K_2Cr_2O_7$ on heating.

Ans: $K_2Cr_2O_7 + 2C(charcol) \xrightarrow{\Delta} Cr_2O_3 + K_2CO_3 + CO \uparrow$ 20. What is the composition of German silver and Bell metal alloys ?

Ans: German silver : Cu, Zn, Ni ; Bell metal : Cu, Sn

21. Name the lightest and heaviest element (in terms of density) among the transition elements.

Ans: (i) Sc, (ii) Os and Ir (nearly same)

22. Which element among the d-block elements has (i) the lowest melting point (ii) the highest melting point ?

Ans: (i) Hg (ii) W

23. Why zinc has lowest melting point in 3d series ?

Ans: In case of zinc, no electrons from 3d-orbitals are involved in the formation of metallic bonds.

24. Why are ionization energies of 5d-elements in general greater than those of 3d-elements?

Ans: In the 5d-series, after lanthanum (Z = 57), there is lanthanide contraction (poor shielding of nuclear charge by f-orbital electrons). As a result, in each group the atomic size of 5d elements is small and its nuclear charge is large. Hence, the ionisation energies of the 5d elements are larger than 3d elements.

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25. Name the three factors which determine the stability of a particular oxidation state of a metal in solution.

Ans: (i) Enthalpy of sublimation (ii) Ionisation energy

(iii) Enthalpy of hydration

26. The E^0 value for the Mn^{+3}/Mn^{+2} couple is much more positive than that of Cr^{+3}/Cr^{+2} or Fe^{+3}/Fe^{+2}. This is because Mn has much higher value of 3rd ionization energy ($Mn^{+2} \longrightarrow Mn^{+3}$). In other words, Mn^{+3} have high tendency to revert back to Mn^{+2}.

27. Why $KMnO_4$ is stored in dark bottle and what happens to its acidic solution? 2 marks

Ans: MnO_4^- solutions are intrinsically unstable in acidic solutions and decomposes slowly. This decomposition is catalyzed by sun light. This is why $KMnO_4$ solution is kept in dark bottles.

28. Why does $AgNO_3$ produce a black stain on the skin ?

Ans: In presence of organic matter (skin) and light, $AgNO_3$ decomposes to produce a black stain of metallic silver.

29. Why it is not advisable to dissolve $KMnO_4$ in cold and conc. H_2SO_4 ? Give reactions.

Ans: Because explosive Mn_2O_7 is formed.

$$2KMnO_4 + 2H_2SO_4 \longrightarrow Mn_2O_7 + 2KHSO_4 + H_2O$$

$$2Mn_2O_7 \longrightarrow 4MnO_2 + 3O_2$$

30. Write balanced chemical equations for:

(i) Mixture of $K_2Cr_2O_7$ and $NaCl$ solids is heated with concentrated H_2SO_4.

(ii) Potassium permanganate is added to a hot solution of manganous sulphate.

(iii) Potassium dichromate and concentered HCl are heated together.

Ans: (i) $K_2Cr_2O_7 + 4NaCl + 6H_2SO_4 \longrightarrow 2CrO_2Cl_2 \uparrow + 4NaHSO_4 + 2KHSO_4 + 3H_2O$

(ii) $2KMnO_4 + 3MnSO_4 + 2H_2O \longrightarrow 5MnO_2 + K_2SO_4 + 2H_2SO_4$

(i) $K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 3Cl2 + 7H_2O$

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31. Actinoid contraction is greater from element to element than lanthanoid contraction. This is because the 5f electrons are more effectively shielded from nuclear charge. In other words the 5f electrons themselves provide poor shielding from element to element in the series.

32. Sm^{+2}, Eu^{+2} and Yb^{+2} ions in solutions are good reducing agents but an aqueous solution of Ce^{+4} is a good oxidizing agent. This is because the most stable oxidation state of lanthanoides is +3. Hence, ions in +2 state tend to oxidize into +3. Similarly ions in +4 state tend to get reduced into +3.

33. Pyrolusite ore on oxidation with $KClO_3/KNO_3$ in basic medium produces dark green colored compound (A), which on electrolysis produces a purple colored compound (B). The purple colored compound can be crystallized to deep purple rhombic prisms. It shows different reactions in different mediums. Excess of compound (B) on heating with concentrated H_2SO_4 gives an explosive oil (C), which on heating decomposes to give another compound (D) along with oxygen. Identify (A) to (D).

Ans: (A) = K_2MnO_4; (B) = $KMnO_4$; (C) = Mn_2O_7; (D) = MnO_2

34. Write the reaction of excess of $AgNO_3$ with I_2 and excess of I_2 with $AgNO_3$.

Ans: $6AgNO_3\ (excess) + 3I_2 + 3\,H_2O \longrightarrow AgIO_3 + 5AgI + 6HNO_3$

$5AgNO_3 + 3I_2(excess) + 3\,H_2O \longrightarrow HIO_3 + 5AgI + 5HNO_3$

35. $FeSO_4 + H_2O_2$ is known as **Fenton's reagent** & is used as catalyst for oxidizing benzene to phenol.

36. **Ziegler-Natta** catalyst is $Al_2(C_2H_5)_6 + TiCl_4$. It is used in the production of **Polyethene**.

36. $MCl_4 \xrightarrow{Zn}$ (B) Purple color compound

(A) Colourless

Liquid at room

Temperature, transition metal

↓

Moist air

White fumes (C).

Identify (A), (B) and (C). Also explain color difference between MCl

Ans: (A) is $TiCl_4$ as it has no unpaired electrons and is liquid at room temperature on account of covalent character because of high polarizing power of Ti^{+4}. $TiCl_4$ being covalent gets hydrolyzed forming $TiO_2(H_2O)_2$ and HCl(C) which fumes in air.

In $[Ti(H_2O)_6]Cl_3$ (B) complex $Ti(III)$ has one unpaired electron ($3d^1$) which gives violet/puple color due to d-d transition.

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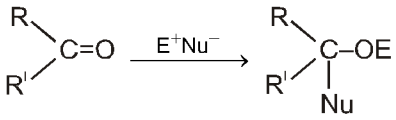
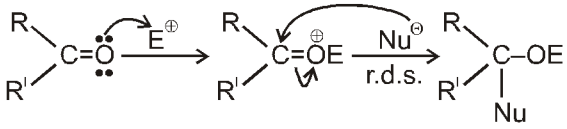
Aldehydes, Ketones, Carboxylic acids & G.R.

Nucleophilic addition reactions of Carbonyl compounds :-

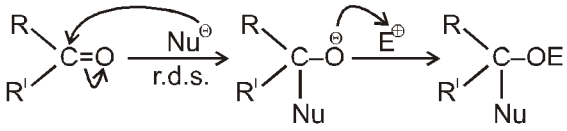
It is a characteristic reaction of carbonyl compounds (aldehydes and ketones).
This reaction involves addition of a nuceophile and a electrophile across the (C=O) double bond.
The general nucleophilic addition reaction can be represented as followed :



The mechanism can be carried out in following two manners.
(a) acid catalysed (first attack is of electrophile)



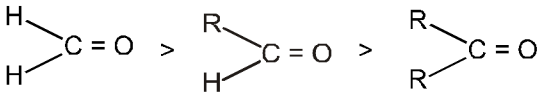
(b) Base catalysed (first attack is of nucleophile)



acid catalysed nucleophilic addition is generally faster than base catalysed.

The reactivity of the carbonyl group arises from the electronegativity of the oxygen atom and the resulting polarization of the carbon-oxygen double bond. The electrophilic carbonyl carbon atom is sp^2 hybridized and flat, hence it is relatively unhindered and open to attack from either face of the double bond.

Reactivity : Aldehydes are more reactive than ketones in nucleophilic addition reactions.

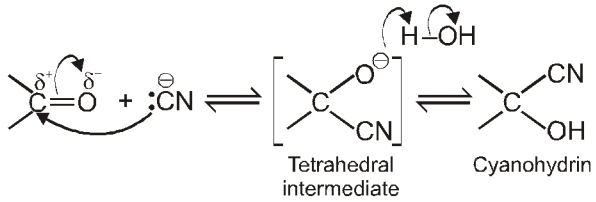


There are two factors which influence the reactivity of ketone and aldehyde.

(i) Inductive effect (ii) steric factor
(i) + I effect of alkyl group decrease the amount of charge on C^+ ($C^+ - O^-$). in ketones.
(ii) Steric effect also causes the less reactivity of carbonyl group.

(i) Addition of hydrogen cyanide (HCN)

$HCN + OH^- \longleftrightarrow :\bar{C}N + H_2O$



Note : (i) Addition of HCN over aldehyde and ketones gives cyanohydrin.
(ii) Cyanohydrin on acid hydrolysis gives α-hydroxy acid.
(iii) Cyanohydrin on treating with $NH_3(l)$ followed by acid hydrolysis gives α-amino acid.
(iv) In case of ketone cyanohydrin formation is reversible due to bulky groups of ketone which hinder the formation.

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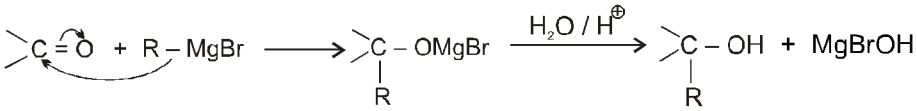
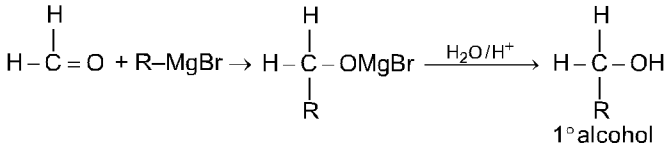
e.g. $CH_3-CHO \xrightarrow[H_2SO_4]{NaCN} CH_3-\underset{\underset{OH}{|}}{CH}-CN$

Acetaldehyde — Cyanohydrin

(ii) Addition of Grignard reagent :



(a) When formaldehyde is treated with Grignard reagent followed by acid hydrolysis primary alcohol is obtained.

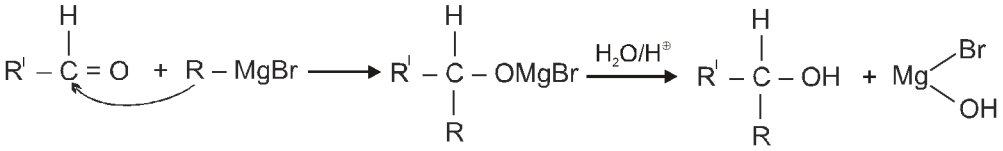


1° alcohol

e.g. $HCHO + CH_3MgBr \xrightarrow{H_2O/H^+} CH_3-CH_2-OH$

Ethanol

(b) When aldehyde except formaldehyde is treated with Grignard reagent followed by hydrolysis 2° alcohol is obtained.



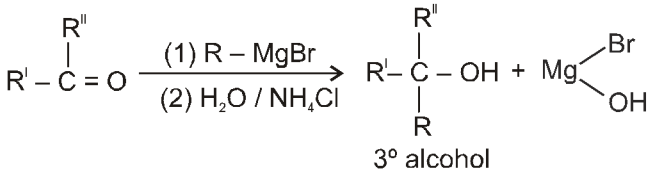
2° alcohol

e.g. $CH_3-CHO + CH_3MgBr \xrightarrow{H_2O/H^+} CH_3-\underset{\underset{OH}{|}}{C}-CH_3$

Propan-2-ol

e.g. $CH_3-CHO + PHMgBr \xrightarrow{H_2O/H^+} PH-\underset{\underset{OH}{|}}{CH}-CH_3$

(c) When ketone is treated with Grignard reagent followed by acid hydrolysis 3° alcohol is obtained.



3° alcohol

e.g. $CH_3-\overset{\overset{O}{||}}{C}-CH_2-CH_3 + C_2H_5 \xrightarrow{H_2O/H^+} CH_3-\underset{\underset{OH}{|}}{\overset{\overset{C_2H_5}{|}}{C}}-C_2H_5$

3-Methylpentan-3-ol

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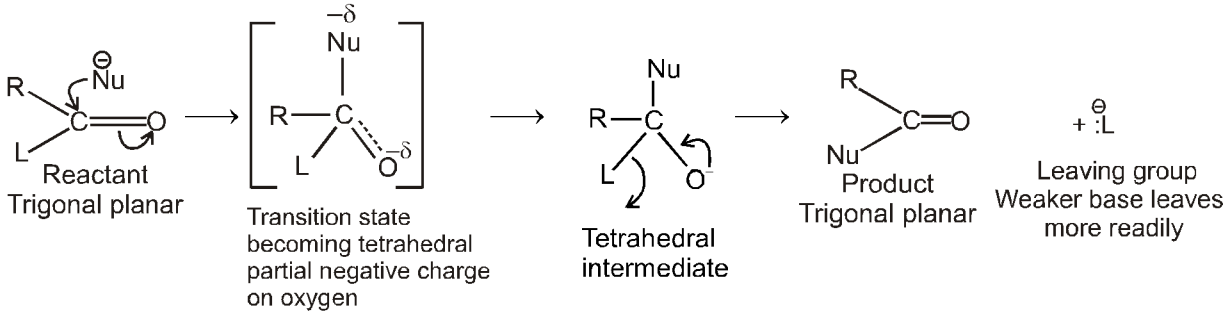
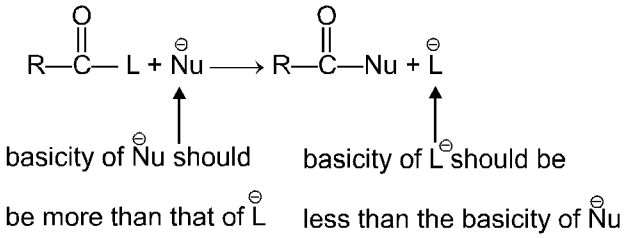
Bimolecular nucleophilic substitution reaction with tetrahedral intermediate (S_N2Th) :-

Characteristic reaction of acid and it's derivatives (acid halide, anhydride, ester and amide) is S_N2Th. Nucleophilic acyl substitution usually takes place by an **addition-elimination** mechanism.The incoming nucleophile adds to the carbonyl to form a tetrasubstituted intermediate with a tetrahedral carbon.



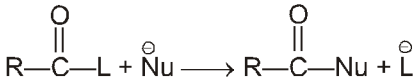
The tetrahedral intermediate formed, when a nucleophile attacks the carbonyl carbon of a carboxylic acid derivative, is not stable and can not be isolated.

A pair of nonbonding electrons on the oxygen reforms the π bond, and either $L^{\ominus}$ or $Nu^{\ominus}$ is eliminated with its bonding electrons. Whether $L^{\ominus}$ or $Nu^{\ominus}$ is eliminated depends on their relative basicities. The weaker base is preferentially eliminated because the weaker base is the better leaving group.



Thus carboxylic acid derivatives will undergo a nucleophilic acyl substitution reaction provided that the incoming nucleophile is a stronger base than the group that is to be replaced. If the incoming nucleophile and the group attached to acyl group in the starting material have similar basicities, the tetrahedral intermediate can expect either group with similar ease. A mixture of starting material and substitution product will result.

Conditions for acyl nucleophilic substitution reactions :



(i) $L^{\ominus}$ must be better leaving group than $Nu^{\ominus}$, i.e., basicity of $Nu^{\ominus}$ should be more than that of $L^{\ominus}$.
(ii) $Nu^{\ominus}$ must be a strong enough nucleophilic to attack RCOL.
(iii) Carbonyl carbon must be enough electrophilic to react with $Nu^{\ominus}$.

Reactivity order : $R-\overset{O}{\overset{\|}{C}}-Cl > R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R > R-\overset{O}{\overset{\|}{C}}-OR > R-\overset{O}{\overset{\|}{C}}-NH_2$

(A) S_N2Th Reaction of carboxylic acid :

(i) Formation of acid chlorides :

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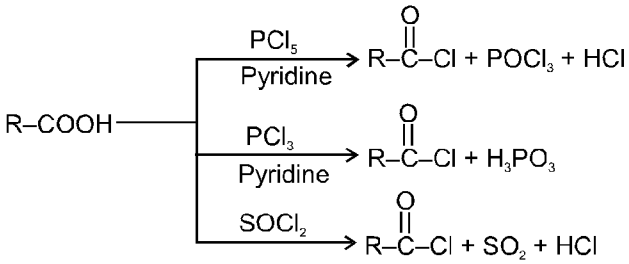
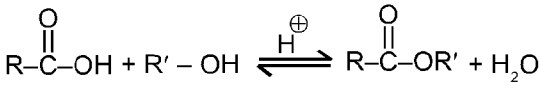


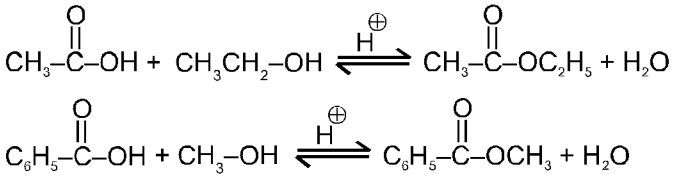
(ii) Fisher's Esterification :

Carboxylic acids reacts with alcohol to form esters through a condensation reaction known as esterification.

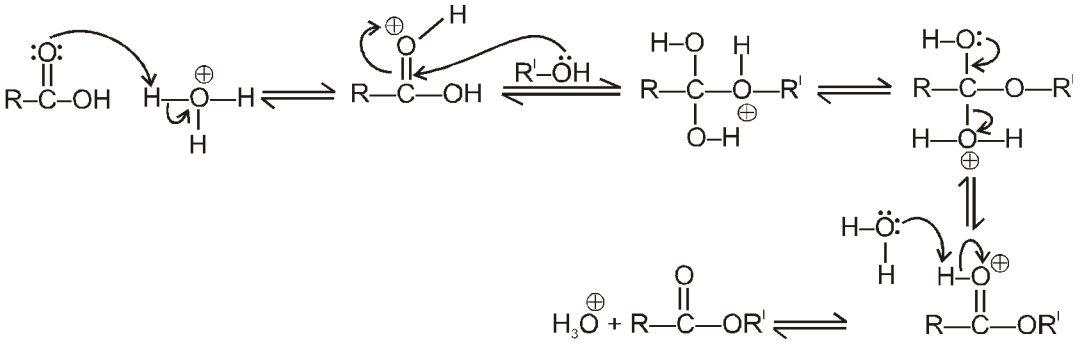
General Reaction :



Specific Examples:

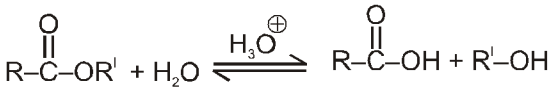


Mechanism : (Acid catalysed esterification)



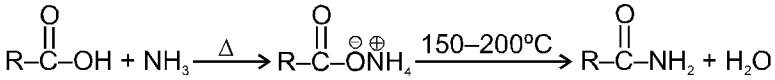
If we follow the forward reactions in this mechanism, we have the mechanism for the acid catalysed esterification of an acid. If however, we follow the reverse reactions, we have the mechanism for the acid catalysed hydrolysis of an ester.

Acid catalysed ester hydrolysis.



which result we obtain will depend on the conditions we choose. If we want to esterify an acid, we use an excess of the alcohol and, if possible remove the water as it is formed. If we want to hydrolyse an ester, we use a large excess of water that is we reflux the ester with dilute aqueous HCl or dilute aqueous H_2SO_4.

(iii) Formation of amides :



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In fact amides can not be prepared from carboxylic acids and amines unless the ammonium salt is heated strongly to dehydrate it. This is not usually a good method of preparing amides.

e.g. $Ph-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[\Delta]{NH_3} Ph-\overset{O}{\overset{\|}{C}}-NH_2$

Benzoic acid → Benzamide

(iv) Formation of acid anhydride :

$$2R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow[\Delta]{P_2O_5} R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R + H_2O$$

e.g. $CH_3-COOH + HOOC-CH_3 \xrightarrow[\Delta]{P_2O_5} CH_3-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-CH_3 + H_2O$

(B) S_N2Th Reaction of acid derivatives :

(i) Conversion of acid derivatives to other acid derivatives.

$$R-\overset{O}{\overset{\|}{C}}-Cl + 2NH_3 \longrightarrow R-\overset{O}{\overset{\|}{C}}-NH_2 + NH_4Cl$$

$$R-\overset{O}{\overset{\|}{C}}-O-R' + NaNH_2 \longrightarrow R-\overset{O}{\overset{\|}{C}}-NH_2 + R'-ONa$$

$$R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R + R'-OH \longrightarrow R-\overset{O}{\overset{\|}{C}}-OR' + R-\overset{O}{\overset{\|}{C}}-OH$$

$$R-\overset{O}{\overset{\|}{C}}-NH_2 + NaCl \longrightarrow \text{No reaction}$$

Note: An S_N2Th reaction is feasible when nucleophile is more basic than the leaving group.

(ii) Hydrolysis of acid derivatives :

$$R-\overset{O}{\overset{\|}{C}}-G \xrightarrow[H_2O/\bar{O}H]{H_2O/H^{\oplus}} R-\overset{O}{\overset{\|}{C}}-OH + GH$$

e.g. $R-\overset{O}{\overset{\|}{C}}-O-R \xrightarrow{H_2O/H^{\oplus}} R-\overset{O}{\overset{\|}{C}}-O-H + R-OH$

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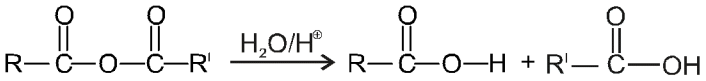
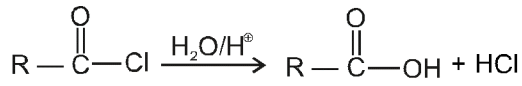
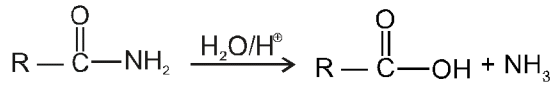
e.g. 

e.g. 

e.g. 

Note: Hydrolysis of amides and esters are not possible in neutral medium.

* * * * *

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Preparation methods of Aldehydes and Ketones :

By oxidation of alcohols :

Primary alcohols $\xrightarrow{[O]}$ Aldehydes

For oxidising primary alcohols to aldehydes PCC (Pyridiniumchloro chromate ($C_5H_5NH^+CrO_3Cl^-$)) may be used.

Secondary alcohols $\xrightarrow{[O]}$ Ketones

For oxidising 2° alcohols to ketones $CrO_3/K_2Cr_2O_7$ may be used.

By dehydrogenation of (1°&2°) alcohols :

Dehydrogenation means removal of hydrogen and reagent used is heated copper or silver. The aclohol vapours are passed over these heavy metal catalyst to yield the following products.

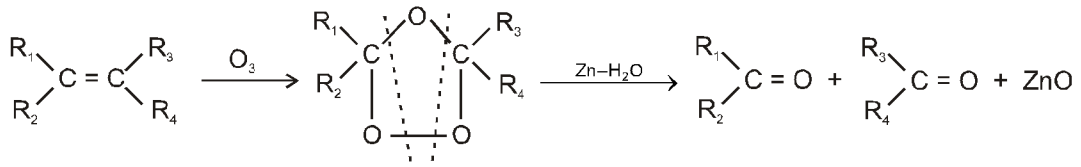
$$1^\circ \text{ alcohol } (RCH_2OH) \xrightarrow[-H_2]{Cu/300^\circ C} \text{Aldehyde } (R-CH=O)$$

$$2^\circ \text{ alcohol } (R_2CHOH) \xrightarrow[-H_2]{Cu/300^\circ C} \text{Ketone } (R_2C=O)$$

$$3^\circ \text{ alcohol} \xrightarrow[-H_2O]{Cu/300^\circ C} \text{Alkene (Dehydration)}$$

Ozonolysis of alkene :

It is used to get carbonyl compounds from alkene. The reaction is



Mono-ozonide

Note : (i) During the cleavage of ozonide Zn is used to check further oxidation of aldehyde into acid.
(ii) By this method we can locate double bond in olefin or exact structrue of hydrocarbon can be determined by knowing ozonolysis product i.e. by placing double bond at the place of two carbonyl oxygen atoms of two carbonyl compounds.
(iii) Among the three molecules of carbonyl compounds.
(a) If one molecule contains two carbonyl groups, then hydrocarbon will be alkadiene.
(b) If all the three molecules contain two carbonyl group then hydrocarbon will be cycloalkatriene.
(iv) This method is used for aliphatic carbonyl compounds.

Wacker process :

Alkenes can directly be oxidised to corresponding aldehydes or ketones by treating them with a solution of $PdCl_2$ containing a catalytic amount of $CuCl_2$ in presence of air or O_2. Except ethene any higher alkene will give ketone.

$$CH_2=CH_2 + H_2O + PdCl_2 \xrightarrow[\text{air or } O_2]{CuCl_2} CH_3-CH=O + Pd + 2HCl$$

$$R-CH=CH_2 + H_2O + PdCl_2 \xrightarrow[\text{air or } O_2]{CuCl_2} R-\overset{O}{\overset{\|}{C}}-CH_3 + Pd + 2HCl$$

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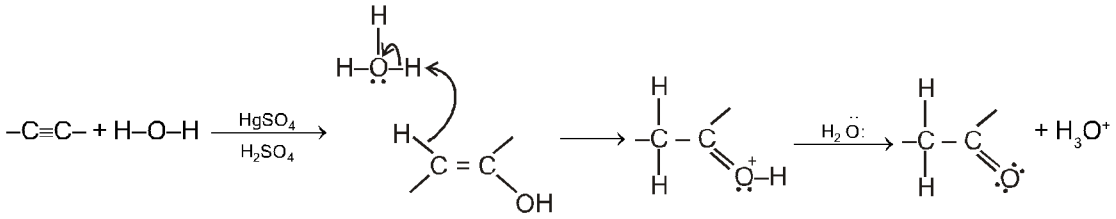
Note : During the reaction $PdCl_2$ is reduced to Pd and $CuCl_2$ is reduced to Cu(I)

Hydration of alkynes :

$$\underset{\text{Ethyne}}{CH \equiv CH} \xrightarrow[H_2SO_4/HgSO_4]{H_2O} \underset{\text{Ethanal}}{CH_3CHO}$$

Mechanism :



This is used mainly for terminal alkynes.
This method is not useful for internal alkynes as they produce mixed products in the reactions.
Other alkynes give ketones in this reaction.

$$R - C \equiv C - H \xrightarrow[H_2O]{Hg^{++}/H_2SO_4} R-\underset{\underset{O}{\|}}{C}-CH_3$$

Hydroboration of alkyne :

It is used to get aldehyde from terminal alkyne. Here reagent is (i) diborane (B_2H_6) (ii) H_2O_2,OH^-

$$R - C \equiv C - H \xrightarrow[(ii)H_2O_2/OH^-]{(i)\ B_2H_6} R - CH_2 - CH = O$$

In this reaction Borane (BH_3) is act as electrophile.

Dry distillation of calcium salt of acid :

$$(RCOO)_2Ca \xrightarrow{\Delta} \underset{\text{Ketone}}{R-\overset{\overset{O}{\|}}{C}-R} + CaCO_3$$

Ex. $(CH_3COO)_2Ca \xrightarrow{\Delta} CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + CaCO_3$

calcium acetate

On dry distillation of calcium salt of acetic acid with calcium salt of formic acid we get a mixture of aldehyde, ketone and formaldehyde.
Calcium salt of dibasic acid (1,4 & higher) on distillation) gives cyclic ketones.

On passing vapours of fatty acids over Mangnous oxide at 300°C :

$$\underset{\text{(Vap)}}{2RCOOH} \xrightarrow{MnO\ /\ 300^oC} \underset{\text{Ketone}}{R-\overset{\overset{O}{\|}}{C}-R} + CO_2 + H_2O$$

$$Ph - CH_3 \xrightarrow[500^oC]{MnO/Air} Ph - CHO$$

On passing mixture of vapours of fatty acid with formic acid we get a mixture of aldehyde, ketone and formaldehyde.

On aqueous alkali hydrolysis of gem-dihalides :

Terminal gemdihalides will give aldehyde while non-terminal will give ketone as follows

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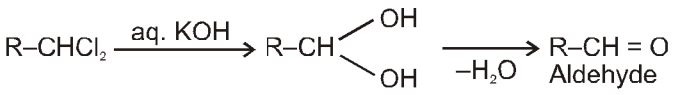
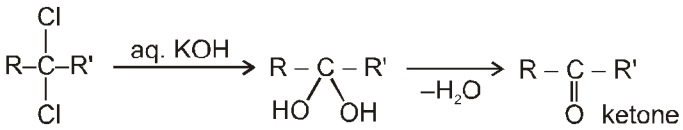
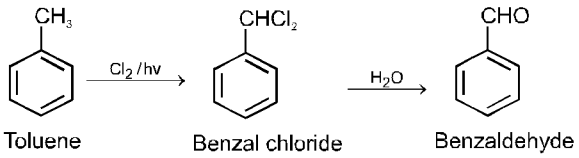


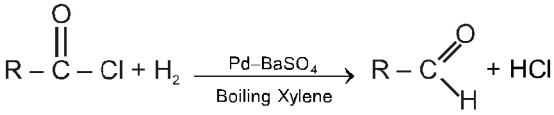




Methods used for the preparation of Aldehydes only.

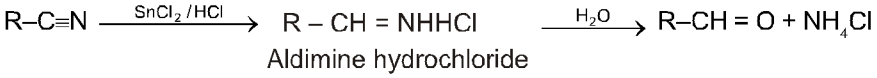
Rosenmund's reduction :

Here acid chlorides are reduced to aldehyde with H_2 in boiling xylene using palladium as a catalyst supported on barium sulphate.



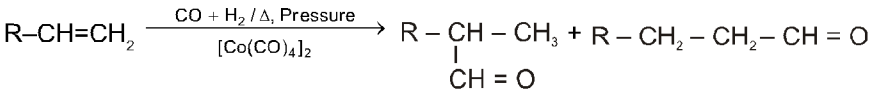
Note : (a) Pd Catalyst is poisoned by $BaSO_4$ (Lindlars catalyst) to check further reduction of aldehyde to alcohol.
(b) Formaldehyde cannot be obtained by this method because HCOCl is unstable at common temperature.
(c) Reaction with acid chloride and dialkyl cadmium or Lithium dialkyl cuprate we can obtain ketone.

Stephen's reduction :



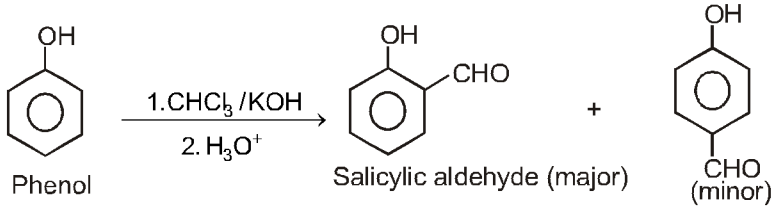
Oxo-process :

It is also called as carbonylation here terminal alkene reacts with water gas at high temperature and pressure in the presence of cobalt carbonyl catalyst to give aldehyde.
Wilkinson catalyst $[RhH(CO)(Ph_3P)_3]$ can also be used in promoting the oxo-process.

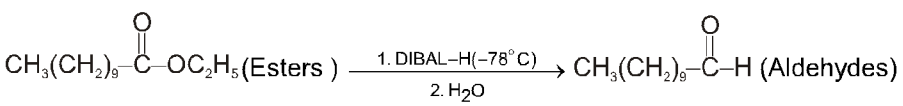


Reimer-Teimann Reaction :

By this method phenolic aldehyde is prepared



From esters or nitrile :



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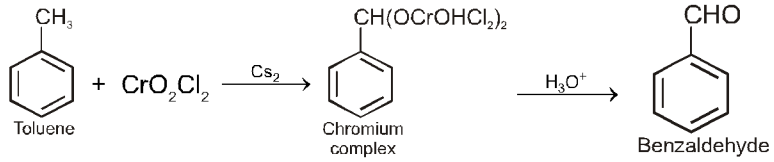
$$R{-}CN \xrightarrow[2.\ H_2O]{1.\ DIBAL-H} R{-}CH{=}O$$

DIBAL–H : Diisobutyl aluminium hydride [AlH(i-Bu)$_2$] is a reducing agent.

This is less reactive than LAH, LiAlH$_4$ and is more sterically hindered and has difficulty in transferring hydride ions.

From hydrocarbons :

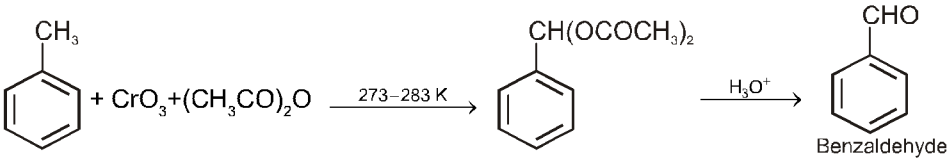
By oxidation of methyl benzene and its derivative using chromyl chloride (CrO_2Cl_2)



This reaction is called **Etard reaction.**

→ $Ph - CH_3 \xrightarrow[500°C]{MnO/Air} Ph - CHO$

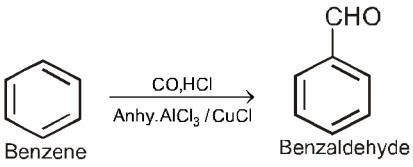
By oxidation of methyl benzene and its derivative using chromic oxide (CrO_3) in acetic anhydride:



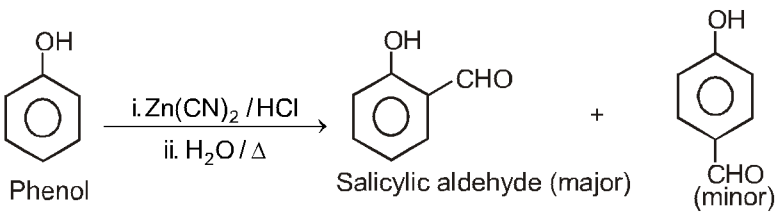
By Gatterman-Koch reaction :

This reaction is mainly given by aromatic hydrocarbons & Halobenzene

Benzene or its derivative $\xrightarrow[Anhy.AlCl_3/CuCl]{CO,HCl}$ Benzaldehyde or substituted benzaldehyde



Gattermann formylation :



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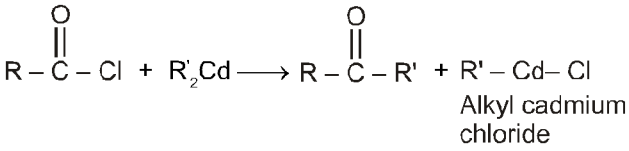
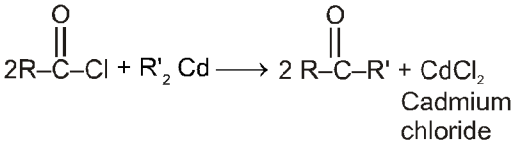
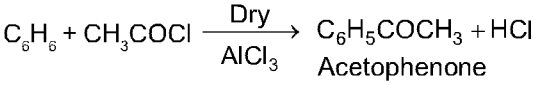
Methods used for the preparation of Ketones only

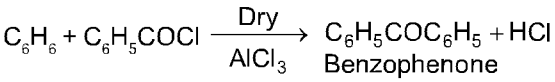
Using alkanoylchloride and dialkyl cadmium :

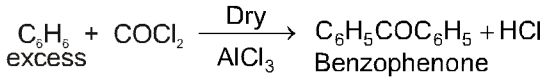




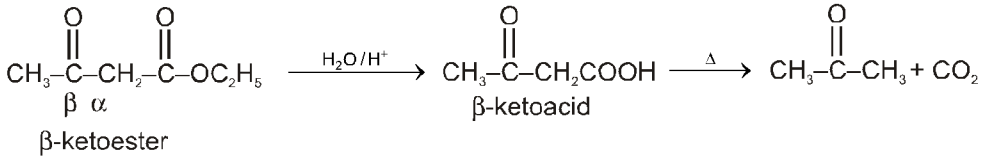
By acylation or benzoylation of aromatic hydrocarbon (Friedel-Craft Reaction)



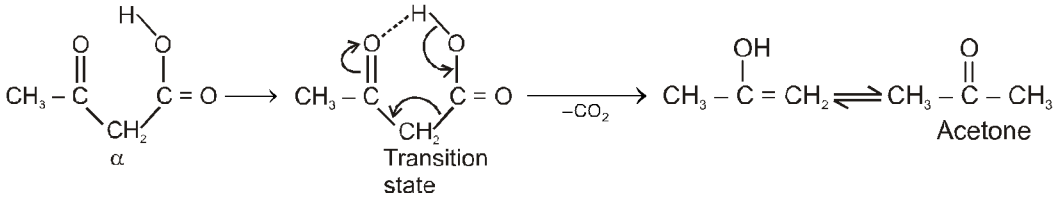




By acid hydrolysis followed by heating of β-Ketoester :

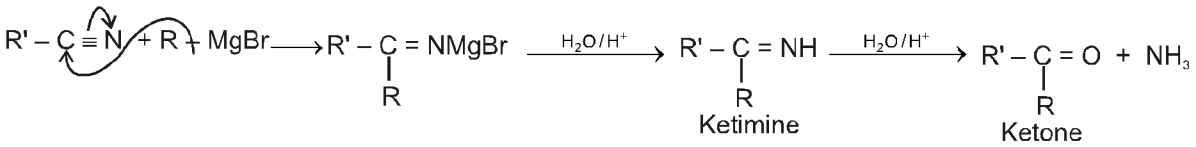


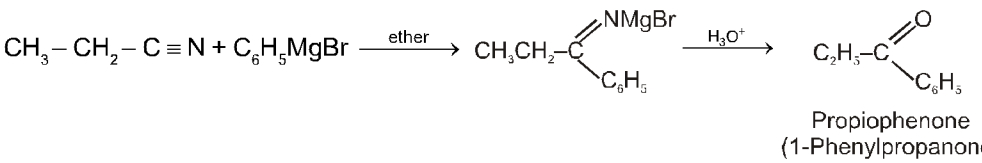
Note : It is β-ketoacid which decarboxylate more readily as it proceeds via six membered cyclic transition-state.



From nitriles :

Treatment of nitrile with Grignard reagent followed by hydrolysis gives a ketone.





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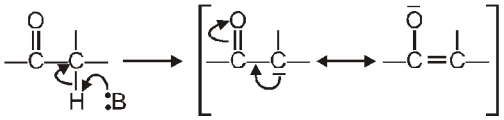
Reactions due to α-Hydrogen

α-Hydrogen of aldehydes and ketones are acidic: They undergo a number of reactions due to the acidic nature of α-hydrogen.

Reason for the acidity of α-hydrogen: Strong electron-withdrawing effect of the carbonyl group, and resonance stabilisation of the conjugate base



Resonating structure of conjugate base

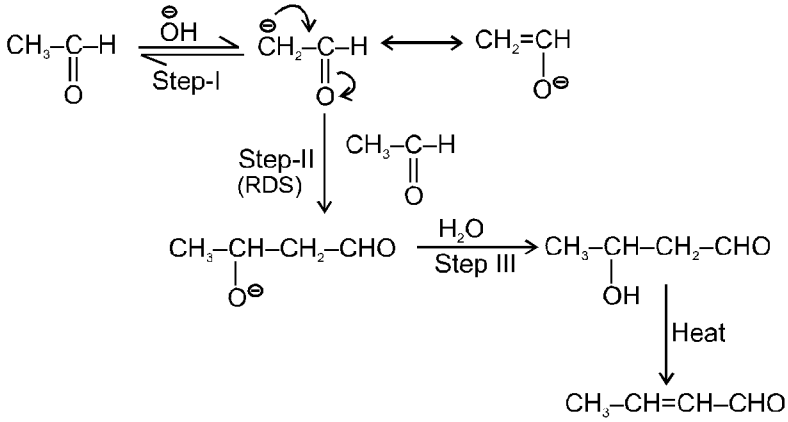
(I) Aldol condensation (or aldol reaction)

Aldehydes and ketones with at least one α-hydrogen undergo a reaction in the presence of dilute alkali as catalyst.

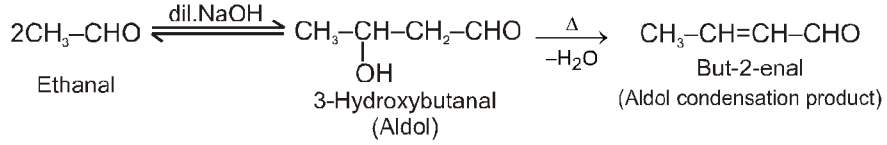
Aldol is derived from two functional groups aldehyde(s) and alcohol.

Ketones gives ketols (keto + alcohols) groups. The name aldol condensation still applies to their reactions.

Mechanism

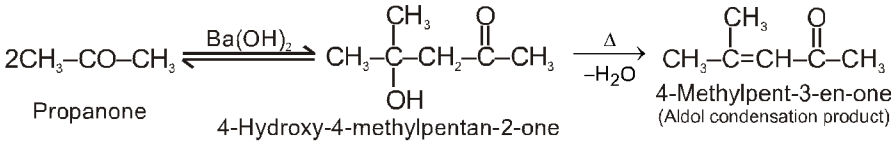


Ex. **(a)**



Ethanal — 3-Hydroxybutanal (Aldol) — But-2-enal (Aldol condensation product)

(b)



Propanone — 4-Hydroxy-4-methylpentan-2-one — 4-Methylpent-3-en-one (Aldol condensation product)

(II) Cross-Aldol condensation :

On using two types of carbonyl compounds both having α-hydrogen atoms we get a mixture of four condensed product because two types of carbonyl compounds will give two type of carbanions which will be nucleophile for itself and other molecule.

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$CH_3CHO + CH_3CH_2CHO \xrightarrow[(ii)\ \Delta]{(i)\ NaOH}$

$CH_3–CH=CH–CHO$ (Simple or self aldol products)

+

$CH_3–CH_2–CH=C(CH_3)–CHO$ (Simple or self aldol products)

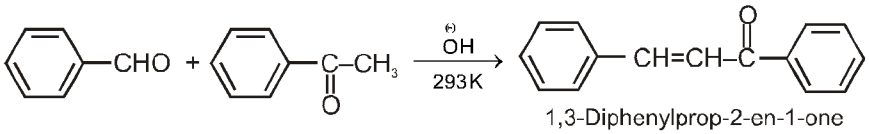
+

$CH_3–CH=C(CH_3)–CHO$ (Cross aldol products)

+

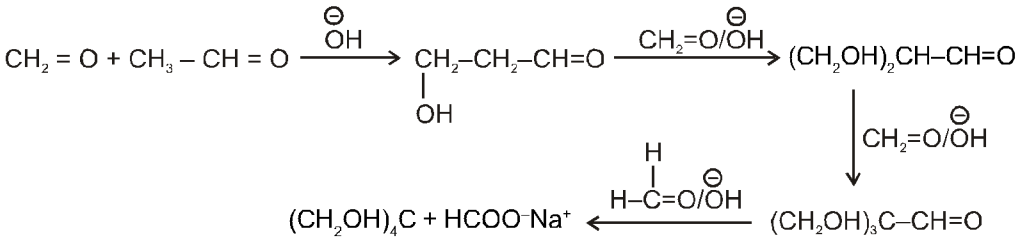
$CH_3–CH_2–CH=CH–CHO$ (Cross aldol products)

Ketones can also be used as one component in cross-aldol reactions.



1,3-Diphenylprop-2-en-1-one
{Benzalacetophenone}
{Major product}

Note : On using formaldehyde and acetaldehyde during crossed aldol all the α-hydrogen atom of acetaldehyde are replaced one by one by hydroxymethyl group because of smaller size of formaldehyde to give trihydroxymethylacetaldehyde which undergoes crossed cannizaro's reaction with formaldehyde to give tetrahydroxymethyl methane and formate ion as a final product.

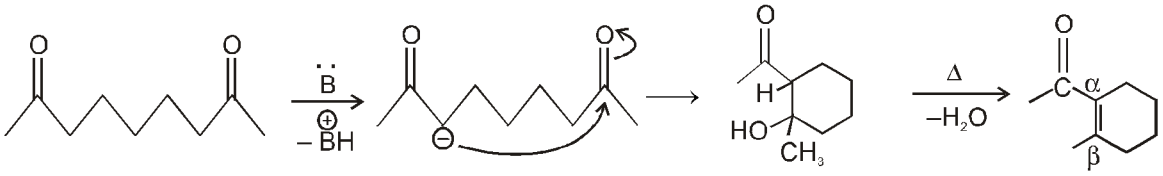


Ex. Show how cinnamaldehyde is prepared by crossed aldol condensation ?

Sol. $C_6H_5CHO + CH_3CHO \xrightarrow[(-H_2O)]{OH^{\ominus}} C_6H_5CH=CH–CHO + H_2O$

(III) Intramolecular aldol condensation :

If two carbonyl groups with α-hydrogen atoms are present within the same molecule, then we get cyclic α, β-unsaturated aldehyde / ketones via the formation of cyclic-β-hydroxy aldehyde / ketone in presence of basic medium.



Cyclic β-hydroxy-ketone

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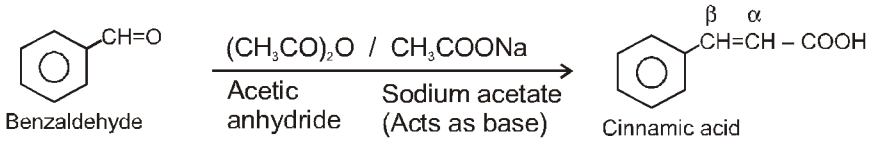
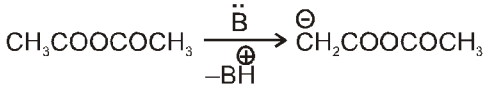
By knowing product we can get reactant as in case of intermolecular aldol condensation :

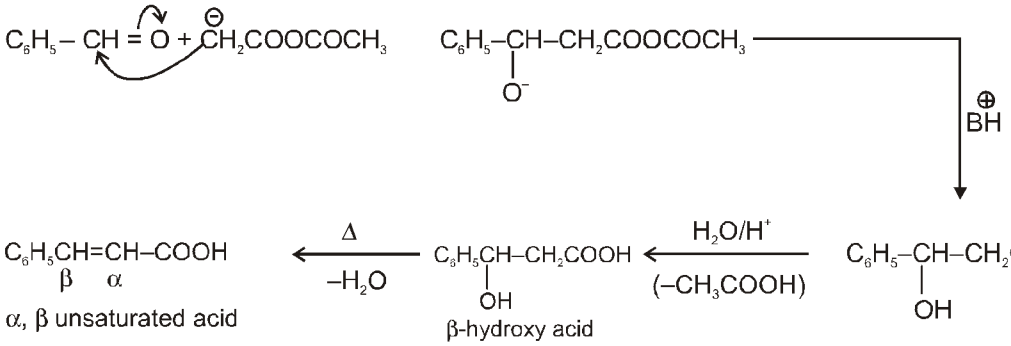
Perkin reaction :

When aromatic aldehyde like benzaldehyde is treated with anhydride in the presence of sodium salt of acid from which anhydride is derived we get α, β-unsaturated acid.



Mechanism :

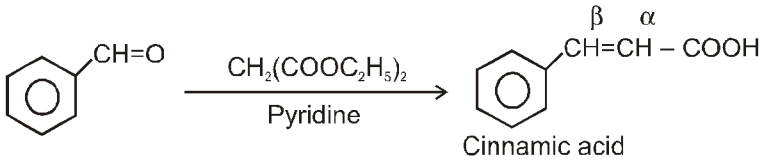




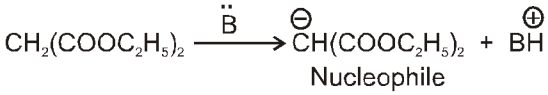
Note : By knowing α, β-unsaturated acid we can get idea about the anhydride used in perkin reaction. This can be done by keeping 'H' at α and –OH at β-carbon atom followed by breaking α, β carbon. By this we can know about acid and it will be anhydride of this acid only.

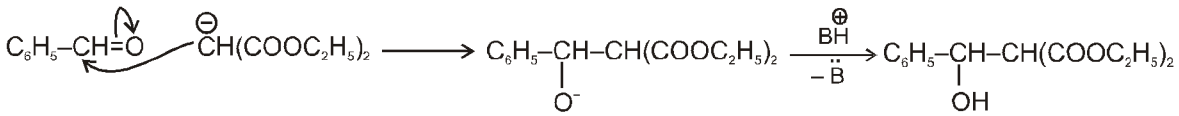
Knoevenagel reaction :

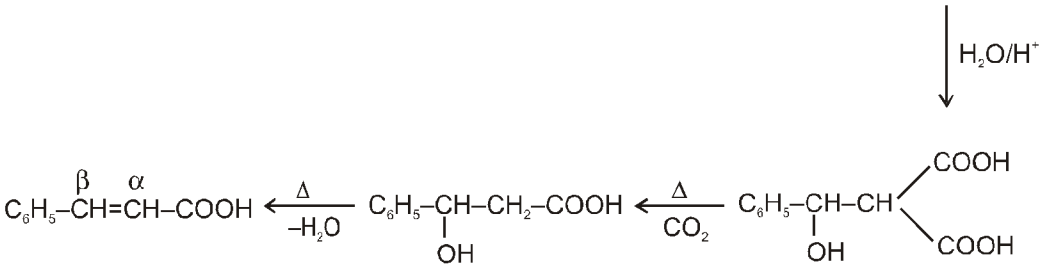
It is preparation of α, β-unsaturated acid with carbonyl compound using malonic ester in the presence of pyridine base.



Mechanism :







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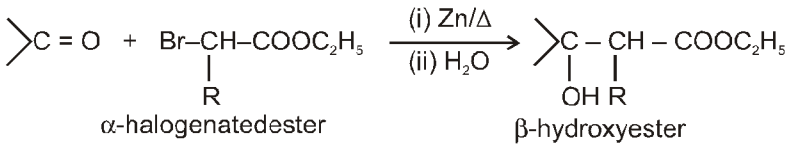
that is we reflux the ester with dilute aqueous HCl or dilute aqueous H_2SO_4.

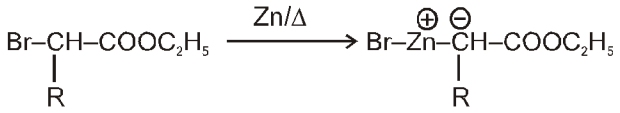
Reformatsky reaction :

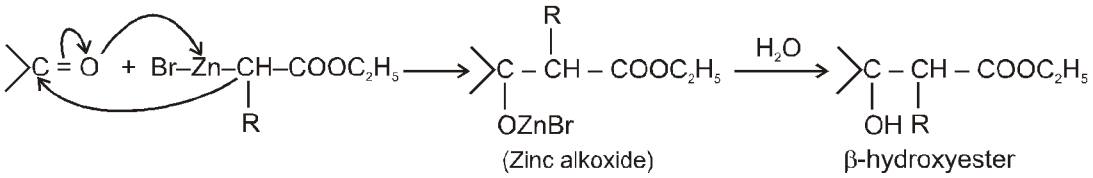
When carbonyl compound and α-halogenated ester are heated with zinc followed by treating with water we get β-hydroxyester.

The organozinc reagent generated is less reactive than a grignard reagent, it does not add to the ester group.



This reaction can be represented as –

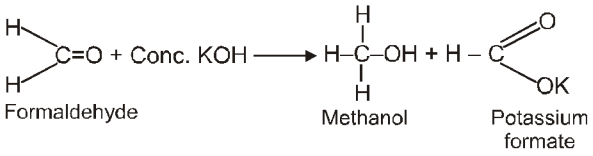




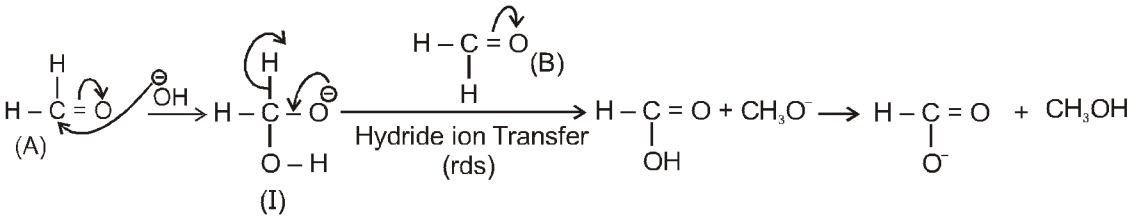
Cannizzaro reaction :

Aldehydes which do not have an α-hydrogen atom, undergo self oxidation and reduction (disproportionation) reaction on treatment with a concentrated alkali.

Ex.



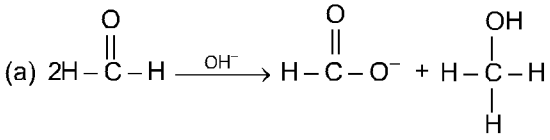
Mechanism :



By this mechanism it is clear that acid corresponds to that carbonyl compound over which $\overset{\ominus}{O}H$ can attack easily as nucleophile.

Note : It is observed that hydride ion transfer from (I) to Carbonyl compound (B) is rate determining step.

Predict the condensation products of these reactions :



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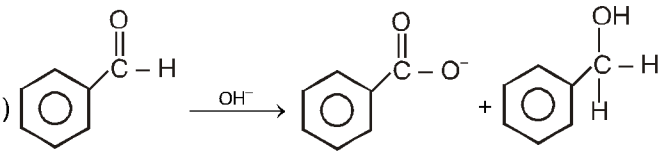
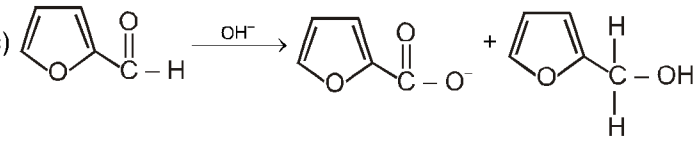
(b) 

(c) 

Crossed Cannizzaro reaction :

On using two types of carbonyl compounds not having α-hydrogen atom, acid salt will be corresponding to that aldehyde over which $\overset{\ominus}{O}H$ will approach without any hindrence.

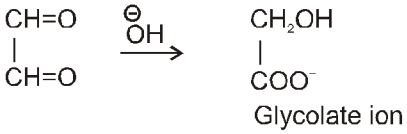
(i) $\underset{(A)}{H-C(=O)-H} + \underset{(B)}{C_6H_5-CH=O} \xrightarrow{\overset{\ominus}{O}H} H-COO^- + C_6H_5CH_2OH$

(ii) $\underset{(A)}{(CH_2OH)_3C-CH=O} + \underset{(B)}{H-C(=O)-H} \xrightarrow{\overset{\ominus}{O}H} (CH_2OH)_3C-CH_2OH + H-COO^-$

In case (i) $\overset{\ominus}{O}H$ will easily go to (A) and in case (ii) it will go to (B) hence acid salt will be formate ion in both the cases.

Intramolecular Cannizzaro reacion :

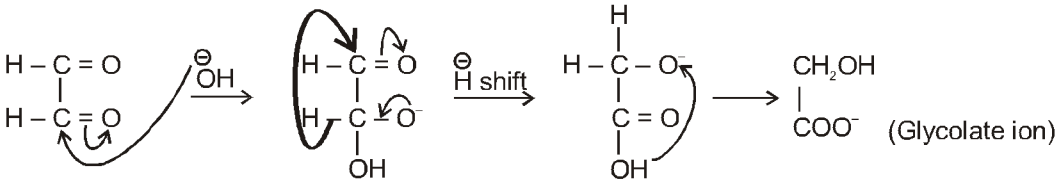
Here two carbonyl groups (without α-hydrogen atom) are present within the same molecule.



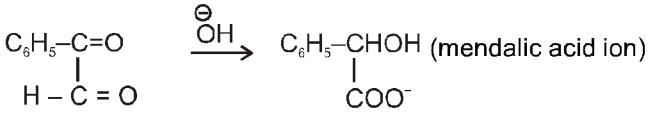
Glycolate ion

Mechanism :

$\Rightarrow$ α-keto aldehyde gives internal cannizaro reaction.



When the reaction is caried out in D_2O, the obtained alcohol contains no α deuterium.

Ex. 

(mendalic acid ion)

Note : 1. CX_3CHO does not gives cannizaro reaction. 2. $(CH_3)_2CH-CH=O$ gives cannizaro reaction.

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Reduction reactions

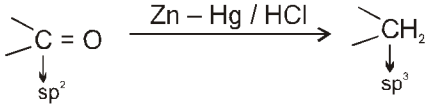
(I) Reduction to alcohols :

Aldehydes $\xrightarrow[\text{or Catalytic hydrogenation}]{NaBH_4 \text{ or } LiAlH_4}$ Primary alcohols

Ketones $\xrightarrow[\text{or Catalytic hydrogenation}]{NaBH_4 \text{ or } LiAlH_4}$ Secondary alcohols

(II) Clemmensen reduction :

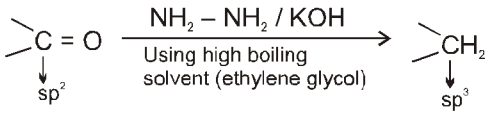
Used to get alkane from carbonyl compounds.



Note : Generally this reaction is avoided if **acid sensitive** groups are present in the carbonyl compounds.

(III) Wolf-Kishner reduction :

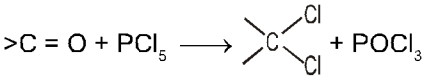
Used to get alkane from carbonyl compounds.



Note : Generally this reaction is avoided if **base sensitive** groups are present in the carbonyl compounds

(IV) Reaction with PCl_5 :

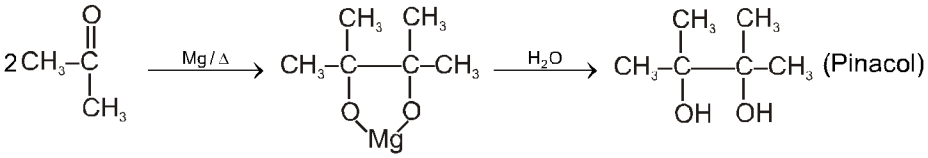
Carbonyl compounds give gemdihalides



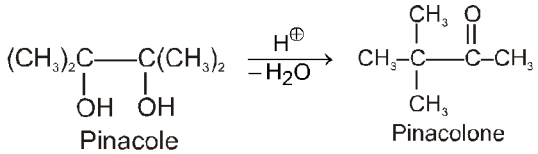
(i) $CH_3CH = O + PCl_5 \longrightarrow CH_3–CHCl_2 + POCl_3$ (ii) $CH_3–\underset{\displaystyle O}{\overset{\displaystyle CH_3}{\overset{|}{C}}}=O + PCl_5 \longrightarrow CH_3–\underset{\displaystyle Cl}{\overset{\displaystyle CH_3}{\overset{|}{C}}}–Cl + POCl_3$

(V) Preparation of Pinacol:

Pinacole is obtained when 2 moles of acetone are heated with magnesium metal followed by treating with water.



Pinacole undergoes rearrangement in acidic medium to give pinacolone



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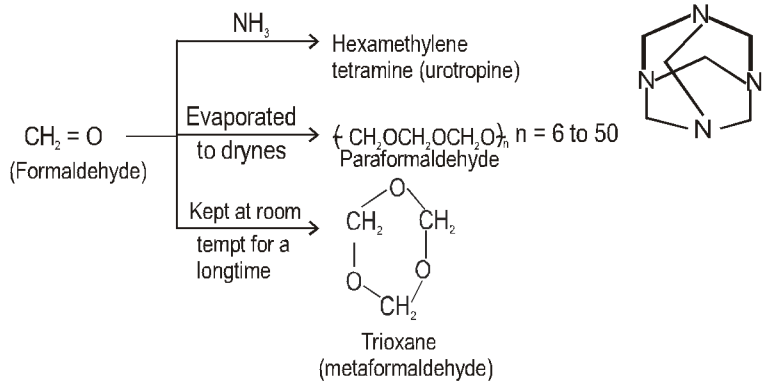
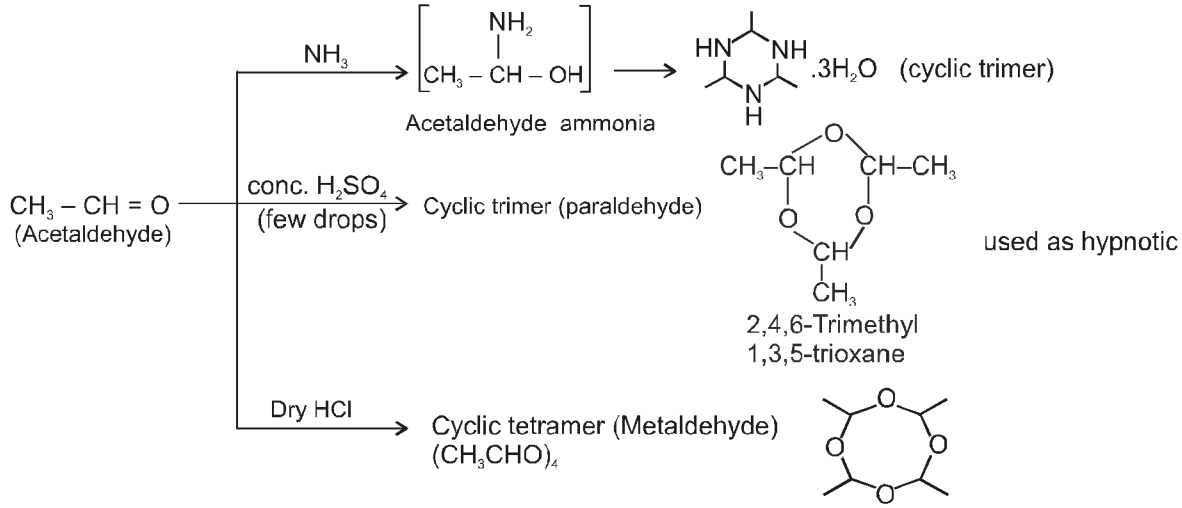
Other miscellaneous reactions :

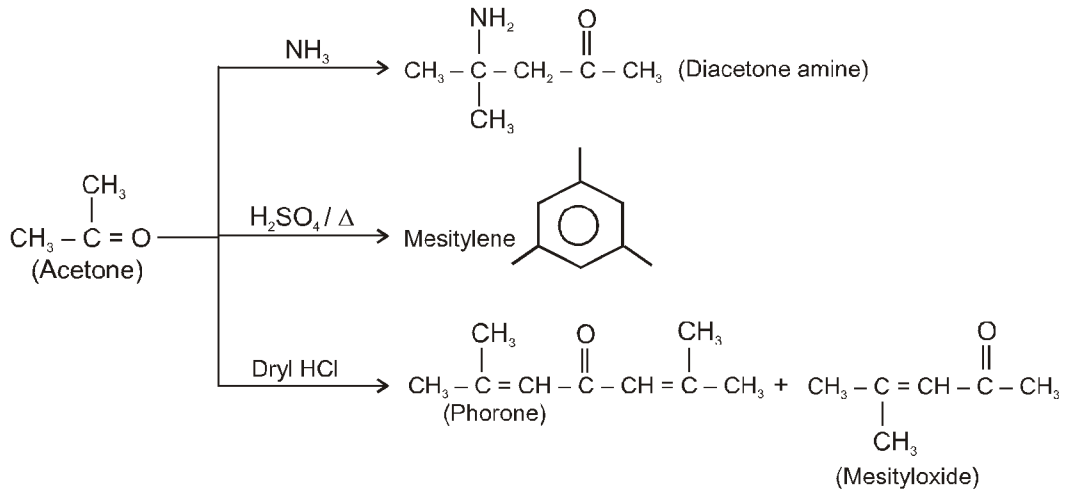
(I)



(II)



(III)



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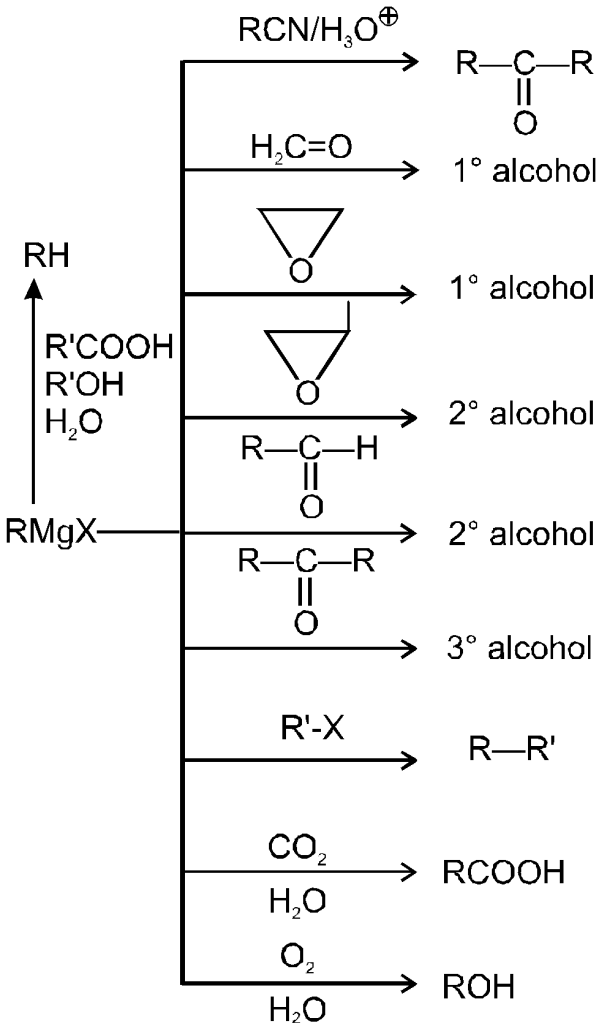
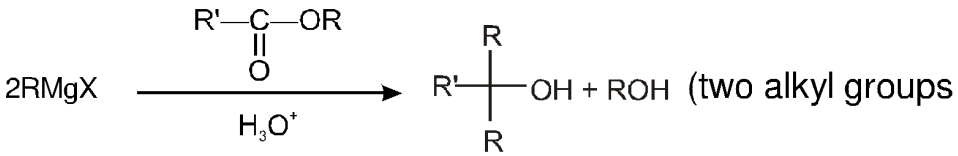
• Points to remember in Grignard reagents

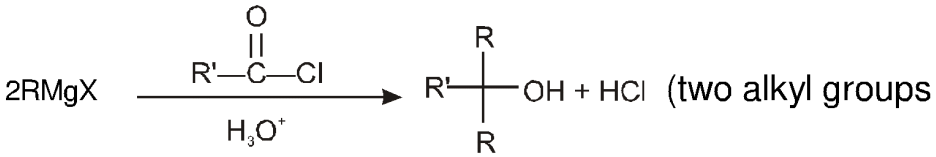
Grignard's Reagent : RMgX (alkyl magnesium halide)

* Active – H Containing compounds $\xrightarrow{RMgX}$ R – H

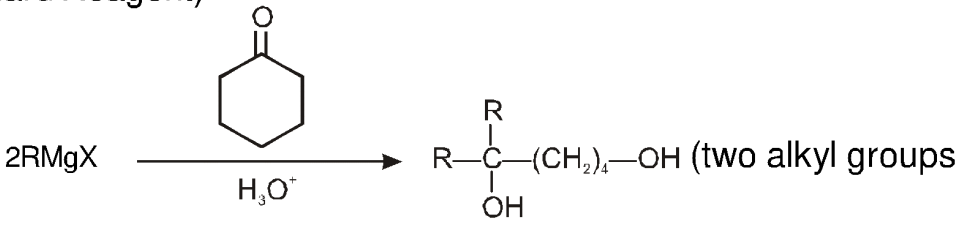




(two alkyl groups come from Grignard Reagent)



(two alkyl groups come from Grignard Reagent)



(two alkyl groups come from Grignard Reagent)

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THERMODYNAMICS

THERMODYNAMICS

Study of energy transfer between the system & the surrounding is called thermodynamics.

Application of thermodynamics	Limitations of thermodynamics
(1) **Spontaneity** of a given Process/Reaction can be predicted.	(1) It does not deal with **speed of reaction**. (time taken)
(2) **Equilibrium constant** of a reaction can be calculated.	(2) Laws of thermodynamics are not applicable on individual particle i.e. valid for **macro system, not for micro system**.

Some basic terminology:

(1) System: anything which is under observation.

(2) Surrounding: everything except system. (**System + surrounding = universe**)

(3) Boundary: anything which separates system & surrounding is called boundary

Boundary can be real or imaginary

Boundary can be flexible or rigid

Boundary can be adiabatic (**non-conducting**) or diathermic (**conducting**).

(4) Type of system

S.No.	System	Exchange with system & surrounding		Example
		Mass transfer	Energy transfer	
1	Open	Yes	Yes	Tea in open cup, living system
2	Close	No	Yes	Tea in closed cup
3	Isolated	No	No	Tea in thermo flask.

Note: (1) In real life, **no perfect isolated** system exists.

(2) Universe is a perfect example of an isolated system.

(5) State of system: The condition in which the **system is present**. It can be defined by measuring some observable properties of system.

(a) State function	(b) Path function
Dependent only on the state of the system and not only the way the state has been achieved.	Dependent on the path/way the system has achieved a particular state
e.g. T, P, V, total energy (E or U), enthalpy (H), Gibbs energy (G), entropy (S)	e.g. Work, heat, molar heat capacity, specific heat capacity

Note: For cyclic process: Change in state function = 0 or **Δ(state Function) = 0**

Change in path function $\neq$ 0 or **Δ(path Function) $\neq$ 0**

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(6) Intensive & extensive property

Sr .No.	Intensive Property	Extensive Property
1	**Non additive** Property	**Additive** Property
2	**Independent** of mass or size of system	**Dependent** on mass & size of system
	E.g. Electro motive force (E_{cell}), Pressure, temperature, specific/molar heat capacity	E.g. entropy, Heat capacity, work

Note: (1) If molar (1 mole) or specific (1 gm.) word is used before the **extensive quantity,** then it becomes **intensive quantity.**

(2) Ratio of two **extensive properties** becomes **intensive** property.

E.g. $(D = \frac{m}{v})$ Density is Intensive but mass & volume are **extensive**.

(3) All energy terms are **extensive properties**

(7) Thermodynamic equilibrium (state of constancy)

A state where there is no change in any observable property of a system with time

Sr. no	Type of Equilibrium	Condition at equilibrium
1.	**Mechanical equilibrium**	There should **not** be any **pressure gradient** (P = Constant)
2.	**Thermal equilibrium**	There should **not** be any **temperature gradient** (T = constant)
3.	**Chemical equilibrium**	There should **not** be any **concentration gradient** of any of the species in the system.

When all these **three equilibriums** are established **at a time,** then system & surrounding are in **thermodynamic equilibrium**.

(8) Types of thermodynamic process on basis of state/conditions

Thermodynamic process: Any method/process by which system can change its state from one state of Thermodynamic equilibrium to another state of Thermodynamic equilibrium.

Sr. No	Type of Process	Name of Parameter to be kept constant	Change in parameter
1	Isothermal Process	Temperature	dT = 0
2	Isochoric Process	Volume	dV = 0
3	Isobaric Process	Pressure	dP = 0
4	Adiabatic Process	Heat exchange with surrounding is zero	dq = 0

Classification of thermodynamic process on the basis of the way these processes are carried out

Reversible Process (Quasistatic)	Irreversible(Spontaneous/Natural/Real) Process
(i) This process is carried out by removing / adding **sand particles** one by one.	(i) This process is carried out by removing / adding **finite mass** one by one. & it is fast process.
(ii) $F_{Driving} - F_{opposing} = dF \approx 0$	(ii) $F_{Driving} - F_{opposing} = \Delta F \neq 0$
(iii) This process occur with **infinitesimally slow rate**. Thus this process completes in **infinite time**.	(iii) This process occur with **finite rate** & takes **finite time** for completion.
(iv) System & surrounding **always remain in equilibrium during the complete process**.	(iv) System & surrounding **remain at thermodynamic equilibrium only at initial & final state**. (Not during the process).

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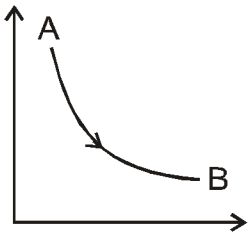
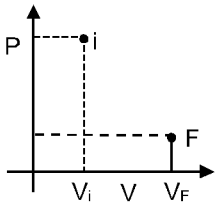
(v) State parameters are defined **throughout the process**. 	(v) State of system is defined **only at initial & final state and may be at some finite number of points.** 
vi) During the process, $P_{ext} = P_{int} = P_{gas}$	(vi) During the process P_{ext} **= constant ;** P_{gas} **= not defined**
Note: In a reversible Process, if opposite change is made reversibly, then both system & surrounding **get restored back to their original states.**	**Note**: In a Irreversible process if opposite change is made irreversibly, both system & surrounding will **not get restored back to their original states.**

(9) Heat & work: both are modes of energy transfer.

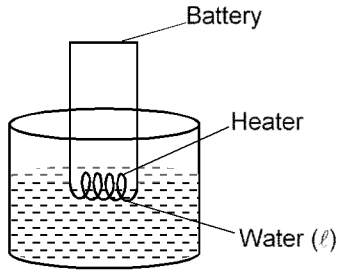
Heat (q): energy transfer **due to the temperature difference** between the system & the surrounding.

Work (w): energy transfer **which is not because of temperature difference** between system & surrounding.

E.g. Electrical work, mechanical work, Gravitational work, magnetic work (all are non **PV** work)

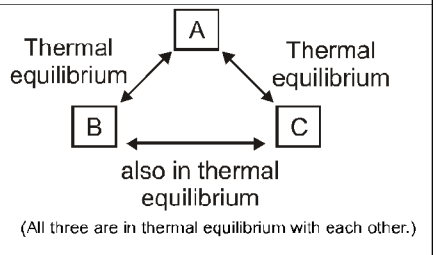
Note: Same energy transfer can be called as heat or work depending upon choice of system.

E.g.



Condition	**System**	**Surrounding**	**Energy term**
Case 1	Water	Rod + Battery	**Heat**
Case 2	Water + Rod	Battery etc.	**Work**

(10)

Zero Law of Thermodynamics	**Diagram**
If A system is in thermal equilibrium with B and B system is in thermal equilibrium with C, then A must be in thermal equilibrium with C. All systems are in thermal equilibrium wrt each other $T_1 = T_2$ & $T_2 = T_3 \Rightarrow T_1 = T_3$ or $(T_1 = T_2 = T_3)$	

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Enthalpy (H): Summation of **internal energy** and **pressure volume energy** of a system is known as **enthalpy**.

$H = U + PV$

Absolute value of H cannot be determined

differential form : $\Delta H = \Delta U + \Delta(PV) \Rightarrow \Delta H = \Delta U + (P_2V_2 - P_1V_1)$ (valid for ideal gas, real gas, solid, liquid etc.)

at constant pressure: $\Delta H = \Delta U + P.\ \Delta V$

at constant volume: $\Delta H = \Delta U + V.\ \Delta P$

for ideal gas (No reaction involved) : $\Delta H = \Delta U + nR\Delta T$

In case of chemical reactions and phase transition at constant temperature (T) (involving ideal gases only)

$\Delta H = \Delta U + \Delta n_g.RT$ $\quad [\Delta n_g = \Sigma$ S.C. of gaseous product(s) $- \Sigma$ S.C. of gaseous reactant(s)]

Heat transfer between system & surrounding:

at constant pressure, $q = \Delta H$	$\Delta H = nC_p\Delta T$
at constant volume, $q = \Delta U$	$\Delta U = nC_v\Delta T$

Note: Above formulae are valid for a process undergone by a fixed amount of ideal gas.

Relation between C_P & C_V (for ideal gas)

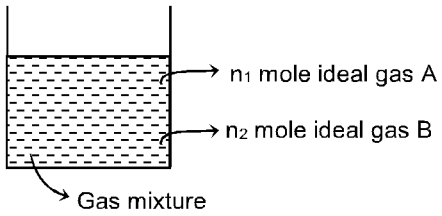
$C_P = \left(\frac{f+2}{2}\right)R$, $\quad C_V = \frac{fR}{2}$, $\quad C_P - C_V = nR$ for n mole(s) of ideal gas $\quad \therefore$ **$C_p > C_v$**

Poisson ratio: $\gamma = \frac{C_p}{C_v} = \frac{f+2}{f}$, $\quad C_p = \frac{\gamma R}{\gamma - 1}$, $\quad (C_v) = \frac{R}{\gamma - 1}$

Gas nature	DOF (f)	$C_p = \left(\frac{f+2}{2}\right)R$	$C_v = \left(\frac{fR}{2}\right)$	$\gamma = \left(\frac{f+2}{f}\right)$
(1) **Monoatomic gas**	3	5 R/2	3 R/2	5/3 = 1.67
(2) **Diatomic/Linear polyatomic**	5	7 R/2	5R/2	7/5 = 1.4
(3) **Polyatomic (Non-Linear)**	6	4 R	3R	4/3 = 1.33

Eg. $\begin{matrix} \text{gas} :: & He : & H_2 : & SO_2 \\ \gamma :: & 5/3 : & 7/5 : & 4/3 \end{matrix}$

Calculation of C_{Vmix}, C_{Pmix}, γ_{mix}



$$(C_V)_{mix} = \frac{n_1C_{V1} + n_2C_{V2}}{n_1 + n_2}$$

$$(C_P)_{mix} = \frac{n_1C_{P1} + n_2C_{P2}}{n_1 + n_2}$$

$$\gamma_{mix} = \frac{(C_P)_{mix}}{(C_V)_{mix}} = \frac{n_1C_{P1} + n_2C_{P2}}{n_1C_{V1} + n_2C_{V2}}$$

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Calculation for different processes on a fixed amount of ideal gas :

Isothermal process (T = constant) :

Reversible Isothermal	Irreversible Isothermal process
$\Delta U = nC_V.\Delta T = 0$, $\Delta U = q + w = 0$ $q = -w = nRT\ell n\left(\frac{V_2}{V_1}\right)$ $\Delta H = nC_P.\Delta T = 0$	$W = -P_{ext}\Delta V = -P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT}{P_1}\right)$ $\Delta U = q + w = 0 \quad \Rightarrow q = -w$ $\Delta H = nC_P.\Delta T = 0$

Isobaric process (P = Constant) :

$W = -P.\ \Delta V = = -nR\Delta T$, $\Delta U = nC_V\Delta T$, $q = \Delta H = nC_P\Delta T$, $\Delta H = \gamma\Delta U$

Isochoric process (V = constant) :

$q = \Delta U = nC_V\Delta T$, $\quad w = \int -P.dV = 0$, $\quad \Delta H = nC_P\Delta T = \gamma\Delta U$

Adiabatic process:

(a) Reversible adiabatic: dq = 0 $\Rightarrow$ q = 0 (for complete process)

$T.V^{\gamma-1}$ = constant or $P.V^{\gamma}$ = constant (valid for a fixed amount of ideal gas)

Note: (1) Both expressions are valid **only for reversible adiabatic process**.
(2) Temperature of a system can change even if heat exchange between system and surrounding is zero (q = 0)

Work of adiabatic process:

$$W = \Delta U = nC_V\Delta T = \frac{nR\Delta T}{\gamma - 1} = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$$

Note: (1) Work formula for both **Reversible** and **Irreversible adiabatic** processes is **same**.

(2) $\Delta H = \gamma.\Delta U$

(b) Irreversible adiabatic : (P_{ext} = constant, q = 0)

$\Delta U = q + w = w$

$$\Delta U = W_{irr} \quad \Rightarrow nC_V.(T_2 - T_1) = -P_{ext}(V_2 - V_1) = -P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

$$W = \Delta U = nC_V\Delta T = \frac{nR}{\gamma - 1}(T_2 - T_1) \ \&\ \Delta H = nC_P(T_2 - T_1)$$ **(Same for both reversible & irreversible process)**

Free expansion of an ideal gas :

Adiabatic expansion against vacuum (p_{ext} = 0)

$q = 0$ & $W = -p_{ext}.\ \Delta V = 0 \Rightarrow \Delta U = 0 \Rightarrow \Delta H = 0 \Rightarrow \Delta T = 0$

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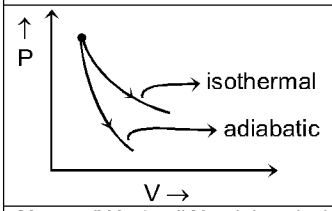
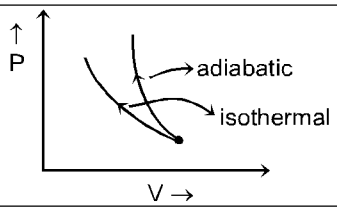
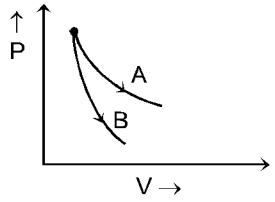
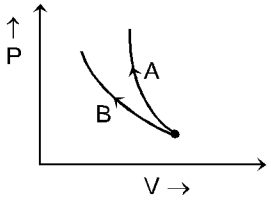
Comparison of slope of Reversible adiabatic & Reversible isothermal process on P–V diagram

(a)

Reversible isothermal	Reversible Adiabatic
PV = constant	PV^{γ} = constant
Slope : $-\frac{P}{V}$	Slope : $-\gamma\frac{P}{V}$
Slope of adiabatic (S_1) = γ. Slope of isothermal (S_2) ; $\lvert S_1\rvert > \lvert S_2\rvert$	

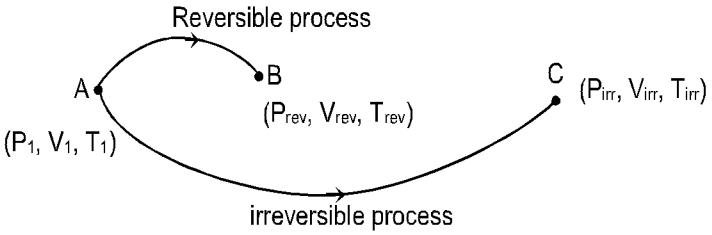
Note: Out of the two graphs, the more steeper one will be of reversible adiabatic.

(b) Comparison Graphs of Expansion & Compression for same & different ideal gas:

Expansion (for same gas)	**Compression (for same gas)**
	
Note : $\lvert W_{iso}\rvert > \lvert W_{adia}\rvert$ (work done by gas)	**Note**: $W_{iso} < W_{adia}$ (work done on gas)
Adiabatic Expansion (for different gases)	**Adiabatic Compression (for different gases)**
 Gas A → lesser γ Gas B → greater γ	 Gas A → greater γ Gas B → lesser γ

Graphical representation of Reversible & Irreversible adiabatic process :

Two states A and B can **never lie** both on a reversible as well as irreversible adiabatic path. There **lies only one unique** adiabatic path linkage between two states A and B.



$q = 0$, $\Delta U = q + w \Rightarrow \Delta U = w$ (for both **irreversible & reversible** path)

$(W_{gas})_{rev} < (W_{gas})_{irrev}$ (for expansion or compression for same volume change). So $\Delta U_{Irr} > \Delta U_{rev}$

$\therefore$ $(T_{final})_{irr} > (T_{final})_{rev}$ and hence $(P_F)_{irr} > (P_F)_{rev}$ (if **starting** from **same initial state and bringing final volume same**).

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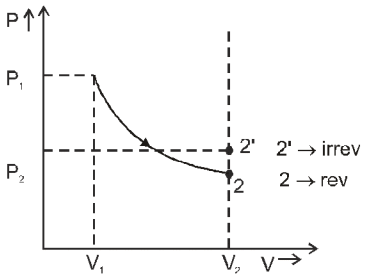
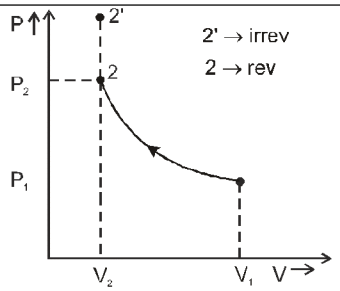
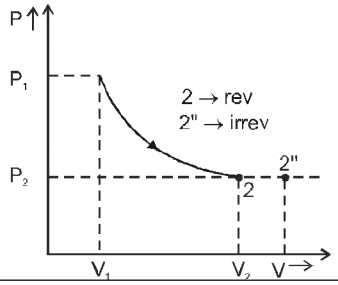
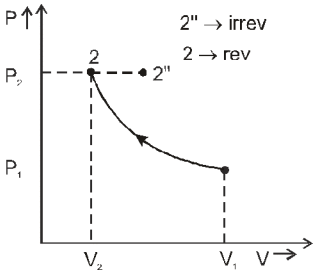
For expansion	For compression
Adiabatic expansion of an ideal gas causes **Cooling**.	**Adiabatic compression** of an ideal gas causes **Heating**.
$q = 0, \Delta U = q + w, \Delta U = w = -ve\ (T_f < T_i)$	$q = 0, \Delta U = q + w, \Delta U = w = +ve\ (T_f > T_i)$
	
	

On increasing the number of stages in irreversible process, points 2' and 2'' move down and left respectively trying to approach point 2.

Note : Starting from the same initial state, the same final state can be achieved following a reversible isothermal process & an irreversible isothermal process

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Polytropic process: (A reversible process for an ideal gas): PV^X **= constant (X ≠ 1)**

Special cases:

If X = 0	P = constant, isobaric process.
If X = γ	PV^γ = constant, Rev adiabatic process.

Work for Polytrophic process: $w = \frac{P_2V_2 - P_1V_1}{X-1} = \frac{nR\Delta T}{X-1}$ (X ≠ 1, isothermal process).

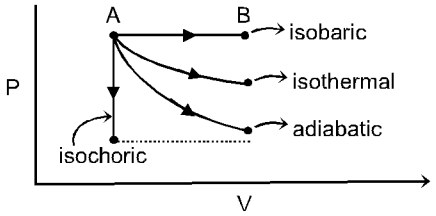
Molar heat capacity:

$C = (C_V) + \frac{R}{1-X}$

$q = nC\Delta T$; $\Delta U = nC_V\Delta T$;

$\Delta H = \gamma\Delta U$

Comparison of work for different type of Processes:



According to graph :

$|W_{isobaric}| > |W_{isothermal}| > |W_{adiabatic}| > |W_{isochoric}|$

Summary table:

S.No.	Process	w	q	ΔU	ΔH	Molar heat capacity
1.	Isochoric	0	$nC_v(T_f - T_i)$	$nC_v(T_f - T_i)$	$nC_p(T_f - T_i)$	C_v
2.	Isobaric	$W_{rev} = -P_{gas}(V_f - V_i)$	$nC_p(T_f - T_i)$	$nC_v(T_f - T_i)$	$nC_p(T_f - T_i)$	C_p
3.	Isothermal	$W_{rev} = -nRT\ln\left(\frac{V_f}{V_i}\right)$ $W_{irrev} = -P_{ext}(V_f - V_i)$	−w	$nC_v(T_f - T_i) = 0$	$nC_p(T_f - T_i) = 0$	± ∞
4.	Adiabatic	$\frac{P_fV_f - P_iV_i}{\gamma - 1}$	0	$nC_v(T_f - T_i) = w$	$nC_p(T_f - T_i)$	0
5.	Polytropic	$\frac{P_fV_f - P_iV_i}{X - 1}$	$nC(T_f - T_i)$	$nC_v(T_f - T_i)$	$nC_p(T_f - T_i)$	$C_v + \frac{R}{1-X}$

Determination of q, w, ΔU & ΔH during phase transition

Phase transition occurs at constant T & P (q = ΔH)

Sublimation: $M_{(s)} \longrightarrow M_{(g)}$	**Fusion:** $M_{(s)} \longrightarrow M_{(\ell)}$
Vaporisation : $M_{(\ell)} \longrightarrow M_{(g)}$	**Allotropic transition:**

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	M(**one allotrope**) ⟶ M (**other allotropic form**)

$\Delta T = 0$, during the transition. **K.E. remains same** but **P.E. gets changed (∵ intermolecular forces change),** so internal energy **(U)** get **changed**.

$\Delta U \neq 0$, $\Delta(PV) \neq 0$, $\Delta H \neq 0$

$q = \Delta H$ = Latent heat = mL ; $w = -P_{ext}(V_f - V_i)$

For ΔU calculation, we use ($\Delta U = q + w$).

Eg. For $X_{(\ell)} \longrightarrow X_{(V)}$, Vaporisation process

$w = -P_{ext}.(V_{vap} - V_{liq.})$ ($V_{liq.}$ is neglected, if not given)

$$= -P_{ext}(V_{vap}) = -P\left(\frac{nRT}{P}\right) = -nRT$$

For processes on solid & liquid:

$w = -P_{ext}\,\Delta V$, $\Delta H = \Delta U + \Delta(PV) = \Delta U + (P_2V_2 - P_1V_1)$

Second and third law of Thermodynamics:

Limitations of first law: First law does **not give any information** regarding **feasibility of a process** that means, whether the **process will occur or not in given direction**.

Process: can be Spontaneous or non-Spontaneous

Spontaneous Process (Self-accruing)	**Non Spontaneous process** (Not self-accruing, external work needed)
The process which has a **natural tendency** to occur **with or without initiation** under the **given set of conditions** is called **a Spontaneous Process** It is an **irreversible process**	Process which does **not have a tendency** to occur by itself under the given set of conditions is called **non-Spontaneous process**.
E.g. (1) Heat flow from high temperature to low temperature (2) Flow of gas from high pressure to low pressure. (3) Mixing of gases (4) Rusting of iron (5) $H_2O(s) \rightarrow H_2O(\ell)$ (at 1 atm, T > 273 K)	E.g. (1) Heat flow from low temperature to high temperature (2) Flow of gas from low pressure to high pressure (3) Separation of gas (4) Formation of fresh iron from rust. (5) $H_2O(s) \rightarrow H_2O(\ell)$ (at 1 atm, T < 273 K)

Entropy (S):

This is a property of a system which measures **degree of randomness** or **degree of disorderness** of a system.

It is **state function**, **extensive property**, can be **determined absolutely.**

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Comparison of entropy:

In general:

(1) $S_{solid} < S_{liquid} < S_{gas}$	(2) No. of mole of gas increase, **S** ↑	(3) **n** & **V** same, as **T** ↑ , **S** ↑
(4) **n** & **T** same, as **V** ↑ or **P** ↓ , **S** ↑	(5) **n** same, **P,V,T** same $S_{mixture} > S_{pure}$	(6) **n** same, **P,V,T,** pure gas samples **molar mass** ↑ , **S** ↑
(7) **n,P,V,T, atomicity same,** as **molar mass** increase ,**S** increase	(8) $S_{perfect\ crystal} < S_{imperfect\ crystal}$	(9) On increasing **temperature, entropy** ↑ **but** ΔS ↓
Heat **added** / **removed** at low temperature will cause more change in entropy	(10) Entropy of **gas decreases** when it is dissolved in **a solid or liquid.**	**Comparison Order of entropy: Mole > Mixture > atomicity > molar mass**

Example: (1) During **boiling** of egg, Entropy **increases.**

(2) **During stretching** of rubber, Entropy **Decreases.**

(3) Entropy of solution containing nonvolatile solute is greater than that of pure solvent.

Calculation of entropy: $dS = \frac{dq_{Rev}}{T}$

For System: $dS_{sys} = \frac{dq_{Rev}}{T_{sys}}$, $\Delta S_{sys}(rev/irrev) = \int \frac{dq_{Rev}}{T_{sys}}$;

For Surrounding: $dS_{surr} = \frac{-dq_{sys}}{T_{surr}}$, $\Delta S_{surr} = \frac{-q_{sys}}{T_{surr}}$ **(∵ T_{surr} remains constant)**

Determination of ΔS_{system} for a system containing ideal gas:

$$\Delta S = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) \quad \text{or} \quad \Delta S = nC_p \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{P_1}{P_2}\right)$$

Note: Above formulae are applicable for both **reversible & irreversible** processes because Entropy is a **state function**.

Entropy calculation for different processes on a system involving fixed amount of ideal gas:

Sr. No.	Process	Reversible / Irreversible	ΔS_{system}	$\Delta S_{surrounding}$	$\Delta S_{Total} = \Delta S_{sys} + \Delta S_{surr.}$
1	Isothermal	Reversible	$nR\ln\left(\frac{V_2}{V_1}\right)$	$-nR\ln\left(\frac{V_2}{V_1}\right)$	0
		Irreversible	$nR\ln\left(\frac{V_2}{V_1}\right)$	$\frac{-P_{ext}(V_2 - V_1)}{T_{2(surr)}}$	Positive
2.	Adiabatic	Reversible	**0**	**0**	**0**

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		Irreversible	$nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right)$	0	Positive
3.	Isobaric	Reversible	$nC_p \ln\left(\frac{T_2}{T_1}\right)$	$-nC_p \ln\left(\frac{T_2}{T_1}\right)$	0
4.	Isochoric	Reversible	$nC_v \ln\left(\frac{T_2}{T_1}\right)$	$-nC_v \ln\left(\frac{T_2}{T_1}\right)$	0
		Irreversible	$nC_v \ln\left(\frac{T_2}{T_1}\right)$	$\frac{-nC_v(T_2-T_1)}{T_{2(surr)}}$	Positive

Sr. No.	Process	Reversible / Irreversible	ΔS_{system}	$\Delta S_{surrounding}$	$\Delta S_{Total} = \Delta S_{sys} + \Delta S_{surr.}$
1.	Heating of solid and liquid	Reversible	$nC_m \ln\left(\frac{T_2}{T_1}\right)$	$-nC_m \ln\left(\frac{T_2}{T_1}\right)$	0
		Irreversible	$nC_m \ln\left(\frac{T_2}{T_1}\right)$	$\frac{-nC_m(T_2-T_1)}{T_{surr}}$	Positive
2.	Phase transition (at constant Temperature & Pressure)	Reversible	$\frac{\Delta H_{Phase\ transition}}{T_{at\ reversible}}$	$-\frac{\Delta H_{Phase\ transition}}{T_{at\ reversible}}$	0
		Irreversible	$\frac{\Delta H_{Phase\ transition}}{T_{at\ reversible}}$	$\frac{\Delta H_{Phase\ transition}}{T_{surrounding}}$	Positive

Note: (1) All **reversible processes** have **total entropy change** (of universe) as **zero**.
(2) All **irreversible processes** have **total entropy change** (of universe) as **positive**.
(3) **Reversible Adiabatic process** is also called **isoentropic** process.
(4) For solid and liquid: $C_{p,m} \simeq C_{V,m} \simeq C_m$

(5) $\Delta S_{vap} = \frac{\Delta H_{vap}}{T_{B.P}}$, $\Delta S_{Fusion} = \frac{\Delta H_{Fusion}}{T_{M.P}}$

Example: $H_2O(s) \longrightarrow H_2O(\ell)$, given: ΔH_{Fusion} = 50 KJ/mole. Calculate ΔS_{system}, $\Delta S_{surrounding}$ and ΔS_{total} at given temperature and pressure conditions for 1 mole.

Pressure and surr temperature	ΔS_{system} (kJ/K)	ΔS_{surr}(kJ/K)	ΔS_{Total}(kJ/K)
1 atm, 250 K	50/273	–50/250	Negative
1 atm, 273 K	50/273	–50/273	0
1 atm, 300 K	50/273	–50/300	Positive

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Second law of thermodynamics:

Entropy of an isolated system **must increases** during **spontaneous change** and it goes to **maximum at equilibrium**.

Spontaneity of process can be determined by **calculation of change in entropy** (ΔS)

For Spontaneous process	$\Delta S_{isolated\ system} > 0$ **or** $\Delta S_{universe} > 0$ **or** $\Delta S_{system} + \Delta S_{surrounding} > 0$
For Reversible process	$\Delta S_{system} + \Delta S_{surrounding} = 0$
For Non spontaneous process	$\Delta S_{system} + \Delta S_{surrounding} < 0$

Gibb's energy (G) : a system parameter to predict the spontaneity of a process/chemical reaction.

$G = H - T.S$ **Gibbs Helmholtz equation**

It is **state Function**, **extensive property** and **can't be determined absolutely.**

At constant temperature, $\Delta G = \Delta H - T.\Delta S$

Physical significance of ΔG:

$$dG = dw_{rev(non\ PV)} + V.dp - S.dT$$

At constant P & T $dG = dw_{rev(non\ PV)}$ (**useful work**)

Decrease in Gibb's energy at constant P & T = Maximum useful work that can be done by the system.

If No Non-PV work is involved, then at **constant temperature for ideal gas,** $\Delta G = nRT\ln\left(\frac{P_2}{P_1}\right)$

Other Criteria of Spontaneity:

At **constant temperature & pressure** $(\Delta G_{system})_{T,P}$ < 0 for spontaneous process/reaction

> 0 for non-spontaneous process/reaction

$= 0$ for process/reaction at equilibrium

Sr. No	ΔH	ΔS	Temperature condition & ΔG
Case-1	Endothermic reaction (ΔH = +ve)	Positive	Reaction is always Spontaneous at **all temperature** (ΔG= negative)
		Negative	At T_{Low} Reaction is **Spontaneous** (**Forward**) At T_{High} Reaction is **Non Spontaneous** (**backward**)
Case-2	Endothermic reaction (ΔH = +ve)	Positive	At T_{low} Reaction is **Non Spontaneous** (**backward**) At T_{High} Reaction is **Spontaneous** (**Forward**)
		Negative	Reaction is always Non Spontaneous at **all temperature** (ΔG=Positive)

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Relationship between ΔG, ΔG°, Q & K_{eq}

Standard condition (**activity = 1, Temperature: Any fixed value, normally 25°C**)

$\Delta G = \Delta G^\circ + RT \ln Q$

Under **standard condition**: **$\Delta G = \Delta G^\circ$ (as Q =1)**

For **spontaneous Reaction**: **$\Delta G < 0$ (ΔG° may be positive, negative or zero)**

For **spontaneous reaction under standard conditions**: ΔG or $\Delta G^\circ < 0$

At **equilibrium** $\Delta G = 0$, $\Delta G^\circ = -RT \ln K_{eq}$

$\Delta G = RT \ln \left(\frac{Q}{K_{eq}}\right)$ $\therefore Q < K_{eq} \Rightarrow \Delta G =$ **negative (spontaneous in forward direction)**

$\therefore Q = K_{eq} \Rightarrow \Delta G =$ **0 (at equilibrium)**

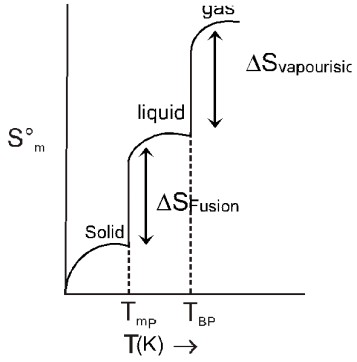
$\therefore Q > K_{eq} \Rightarrow \Delta G =$ **positive (non-spontaneous in forward direction; spontaneous in backward direction)**

Third law of thermodynamics:

Standard molar entropy of a **perfect crystalline** pure substance at **'Zero Kelvin'** is taken **to be zero**.

$$\int_{S_{O'K}}^{S_{T'K}} dS = \int_{'O'K}^{'T'K} \frac{dq_{rev}}{T} \Rightarrow S_T - S_0 = \int_{'O'K}^{'T'K} \frac{dq_{Rev}}{T} = \mathbf{S_T} \ (\because S_{O'K'} \text{ is zero}) \therefore \boxed{S^\circ_m = \int_0^T \frac{nC_p dT}{T}}$$

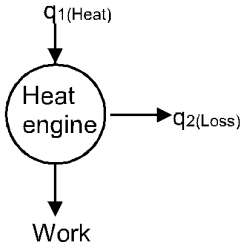
$$S_{T(K)} = \int_{'O'K}^{M.P} \frac{n_{Solid} C_m . dT}{T} + \frac{n.\Delta H_{fus}}{T_{M.P}} + \int_{M.P}^{B.P} \frac{n_{liquid} C_m . dT}{T} + \frac{n.\Delta H_{vap}}{T_{B.P}} + \int_{B.P}^{T} \frac{n_{gas}\, C_m dT}{T}$$



Reversible cyclic process: $\Delta S_{sys} = 0$, $\Delta S_{surr} = 0$

Irreversible Cyclic process (spontaneous): $\Delta S_{sys} = 0$, $\Delta S_{surr} > 0$ (As $\Delta S_{Total} > 0$)

Efficiency of reaction = $\boxed{\frac{\Delta G}{\Delta H} \times 100}$



$\boxed{\text{Efficiency} = \frac{w}{q_1} \times 100}$

E.g. $H_2O(\ell)_{1atm,373K} \longrightarrow H_2O(g)_{2atm,373K}$ Calculate ΔG for the conversion of 1 mole of water into steam under given

Sol: $H_2O(\ell)_{1atm,373K} \longrightarrow H_2O(g)_{1atm,373K} \longrightarrow H_2O(g)_{2atm,373K}$ $\Delta G_1 = 0$, ΔG_2 $nRT\ell n\left(\frac{P_2}{P_1}\right) = 1 \times R \times 373 \ell n\left(\frac{2}{1}\right)$

$= R \times 373\ \ell n 2$

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E.g. $H_2O(\ell) \longrightarrow H_2O(g)$ Comment on sign of ΔS system, surrounding, total and $\Delta G_{(System)\ T.P}$ for shown process.

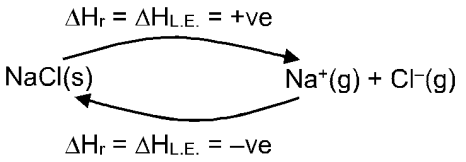
Condition	ΔS_{system}	$\Delta S_{surrounding}$	ΔS_{Total}	$\Delta G_{(System)\ T.P.}$
1 atm, 350 K	Positive	Negative	Negative	Positive
1 atm, 373 K	Positive	Negative	Zero	Zero
1 atm, 400 K	Positive	Negative	Positive	Negative
2 atm, 373 K	Positive	Negative	Negative	Positive
0.5 atm, 373 K	Positive	Negative	Positive	Negative

Born Haber Cycle (an application of Hess Law) :

Standard Lattice Enthalpy of a crystal (ΔH^{o}_{LE}) : The standard enthalpy change to completely separate one mole of a solid ionic compound into gaseous ion (Lattice Enthalpy : Positive) and vice versa (Lattice Enthalpy: negative)

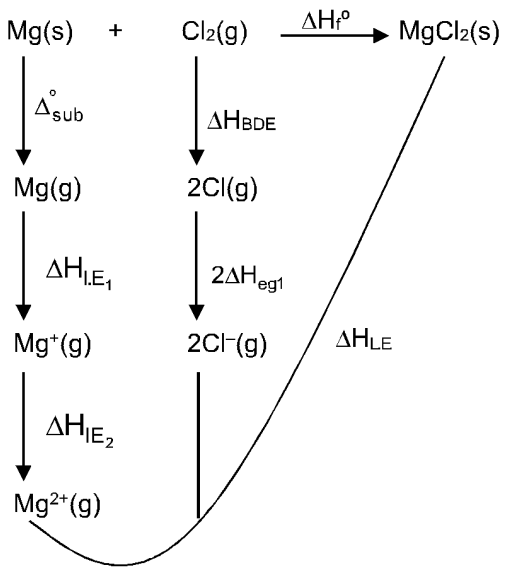
Note:

The large the magnitude of Lattice Enthalpy, the more stable is ionic compound.



Formation of ionic compounds:

E.g. Setup Born Haber cycle for formation of $MgCl_2$ crystal.



$$\Delta H^{o}_{F} = \Delta H_{sub} + \Delta H_{IE_1} + \Delta H_{IE_2} + \Delta H_{BDE} + 2.\Delta H_{eg1} + \Delta H_{L.E}$$

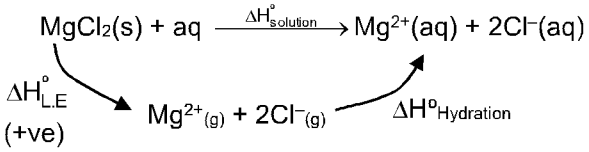
Note: (1) All enthalpies should be put with sign.

(2) If $\Delta H_{L.E}$ is given and any of above other quantity is to be determined, then $\Delta H_{L.E}$ should be put in above expression with –ve sign.

Enthalpy of Solution (of ionic compounds) (ΔH_{sol})

The enthalpy change upon dissolving 1 mole of a ionic solid compound in excess solvent and to be broken into aqueous ions.

Eg.



$|\Delta H_{L.E}| > |\Delta H_{Hyd}|$ then compound is insoluble

$|\Delta H_{L.E}| < |\Delta H_{Hyd}|$ then compound is soluble

$\Delta H^{\circ}_{solution} = \Delta H^{\circ}_{L.E.} + \Delta H^{\circ}_{Hydration}\ (Mg^{2+}_{(g)}) + 2\Delta H^{\circ}_{Hydration}\ Cl^{-}_{(g)}$

Note: If $\Delta H_{L.E}$ is given and any of above other quantity is to be determined, then $\Delta H_{L.E}$ should be put in above expression with +ve sign.

Bond enthalpy / Bond dissociation enthalpy: Defined only in gaseous state (for covalent compounds)

The average enthalpy change to break one mole of a bond in gaseous state into gaseous atoms.

$$\Delta H^{\circ}_{Reaction} = \sum (B.E.)_{Reactants} - \sum (B.E.)_{Products}$$

E.g $H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$ $\Delta H^{\circ}_{rxn} = BE_{H-H} + BE_{Cl-Cl} - 2.BE_{HCl}$	$CH_3OH(\ell) + \frac{3}{2} O_2(g) \longrightarrow CO_2(g) + 2H_2O(\ell)$ ΔH°_{rxn} = (ΔH° vap of $CH_3OH(\ell)$ + 3BE of C–H + BE_{C-O} + BE_{OH}) + $\frac{3}{2} BE_{O=O}$ – (2$BE_{C=O}$ + 2.ΔH°_{vap} of $H_2O(\ell)$+ 4BE_{O-H})

Standard enthalpy of atomization: The standard enthalpy change during breakage of one mole of a substance into gaseous atoms.

Eg. **C(grap,s) ⟶ C (g) ;** ΔH°_{r} = $\Delta H^{\circ}_{atomisation}$ **of C(grap,s) =** $\Delta H^{\circ}_{sublimation}$ **of C(grap) =** ΔH°_{f} **C(g)**

$H_2(g) \longrightarrow 2H(g)$; ΔH°_{r} = $\Delta H^{\circ}_{atomisation}$ **of $H_2(g) = BE_{H-H}$ or enthalpy of dissociation of $H_2(g) = \Delta H^{\circ}_{f}$ H(g)**

$O_2(g) \longrightarrow 2O(g)$; ΔH°_{r} = $\Delta H^{\circ}_{atomisation}$ **of $O_2(g) = BE_{O=O}$ or enthalpy of dissociation of $O_2(g) = \Delta H^{\circ}_{f}$ O(g)**

$CH_4(g) \longrightarrow C(g) + 4H(g)$, $\Delta H^{\circ}_{atomisation}$ **of $CH_4(g) = 4(BE_{C-H})$**

$C_2H_6(g) \longrightarrow 2C(g) + 6H(g)$, $\Delta H^{\circ}_{atomisation}$ **of $C_2H_6(g) = (BE_{C-C}) + 6(BE_{C-H})$**

$C_{(Dia)} \longrightarrow C(g)$, $\Delta H^{\circ}_{atomisation}$ **(C, Diamond) = 4 × $\frac{1}{2}$ (BE_{C-C}) = 2(BE_{C-C})**

$SiO_2(s) \longrightarrow Si(g) + 2O(g)$, $\Delta H^{\circ}_{atomisation}$ **(SiO_2) = 4(BE_{Si-O})**

Resonance Enthalpy (ΔH°_{RE} **, always –ve) :**

When one mole of most stable resonating structure is converted into resonance hybrid, then enthalpy change is known as resonance enthalpy.

Resonance enthalpy = $\Delta H^{\circ}_{f(obs)} - \Delta H^{\circ}_{f\ (Theoretical)}$

$= \Delta H^{\circ}_{Hydrogenation(Theoretical)} - \Delta H^{\circ}_{Hydrogenation(observed)}$

Standard enthalpy of neutralization (ΔH°_{Neut} = –ve always):

The standard enthalpy change when 1 equivalent of a strong acid (1 mole of H^{+}_{aq}) combines with 1 equivalent of a strong base (1 mole of OH^{-}) in dilute aqueous solution

$H^{+}_{(aq)} + OH^{-}(aq) \longrightarrow H_2O(\ell)$, ΔH°_{Neu} = – 57.1 kJ/mole

or $HCl_{(aq)} + NaOH(aq) \longrightarrow H_2O(\ell) + NaCl(aq)$ ΔH°_{Neu} = – 13.7 kCal/mole

Standard enthalpy of Ionization of weak acid/base (ΔH°_{ion} = +ve always):

If acid/base or both are weak, then released energy upon neutralisation will be less than the 57.1 kJ/mole because some energy is consumed to ionize the weaker component (acid/base or both).

It is known as enthalpy of ionization of weak component (acid/base or both).

Note: 1. Weaker is the acid/base, $\Delta H_{ionisation}$ will be higher, $|\Delta H_{Neut}|$ will be less

2. $\Delta H_{dissociation}$ of weak acid/base = 57.1 (kJ) – $|\Delta H_{Neu}|$ weak acid/base

3. All strong acids & bases have $\Delta H_{ionisation}$ = 0

Standard enthalpy of transition (ΔH°_{trans}) :

Standard enthalpy change when one mole of one allotropic form changes to another.

Eg. $C_{(graphite)} \longrightarrow C_{(diamond)}$, $\Delta H^{\circ}_{transition}$ = 1.9 kJ/mole (can be determined from ΔH°_{C} data of both).

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ATOMIC STRUCTURE

1. All cathode rays **irrespective** of the gas used have specific charge (e/m) **= 1. 76×10^{11} C/kg.**

2. Specific charge for anode rays is maximum for **hydrogen.**

3. For discovery of neutron : $^{9}_{4}Be + \underset{(\alpha\text{-partical})}{^{4}_{2}He} \longrightarrow {}^{12}_{6}C + {}^{1}_{0}n$

4.

(i) Order of radius of atom $\approx 10^{-10}$ **m**
(ii) Order of radius of nucleus $\approx 10^{-15}$ **m**

5. Radius of nucleus of any atom : $R_{nucleus} = R_0(A)^{1/3}$, A → Atomic mass no. $R_0 = 1.3\times10^{-15}$ **m**

6. Calculate the radius of nucleus of $^{27}_{13}Al$, $R_0 = \frac{4}{3}\times10^{-15}$m

Sol. $R = R_0(A)^{1/3} = \frac{4}{3}\times10^{-15}(27)^{1/3} = 4\times10^{-15}$m

7. Distance of closest approach (R_{min}) of α-particle ($^{4}_{2}He$).

$$\left[R_{min} = \frac{4KZe^2}{M_\alpha v_\alpha^2}\right]$$

Z → No of proton in nucleus of atom

$K = 9\times10^{9}$ Nm^2/C^2

8. For α particle closest distance of approach is R when v_α = v. Then calculate % change in R_{min} when $v_\alpha = v/2$

Sol. $R_{min} = \frac{4kze^2}{m_\alpha v_\alpha^2}$ $\quad R_i = \frac{4kze^2}{m_\alpha v^2}$ .....(1)

$$R_f = \frac{4kze^2}{m_\alpha\left(\frac{v}{2}\right)^2} = 4R \qquad .....(2)$$

$$\% \text{ change in } R = \frac{R_f - R_i}{R_i}\times100 = \frac{4R - R}{R}\times100 = 300\%$$

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9. Charge is quantised and $q_T = ne$ (n is integer)

$q_T = ne = i.t.$ i = current (amp.), t = time (sec.), n = No. of electron

e = charge on electron = 1.6×10^{-19} C

10. Force acting between two charged particle q_1 & q_2 at distance R

$$F = \frac{Kq_1q_2}{R^2}$$

11. Potential energy between two charged particle q_1 & q_2 at distance R

$$\textbf{P.E.} = \frac{Kq_1q_2}{R}$$

12. **For opposite charge system, PE= –ve,** **i.e. stable system**

for same charge system, PE = +ve **i.e. unstable system**

13. **Calculate PE in Na^{10+} ion with R separation between electron and nucleus.**

Sol. $q_1 = (-e), q_2 = Ze = 11e^-$, $PE = \frac{K.q_1q_2}{R} = \frac{K(-e)(11e)}{R} = \left(\frac{1}{4\pi\varepsilon_0}\right)\left(\frac{-11e^2}{R}\right)J$

14. **Total energy = K.E. + P.E.**

15. **Potential energy variation for same / appositive charges system when approaching / separating.**

For same charge		**For opposite charge**	
When going away (R↑)	**when charges approaching, (R↓)**	**When going away(R↑)**	**when charges approaching, (R↓)**
$PE = \frac{K.q_1q_2}{R}$ **PE↓,KE↑, T.E = constant**	$PE = \frac{K.q_1q_2}{R}$, **PE↑,KE↓, T.E = constant**	$PE = \frac{K.q_1q_2}{R}$ **PE↑,KE↓, T.E = constant**	$PE = \frac{K.q_1q_2}{R}$, **PE↓,KE↑, T.E = constant**

16. **If a charge q is placed on a surface having potential 'V', then value of PE is**

$\boxed{PE = q.V}$ q→ in coulombs , V → in volts, PE → in J

17. 1 ev= **1.6×10^{-19}** J= **96** kJ/mole = **23** K.Cal/mole

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18. Isotopes: **Having same no. of proton.** (Z → same, A → Different) eg. → $^{14}_{6}C$, $^{12}_{6}C$

19. Isotons : **Having same no. of neutron.** $(A_1-Z_1) = (A_2-Z_2)$

eg. → $^{39}_{19}K$, $^{40}_{20}Ca$

20. Isobars : **Having same no. of nucleons or mass no.** (A→ same, Z →different)

eg. → $^{40}_{18}Ar$, $^{40}_{20}Ca$

21. Isoelectronic : **Having same no. of total electron.**

eg. → N^{3-}, O^{2-},F^-,Ne,Na^+,Mg^{2+},Al^{3+}

eg. → C_2^{2-},N_2,NO^+,CO

22. Wave no. $(\bar{\nu}) = \frac{1}{\lambda}$ λ→ wavelength ; SI unit : m^{-1}

23. Time period (T) = $\frac{1}{\nu}$, ν → frequency; SI unit : Cycle/sec or Sec^{-1} or Hz

24. Relation between v, ν & λ $\nu = \frac{v}{\lambda}$, v → velocity , ν → frequency, λ → wavelength

25. For Electromagnetic radiation (v), speed of light (c) = 3×10^8 m/s

26. An electromagnetic radiation has wavelength 500 nm. Calculate wave number, frequency.

Sol. λ = 5000Å , wave number $(\bar{\nu}) = \frac{1}{\lambda} = \frac{1}{5000}\times10^{10} = 2\times10^6 m^{-1}$

$\boxed{1\,nm = 10Å}$ $c = \nu.\lambda \Rightarrow 3\times10^8 = \nu \times 5000\times10^{-10} \Rightarrow \nu = 6\times10^{14}$ Hz (sec^{-1})

27. Order of wavelength in electromagnetic spectrum

γ – rays < x-rays< ultraviolet rays< visible rays < infrared rays < Micro wave < radio wave

λ – **min** λ – **max**

ν – **max** ν – **min**

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28. Range of wavelength of visible region : **3800Å to 7600Å** (V I B G Y O R)
(**violet color** having **minimum** wavelength and **red color** having **maximum** wavelength in visible region)

29. Energy of one photon = $E_P = h\nu = \frac{hc}{\lambda}$, h-planck's constant : **6.626×10^{-34} J.sec.**

λ– wave length of light (m), c– speed of light (m/s), ν– frequency (sec^{-1})

$$\boxed{E_p = \frac{hc}{\lambda}J = \frac{12400}{\lambda(Å)}ev}$$

30. An EMR having wavelength 4000 Å. Calculate energy of one photon ?

Sol. $E_P = \frac{hc}{\lambda} = \frac{12400}{\lambda(Å)} = \frac{12400}{4000} = 3.1 \text{ ev} = 3.1\times1.6\times10^{-19} \text{ J} = 4.96\times10^{-19} \text{ J}$

31. For a light source having input power 'p' watt and efficiency (η) emitting N photon in t sec.

$$\boxed{\frac{\eta}{100}P\times t = \frac{N.hc}{\lambda}}$$

32. **A light source of 50 watt having efficiency of 75% calculate number of photon in 4 sec. if wavelength of light source 620 nm.**

Sol. **From energy conservation :**
Electrical energy = Light energy

$$E = \frac{\eta}{100}P\times t = \frac{N.hc}{\lambda} \Rightarrow \frac{50\times3}{4}\times4 = N\left(\frac{12400}{6200}\right)\times1.6\times10^{-19}$$

$$150 = N\,(2)\times1.6\times10^{-19} = N(3.2)\times10^{-19}$$

$$N = 4.6\times10^{20}$$

33. **Photoelectric effect : When certain metal is exposed to beam of light e^- are ejected and this phenomenon is called photoelectric effect.**

34. **Work function of metal : The minimum energy required to ionized the e^- from metal surface and it is fixed for each metal.**

35. **w = work function = $\phi = h\nu_0 = \frac{hc}{\lambda_0}$,** ν_0 = Threshold frequency

λ_0 = Threshold wavelength

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36. **Threshold frequency (ν_0) :** The **minimum frequency** of radiation required to eject the e^- from metal surface

37. **Threshold wavelength (λ_0)** : The **maximum Wavelength** of radiation required to eject the e^- from metal surface

38. **Light intensity:** No of photon falling per unit area per unit time.

39. The no of ejected electron α **intensity or brightness of light**

40. If frequency increases ($\nu > \nu_0$), then maximum **kinetic energy of electron (KE_e) will increase.**

41. Kinetic energy equation of ejected photo electron :

$$(KE)_{e_{max}} = E_p - W = h\nu - h\nu_0 = h(\nu - \nu_0) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

42. Conditions of Photo electric effect , $E_{photon} \geq W$ $(\nu \geq \nu_0)$ $(\lambda \leq \lambda_0)$

43. For a metal (x) having work function 3.1 eV. calculate threshold frequency (ν_0) & wavelength (λ_0).

Sol. $W = \frac{hc}{\lambda_0} = \frac{12400}{\lambda_0(\text{Å})} = 3.1\ eV \quad \Rightarrow \quad \lambda_0(\text{Å}) = 4000\text{Å}\ ,\ \nu_0 = \frac{c}{\lambda_0} = \frac{3\times10^8}{4000\times10^{-10}} = 7.5\times10^{14}\text{Hz}$

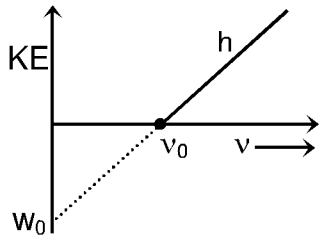
44. A light having Wavelength 3000Å. Fall on metal surface having work function 2 eV. calculate $(KE)_{max}$ of ejected photo electron.

Sol. $E_P = \frac{12400}{3000} = 4.13\ eV\ ,(KE)_e = E_P - \phi = 4.13 - 2 = 2.13\ eV$

Diagram related to Photoelectric effect.

45. **KE v/s frequency**

$KE = h\nu - w_0$



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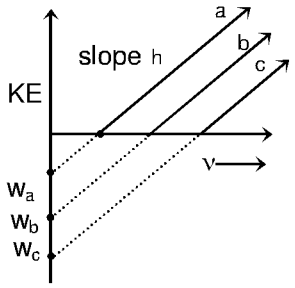
46. **KE v/s frequency**



$KE=h\nu-w=h(\nu-\nu_0)$

work function of metal a,b,c,$\rightarrow w_c>w_b>w_a$

47. **Stopping Potential (Vs):**

The potential difference applied across the metal surface to reduce the **maximum KE** of ejected electron to **zero** is called stopping potential

$\boxed{KE_{max} = eV_s}$ KE_{max} in eV and Vs in V have some numerical value.

48. **Orbit angular momentum** $\boxed{mvr = \frac{nh}{2\pi}}$

$m \rightarrow$ mass of e^-

$v \rightarrow$ velocity of e^-

$r -$ radii of orbit

$h \rightarrow$ Plank constant $(6.6 \times 10^{-34}$ Js)

$n \rightarrow$ orbit number

49. Possible values of angular momentum for single electron system :

(A*) $\frac{h}{\pi}$ (B*) $\frac{1.5h}{\pi}$ (C*) $\frac{h}{2\pi}$ (D*) $\frac{6h}{4\pi}$

Sol. Angular momentum = $\frac{nh}{2\pi}$, n= 1,2,3.....

50. Bohr model is applicable only for **single electron system** (H-like species)

Like: H, He^+ , Li^{2+}, Be^{3+}

51. Radius of n^{th} orbit $\boxed{r_n = r_0 \frac{n^2}{Z} Å}$

$r_0 \rightarrow 0.529Å$

$z \rightarrow$ atomic no, $n \rightarrow$ orbit no

52. Calculate radius ratio for any single electron system $r_1 : r_2 : r_3 : r_4$

Sol. $r_n = \frac{r_0 n^2}{z}$ $r_1 : r_2 : r_3 : r_4 = n_1^2 : n_2^2 : n_3^2 : n_4^2 = 1^2 : 2^2 : 3^2 : 4^2 = 1 : 4 : 9 : 16$

53. $r_H : r_{He^+} : r_{Li^{2+}}$ (for n^{th} orbit)

Sol. $\frac{r_0 n^2}{z_1} : \frac{r_0 n^2}{z_2} : \frac{r_0 n^2}{z_3} = \frac{1}{1} : \frac{1}{2} : \frac{1}{3} = 6 : 3 : 2$

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54. **Velocity of an electron in n^{th} orbit :** $v_n = v_0 \frac{Z}{n}$ m/s , $v_0 = 2.18\times10^6$ m/s

55. **Calculation of energy of e^- in any orbit :**

(1) $KE = \frac{1}{2}M_e v^2 = \frac{1}{2}\frac{KZe^2}{r}$

(2) $PE = \frac{-KZe^2}{r}$

(3) $TE = KE + PE = -\frac{1}{2}\frac{KZe^2}{r}$

56. **Relation between KE & PE & Total energy (TE)**

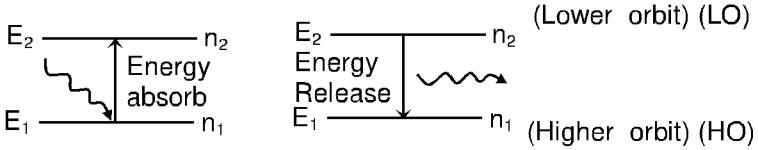
$\boxed{TE = -KE = \frac{PE}{2}}$ for Bohr model$\rightarrow$ $\left.\begin{matrix} KE = +ve \\ PE = -ve \\ TE = -ve \end{matrix}\right]$ always

When electron moves away from nucleus PE increases, KE decreases & TE increases

E_n : Energy of e^- in n^{th} orbit T.E. $E_n = -\frac{13.6Z^2}{n^2}$ eV/species

57. On moving from n =1 to ∞, Total energy of e^- becomes zero.

58. Energy change during any transition :



(Lower orbit) (LO)

(Higher orbit) (HO)

$\Delta E = E_{H.O} - E_{L.O} = 13.6\, Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ eV $(n_1 < n_2)$

59. Calculate total energy, kinetic energy and potential energy of hydrogen atom in orbit number 1,2,3,4,5 ?

Sol. Total energy $(E_n) = \frac{-13.6 \times Z^2}{n^2}$ eV, for hydrogen Z = 1

Orbit number	Total energy (eV) $TE = \frac{-13.6\times1^2}{n^2}$	Kinetic energy (eV) Total Energy = Kinetic energy (TE= –KE)	Potential energy (eV) PE = 2×TE
n=1	–13.6	13.6	–27.2
n=2	–3.4	3.4	–6.8
n=3	–1.51	1.51	–3.02
n=4	–0.85	0.85	–1.7
n=5	–0.54	0.54	–1.08

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60. Time period of revolution of e^- in orbit = $T_n = T_0 \frac{n^3}{Z^2}$ sec

61. Frequency of revolution of e^- in orbit $= F_n = \frac{1}{T_n} = \frac{f_0 . Z^2}{n^3}$ sec^{-1}

Definitions for single e^- system :

62. **Ground state** : Lowest energy state of atom /ion ($n_1 = 1$)

63. **Excited state**: Energy state other than ground state (n=2,3,4,5,.....)

n^{th} excited state $\Rightarrow$ orbit number = n+1 (i.e. 1^{st} excited state: n= 1+1=2)

64. **Ionization energy (IE):** Minimum energy required to remove the electron to infinity from ground sate

$n_1 = 1, n_2 = \infty$

$$\Delta E = 13.6 \times Z^2 \left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) = 13.6 \times Z^2 \text{ eV}$$

65. **Excitation energy (EE)** : Energy required to excite the electron from ground state to an other excited state is known excitation energy of that state $n_1 = 1$, n_2=2,3,4, ...∞

$$EE = 13.6 \times Z^2 \left(\frac{1}{1^2} - \frac{1}{n_2^2}\right)$$

66. **Excitation energy of 2^{nd} orbit = Excitation energy of 1^{st} excited state**

67. **Separation/ binding energy** : Energy required to separate the electron to ∞ from any orbit is known as Separation/ binding energy of that state.

$n_1 = 1, 2, 3, 4.....$, $n_2 = \infty$

Binding energy of n^{th} orbit $= 13.6 \times Z^2 \left(\frac{1}{n^2} - \frac{1}{\infty^2}\right)$ SE or BE= $\frac{13.6 \times Z^2}{n^2}$ eV

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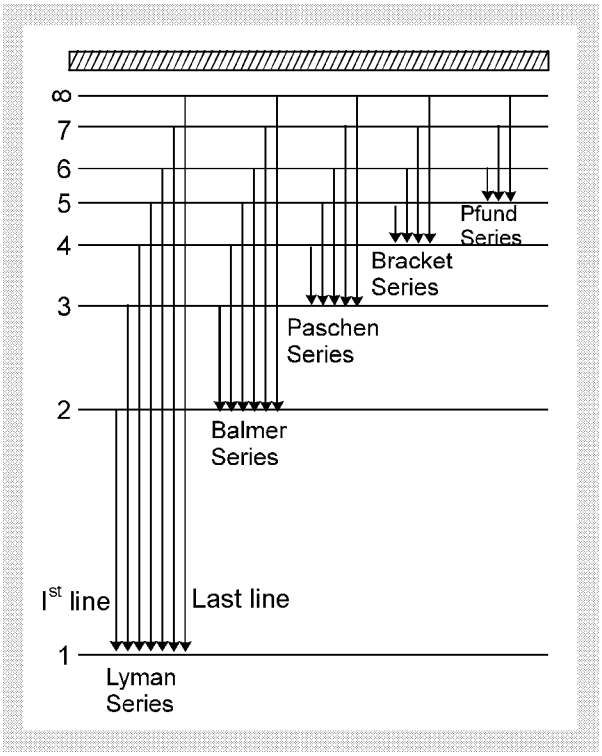
1. **Spectrum:** Collection of wavelength or frequency arranged in increasing or decreasing order called spectrum.

2. **Hydrogen spectrum : (Atomic Spectrum of hydrogen) (Emission, Discontinuous, line spectrum)**



3.

Line terminate at	Value of n_1 and n_2	Region of spectrum for H-atom
n=1, Lyman series	n_1=1, n_2= 2,3,4,5.....	Ultra violet region
n=2, Balmer series	n_1=2, n_2 = 3,4,5....	I st four line in visible region. Rest of line in U.V. region. 3→2 (Red), 4→2 (aqua),5→2 (Blue), 6→2 (Voilet)
n=3, Paschen series	n_1=3, n_2 = 4,5,6....	IR region
n=4, Bracket series	n_1=4, n_2 = 5,6,7....	IR region
n=5 Pfund series	n_1=5, n_2 = 6,7,8....	IR region

4. Wave number, $\frac{1}{\lambda} = \bar{\nu} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$ λ = wave length of emitted photon

$[n_1 < n_2]$

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R = Rydberg constant = $1.1 \times 10^7\ m^{-1}$; R ch = 13.6 eV

5. First line of any series has highest wave length & lowest energy.

6. Last line of any series has highest energy & lowest wave length.

7. Last line of any series is called series limit

11. **Maximum possible number of different photon emitted by H-atom/ H-like species or maximum possible number of different spectral line that can be observed.**

(i) **From a sample of 'H' like species**:

If an electron is in any higher state $n = n_2$ and makes a transition to another excited state $n = n_1$, then

No. of different photons that can be emitted = $\frac{\Delta n \quad (\Delta n + 1)}{2}$, where $\Delta n = n_2 - n_1$

(ii) **From a single isolated one atom**: Number of Maximum line = n–1

Number of minimum line = 1

12 **Equal energy – Energy level in different H like spectrum**

$$\boxed{\frac{z_1}{z_2} = \frac{n_1}{n_2}}$$

For H	$2 \rightarrow 1$
For He^+	$4 \rightarrow 2$
For Li^{2+}	$6 \rightarrow 3$

13 Dual nature of electron (de-Broglie Hypothesis):

$$\lambda = \frac{h}{p} = \underset{v = \text{Velocity}}{\frac{h}{mv}} = \frac{h}{\sqrt{2m.KE}} = \frac{h}{\sqrt{2m.q.V}}$$

V = Potential difference through which charge particle is accelerated from rest.

$(\lambda_e) = \sqrt{\frac{150}{V}}\text{Å} = \frac{12.3}{\sqrt{V}}\text{Å}$ $(\lambda_p) = \frac{0.286}{\sqrt{V}}\text{Å}$, $(\lambda_\alpha) = \frac{0.101}{\sqrt{V}}\text{Å}$ V = Potential deference

Note: de-Broglie Hypothesis applicable only for microscopic world not for microscopic world.

14. **The circumference of n^{th} orbit is equal to n times of wave length of electron.**

$2\pi r_n = n\lambda$ or $\lambda = 2\pi r_0 \frac{n}{Z}$ n = orbit number , r_n= radius of orbit , Z = atomic number, r_0 = 0.529Å

λ= de Broglie wavelength

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15. Heisenberg uncertainity principle (HUP) (Applicable for pair of terms) (P,x), (E,t)

The exact position and momentum of fast moving particle can't be calculated precisely as the same moment of time.

(I) $\Delta x.\Delta p \geq \frac{h}{4\pi}$ or $\Delta x.(m\Delta v) \geq \frac{h}{4\pi}$

where, Δx = uncertainty in position, Δp = uncertainty in momentum
h = Plank's constant, m = mass of the particle Δv = uncertainty in velocity

(II) $\Delta E.\Delta t \geq \frac{h}{4\pi}$

where, ΔE = uncertainty in energy, Δt = uncertainty in time

16. Orbital: An orbital may be defined as the region of space around the nucleus where the probability of finding an electron is maximum (90% to 95%)

19. Node & Nodal plane:

Node: It is region of space around the nucleus where probability of finding an electron is zero.
Nodal plane: Plane where probability of finding an electron is zero.

20. Type of Node :

(i) **Radial or Spherical node** : Node which are spherical in shape.

$\boxed{RN = n-\ell-1}$ where n and ℓ are principal and azimuthal quantum numbers.

(ii) **Angular node or nodel plane:** Node which are angular in nature that is at any angle or along any particular plane.

$\boxed{AN = \ell}$

Total node = RN + AN = (n–1)

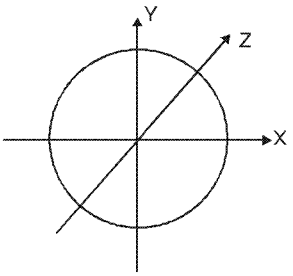
Note: s-orbital (ℓ=0) does not contain any **angular node** or **nodal plane**.

e.g. In 3p-orbital, Radial nodes = 3 – 1 – 1 = 1 (= n – ℓ – 1)
Angular nodes = 1 (= ℓ)
Total nodes = 2 (one radial, one angular)

17. Shape of the orbitals :

Shape of the orbitals are related to the solutions of Schrodinger wave equation, and gives the space in which the probability of finding an electron is maximum.

(i) s-orbital : Shape → spherical



s-orbital is non directional and it is closest to the nucleus, having lowest energy.
s-orbital can accommodate maximum of two electrons.

n s

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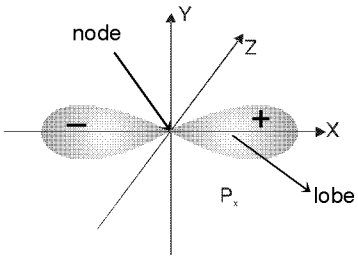
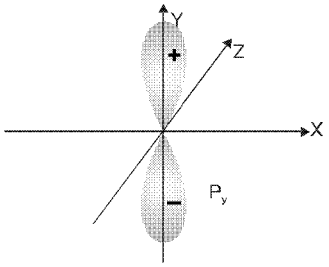
(ii)p-orbital : Shape → dumb bell

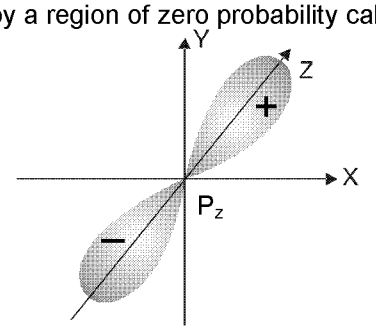
Dumb bell shape consists of two lobes which are separated by a region of zero probability called node.



Nodal plane = YZ plane

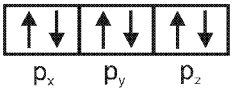


Nodal plane = XZ plane



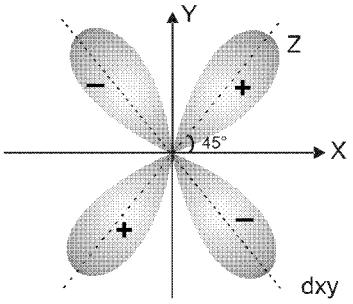
Nodal plane = XY plane

p-subshell can accomodate maximum no. of six electrons.

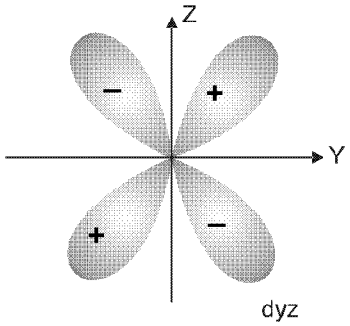


(III) d-Orbital : Shape → double dumb bell

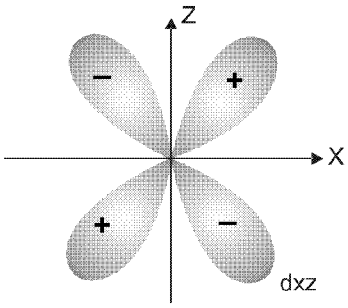
(a) Non Axial orbital (d_{xy}, d_{yz}, d_{xz}): Orbital in which lobes are present in between axis.
Electron density maximum between the axis and zero around the axis.



Nodal plane: YZ & ZX plan



Nodal plane: XY & ZX plane



Nodal plane: XY & YZ plane

(b) Axial orbital $\left(d_{x^2-y^2}, d_{z^2}\right)$ **:** Orbital in which lobes are present in along the axis.
Electron density maximum along the axis

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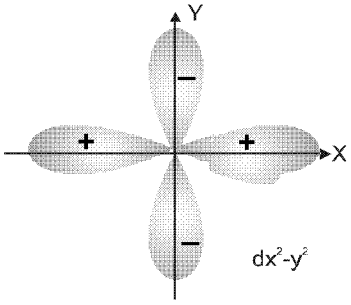
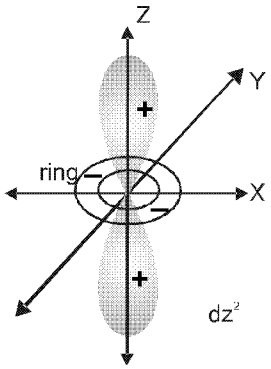
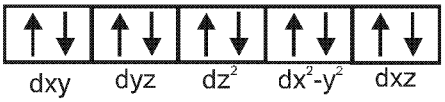


(Nodal plane lies at 45º) from X,Y axis Perpendicular to XY plane

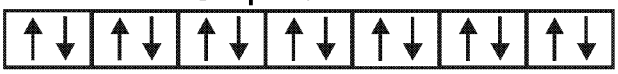


Two Conical node
(No nodal plane)

d-subshell can accommodate maximum no. of 10 electrons.



(iv) **f-orbital : Shape → leaf like**



f-subshell can accommodate maximum no. of 14 electrons.

Note: +/- is not charge. It is sign of wave.

21 QUANTUM NUMBERS :

The set of four numbers required to define an electron completely in an atom are called quantum numbers. The first three have been derived from Schrodinger wave equation.

(i) Principal quantum number (n) :

It describes the shell in which electron is presents, size of the electron wave and the total energy of the electron.
It has integral natural number values 1, 2, 3, 4 ...., etc., and is denoted by K, L, M, N. ..., etc.
Number of orbitals present in n^{th} shell = n^2
Number of subshell present in n^{th} shell = n

n	**subshell**	**Orbital**
1	s	1s,
2	s, p	2s,$2p_x$, $2p_y$, $2p_z$
3	s, p, d	3s,$3p_x$, $3p_y$, $3p_z$, $3d_{xy}$, $3d_{yz}$, $3d_{xz}$, $3d_{z^2}$, $3d_{x^2-y^2}$
4	s, p, d, f	

The maximum number of electrons which can be present in a principal energy shell = $2n^2$

Angular momentum of any orbit = $\frac{nh}{2\pi} = n\hbar$ $\qquad \hbar = \frac{h}{2\pi}$

(ii) Azimuthal quantum number (ℓ) :

It describes the shape of electron cloud and subshell in which electron is present.
It can have whole number values from 0 to (n – 1)

value of ℓ **subshell**

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0	s
1	p
2	d
3	f
4	g

Number of orbitals in a subshell = $2\ell + 1$

Maximum number of electrons in particular subshell = $2 \times (2\ell + 1)$

Orbital angular momentum L = $\frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \hbar\sqrt{\ell(\ell+1)}$ $\left[\hbar = \frac{h}{2\pi}\right]$

Eg. Orbital angular momentum of s (1s,2s,3s,4s....) (ℓ=0) orbital electron = 0

Orbital angular momentum of p (2p,3p,4p,5p.....) (ℓ=1) orbital electron = $\sqrt{2}\frac{h}{2\pi}$

Orbital angular momentum of d (3d,4d,5d,6d.....) (ℓ=2) orbital electron = $\frac{\sqrt{6}h}{2\pi}$

Ex.

Value of n	No. of value of ℓ (ℓ =n)	Value of ℓ (0 to n–1)	Subshell	Total Electron ($2n^2$ or 2(2l+1)
1	1	0	1s	2
2	2	0 1	2s 2p	2+6=8
3	3	0 1 2	3s 3p 3d	2+6+10= 18
4	4	0 1 2 3	4s 4p 4d 4f	2+6+10+14=32

Note: 1p, 2d, 3f,4g subshell do not exist because (n > ℓ) but here n = ℓ in all cases.

(iii) Magnetic quantum number (m) :

It describes the orbital in which electron is present and orientations of orbitals in a subshell with respect to standard set of coordinate axes.

It can have integral values from –ℓ to + ℓ including zero.

Total number of values of m or Total no of orientation =$(2\ell + 1)$

s-subshell	one orbital
p-subshell	three orbitals (p_x, p_y and p_z)
d-subshell	five orbitals (d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$, d_{z^2})
f-subshell	seven orbitals.

The total number of orbitals present in a main energy level is 'n^2'

Ex.

Value of ℓ	Total number of value of m=$(2\ell +1)$	Value of m
ℓ = 0 (s–subshell)	1	0
ℓ =1 (p–subshell)	3	–1,0,+1
ℓ =2 (d-subshell)	5	–2,–1,0,+1,+2
ℓ =3 (f-subshell)	7	–3,–2,–1,0,+1,+2,+3

(iv) Spin quantum number (s) :

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It describes the direction of spin of the electron in an orbital.
It has values +1/2 and –1/2 signifies clockwise and other anticlockwise spin.

spin angular momentum = $\frac{h}{2\pi}\sqrt{s(s+1)} = \sqrt{\frac{3}{4}}\times\frac{h}{2\pi}$ $\left(s= +\frac{1}{2}\right)$

22. Electronic configuration :

Rules

Pauli's exclusion principle :

No two electrons in an atom can have the same set of all the four quantum numbers.
An orbital cannot have more than 2 electrons

eg. $^{4}_{2}He$ for 1st Electron n=2 , l=0 , m=0 ,s= $+\frac{1}{2}$

for 2nd Electron n=2 , l=0 , m=0 ,s= $-\frac{1}{2}$

I^{st} 3 Q.N. same but 4^{th} is different.

Aufbau principle :

The electrons are filled in various orbitals in order of their increasing energies.
An orbital of lowest energy is filled first.
The energy order of the orbitals is governed by (n + ℓ) rule.

Case-1, Value of (n +l) is different

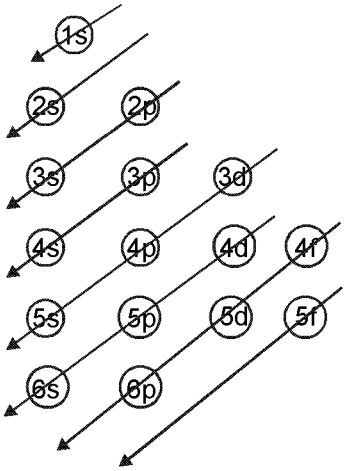
❖ The sub-shell with lower value of (n + ℓ) has lower energy and it should be filled first.

eg. 3d 4s

(n + ℓ) = 3 + 2 = 5 (n + ℓ) = 4 + 0 = 4

Since, (n + ℓ) value of 3d is more than 4s therefore, 4s will be filled before 3d.

MEMORY MAP :



Order of electron filling : 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, ...

Case-2, Value of (n +l) is same.

If value of n+l is same then electron is filled I^{st} in that orbital having low value of n.

eg. 2p 3s

(n + ℓ) = 2 + 1 = 3 (n + ℓ) = 3 + 0 = 3

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Since, $(n + \ell)$ value is same for both. But value of n is less for 2p so electron 1^{st} filled in 2p then 3s.

Note: Above rule is applicable only for multi electron system. But single electron system follows the Bohr model. If n is same then energy is also same.

1s< 2s= 2p< 3s=3p=3d < 4s=4p=4d=4f < 5s=5p=5d= 5f....

Ex. $4p_x=4p_y=4p_z=4s=4d_{xy}$ for (H, He^+, Li^{2+})

Hund's rule :

No electron pairing takes place in the orbitals in a sub - shell until each orbital is occupied by one electron with parallel same spin.

Exactly half filled and fully filled orbitals are observed to be more stable, i.e., p^3, p^6, d^5, d^{10}, f^7 and f^{14} configuration are most stable probably because of the following reasons:
(i) larger exchange energy
(ii) Symmetrical distribution of charge.

Note : **Exceptional** Electronic configuration : $_{24}Cr$: [Ar] $4s^1. 3d^5$
$_{29}Cu$: [Ar] $4s^1. 3d^{10}$

Magnetic behavior :

Diamagnetic : having no unpaired electron.

Peramagnetic : having unpaired electron

Spin magnetic moment $\mu_s = \sqrt{x(x+2)}$ B.M. (x = no. of unpaired electrons)

Total/ Maximum spin of atom/ion = $\frac{1}{2}$ × No. of unpaired electrons or $\left(\pm\frac{1}{2}\right)$ × number of unpaired electron.

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PHYSICAL CHEMISTRY (Max. Marks : 108)

1. **Shell:** In QMMA, the 2-D circular orbits proposed by Bohr were replaced by 3-D spherical shells having maximum electron filling capacity same as that of an orbit (= $2n^2$)

2. **Orbital:** An orbital may be defined as the region of space around the nucleus where the probability of finding an electron is maximum (90% to 95%).
An orbital can accommodate **maximum no. of two electrons**.
n^{th} shell has a total of **n^2 number of orbitals**.

3. **Subshell:** A collection of similar shaped / same energy orbitals of a shell is called subshell.

4. **Node & Nodal plane:**

Node: Surface around the nucleus where probability of finding an electron is zero.

Nodal plane: Plane where probability of finding an electron is zero.

5. **Types of Node:**
(i) Radial or Spherical node: Spherical surface at non-zero finite distance from nucleus having zero electron finding probability.

$\boxed{\text{No. of RN} = n-\ell-1}$ where n and ℓ are **principal and azimuthal quantum numbers**.

(ii) Angular or non-spherical node: A surface / plane at a particular angle from coordinate axes having zero electron finding probability on it at any distance from nucleus.

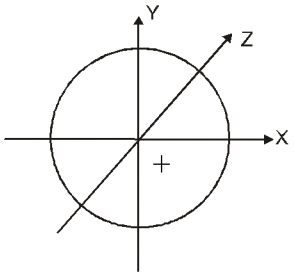
$\boxed{\text{No. of AN} = \ell}$

Total nodes = **(n – 1)**

Note: Angular node and nodal plane are **same** except **for d_{z^2} where the two angular nodes are two conical nodal surfaces.**
1s-orbital does not has any radial & angular node.
e.g. In $3p_y$-orbital, Radial nodes = 3 – 1 – 1 = 1
Angular nodes = 1 (ZX plane)
Total nodes = 2

6. **Shape of the orbitals:**

Shape of the orbitals are related to the solutions of **Schrodinger wave equation**, and gives the space in which the **probability of finding an electron is maximum**.
(i) s-orbital : Shape → spherical



s-orbital is **non-directional** and it is closest to the nucleus, having **lowest energy**.
s-orbital can accommodate maximum of **two electrons**.

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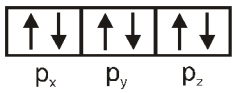
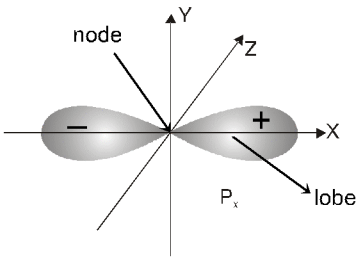
(ii) p-orbital : Shape → dumb bell

Dumb bell shape consists of **two lobes** connected at origin (nucleus).

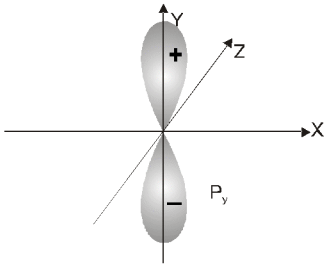
p-orbital is **directional**.

p-subshell has **three orbitals (p_x, p_y, p_z)** corresponding to **orientation of dumb bell along different directions** & can accommodate **maximum no. of six electrons**.

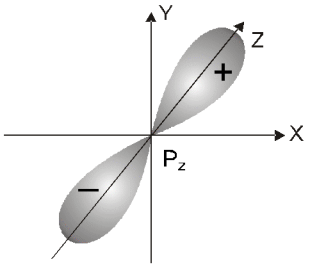




Nodal plane = YZ plane



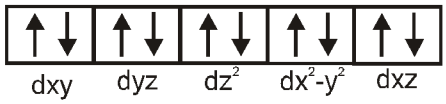
Nodal plane = XZ plane



Nodal plane = XY plane

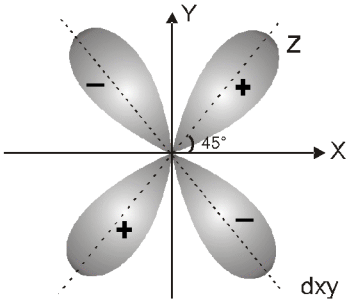
(III) d-Orbital: Shape → double dumb bell

d-subshell has **five orbitals (d_{xy}, d_{yz}, d_{xz}, $d_{x^2-y^2}$, d_{z^2})** & can accommodate **maximum no. of 10 electrons.**

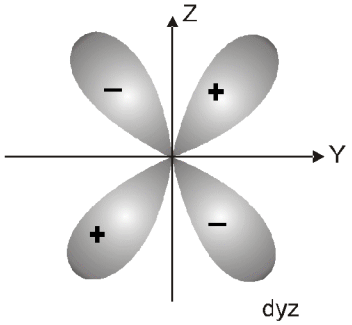


Orbitals of d-subshell are classified into:

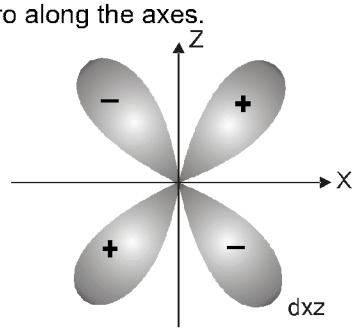
(a) Non-Axial orbitals (d_{xy}, d_{yz}, d_{xz}): Orbitals in which lobes are present in between axes & not along axes. Electron density is maximum between the axes along 45° and zero along the axes.



Nodal plane: YZ & ZX plan



Nodal plane: XY & ZX plane



Nodal plane: XY & YZ plane

(b) Axial orbitals $\left(d_{x^2-y^2}, d_{z^2}\right)$: Orbitals in which lobes are present along the axes.

Electron density is maximum along the axes.

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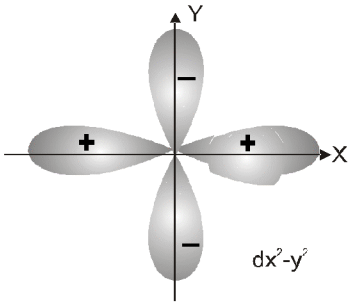
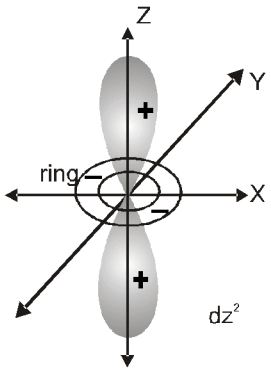


(Nodal plane lies at 45°) from X,Y axis Perpendicular to XY plane



Two Conical node
(No nodal plane)

(iv) **f-orbital :** **Shape → leaf like / complicated**

f-subshell has **seven orbitals** and can accommodate **maximum no. of 14 electrons.**

Note: + / – is not **charge**, it is **sign of wave**.

21. QUANTUM NUMBERS :

The set of four numbers required to define an electron completely in an atom are called quantum numbers.
The **first three** have been derived from **Schrodinger wave equation.**

(i) Principal quantum number (n) :

It describes the **shell in which electron is present**, **size of the electron wave** and **is mainly responsible for the total energy of the electron.**
It has **natural number values 1, 2, 3, 4 ...., etc.**, and is denoted by **K, L, M, N. ... shells etc.**
Number of orbitals present in n^{th} shell = $\mathbf{n^2}$
Number of subshell present in n^{th} shell = **n**

n	**subshell**	**Orbital**
1	s	1s
2	s, p	2s, $2p_x$, $2p_y$, $2p_z$
3	s, p, d	3s, $3p_x$, $3p_y$, $3p_z$, $3d_{xy}$, $3d_{yz}$, $3d_{xz}$, $3d_{z^2}$, $3d_{x^2-y^2}$
4	s, p, d, f	4s, $4p_x$, $4p_y$, $4p_z$, $4d_{xy}$, $4d_{yz}$, $4d_{xz}$, $4d_{z^2}$, $4d_{x^2-y^2}$ & seven 4f-orbitals.

Maximum number of electrons which can be present in a shell = $\mathbf{2n^2}$

(ii) Azimuthal quantum number or Subsidiary quantum number (ℓ) :

It describes the shape of electron cloud (orbital) and subshell in which electron is present.
It can have whole number values from 0 to (n – 1)

value of ℓ	**subshell**	**shape of orbital**
0	s	spherical
1	p	dumb bell
2	d	double dumb bell
3	f	leaf like / complicated
4	g	-

Number of orbitals in a subshell = $\mathbf{2\ell + 1}$

Maximum number of electrons in particular subshell = $\mathbf{2 \times (2\ell + 1)}$

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Orbital angular momentum $L = \frac{h}{2\pi}\sqrt{\ell(\ell+1)} = \hbar\sqrt{\ell(\ell+1)}$ $\left[\hbar = \frac{h}{2\pi}\right]$

Eg. Orbital angular momentum of s (1s, 2s, 3s, 4s....) (ℓ=0) orbital electron = 0

Orbital angular momentum of p (2p, 3p, 4p, 5p.....) (ℓ=1) orbital electron = $\sqrt{2}\frac{h}{2\pi}$

Orbital angular momentum of d (3d, 4d, 5d, 6d.....) (ℓ=2) orbital electron = $\frac{\sqrt{6}h}{2\pi}$

Ex.

Value of n	No. of values of ℓ	Value of ℓ	Subshell	Max. Electron
1	1	0	1s	2
2	2	0 1	2s 2p	2 + 6 = 8
3	3	0 1 2	3s 3p 3d	2 + 6 + 10 = 18
4	4	0 1 2 3	4s 4p 4d 4f	2 + 6 + 10 + 14 = 32

Note: 1p, 2d, 3f, 4g subshell **do not exist** because **(n > ℓ) but here, n = ℓ in all cases.**

(iii) Magnetic quantum number (m or m_ℓ) :

It describes the **orbital in which electron is present** and **orientations of orbitals in a subshell** with respect to standard set of coordinate axes.

It can have **integral values from –ℓ to + ℓ including zero.**

Total number of values of m or **Total no. of orientation or Total no. of orbitals in a subshell**=(2ℓ + 1)

Subshell	**Orbitals with no.**
s-subshell	one orbital
p-subshell	three orbitals (p_x, p_y and p_z)
d-subshell	five orbitals (d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$, d_{z^2})
f-subshell	seven orbitals

The total number of orbitals present in a main energy level is '**n^2**'

Ex.

Subshell	Total number of value of m=(2ℓ +1)	Value of m
s-subshell	1	0
p-subshell	3	–1,0,+1
d-subshell	5	–2,–1,0,+1,+2
f-subshell	7	–3,–2,–1,0,+1,+2,+3

For a particular ℓ value, each possible value of m represents **one orbital of that subshell.**

All of the orbitals of a subshell have **equal energies** in the **absence of electric & magnetic field** and are called **degenerate orbitals**.

(iv) Spin quantum number (s or m_s) :

This quantum number is **not derived from Schrodinger equation**. It takes into consideration **particle nature of electron**.

It describes the **direction of spin of the electron in an orbital.**

It has values **+1/2 and –1/2,** one signifies clockwise and other anticlockwise spin.

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Spin angular momentum of electron = $\frac{h}{2\pi}\sqrt{s(s+1)} = \sqrt{\frac{3}{4}} \times \frac{h}{2\pi}$ $\left(s = +\frac{1}{2}\right)$

= constant for every electron

22. Electronic configuration :

Rules

Pauli's exclusion principle :

No two electrons in an atom can have the same set of all the four quantum numbers.

Alternatively, **an orbital cannot have more than 2 electrons**

eg. $_2^4He$ 2s

for 1st Electron, **n=2 , ℓ =0 , m=0 ,s= +** $\frac{1}{2}$

for 2nd Electron, **n=2 , ℓ =0 , m=0 ,s= -** $\frac{1}{2}$

Ist 3 Q.N. are same but 4th is different.

Aufbau principle :

The electrons are filled in various orbitals in order of their increasing energies. An orbital of lowest energy is filled first.

The energy order of the orbitals is governed by **(n + ℓ) rule.**

Case-1 : Value of (n +ℓ) is different

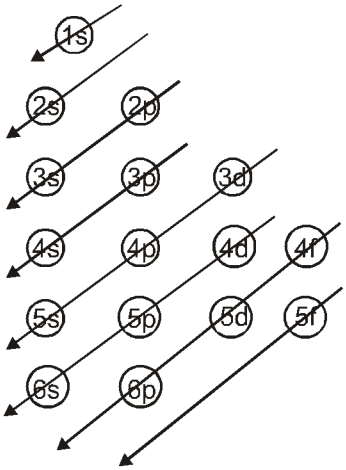
The sub-shell with lower value of (n + ℓ) has lower energy and it should be filled first.

eg. 3d 4s

(n + ℓ) = 3 + 2 = 5 **(n + ℓ) = 4 + 0 = 4**

Since, (n + ℓ) value of 3d is more than 4s therefore, 4s will be filled before 3d.

MEMORY MAP :



Order of electron filling : **1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p**

Case-2 : Value of (n +ℓ) is same.

If value of n+ℓ is same, then electron is filled Ist in that orbital having lower value of n.

eg. 2p 3s

(n + ℓ) = 2 + 1 = 3 **(n + ℓ) = 3 + 0 = 3**

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Since, (n + ℓ) value is same for both, but value of n is less for 2p, so electron 1st filled in 2p then 3s.

Note: Above rule is applicable only for **multi electron system**. But **single electron system follows the Bohr model**. If n is same, **then energy is also same.**

1s< 2s= 2p< 3s=3p=3d < 4s=4p=4d=4f < 5s=5p=5d= 5f...

Ex. **$4p_x=4p_y=4p_z=4s=4d_{xy}$ for (H, He^+, Li^{2+})**

Hund's rule :
No electron pairing takes place in the orbitals in a sub - shell until each orbital is occupied by one electron with parallel/same spin.

Exactly half filled and fully filled orbitals are observed to be more stable, i.e. **s^2, p^3, p^6, d^5, d^{10}, f^7 and f^{14}** configuration are most stable probably because of the following reasons:
(i) Larger exchange energy
(ii) Symmetrical distribution of charge

Note : Exceptional Electronic configuration : **$_{24}Cr$: [Ar] $4s^1$ $3d^5$**
$_{29}Cu$: [Ar] $4s^1$ $3d^{10}$

Magnetic behaviour :
Diamagnetic : having no unpaired electron

Paramagnetic : having unpaired electron(s)

Spin magnetic moment $\mu_s = \sqrt{x(x+2)}$ B.M. (x = no. of unpaired electrons)

Total/ Maximum spin of atom/ion = $\frac{1}{2}$ × No. of unpaired electrons or $\left(\pm\frac{1}{2}\right)$ × number of unpaired electrons.

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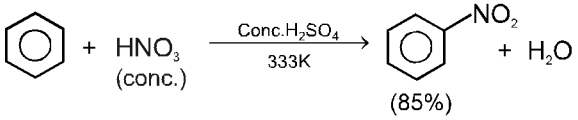
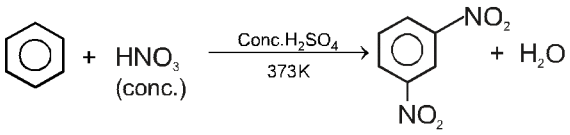
NITROGEN CONTAINING COMPOUNDS

Nitrobenzene

(A) Preparation of Nitrobenzene :

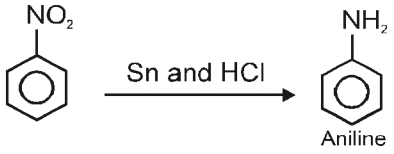


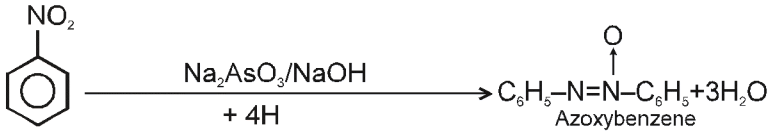


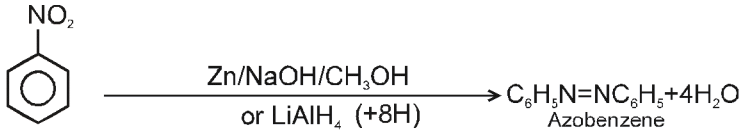
(B) Chemical Reactions of Nitrobenzene :

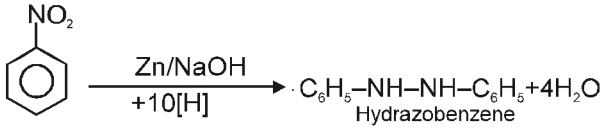
(1)

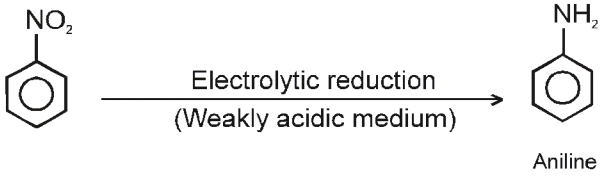
(Catalytic reduction)

(2) 

(3) 

(4) 

(5) 

(6) 

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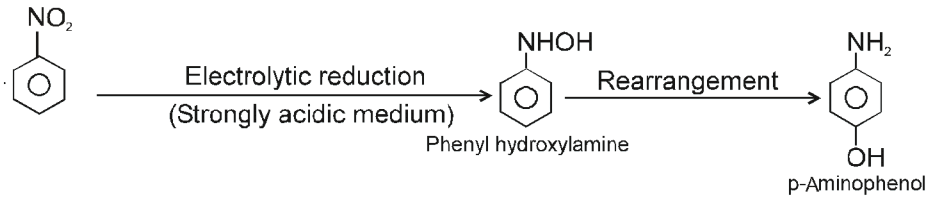
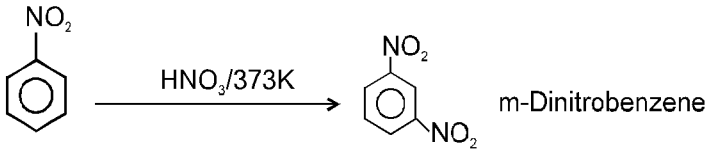
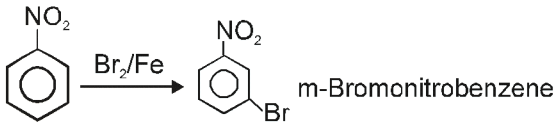
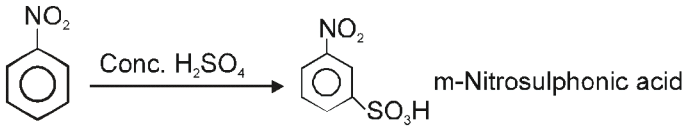
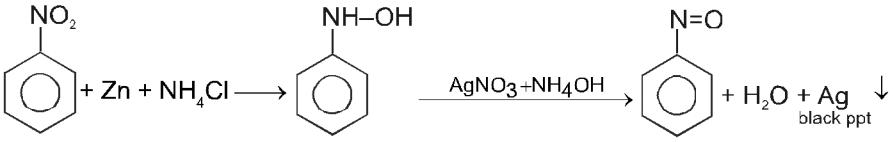
(7) 

(8) 

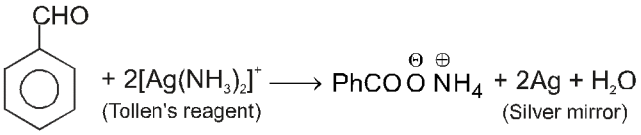
(9) 

(10) 

(C)Test of Nitro Benzene : (Mulliken- Barker's Test)

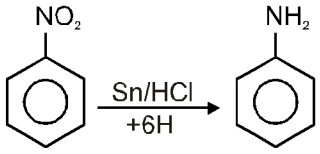


(D) Test of Benzaldehyde (Aromatic aldehyde)



Aniline

(A) Prepartion of Aniline

(1) 

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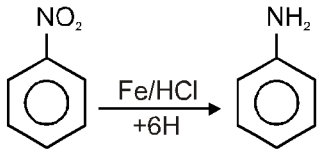
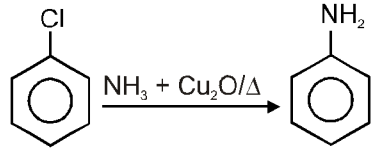
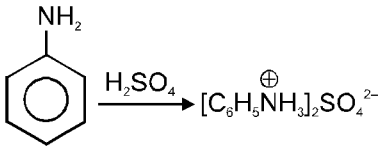
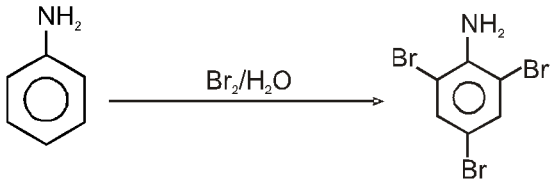
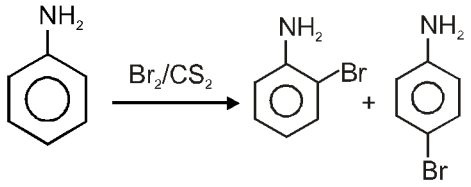
(2) 

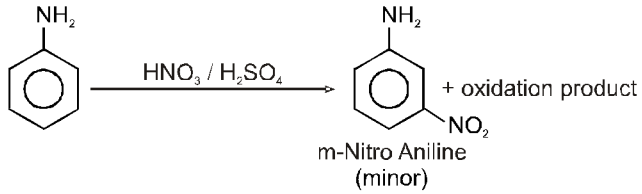
(3) 

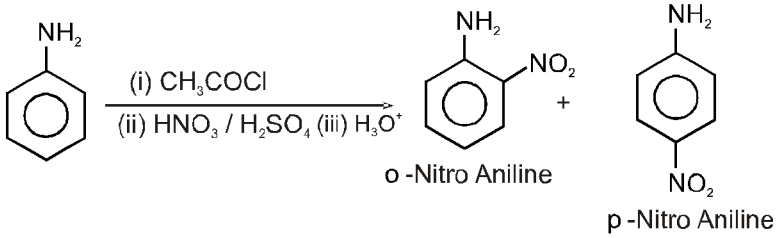
(B) Chemical Reactions of Aniline

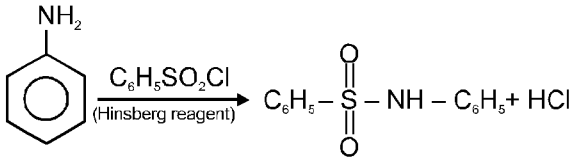
(1) 

(2) 

(3) 

(4) 

(5) 

(6) 

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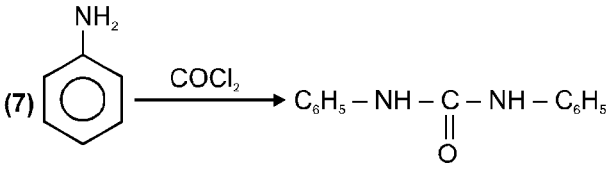
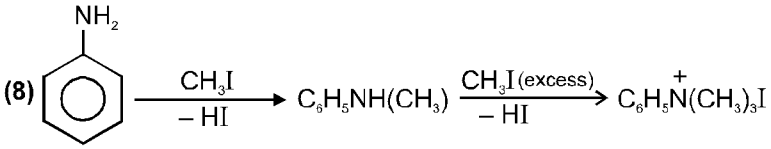
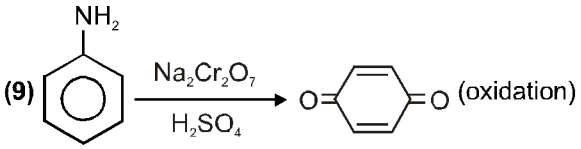
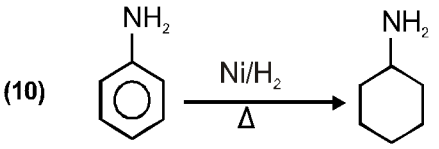




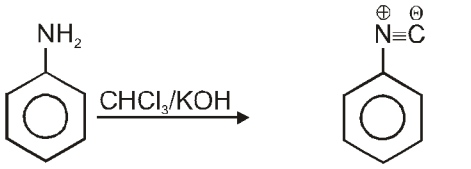




(C) Test of Aniline :

(i) Carbylamine Reaction

Primary amines (alphatic as well as aromatic) react with chloroform ($CHCl_3$) on heating in the presence of ethanolic solution of KOH to form isocyanides (also called carbylamines) which are foul smelling substances. Secondary and tertiary amines do not undergo this reaction, therefore this reaction is used as a test for Primary amines (alphatic as well as aromatic).

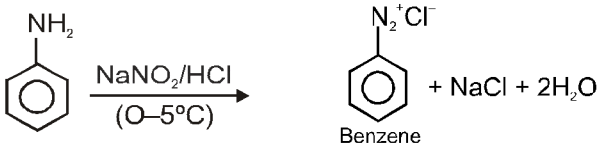


(Phenyl isocyanide)

(ii) Reaction with Nitrous Acid

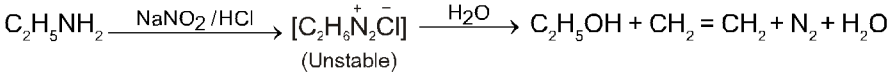
Amines of different classes react with nitrous acid (unstable acid). HNO_2 is prepared by the reaction of sodium nitrite ($NaNO_2$) and dilute hydrochloric acid.

Primary aromatic amines react with nitrous acid at low temperature (273-278 K) to give aromatic diazonium salts. This reaction is known as **diazotisation.**



Benzene diazonium chloride

Primary aliphatic amines also react with nitrous acid to form diazonium salt, however, the aliphatic diazonium salts being unstable, decompose to yield mixture of alcohols, alkenes along with nitrogen gas.So this reaction is used to distinguish between aromatic and aliphatic 1° amines.



(Unstable)

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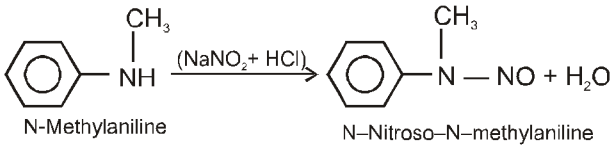
Secondary aliphatic and aromatic amines react with nitrous acid to produce nitroso-amines that are insoluble in the aqueous solution and separate out as a yellow oily layer.



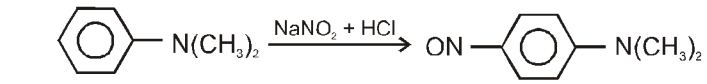
N-Methylaniline — N–Nitroso–N–methylaniline

$(C_2H_5)_2NH + HNO_2 \longrightarrow (C_2H_5)_2N-NO + H_2O$

Diethyl amine — N-Nitrosodiethylamine

Tertiary aliphatic amines react with nitrous acid to form salts, which decompose on warming to nitrosoamine and alcohol while **tertiary aromatic amines** undergo electrophilic substitution at the ring.

$R_3N + HNO_2 \longrightarrow \underset{\text{Nitrite salt}}{R_3N.HNO_2} \xrightarrow{\Delta} \underset{\text{Nitrosoamine}}{R_2N-N=O} + ROH$



N,N-Dimethylaniline — p-Nitroso-N, N dimethylaniline

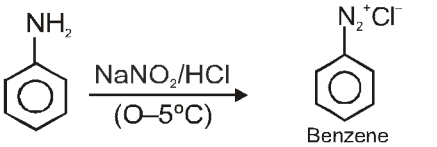
(iii) Hofmann mustard oil reaction

$CH_3CH_2NH_2 + S=C=S \xrightarrow{\Delta} CH_3CH_2NH-\underset{\underset{S}{\|}}{C}-SH \xrightarrow[\Delta]{HgCl_2} CH_3CH_2-N=C=S + HgS + 2HCl$

(1° amine) — Ethyl isothiocyanate

Benzenediazonium chloride

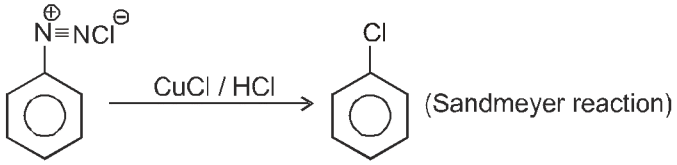
(A) Preparation of diazonium salt



Benzene diazonium chloride

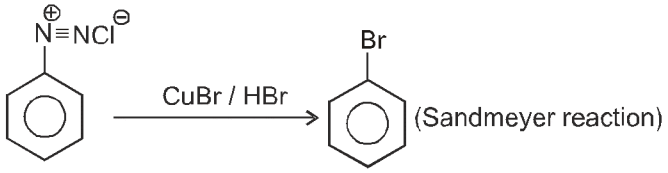
(B) Chemical reactions

(1)



(Sandmeyer reaction)

(2)



(Sandmeyer reaction)

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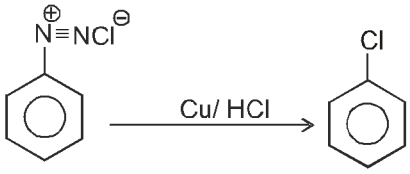
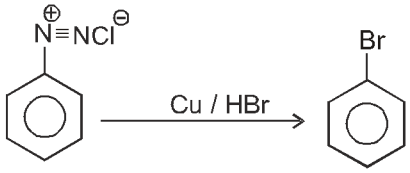
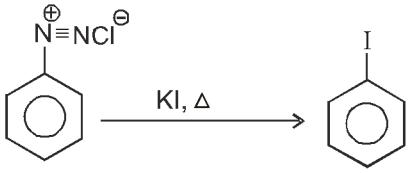
(3) 

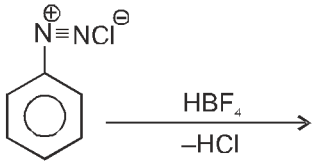
(Gattermann reaction)

(4) 

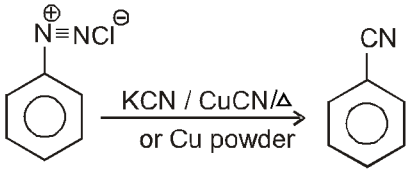
(Gattermann reaction)

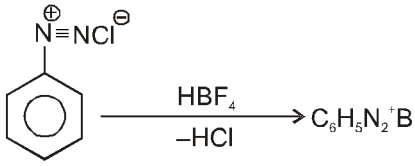
(5) 

+ N_2 + KCl

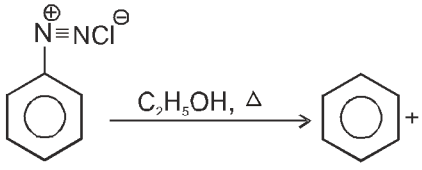
(6) 

$C_6H_5N_2^{\oplus}BF_4^{\ominus} \xrightarrow{\Delta} C_6H_5–F + BF_3 + N_2$ (Balz–Schiemann reaction)

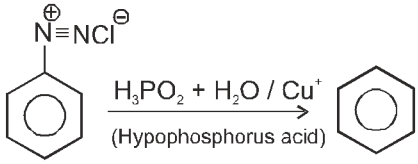
(7) 

(8) 

$C_6H_5N_2^+BF_4^- \xrightarrow[Cu\ powder/\Delta]{NaNO_2} C_6H_5NO_2 + NaBF_4 + N_2$

(9) 

+ N_2 + HCl + CH_3CHO

(10) 

+ N_2 + H_3PO_3 + HCl (Deamination)

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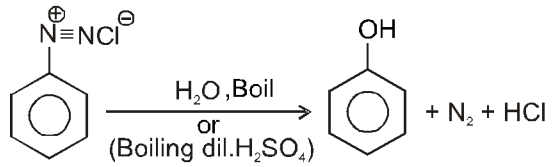
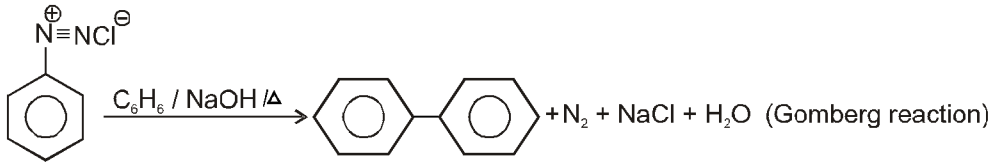
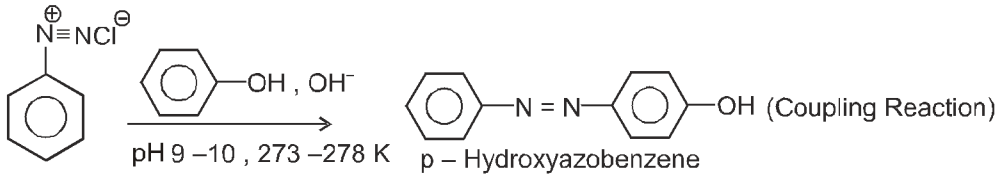
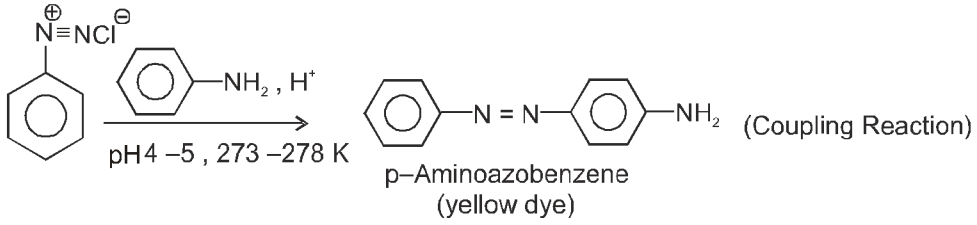
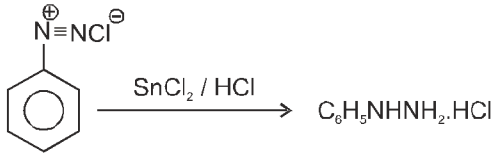
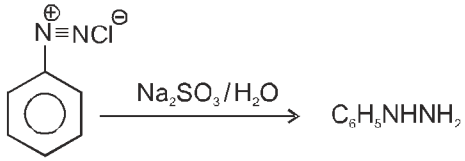
(11) 

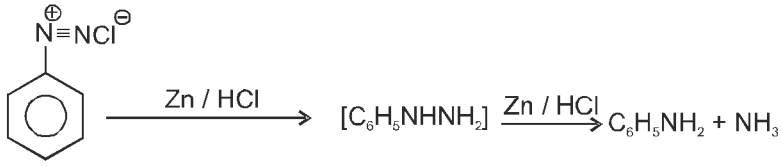
(12) 

(13) 

(14) 

(15) 

(16) 

(17) 

* * * * *

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SOLID STATE

Tab-1

Properties	Solid	Liquid	Gases
(i) Motion of particles	No free motion only vibration allow.	Random motion to a limited extent is allowed.	Totally random.
(ii) Inter molecular forces	Very strong	Intermediate strength	Very weak (~ zero)
(iii) Average separation (volume)	Average separation is fixed so volume is fix.	Average separation is almost constant so almost fixed volume.	No fixed volume.
(iv) Shape	Definite shape as location of partical are fixed.	Average separation is fixed but location of partical is not fixed so no definite shape..	No fixed shape.
(v) Effect of change in pressure & temperature.	Are incompressible.	Liquid are also almost incompressible.	Highly compressible.
(vi) Heat capacities	Heat capacity is almost independent of process.	Same as solid.	Heat capacity is dependent on process.

Types of Solid :

Tab-2 ***Classification on the basis of forces among constituting particles***

Type of solid	Constituent partical	Force of interaction	Example	Physical state	Melting point
(i) Molecular solid (non conducting)	Molecules	(i) Non polar → dispersion force. (ii) Polar → dipole-dipole. (iii) Polar & H-bonding.	I_2, Xe(s), C_6H_6, CCl_4, H_2 HCl, SO_2, SF_4 H_2O (s), H_3BO_3 (s)	Very soft Soft Hard	Very low Low Low
(ii) Ionic solid. solid → insulator Molten & aqueous → conducting.	Ions	Coulombic non directional long range.	NaCl, ZnS, CaF_4, CsCl	Very hard brittle	Very high
(iii) Metallic solid good conductor in solid & molten state.	Metal ion at fixed locations in sea of delocalised electrons.	Metallic bond.	Cu, Al, Zn, Ag, etc.	Soft → Hard depending on metallic bond	Low → High
(iv) Covalent or network. Insulator except C (graphite).	Atoms	Covalent bond.	C(diamond), SiC, SiO_2, AlN, graphite.	Very hard Graphite → Soft only conducting.	Very high

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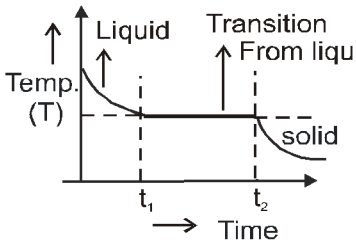
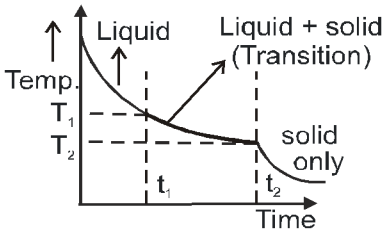
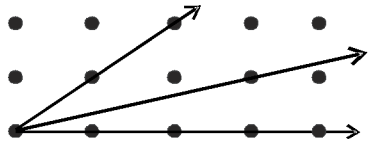
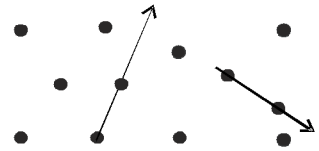
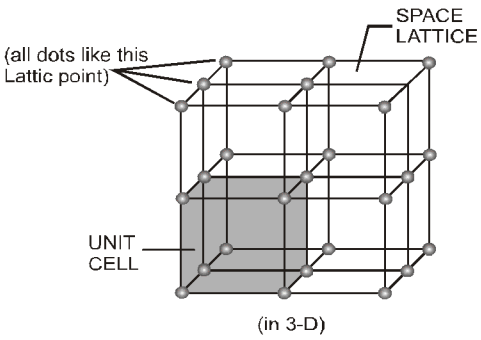
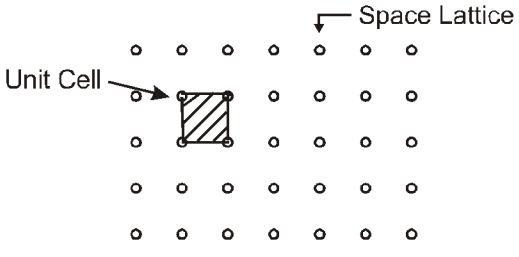
Tab-3 Types of SOLIDS on the basis of arrangement of particles :

	Crystalline solid		Amorphous solids
	True solid		Pseudo solids, super cooled liquid [In between solid & liquid]
1	The constituent partical (atoms, molecule, ion) follow a definite repetiting arrangement.	1	No particular pattern is followed partical are random arranged.
2	These have long range order.	2	They have short range order no long range order are found.
3	These are produced by slow cooling under controlled condition of liquid. The crystalline structure is also dependent on conditions. Same substance can have different crystalline structure in different condition. ***Different crystalline structure of the same substance are called its polymorphic forms & this is known as polymorphism.***	3	Rapid or suddenly cooling of the liquid generate the ***amorphous solid***.
4	These have fixed or ***sharp melting*** point and enthalpy of fusion. **Cooling Curve :** 	4	These have a range of temperature in which they melts as. There melting point and enthalpy of fusion is not fixed. 
5	***These are anisotropic*** : Physical properties will have different values in different direction.  Ex. : Ag, Fe, Cu, NaCl, H_2O (s), Dimond, Quartz, Sucrose (Sugar)	5	***These are isotropic :*** All different physical properties are same in all different direction. **Reason :** Due to random arrangement of partical.  **Ex. :** Glass, Plastic, Amorphous silica, Rubber, Starch.

Lattice :



(in 3-D)



(in 2-D)

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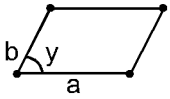
2-Dimensional Space Lattice : Regular arrangement of point in plane 3 parameter required → Two edge lengths & angle between these two edge.



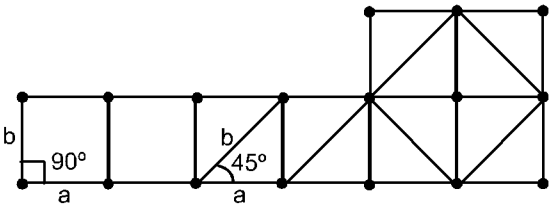
Parallelogram

We can only shift the unit cell parallel not rotate it.

(i) Square unit cell → a = b, $\gamma = 90^\circ$
(ii) Rectangle unit cell → a ≠ b, $\gamma = 90^\circ$
(iii) Hexagonal unit cell → a = b, $\gamma = 120^\circ$
(iv) Rhombic unit cell → a = b, $\gamma \neq 90^\circ$ & $\gamma \neq 60^\circ$
(v) Parallelogram → a ≠ b, $\gamma \neq 90^\circ$

Most symmetrical → square unit cell.

Square unit cell :

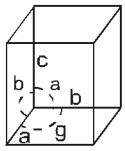


Primitive unit cell : Unit cell having lattice point only at the corner.
Non Primitive or centered unit cell : Unit cell having lattice point at corner as well as with in the unit cell.

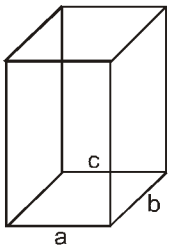
3-Dimensional Space Lattice :

Seven Primitive unit cells in crystals

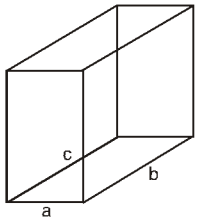
Cubic



; ***Tetragonal***



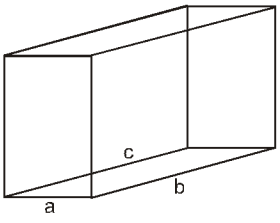
; ***Orthorhombic***



;

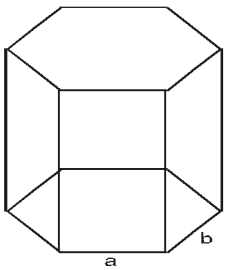
($a = b = c$, $\alpha = \beta = \gamma = 90^\circ$) ($a = b \neq c$, $\alpha = \beta = \gamma = 90^\circ$) ($a \neq b \neq c$, $\alpha = \beta = \gamma = 90^\circ$)

Monoclinic



;

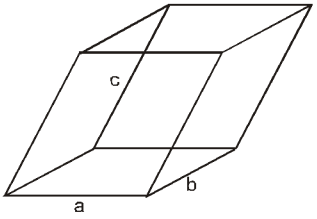
Hexagonal



;

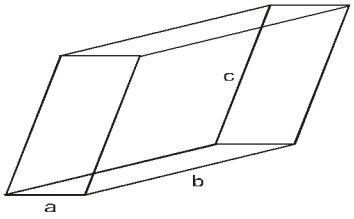
($a \neq b \neq c$, $\alpha = \gamma = 90^\circ$, $\beta \neq 120^\circ, \neq 90^\circ, \neq 60^\circ$) ($a = b \neq c$, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$)

Rhombohedral



;

Triclinic



($a = b = c$, $\alpha = \beta = \gamma \neq 90^\circ$) ($a \neq b \neq c$, $\alpha \neq \beta \neq \gamma \neq 90^\circ$)

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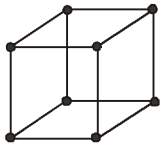
The type of unit cells found in different types of crystals.

(Total number of atoms in unit cell)

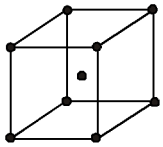
1. Primitive /simple unit cell :



$= 8 \times \frac{1}{8} = 1$

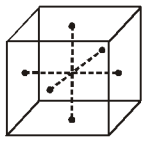
2. Non primitive :

(a) Body centered (B.C.) :



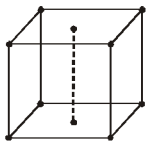
$= 8 \times \frac{1}{8} + 1 \times 1 = 2$

(b) Face centered (F.C.) :



$= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

(c) End centered (E.C.) :



$= 8 \times \frac{1}{8} + \frac{1}{2} \times 2 = 2$

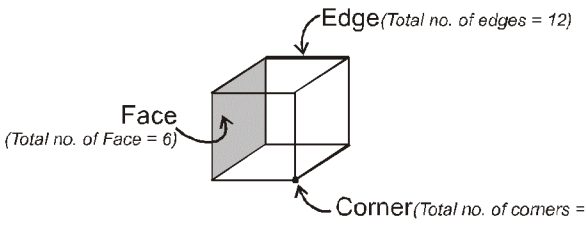
Tab-4 *Seven Crystal System*

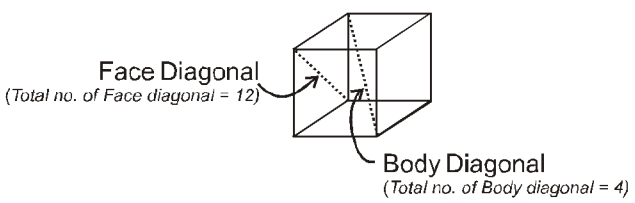
S.No.	Crystal System	Edge length	Angles	Unit cell found	Examples
1	**Cu**bic	a = b = c	$\alpha = \beta = \gamma = 90^o$	SC, BCC, FCC (3)	NaCl, ZnS, Fe, Al, Cu, C (diamond), CsCl, Na_2O, CaF_2, KCl, Pb, Alum.
2	**Te**tragonal	a = b ≠ c	$\alpha = \beta = \gamma = 90^o$	SC, BC (2)	Sn (white tin), SnO_2, ,TiO_2, ZnO_2, $NiSO_4$, urea.
3	**Or**thorhombic	a ≠ b ≠ c	$\alpha = \beta = \gamma = 90^o$	SC, BC, FC, EC (4)	Rhombic sulphur, $BaSO_4$, KNO_3, $PbCO_3$, , $CaCO_3$ (aragonite)
4	**Mo**noclinic	a ≠ b ≠ c	$\alpha = \gamma = 90^o$ $\beta \neq 120^o$, $\neq 90^o, \neq 60^o$	SC, EC (2)	Monoclinic sulphur, $PbCrO_4$, Na_2SO_4, $10H_2O$, $Na_2B_4O_7.10H_2O$
5	**Tr**iclinic	a ≠ b ≠ c	$\alpha \neq \beta \neq \gamma \neq 90^o$	SC (1)	$K_2Cr_2O_7$, $CuSO_4.5H_2O$, H_3BO_3
6	**He**xagonal	a = b ≠ c	$\alpha = \beta = 90^o$ $\gamma \neq 120^o$	SC (1)	Graphite, ZnO, CdS, Mg, PbI_2, SiC.
7	**R**hombohedral or **T**rigonal	a = b = c	$\alpha = \beta = \gamma \neq 90^o$	SC (1)	$CaCO_3$ (Calcite), HgS(Cinnabar), $NaNO_3$, ICl.

Hint for memorise : CuTe Or MoTHeR

Note : In 3-D 14 different types of unit cell are found and these are also known as 14 **Bravais lattice**.

Properties of a cube :





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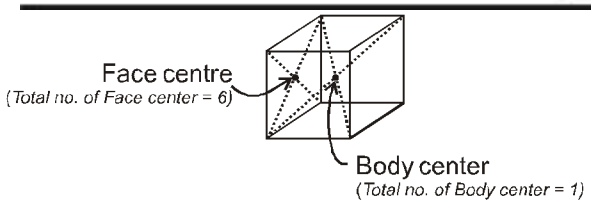


Some Definations

Coordination number : *The number of nearest neighbours sphere in a packing is called coordination number.*

Density of unit cell : *It is the ratio of mass of the spheres present in unit cell and total volume of unit cell.*

$$\text{Density of the unit cell} = \frac{\text{Mass of total atoms present in a unit cell}}{\text{Volume of that unit Cell}}$$

$$d = \frac{Z(M/N_A)}{a^3 \times 10^{-30}} gcm^{-3} \quad \Rightarrow \quad \boxed{d = \frac{ZM}{N_A\,(a^3 \times 10^{-30})} gcm^{-3}}$$

Where Z = no. of atoms in a unit cell

M/N_A = mass of a single atom

M = molar mass

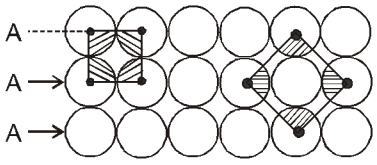
N_A = Avogadro number (6.023×10^{23})

Packing efficiency : *The perecentage of total space filled by the particles is called packing efficiency. Different types of packing arrangements have different packing efficiency.*

$$\boxed{\text{Packing efficiency} = \frac{\text{Volume of atoms in a unit cell}}{\text{Total volume of a unit cell}} \times 100\%}$$

Tab-5

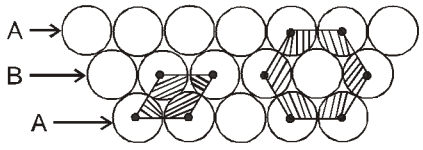
(i) Square arrangement (poor arrangement)
Array are arranged such that sphere of one array are exactly above the sphere of another array



A–A–A type packing

Square packing in two dimension.
Two type of unit cell : = [primitive & non-primitive]
(i) Effective no. of atom (Z) = [1 atom & 2 atoms.]
(ii) Edge length (a = b) = 2R.
(iii) Packing efficiency (Area wise) = $\frac{1 \times \pi R^2}{4R^2} = \frac{\pi}{4} = 78.5\%$
(iv) Co-ordination no. (CN) = 4.

Hexagonal Arrangement (Better arrangement)
One dimensional array are arranged such that sphere of one array occupy the depression of other array.



AB–AB type packing

Hexagonal close packing in two dimension.
Two type of unit cell : = [primitive & non-primitive]
(i) Effective no. of atom (Z) = [1 atom & 3 atoms.]
(ii) Edge length (a = b) = 2R.
(iii) Packing efficiency (Area wise) = $\frac{3(\pi R^2)}{6 \times \frac{\sqrt{3}}{4}(2R)^2} = 90.6\%$
(iv) Co-ordination no. (CN) = 6.

3-D Lattice :

When 2-D close packed layers are kept on each other 3-D close packing will be generated.

(A) Square packed sheets :

(i) Square packed sheets are kept on another such that -
atoms / spheres of one sheet are exactly above spheres of other sheet.
→ A A A A ..... pattern repeat.
→ **Simple cube can be taken as unit cell of this particular lattice :**

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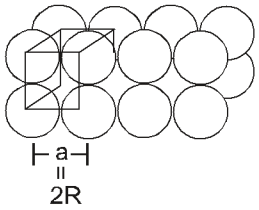
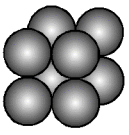
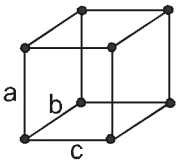




Simple cubic



a = b = c

(i) Relation between a & R.

corner atoms are touching each other so, **a = 2R.**

(ii) Effective no. of atom (Z) (per unit cell).

$$Z = 8\ [\text{corner}] \times \frac{1}{8} = \mathbf{1}\ \text{atom.}$$

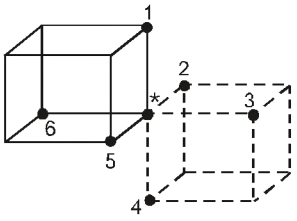
(iii) Packing efficiency : $\dfrac{1\times\frac{4}{3}\pi R^3}{(2R)^3} = \dfrac{\pi}{6}$ **= 52.33%.**

(iv) Density = $\dfrac{\text{Mass of unit cell}}{\text{volume of unit cell}} = \left(\dfrac{Z\times M}{N_A \times a^3}\right)$.

(v) Co-ordination number :

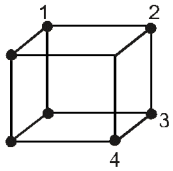
Number of nearest neighbors's or (no. of sphere which are touching any particular sphere). **CN = 6.**

(The * marked atom touches atom number 1, 5, 6 in its own unit cell and 2, 3, 4 in other unit cells hence total number of nearest atom = 6)



Tab-6

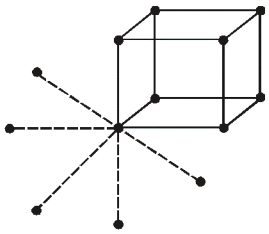
Type of neighbour	Distance	No. of neighbours
(i) Nearest	a (distance b/w 1 & 2)	$\left\{\frac{3\times 8}{4} = 6\right\} \Rightarrow 6$
(ii) $(\text{next})^1$	$\sqrt{2}$ a (distance b/w 1 & 3)	$\left\{\frac{3\times 8}{2} = 12\right\} \Rightarrow 12$



Reference figure

(2) Square packed layer are placed such that sphere of one layer occupy the depression of other layer.

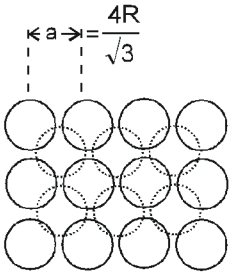
→ AB–AB type of arrangement of square sheet in 3-D.

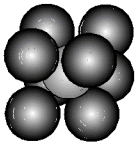


a = b = 2R
c = $2\sqrt{2}$ R

a =edge length of cube ; R = radius of sphere

Body centered cubic (BCC) :





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* ***Not so close pack arrangement.***
* *The lattice point in 2-D array do not touch each other. The sphere start touching each other only upon moving from 2-D to 3-D.*
* *2-D array placed on top of each other such that sphere of next plane are into cavities of first plane of sphere.*

(i) Relation between a & R : (a ≠ 2 R)

Sphere are not touching along edge they touching along the body diagonal.

So $\frac{\sqrt{3}}{2}a = 2R.$

(ii) ***Effective no. of atom (Z) = 8 × [corner] ×*** $\frac{1}{8}$ ***+ 1 = 2***

(iii) Packing fraction

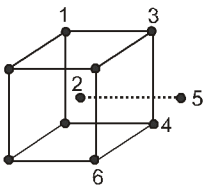
$$= \frac{2 \times \frac{4}{3}\pi R^3}{\left(\frac{4R}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{8} = 68\%.$$

(iv) Co-ordination No. (CN) = 8. *(Body centered atom have only 8 nearest neighbouring atoms so* ***CN = 8****)*

(v) Density = $\frac{Z \times M}{N_A (a)^3}$ where **Z = 2**.

Tab-7

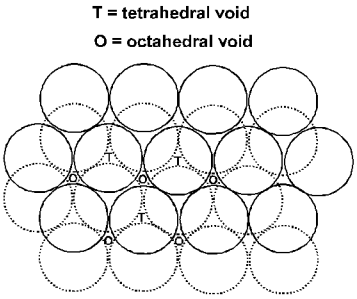
Type of neighbour	Distance	No. of neighbours
(i) Nearest	$\frac{\sqrt{3}a}{2}$ (distance b/w 1 & 2)	8
(ii) $(next)^1$	a (1 & 3)	6



Reference figure

(Note : In bcc crystal structure, the co-ordination no. is 8 because each atom touches four atom in the layer above it, four in the layer below it and none in its own layers.)

(B) Arrangement of hexagonal closed packed sheets :



* *To generate close packing two dimensions arrangement must be hexagonal.*
* *Sheet are arranged such that void or depressions of one sheet are occupied by sphere of other sheet.*
* ***Only 50% void of one layer can be occupied by sphere of other layer (II layer).***
* *Now there will be two method to place III layer on the II layer.*

(1) Hexagonal close packing (HCP) AB–AB–AB- - - -Type.

* ***Sphere of III layer occupy those voids of II layer under which there are sphere of I^{st} layer. So third layer is exactly identical to I^{st} layer.***

That why this generate ABAB- - - -AB pattern.

One type of void always remain on occupied.

Unit cell : a = 2r = b; $\gamma = 120^\circ$

(i) Relation between a, b, c and R :

a = b = 2R

Area = 6 $\left[\frac{1}{2}a \times \frac{\sqrt{3}}{2}a\right] = \frac{6\sqrt{3}a^2}{4}$

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volume of hexagon = area of base × height

$$= \frac{6.\sqrt{3}}{4} \times a^2 \times 2\sqrt{\frac{2}{3}}\, a = \frac{6.\sqrt{3}}{4} \times (2R)^2 + 2\sqrt{\frac{2}{3}} \times (2R) = \mathbf{24\sqrt{2}\, R^3}$$

(ii) Effective no. of atoms (Z) $= 3 + 2 \times \frac{1}{2} + 12 \times \frac{1}{6} = 3 + 1 + 2 = \mathbf{6}$.

(iii) Packing efficiency $= \frac{6 \times \frac{4}{3}\pi R^3}{24\sqrt{2}\, R^3} = \frac{\pi}{3\sqrt{2}} = \mathbf{74\%}$.

(iv) Coordination number, (CN) = 12

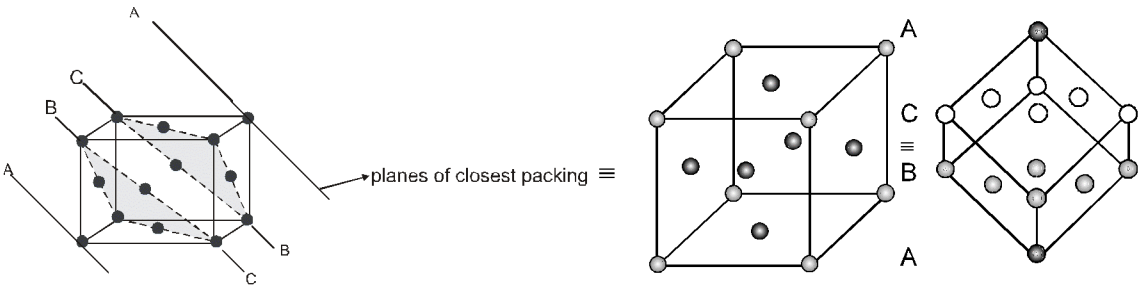
(Each sphere touch 6 sphere in its layer 3 above and 3 below)

(v) Density (d) $= \frac{\text{mass}}{\text{volume}} = \left[\frac{Z \times M}{N_A \times \text{volume}}\right]$ (z = 6)

(2) ABC–ABC arrangement (Cubic close packing (CCP) or Face centred cube (FCC))

Third layer sphere are placed such that these occupy those 50% voids of II layer under which there are void of Ist layer.

- ***IIIrd layer will be different from Ist layer as well as IInd layer.***
- ***ABC–ABC type of arrangement.***
- ***It is also known as cubical close packing, unit cell chosen is face centered unit cell (FCC).***



(i) Relation between 'a' and 'R' :

$a \neq 2R \quad \Rightarrow \quad \sqrt{2}a = 4R$ **(sphere are touching along the face diagonal)**

(ii) Effective no. of atoms per unit cell (Z) $= \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$

(iii) Packing fraction : $\text{P.F.} = \frac{4 \times \frac{4}{3}\pi R^3}{4 \times 4 \times 4\, R^3} \times \sqrt{2} \times 2 = \frac{\pi}{3\sqrt{2}} = \mathbf{0.74(\ 74\%\ )}$

(iv) Coordination number, (CN) = 12

(v) Density (d) $= \frac{Z \times M}{N_A . a^3}$ g/cm^3 (a = Edge length of FCC unit cell)

Note : *HCP and CCP are the only two type of close packed lattice (because of their effective efficiency 74%).*

Types of voids found in close packings :

R = Radius of sphere **(Bigger atom or anion)**

r = Radius of sphere placed in void. **(Smaller atom or cation)**

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2-Dimensions void :

(i) Triangular void [2D–3 coordinate void]

* *This type of void found in close packed structure of plane when three sphere are in contact.*

$$\frac{r}{R} = 0.155.$$

3-Dimensions void :

Tetrahedral void [3D-4 coordinate void]

* *If IInd layer sphere are placed over a triangular void then new void is tetrahedral.*

$$\frac{r}{R} = 0.225$$

3-Octahedral void : [3D–6 coordinate void]

Octahedral void is formed when ever two sphere are placed, one on top and the other below a square arrangement of sphere.

$$\frac{r}{R} = 0.414$$

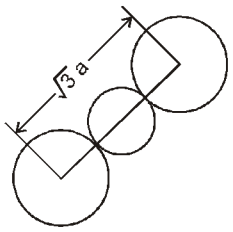
Cubical void (3D-8-coordinate void)

Sphere are touch along body diagonal :

$\sqrt{3}\,a = 2\,[R + r]$

$2\sqrt{3}\,R = 2\,[R + r]$

$$\frac{r}{R} = 0.732$$

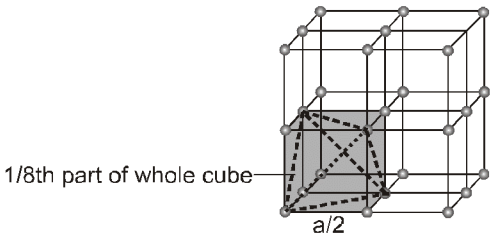


Location of void :

(i) FCC or CCP unit cell :

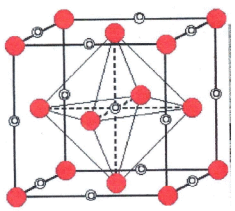
(a) Tetrahedral void :

* *FCC unit cell has **8 tetrahedral void** per unit cell. Just below every corner of the unit cell.*



* *As unit cell has **8 corner that's why FCC unit cell has 8 tetrahedral void** per unit cell.*
* *Distance of tetrahedral void form corner is $\frac{\sqrt{3}a}{4}$ and distance between two tetrahedral void is = $\frac{\sqrt{3}a}{2}$.*
* *FCC unit cell has 4 body diagonal and each body diagonal contain **2 tetrahedral void** that's why total **tetrahedral void = 8**.*
* *If a large size cube is divided in 8 minicube than center of each minicube contain one tetrahedral void and distance of center of minicube from corner is $\frac{\sqrt{3}a}{4}$.*
* ***For FCC unit cell Z = 4 and tetrahedral void = 8 so tetrahedral void = 2 Z.***

(b) Octahedral void (OV) :



(Circles labeled O represent centers of the octahedral interstices in the ccp arrangement of anions (fcc unit cell). The cell "owns" 4 octahedral sites.)

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* ***Each edge center of FCC unit cell have one octahedral void and body center also contain 1 OV.***

* ***Number of octahedral void per unit cell (Z) = 12 [Edge center] × $\frac{1}{4}$ + 1[Body center] = 4.***

* ***For FCC unit cell (Z) = 4 and octahedral void = 4.So octahedral void = Z.***

Structure of some ionic compound :

(1) NaCl type of structure (Rock salt structure) :

6 : 6

"coordination number of cation" — "coordination number of anion"

* ***Experimental*** $\frac{r_{Na^+}}{r_{Cl^-}}$ ***= 0.51*** ***(0.414 <*** $\frac{r_{Na^+}}{r_{Cl^-}}$ ***< 0.732)***

(i) Cl^- ions form the FCC lattice while Na^+ ion occupy all the octahedral void.

(ii) Effective number of ion per unit cell

(iii) Effective number of Cl^- ion per unit cell = 4
Effective number of Na^+ ion per unit cell = 4

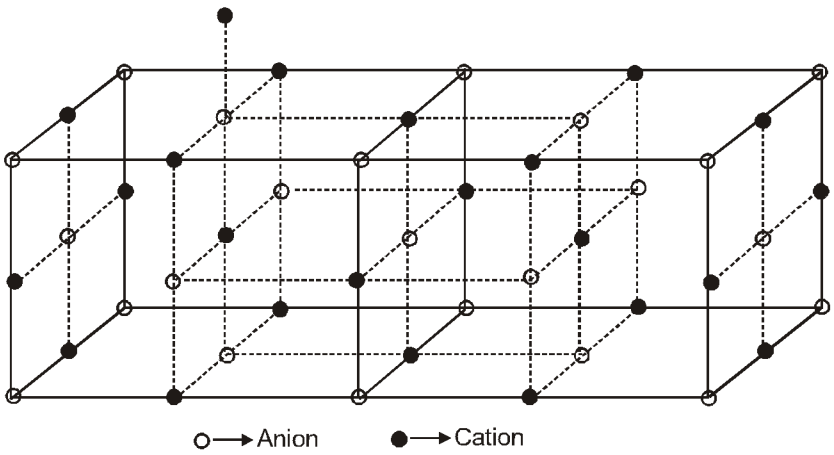
(iv) Formula of unit cell Na_4Cl_4
Formula of ionic compound = NaCl
Effective number of formula unit (Z) = 4

(v) Density (d) = $\frac{Z \times M}{N_A \times a^3}$

→ Here $\sqrt{2}\,a \neq 4R_{Cl^-}$ → ***as anion–anion are not in contact.***

→ $\boxed{[r_{Cl^-} + r_{Na^+}] = a/2}$

(vi) **Coordination number :**
Coordination number of Cl^- = Number of Na^+ ion touching it = 6.
Coordination number of Na^+ = 6 [placed in octahedral void of Cl^- ions].



Example : ***Halide of Li, Na, K and Rb, AgCl, AgBr, NH_4Cl.***

Note : *Lattice of NaCl is FCC of Cl^- in which all octahedral void are occupied by Na^+.*
or → FCC of Na^+ in which octahedral void are occupied by Cl^-.

(2) ZnS type (Zinc blende) (AB type) (4 : 4 coordination number compound) :

ZnS → ***Zinc blende → FCC***
ZnS → ***Wurtzite → HCP***

4 : 4

"coordination number of cation" — "coordination number of anion"

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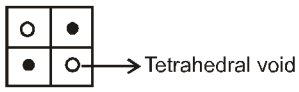
Zinc blende type :

(i) S^{2-} ion form the FCC lattice.

* ***Zn^{2+} ion occupy alternate (non adjacent) four tetrahedral void.***



(ii) Effective number of ion

* *Effective number of S^{2-} ion per unit cell = 4.*
* *Effective number of Zn^{2+} ion per unit cell = 4.*

(iii) Formula of unit cell = Zn_4S_4

Formula of ionic compound = ZnS

(iv) ***Experimental*** $\frac{r_{Zn^{2+}}}{r_{S^{2-}}}$ **= 0.3** ***(0.225 <*** $\frac{r_{Zn^{2+}}}{r_{S^{2-}}}$ ***< 0.414)***

so anion–anion contact is not there.

so $\boxed{[r_{Zn^{2+}} + r_{S^{2-}}] = \frac{\sqrt{3}}{4} a_{FCC}}$

(v) Density (d) = $\left[\frac{Z \times M}{N_A \times a^3}\right]$ M = 97 gram/mole. (M_{Zn} = 65.3) + (M_S = 32)

(vi) Coordination number :

Coordination number of Zn^{2+} ions = 4.

Coordination number of S^{2-} ions = 4.

Ex : ***ZnS, CuCl, CuBr, CuI, AgI.***

(3) CaF_2 [Flourite structure] [AB_2 type] (It is a different case)

Coordination number of compound

8 : 4

"coordination number of cation" *"coordination number of anion"*

(i) Ca^{2+} ion form the FCC lattice.

* ***F^- ion occupies all tetrahedral voids.***

(ii) Coordination number of F^- ion = 4 ∴ ***It occupies tetrahedral void.***

* ***Coordination number of Ca^{2+} ion = 8***

(iii) Effective number of Ca^{2+} ion = 4.

* ***Effective number of F^- ion = 8.***

(iv) Formula of unit cell = Ca_4F_8

* ***Formula of ionic compound = CaF_2.***

(v) Effective number of formula unit cell (Z) = 4.

(vi) Density = $\left[\frac{Z \times M}{N_A \times a^3}\right]$

(vii) ***0.225 <*** $\frac{r_{F^-}}{r_{Ca^{2+}}}$ ***< 0.414*** ; $[r_{Ca^{2+}} + r_{F^-}] = \frac{\sqrt{3}a}{4}$

Other examples : ***CaF_2 , SrF_2 , BaF_2, $BaCl_2$.***

(4) Na_2O [Sodium oxide] [A_2B type]

* ***Anti-fluorite structure [4 : 8 coordination compound]***

⇒ (O^{2-} ion at FCC lattice).

Na^+ = All the tetrahedral void.

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(5) CsCl type of structure [8 : 8 coordination compound] :

$$\frac{r_{Cs^+}}{r_{Cl^-}} \simeq 0.93$$

(i) Cl^- form simple cubic lattice.

* ***Cs^+ ion occupies cubical void.***

(ii) Effective number of ion :

* ***Effective number of Cs^+ ion = 1. & Effective number of Cl^- ion = 1.***

(iii) Formula of ionic compound = CsCl. So Z = 1.

(iv) Density (d) = $\left[\frac{Z \times M}{N_A \times a^3}\right]$

* ***anion–anion contact is not there so,***

$$[r_{Cl^-} + r_{Cs^+}] = \frac{\sqrt{3}\, a_{SC}}{2}$$

(v) Coordination number of Cs^+ ion = 8.

(vi) Coordination number of Cl^+ ion = 8.

Other example : *CsCl, CsBr, CsI.*

Note : On increasing pressure → Coordination number tends to increase.
On increasing temperature → Coordination number tends to decrease.

$4:4 \xrightarrow{P\uparrow} 6:6$; $8:8 \xrightarrow{T\uparrow} 6:6$

Other important structure :

(1) Spinel structure [AB_2O_4] : Spinel is an oxide consisting of two type of metal ions with the oxides ion arranged in CCP layers in normal spinel one eight of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral holes occupied by another type of metal ion. eg : A spinel is formed by Zn^{2+}, Al^{3+} and O^{2-} with Zn^{2+} ion in the tetrahedral holes. The formula of the spinel is $ZnAl_2O_4$.

(2) Perovskite structure [ABO_3] : This structure may be described as a cubic lattice, with barium ions occupying the corners of the unit cell, oxide ions occupying the face centers and titanium ions occupying the centres of the unit cells. eg : ***$BaTiO_3$ or $MgTiO_3$.***

(3) Lattice of diamond : ZnS types in which all S^{2-} location and all Zn^{2+} location are occupied by C atoms.
So, Z = 8 atom per unit cell

$$d_{c-c} = \frac{\sqrt{3}}{4} a_{FCC}$$

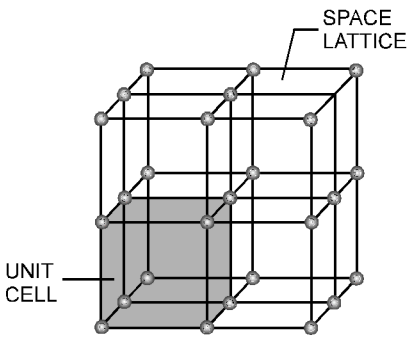
$d_{C-C} = 2r_{C-C}$ where r_{C-C} is radius of C-atom.

* ***Packing efficiency = 34%.***
* ***In crystal lattice of diamond, carbon atoms adopt FCC arrangement with occupancy of 50% tetrahedral voids.***

Note : (1) Distance between two plane in FCC or HCP arrangement is $\sqrt{\frac{2}{3}}\, a$ or $a = 2R = 2\sqrt{\frac{2}{3}}\, R$.

(2) If number of unit cell along one edge are 'x' then total number of unit cell in cube = x^3.

e.g.



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Number of unit cell along one edge = 2, then total number of unit cell in cube = $2^3 = 8$.

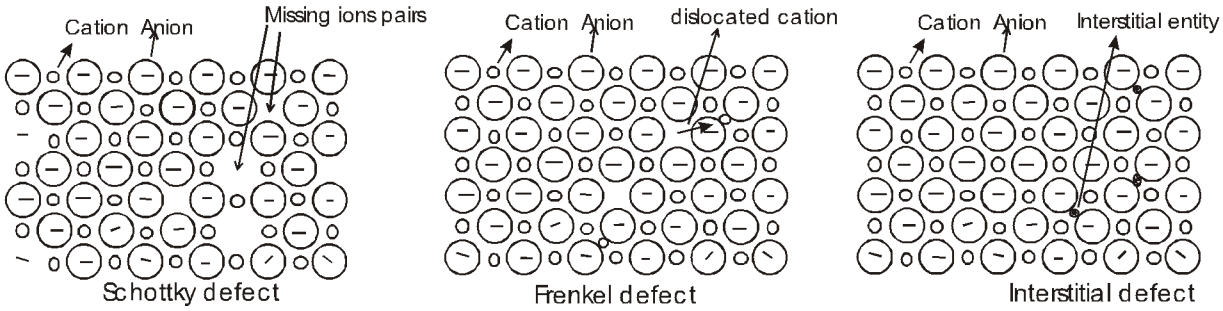
Point Defects :

1. Stoichiometric -The formula of compound remains same even after presence of these defects.

2.Non-stoichiometric -The formula of compound will get modified because of the presence of these defects.

(i) Stoichiometric defects :

(a) Schottkey (b) Frenkel (c) Interstitial



Schottkey : When atoms/point are totally missing from the lattice.

net density of crystal will get decreased

$\rho_{exp} < \rho_{theoritical}$

$$\% \text{ missing units} = \left(\frac{\rho_{th} - \rho_{exp}}{\rho_{th}}\right) \times 100\%$$

Frenkel : When atoms/ions displaced from normal lattice positions and are present in some interstitial voids. Density remains same

Interstitial : When some small foreign atoms are trapped in interstitial voids of the lattice without any chemical reaction. Formula remains the same $\rho_{exp} > \rho_{theortical}$

(ii) Non-stoichiometric defects :

(a) metal excess or cation excess defect :

- If no. of missing anion is more than no. of missing cations.
- To maintain electrical neutrality some electron are trapped at anionic vacancies.

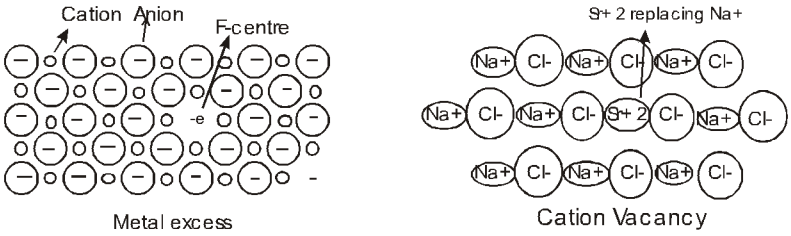
Because of these extra elecorons the electrical and optical (colour) properties of the compound get modified. So these locations of elecoron are also known as color centres (F-centres).

ZnO - white in color at room temperature.

- on heating some O^{2-} ion get released in the form of O_2 and e^- are trapped at their locations. Because of this it because yellow in color.

(b) metal deficiency (cation deficiency) defects :

- no. of missing cations should be more than no. of missing anions.
- oxidation no. of cation will get modified to maintain electrical neutrality.



Properties of Solids :

(i) Electrical Properties

(ii) Magnetic Properties

(i) Electrical Properties : Solids exhibit an amazing range of electrical conductivities, the range of electrical

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conductivities from 10^{-20} to 10^{7} ohm^{-1} m^{-1}. Solids can be classified into three types on the basis of their conductivities.

(1) Conductors : *Metals are goods condcutors and have conductiveites in the order 10^{7} $(\Omega m)^{-1}$.*

(2) Insulators : *Those solids which have very low conductivities ranging from 10^{-20} to 10^{-10} $(\Omega m)^{-1}$ are electrical insulators* e.g.; MnO, CoO; NiO, CuO, Fe_2O_3, TiO_2.

(3) Semicondutors : *Those solids which have intermideidate conductivities generally from 10^{-6} to $10^{4}(\Omega m)^{-1}$ are termed as semicondcutors.*

Intrinsic Semicondcutors : *The conduction by pure subtances such as silicon and germanium is called intrisic conduction and these pure substances exhibiting electrical condcutivity are called intrinsic semicodcutors.*

Super Conductivity : *The electrical resistance of metals is found to depend on temperature. Electrical resistance decreases with decrease in temperature and becomes almost zero near the absolute zero. Materials in this state are said to possess. superconductivity.* The phenonmenon of superconductivity was first discovered by kammerlingh Onners in 1913 when he found that mercury becomes superconducting at 4.0 K temperature.

Transition Temperature : *The temperature at which a substance starts behaving as super-condcutor is called transition temperature.*

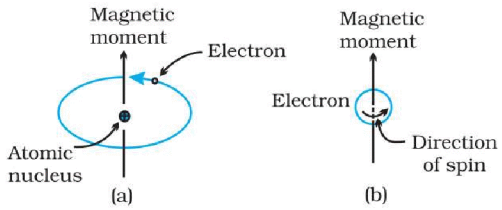
(ii) Magnetic Properties :

Every substance has some magnetic properties associated with it. The origin of these properties lies in the electrons. Each electron in an atom behaves like a tiny magnet.

Its magnetic moment originates from two types of motions.

(i) its orbital motion around the nucleus.

(ii) its spin around its own axis.



Demonstration of the magnetic moment associated with
(a) an orbiting electron and (b) a spinning electron.

<u>On the basis of their magnetic properties, substances can be classified into five categories:</u>

(i) paramagnetic **(ii) diamagnetic**

(iii) ferromagnetic **(iv) antiferromagnetic and**

(v) ferrimagnetic.

(1) Paramagnetism : *When substances which are attracted by the external magnetic field are called paramagnetic substances and the phenomenon is called as paramagnetism.* Atoms ion or molecules containing unpaired electron show this property, eg. O_2 Cu^{2+}, Fe^{3+} etc. these substances lost their magnetism in the absence of magnetic field.

(2) Diamagnetic materials : *Those materials which are repelled by magnetic field are called diamagnetic materials* e.g. Cu^{+}, TiO_2, NaCl and C_6H_6. They do not have unpaired electrons.

(3) Ferromagnetism : *When substances show permanent magnetism even in the absence of the magnetic field this phenomenon is called as Ferromagnetism and such substances as called as Ferromagnetic substances* e.g. Fe Ni Co and CrO_2.

This type of magnetism arise due to the spontaneous alignment of magnetic moments due to unpaired electron in the same direction.

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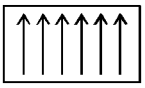
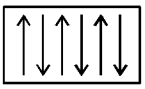
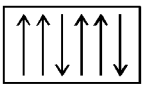


(4) Anti Ferro Magnetism : *Substances which are expected to possess paramagnetism or ferro magnetism on the basis of unpaired electrons but actually they possess zero net magnetic moment are called Anti-ferromatic substances and the phenomenon is called as Anti-ferromagnetism.* eg. MnO, Anti-ferromagnetism is due to the presence equal number of magnetic moments in the opposite direction.



(5) Ferrimagnetism : *Substances which are expected to possess large magnetism on the basis of unpaired electrons, but actually have small magnetic momentum are called ferrimagnetic substances* eg. Fe_3O_4, ferrites of the formula M^{2+}, Fe_2O_4 where M = Mg, Cu, Zn etc. Ferrimagnetism arises due to the unequal moments in opposite direction resulting in same net magnetic moment. On heating these substance loss their magnetism and convert in to paramagentic substance



Curie Temperature : *The temperature at which a ferromagnetic substance loses its ferromagnetism and becomes only paramagnetic.* For iron the curie temperature is 1033 K and for nickel 629 K, for Fe_3O_4 850 K. Below this temperature paramagnetic solid becomes ferromagnetic.

Domain : *In solid state the metal ions of ferromagnetic substances are grouped together into small regions called domains.*

Effect of Temperature on Metal (Conductor) Semiconductor or Insulator :

1. The conductivity of semiconductors and insulators increases with increase in temperature

2. The conductivity of metal (conductors) decreases with increase in temperature.

* * * * *

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s-BLOCK & Boron and Carbon Family

Group 1 of the periodic table consists of the elements : lithium, sodium, potassium, rubidium, caesium and francium .
The elements of Group 2 include beryllium, magnesium, calcium, strontium, barium and radium.

Hydration Enthalpy :

The hydration enthalpies of alkali metal ions decrease with increase in ionic sizes.Li^+ has maximum degree of hydration and for this reasons lithium salts are mostly hydrated e.g., $LiCl \cdot 2H_2O$

Physical properties :

All the alkali metal are silvery white, soft and light metals. Because of the larger size, these element have low density. The melting and boiling point of the alkali metals are low indicating weak metallic bonding alkali metals and their salts impart characteristic colour to an oxidizing flame.

Metal	Li	Na	K	Rb	Cs
Colour	Crimson red	Yellow	Violet/ Lilac	Red violet	Blue

Chemical Properties:

The alkali metal are highly reactive due to their larger size and low ionization enthalpy.

- **Reactivity towards air :** They burn vigorously in oxygen forming oxides. Lithium forms monoxide, sodium forms peroxide, the other metals form superoxide.
- **Reducing nature:** The alkali metals, are strong reducing agents, lithium being the most and sodium the least powerful.
- **Solution in liquid ammonia:** The alkali metals dissolve in liquid ammonia giving deep blue solution which are conducting in nature.

$$M + (x + y)\, NH_3 \longrightarrow [M(NH_3)_x]^+ + [e(NH_3)_y]^-$$

The blue colour of the solution is due to the ammoniated electron and the solutions is paramagnetic.

$$M^+(am) + e^- + NH_3(\ell) \xrightarrow{\text{on standing}} MNH_2(am) + 1/2\, H_2(g)$$

In concentrated solution, the blue colour changes to bronze colour and becomes, diamagnetic.

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ANOMALOUS PROPERTIES OF LITHIUM

(i) exceptionally small size of its atom and ion, and (ii) high polarising power (i.e., charge/ radius ratio).

The similarity between lithium and magnesium is particularly striking and arises because of their similar size: atomic radii, Li = 152 pm, Mg = 160 pm; ionic radii : Li^+ = 76 pm, Mg^{2+} = 72 pm.

GROUP 2 ELEMENTS : ALKALINE EARTH METALS

The first element beryllium differs from the rest of the member and shows diagonal relationship to aluminium.

Hydration Enthalpies

Hydration enthalpies of alkaline earth metal ions. $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$. The hydration enthalpies of alkaline earth metal ions are larger than those of alkali metal ions. Thus, compounds of alkaline earth metals are more extensively hydrated than those of alkali metals , e.g., $MgCl_2$ and $CaCl_2$ exist as $MgCl_2.6H_2O$ and $CaCl_2.6H_2O$ while NaCl and KCl do not form such hydrates.

Physical Properties

The alkaline earth metals, in general, are silvery white, lustrous and relatively soft but harder than the alkali metals. The melting and boiling point of these metals are higher due to smaller sizes. Because of the low ionisation enthalpies they are strongly electropositive in nature. The electrons in beryllium and magnesium are too strongly bound to get excited by flame. Hence these elements do not impart any colour to the flame.

Calcium, strontium and barium impart characteristic colour to the flame.

Metal	Be	Mg	Ca	Sr	Ba
Colour	No colour	No colour	Brick red	Crimson	Apple green

Chemical Properties

❍ **Reactivity towards air and water :** Beryllium and magnesium are inert to oxygen and water. Magnesium is more electropositive and burns with dazzling brilliance in air to give MgO and Mg_3N_2 .Calcium, strontium and barium are readily attacked by air to form the oxide and nitride.

❍ **Reducing nature :** The alkaline earth metals are strong reducing agent. This is indicated by large negative value of their reduction potentials.

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❍ **Solution in liquid ammonia:** The alkaline earth metals dissolve in liquid ammonia to give deep blue black solution forming ammoniated ions.

$$M + (x + y)\ NH_3 \rightarrow [\ M(NH_3)_x]^{2+} + 2\ [e(NH_3)_Y\ ]^-$$

From these solutions, the ammoniates, $[M(NH_3)_6]^{2+}$ can be recovered.

ANOMALOUS BEHAVIOUR OF BERYLLIUM

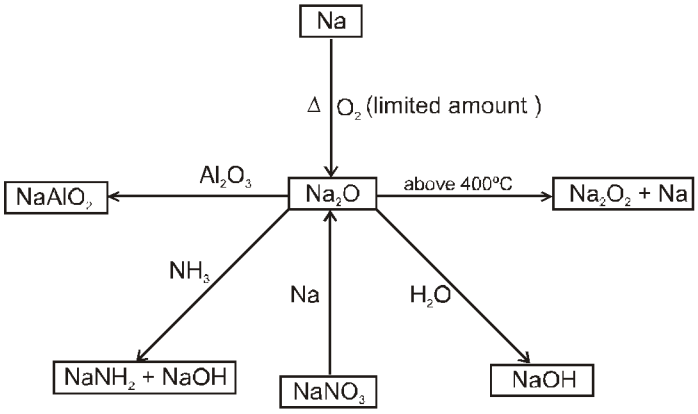
Beryllium the first member of the Group 2 metals, shows anomalous behaviour as compared to magnesium and rest of the members. Further, it shows diagonal relationship to aluminium.

Diagonal Relationship between Beryllium and Aluminium

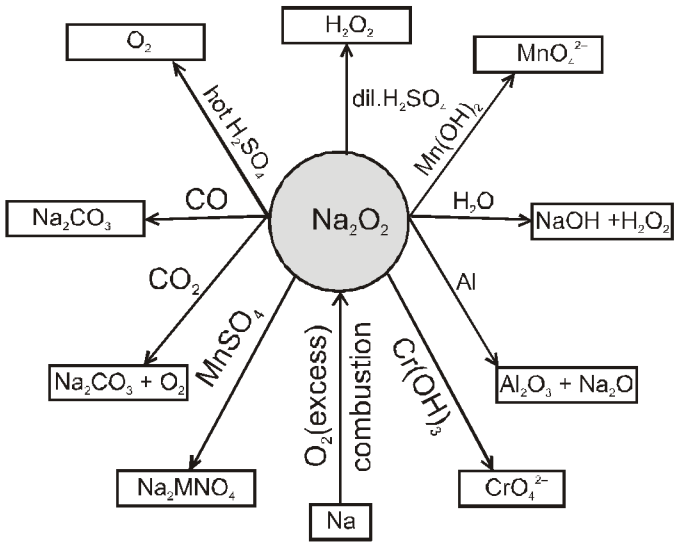
The ionic radius of Be^{2+} is estimated to be 31 pm; the charge/ radius ratio is nearly the same as that of the Al^{3+} ion. Hence beryllium resembles aluminium in some ways.

Compounds of s-block elements :

1. Sodium Oxide (Na_2O) :



2. Sodium peroxide (Na_2O_2) :



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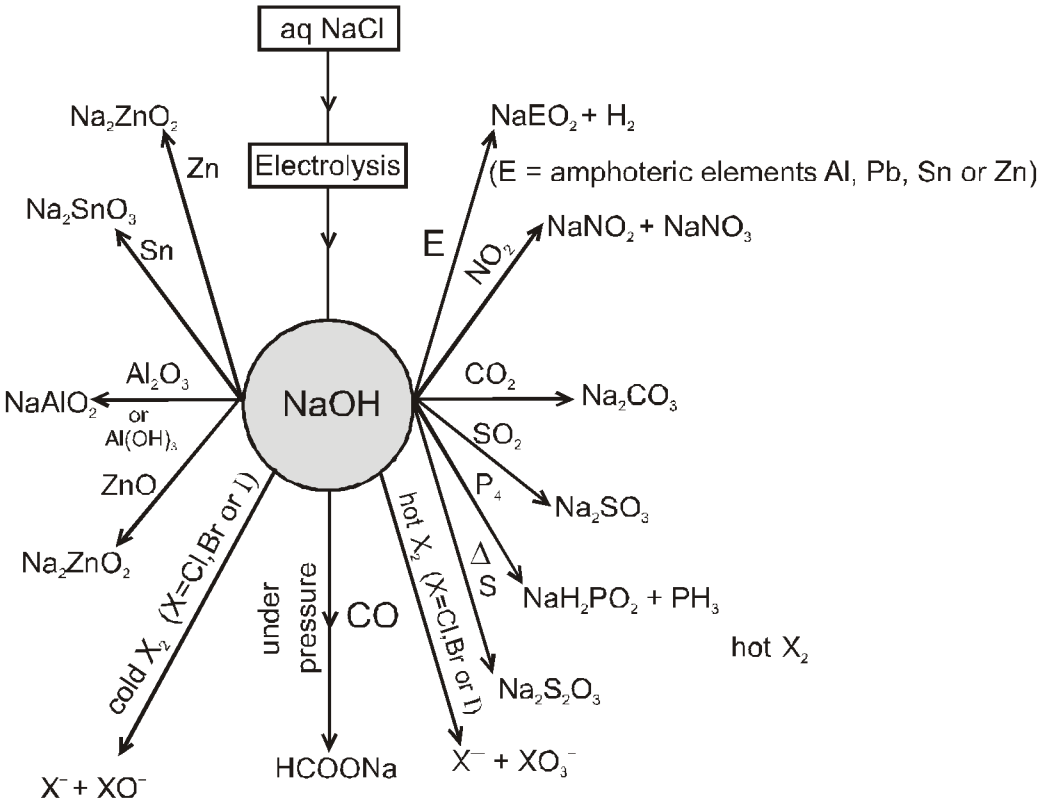
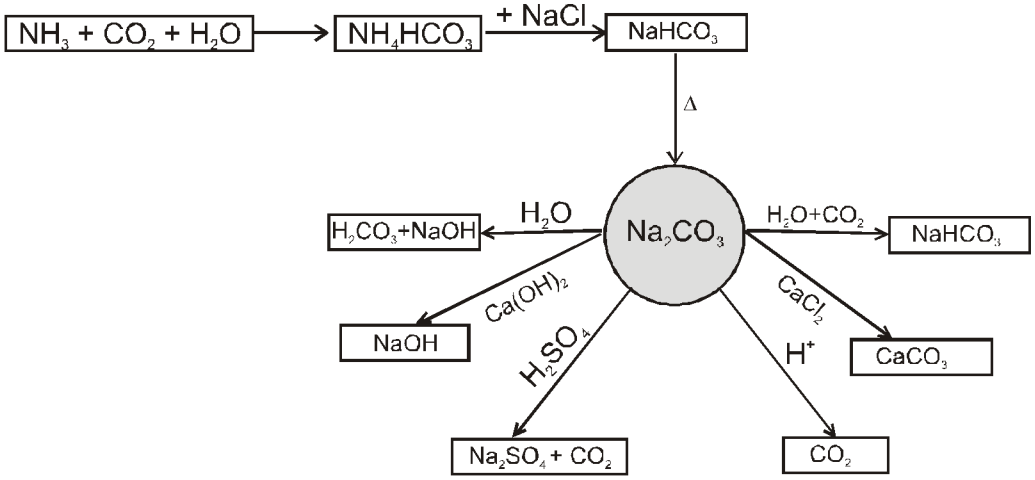
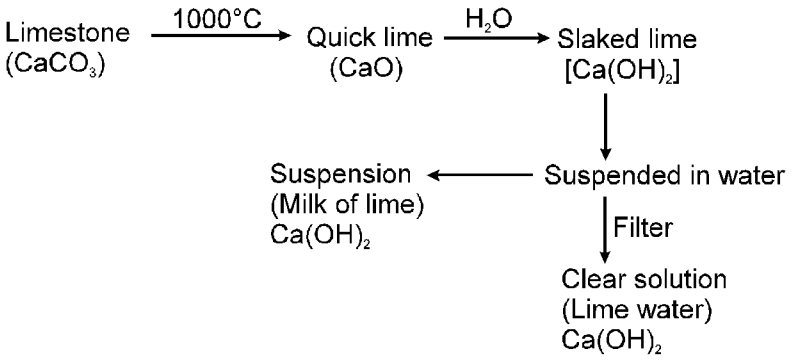
3. Sodium Hydroxide (NaOH) :



4. Sodium Carbonate (Na_2CO_3) :



5. Quick Lime, Slaked Lime and Lime Water :



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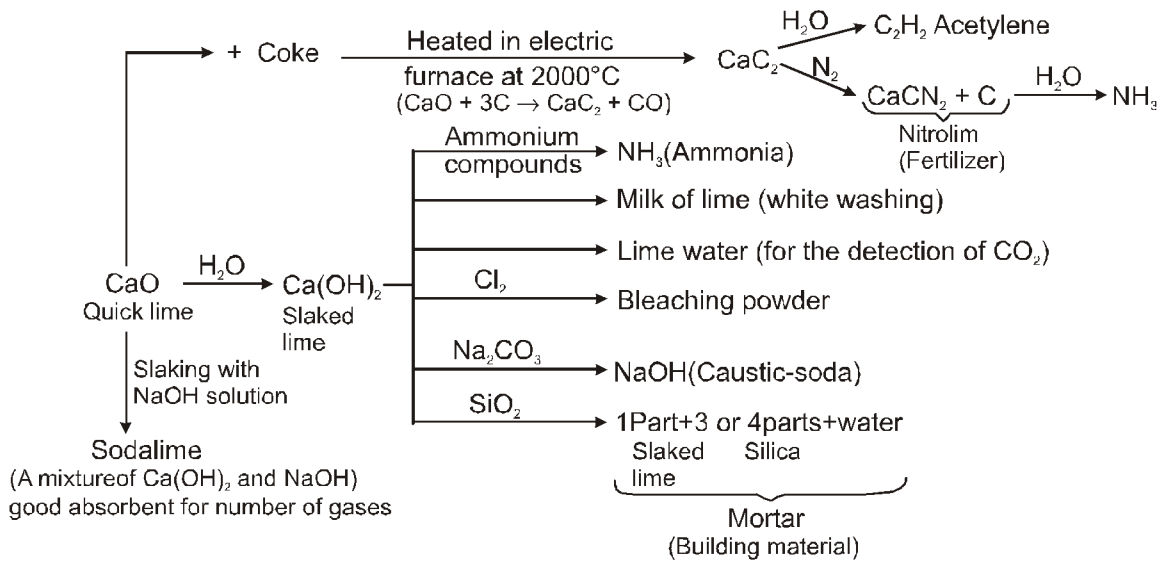
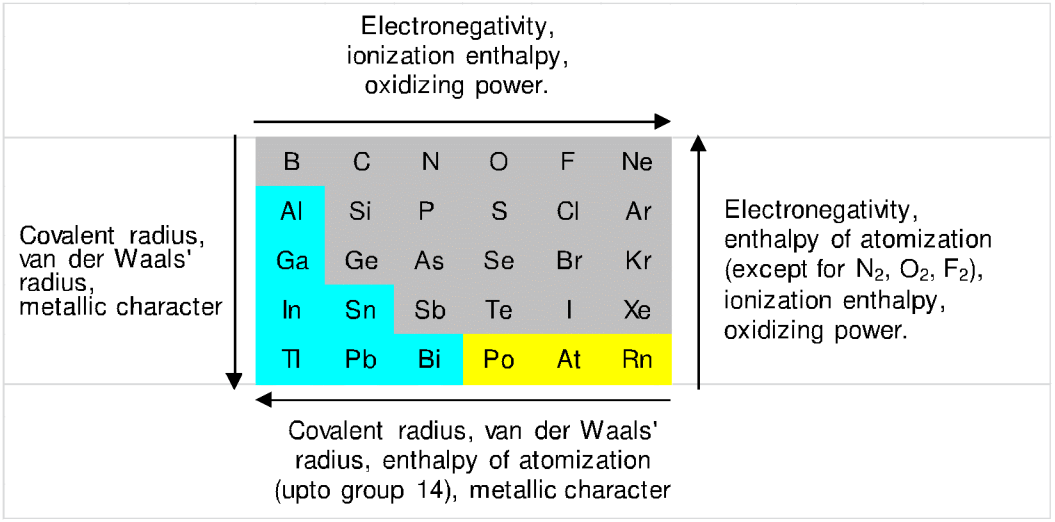


$$3Ca(OH)_2 + 2Cl_2 \longrightarrow Ca(OCl)_2.Ca(OH)_2.CaCl_2.2H_2O \text{ (bleaching powder).}$$

p-BLOCK ELEMENTS & THEIR COMPOUNDS

TRENDS IN PROPERTIES OF p-BLOCK ELEMENTS.



(A) GROUP 13 ELEMENTS : THE BORON FAMILY

Oxidation state and trends in chemical reactivity :

General Oxidation State = + 3.

Reactivity towards acids and alkalies

$$2\,Al(s) + 6\,HCl(aq) \longrightarrow 2\,Al^{3+}(aq) + 6\,Cl^{-}(aq) + 3\,H_2(g)$$

$$2Al(s) + 2NaOH\,(aq) + 6H_2O\,(1) \longrightarrow 2Na^{+}\,[Al(OH)_4]^{-}\,(aq) + 3H_2(g)$$

Sodium tetrahydroxoaluminate (III)

Reactivity towards halogens

$$2E(s) + 3X_2\,(g) \rightarrow 2EX_3\,(s) \qquad (X = F, Cl\ Br, I)$$

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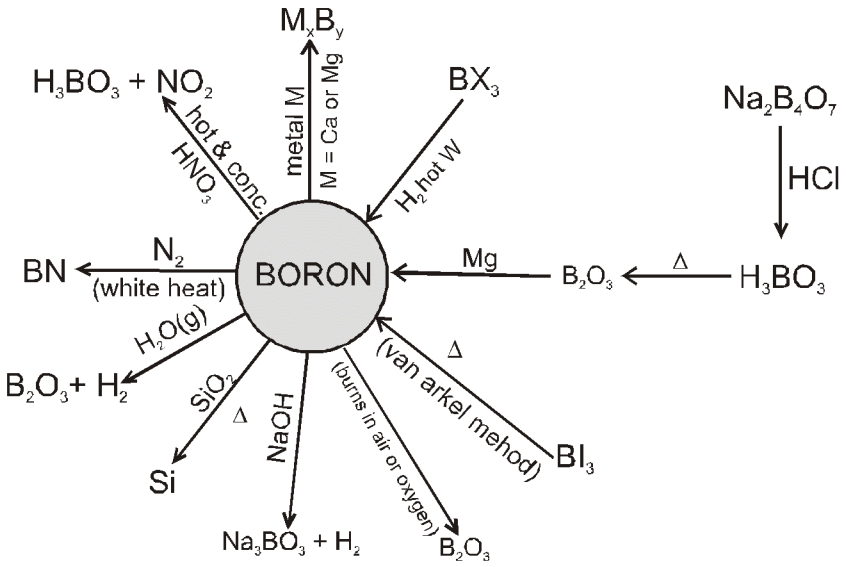
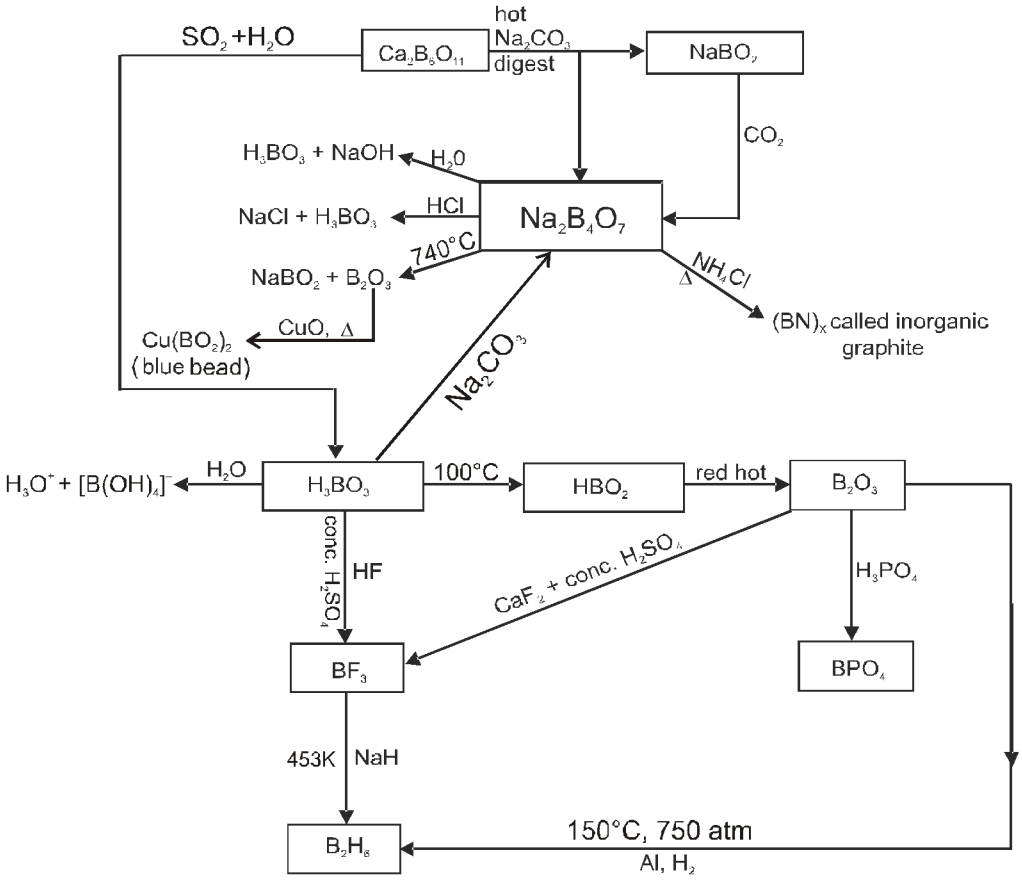
BORON (B):

Some Important Reactions of Boron and its compounds :





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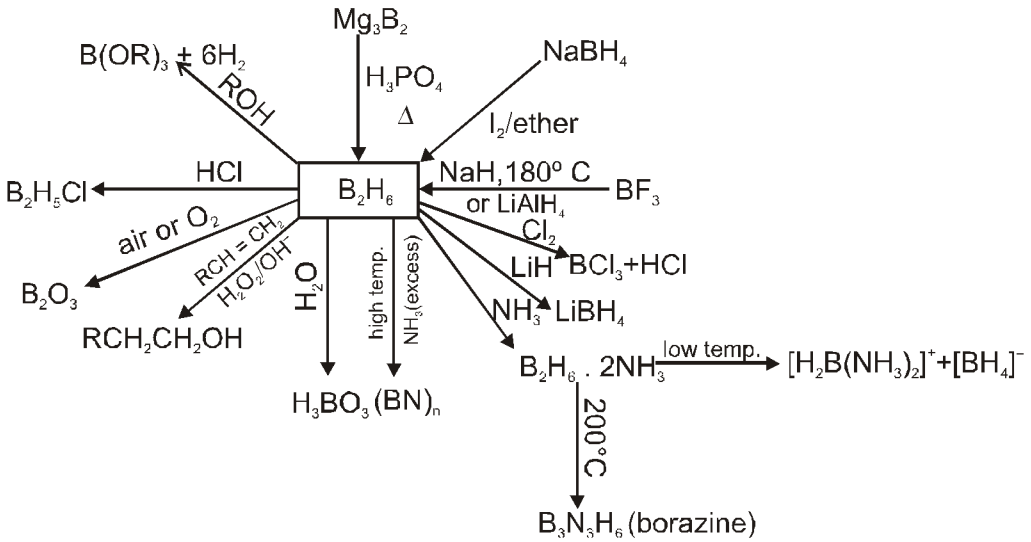

❍ Small amines such as NH_3, CH_3NH_2 and $(CH_3)_2NH$ give unsymmetrical cleavage of diborane.

$$B_2H_6 + 2NH_3 \longrightarrow [H_2B(NH_3)_2]^+ + [BH_4]^-$$

❍ Large amines such as $(CH_3)_3N$ and pyridine give symmetrical cleavage of diborane.

$$2(CH_3)_3N + B_2H_6 \longrightarrow 2H_3B \longleftarrow N(CH_3)_3$$

❍ $B_2H_6 + 2CO \xrightarrow{200°C,\ 20\ atm} 2BH_3CO$ (borane carbonyl)

(B) GROUP 14 ELEMENTS : THE CARBON FAMILY

Carbon (C), silicon (Si), germanium (Ge), tin (Sn) and lead (Pb) are the members of group 14.

Electronic Configuration = $ns^2 np^2$.

Oxidation states and trends in chemical reactivity

Common oxidation states = +4 and +2. Carbon also exhibits negative oxidation states. In heavier members the tendency to show +2 oxidation state increases in the sequence Ge < Sn < Pb.

(i) Reactivity towards oxygen :

All members when heated in oxygen form oxides. There are mainly two types of oxides, i.e. monoxide and dioxide of formula MO and MO_2 respectively.

(ii) Reactivity towards water :

Tin decomposes steam to form dioxide and dihydrogen gas.

(iii) Reactivity towards halogen :

These elements can form halides of formula MX_2 and MX_4 (where X = F, Cl Br, I). Stability of dihalides increases down the group.

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ANOMALOUS BEHAVIOUR OF CARBON :

Catenation :

The order of catenation is C > > Si > Ge ≈ Sn. Lead does not show catenation. Due to the property of catenation and pπ-pπ bonds formation, carbon is able to show allotropic forms.

Bond	Bond enthalpy (kJ mol^{-1})	Bond	Bond enthalpy (kJ mol^{-1})
C—C	348	Si—Si	297
Ge—Ge	260	Sn—Sn	240

Allotropes of Carbon

Diamond :

Crystalline lattice sp^3 hybridisation and linked to four other carbon atoms by using hybridised orbitals in tetrahedral manner. The C–C bond length is 154 pm. and produces a rigid three dimensional network of carbon atoms.

Graphite :

Graphite has layered structure. Layers are held by van der Waal's forces and distance between two layers is 340 pm. Each layer is composed of planar hexagonal rings of carbon atoms. C – C bond length within the layer is 141.5 pm. Each carbon atom in hexagonal ring undergoes sp^2 hybridisation graphite conducts electricity along the sheet. Graphite cleaves easily between the layers and therefore, it is very soft and slippery. For this reason graphite is used as a dry lubricant in machines running at high temperature.

Fullerenes :

C_{60} molecule has a shape like soccer ball and called **Buckminsterfullerene.** It contains twenty six -membered rings and twelve five membered rings. This ball shaped molecule has 60 vertices and each one is occupied by one carbon atom and it also contains both single and double bonds with C – C distance of 143.5 pm and 138.3 pm respectively.

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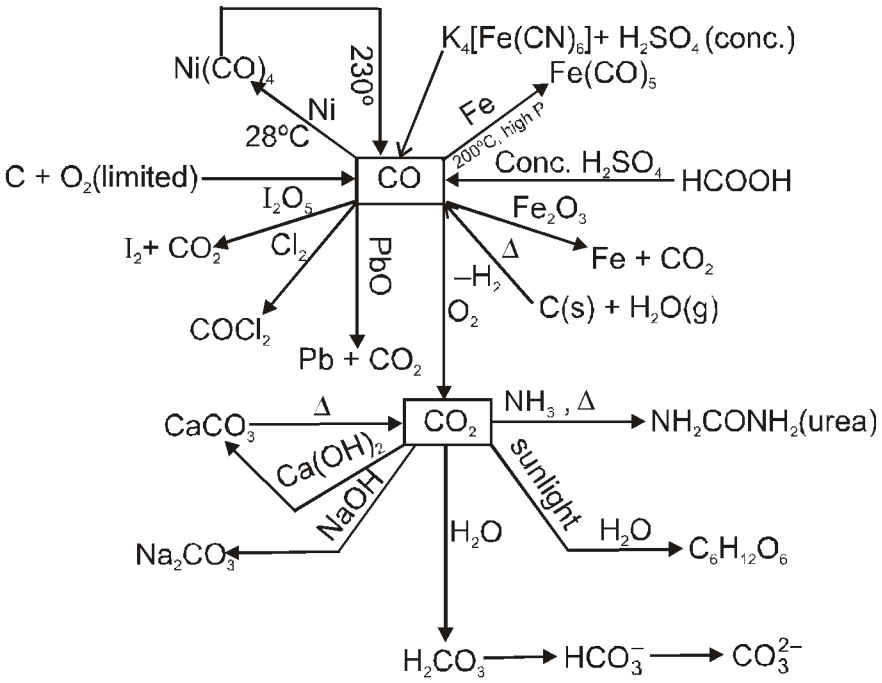
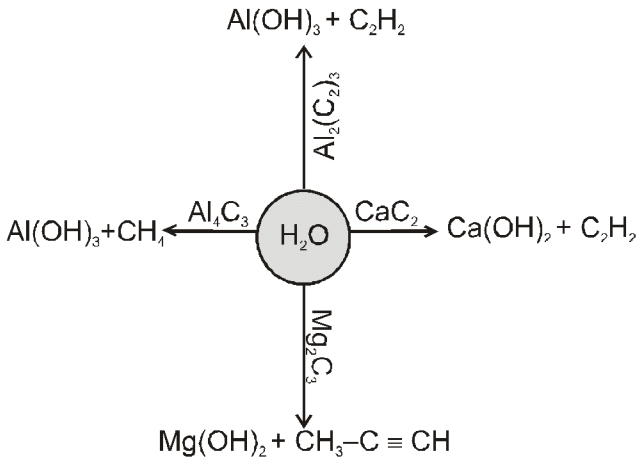
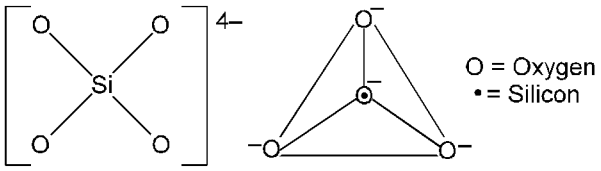
SOME IMPORTANT REACTIONS OF CO, CO_2 AND METAL CARBIDES :



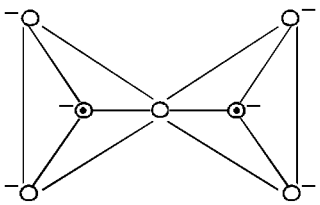


● CLASSIFICATION OF SILICATES :

(A) Orthosilicates :



(B) Pyrosilicate :



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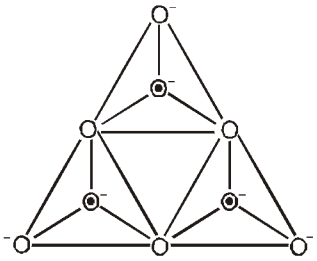
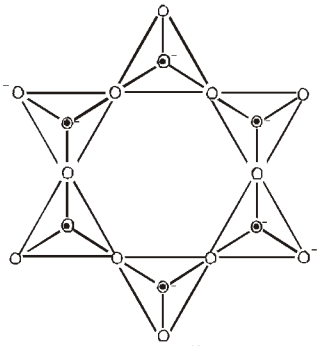
(C) Cyclic silicates :

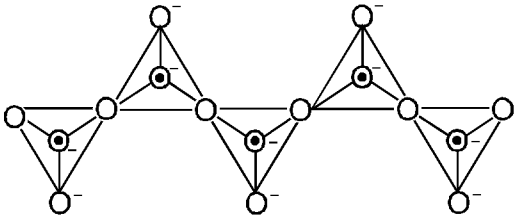


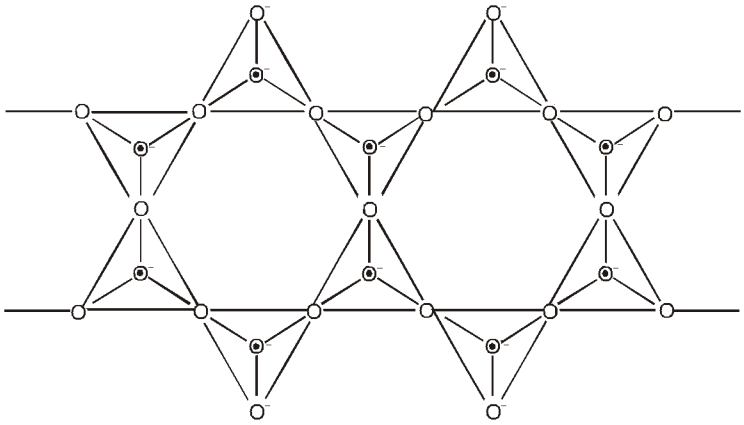
$Si_3O_9^{6-}$



$Si_6O_{18}^{12-}$

(D) Chain silicates :





(E) Two dimensional sheet silicates :

In such silicates, three oxygen atoms of each tetrahedral are shared with adjacent SiO_4^{4-} tetrahedrals. Such sharing forms two dimension sheet structure with general formula $(Si_2O_5)_n^{2n-}$

(F) Three dimenstional sheet silicates :

These silicates involve all four oxygen atom in sharing with adjacent SiO_4^{4-} tetrahedral units.

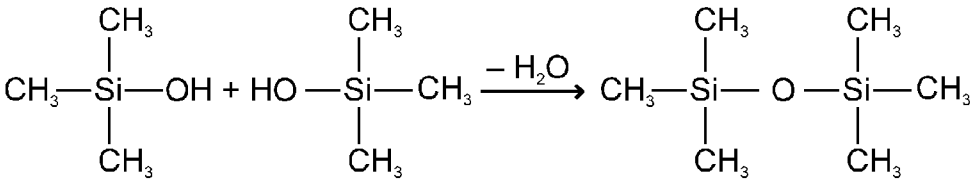
● SILICONES :

☞ Silicones can be prepared from the following types of compounds only.

(i) R_3SiCl (ii) R_2SiCl_2 (iii) $RSiCl_3$

☞ Silicones from the hydrolysis of $(CH_3)_3SiCl$

$$2\,(CH_3)_3SiCl \xrightarrow{H_2O} 2(CH_3)_3Si(OH) \longrightarrow$$



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BIOMOLECULES & POLYMERS

Classification of Carbohydrate

(A) Classification on the basis of number of hydrolysed products

S.No.	Carbohydrate	No. of units (on hydrolysis)	Examples
1	Monosaccharides	1 or single unit (cannot be hydrolysed)	Glucose, Fructose, Galactose
2	Oligosaccharides	2 to 10 units	Sucrose, Maltose, Lactose
3	Polysaccharides	Many units	Starch, Cellulose, Gums, Resins etc

(B) Classification on the basic of functional groups

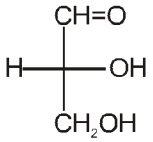
S.No.	Carbohydrate	No. of functional group	Examples
1	Aldose	Aldehyde CH = O | $(CHOH)_n$ | CH_2OH	Glyceraldehyde, Erythrose, Threose, Ribose & 2-Deoxyribose Glucose, Mannose, Allose.
2	Ketose	Ketone CH_2OH | C = O | $(CHOH)_n$ | CH_2OH	n = 0; Ketotriose, n = 1; Ketotetroses, n = 2; Ketopentoses, n = 3; Ketohexoses

(C) Classification of monosaccharides on basis of carbon atoms in hydrolysed product.

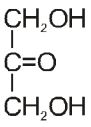
S.No.	Carbon atoms	General term	Aldehyde	Ketone
1	3	Triose	Aldotriose	Ketotriose
2	4	Tetrose	Aldotetrose	Ketotetrose
3	5	Pentose	Aldopentose	Ketopentose
4	6	Hexose	Aldohexose	Ketohexose
5	7	Heptose	Aldoheptose	Ketoheptose

Some facts :

1. Number of carbons in monosaccharides are generally 3 to 7.
2. Simplest aldose is Glyceraldehyde and simplest Ketose is Dihydroxyacetone.

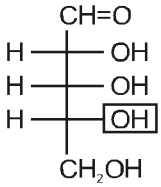


(D-Glyceraldehyde)

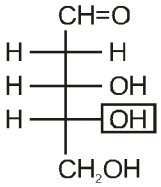


(Dihydroxyacetone)

3. Most naturally occuring monosacharides are :
   (a) Pentoses E.g. D-Ribose (present in RNA) and 2-Deoxyribose (present in DNA)
   (b) Hexoses E.g. D-Glucose, D-Mannose, D-Allose, D-Galactose and D-Fructose



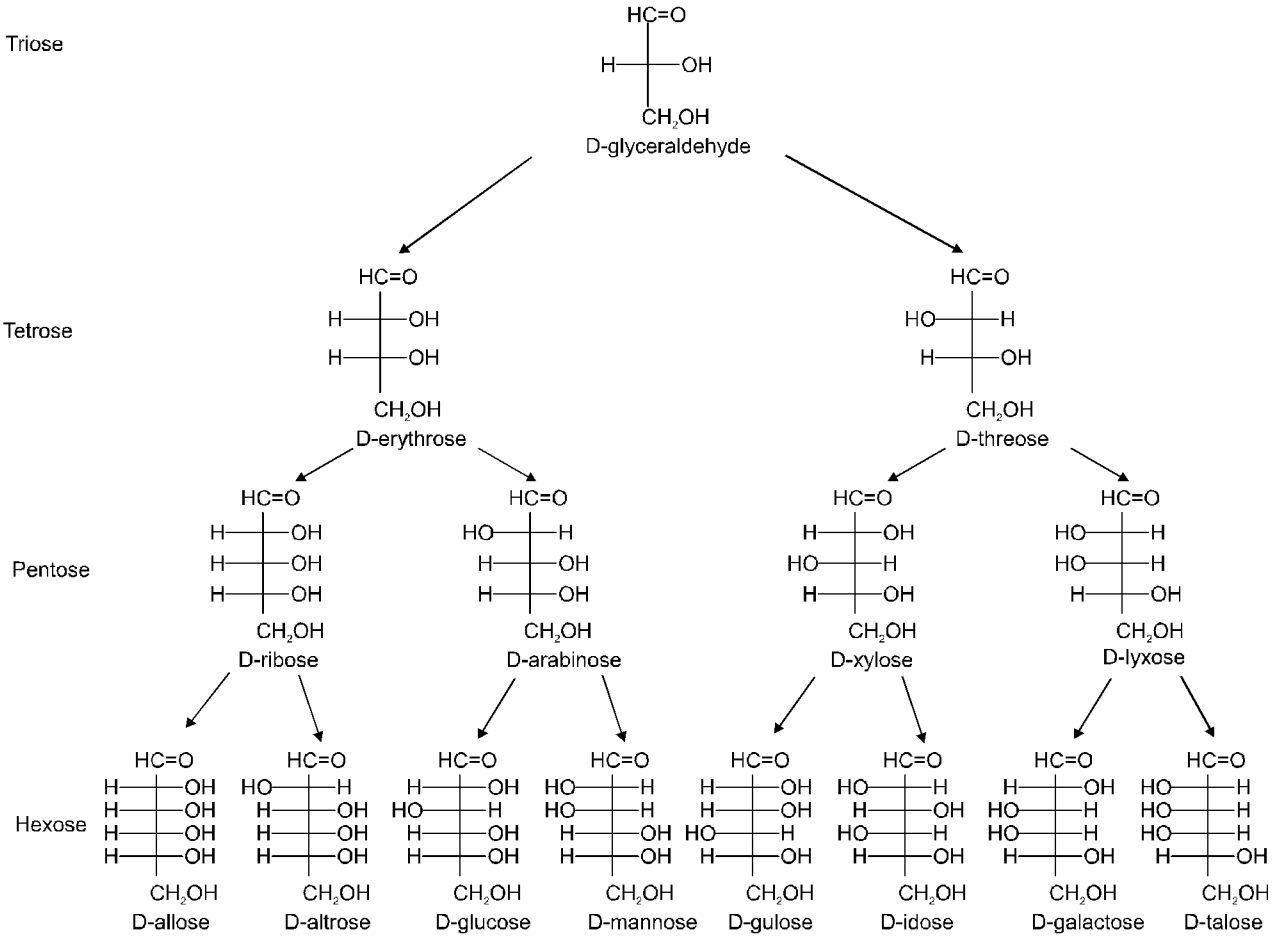
(D-Ribose)



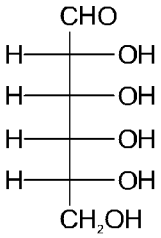
(D-2-Deoxyribose)

3. Stereochemistry of Aldoses :

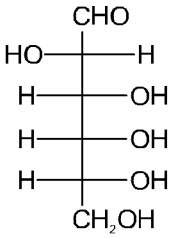
The monosaccharides are chiral and may have D or L configuration. A simple illustration of all 'D' forms is given below.



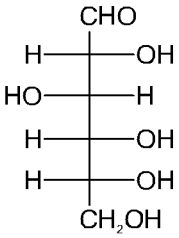
Optical isomers of Aldohexoses : Aldohexoses have four asymmetric carbon atoms, therefore they have sixteen optical isomers out of which 8 are D and 8 are L variety (overall Eight pairs of enantiomers). D-variety of them are as follows



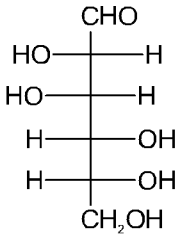
D(+)-allose
(I)



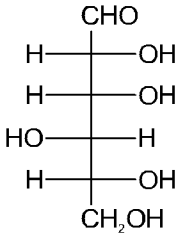
D(+)-altrose
(II)

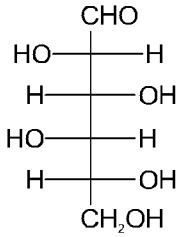


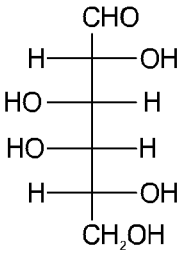
D(+)-glucose
(III)

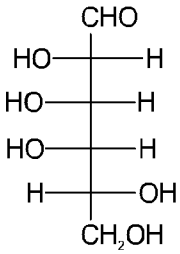


D(+)-mannose
(IV)


D(–)-gulose
(V)


D(–)-idose
(VI)

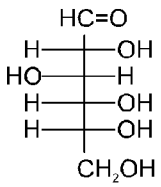
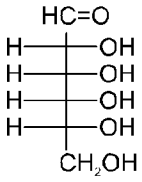
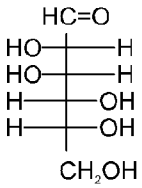
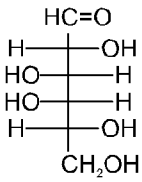
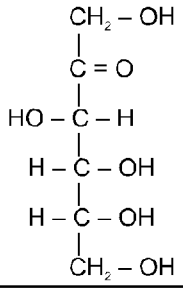

D(+)-galactose
(VII)


D(+)-talose
(VIII)

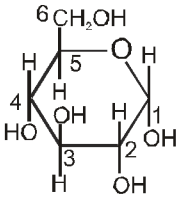
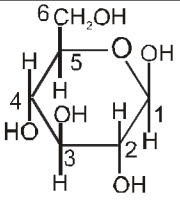
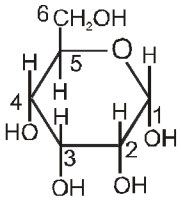
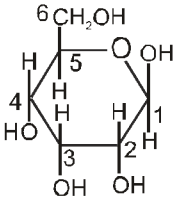
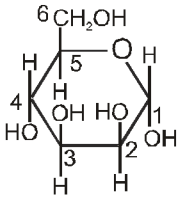
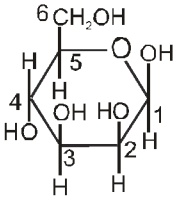
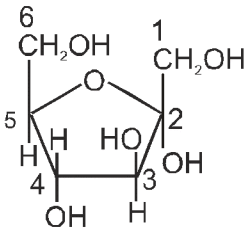
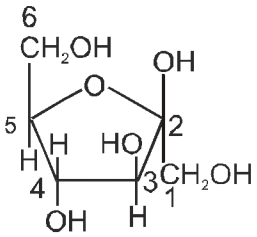
Note : **1.** D-aldohexoses shown above have diastereomeric relationship with each other
2. D-aldohexoses can be either dextro (+) or laevo (-).

4. **Structure of aldohexoses :** All form of Aldose or ketose may exist in open chain form as well as in cyclic pyranose or furanose form.

(i) **Open chain structure of mono saccharide**

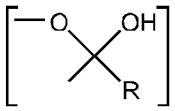
Carbohydrate	Structure	Functional Group	Typical nature
D - Glucose		aldehyde	3rd (L)
D - Allose		aldehyde	No (L)
D - Mannose		aldehyde	2, 3 (L)
D - Galatose		aldehyde	3, 4 (L)
D - Fructose		Ketone	3 (L)

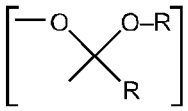
(ii) **Cyclic structure of monosaccharide**

Carbohydrate	Cyclic structure	
Glucose		
	α^{o} = (112°) α-D-glucopyranose	α^{o} = (19°) β-D-glucopyranose
Allose		
	α-D-Allopyranose	β-D-Allopyranose
Mannose		
	α-D-Mannopyranose	β-D-Mannopyranose
Fructose		
	α-fructofuranose	**β-fructofuranose**

- **Anomers :**

Anomers are diastereomers that differ in the configuration at the acetal or hemiacetal C atom of a sugar in its cyclic form (Anomeric carbon: A carbon bonded with two 'O' atoms). For example, α D(+) and β -D(+) glucose are anomers. α-D(–) and β-D(–) fructose are anomers.

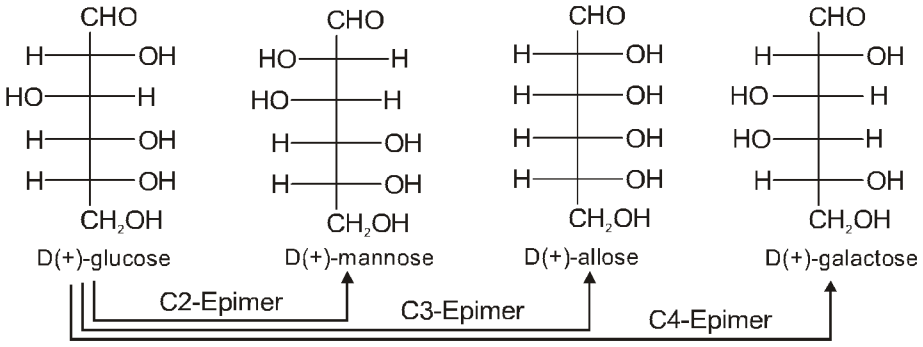

Hemiacetal functional group


Acetal functional group

- **Epimers :** Diastereomers with more than one stereocentre that differ in the configuration about only one stereocentre (other than anomeric carbon) are called epimers.

i. D-Erythrose and L-threose are epimers.

ii. D-glucose and D-galactose are C-4 epimers and
iii. D-idose and D-talose are C-3 epimers.
iv. D-glucose and D-mannose are C-2 epimers.
v. Epimerisation of glucose at C-2 gives mannose.
vi. Epimerisation of glucose at C-3 gives allose.
vii. Epimerisation of glucose at C-4 gives galactose.



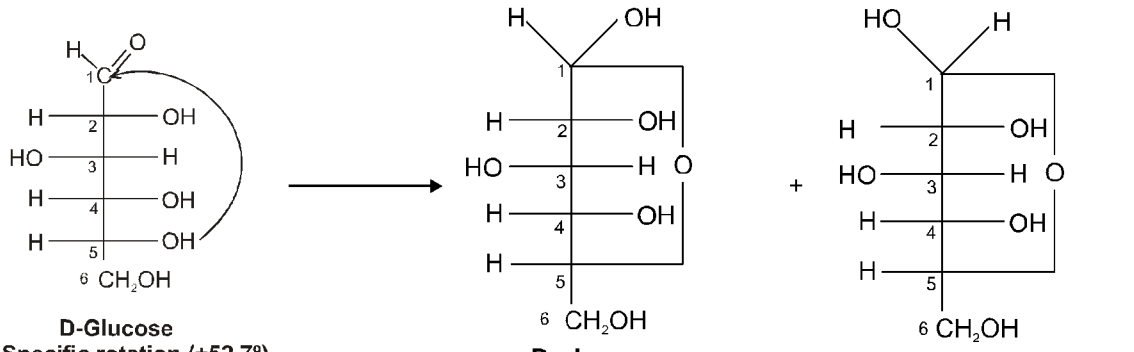
Reducing and non Reducing properties of (Sugars) :

	(I) Reducing sugars	**(II) Non Reducing sugars**
1.	Reduces Tollen's reagent, Fehling's solution & Benedicts's solution.	Don't reduce Tollen's, Fehling's & Benedict's solution.
2.	Should have atleast one hemiacetal or hemiketal functional group.	Should have acetal linkage.
Ex.	All Mono and Oligosaccharides except Sucrose	Ex. All Polysaccharides and few Oligosaccharides (Ex. Sucrose)

Structure of Glucose :

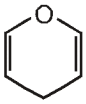
Open chain structure (Fisher projection) and Cyclic structure (Haworth projection) :

(1) Despite having aldehyde group , glucose does not give schiff's test & it does not form the hydrogen sulphite (bisulphite) addition product with $NaHSO_3$.
(2) The pentaacetate of glucose does not react with hydroxylamine indicating the absense of free –CHO group.
This behaviour could not be explained by open chain structure. It was proposed that one of –OH group add to CHO group , forms a cyclic structure.
These two cyclic hemiacetal form of glucose differs only in configuration of the hydroxyl group at C1, called anomeric carbon. Such isomers i.e. α-form & β-form, are called anomers.



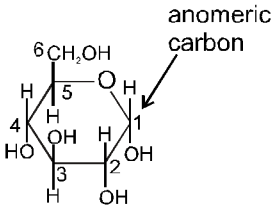
Note : β-form of D-glucose is more stable than α-D-Glucose.
Haworth projection

The six membered cyclic structure of glucose is called pyranose structure (α- or β-), in analogy with pyran. and five membered cyclic structure of monosaccharides is called furanose structure (α or β) in analogy with furan

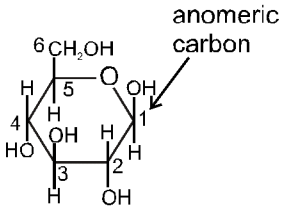


Pyran

Furan

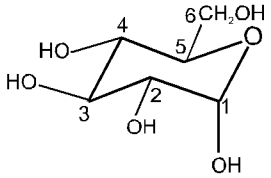


Specific rotation (112°)
α-D-glucose
α-D-glucopyronose

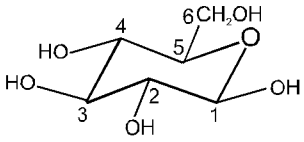


Specific rotation (19°)
β-D-glucose
β-D-glucopyranose

α and β-Glucose are anomer's of each other (i.e change in configuration of 1st carbon atom only)



α-D Glucose

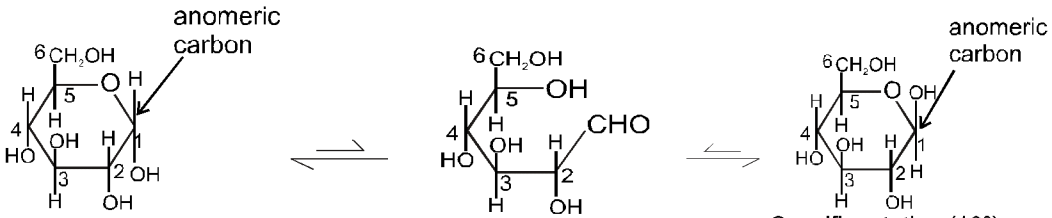


β-D-Glucose

4.1 Properties of Anomers : Mutarotation

When one of the pure glucose anomers dissolve in water, an intersting change in the specific rotation is observed. When the α-anomer dissolves, its specific rotation gradually decreases from an initial value of +112° to +52.7°. When the pure β anomer dissolves, its specific rotation gradually increases from +19° to the same value of +52.7°. **This change (mutation) in the specific rotation is called mutarotation**. What is happening to each solution ?

Initially solution with only one anomeric form, undergoes equilibrium to the same mixture of α-and β-forms. The open chain form is in intermediate in the process of equilibrium. For mutarotation atleast one hemiacetal group must be present in the sugar therefore **all reducing sugars will mutarotate.**



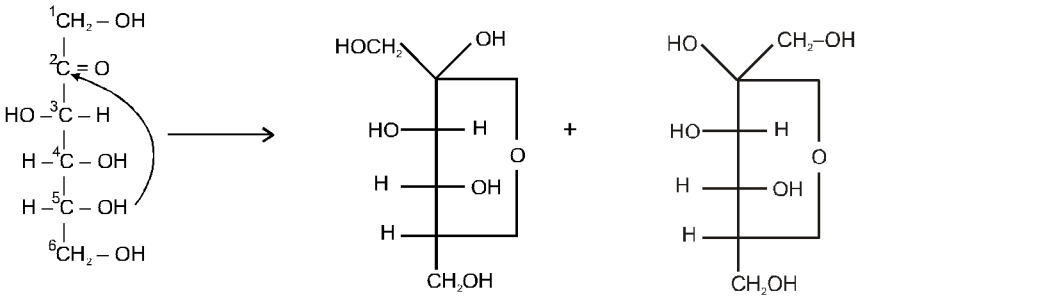
Specific rotation (112°)
α-D-glucose
α-D-glucopyranose

Open chain form
of -D-glucose
α specific (+ 52.7°)

Specific rotation (19°)
β-D-glucose
β-D-glucopyranose

Structure of fructose :

It also exist in two cyclic forms which are obtained by the addition of –OH at C5 to the (>C=0) group.



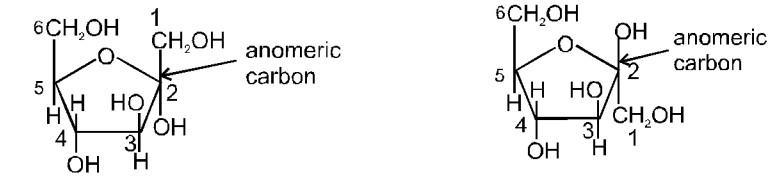
D-fructose
Specific Rotation (– 92°)

α-D-fructose
Specific Rotation (– 21°)

β-D-fructose
Specific Rotation (– 133°)

Haworth projection : Fructofuranose

The five membered ring & is named as furanose with analogy to the compound furan.



Specific rotation (-21°)
α-fructose
α-fructofuranose

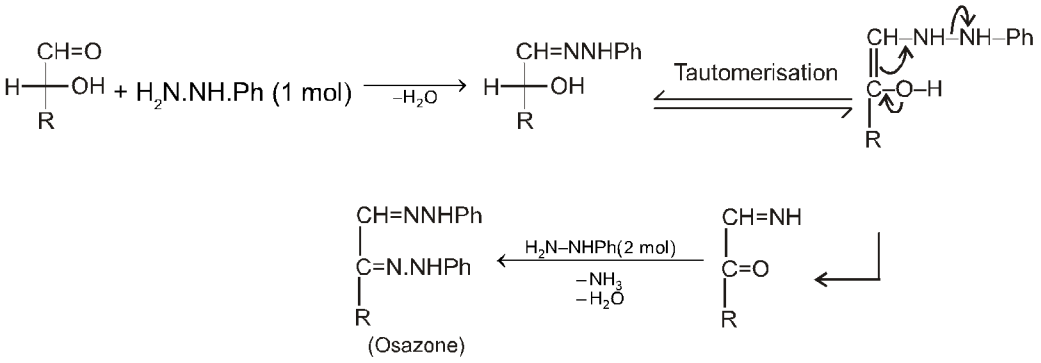
Specific rotation (-133°)
β-fructose
β-fructofuranose

Some Highlights :

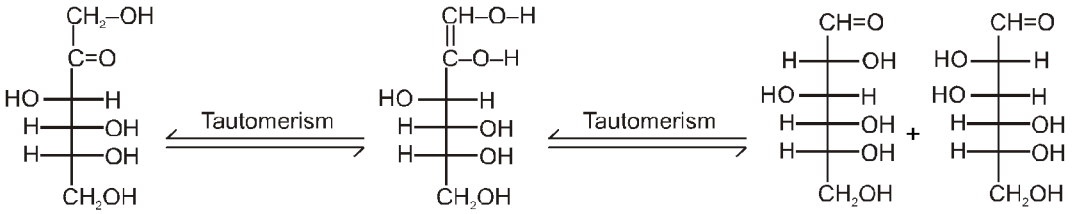
1. Since glucose & fructose (Aldoses/Ketoses) reacts with HCN, H_2NOH, H_2NNHPh which indicates the presence of carbonyl group but they don't react with DNP, $NaHSO_3$ & Schiff's reagent (weak reagents) therefore we can conclude that carbonyl group is not free, but remains in the form of cyclic structures.

2. In the formation of osazone, C_1 & C_2 are only involved so glucose, fructose and C-2 epimers (Glucose & Mannose, Threose and Erythrose) give same osazone.

3. Osazone are crystalline solid having sharp melting point so used for identifying the carbohydrates. In the osazone formation three molecules of NH_2NHPh is overall consumed out of which two molecule react with nucleophilic addition/elimination reaction forming hydrazone whereas one molecule undergoes redox reaction.



3. Both glucose and fructose gives test with Tollen's reagent, Fehling's solution and Benedict's solution because in basic medium, ketoses remains in the form of dynamic equilibrium with Aldoses (C-2 epimers) by the process of **tautomerisation/enediol rearrangement as below**.



D-fructose

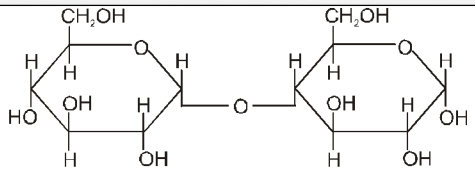
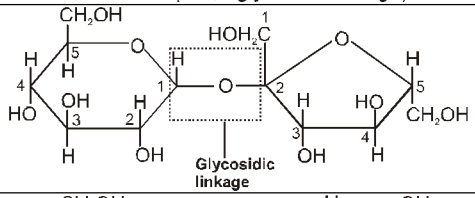
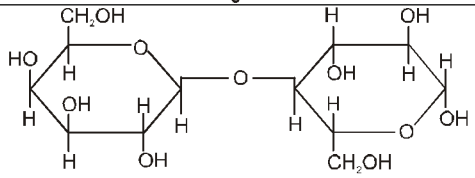
[Enediol of D-fructose
Enediol of D-Glucose
Enediol of D-Mannose]

Intermediate

D-Glucose D-Mannose

4. Only Br_2/H_2O is used for the identification of Aldoses & Ketoses. (Mild oxidising agent like bromine water (Neutral) Oxidises only aldehydic group).
5. Oxidation with HNO_3, gives information that one primary alcohol is present in aldoses and two primary alcohols are present in ketoses.
6. Reduction product with Na/Hg and H_2O gives only one alcohol with aldoses and two alcohols with ketoses (C-2 epimers)
7. Reduction product with Red P & HI, gives n-Hexane which indicates that all the six carbon atoms are linearly arranged.

DISACCHARIDE

Disaccharide	Structure	Monomeric unit linkage	Properties
Maltose	 Maltose (α-1,4-glycosidic linkage)	D (+) Glucose + D(+) Glucose (α-1, 4-glycosidic linkage)	Produced by action of malt on starch. Undergoes mutarotation.
Sucrose	 Glycosidic linkage	α-D-glucose + β-D-fructose (α-1,β-2 glycosidic linkage)	White crystalline solid , soluble in water. dextrorotatory specific rotation + 66.5°
Lactose	 Lactose (β-1,4-glucosidic linkage)	D(+) Glucose + D (+) galactose (β-1,4- glycosidic linkage)	dextrorotatory

POLYSACCHARIDES

Polysaccharides : It contains large number of monosaccharide units joined together by glycosidic linkage (acetal bond). They are food storage or structural material.

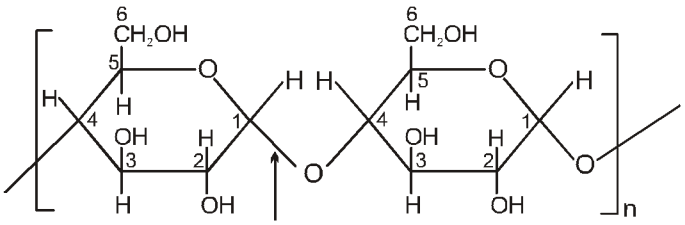
(i) Starch, $(C_6H_{10}O_5)_n$

(i) Starch is the main contributor of carbohydrates in our diet. It exists exclusively in plants, stored in the seeds, roots, and fibres as food reserve. Example rice, potato.

(ii) Starch is actually a mixture of two structurally different polysaccharides, Amylose (15-20%) and Amylopectin (80-85%).

(iii) When starch is heated with hot water, it can be separated into its components. The part that is soluble

in water is amylose and remaining fraction is amylopectin.
(iv) Both amylose and amylopectin are composed of D-glucose units.
(v) The **amylose** molecule is made up of D-glucose unit joined by α-glycosidic linkages between C-1 of one glucose unit and C-4 of the next glucose unit. The number of D-glucose units in amylose range from 60-300.

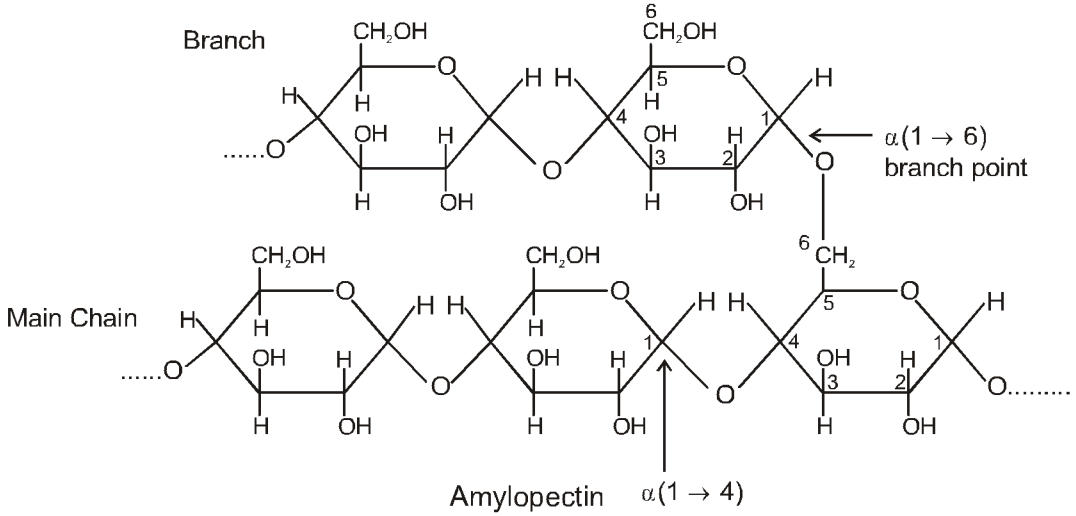


α(1 → 4) Glucosidic linkage

Amylose

1,4-Glycosidic linkage of α-D-Glucose

(vi) **Amylopectin** has a branched-chain structure. It is composed of chains of 25 to 30 D-glucose units joined by α-glycosidic linkages between C-1 to one glucose unit and C-4 of the next glucose unit. These chains are in turn connected to each other by 1, 6-linkages.



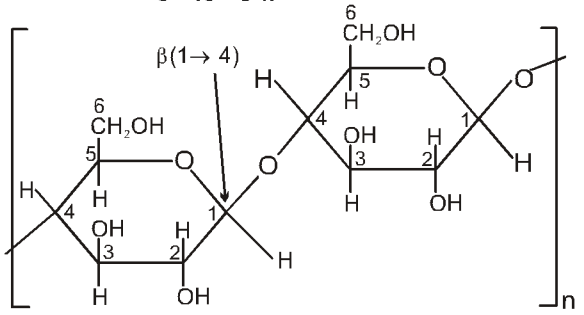
1,4-Glycosidic linkage of α-D-Glucose with Branching at 1, 6

α-amylose soluble in water, and the solution gives a blue colour with iodine. Amylopectin is insoluable in water, is stable in contact with water, and gives a violet colour with iodine.

(ii) Glycogen (Animal Storage)

It is also like amylopectin but branching will take place after every 5 to 6 glucose unit. (highly branched)

(iii) Cellulose, $(C_6H_{10}O_5)_n$



Cellulose

1,4-Glycosidic linkage of β-D-Glucose

1. Cellulose is linear chain natural polymers of β-D glucose units joined by 1, 4-glycosidic linkage (Natural linear polymers).
2. Cellulose is the main structural material of tree and other plants. Wood is 50% cellulose, while cotton wool is almost pure cellulose.
3. Artificial silk, rayon, is used collectively to cover all synthetic or manufactured fibres from cellulose.
4. The nitrates are prepared by the reaction of cellulose with a mixture of nitric and sulphuric acids, and the degree of 'nitration' depends on the concentrations of the acids and the time of the reaction. Cellulose trinitrate (12.2 – 13.2%N) is known as **gun-cotton** and is used in the manufacture of blasting explosives and smokeless powders.

Starch : **(Plant Storage, Polymer of α–D-glucose),**
Glycogen : **(Animal Storage, Polymer of α–D-glucose),**
Cellulose : **(Plant Skeleton, Polymer of β–D-glucose)**

• Points to remember in Polymers

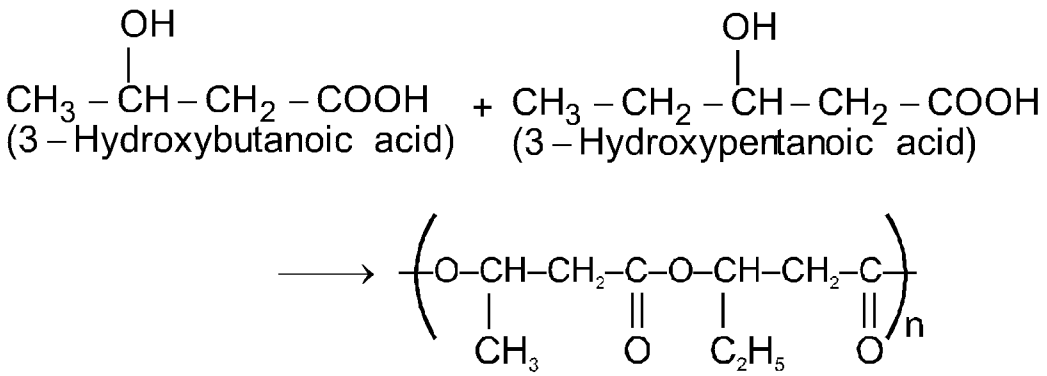
Biodegradable Polymers :

A large number of polymers are quite resistant to the environmental degradation processes and are thus responsible for the accumulation of polymeric soild waste materials. These soild wastes cause acute environmental problems and remain undegraded for quite a long time. In view of the general awareness and concern for the problems created by the polymeric soild wastes, certain new biodegradable synthetic polymers have been designed and developed. These polymers contain functional groups similar to the functional groups present in biopolymers.

Aliphatic polyesters are one of the important classes of biodegradable poylmers. Some examples are given below :

(a) Poly β-hydroxybutyrate – co–β-hydroxy valerate (PHBV) :

It is obtained by the copolymerisation of 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.

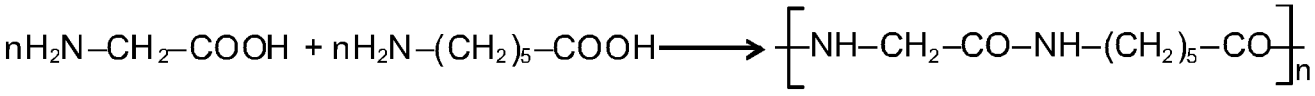


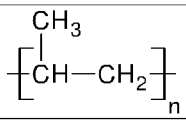
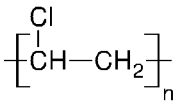
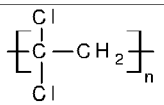
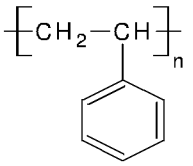
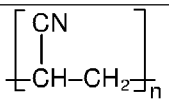
PHBV

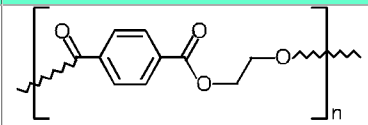
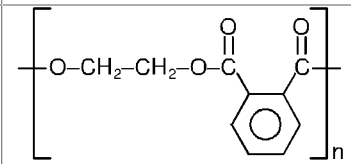
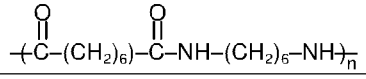
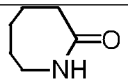
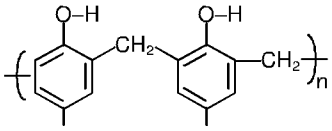
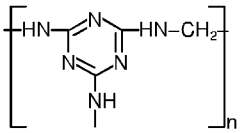
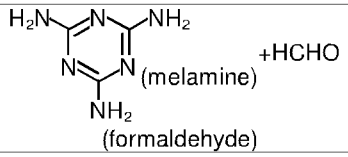
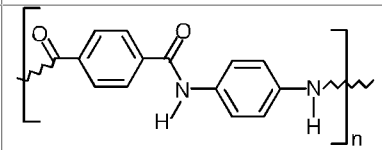
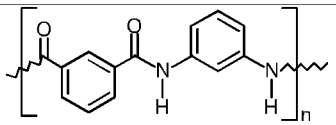
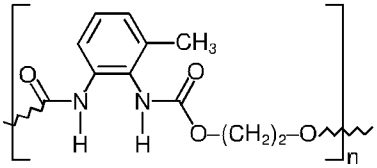
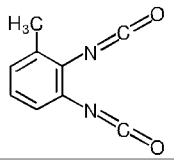
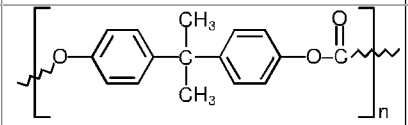
PHBV is used in speciality packaging, orthopaedic devices and in controlled release of drugs. PHBV undergoes bacterial degradation in the environment.

(b) Nylon–2–nylon–6 :

It is an alternating polyamide copolymer of glycine ($H_2N–CH_2–COOH$) and amino caproic acid ($H_2N(CH_2)_5COOH$) and it is also biodegradable polymer.

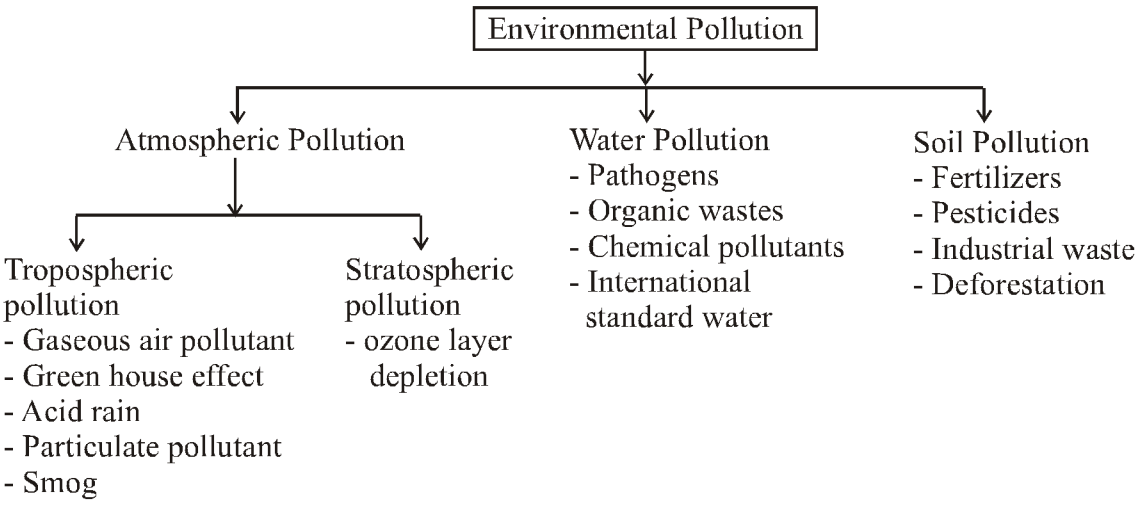


Some common addition polymers/chain growth polymer				
S. No.	Name(s)	Formula	Monomer	Uses
1.	Polyethylene (low density (LDPE))	$-(CH_2\text{-}CH_2)_n-$	$CH_2=CH_2$ (ethylene)	Film wrap, Plastic Bags
2.	Polyethylene (high density (HDPE))	$-(CH_2\text{-}CH_2)_n-$	$CH_2=CH_2$ (ethylene)	Electrical insulation bottles, toys
3.	Polypropylene (PP) different grades		$CH_2=CHCH_3$ (propylene)	Manufacture of ropes, toys, pipes, fibres etc.
4.	Poly vinyl chloride (PVC)		$CH_2=CHCl$ (vinyl chloride)	Manufacture of rain coats, hand bags, vinyl flooring, water Pipes etc.
5.	Poly vinylidene chloride (Saran A)		$CH_2=CCl_2$ (vinylidene chloride)	Seat covers, films & fibers
6.	Polystyrene (Styron)		$CH_2=CHC_6H_5$ (styrene)	As insulator, wrapping material, manufactures of toys, radio and Television cabinets
7.	Polyacrylonitrile (PAN, Orlon, Acrilan)		$CH_2=CHCN$ (acrylonitrile)	Rugs, Blankets clothing
8.	Polytetrafluoroethylene (PTFE, Teflon)	$-(CF_2\text{-}CF_2)_n-$	$CF_2=CF_2$ (tetrafluoroethylene)	Non-stick surfaces electrical insulation
9.	Poly methyl methacrylate (PMMA, Lucite, Plexiglas, perspex)	$-[CH_2C(CH_3)CO_2CH_3]_n-$	$CH_2=C(CH_3)CO_2CH_3$ (methylmethacrylate)	Lighting covers, signs skylights
10.	Poly vinyl acetate (PVAc)	$-(CH_2\text{-}CHOCOCH_3)_n-$	$CH_2=CHOCOCH_3$ (vinyl acetate)	Latex paints, Adhesives
11.	Natural Rubber	$-[CH_2\text{-}CH=C(CH_3)\text{-}CH_2]_n-$ (cis)	$CH_2=CH-C(CH_3)=CH_2$ (isoprene)	Requires vulcanization for practical use
12.	Neoprene	$-[CH_2\text{-}CH=CCl\text{-}CH_2]_n-$	$CH_2=CH\text{-}CCl=CH_2$ (chloroprene)	Synthetic rubber, oil resistant seal, gaskets, hoses & conveyor belts
13.	SBR styrene butadiene rubber (Buna-S)	$\text{-}[CH_2\text{-}CH(Ph)\text{-}CH_2\text{-}CH=CH\text{-}CH_2]\text{-}$	$H_2C=CHC_6H_5$ and $H_2C=CH\text{-}CH=CH_2$	Tyres, floortiles, foot wear & cable insulation
14.	Nitrile Rubber (Buna-N)	$\text{-}[CH_2\text{-}CH(CN)\text{-}CH_2\text{-}CH=CH\text{-}CH_2]\text{-}$	$H_2C=CHCN$ and $H_2C=CH\text{-}CH=CH_2$	Making oil seals, tank lining and hoses

Some condensation polymers/step growth polymers				
S.	Name(s)	Formula	Monomer	Uses
1.	Polyester/Dacron/ Terylene/Mylar		$HO_2C\ C_6H_4\ CO_2H$ (Terephthalic acid) $HO–CH_2CH_2–OH$ Ethylene glycol	Fabric, Tyrecord
2.	Glyptal or Alkyds resin		$HO_2C–C_6H_4–CO_2H$ (Phthalic acid) $HO–CH_2CH_2–OH$	Paints and Lacquers
3.	Polyamide (Nylon 6,6)	$\sim[CO(CH_2)_4CO–NH(CH_2)_6NH]_n\sim$	$HO_2C–(CH_2)_4–CO_2H$ $H_2N–(CH_2)_6–NH_2$	Parachutes & Clothing
4.	Nylon 6,10		$HOOC–(CH_2)_8–COOH$ $H_2N–(CH_2)_6–NH_2$	
5.	Polyamide Nylon 6, Perlon-L	$\sim[CO(CH_2)_5NH]_n\sim$		Rope & Tyrecord
6.	Bakelite		PhOH + HCHO in (excess)	Electrical Switch, combs, Handle of Utensils, computer discs and Bowling Balls
7.	Urea-formaldehyle resin	$(–NH–CO–NH–CH_2–)_n$	$H_2N–CO–NH_2$ (Urea) HCHO (Formaldehyde)	Making unbreakable cups and laminated sheets.
8.	Melamine formaldehyde resin		 +HCHO (melamine) (formaldehyde)	Unbreakabl e crockery
9.	Polyamide Kevlar		Para $HO_2C–C_6H_4–CO_2H$	Tyre
10.	Polyamide Nomex		Meta $HO_2C–C_6H_4–CO_2H$ Meta $H_2N–C_6H_4–NH_2$	
11.	Polyurethane Spandex		$HOCH_2CH_2OH$ 	Foams, Shoes, Automobile seats and components
12.	Polycarbonate Lexan		$(HO–C_6H_4–)_2C(CH_3)_2$ (Bisphenol A) $X_2C=O$ (X = OCH_3 or Cl)	Bike helmet, goggles, bullet proof glass

Environmental Chemistry & Chemistry in Everyday Life

CLASSIFICATION OF ENVIRONMENTAL POLLUTION :



ENVIRONMENTAL POLLUTION :

It is the effect of undesirable changes by humans in our surroundings that have harmful effects on plants, animals and human beings.

Environmental Pollutants : Agents

(i) Chemical agents

- Gases - Pesticides - Particulates

(ii) Physical agents

- Heat - Noise - Radiation

(iii) Biological agents

- Micro-organisms

COMMON TERMS

(i) Pollutants :

Substance which causes pollution is known pollutant & produced by nature or human.

- Pollutants can be solid, liquid or gaseous substances.
- Pollutants can be bio-degradable or non-biodegradable.

(ii) Contaminant :

Substance that doesn't occur in nature but is introduced by human activity into atmosphere affecting its composition are called contaminant. Contaminant may be pollutant, if it is harmful to nature. It may not be harmful too.

(iii) Source :

Site from which pollutant originate are called sources.

(iv) Sink :

Material or medium that consumes pollutant are called sink.

- Marble wall is the sink for atmospheric sulphuric acid.
- Trees are sink for carbon dioxide.
- Ocean is sink for carbondioxide and other water soluble gases.
- Ground water is sink for pesticides used in agriculture.

(v) Receptors : Anything that is affected by pollutants is called Receptors.

(vi) Threshold Limits :

It is permissible limit of a pollutants in environment that will not cause adverse affect.

AIR POLLUTION :

Condition in which chemical substance in air gradually builds up their concentration in atmosphere to such an extent that it is undesirable and produce harmful effects on nature.

PRIMARY & SECONDARY AIR POLLUTANT :

(i) Primary pollutant :

These pollutants are emitted directly from sources and remain as such after they enter into environment. **e.g.** Ash, Smoke, Dust, Sulphur oxide, Carbon monoxide, Ammonia, Hydrocarbons and radioactive substances etc.

(ii) Secondary Pollutant :

These pollutants are formed in atmosphere by chemical interaction among primary pollutants and atmospheric constituents.

e.g. Sulphur trioxide, Nitrogen dioxide, Ozone, PAN, H_2SO_4 etc.

$$O_2(g) \xrightarrow{UV} O(g) + O(g)$$

$$O(g) + O_2(g) \underset{}{\overset{UV}{\rightleftharpoons}} O_3(g)$$

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

GASEOUS AIR POLLUTANTS :

(i) Sulphur dioxide

Sources : Most of SO_2 comes from volcanic eruption rest is produced when sulphur containing fossil fuel is burnt.

$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

Effect :

It has been reported that even a low concentration of SO_2 causes respiratory disease.

e.g. : Asthma, Bronchitis in human beings.

- It also causes irritation to the eyes, resulting in tears and redness.
- It also causes deterioration of fabrics, paper and leather.
- High concentration of SO_2 leads to stiffness of flower buds which eventually fall off from plants.

Sinks : Oxides of sulphur is converted to sulphuric acid, which then combines with limestones materials.

(ii) Oxides of Nitrogen :

Source :

NO and NO_2 are atmospheric pollutants. Dinitrogen and dioxygen are the main constituents of air. These gases do not react with each other at a normal temperature. At high altitudes when lighting strikes, they combine to form oxides of nitrogen.

In an automobile engine, when fossil fuel is burnt, dinitrogen and dioxygen combine to yield significant quantities of NO and NO_2.

Effect :

NO like CO binds with haemoglobin and decreases oxygen transport efficient of blood. High concentration of NO_2 damage leaves of plants and retard rate of photosynthesis.

Sink :

Mainly natural process acts as sink for oxide of nitrogen. These oxides are broken into nitrogen and oxygen.

(iii) Hydrocarbons :

Hydrocarbon are composed of hydrogen and carbon only and are formed by incomplete combustion of fuel used in automobiles. Hydrocarbons are carcinogenic it they cause cancer.

They harm plants by causing ageing, breakdown of tissues and shedding of leaves, flowers and twigs.

(iii) Carbon monoxides :

Source :

- It is mainly produced by incomplete combustion of carbon.
- It is produced from forest fire and volcanic activity.
- Also formed by bacterial decay or organic matter in CO.
- Released into air by automobile exhaust.

Effect :

It is poisonous. It binds to haemoglobin to form carboxyhaemoglobin, which is about 300 times more stable than the oxygen haemoglobin complex.

In blood, when concentration of carboxyhaemoglobin reaches about 3-4 percent, oxygen carrying capacity of blood is greatly reduced and results into Headache, Weak eye sight, Nervousness and Cardiovascular disorder.

Sink :

Soil is a major sink for CO. Microorganism in soil removed CO.

(iv) Carbon dioxide :

Source :

- It is released into atmosphere by respiration.
- Also by burning of fossil fuels for energy and also emitted during volcanic eruptions.

Effect :

Increased amount of CO_2 in the air is mainly responsible for global warming.

Sink :

- Ocean is the major sink for CO_2
- Also it is removed by green plants.

GREEN HOUSE EFFECT :

In cold places flowers, vegetables and fruits are grown in glass covered areas called green house. In a green house, visible light passes through the transparent glass and heats up the soil and the plants. The warm soil and plants emit infrared radiations.

Since glass is opaque to infrared (heat) radiations, it partly reflects and partly absorbs these radiations. This mechanism keep the energy of the sun trapped in the green house.

GLOBAL WARMING AND GREEN HOUSE EFFECT OF EARTH :

About 75% of the solar energy reaching the earth is absorbed by the earth's surface, which increase its temperature. The rest of the heat radiates back to the atmosphere.

Some of the heat is trapped by gases such as Carbon dioxide, Methane, Ozone, CFCs and Water vapour in the atmospheres. Thus, they add to the heating of the atmosphere. This cause global warming. This is called natural green house effect.

Normally rain water has a pH of 5.6 due to the presence of H^+ ions formed by the reactions of rain water with CO_2 present in the atmosphere. When the pH of the rain water drops below 5.6, it is called acid rain.

$$H_2O\ (l) + CO_2 \rightleftharpoons H_2CO_3\ (aq)$$

$$H_2CO_3\ (aq.) \rightleftharpoons H^+\ (aq) + HCO_3^-\ (aq)$$

SMOG(Smoke + Fog) :

Two Types

Classical/London smog/Sulphurous smog

- First seen in London 1905 and resulted in loss of human life.
- It occurs in cold humid climate especially in morning hours in winters.
- It is a mixture of smoke, fog, SO_2.
- Source is industrial and household fuels.
- Due to SO_2, it has reducing mixture and it is also called as reducing smog.

Photochemical smog/Los Angles smog

- First seen in Los Angles 1943.
- It occurs in warm, dry and sunny climate. Formed around midday of summer when solar radiations is high.
- It occurs due to photochemical reaction between unsaturated hydrocarbon and nitrogen oxide produced by automobiles and factories in presence of sun light.

Nitrogen oxides + hydrocarbon $\xrightarrow{\text{UV-radiation of sunlight}}$ PAN + Ozone

- Main constituents are ozone and oxides of nitrogen.
- Due to O_3 and NO_2 it is oxidising smog.

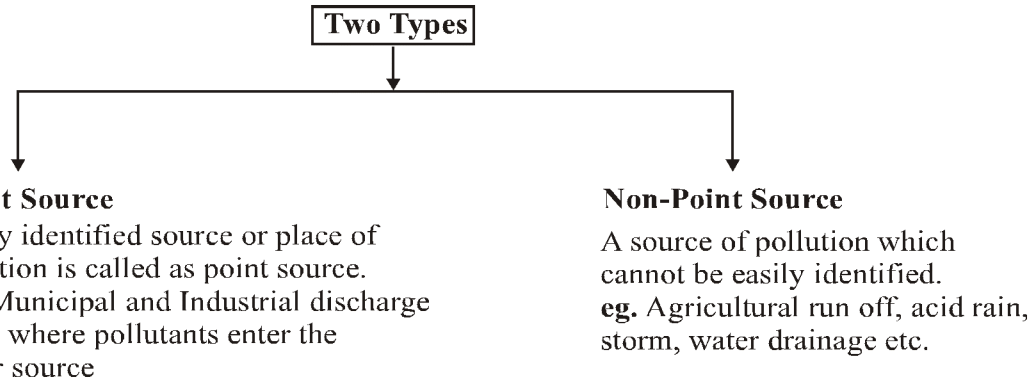
STRATOSPHERIC POLLUTION :

The upper stratosphere consists of considerable amount of ozone which protects us from the harmful ultraviolet(UV) radiations coming from the sun. These radiations cause skin cancer in humans.(Melanoma)

WATER POLLUTION :

Source

Pollution of water originates from human activities. Through different plants, pollution reaches surface or ground water.



INDUSTRIAL EFFLUENTS (or industrial discharges)

Industries usually discharge waste water into ponds, lakes and rivers. Industrial waste water contains heavy metals (mercury, lead, copper, arsenic and cadmium), inorganic pollutants (acids, alkalies and bleaching liquors), organic pollutants (phenol, naphtha, proteins, aromatic compounds, cellulose fibres etc.) Industrial effluents are the most hazardous pollutants on land and water.

THERMAL POLLUTION

(i) Hot water is produced by many industries , power generation plants and thermal power plants.

(ii) Thermal pollution is caused by addition of hot water effluents in water bodies, it bring about rise in water temperature.

(iii) Warmer water contains less oxygen. Therefore, there is decrease in the rate of decomposition of organic matter.

(iv) In hot water green algae are replaced by less desirable blue green algae.

(v) Many organisms fail to reproduce in hot water e.g. Salmon, Trout.

SOIL POLLUTION

(i) Unfavourable alteration of soil by addition or removal of substances and factors which decrease soil productivity, quality of plant products and ground water is called soil pollution.

(ii) The soil pollutants include pesticides, fertilizers, industrial wastes, salts, radio-nuclids , tin, iron, lead, copper, mercury, aluminium, plastics, paper, glass, broken bottles, discarded food etc.

Types of Soil pollution

Soil pollution is of two main types

(i) Negative soil pollution

(ii) Positive soil pollution

NOISE POLLUTION

(i) Various kinds of undesirable loud sounds, which disturb our environment are called noise pollutants.

(ii) Noise pollution is produced by loud sounds of various machines, loudly played radio, automobiles, thundering of jet planes, loud speakers etc.

(iii) The intensity of noise is measured in decibels (or dB).

(iv) The sound above 80 dB causes noise pollution.

(v) Moderate conversation produces 60 dB sound, loud conversation 70dB.

GREEN CHEMISTRY :

Green chemistry is about utilizing existing knowledge to reduce adverse impact on environment.

ASPECTS COVERED :

(i) Use less hazardous starting material.

(ii) Minimize waste generation.

(iii) Make sure final and by products is not toxic.

(iv) Search for more ecofriendly alternatives.

Fuel obtained from plastic waste has high octane rating. It contains no lead and is known as green fuel. Technology has been developed to produce electricity from the garbage.

Chemistry in Everyday Life

(1) DRUG & THEIR CLASSIFICATION :

Drug's are chemicals of low molecular masses (–100–500 μ). These interact with macromolecular targets and produce a biological response.

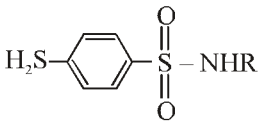
Classification of drugs :

Drugs can be classified in a number of ways as explained below.

(a) *On the basis of pharmacological effect* : For example. Analgesics have pain killing effect, antiseptics kill or arrest the growth of microorganisms.

(b) *On the basis of drug action*: It is based on the action of a drug on a particular biochemical process. For example, all antihistamines inhibit the action of the compound histamine, which causes inflammation in the body. There are various ways in which action of histamine can be blocked.

(c) *On the basis of chemical structure*: It is based on the chemical structure of the drug. For example, sulphonamides have common structural feature as given below.



Structural features of sulphonamides

(d) *On the basis of molecular targets:* Drugs usually interact with biomolecules such as carbohydrates, lipids, proteins and nucleic acids. These are called target molecules or drug targets.

Antagonist and Agonists

Drugs that bind to the receptor site and inhibit its natural function are called **antagonists**. These are useful when blocking of message is required. There are other type of drugs that mimic or copy the natural messenger by switching on the receptor, these are called **agonists**.

1.1 MEDICINES :

Chemotherapy :

The branch of science which deals with the treatment of diseases using suitable chemicals is known as **chemotherapy**.

Classification of Medicines : Medicines are generally classified according to the purpose for which they are used. The different terms thus used along with examples are given below :

(a) Antipyretics :

Chemicals which are used to bring down the body temperature during high fever are called antipyretics. On taking these medicines, the person gets a lot of perspiration. A few well known examples of antipyretics are Aspirin, Phenacetin and Paracetamol,

Novalgin is another well known antipyretic. Quinine has also some antipyretic effect. Paracetamol is preferred over asprin as an antipyretic since asprin gets hydrolysed to salicylic acid in the stomach. The salicylic acid thus produced may cause ulcer in the stomach walls where from the bleeding may take place.

(b) Antimalarials :

Chemical substances which are used to bring down the body temperature during malaria fever are called antimalarials. Originally quinine (an alkloid) was the only drug known to be effective against malaria. Nowadays, a number of synthetic drugs are used for the purpose. These are Chloroquine, Paraquine, Primaquine etc.

(c) Analgesics :

Medicines used for getting relief from pain are called analgesics. These are of two types
(i) Narcotics and (ii) Non-narcotics

(i) Narcotics : Drugs which produce sleep and unconciousness are called narcotics. These are mostly opium products. Opium contains alkaloids such as morphine and codeine which are very effective analgesics. Morphine diacetate (commonly known as Heroin) is the most widely used analgesic.

(ii) Non-narcotics : The most important non-narcotic analgesics are butazolidine and brufen or ibuprofen. Aspirin, Phenacetin, and Novalgin act both as antipyretic as well as analgesic. They are quite effective and give immediate relief from pain and fever.

(d) Tranquilizers :

The chemical substances used to cure mental diseases are called **tranquilizers. These** are used to release mental tension and reduce anxiety. These are the constituents of sleeping pills. They act on higher centres of nervous system. These are also called **psychotherapeutic drugs**. These drugs make the patient passive and help to control their emotional distress or depression. These also help to restore confidence and the patients work with full capacities which they already have.

(i) Hypnotics : These are also known as tranquilizers and are used to reduce mental tension and anxiety. These induce sleep. These are components of sleeping pills (sedatives)

(ii) Non Hypnotics : They reduce tension and anxiety. These do not induce sleep. Equanil is also an important tranquilizers used in depression and hypertension.

Ex. Chlordiazepoxide, Meprobamate.

It may be noted that different terms are used for drugs which are used to cure mental diseases.

(e) Antidepressants : These drugs are given to patients with shattered confidence. These produce a felling of well being and confidence in the person of depressed mood. Therefore, these are also called mood booster drugs. The common examples are Vitalin, Cocaine, Methedrine etc.

(f) Antimicrobials : Diseases in human beings and animals may be caused by a variety of microorganisms such as bacteria. virus, fungi and other pathogens. An antimicrobial tends to destroy/prevent development or inhibit the pathogenic action of microbes such as bacteria (antibacterial drugs), or fungi (antifungal agents), virus (antiviral agents), or other parasites (antiparasitic drugs) selectively.

Type of Antimicrobials :

(ii) Antiseptics and Disinfectants :

(i) Antiseptics : The chemicals which kill or prevent the growth of micro-organisms are called **antiseptics**. These are not harmful to living tissues and can be safely applied on wounds, cuts, ulcers, diseased skin surfaces. These are also used to reduce odours resulting from bacterial decomposition

of the body or in the mouth. They are, therefore mixed with deodrants, face powders and breath purifiers.

(ii) Disinfectants : The chemical substances which are used to kill mircoorganisms but they cannot be applied on living tissues are called **disinfectants**. These are commonly applied to inanimate objects such as floors, instruments, etc.

(g) Anti–fertility drugs :

Chemical substances which are used to check pregnancy in women are called anti-fertility drugs or birth control drugs or oral contraceptives.

All these drugs contain chemicals related to female sex hormones having a steroid ring structure. Most of these contain a combination of an estrogen and a progesterone. For example, a common brand name Enovid E contains norethindrone (a progestin) mestranol or ethinylestradiol monomethyl ether (an estrogen). All such drugs are expected to have side effects and hence should be used only under proper medical advice.

(h) Antihistamines :

The drugs which have been used to fight allergy are called **antihistamines**. These are so called because they check the production of histamines. Thus, antihistamines are widely used for treatment to hay fever, conjunctivitis, nasal discharges, irradation sickness, motion sickness (air, sea, road), nausea in pregnancy and post operative vomiting. The antihistamine drugs which are widely used are diphenhydramine hydrochloride, Cetrizine, Chloropheniramine, Promethazine hydrochloride etc. The structure of some of these are given below :

(i) Antacids : These are used to treat over production of acid in the stomach. Conventional antacids like $NaHCO_3$, $Mg(OH)_2$, $Al(OH)_3$ neutralise the excess acid in the stomach. However, modern antacids like cimetidine (Tegamet) or ranitidine (Zantac), which are infact antihistamines, prevent the release of excess acid in the stomach. This is because histamine stimulates the secretion of pepsin and HCl in the stomach. These drugs prevent the interaction of histamine with the receptors present in the stomach wall.

1.2 CHEMICAL IN FOOD

Chemicals are added to food for (i) their preservation (ii) enhancing their appeal, and (iii) adding nutritive value in them. Main categories of food additives are as follows :

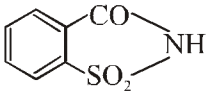
(i) Food colours
(ii) Flavours and Sweeteners
(iii) Fat emulsifiers and Stabillising agents
(iv) Flour improvers - Antistaling agents and Bleaches
(v) Antioxidants
(vi) Preservatives
(vii) Nutritional supplements such as minerals, vitamins and amino acids.

Except for chemicals of category (vii), none of the above additives have nutritive value. These are added either to increase the shelf life of stored food or for cosmetic purposes. In this section we will discuss only sweeteners and food preservatives.

(I) Artificial Sweeteners : Natural sweeteners, **e.g.,** sucrose add to calorie intake and therefore many people prefer to use artificial sweeteners. Some important artificial sweeteners are discussed below.

(a) Saccharin : Ortho-sulphobenzimide, also called saccharin, is the first popular artificial sweetening agent. It is about 550 times as sweet as cane sugar. It is excreted from the body in urine unchanged. It appears to be entirely inert and harmless when taken. Its use is of great value to

diabetic persons and people who need to control intake of calories.



Saccharin

(b) Aspartame : It is the most successful and widely used artificial sweetener. It is roughly 100 times as sweet as can sugar. It is methyl ester of dipeptide formed from aspartic acid and phenylalanine. Use of aspartame is limited to cold foods and soft drinks because it unstable at cooking temperature.

(c) Alitame : It is high potency sweetener. It is nearly 2000 times as sweet as cane sugar. Although it is more stable than aspartame, the control of sweetness of food is difficult while using it.

(d) Sucrolose : It is trichloro derivative of sucrose. Its appearance and taste are like sugar. It is stable at cooking temperature. It does not provide calories.

(II) Food preservations : Food preservatives prevent spoilage of food due to microbial growth. The most commonly used food preservative include table salt, sugar vegetable oils and sodium benzoate, C_6H_5COONa. Sodium benzoate is used in limited quantities and is metabolised in the body.

2. SOAPS & DETERGENTS

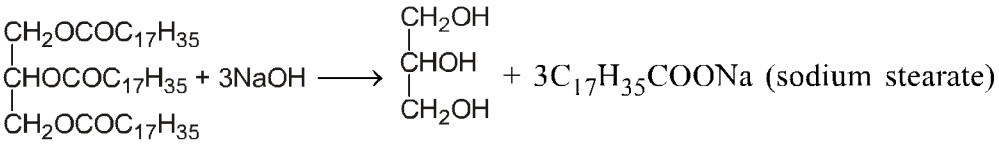
Introduction :

(a) Sodium salts of higher fatty acids like stearic acid, oleic acid, palmitic acid etc. are called hard soaps and the potassium salts of these fatty acids are called soft soaps.

(b) Soaps are prepared by hydrolysis of higher fatty acids.

(c) A higher proportion of salts of saturated acids (palmitic, stearic etc.) gives Hard soaps, while a higher proportion of salts of unsaturated acids (oleic acid) yields soft soaps.

(d) Alkaline hydrolysis of oils or fats by NaOH or KOH gives glycerol and sodium or potassium salt of the fatty acid. This reaction is known as **saponification**.



(I) SOAPS

Types of soaps :

(a) Hard soaps – These are obtained from cheap oils and fats using sodium hydroxide. These contains free alkali and are used for washing purposes.

(b) Soft soaps – These are obtained from good oils using potassium hydroxide. These do not contain free alkali and are used as toilet soaps, shaving cream, in shaving sticks and shampoo.

(c) Transparent soaps : These are formed by dissolving toilet soaps in alcohol and evoporating the filtrate. They contain glycerol.

(d) **Medicated soaps :** Toilet soaps containing some medicinal important substance are called medicated soaps.

(e) **Metallic soaps :** These are soaps of metals other than sodium and potassium.

(II) DETERGENTS

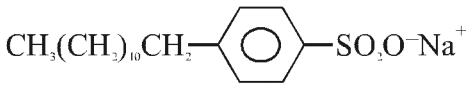
The synthetic detergents, soapless, deter-gents, soapless soaps or syndets are substitutes of soaps. Unlike soaps, they are derived from purely synthetic chemicals rather than from chemicals obtained from natural sources like oils/fats. However, like soaps they contain both hydrophilic (water-soluble) and hydrophobic (oil-soluble) parts. Synthetic detergents are mainly of two types.

(a) **Sodium alkyl sulphates :** These are the half esters of an inorganic acid (H_2SO_4) rather than an organic acid and a higher primary alcohol, *e.g.* lauryl acohol ($C_{12}H_{25}OH$). Sodium lauryl sulphate is the most important detergent of this type.

Ex. $CH_3(CH_2)_{10}CH_2OSO_2O^-Na^+$

sodium lauryl sulphate

(b) **Sodium alkyl aryl sulphates :** These are the sodium salts of long chain (linear) alkyl substituted benzene sulphonic acids and are most widely used. The most common is *sodium dodecylbenzene sulphate.*

Ex. 

Since in these detergents, the active portion is negatively charged, these are commonly known as anionic detergents.

(c) **Cleansing Action of Detergents :** Same as soaps detergent also contain two parts – one is long chain hydrocarbon part and other is ionic part. Hydrocarbon part is soluble in oils or greasy materials and insoluble in water while the ionic part is insoluble in oily or greasy substances and soluble in water. Detergents form colloidal solution in water, the hydrocarbon part of which forms big clusters with dirt particles and the ionic part goes into water. In this way on washing with water the whole part washes away with water.

DYES

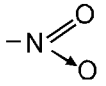
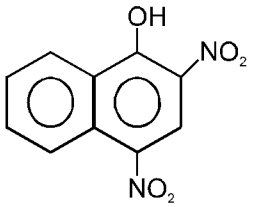
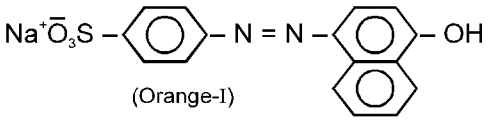
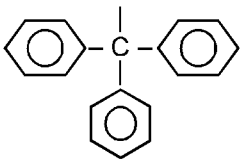
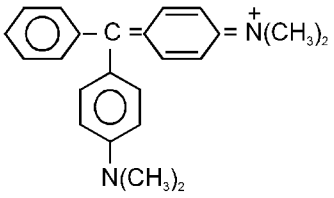
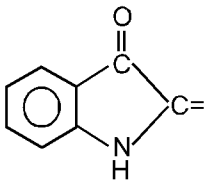
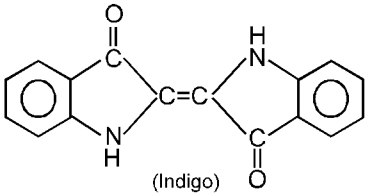
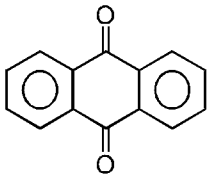
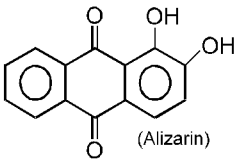
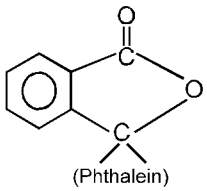
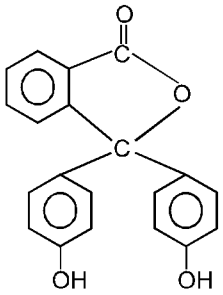
Dyes are chemical substances used to impart colours to the textiles, foodstuffs, silk, wool and other objects. The discovery of dyes has provided glamour in our modern life. All coloured substances are, however, not dyes. A substance can act as a dye if it satisfies the following conditions :

(i) It must have a suitable colour.

(ii) It must be able to fix itself permanently to the material being dyed.

(iii) When fixed, it must be fast to light and washing.

(iv) It must be resistant to the action of water, soap, acids and alkalies or other solvents used in dry cleaning etc.

	Type	Structural unit	Example
1.	Nitro dyes		 (Martius yellow)
2.	Azo dyes	– N=N – (Azo)	 (Orange-I)
3.	Triphenylmethane dyes	 (Triphenylmethane)	 (Malachite green)
4.	Indigoid dyes	 (Indigoid)	 (Indigo)
5.	Anthraquinone dyes	 (Anthraquinone)	 (Alizarin)
6.	Phthalein dyes	 (Phthalein)	 (Phenolphthalein)